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Chapter 1. Manuscript frames 1–139. Catalogue snapshot: 5 October 2026.

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# Chapter 1

## Gelfand Seminar

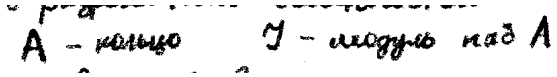
Diplomatic translation. Source frames 1–139.

GELFAND-P000001

Monday, 21/IX. Cartesian and Boolean products of groups.

“Hironaka: on resolution of singularities.” Gelfand

$A$ is a ring, $\mathcal{J}$ is a module over $A$.

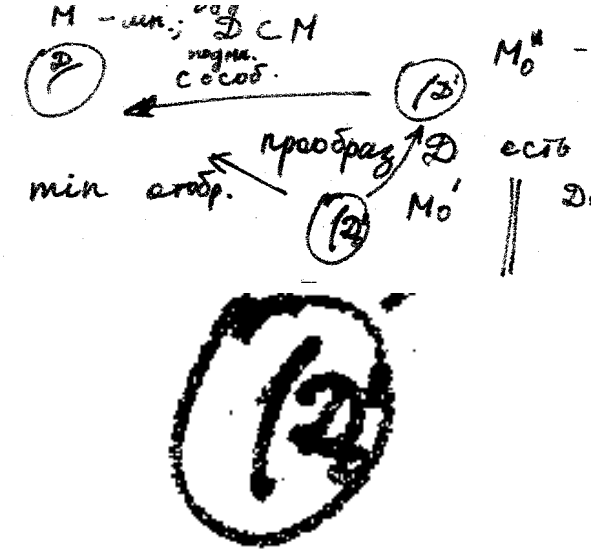


Original source details: GELFAND-P000001

**Uncertain reading.** The letter for the module and ideal is rendered as $\mathcal{J}$; $\mathcal{I}$ or $J$ are possible.

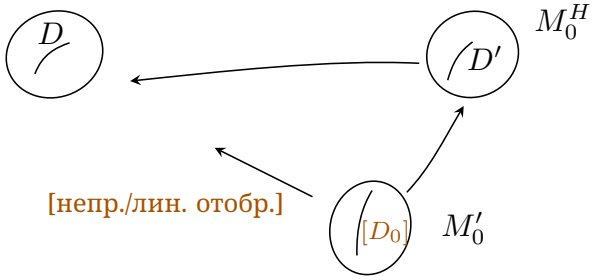
Blow-up — ? $M$ is a manifold; $D \subset M$. [Beneath $D$:] a subset with singularities.

$M_0^H$ is the blow-up. Blowing-up. The inverse image of $D$ is $D'$, a manifold of dimension $n-1$. $D_0$ is $(n-1)$-dimensional.



Original source details: GELFAND-P000001

**Diagram and typed description.** Three enclosed regions: $D$ at left, $D'$ and $M_0^H$ at upper right, and a region labelled $M_0'$ below. The upper and lower oblique arrows point left; a curved arrow rises from the lower region toward the upper right one. The lower-left arrow’s caption reads непр. / лин. отобр. [continuous/linear map]; the lower interior sign reads $D_0$ / $D$, with overlapping strokes. Some curve endings remain indistinct.




Typographic recreation; square brackets mark uncertain readings.



Original source details: GELFAND-P000001

**Uncertain reading.** The upper index on $M_0$ resembles $H$; the lower region’s label resembles $M_0'$.

The construction is Grothendieck’s. $A$ is a ring, $\mathcal{J}$ is a module. Example. $A$ is a ring of functions; $\mathcal{J}$ is the ideal of functions taking the value $0$ on $D$.

One obtains a sheaf of rings and a sheaf of modules. $A$ is a ring, $\mathcal{J}$ is an ideal. Construct $X$, $O_X$ a sheaf of rings, $\mathcal{J}_X$ a sheaf of ideals, so that

$$(A, \mathcal{J}) \longleftarrow (X, O_X, \mathcal{J}_X).$$

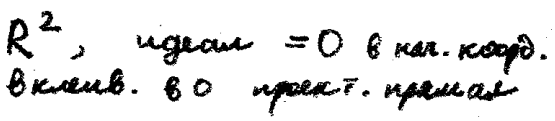
[Beneath the arrow:] linear map.

$(A, \mathcal{J})$ is a sheaf generated by $1$ ideal.

$\mathcal{J}_X$ is invertible, i.e. an ideal of the ring: $\exists f \in \mathcal{J}_X$ such that $O_X f = \mathcal{J}_X$. [Above the formula, an arrow to $\exists$:] generator.

(The ideal is principal if $\mathcal{J}$ is an ideal.) Then the zero variety is $1$ equation.

$A$ is a ring, $\mathcal{J}$ is an ideal. Typical example: $R^2$, the ideal $= 0$ at the origin.




Original source details: GELFAND-P000001

**Uncertain reading.** The next line: “Gluing a projective line into $0$” [source: Вклеив. в $0$ проект. прямой]; the beginning may read “Оклеив.”.

$A$, $\mathcal{J}$ are modules.

$$\mathcal{J} \otimes_A \mathcal{J}, \qquad \mathcal{J} \otimes_A \mathcal{J} \otimes_A \mathcal{J}, \dots$$

[Above the groups of tensor products:] symmetric; symmetric.

Consider formal sums

$$\sum_{k=0}^{N} a_k z^k.$$

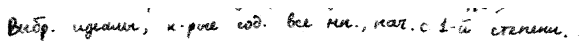
Add and multiply. $x \otimes y = y \otimes x$.

$$a_0 \in A, \qquad a_1 \in \mathcal{J}, \qquad a_2 \in \mathcal{J} \otimes_A \mathcal{J} \quad —$$

GELFAND-P000002

The ring of these sums is $S_A(\mathcal{J})$. Consider homogeneous ideals [inserted above the line:] (graded). (Together with a polynomial it contains all its homogeneous components.)

Consider all homogeneous prime ideals in $S_A(\mathcal{J})$. Выбр. [source abbreviation] ideals which contain all polynomials, starting with degree $1$.




Original source details: GELFAND-P000002

**Uncertain reading.** The end of the line after “contain all” is better read as “polynomials, starting with degree $1$”; the middle strokes of the abbreviation “нач.” remain indistinct.

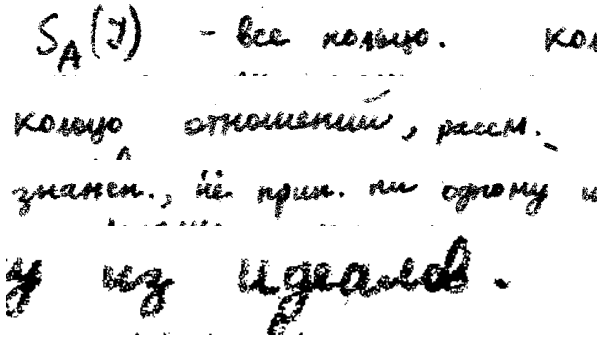
The collection of ideals is $X$. (The Zariski topology.) $O_X$, $\mathcal{J}_X$. If $U$ is an open set, then $O_U$, $\mathcal{J}_U$.




Original source details: GELFAND-P000002

**Uncertain reading.** After “then”, the first letter has the same form as $O_X$ above, the second as the ideal letter: $O_U$, $\mathcal{J}_U$; the latter may be $\mathcal{I}_U$.

$S_A(\mathcal{J})$ is the whole ring. The ring of fractions, numerators, denominators, not belonging to any of the ideals. Localize the ring at this set. Fractions of degree $1$: $\mathcal{J}_U$.

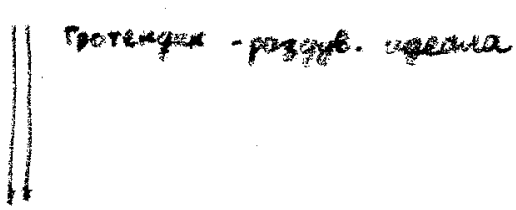



Original source details: GELFAND-P000002

**Uncertain reading.** In the line about fractions, the abbreviations “numerator–denominator” and whether “not belonging” applies only to the denominator are unclear. No words are supplied from the meaning.

$$\Downarrow$$

[At right:] Grothendieck — blow up ideally.



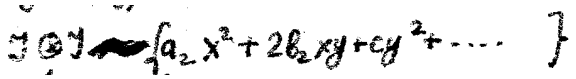
Original source details: GELFAND-P000002

**Uncertain reading.** The right-hand note reads “Grothendieck — blow up ideally”; the last word may be “идеально” [ideally] or “идеалы” [ideals].

$A$ is the ring of polynomials in $2$ variables. $\mathcal{J}$ is the ideal of polynomials with $P(0,0)=0$.

$$A=\{P(x,y)\},\qquad \mathcal{J}=P(x,y)=a_1x+b_1y+\cdots.$$

$$\mathcal{J}\otimes_A\mathcal{J}\ \longleftrightarrow\ \{a_2x^2+2b_2xy+cy^2+\cdots\}.$$



Original source details: GELFAND-P000002

**Uncertain reading.** The coefficients in the quadratic polynomial read $a_2$, $2b_2$, $c$; the index on $c$ and the shape of the thick two-headed arrow are uncertain.

Because of the interchange, one obtains the same as the square of the ideal. Then — the ideal of polynomials equal to $0$ at this point. What will it blow up into? ?

$$\sum P_k(x,y)z^k,\qquad P_k(x,y)=ax^k+\cdots.$$

[At right:] Graded $\Sigma$ of tensor products.

$$A=\{P_0(x,y)\}\longrightarrow S_A(\mathcal{J}),\qquad P_0(x,y)\sim P_0(x,y).$$

The set of prime ideals obtained from the ideal $P(0,0)=0$.

$$S_A\sim\sum P_kz^k,\qquad P_k\in\mathcal{J}^k,\qquad K\sim\sum P_kz^k.$$

It suffices to find the max. ideals of the quotient ring — (to find the ideals containing the given one).

$$S_A/K\ \longleftrightarrow\ a_0+(a_1x-b_1y)z+\cdots.$$



Original source details: GELFAND-P000002

**Uncertain reading.** In the last formula a minus is visible before $b_1y$, although the next scan has a plus. The visible minus is retained; the direction of the thick arrow is also uncertain.

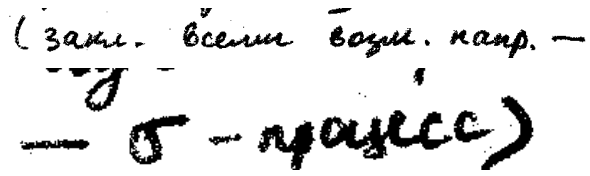
GELFAND-P000003

All homogeneous max. ideals in it will be the points of the extension. There is no constant term (then also $1$).

$$(a_1x+b_1y)z.$$

Only $1$ linear combination of $x$ and $y$. (Not $2$, since then all.)

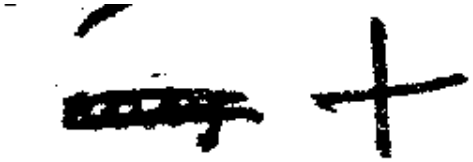
The ideals determine directions in the plane (замк. [source abbreviation] with all possible directions — the $\sigma$-process).



Original source details: GELFAND-P000003

**Uncertain reading.** The abbreviations in parentheses read “замк. всеми возм. напр.”; neither the first word nor the meaning of the sign $\sigma$ is asserted.

Problem. In the space [cancelled:]



Original source details: GELFAND-P000003

**Uncertain reading.** The cancelled word before the drawing is illegible.



Original diagram



Typeset recreation

What results from gluing it in?

$$z = 0, \qquad xy = 0, \qquad \mathcal{J} = (z, xy).$$

[A line is drawn beneath the system.]

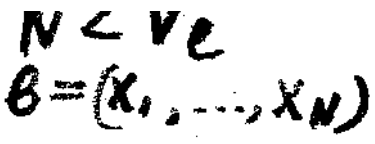
Dobrushin. [Cancelled before “Find”:]



Original source details: GELFAND-P000003

**Uncertain reading.** The cancelled beginning of the line cannot be read confidently.

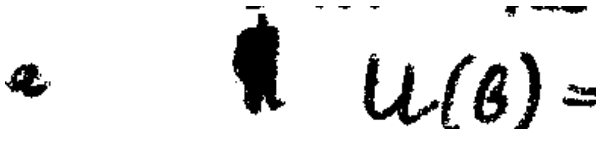
Find an example of a gas in which phase transitions occur. A cube of the integer lattice. $\ell$ is the length. $V_\ell = \ell^\nu$. $N$ points, $N < V_\ell$. They can occupy positions on the lattice $\theta = (x_1, \ldots, x_N)$.



Original source details: GELFAND-P000003

**Uncertain reading.** The configuration sign is rendered as $\theta$; the handwritten sign also resembles $b$ or Cyrillic “в”.

The Ising model. [Before the formula, a densely cancelled sign:]

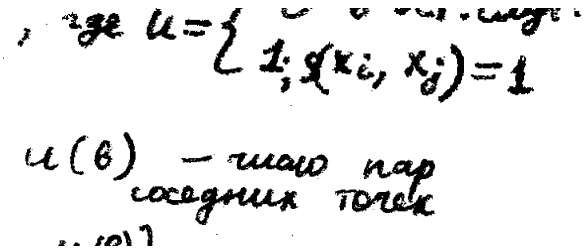


Original source details: GELFAND-P000003

**Uncertain reading.** The separate cancelled sign before $U(\theta)$ is illegible.

$$U(\theta) = -\frac{1}{2}\sum_{i\neq j} u(x_i, x_j), \qquad u = \begin{cases} 0 & \text{in the other cases,} \\ 1; & \rho(x_i, x_j) = 1. \end{cases}$$

$u(\theta)$ is the number of pairs of neighboring points.



**Uncertain reading.** In the condition for the second branch a sign resembling $\rho$ is visible; the semicolon after $1$ is retained. In the caption below, the first letter resembles lowercase $u$ rather than $U$.

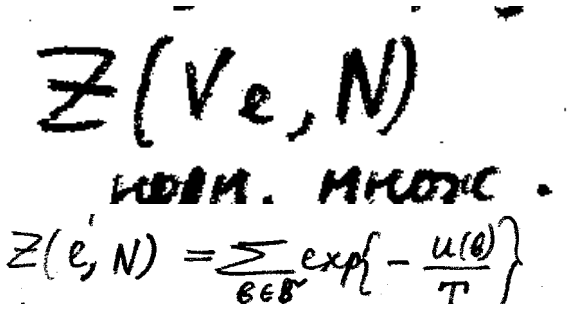
Original source details: GELFAND-P000003

The basic axiom is the Gibbs distribution.

$$\mathsf{P}(B) = \frac{\sum\limits_{\theta\in B} \exp\{-U(\theta)/T\}}{Z(V_\ell, N)}, \qquad T > 0 \text{ — temperature.}$$

[Beneath the denominator:] normalizing factor.

$$Z(\ell, N) = \sum_{\theta\in\mathcal{B}} \exp\{-U(\theta)/T\}.$$



**Uncertain reading.** The denominator in the first formula reads $Z(V_\ell, N)$, the next line $Z(\ell, N)$; the set sign beneath the sum resembles $\mathcal{B}$.

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The molecules attract one another. ([Two filled signs with arrows:] neighbors $>$ their probability.)



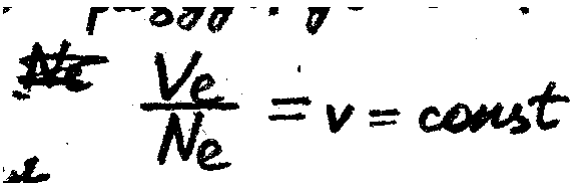
**Uncertain reading.** In parentheses after “The molecules attract”, there are filled signs and arrows; the reading “neighbors $>$ their probability” is uncertain.

Original source details: GELFAND-P000003

$B \subset \mathcal{B}$ is a collection of configurations.

$$p(\theta) = \frac{\exp\{-U(\theta)/T\}}{Z(\ell, N)}.$$

Take the limit $\ell \to \infty$; the cube expands to $\infty$. $N_\ell$ depends on $\ell$. [A cancelled sign before the fraction:]



**Uncertain reading.** The cancelled sign before $V_\ell/N_\ell$ is illegible.

Original source details: GELFAND-P000003

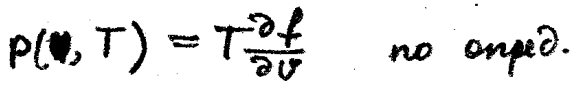
$$\frac{V_\ell}{N_\ell} = v = \text{const.}$$

The lattice grows. One can show that

$$\frac{1}{N_\ell} \log Z(\ell, N_\ell) \longrightarrow f(v, T).$$

GELFAND-P000004

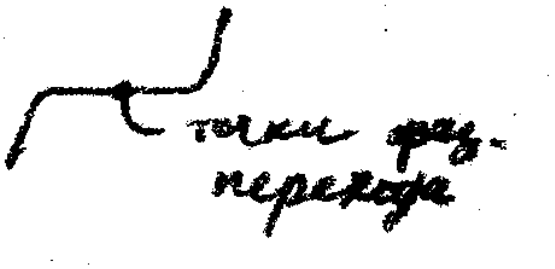
$$p(v,T) = T\frac{\partial f}{\partial v} \qquad \text{by definition.}$$




Original source details: GELFAND-P000004

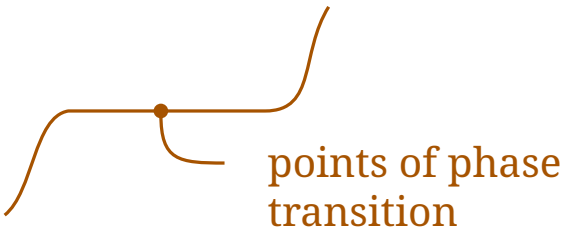
**Uncertain reading.** The heavily written first argument of $p$ resembles $v$; no minus is visible before the derivative.

Phase transition. As for $f$: it is a monotone, convex function of $v$. Therefore $p > 0$, and $p$ is a monotone function of $v$. Phase transition: $p(v, T)$ has a flat segment.



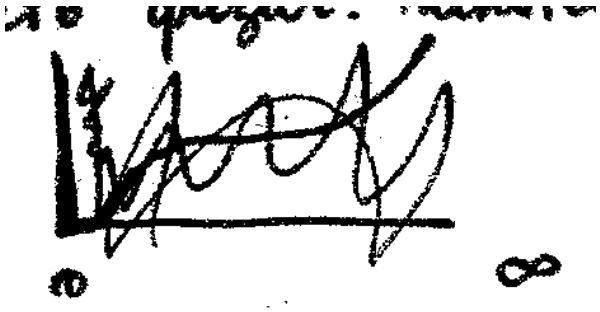

Original source details: GELFAND-P000004

**Diagram and typed description.** In the upper right-hand field is a bent, S-like curve with an almost horizontal middle segment and an arrow toward it; the caption reads “points of phase transition” [source abbreviation: точки фаз. перехода]. Precise axes and points in the original drawing are not identified confidently.



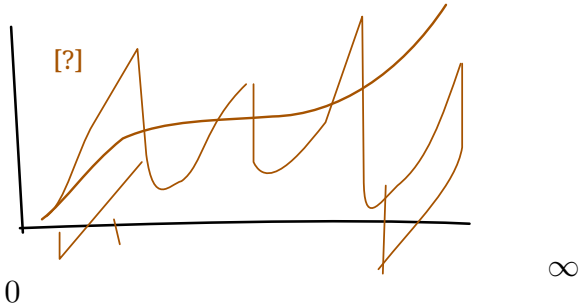

Typeset sketch; limitations are recorded in the note.

There is a physical hypothesis: 1 transition segment.

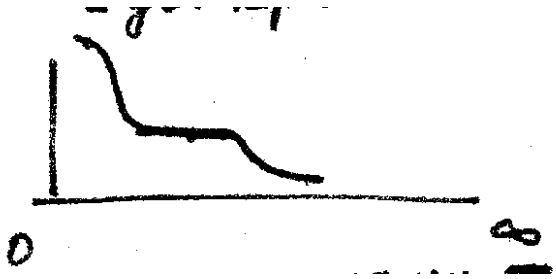



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**Diagram and typed description.** Intersecting wavy and oblique lines, horizontal and vertical axes, $0$ and $\infty$. The strokes overlap; their exact separation into curves and the vertical label are unresolved.

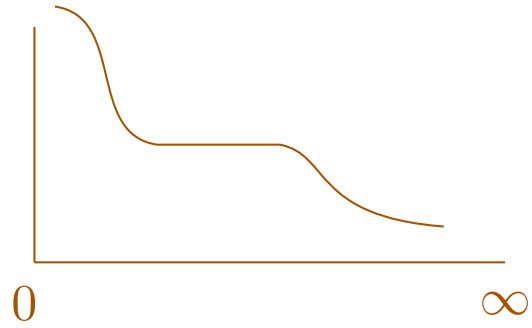



Visible strokes; [?] marks the unreadable vertical label.




Original source details: GELFAND-P000004

**Diagram and typed description.** Neighboring drawing at right: a clean descending curve with a flat segment. This is a separate graph, not a reconstruction of the cancelled drawing at left.

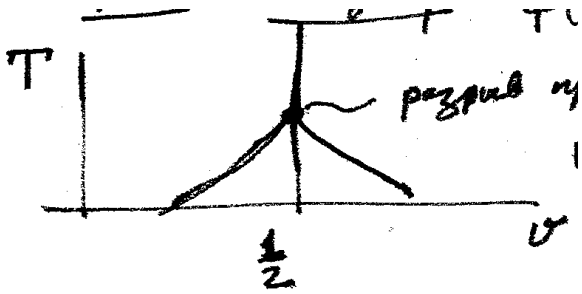



Typeset recreation.

A flat segment occurs for sufficiently small $T$ ($< T_{\text{crit}}$).

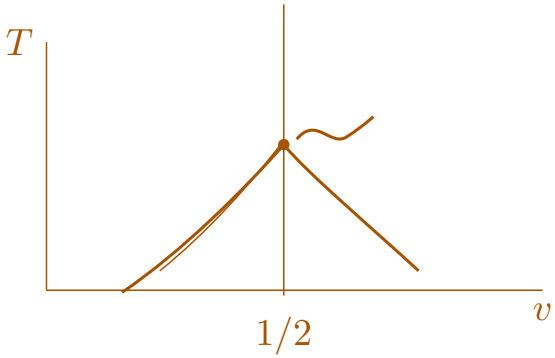
It admits an explicit solution for $n = 1$. $Z$ and $f(v, T)$ are calculated. There are no phase transitions, which is considered physically natural.

$n = 2$. Onsager. $f(v, T)$ is calculated for particular $v$ and $T$. A discontinuity of the derivative corresponds to phase transitions.



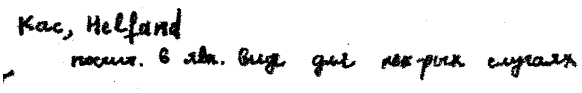

Original source details: GELFAND-P000004

**Diagram and typed description.** A graph with the $T$ axis upward, the $v$ axis to the right, and the label $1/2$ beneath a vertical through the vertex. The curve rises from the left to the vertex and descends to the right; the vertical line passes through the vertex, with a short wavy segment to its right. The precise geometry and the meaning of the wavy branch have not been established.



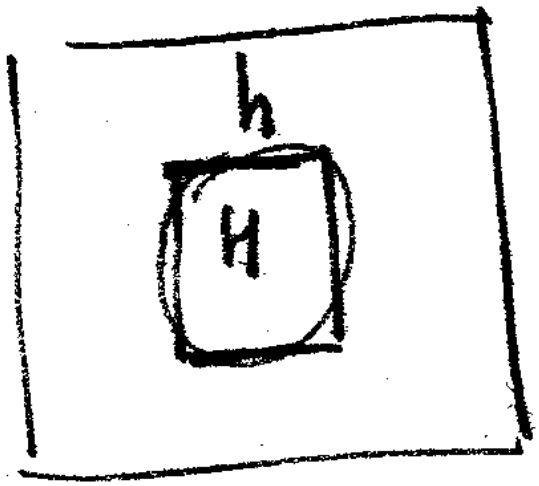

Typeset sketch; limitations are recorded in the note.

Kac, Helfand. Exact, in explicit form, for some cases [source endings: точн. в яв. виду для нек-рых случаях].



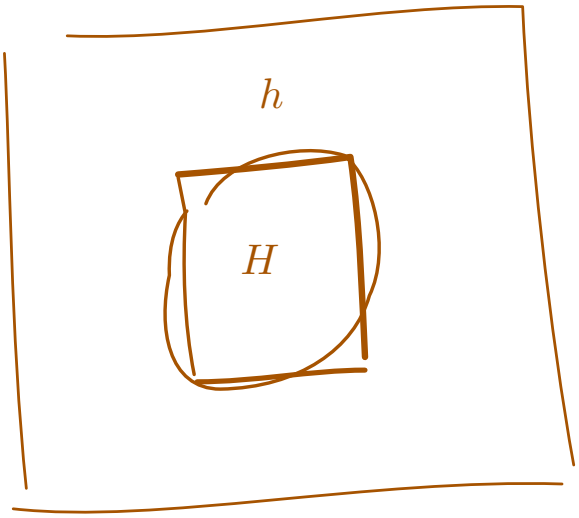

Original source details: GELFAND-P000004

**Uncertain reading.** The line reads “точн. в яв. виду для нек-рых случаях”; the endings may instead be “виде” and “случаев”.




Original source details: GELFAND-P000004

**Diagram and typed description.** Within the large frame, a small rectangular contour overlaps a rounded one; $H$ lies inside and $h$ above. The rounded contour protrudes to the right and left. Individual lines merge at the heavy intersections.




Visible contours and strokes; uncertain connections have not been completed.

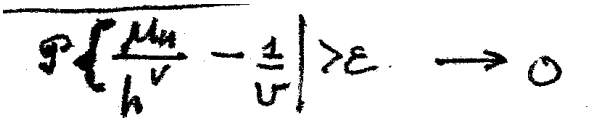
$$\frac{\mu_H}{h^\nu}.$$

$\mu_H$ is the number of particles in $H$ — a subcube. $\nu$ is the dimension.

$$\frac{\mu_H}{h^\nu} = \frac{1}{v} \qquad \text{as in the whole space.}$$

Law of large numbers.

$$\mathsf{P}\left\{\left|\frac{\mu_H}{h^\nu} - \frac{1}{v}\right| > \varepsilon\right\} \longrightarrow 0.$$



Original source details: GELFAND-P000004

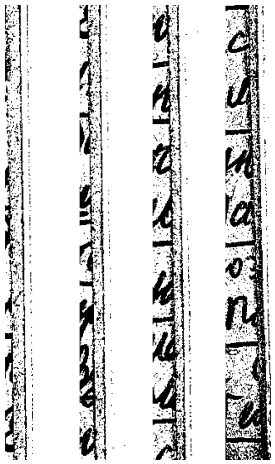
**Uncertain reading.** In the probability formula the right vertical bar before $> \varepsilon$ is visible; a left bar before the fraction has not been read confidently.

$$\ell \to \infty, \qquad H \to \infty, \qquad d < \frac{h}{\ell} < 1 - d, \qquad d > 0.$$

**Uncertain reading.** At the lower right edge a small sign resembles “2” or a broken-off letter; it has not been adopted as a page number.



Original source details: GELFAND-P000004

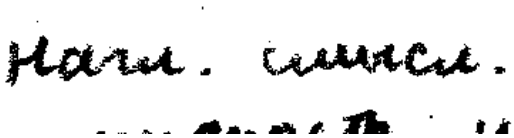


Original source details: GELFAND-P000004

**Undeciphered passage.** Along the extreme left edge is a narrow strip of neighboring handwritten text, cropped to isolated endings and strokes; no connected text can be read.

GELFAND-P000005

If this does not hold, a phase transition takes place. Нагл. meaning. 2 phases coexist; the particles are distributed unevenly — — liquid and gas.

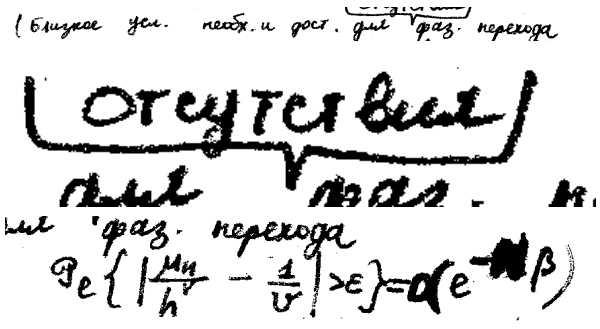


Original source details: GELFAND-P000005

**Uncertain reading.** Нагл. [candidate abbreviation; the middle strokes are not fully distinguishable]

[Above the line:] (law of large numbers). [Inserted above “phase transition”, with a bracket:] absence. (A closely related condition is necessary and sufficient for a phase transition

$$\mathsf{P}_\ell\left\{\left|\frac{\mu_H}{h^\nu}-\frac{1}{v}\right|>\varepsilon\right\}=O(e^{-N\beta}).$$



Original source details: GELFAND-P000005

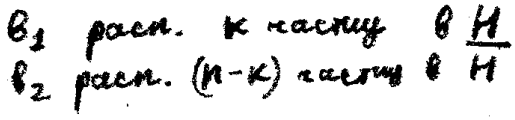
**Uncertain reading.** The exponent of $e$ contains a heavily written sign before $\beta$; the candidate is $-N\beta$, but $N$ is obscured by overlapping strokes. The parenthesis opened before “A closely related” is not clearly closed in the scan.

[At right:] $N$ is the number of particles.

Calculate the probability that

$$\mathsf{P}\{\mu_H=k\}=Z^{-1}\sum_{\theta\in\mathcal{B}_k}\exp\{-U(\theta)/T\}.$$

Over configurations in which $H$ contains $k$ particles. $\theta$ is a configuration of $N$ particles. $\theta\in\mathcal{B}_k$. Then $\theta=(\theta_1,\theta_2)$. $\theta_1$ is a configuration of $k$ particles in $H$. $\theta_2$ is a configuration of $(N-k)$ particles in $\overline{H}$.

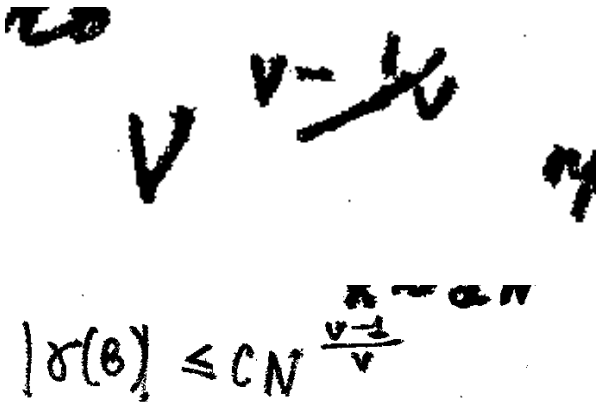


Original source details: GELFAND-P000005

**Uncertain reading.** The upper bar on $H$ in the line about $\theta_2$ is not completely distinct; the candidate is $\overline{H}$.

$$U(\theta) = U(\theta_1) + U(\theta_2) + \gamma(\theta).$$

$U(\theta_1)$ is the interaction inside. $U(\theta_2)$: neighbors outside. $\gamma(\theta)$ is the interaction across the surface. $\gamma(\theta)$ can be neglected asymptotically. (Across the boundary: of the order of the number of boundary points) $V^{\nu-1/\nu}$ as $N \to \infty$. $k \sim dN$.

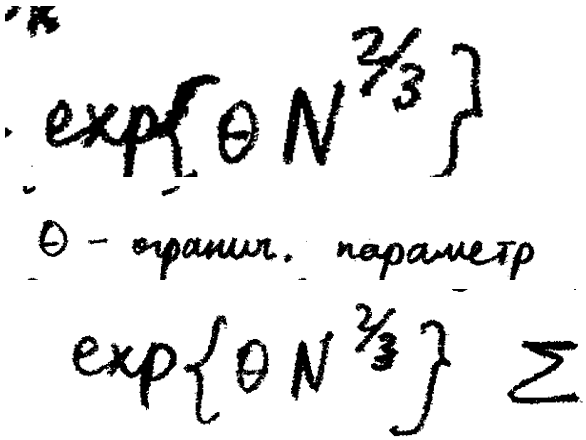


**Uncertain reading.** The exponent on $V$ looks like $\nu - 1/\nu$, without a confidently distinguishable common numerator. A possible grouping is $(\nu - 1)/\nu$.

Original source details: GELFAND-P000005

$$|\gamma(\theta)| \le CN^{\frac{\nu-1}{\nu}}.$$

$$\sum \exp\{-U(\theta)/T\} = \sum_{\theta \in \mathcal{B}_k} \exp\{-U(\theta_1)/T\} \exp\{-U(\theta_2)/T\}$$
$$\times \exp\{\theta N^{2/3}\} =$$



**Uncertain reading.** The letter before $N^{2/3}$ resembles $\theta$ or $\Theta$; below, the same letter is glossed as “bounded parameter”. It may differ from the configuration sign.

Original source details: GELFAND-P000005

$\theta$ is a bounded parameter.

$$= \exp\{\theta N^{2/3}\} \sum_{\theta_1} \exp\{-U(\theta_1)/T\} \sum_{\theta_2} \exp\{-U(\theta_2)/T\}.$$

[Beneath the first sum:] configurations.

GELFAND-P000006

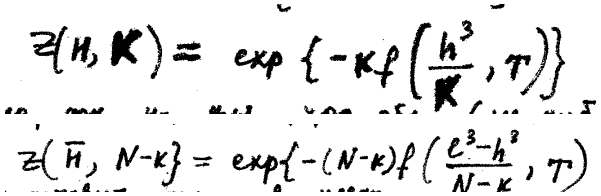
Factor 1: $Z$, calculated for the subcube. Factor 2 — ditto — $Z$ for the outer cube.

$$\mathsf{P}\{\mu_H = k\} = \frac{Z(H,k)Z(\overline{H}, N-k)}{Z(V,N)} \exp\{\theta N^{2/3}\}.$$

[Asterisk.] Apply to them the asymptotic formula

$$Z(V,N) \approx \exp\{-Nf(v,T)\},$$

$$Z(H,k) = \exp\{-kf(h^3/k,T)\}.$$



**Uncertain reading.** The numerator of the argument of $f$ in this line visibly has exponent $3$ on $h$, although $\nu$ is used later.

Original source details: GELFAND-P000006

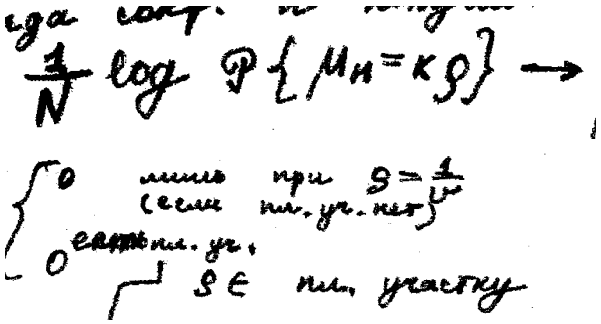
It can also be proved for well-behaved regions (not a cube). The limiting formula does not depend on the region's shape.

$$Z(\overline{H},N-k) = \exp\{-(N-k)f((\ell^3-h^3)/(N-k),T)\}.$$

Substitute there under the assumption that $f$ has a linear segment. If $f$ has a linear segment (the derivative has a horizontal one), then terms cancel and we obtain

$$\frac{1}{N}\log \mathsf{P}\{\mu_H = k\rho\} \longrightarrow \begin{cases} 0 & \text{only at } \rho = 1/v \text{ (if there is no linear segment)}, \\ 0 & \rho \in \text{ the linear segment}. \end{cases}$$

[Cancelled by the second branch:]



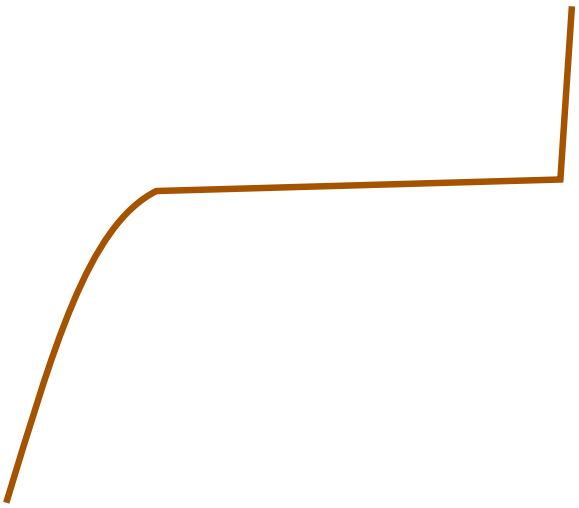
**Uncertain reading.** The event's argument looks like $\mu_H = k\rho$, but the final sign may be $\theta$ or $\delta$. The density letter at right resembles $\rho$. The cancelled text above the second branch has not been read.

Original source details: GELFAND-P000006



Original source details: GELFAND-P000006

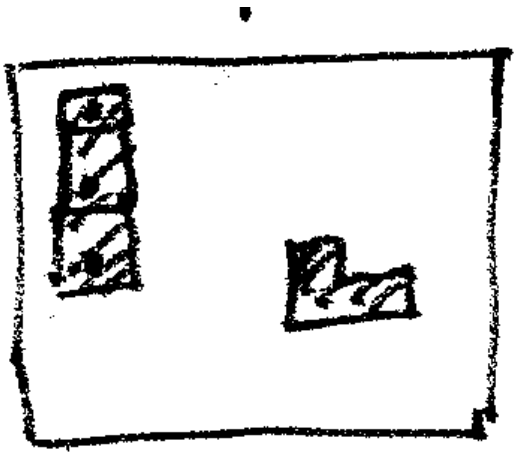
**Diagram and typed description.** At right of the second branch of the limit is a small stroke: from the lower left, a line rises almost vertically, then runs almost horizontally to the right, and rises again at the right end. Separate axes, arrows and labels have not been established; the sketch's mathematical meaning requires the original image.



Visible stroke; no axes or values supplied.

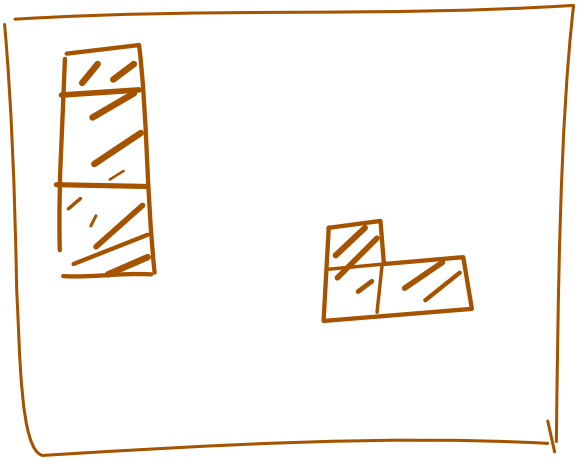
$\mathsf{P} \to 0$ exponentially. ($1/v$ follows from the convexity of $f$.)

Arrangement of particles. Its boundary; around each particle a cube of side $1$. $\Gamma'(\theta)$ is the boundary of the occupied region. $\Gamma(\theta)$ is the length [inserted below:] area.



Original source details: GELFAND-P000006

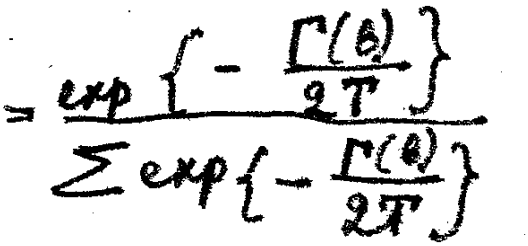
**Diagram and typed description.** At left is a vertical outline with two strong transverse strokes, diagonal hatching and separate short marks. At right, three cells form an L: two below and one above the left cell. In the left column, divisions merge with the hatching; the exact cell count is uncertain.



Visible contours and strokes; uncertain connections have not been completed.

Replace the Gibbs probability.

$$\mathsf{P}(\theta) = \frac{\exp\{-U(\theta)/T\}}{\sum \exp\{-U(\theta)/T\}} = \frac{\exp\{-\Gamma'(\theta)/(2T)\}}{\sum \exp\{-\Gamma'(\theta)/(2T)\}}.$$

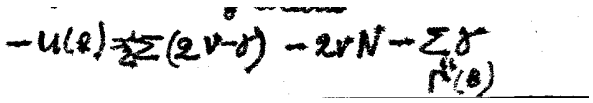


Original source details: GELFAND-P000006

**Uncertain reading.** The prime on $\Gamma$ in the two exponents after the equality is not completely clear; the visible letter resembles $\Gamma'$ or $\Gamma$.

[At right:] $2\nu - \delta$. Unoccupied squares $\Gamma'(\theta)$. [Inserted:] at neighbors.

$$-U(\theta) \geq \sum(2\nu - \delta) = 2\nu N - \sum_{\Gamma'(\theta)} \delta.$$



Original source details: GELFAND-P000006

**Uncertain reading.** The sign after $-U(\theta)$ is written over another sign: candidates are $\geq$, $\leq$, and $=$. The sign before the final sum and the sum's index may belong to another term.

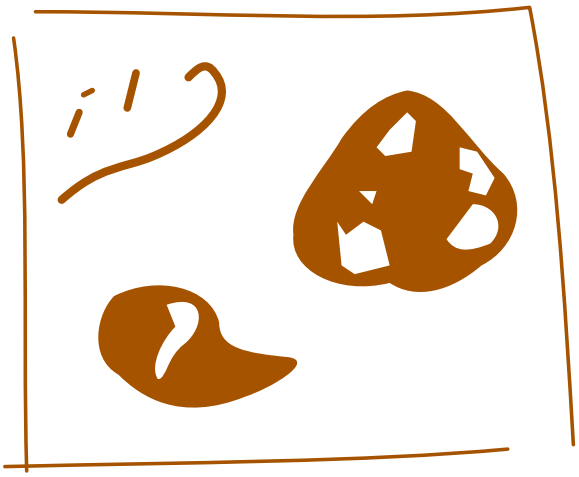
GELFAND-P000007

($U(\theta)$ is minus the number of neighbors.) A smaller boundary makes an arrangement more favorable. Force them into one corner, but this is unfavorable (there are few such arrangements in all). A finite number of clumps. (Inside the gas: liquid.)



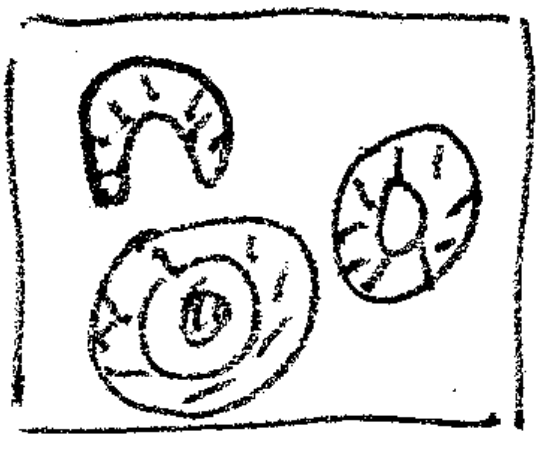
Original source details: GELFAND-P000007

**Diagram and typed description.** The small frame contains two dark regions with light interior gaps. At upper left are a bent heavy stroke and separate short oblique marks. These marks have not been identified as letters or arrows.



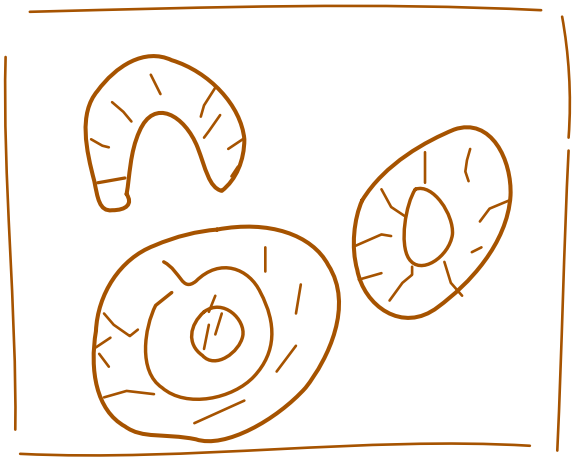
Visible contours and strokes; uncertain connections have not been completed.

And this contradicts the law of large numbers.



Original source details: GELFAND-P000007

**Diagram and typed description.** Inside the large frame, the upper-left horseshoe-shaped region has an open lower gap and a small outlined part at its left tip. At right is a slanted ring. The large lower-left region contains an open or broken inner contour and a small central contour. Individual hatch marks are visible; some connections are uncertain.

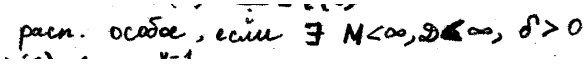


Visible contours and strokes; uncertain connections have not been completed.

The whole boundary splits into connected cycles. Each arrangement has its own number. Number them

$$\Gamma_1(\theta) \geq \Gamma_2(\theta) \geq \cdots, \qquad \Gamma(\theta) = \sum_i \Gamma_i(\theta).$$

An arrangement is special if $\exists N < \infty$, $D < \infty$, $\delta > 0$.

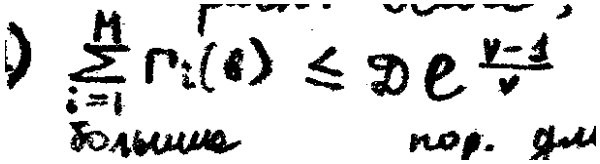


Original source details: GELFAND-P000007

**Uncertain reading.** In this line the first finite parameter resembles $N$, whereas the later summation limit resembles $M$.

$$1) \quad \sum_{i=1}^{M} \Gamma_i(\theta) \leq D\ell^{\frac{\nu-1}{\nu}}.$$

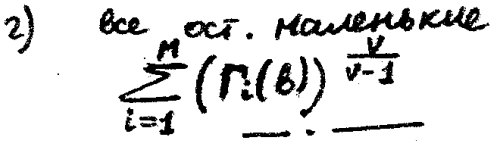
[Beneath the sum:] Large. Of the order of the boundary length. (Almost the whole cube.)



Original source details: GELFAND-P000007

**Uncertain reading.** The exponent on $\ell$ in the first condition is written as a fraction, read as $(\nu-1)/\nu$; in GELFAND-P000010 the analogous line has $\nu - 1$.

$$2) \quad \text{all the others small} \qquad \sum_{i=1}^{M} \left(\Gamma_i(\theta)\right)^{\frac{\nu}{\nu-1}}.$$



Original source details: GELFAND-P000007

**Uncertain reading.** The visible limits of the second sum look like $1$ and $M$, although the caption says “all the others small”. No condition is visible after the sum.

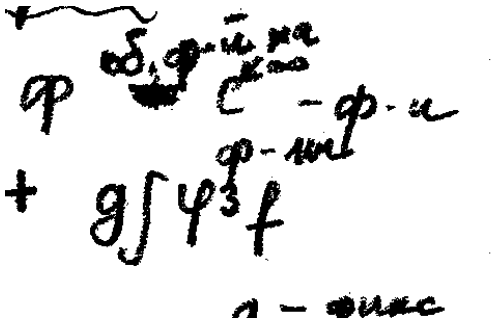
[Dividing line.]

24/IX. Thursday. 16 h, room 13-11. Prof. Segal (USA). Theory of functions in infinite-dimensional space.

$$+\Box\varphi = m^2\varphi + g\varphi^3.$$

$$F_f(\varphi) = \int (-\Box + m^2) f\varphi + g \int \varphi^3 f.$$

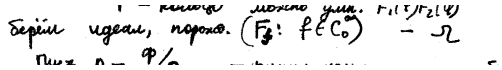
[Above $\varphi^3$:] $\varphi$ — мн. [source abbreviation]. $\varphi$ [arrow] $C_0^\infty$ — functions.



Original source details: GELFAND-P000007

**Uncertain reading.** Above the note at right are the visible words “б, ф-й на” [fragment: functions on] and an arrow toward $C_0^\infty$.

$g$ is fixed. $F_f \in \Phi$ is a nonlinear functional of $\varphi$. $\Phi$ is a ring; one can multiply $F_1(\varphi)F_2(\varphi)$. Take the ideal generated by $(F_f : f \in C_0^\infty)$ — $\Omega$.



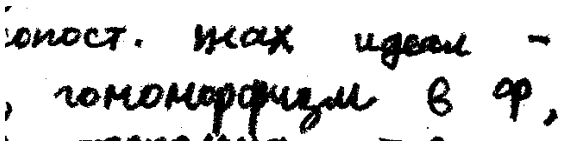
Original source details: GELFAND-P000007

**Uncertain reading.** The index on $F$ in the ideal’s definition resembles $f$; the source sign after the parenthesis is $\Omega$ or a calligraphic $J$.

Let $p = \Phi/\Omega$ be the function ring on the manifold of solutions of the given partial differential equation. To obtain a solution $(1) \stackrel{\text{def}}{=}$ a point of the space of maximal ideals of the ring $p$. An ordinary solution $\varphi_0$ [a check mark above the word] is assigned a maximal ideal: consider the map

$$F \longrightarrow F(\varphi_0),$$

a homomorphism into $\Phi$, $F_f$ is zero on the ideal $\Omega$. This is a homomorphism of the quotient ring as well, i.e. a maximal ideal.



Original source details: GELFAND-P000007

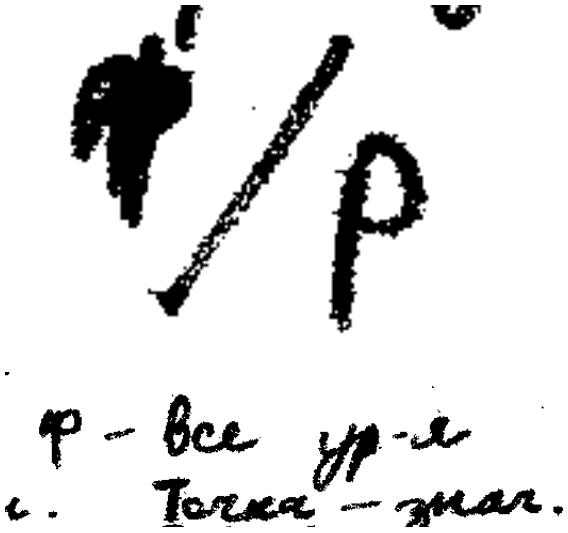
**Uncertain reading.** After “a homomorphism into”, the visible letter resembles $\Phi$.

GELFAND-P000008

A maximal ideal is a generalized solution of a nonlinear equation. The algebraic case.

$$f_1(x_1, \ldots, x_n) = 0, \qquad f_k(x_1, \ldots, x_n) = 0.$$

Solutions in terms of a ring: in the ring of all polynomials. The ideal generated by these functions is $p$, and the quotient ring is $\Phi/p$.

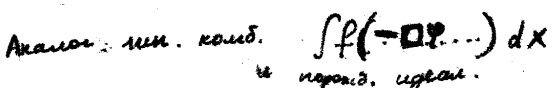


Original source details: GELFAND-P000008

**Uncertain reading.** In the fraction, the numerator resembles $\Phi$, $P$, or a heavily written $A$; the candidate $\Phi$ is retained.

$\Phi$ — all equations. This is the ring of algebraic functions. A point, the value at the point, is precisely a maximal ideal. In the algebraic case, conversely as well. Analogy: linear oscillations.

$$\int f(-\Box\varphi\ldots)\,dx.$$

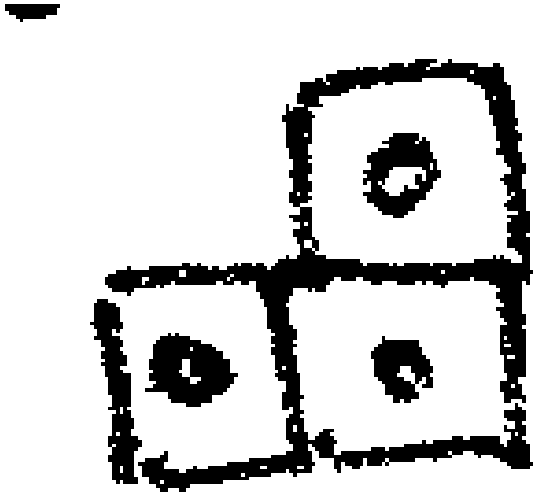


Original source details: GELFAND-P000008

**Uncertain reading.** The part of the integral after $-\Box$ looks like $\varphi\ldots$, but the letter is partly inked over.

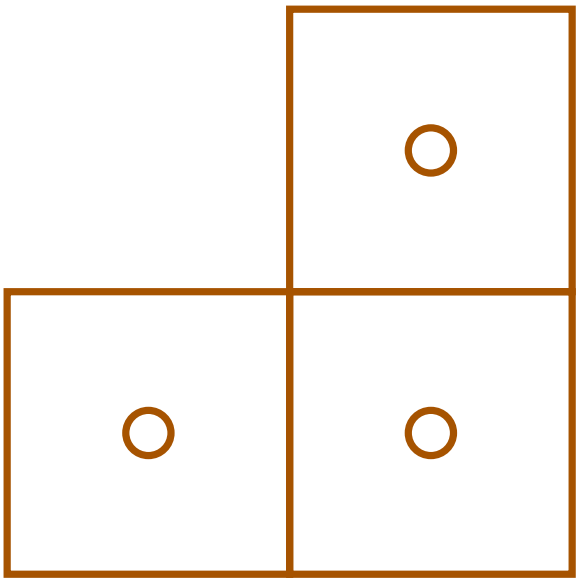
And the corresponding ideal.

Dobrushin. $\nu = 2, 3$. $V_\ell = \ell^\nu$, $N < V_\ell$, $\theta = (x_1, \ldots, x_N)$.



Original source details: GELFAND-P000008

**Diagram and typed description.** Original diagram and typographic recreation.



$\Gamma(\theta)$ is the boundary length. $U(\theta)$ is the number of neighbors (pairs of neighbors). The Gibbs distribution

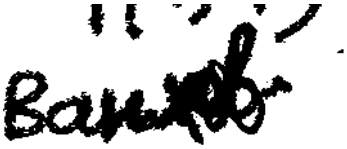
$$p(\theta) = \frac{\exp\{U(\theta)/T\}}{\sum\limits_{\theta\in\mathcal{B}(V_\ell,N)} \exp\{U(\theta)/T\}} = \frac{\exp\{-\Gamma(\theta)/(2T)\}}{\sum\limits_{\theta} \exp\{-\Gamma(\theta)/(2T)\}}.$$

[Beneath the first sum:] all arrangements.

$$(\text{since } 2U(\theta) = 2\nu N - \Gamma(\theta)).$$

Phase transition. Let the lattice grow, $\ell \to \infty$, and let $N_\ell \to \infty$. The limit $V_\ell/N_\ell \to v$ is the specific volume. It turns out that

$$\frac{1}{N} \log \sum_\theta \exp\{U(\theta)/T\} \longrightarrow f(v,T).$$



Original source details: GELFAND-P000008

**Uncertain reading.** Beneath the limit the name reads “Ван Хов” or “Ванхов” [Van Hove]; the middle and ending are heavily overwritten.

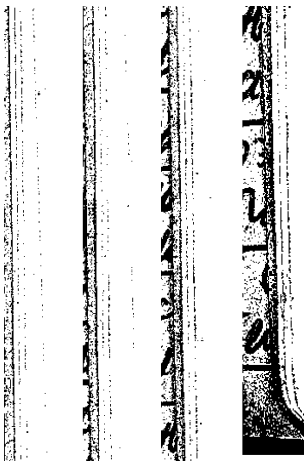
$(v,T)$ is called a point of phase transition if

$$\frac{\partial f(v,T)}{\partial v} = 0 \quad \text{in some neighborhood at } T = \text{const}.$$



Original source details: GELFAND-P000008

**Uncertain reading.** Between $T$ and const, the sign resembles $=$ or $\neq$; the candidate $=$ is retained, but the reading is not asserted.



Original source details: GELFAND-P000008

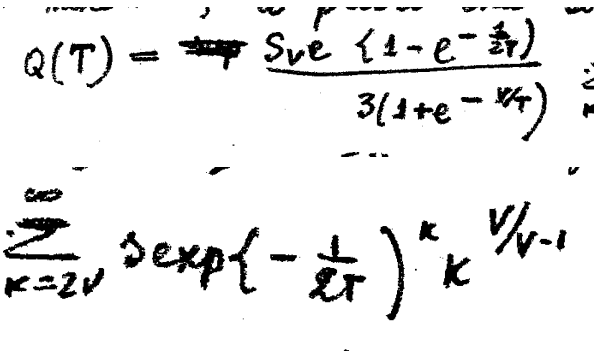
**Undeciphered passage.** At the extreme left edge is a narrow cropped column of handwritten endings from neighboring text; continuous words cannot be recovered.

GELFAND-P000009

(Since the function is linear; a priori it is convex.)

$$Q(T) = \frac{S_\nu e(1 - e^{-1/(2T)})}{3(1 + e^{-\nu/T})} \sum_{k=2\nu}^{\infty} 3 \exp\{-1/(2T)\}^k k^{\frac{\nu}{\nu-1}}.$$

[After the first equality, a cancelled fragment:]

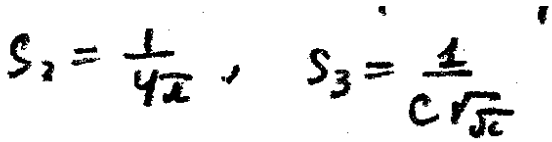


**Uncertain reading.** The cancelled sign after $=$ is illegible. In the numerator, the factor after $S_\nu$ resembles $e$; the upper summation limit resembles $\infty$ or a double stroke. The exponent $k$ may apply to the factor $3$ as well, but in the scan it stands by the closing brace.

Original source details: GELFAND-P000009

$S_\nu$ is the volume of a $\nu$-dimensional sphere with surface $1$.

$$S_2 = \frac{1}{4\pi}, \qquad S_3 = \frac{1}{C\sqrt{\pi}}.$$



**Uncertain reading.** The first factor in the denominator of $S_3$ resembles $C$, $c$, or $6$; the written $C$ is only a candidate.

Original source details: GELFAND-P000009

If

$$Q(T) < \frac{1}{v} < 1 - Q(T),$$

then a phase transition takes place.

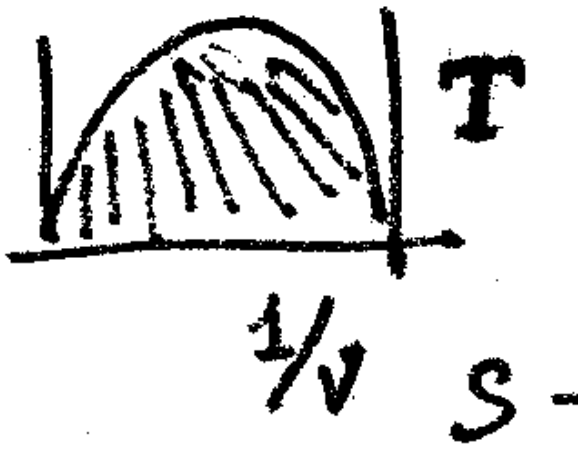
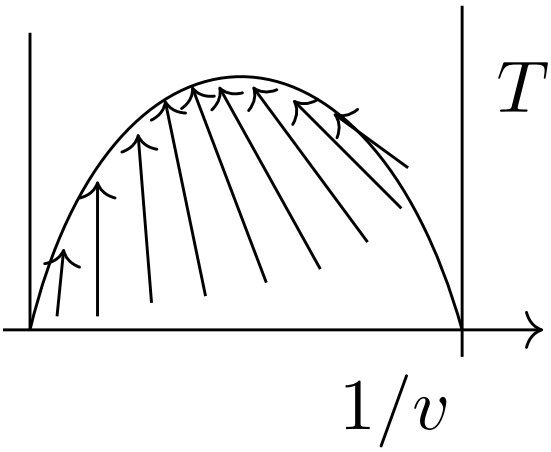


**Diagram and typed description.** Original diagram and typographic recreation.




Original source details: GELFAND-P000009

[Dividing line.]

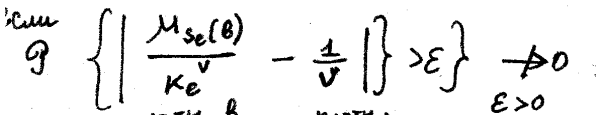
$S$ is a subcube. $S_\ell \subset V_\ell$. $K_\ell$ is the side of $S_\ell$.

$$0 < d < \frac{K_\ell}{\ell} < 1 - d.$$

Introduce a random variable. $\mu(S_\ell)$ is the number of points of the arrangement $\theta$ that fall in $S_\ell$. $\mu_{S_\ell}(\theta)$. If

$$\mathsf{P}\left\{\left|\frac{\mu_{S_\ell}(\theta)}{K_\ell^\nu} - \frac{1}{v}\right|\right\} > \varepsilon\} \not\longrightarrow 0, \qquad \varepsilon > 0,$$

then there is a phase transition.



Original source details: GELFAND-P000009

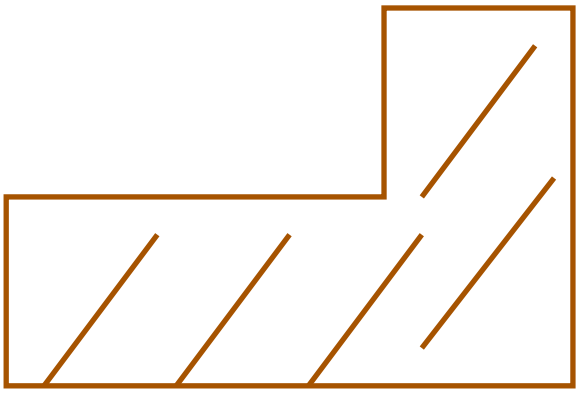
**Uncertain reading.** After the right absolute-value bar, the scan shows a brace before $> \varepsilon$ and another after $\varepsilon$.

[Beneath the first fraction:] density in the subcube. [Beneath $1/v$:] density in the whole cube. And a closely related condition is necessary. (Physical meaning: coexistence of $2$ phases.)



Original source details: GELFAND-P000009

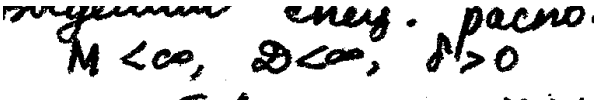
**Diagram and typed description.** Original diagram and typographic recreation.



The boundary splits into components — — but a connection through corners ($\nu = 2$) or an edge ($\nu = 3$) is not counted.

$$\Gamma(\theta) = \sum_i \Gamma_i(\theta).$$

$\Gamma_i(\theta)$ is the length or area of a boundary component. Definition. Single out a special arrangement. It depends on $3$ numbers, $N < \infty$, $D < \infty$, $\delta > 0$.



Original source details: GELFAND-P000009

**Uncertain reading.** The first letter resembles $N$, but admits $M$; the enlarged fragment does not resolve the distinction.

Number the cycles in decreasing order of length

$$\Gamma_1(\theta) \geq \Gamma_2(\theta) \geq \cdots .$$

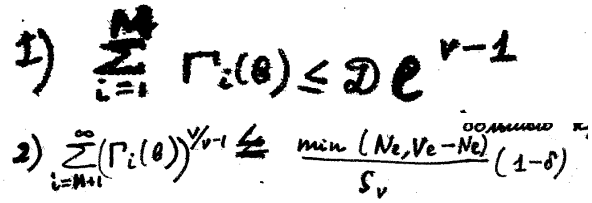
GELFAND-P000010

Conditions.

$$1) \quad \sum_{i=1}^{M} \Gamma_i(\theta) \leq D\ell^{\nu-1}.$$

(Of the order of the surface of the large cube.)

$$2) \quad \sum_{i=M+1}^{\infty} \big(\Gamma_i(\theta)\big)^{\frac{\nu}{\nu-1}} \leq \frac{\min(N_\ell, V_\ell - N_\ell)}{S_\nu}(1-\delta).$$

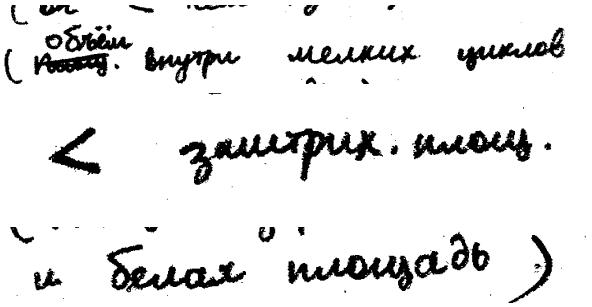


**Uncertain reading.** The inequality sign before the fraction is heavily overwritten: candidates are $\leq$ and $<$. The parameter in the upper limit of the first sum resembles $M$ or $N$.

Original source details: GELFAND-P000010

$$S_\nu \left(\Gamma_i(\theta)\right)^{\frac{\nu}{\nu-1}}$$

Area of a circle with length $\Gamma_i$. A ball has maximal volume; this is an upper bound on the volume of the region inside the cycles. (It is $<$ what is needed to contain all occupied points.) ([A word is cancelled; inserted above:] volume. Inside small cycles $<$ the shaded area and the white area.)

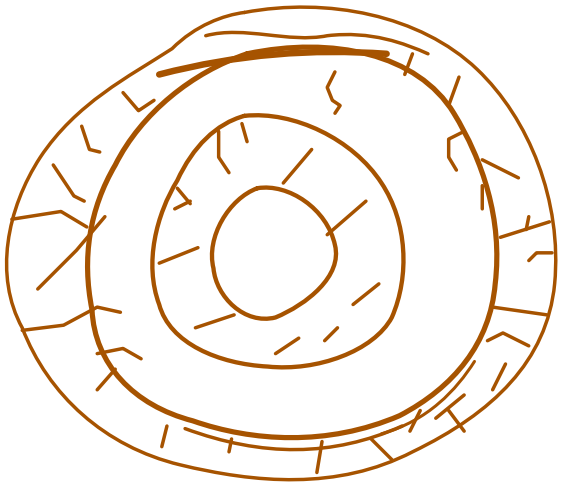


**Uncertain reading.** The word beneath the inserted “volume” is heavily cancelled and illegible. The two lines write “Area”.

Original source details: GELFAND-P000010



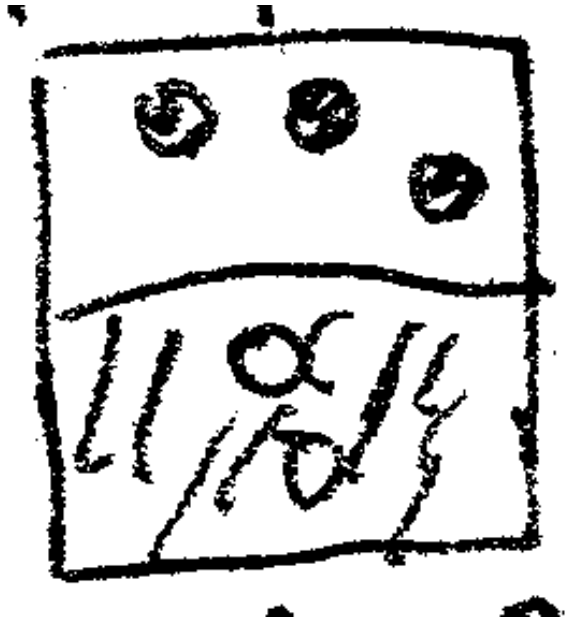
**Diagram and typed description.** Four nested irregular rounded contours. The outer ring has uneven thick and thin strokes, retraced arcs above and at lower right, and separate ends bending inward. The inner ring has radial and oblique strokes. Lines merge in heavy areas; no additional closed boundaries or arrows have been established.



Visible irregular contours and strokes; no directions assigned.

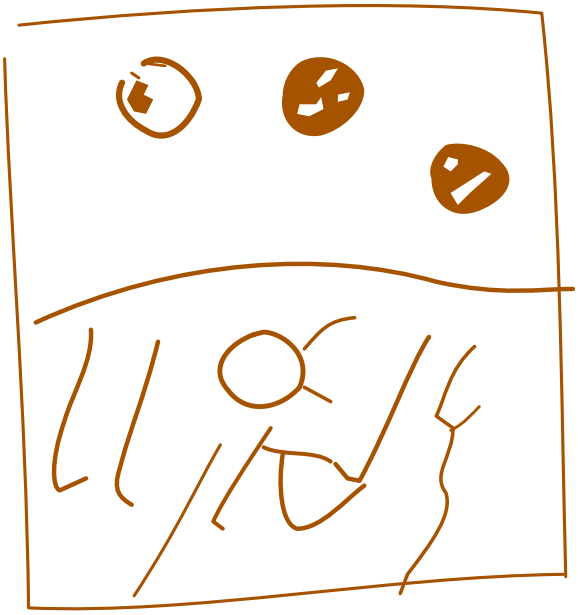
Original source details: GELFAND-P000010

Neither all the points nor all the empty sites can be forced into it. An example of an arrangement where the conditions do not hold.



Original source details: GELFAND-P000010

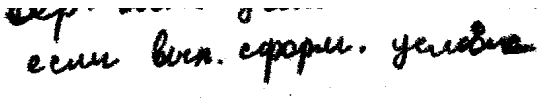
**Diagram and typed description.** The three upper dark regions have distinct light centres. Below the curved divider are a rounded contour with two right branches, an incomplete lower curve, several oblique and bent strokes, and a short branch on the right. Some connections at the lower intersections and gaps are uncertain.



Visible contours and strokes; uncertain connections have not been completed.

This is a phase transition. Everything reduces to 2 lemmas.

1) $\exists M, D, \delta$ such that the probability that the conditions hold $\longrightarrow 1$, if the stated condition holds. [Beneath the parameters:] independent of everything.

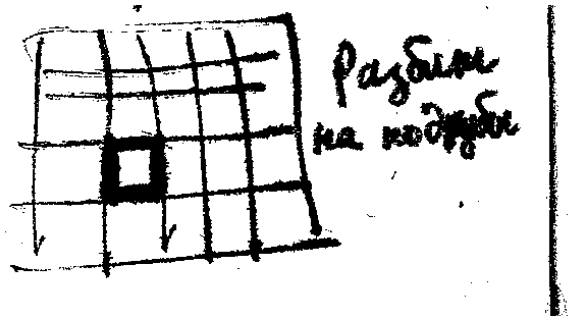


Original source details: GELFAND-P000010

**Uncertain reading.** In the line after "if ... holds", the word looks like "сформ." [stated] or "строгое" [strict]; nothing is supplied from the meaning.

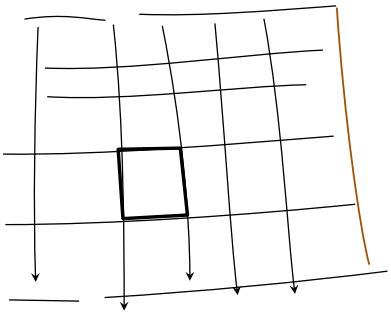
2) Lemma 2. If almost all arrangements are special, then there is a phase transition. Idea of the proof of Lemma 2. $G$ is the volume of the region enclosed by the small cycles.

$$G \leq N_\ell(1-\delta), \qquad \frac{G}{V_\ell} \leq \frac{1}{v}(1-\delta).$$




Original source details: GELFAND-P000010

**Diagram and typed description.** A grid of five columns and five rows; the second cell from the left in the fourth row from the top is marked. The left edge and four internal vertical lines have downward tips. Two upper horizontals stop short of the right edge; its lower endpoint is thickened and its exact shape is unclear. Caption: “Divided into subcubes”.

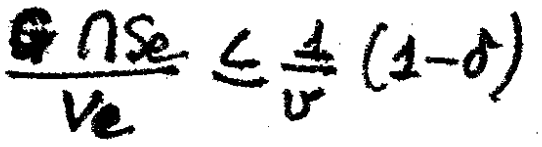


Divided into subcubes

Five downward tips; the uncertain right endpoint is a separate stroke.

There exists a subcube $S_\ell$:

$$\frac{G \cap S_\ell}{V_\ell} \leq \frac{1}{v}(1-\delta).$$

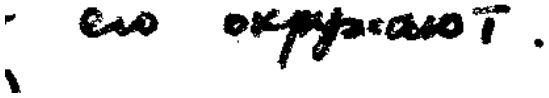


Original source details: GELFAND-P000010

**Uncertain reading.** In the numerator, the heavy sign before $\cap S_\ell$ resembles $G$; the denominator clearly resembles $V_\ell$.

GELFAND-P000011

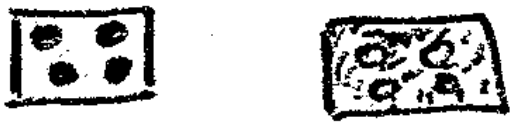
Let all large cycles surround co / $\infty$.



Original source details: GELFAND-P000011

**Uncertain reading.** surround co / $\infty$. [The two-loop sign before the verb is uncertain.]

2 cases.



Original source details: GELFAND-P000011

**Diagram and typed description.** Two small rectangular frames, each with four dark rounded marks. The right frame has light centres, an irregular chain of strokes above, and detached short and curved lines between the rounded marks and near the edges. Their exact connections and possible arrowheads are indistinct.



Visible interior strokes; connections and directions remain uncertain.

Lemma. [Непрс.] a boundary of $k$ [cancelled: точк.; inserted above/below: links] $\leq 3^k$.



**Uncertain reading.** Непрс. [uncertain source abbreviation]

Original source details: GELFAND-P000011



**Uncertain reading.** точк. [points; uncertain cancelled reading]

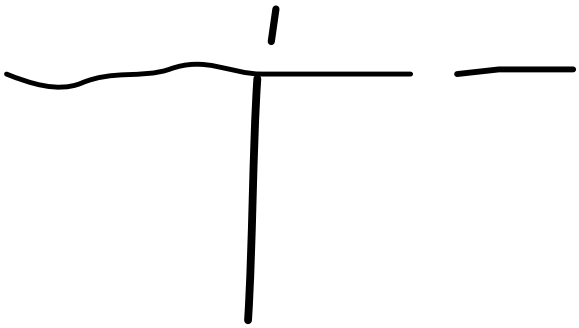
Original source details: GELFAND-P000011

Gain $3$, loss.



**Diagram and typed description.** Beneath the words about gain and loss is a wavy, almost horizontal line, a gap before a separate right segment, and a long vertical segment extending downward. A detached short stroke lies above their junction; it has not been identified as a letter or arrow.

Original source details: GELFAND-P000011



The detached upper stroke is retained without assigning a letter or arrow.

$$\exp\left\{-\frac{1}{2T}\right\}, \qquad \left(3\exp\left\{-\frac{1}{2T}\right\}\right)^k.$$

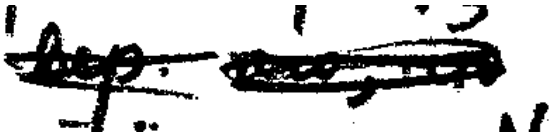
Problem. (unsolved) Let $N$ be large. Turn with probability $\frac{1}{3}$, but returning is forbidden [Cancelled: not].



**Uncertain reading.** *Cancelled.* не [not; uncertain candidate for a short word]

Original source details: GELFAND-P000011

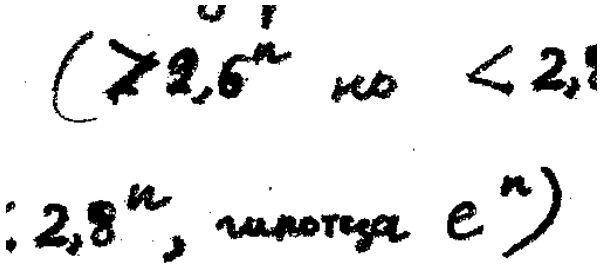
What is [Cancelled: prob. ...] the expected distance he will travel in $N$ steps?



Original source details: GELFAND-P000011

**Uncertain reading.** *Cancelled.* вер. ... [prob.; candidate for the first abbreviation; the following cancelled writing is unreadable]

(If returning is allowed, then $\sim \sqrt{n}$.) If returning is allowed, then
($\geq 2{,}6^n$ but $< 2{,}8^n$, conjecture $e^n$).



Original source details: GELFAND-P000011

**Uncertain reading.**
($\geq 2{,}6^n$ but $< 2{,}8^n$, conjecture $e^n$); the first sign may also read $>$; additional strokes are uncertain

[Horizontal divider.]

28/IX Segal. Theory of functions on the set of solutions of a relativistic differential equation. [In the left margin: They are all written this way.]

$$u' = Au + K(u) \qquad (A) \qquad u(t), \quad -\infty < t < +\infty,$$

$$A = -A^*, \qquad u \in L^2, \qquad u(t) \in H.$$

$K$ is a given nonlinear operator. In integral form

$$u(t) = e^{A(t-t_0)}u(t_0) + \int_{t_0}^{t} e^{A(t-s)}K(u(s))\,ds \qquad (B).$$

When $K \in C^1$, i.e. $\partial_u K = K_u$ exists in the Fréchet sense and is continuous, every solution of (B) is a solution of A (if $u(t)$ is in the domain of A).

[Horizontal divider.]

Suppose (B) has a global solution for every initial datum at $t_0$ ($u_0$). (And $K \in C^1$.) In quantum mechanics

$$\Box\varphi = f(\varphi)$$

[Beneath □: d'Alembertian.]

$$u(t) = [\varphi(\cdot, t), \dot{\varphi}(\cdot, t)], \qquad \varphi = \varphi(\vec{x}, t).$$

$K$ is a nonlinear operator, Немыц. / Намыц.



Original source details: GELFAND-P000011

**Uncertain reading.** Немыц. / Намыц. [Nemyts. / Namyts.; the final source abbreviation is uncertain]

$$\|[\varphi,\omega]\| \qquad \Box\varphi = m^2\varphi + f(\varphi)$$

[Beneath the formula: A shorter form.]

$$B = \sqrt{m^2 I - \Delta}, \qquad \|[\varphi,\omega]\| = \|B\varphi\|^2_{L^2} + \|\omega\|^2_{L^2}.$$



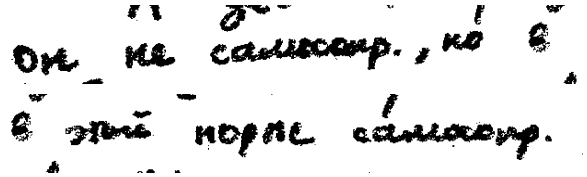
Original source details: GELFAND-P000012

**Uncertain reading.** No square is visible above the norm on the left of the last formula; the right side contains squares.

$A$ is given by a matrix of order 2,

$$\begin{pmatrix} 0 & I \\ -B^2 & 0 \end{pmatrix}.$$

It is not self-adjoint, but in this norm it is self-adjoint.



Original source details: GELFAND-P000012

**Uncertain reading.** Both places read “самосопр.” [self-adjoint]; $A = -A^*$ is written above.

Any equation of relativistic physics can be written this way.

$M$ is the collection of all solutions.

$$M = [u(\cdot);\ (**) \text{ satisfies}]$$

[At right/beneath the formula: of $t$; various B.]



Original source details: GELFAND-P000012

**Uncertain reading.** различные B. [various B.; uncertain reading of the underlined note]

Now let $K \in C^\infty$. $M$ is a Banach manifold (every point has a neighbourhood homeomorphic to a Banach [Cancelled: illegible] space; the transition is a $C^\infty$ function in the Fréchet sense).



Original source details: GELFAND-P000012

**Uncertain reading.** *Cancelled.* unreadable word

$$S_t : M \longrightarrow H, \qquad u(\cdot) \longmapsto u(t).$$

[At right: operator. set of sol. $\longrightarrow H$]



Original source details: GELFAND-P000012

**Uncertain reading.** мн. реш. $\longrightarrow H$ / им. реш. $\longrightarrow H$ [set of sol. / им. sol.; the first source abbreviation is uncertain; the final arrow and $H$ are visible]

Theorem. There exists a unique $C^\infty$ manifold structure for which the map $S_t$ is a $C^\infty$ isomorphism. $u(t, u_0)$ depends on $u_0$ in a $C^\infty$ manner.

A true manifold. There is a tangent space $T_{u(\cdot)}$. The tangent map

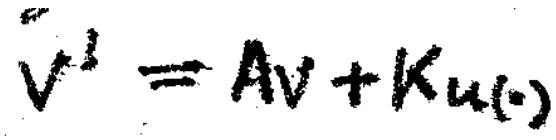
$$dS_t : T_{u(\cdot)} \longrightarrow T_{u(t)}.$$

Of course, at every point it is isomorphic to the space itself.

$$\lambda \in T_{u(\cdot)}, \qquad dS_t(\lambda) = v(t) \in H.$$

Thus a function corresponds to each tangent vector. [Underlined and cancelled: Lemma.] Denote by $v(t)$ (not the value of $v$ at the point $t$); variational equation, an equation of first order:

$$v' = Av + K_{u(\cdot)}.$$



**Uncertain reading.** No separate factor $v$ is visible after $K_{u(\cdot)}$ on this line.

Original source details: GELFAND-P000012

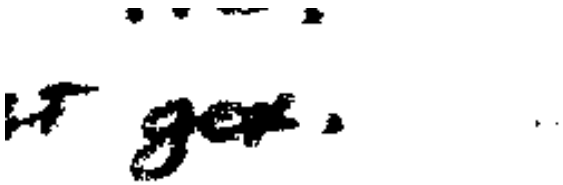
Solutions of this equation arise from tangent vectors $\lambda$, and conversely.

1) Canonical differential form on $M$.

2) Poisson brackets (analogy with the finite-dimensional case).

GELFAND-P000013

3) Vector fields in $M$. In their terms there will be зд.



**Uncertain reading.** зд. [the final source abbreviation is uncertain]

Original source details: GELFAND-P000013

4) Canonical commutation relations.

5) A representation and an explicit definition of the operators will be given.

5/X. Segal. I. Renormalized field equation. 1) Derivation. 2) Existence and uniqueness.

II. Definition of the scattering matrix $S$. A. Classical $S$ operator. B. Extension of generalized probability distributions to the nonlinear case.

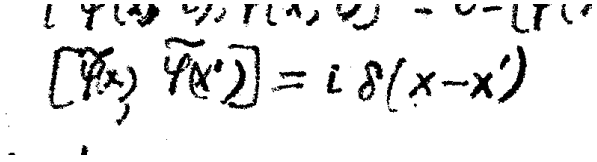
III. Discussion of interpretation in terms of particle theory.

[Horizontal divider.]

Consider the equation

$$\Box\widetilde{\varphi} = m^2\widetilde{\varphi} + \widetilde{\varphi}^3,$$

$$[\widetilde{\varphi}(\vec{x}, t), \widetilde{\varphi}(\vec{x}', t)] = 0 = [\dot{\widetilde{\varphi}}(\vec{x}, t), \dot{\widetilde{\varphi}}(\vec{x}', t)].$$



**Uncertain reading.** Candidates for the source line: $[\widetilde{\varphi}(x), \widetilde{\varphi}(x')] = i\delta'(x - x')$ / $[\widetilde{\varphi}(x), \dot{\widetilde{\varphi}}(x')] = i\delta(x - x')$. The dot above the second letter, the additional stroke after $\delta$, and the exact placement of the arguments are not fully distinguishable.

Original source details: GELFAND-P000013

Loran Schwartz. There are 2 types of measures: Radon measures and simply measures [inserted above the last word: abstract]. For now, restrict to positive measures. An abstract measure on a set $X$, $F$ is a $\sigma$-algebra. $\mu$ is a measure. $\mu$ is $\sigma$-additive on $F$.

[Horizontal line.]

Once a measure is given, one can consider the $\mu$-measurable sets, $F_\mu$ on $X$, and integrate measurable functions.

$$f : X \longrightarrow F, \qquad \int_X f(x)\, d\mu(x).$$

[At right: values in a Banach space; $F$ is a Banach space.]

[Horizontal line.]

Radon measure. $X$ is an arbitrary locally compact space. $C(X)$ consists of continuous functions with compact supports, with the inductive-limit topology, and let $C'(X)$ be the dual space. Its elements are measures on $X$.

GELFAND-P000014

$\mu$ is positive if $\mu(\varphi) \geq 0$ for $\varphi \geq 0$. A measure is bounded if

$$|\mu(\varphi)| \leq C \max |\varphi|.$$

The 2nd definition is a bounded positive Radon measure. The same Lebesgue extension is performed with it. The integral also exists (with values in a Banach space).

The $\sigma$-field constructed from a Radon measure depends on $\mu$, but there exists one independent of $\mu$ (substantially): the field of Borel sets (min containing closed and open sets).



**Uncertain reading.** min containing closed and open sets [the beginning of the parenthetical note is uncertain]

Original source details: GELFAND-P000014

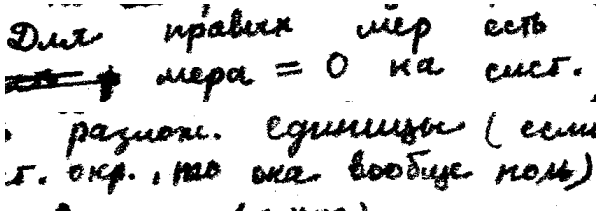
[Horizontal divider.]

Why do various mathematicians prefer Radon measures? Radon measures exist only on locally compact spaces (a restriction in probability theory). But [Cancelled: two illegible words] Radon measures are more convenient.



**Uncertain reading.** *Cancelled.* two short unreadable words

Original source details: GELFAND-P000014



**Uncertain reading.** For “right” measures there is разлож. / различ. единицы [decomp. / различ. of unity] (if [cancelled beginning] a measure $= 0$ on a system of neighbourhoods, then it is zero everywhere). [The first description of the measures and the abbreviation before “unity” are uncertain.]

Original source details: GELFAND-P000014

There are 5–15 theorems true for Radon measures (only).

$$X \xrightarrow{f} Y$$

[Beneath the arrow: topological spaces.] [Cancelled.]

**Uncertain reading.** a short word before “Select”



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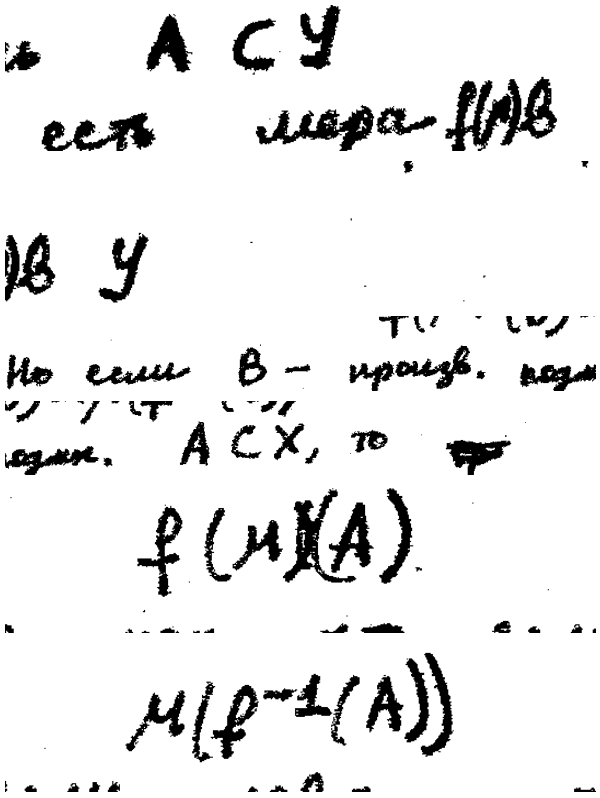
Select the Borel field on $X$ and on $Y$, and an abstract measure $\mu$ on $X$. Let $A \subset Y$. Then there is a measure $f(\mu)$ on $Y$:

$$f(\mu)(B) = \mu(f^{-1}(B)).$$

But if $B$ is an arbitrary subset $A \subset X$, then [one sign cancelled]

$$f(\mu)(A) \qquad \mu(f^{-1}(A)).$$

**Uncertain reading.** The source alternates between $A$ and $B$, and $A \subset X$ is read before the second pair of formulas.




Original source details: GELFAND-P000014

One can show that if the left side makes sense, the right side does too and they are equal; the converse is false. [Cancelled.]

**Uncertain reading.** an unreadable beginning of a line



Original source details: GELFAND-P000014

$$A \text{ — } f(\mu)\text{-measurable} \implies f^{-1}(A) \ \mu\text{-measurable}$$

If $\mu$ is a Radon measure, there is equivalence.

Example (contradictory)

$$X \subseteq Y = [0,1]$$

such that $X$ is a Lebesgue-nonmeasurable set, $Y \setminus X$ has inner measure $0$ (or the outer measure of $X = 1$). A subset of $X$ is Borel $\Longleftrightarrow$ it is an intersection with a Borel set of $Y$. $\mu$ is an abstract measure.

GELFAND-P000015

Definition. $\mu(D)$. $D \subset X$ is Borel.

$$D = X \cap E, \qquad E \text{ — Borel in } Y,$$

$$\mu(D) \underset{\text{def}}{=} dx(E)$$

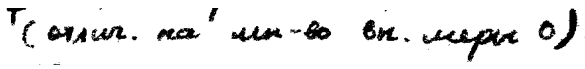
[Beneath $dx(E)$: Lebesgue measure.]



Original source details: GELFAND-P000015

**Uncertain reading.** In $dx(E)$ both letters are on the baseline; the case of the second letter is not fully distinguishable: $dx(E)$ / $dX(E)$.

The measure is the same (differs on a set of вн. measure 0).



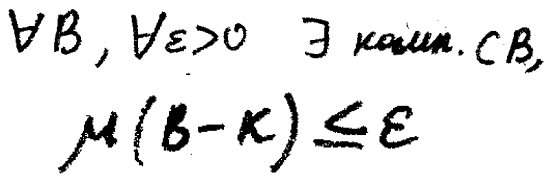
Original source details: GELFAND-P000015

**Uncertain reading.** отлич. на мн-во вн. меры 0 [differs on a set of вн. measure 0; uncertain parenthetical source note]

We have defined an abstract measure on $X$. Its image [The following line repeats: its image] is $dx$. Take some (non-Borel) subset of $Y \setminus X$. It is nonmeasurable, but its inverse image is empty and hence measurable.

The aim of the talk is to show that local compactness is unnecessary. Let $\mu$ be a measure on $F_B$; then $\mu$ is a Radon measure if it is inner compact regular.

$$\mu \text{ Radon} \Longleftrightarrow \forall B,\ \forall \varepsilon > 0\ \exists \text{ compact } \subset B, \qquad \mu(b - k) \leq \varepsilon.$$



Original source details: GELFAND-P000015

**Uncertain reading.** Lowercase $b - k$ is read in the difference, although $\forall B$ precedes it. No separate set letter is visible between “compact” and $\subset B$.

[Horizontal divider.]

$X$ is a topological space. Find a measure with all the properties of locally compact Radon measures. Suppose it is defined (set up) on Borel sets. A definition of a Radon measure valid beyond locally compact spaces. Let $X$ be completely regular (such spaces can be embedded in compact spaces of maximal ideals).

$$X, \qquad \widehat{X} \text{ — Čech completion of } X$$

(maximal ideals of the algebra of continuous functions).

$$X \xrightarrow{i} \widehat{X}.$$

Complete regularity means that this is a homeomorphic map, rather than merely an injective map. Another definition of a Radon measure on $X$: a Radon measure on $\widehat{X}$ for which

$$\mu(\widehat{X} \setminus X) = 0.$$

One can show equivalence. In the locally compact case.

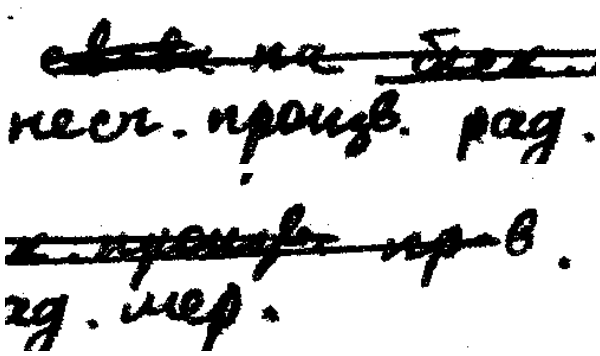
$$\mu \text{ Rad} \iff \forall B \text{ compact } \forall \varepsilon > 0 \; \exists \text{ compact} \quad \mu(B - k) < \varepsilon.$$



Original source details: GELFAND-P000015

**Uncertain reading.** The second sign in the difference resembles lowercase $k$; the first reads as $B$, but its case is uncertain ($B - k$ / $b - k$). No separate set letter or inclusion condition is visible after “compact”.

All the properties of Radon measures hold. The theorem’s proof is valid in $\widehat{X}$, and the rest is almost trivial. There is a single exception: [Cancelled: properties ... Borel ...] spaces.



Original source details: GELFAND-P000015

**Uncertain reading.** *Cancelled.* св-ва ... бор. ... [properties ... Borel ...; visible fragmentary candidates beneath the horizontal stroke; the complete cancelled phrase is unreadable]

— one cannot define the support of arbitrary Radon measures.

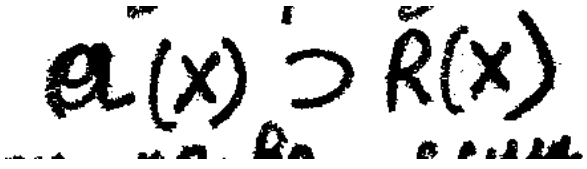
GELFAND-P000016

$X \subset K_0$ into a compact space (any one). Then an analogous definition of Radon measures exists. We obtain the same law.

[Horizontal divider.]

Compare with the definition of a nuclear space (spaces in which the 2 definitions of the tensor product coincide).

$$a(X) \supset R(X).$$



Original source details: GELFAND-P000016

**Uncertain reading.** The first symbol in the inclusion resembles a lowercase $a$ with a large loop; $A$ or a calligraphic form is not wholly excluded.

A Radon space if every abstract measure on it is a Radon measure.

3 properties of Radon spaces. 1) If $X$ is Radon and $Y \subset X$, then $Y$ is Radon $\iff$ it is universally measurable in $X$ (measurable with respect to all Radon measures—for example, Borel, Suslin and analytic).

An important criterion. 2) If $X$ is Radon, then it is universally measurable, i.e. measurable in every containing space and for every measure.



Original source details: GELFAND-P000016

**Uncertain reading.** The source abbreviation after “measurable ∀” reads объемл. / объем.; the ending and a possible superscript stroke are not fully distinguishable. It has not been expanded to a complete source word.

Probability theory—probability is always defined.

3) An example of a broad class of Radon spaces. Distributions, continuous functions, etc.? Let $E$ be a T. L. C.—a separable Fréchet space or their countable inductive limit. $F$ is of the same kind. Then Hom$(E, F)$ is of the same kind (equipped with any topology between pointwise and compact convergence). In particular, the dual space, the space itself ($R \longrightarrow E$). All spaces of generalized functions, of $C^\infty$ functions. And the operator space $\mathcal{L}(S, S)$ is also a Radon space.



Original source details: GELFAND-P000016

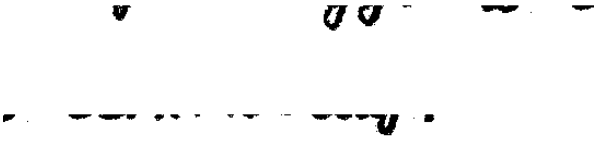
**Uncertain reading.** Above or beside the closing parenthesis is a short note “ф-и” [functions]. A separate stroke after $\mathcal{L}(S, S)$ cannot be securely distinguished from the note and parenthesis; the candidate $\mathcal{L}(S, S)'$ remains uncertain.

Application. Consider a stochastic process given as a probability on the space of all functions with simple (pointwise) convergence (a measure in this space). Take the subspace of continuous functions. It is measurable in $R^R$.

GELFAND-P000017

Hence the probability that a function is continuous always exists. Let it equal $1$. Then there is a probability distribution on this subspace, and it can be defined simply as a Radon measure on $C$. It is concentrated on a countable number of compact sets (equicontinuous). If the probability that a function is continuous is $1$, then with probability $1$ all these functions have a common modulus of continuity. This is true for every stochastic process (there exists a common mod. of continuity with probability $1$).

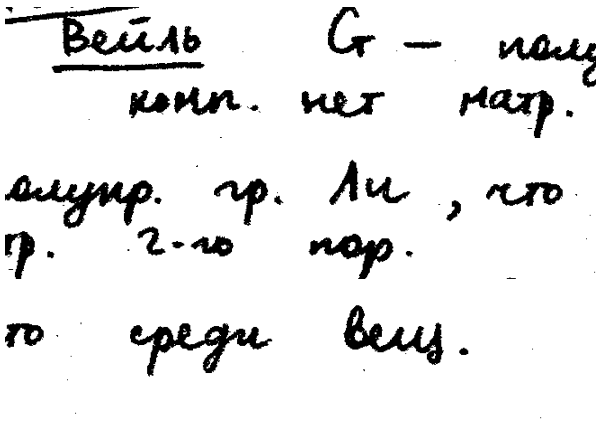
[A large unwritten area; isolated small horizontal or curved strokes are visible along its boundary in the middle of the frame.]



Original source details: GELFAND-P000017

**Uncertain reading.** Only lower parts of isolated letter-like strokes survive along the horizontal boundary of the large unwritten area; no complete word is readable.

$12/$X. Gindikin. Weil. $G$ is a semisimple Lie group such that among the real comp. there are no matrices of order 2.




**Uncertain reading.** The line reads “вещ. комп.” [real comp.] and “матр. 2-го пор.” [matr. of 2nd order].

Original source details: GELFAND-P000017

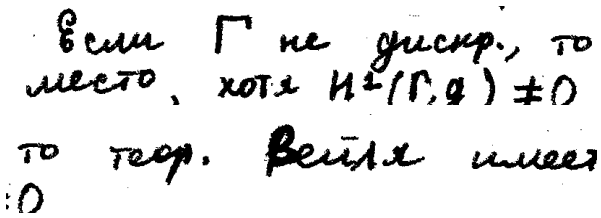
$\Gamma$ is a discrete subgroup; $G/\Gamma$ is compact. Consider deformations of $\Gamma$: $\gamma(t)$, with a discrete group $\Gamma^{(t)}$ for every $t$. A trivial method uses inner automorphisms:

$$g(t)\gamma g(t)^{-1}.$$

The quotient group is the same. Under these conditions, all deformations are trivial. It suffices to prove that

$$H^1(\Gamma, \mathfrak{g}) = 0.$$

$\mathfrak{g}$ is the Lie algebra, carrying a representation of $\Gamma$. If $\Gamma$ is not discrete, Weil’s theorem holds, although $H^1(\Gamma, \mathfrak{g}) \neq 0$.




**Uncertain reading.** The condition reads “не дискр.” [not discrete].

Original source details: GELFAND-P000017

There is a representation $\rho(\gamma)$ in $V^d$. Then the dimension is

$$p = (2g-2)d + i + i'.$$

GELFAND-P000018

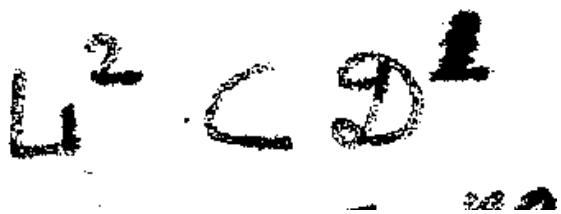
Laurent Schwartz.

$E$ — TVS LCH QC.

[A note leads from $H$ in the abbreviation: Hausdorff.] A locally convex topological vector space, quasi-complete (every closed bounded set is complete). $\mathcal{H}$ is a Hilbert subspace of $E$.

$$\mathcal{H} \subset E.$$

[Beneath the inclusion: the map is continuous; at right: in $\mathcal{H}$ there is a Hilbert metric.] The Hilbert topology does not coincide with, but is stronger than, the topology in $E$. Example.



**Uncertain reading.** The inclusion reads $L^2 \subset \mathcal{D}^1$ / $L^2 \subset \mathcal{D}'$; the heavy upper-right mark above $\mathcal{D}$ resembles both $1$ and a prime.

Original source details: GELFAND-P000018

Another example is the space of constants:

$$(\alpha \mid \beta) = \alpha\overline{\beta}.$$

$\mathcal{H} \in \mathrm{Hilb}(E)$ — all Hilbert subspaces.

[Repeated in the left margin: $\mathrm{Hilb}(E)$, with an arrow to the following heading.] An abstract conical structure. Addition and multiplication by positive numbers must be defined, and an order introduced.

$$\mathcal{H} \in \mathrm{Hilb}(E), \quad \lambda \geq 0, \quad \lambda\mathcal{H}, \quad \mathcal{H}_1 + \mathcal{H}_2, \qquad \mathcal{H}_1 \leq \mathcal{H}_2.$$

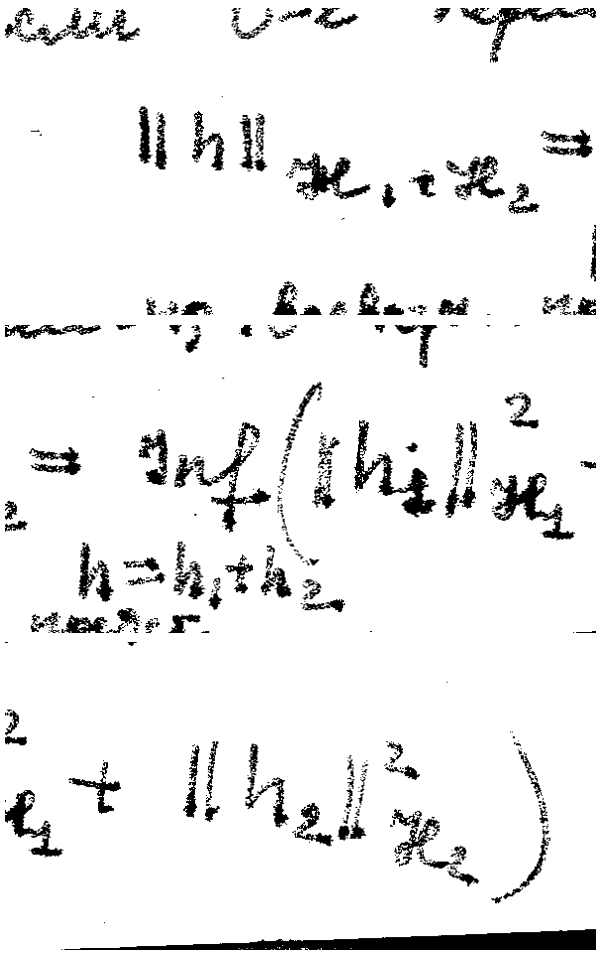
For $\lambda = 0$, $\lambda\mathcal{H}$ is the zero subspace. For $\lambda \neq 0$, $\lambda\mathcal{H}$ is the same space with scalar product

$$(h \mid k)_{\lambda\mathcal{H}} = \frac{1}{\lambda}(h \mid k)_{\mathcal{H}}.$$

$\mathcal{H}_1 + \mathcal{H}_2$. If the intersection is zero, it is a direct sum.

$$\|h\|_{\mathcal{H}_1,\mathcal{H}_2} = \inf\left(\|h_1\|^2_{\mathcal{H}_1} + \|h_2\|^2_{\mathcal{H}_2}\right), \qquad h = h_1 + h_2.$$

[Beneath the formula: over all possible representations.]



**Uncertain reading.** No square is visible on the left of the norm formula, whereas both norms on the right are squared. The subscript has a comma between the two Hilbert-space symbols.

Original source details: GELFAND-P000018

GELFAND-P000019

$$\mathcal{H}_1 \oplus \mathcal{H}_2.$$

A direct sum as abstract Hilbert spaces.

$$(h_1, h_2) \longmapsto h_1 + h_2.$$

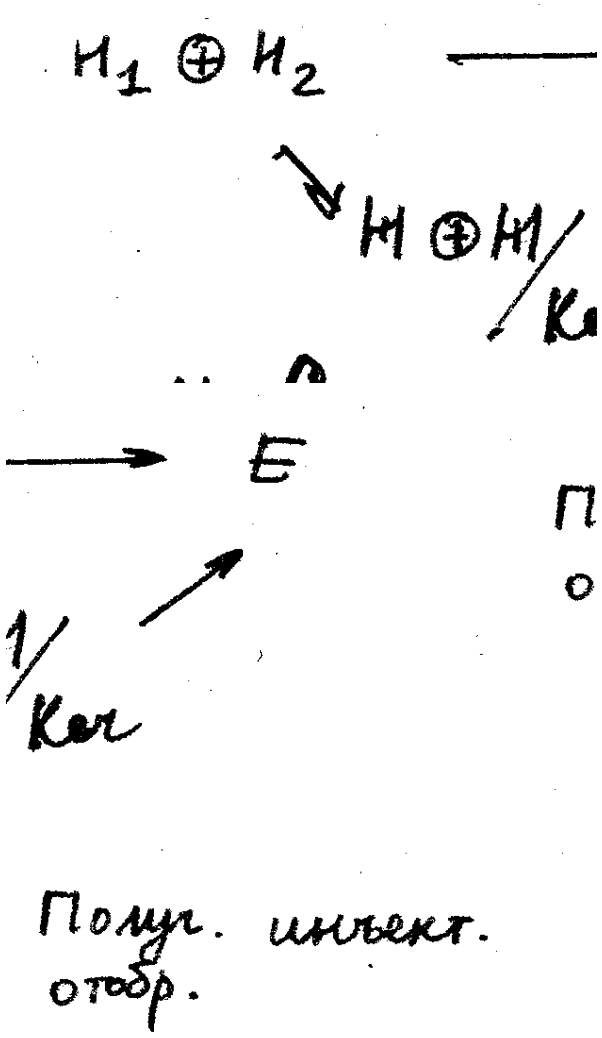
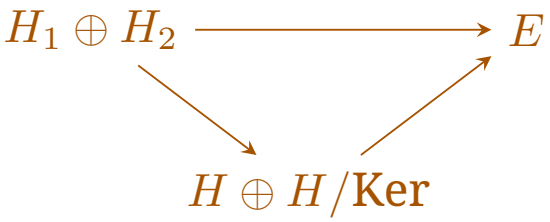



**Diagram and typed description.** A triangular diagram: at upper left $H_1 \oplus H_2$, at upper right $E$, and below the quotient $H \oplus H/\mathrm{Ker}$. The horizontal arrow runs from left to right into $E$; a diagonal arrow runs down from the direct sum to the quotient; the other diagonal arrow runs up from the quotient to $E$. The indices on the letters at the lower vertex are uncertain. At right: “Obtain an injective map.”




The lower Hs retain no reconstructed indices; the stroke above Ker is shown as a slash.

Original source details: GELFAND-P000019

The infimum is precisely the definition of the norm in a quotient space. But a quotient of a Hilbert space is itself a Hilbert space. Thus this is a Hilbert norm. Suppose

$$H_1 \subset H_2 \subset E,$$

then we say that $H_1 \leq H_2$. [Below the first inclusion, an arrow $i$: an operator of norm $\leq 1$.]

$$\|h\|_{H_1} \geq \|h\|_{H_2} > 0.$$

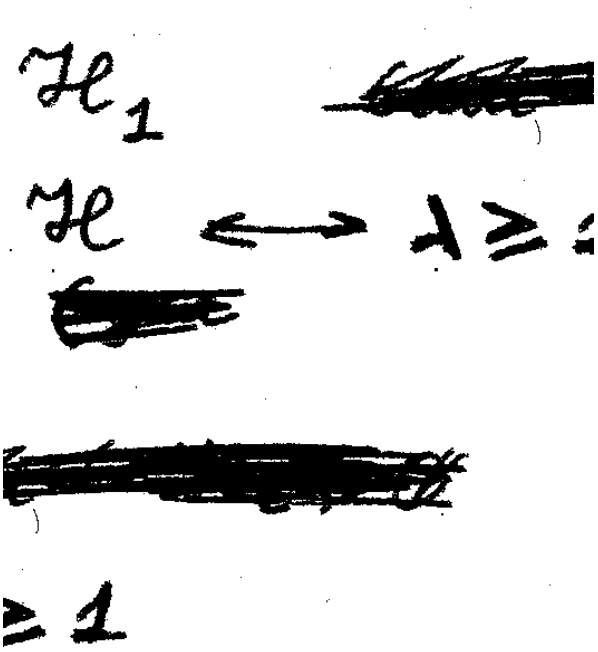
All the properties are easy to check:

$$(\lambda\mu)H = \lambda(\mu H), \qquad (\lambda+\mu)H = \lambda H + \mu H.$$

And all the others.

$$\mathcal{H}_1 + \mathcal{H}_2 \geq \mathcal{H}_1, \qquad \lambda\mathcal{H} \geq \mathcal{H} \iff \lambda \geq 1.$$

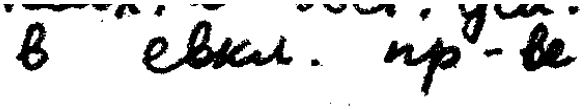
[At right of the first inequality and beneath the second line are heavily cancelled entries.]



**Uncertain reading.** the original letters and signs beneath the dense hatching are unreadable

Original source details: GELFAND-P000019

These are the axioms of a convex cone. There are necessary and sufficient conditions for realizing an abstract conical structure in евкл. / свкл. space.



**Uncertain reading.** евкл. / свкл. space [the initial letter of the source abbreviation is uncertain]

Original source details: GELFAND-P000019

GELFAND-P000020

[The first word of the heading is cancelled: Strict; inserted above: strictly convex.] abstract cone:

$$h \geq h_1 \iff \exists h_2 \text{ unique (!), such that } h = h_1 + h_2.$$

It will be proved that the cone of Hilbert subspaces can be embedded in a linear space; in particular, this condition holds. One must prove that if $H_1 \leq H$, there is a unique $H_2$ such that $H = H_1 + H_2$. Instead, a cone in a linear space will be constructed to which this one is isomorphic.

$$E \text{ — space of kernels}, \qquad H \in \mathcal{L}(E^*, E).$$

$\mathcal{L}$ is the space of linear operators from the antidual space (antilinear [Cancelled: illegible] functionals) to $E$, with the weak topology in $E$.



**Uncertain reading.** *Cancelled.* an unreadable short word

Original source details: GELFAND-P000020

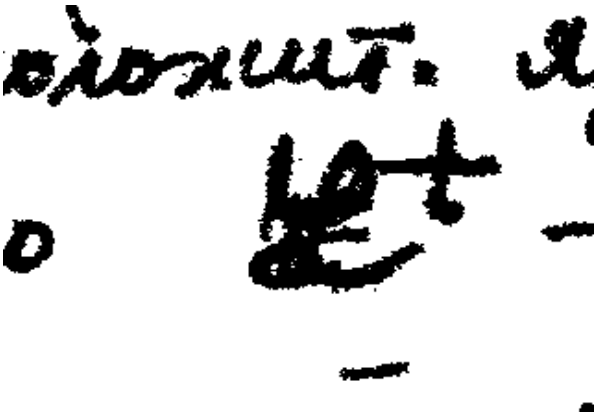
By definition,

$$H \geq 0 \underset{\text{def}}{\iff} \forall e^* \in E^*, \qquad (He^* \mid e^*)_{E,E^*} \geq 0.$$

$$e^* \in E^*, \quad f \in E, \qquad (e^* \mid f)_{E^*,E} = \overline{(f \mid e^*)_{E,E^*}}.$$

$$\mathcal{L}^+(E^*, E) \subset \mathcal{L}(E^*, E).$$

[Beneath the formula: the cone of positive kernels.] Example. $E$ is a Hilbert space; $\mathcal{L}^+$ denotes positive Hermitian operators.



**Uncertain reading.** $\mathcal{L}^+$ [the corrected or overwritten symbol is uncertain]

Original source details: GELFAND-P000020

This is a strictly convex cone. The cone Hilb is isomorphic to it. Let $\mathcal{H} \in \mathrm{Hilb}(E)$. The corresponding kernel must be constructed.

$$e^* \in E^*; \qquad h \longmapsto (h \mid e^*)_{E,E^*}.$$

[Beneath the right side: $e^*$ is antilinear.]

GELFAND-P000021

This is a continuous linear functional on $H$ itself as well, since the embedding is continuous; but then $\exists He^* \in \mathcal{H}$ by means of which this scalar product is expressed.

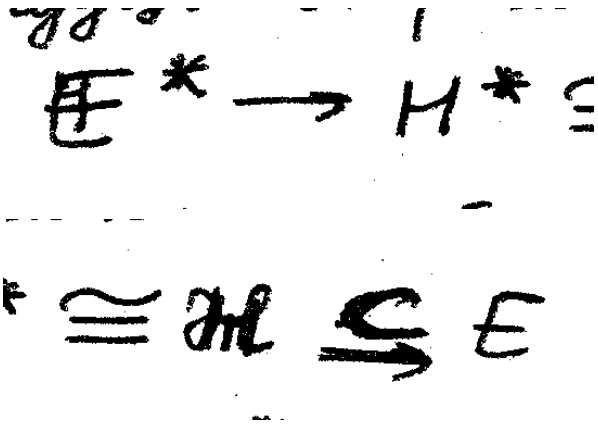
$$(h \mid e^*)_{E,E^*} = (h \mid He^*)_{\mathcal{H}}.$$

We have obtained a map $E^* \longrightarrow \mathcal{H} \longrightarrow E$. It is easy to check that this is a weakly continuous map from $E^*$ to $E$.

$$\mathcal{H} \overset{c}{\longrightarrow} E \qquad \text{injectively embedded.}$$

It induces a map

$$E^* \longrightarrow H^* \cong \mathcal{H} \overset{\subset}{\longrightarrow} E.$$



**Manuscript notation.** The inclusion sign is written above the arrow.

Original source details: GELFAND-P000021

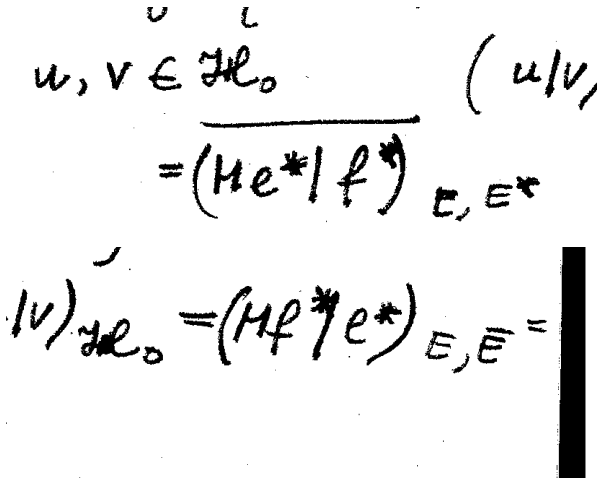
We have a map $E^* \longrightarrow E$, as required. Let $h = Hf^*$; then we obtain

$$(Hf^* \mid e^*)_{E,E^*} = (Hf^* \mid He^*)_{\mathcal{H}},$$
$$(He^* \mid e^*) \geq 0.$$

$$\mathrm{Hilb}(E) \longrightarrow \mathcal{L}^*(E^*, E).$$

One must prove that this is an isomorphism. It will be proved to be one-to-one and onto. Algebraically simple. Topological difficulties. If $H$ is the kernel corresponding to $\mathcal{H}$, then $\forall e^* \in E^*$

$$He^* \in \mathcal{H}, \qquad \mathcal{H}_0 = \{He^* : e^* \in E^*\},$$
$$u, v \in \mathcal{H}_0 \qquad (u \mid v)_{\mathcal{H}_0} = (Hf^* \mid e^*)_{E,E^*} = \overline{(He^* \mid f^*)}_{E,E^*}.$$



**Uncertain reading.** The relation of $u, v$ to $e^*, f^*$ is not written here; a bar is visible above the last scalar product.

Original source details: GELFAND-P000021

GELFAND-P000022

Mackey's theorem: a set is bounded if it is weakly bounded. Prove that the unit ball is weakly bounded. Examples.

$$E = \mathcal{D}', \qquad E^* = \mathcal{D}, \qquad \mathcal{H} = L^2.$$

$\mathcal{L}(\mathcal{D}; \mathcal{D}')$ is the space of kernels, operators from $\mathcal{D}$ to $\mathcal{D}'$. $\mathcal{H} = L^2$ defines a natural map $\mathcal{D} \longrightarrow \mathcal{D}'$. If $\mathcal{H}$ is arbitrary, we obtain an arbitrary positive operator, but by the kernel theorem it is given by a generalized function of 2 variables.

$$H_{x,\xi} \in \mathcal{D}'(R^n \times R^n), \qquad \iint H_{x,\xi}\varphi(x)\overline{\varphi}(\xi)\,dx\,d\xi \geq 0.$$

There is a correspondence between Hilbert subspaces of $\mathcal{D}'$ and such kernels. $\mathcal{H}^1$ is the space of functions, [Cancelled: illegible] $L^2$ with 1st derivatives.

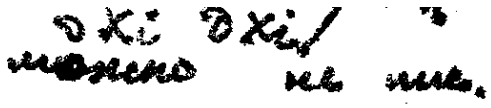


**Uncertain reading.** *Cancelled.* an unreadable word before $L^2$

Original source details: GELFAND-P000022

$$(f \mid g)_{\mathcal{H}^1} = \int \left(\lambda f\bar{g} + \sum \frac{\partial f}{\partial x_i}\frac{\partial \bar{g}}{\partial x_i}\right) dx.$$

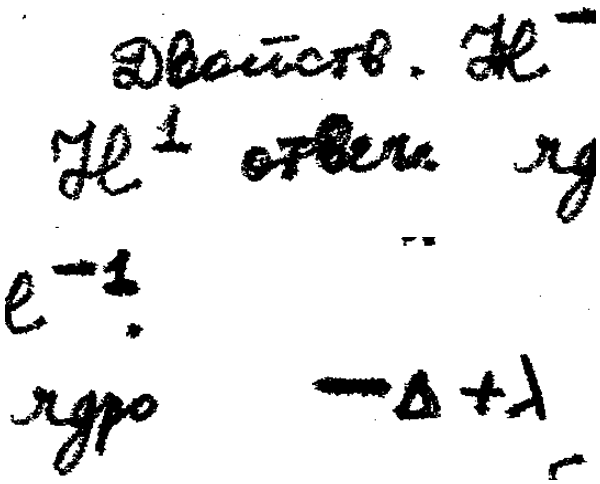
$\lambda > 0$ may be left [unwritten?]



**Uncertain reading.** The final abbreviation resembles «пис.» (written).

Original source details: GELFAND-P000022

The dual is $\mathcal{H}^{-1}$. To $\mathcal{H}^1$ corresponds the kernel $-\Delta + \lambda$.



**Manuscript notation.** The manuscript gives $-\Delta + \lambda$ as the kernel for $\mathcal{H}^1$.

Original source details: GELFAND-P000022

If we consider functions on a domain $\Omega \subset R^n$, these too are examples:

$$\mathcal{H}^{-1}, \quad \mathcal{H}^1, \quad \mathcal{H}_0^1 \quad \text{closure of } \mathcal{D}_0.$$

One can obtain existence theorems for solutions as generalized functions.

GELFAND-P000023



**Uncertain reading.** Several oblique strokes are cut off at the top edge; their meaning is unclear.

Original source details: GELFAND-P000023

Kushnirenko's result on the entropy of dynamical systems. Arnold. [Undeciphered cancellation] Entropy is the rate at which initial data are forgotten.



**Uncertain reading.** Heavy crossing-out obscures the middle of the entry. Its reading is unresolved.

Original source details: GELFAND-P000023

$$0 \qquad h \qquad +\infty.$$

Beneath $0$: with slow mixing. Beneath $+\infty$: strong mixing. Entropy for abstract dynamical systems (a measure-preserving transformation $T$). $M^n$ is a smooth manifold. $M^n$ is compact. $T$ is smooth, the measure of $M = 1$ and is given by a Riemannian metric. Then entropy is always finite and is estimated by local properties of the transformation.

$$h(T) \leq n \log \lambda.$$

$n$ is the dimension of the space. $\lambda$ is the expansion coefficient of $(n-1)$-dimensional area. The bound is attained.

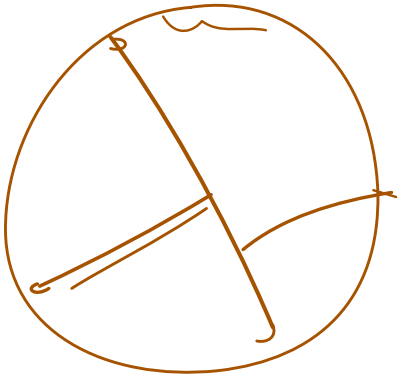
$$\lambda = \sup \frac{dS(Tx)}{dS(x)}.$$

Proof from the isoperimetric inequality. $h$ is estimated by area. Entropy of a partition: $h(\xi) = -\sum p_i \log p_i$.



Original source details: GELFAND-P000023

**Diagram and typed description.** At right is a circle with a diagonal divider from upper left to lower right, a doubled branch towards the lower left and an arc to the right. A narrow light slit separates the two left strokes, with a thickened left endpoint; a separate thick mark projects beyond the circle at the right edge. These thickened marks do not establish arrow directions.

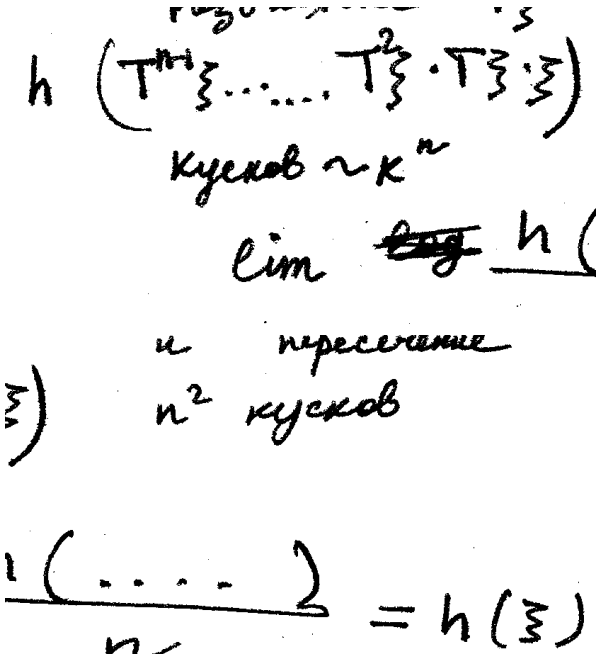


The doubled lower-left trace and terminal hooks are drawn without assigning arrow directions; their exact reading remains unresolved.

The partition $T\xi$ and the intersection.

$$h(T^{n-1}\xi \ldots\ldots T^2\xi \cdot T\xi \cdot \xi) \qquad n^2 \text{ pieces},$$
$$\text{Pieces } \sim k^n,$$
$$\lim \frac{h(\ldots)}{n} = h(\xi).$$

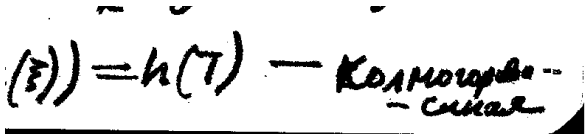
[Before the fraction, log is cancelled.]



**Uncertain reading.** Dots stand between $T^2\xi$, $T\xi$, $\xi$; alongside is written $n^2$ pieces. The limit's result resembles $h(\xi)$.

Original source details: GELFAND-P000023

For equal ones, $\sum \frac{1}{k} \log k = \log k$. Entropy of a dynamical system: $\sup(h(\xi)) = h(T)$—Kolmogoro-skaya [the source's split form].

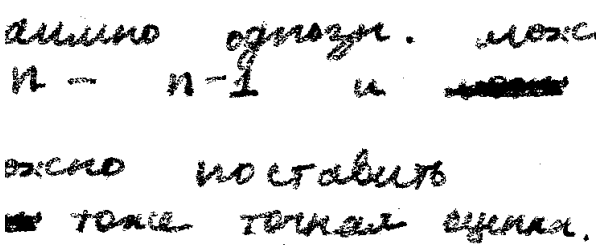


**Uncertain reading.** The final word is divided across lines as Колмогоро- / -ская; letters may be lost in the abbreviation.

Original source details: GELFAND-P000023

GELFAND-P000024

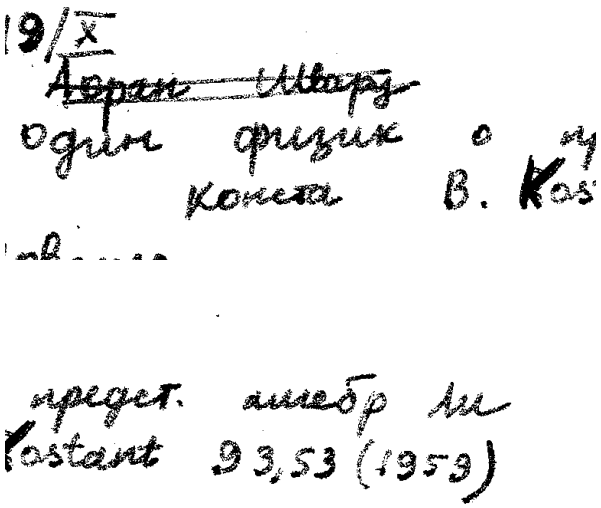
The entropy of a partition is estimated by the area of its boundary. And the boundary areas simply add! For a one-to-one transformation one can put $n - 1$ instead of $n$, and [Cancelled: illegible] there is again a sharp bound.



**Uncertain reading.** *Cancelled.* an unreadable word

Original source details: GELFAND-P000024

19/X. [Cancelled heading: Laurent Schwartz.] A physicist on representations of Lie algebras. [Kone...a?] B. Kostant 93, 53 (1959).

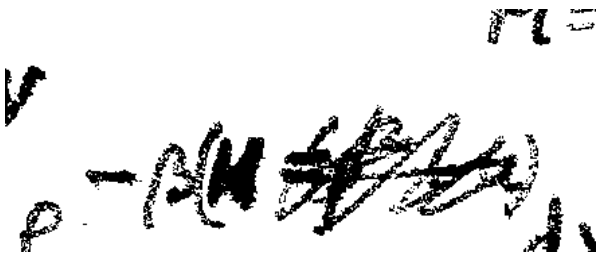


**Uncertain reading.** Before the Latin-script name, Коне is visible, apparently followed later by a. The intervening strokes are undeciphered.

Original source details: GELFAND-P000024

Povzner. Definition of free energy.

$$H = p^2 + u, \qquad \frac{-\ln \frac{1}{N!} \iiint_V^N e^{-\beta(H + \text{[cancelled]})}\, dx}{N}.$$



**Uncertain reading.** After $H$ inside the exponent are several cancelled signs; $+$ and overlapping strokes are visible.

Original source details: GELFAND-P000024

$N$ is the number of integrals.



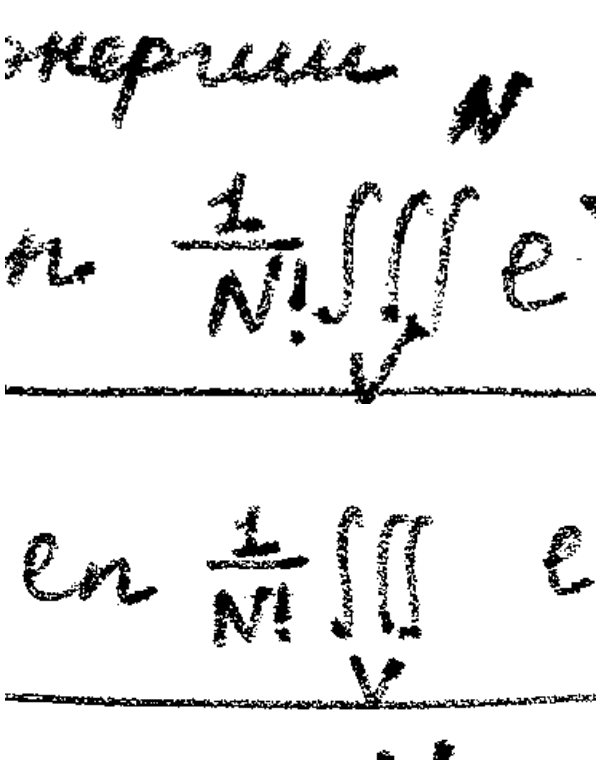
**Manuscript notation.** The manuscript writes «число интегралов» (number of integrals).

Original source details: GELFAND-P000024

$$\frac{V}{N} = \text{const}, \qquad \beta = \frac{1}{T},$$
$$\lim_{N\to\infty} \quad \text{and} \quad V/N = v = \text{const}.$$

free energy

$$\lim_{N\to\infty} \frac{-\ln \frac{1}{N!} \iiint_V e^{-\beta H}\, dx}{N}.$$



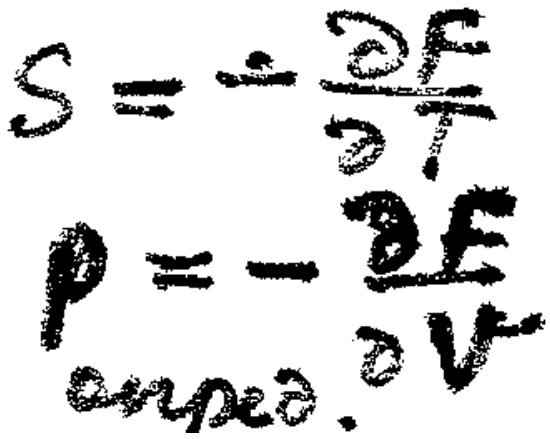
**Uncertain reading.** The first triple integral has $N$ above and $V$ below; only $V$ is distinguishable in the repeated formula.

Original source details: GELFAND-P000024

Entropy from the thermodynamic formulas:

$$S = -\frac{\partial F}{\partial T},$$

$$\text{pressure} \quad p = -\frac{\partial F}{\partial V}.$$

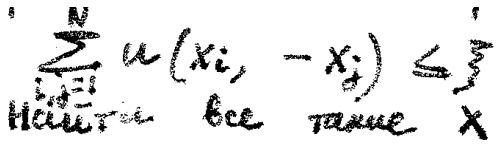


**Uncertain reading.** A dot is written above the minus in the formula for $S$. Its purpose is uncertain.

Original source details: GELFAND-P000024

Entropy can be defined as follows:

$$\frac{1}{N} \sum_{i,j=1}^{N} u(x_i, -x_j) \leq \xi.$$



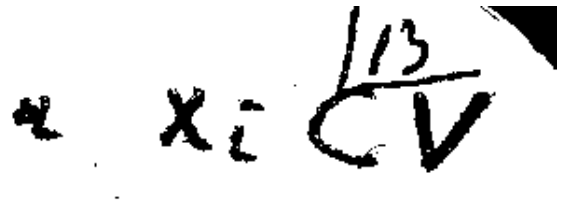
**Manuscript notation.** A comma follows $x_i$ and a minus precedes $x_j$. The letter on the right is $\xi$.

Original source details: GELFAND-P000024

Find all such $x$.

GELFAND-P000025

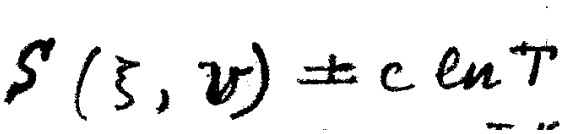
Take the volume of this region in phase space. $P$. In the upper-right margin: and $x_i \subset V$.



**Manuscript notation.** The upper margin has "and" and a curved inclusion sign without a middle bar.

Original source details: GELFAND-P000025

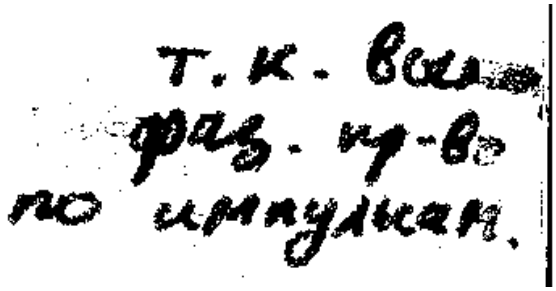
$$\lim_{N\to\infty} \frac{\ln \frac{1}{N!} P(\xi, V/N)}{N} = S(\xi, v) \pm c \ln T.$$



**Manuscript notation.** The manuscript writes $\pm$ before $c \ln T$.

Original source details: GELFAND-P000025

$\xi$ is energy. In the right margin: since весь / всё [all; the source ending is uncertain] phase space in the momenta.

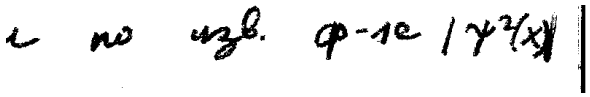


**Uncertain reading.** Весь / всё фаз. пр-во по импульсам [all phase space in the momenta; source grammar retained].

Original source details: GELFAND-P000025

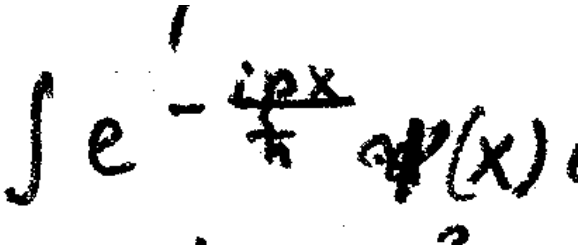
$S$ is a convex function of 2 variables. On Wigner's work (1932) on quantum mechanics. Let $\Psi(x)$ give the probabilities by the familiar formula $|\Psi^2(x)|$ and

$$\widetilde{\Psi}(p) = \int e^{-ipx/\bar{h}} \Psi(x)\, dx.$$



**Manuscript notation.** The first line places the exponent inside the absolute value signs, $|\Psi^2(x)|$.

Original source details: GELFAND-P000025



**Manuscript notation.** The $h$ in the exponent denominator has a cross stroke.

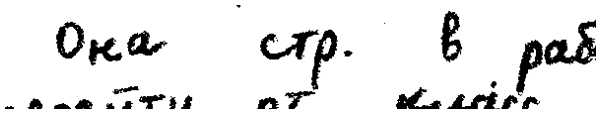
Original source details: GELFAND-P000025

If

$$\int P(x,p)\,dp = |\Psi(x)|^2,$$

$$\text{and} \quad \int P(x,p)\,dx = |\widetilde{\Psi}(p)|^2,$$

then what is $P$? It is [стр.; uncertain source abbreviation] in Wigner's work, and from this one can pass from classical mechanics to statistical mechanics.



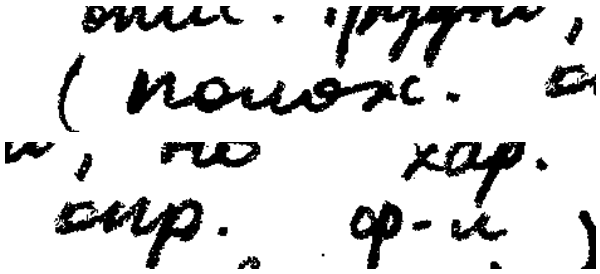
**Uncertain reading.** The manuscript writes «Она стр.». The abbreviation is unresolved.

Original source details: GELFAND-P000025

Kirillov on Thoma's work. Elmar Thoma. Math. Zeitschr. 84 (1964). If $G$ is a discrete noncommutative group, its representations are difficult to describe, but the characters can be found (positive-definite functions).

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}.$$

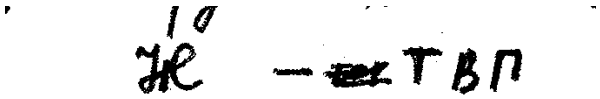
The characters are the same (the same formulas as over the real, finite and $p$-adic numbers).



**Manuscript notation.** The manuscript uses the abbreviation «полож. опр. ф-и».

Original source details: GELFAND-P000025

Laurent Schwartz. $s \in G$—a Lie group. $dx$ is Haar measure (two-sided). $\mathcal{H}$ is a [Cancelled: illegible] topological vector space.

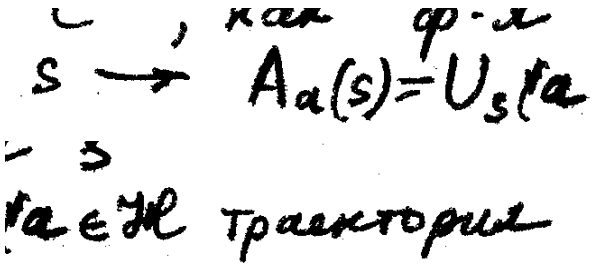


**Uncertain reading.** *Cancelled.* an unreadable word

Original source details: GELFAND-P000025

$U_s$ is a representation in it. $a \in \mathcal{H}$, which is infinitely differentiable with respect to the group; its orbit is $C^\infty$ as a function of $s$.

$$s \longrightarrow A_a(s) = U_s(a \in \mathcal{H} \qquad \text{trajectory.}$$



**Uncertain reading.** Between $U_s$ and $a$ is an opening delimiter stroke: ( / {. No closing delimiter is visible on this line.

Original source details: GELFAND-P000025

$A_a : G \longrightarrow \mathcal{H}$, which is $C^\infty$. Instead of the representation $U_s$, one can consider $U_\mu$ and $U_\varphi$.

$$U_\mu = \int U_s \, d\mu(s).$$

Trajectory

$$U_s A_a(t) = A_a(st).$$

If there is such a curve, then it is an orbit, i.e.

$$\mathcal{H} \supset \mathcal{H}^m \supset \mathcal{H}^\infty.$$

By $\mathcal{H}^m$: the trajectory is $m$ times differentiable. By $\mathcal{H}^\infty$: the trajectory is $C^\infty$. One can prove that $\mathcal{H}^{+\infty}$ is dense in $\mathcal{H}$ (by averaging). Define

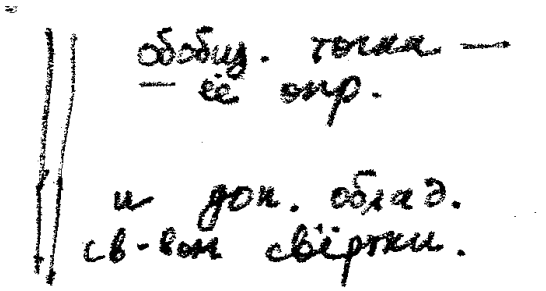
$$\mathcal{H}^{-\infty} \supset \mathcal{H} \supset \mathcal{H}^m \supset \mathcal{H}^{+\infty}.$$

Point generalized functions. We shall consider $U_s A_a(t) = A_a(st)$ as a distribution with respect to Haar measure.

$$U_\varphi A_a(\psi) = A_a(\varphi * \psi).$$

Generalized points of the representation space.

$$A : \mathcal{D}'(G; \mathcal{H}),$$
$$\forall \varphi \in \mathcal{D}(G), \quad A(\varphi) \in \mathcal{H},$$
$$U_\varphi A(\psi) = A(\varphi * \psi).$$

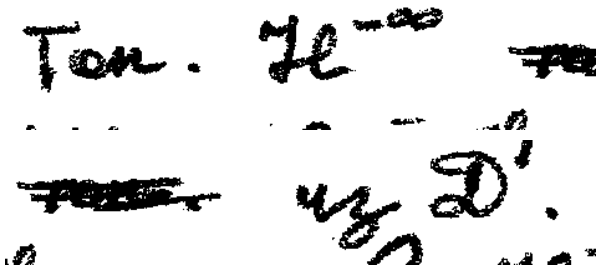


Original source details: GELFAND-P000026

**Diagram and typed description.**

Generalized point—
—its definition.
and до[л. / п.] [uncertain abbrev.] possess
the convolution property.
The last letter of the abbreviation resembles л or п.

In particular, ordinary points possess these properties. Topology of $\mathcal{H}^{-\infty}$ [Cancelled: illegible] from $\mathcal{D}'$. $\mathcal{H}^{-\infty}$ will be complete.

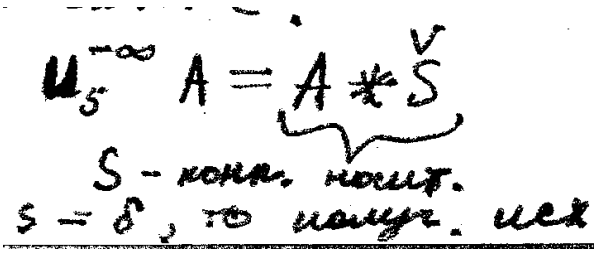


Original source details: GELFAND-P000026

**Uncertain reading.** *Cancelled.* an unreadable word

The group also acts in $\mathcal{H}^{-\infty}$. There the group acts $C^\infty$.

$$U_S^{-\infty} A = \underbrace{A * \check{S}}_{S \text{ — finite supp.}}$$




**Uncertain reading.** The convolution and its caption both use $S$. The index resembles $S$, with a short detached stroke to its right.

Original source details: GELFAND-P000026

If $S = \delta$, then the original representation is obtained.

GELFAND-P000027

In particular, the Lie algebra also acts ($\delta'$).

. . . . . .

Problem. $\Box X = A$, an equation with constant coefficients. This equation is solvable (Ehrenpreis). Let $A$ be Lorentz invariant. Then there exists a Lorentz invariant $X$.

$$\mathcal{D}X = A.$$

The same for an arbitrary group and differential operator—a problem!



**Uncertain reading.** In the repeated formula, the letter before $X$ is rounded: $\mathcal{D}$ / $D$; in the first formula there is a rectangle □.

Original source details: GELFAND-P000027

Solution for the Lorentz group.

$$\Box X = A.$$

One must [Cancelled: illegible] study Lorentz invariant functions.

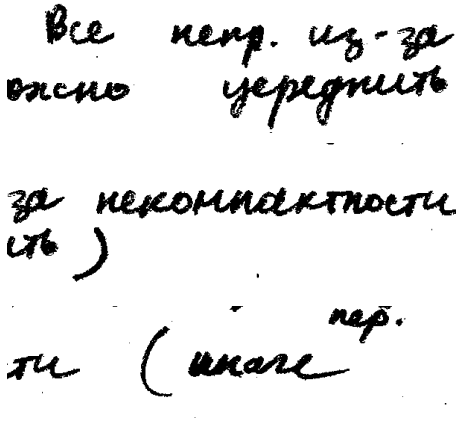


**Uncertain reading.** *Cancelled.* a short unreadable word

Original source details: GELFAND-P000027

Consider $R^n \setminus 0$. This is $T(x_1^2 + \cdots + x_n^2 - x_0^2)$ of one variable.

All непр. [unexpanded source abbreviation] because of noncompactness (otherwise one can average).

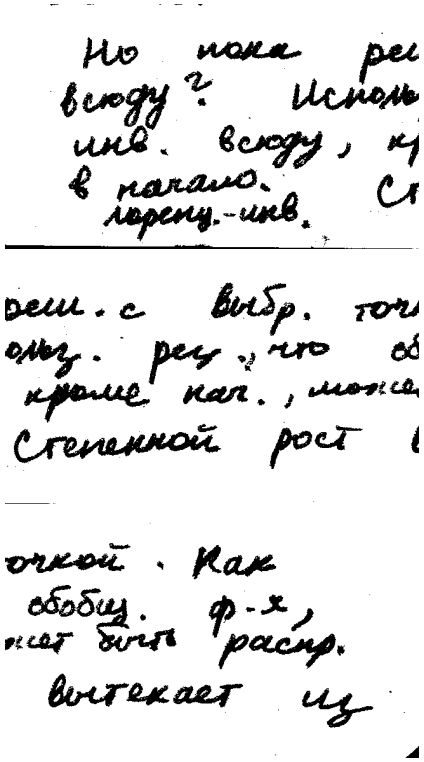


**Uncertain reading.** The manuscript abbreviates the word as непр. Its expansion is unresolved.

Original source details: GELFAND-P000027

For $X$, seek an invariant distribution of one variable. The wave operator turns into an operator of one variable, but with variable coefficients. But for one variable it is solvable. But if there is a group with 2 invariants, then instead of a singularity at 1 point one obtains lines of singularities, etc.

But so far the solution has a chosen point. How everywhere? Use the result that a generalized function, invariant everywhere except at the origin, can be extended to the origin. Power growth follows from Lorentz invariance.

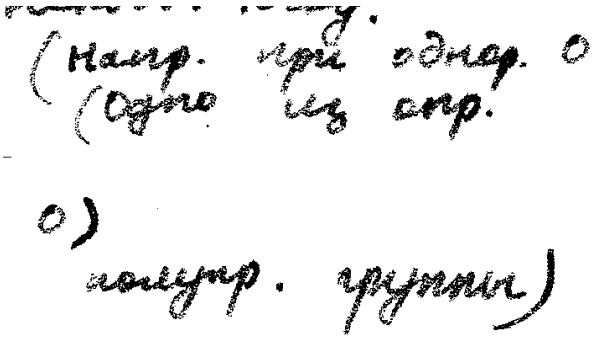


**Manuscript notation.** The manuscript abbreviates “solution,” “chosen,” “result,” “origin,” and “extended.”

Original source details: GELFAND-P000027

GELFAND-P000028

The Lorentz group is semisimple and the space is finite dimensional. A finite dimensional representation of a semisimple Lie group has a fixed point (e.g. at однор. 0 [uncertain source abbreviation]) (one of the definitions of a semisimple group).



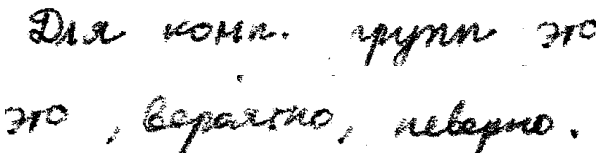
**Manuscript notation.** The manuscript writes «при однор. 0», with an abbreviated adjective.

Original source details: GELFAND-P000028

Another problem. $G$ is a Lie group. A Lie operator $X$ is an infinitesimal translation $Xf(g)$. Consider a polynomial in Lie operators. How to find elementary solutions? (fund.) $PE = \delta$.

$P(X_1, \ldots, X_n)$ a polynomial in Lie operators.

For compact groups this is probably false.

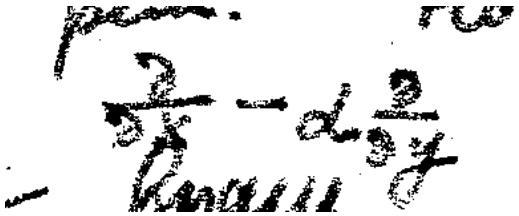


Original source details: GELFAND-P000028

**Manuscript notation.** The manuscript writes «комп.» for compact.

On the torus there is no elementary solution. There is no solution for the torus. But the torus has a fundamental group. And for the rotation group

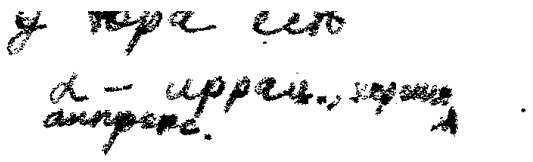
$$\frac{\partial}{\partial x} - \alpha \frac{\partial}{\partial y}.$$



Original source details: GELFAND-P000028

**Manuscript notation.** The manuscript writes $\alpha$ without a separate superscript.

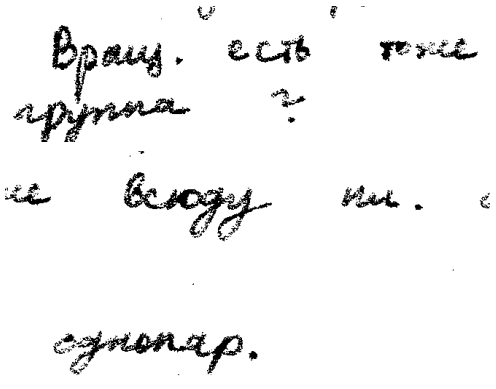
In the right margin: $\alpha$—irrational, good. approx.



Original source details: GELFAND-P000028

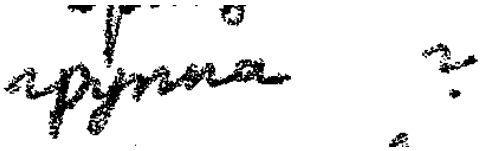
**Uncertain reading.** The last abbreviations in the margin are хорош. / хорошо [good/well]; аппрокс. [approx.]. Below the first is a small detached dark A/1-like stroke whose role is not clear.

For rotations, is there also everywhere an [ин. / им., source abbreviation] one-parameter group?



Original source details: GELFAND-P000028

**Uncertain reading.** Before «однопар.» (one-parameter), the shorter abbreviation resembles ин. or им.



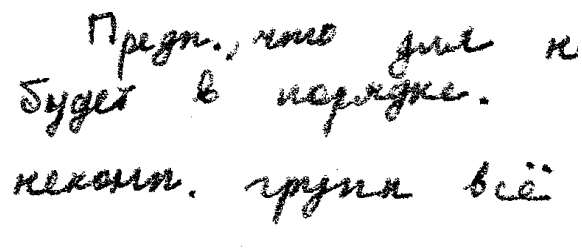
Original source details: GELFAND-P000028

**Manuscript notation.** The word «группа?» (group?) continues on the next line.

Nontrivial even for operators of the 1st order.

$$E(P\varphi) = \varphi(0).$$

Suppose that for a noncompact group everything will be in order. For the circle...



**Manuscript notation.** The manuscript writes «некомп.» for noncompact.

Original source details: GELFAND-P000028

GELFAND-P000029

2/XI. Kirillov. Elmar Thoma. 3 works, representations of countable noncommutative groups [“discrete” written below “noncommutative”]. $G$. There is an explicit form for certain groups of $+$ definite functions constant on conjugacy classes of elements—extreme points of the set of such functions.

$$f(g) \quad + \text{ definite, if } \quad \|f(g_i g_j^{-1})\| \gg 0 \quad + \text{ definite.}$$
$$g_1, \dots, g_n \quad \text{any collection of elements.}$$

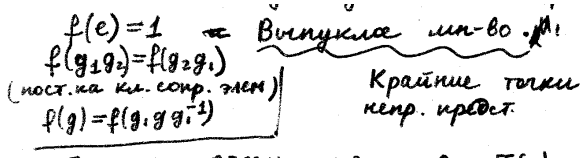


**Diagram and typed description.**

$f(e) = 1 \longleftarrow$ Convex set $M$

$$f(g_1 g_2) = f(g_2 g_1)$$

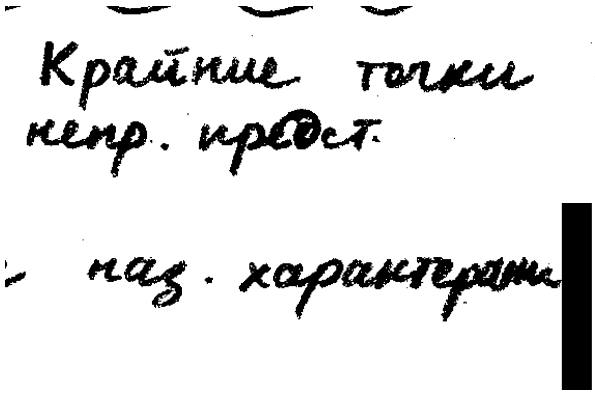
(constant on conjugacy classes of elements)

$$f(g) = f(g_1 g g_1^{-1})$$

The short stroke to the right of $M$ is unresolved.

Original source details: GELFAND-P000029

The extreme points are called characters of irreducible representations.



**Manuscript notation.** The manuscript abbreviates “irreducible” as непр. and uses the fuller неприв. below.

Original source details: GELFAND-P000029

Example: $g \longrightarrow T(g)$, $\frac{1}{n} \operatorname{Sp} T(g)$ is an example of such a function. If the representation is irreducible, such functions form extreme points—easy to check. (The representation is uniquely determined by this function.) There are groups for which describing the representations is impossible (for set theoretic reasons—there are many of them), but the characters exist and are given by formulas.

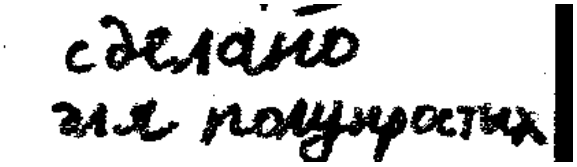
$$\begin{pmatrix} \alpha & \beta \\ 0 & 1 \end{pmatrix} \quad \text{group of affine transformations of the line,}$$
$$x \longrightarrow \alpha x + \beta, \qquad \alpha, \beta \quad \text{rational.}$$

If $\alpha$ and $\beta \in$ a finite field (mod $p$), how to find all traces (irreducible representations). Representations—complex numbers. Not done if the representation is given

with $\alpha, \beta \in (\mathrm{mod}\, p)$.

$$\begin{pmatrix} \alpha & \beta \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \gamma & \delta \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \alpha\gamma & \alpha\delta + \beta \\ 0 & 1 \end{pmatrix}.$$

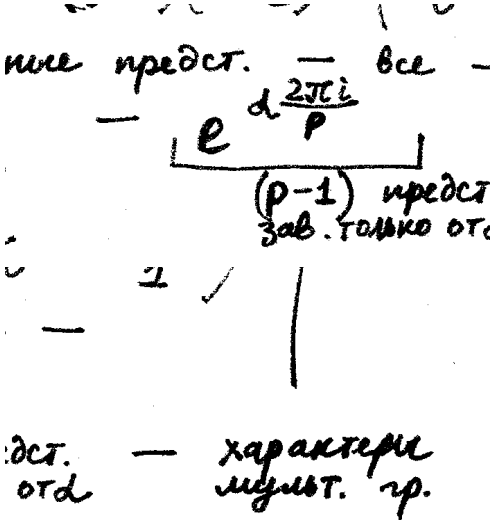
In the right margin: done for semisimple ones.




Original source details: GELFAND-P000029

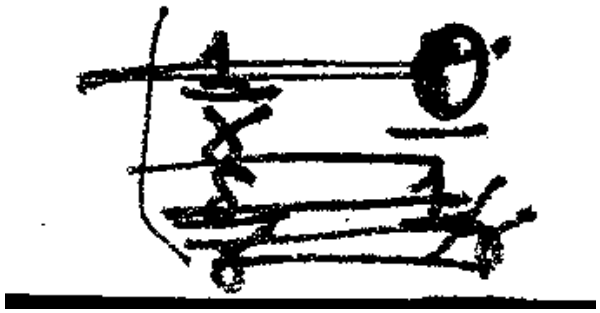
**Manuscript notation.** The margin reads «сделано для полупростых» (done for semisimple ones).

One dimensional representations—all—$e^{\alpha \frac{2\pi i}{p}}$. $(p-1)$ representations depending only on $\alpha$—characters of the multiplicative group.



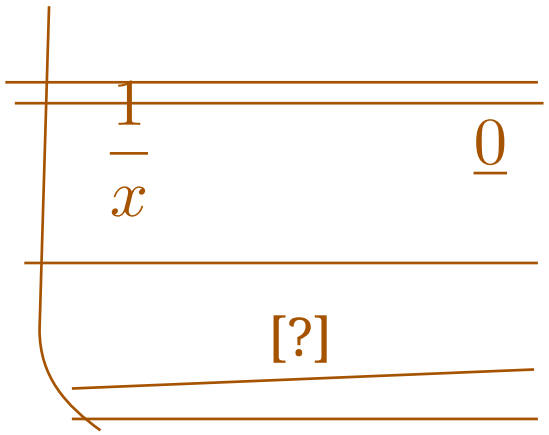

Original source details: GELFAND-P000029

**Uncertain reading.** In the exponent one can see $\alpha\, 2\pi i/p$.



Original source details: GELFAND-P000029

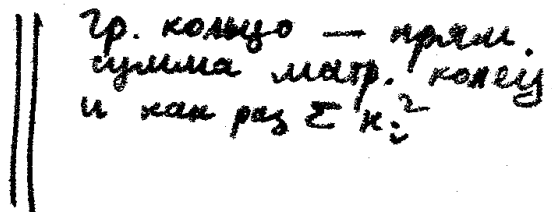
**Diagram and typed description.**




Two fractional expressions, 0 and horizontal strokes are crossed out; bracket-like strokes run down the left and below. The first fraction reads $1/x$. The horizontal arrow and the second fraction cannot be read securely through the cancellations.

$$\sum k_i^2 = p(p-1),$$

$$\sum{}' k_i^2 = p(p-1) - \underbrace{(p-1)}_{\text{one dim.}} = (p-1)^2.$$



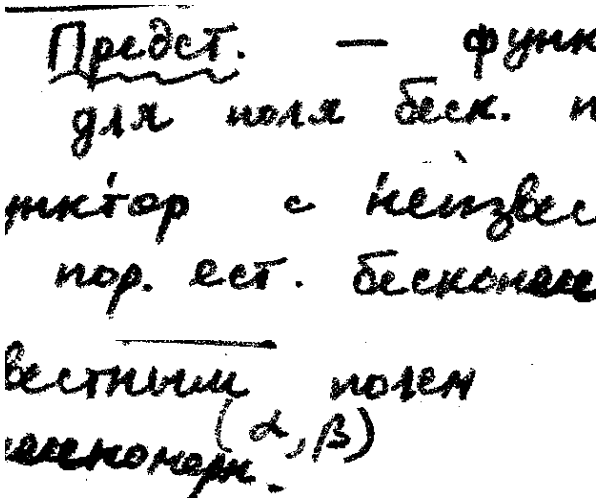

Original source details: GELFAND-P000030

**Diagram and typed description.**

> Group ring—direct sum of matrix rings and precisely $\sum k_i^2$.

Two vertical rules separate the note in the right margin.

Conjecture: another representation of order $p-1$. Representation—a functor with an unknown field. For a field of infinite ord. there is an infinite dimensional $(\alpha, \beta)$.

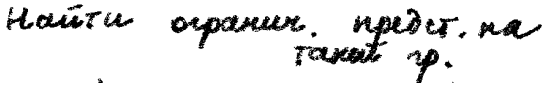



Original source details: GELFAND-P000030

**Uncertain reading.** Беск. пор. ест. бесконечномерн. [infinite ord., there is an infinite dimensional]; the connection of this abbreviated line to “functor with an unknown field” is not explicit.

$\begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix}$ find the restr. of the repr. to such a group,

$$\begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & \beta' \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & \beta+\beta' \\ 0 & 1 \end{pmatrix}.$$




Original source details: GELFAND-P000030

**Uncertain reading.** The manuscript abbreviates the expression as «огранич. предст.». The expansion of огранич. is uncertain.

Its one dimensional representations

$$\begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix} \longrightarrow e^{\frac{2\pi i k}{p}\beta}.$$

$\begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix}$ normal subgroup.

Transformations with [a short word cancelled]

$$\begin{pmatrix}\alpha & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}1 & \beta\\ 0 & 1\end{pmatrix}\begin{pmatrix}\alpha^{-1} & 0\\ 0 & 1\end{pmatrix}=\begin{pmatrix}1 & \alpha\beta\\ 0 & 1\end{pmatrix}.$$



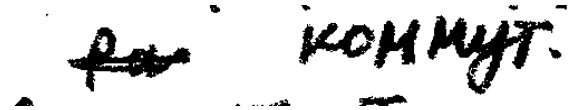

Original source details: GELFAND-P000030

**Uncertain reading.** Between “with” and the matrix is a small obliterated sign or word.

$\beta$ is not needed (it commutes).

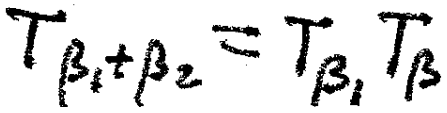
$$\begin{pmatrix}1 & \beta\\ 0 & 1\end{pmatrix}\longrightarrow T_\beta,\qquad T_{\beta_1+\beta_2}=T_{\beta_1}T_\beta.$$

[Cancelled: illegible] commute.




Original source details: GELFAND-P000030

**Uncertain reading.** *Cancelled.* a short word before коммут. [commute]

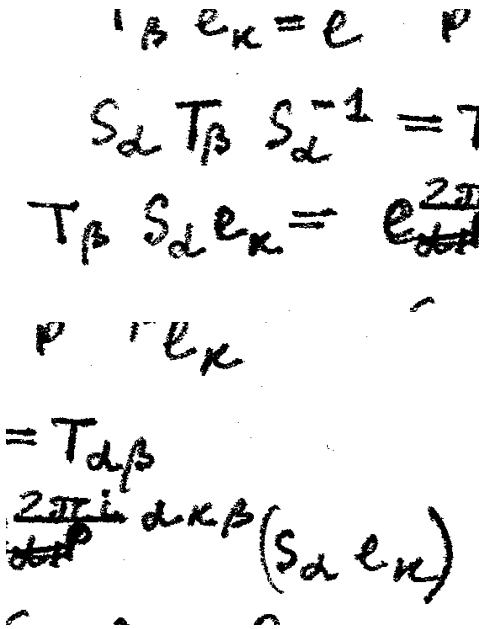


Original source details: GELFAND-P000030

**Manuscript notation.** In the last $T_\beta$, no numeral follows $\beta$.

Decompose the space into eigenvectors with respect to $T_\beta$.

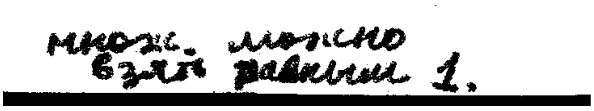
$$T_\beta e_k=e^{\frac{2\pi ik}{p}\beta}e_k,$$
$$S_\alpha T_\beta S_\alpha^{-1}=T_{\alpha\beta},$$
$$T_\beta S_\alpha e_k=e^{\frac{2\pi i}{p}\alpha k\beta}(S_\alpha e_k),$$
$$S_\alpha e_k=e_{\alpha k}.$$



Original source details: GELFAND-P000030

**Uncertain reading.** The source writes $S_\alpha T_\beta S_\alpha^{-1}=T_{\alpha\beta}$ and the factor $\alpha k\beta$ in the exponent beside $T_\beta S_\alpha e_k$. Cancelled signs lie below the denominator of the latter exponent.

At the lower right: multipliers can be taken equal to 1.



Original source details: GELFAND-P000030

**Manuscript notation.** The manuscript writes «множ. можно взять равными 1» (the multipliers can be taken equal to 1).

GELFAND-P000031

There will remain $(p-1)$ vector. It is written how it acts. A more beautiful realization.

$$x \longrightarrow \alpha x + \beta.$$

[At right: aff. line over a finite field.] $f(x)$ — functions on it.

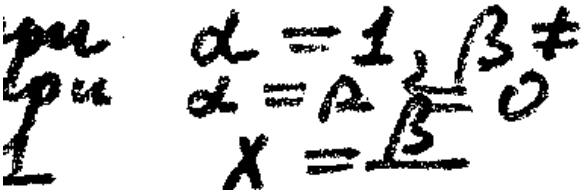
$$T_g f(x) = f(\alpha x + \beta).$$

Const is invariant, so this repr. is reduc. We shall consider functions for which

$$\sum_x f(x) = 0 \qquad (\int = 0).$$

This is a $(p-1)$-dimensional repr.

$$\mathrm{Sp}(T_g) = \begin{cases} 1 & \text{when } \alpha \neq 1, \\ 0 & \text{when } \alpha = 1,\ \beta \neq 0, \\ p & \text{when } \alpha = 1,\ \beta = 0. \end{cases}$$



Original source details: GELFAND-P000031

**Uncertain reading.** In the third condition, the looped sign after $\alpha =$ resembles $1$, $\alpha$ or $p$; the coloured $1$ is a provisional candidate.

[Repeated separately at left beneath the brace: $\alpha \neq 1$. At right: reducible.]

$$x = \alpha x + \beta, \qquad x = \frac{\beta}{1-\alpha}.$$

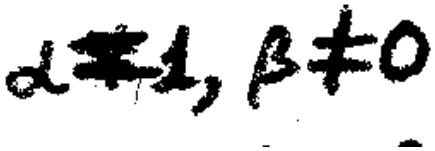
Must subtract the trace [след.; provisional].



Original source details: GELFAND-P000031

**Uncertain reading.** The abbreviation after “subtract” is provisionally read as след. [trace].

$$\mathrm{Sp}\, T_g = \begin{cases} 0 & \alpha \neq 1, \\ -1 & \alpha = 1,\ \beta \neq 0, \\ p-1 & \alpha = 1,\ \beta = 0. \end{cases}$$



Original source details: GELFAND-P000031

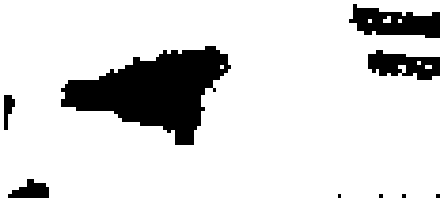
**Uncertain reading.** Heavy overwritten or cancelled strokes stand before $1$ in the second condition. The earlier sign $\neq$ / $=$ cannot be established securely.

How is this related to dyn. systems?

Let $\alpha$ and $\beta$ be real.

$$\beta \longrightarrow T_\beta, \qquad T_{\beta_1+\beta_2} = T_{\beta_1} T_{\beta_2},$$

$$S_\alpha, \qquad S_{\alpha_1\alpha_2} = S_{\alpha_1} S_{\alpha_2}, \qquad S_\alpha^{-1} T_\beta S_\alpha \blacktriangleleft = T_{\alpha\beta}.$$



**Uncertain reading.** Between the last $S_\alpha$ and $=$ is a small separate filled wedge pointing left. Its visible shape is retained; its meaning has not been established.

Original source details: GELFAND-P000031

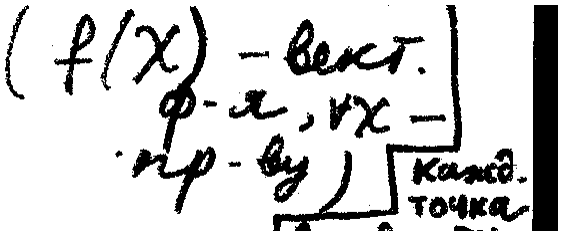
[Horizontal line.] How can they be found? They can be brought to diagonal form (the repr. is unit.).

$$T_\beta f(\chi) = \chi(\beta) f(\chi).$$

[Cancelled at right of the formula: $\chi(\beta)$.] Bring to an operator of multiplication by a function, where

$$\chi(\beta_1 + \beta_2) = \chi(\beta_1)\chi(\beta_2).$$

[In the right brace:] $f(\chi)$ — a vector function, $\forall\chi$ — to a space [пр-ву; abbreviated in the source].



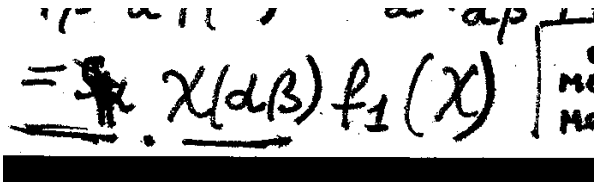

**Uncertain reading.** The sign before $\chi$ resembles $\forall$; пр-ву is an abbreviation for “to a space”. The source does not write out the full word.

Original source details: GELFAND-P000031

[In the left margin, rotated: dyn. syst. on the addit. group — $\longrightarrow$ the mult. group acts on it.]

$$S_\alpha f(\chi) = f_1(\chi),$$

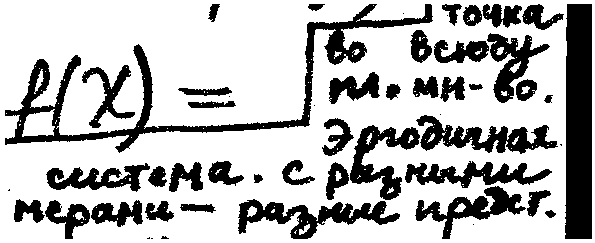
$$T_\beta f_1(\chi) = T_\beta S_\alpha f(\chi) = S_\alpha T_{\alpha\beta} f(\chi) = \chi(\alpha\beta) f_1(\chi).$$




**Uncertain reading.** Before $\chi(\alpha\beta)$ in the last equality, a symbol resembling $S$ / $S_\alpha$ is heavily cancelled. Beneath the cancellation is $\longleftarrow$; beneath $\chi(\alpha\beta)$ is $\longrightarrow$. The cancelled sign remains unreadable.

Original source details: GELFAND-P000031

[In the lower-right margin, within a stepped outline:] Each point — into an everywhere dense [пл.; uncertain abbreviated reading] set.




Original source details: GELFAND-P000031

**Uncertain reading.** The source abbreviation before мн-во [set] resembles пл. [dense]; its middle strokes remain uncertain.

Ergodic system. With different measures — different repr.

Haar measure — [непр. пр.]

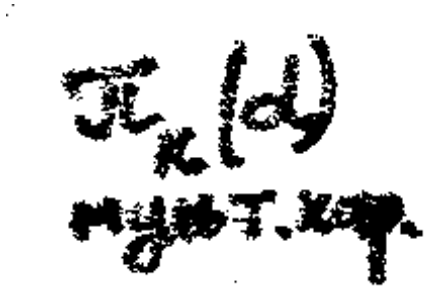



Original source details: GELFAND-P000031

**Uncertain reading.** The expansions of the source abbreviations непр. пр. are uncertain.

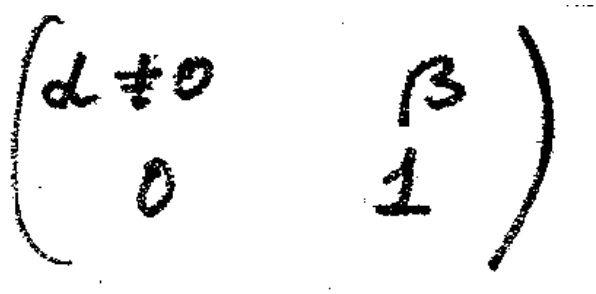
GELFAND-P000032

$$\begin{array}{cc|c|c} \overline{\chi}_K(\alpha) & \begin{pmatrix} \alpha \neq 0 & \beta \\ 0 & 1 \end{pmatrix} & 0 & 0 \\ \text{mult. char.} & & & \\ 1 & \begin{pmatrix} 1 & \beta \neq 0 \\ 0 & 1 \end{pmatrix} & -\dfrac{1}{p-1} & 0 \\ 1 & \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & 1 & 1 \end{array}$$




Original source details: GELFAND-P000032

**Uncertain reading.** The barred notation before the first matrix is provisionally read as $\overline{\chi}_K(\alpha)$. The first sign may be $\pi$, and the index may be Latin $K$ or Cyrillic к. Beneath it is мульт. хар. [mult. char.].




Original source details: GELFAND-P000032

**Uncertain reading.** The upper-left entry reads $\alpha \neq 0$: the final sign is a closed loop. A separate $1$ precedes each of the two lower matrices.

[To the right of the first upper $0$: Char. for finite groups.] If one passes to the lim. as $p \longrightarrow \infty$. [From the line about the limit, $\nearrow$ points to the lower-right $1$.]

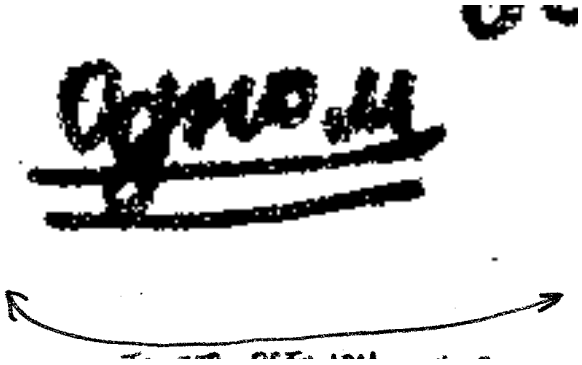


**Diagram and typed description.** The left marginal word has two underlines and resembles Одном / Одному [one / to one, literal candidate readings]. Below it is a long low curve with outward arrowheads at both ends; the right end is beneath the limit. The label remains unresolved.

Одном / Одному

Original source details: GELFAND-P000032

What remains for rational numbers.

Lemma. $G$, $G_0 \subset G$ [beneath $G_0$: norm. div.]. $f$ — pos. def., $f(e) = 1$, const. on conj. $M$ — the set of them. Consider $f|_{G_0}$.

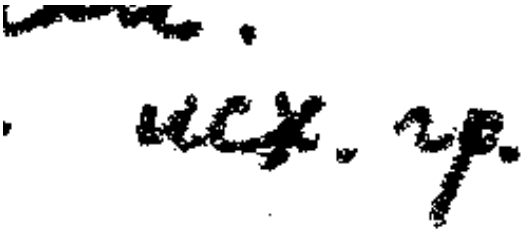
Also consider functions satisfying the same [a short entry is cancelled] conditions on $G_0$.



**Uncertain reading.** The short entry before the source abbreviation усл. is crossed out with heavy horizontal strokes; its letters cannot be established.

Original source details: GELFAND-P000032

Consider $H_{G_0}$ — their extreme points. They coincide with the extreme points of the original / countab. [исх. / сч.; uncertain source abbreviation] gr.



**Uncertain reading.** The joined final source abbreviation resembles исх. гр. [original gr.]; сч. гр. [countab. gr.] remains an alternative. The full source words are not written.

Original source details: GELFAND-P000032

(For countable groups.) One must count how many conj. classes there are and then seek the representing. From + def. [Cancelled; provisional reading: something [чего-то; uncertain reading].] it follows that there are no other repr.



**Uncertain reading.** The crossed-out entry after Из + опр. [From + def.] resembles чего-то [something]; the heavy overlapping strokes do not establish a complete secure word.

Original source details: GELFAND-P000032

[In the left margin: 2nd work.]

Infinite symmetric group: permutations of the natural sequence (only a finite number of elements move — for countability). Conj. classes of elements are specified by cycles. — [A separate horizontal stroke at the left; a short entry is cancelled.]



**Uncertain reading.** The short sign or word before “cycle lengths” remains unreadable beneath two heavy horizontal strokes.

Original source details: GELFAND-P000032

Cycle lengths determine the conj. class of elements.

$$(n_2, n_3, \ldots),$$

only a finite number $\neq 0$.

GELFAND-P000033

Every character

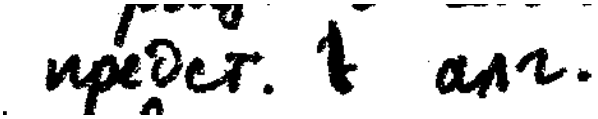
$$\chi(n_2, n_3, \ldots) = \prod a_k^{n_k}.$$

$a_k$ — a sequence of real numbers. For a finite group it is more complicated. [Above “finite”: (arbitr.); between “group” and “more complicated,” inserted above: “char.” with a brace pointer.] (Frobenius, Young, Young diagr. — classification of tensors.) $G_n$ — a subgr. isom. to the original — the first $n$ stay in place. And the char. is the same (depends only on cycle lengths). From this follows multiplicativity of the char. in cycle length, and further. 3rd work.

There are countab. noncompact gr. Practically all are bad. Char. Type 1: there is a norm. div. $G_0$ such that $G/G_0$ is finite. [Inserted beneath norm. div. $G_0$:] isom. $G$. (Necessary and sufficient condition.)

Thoma $\left.\begin{matrix}\text{Math. Zeits.}\\ \text{Math. Ann.}\end{matrix}\right\}$ 1964.

These are his first 3 works. Not done: characters for the gr. of matrices of order 2 over the field of rational numbers.
What is a repr. [mark] of an alg. group with an unknown field?

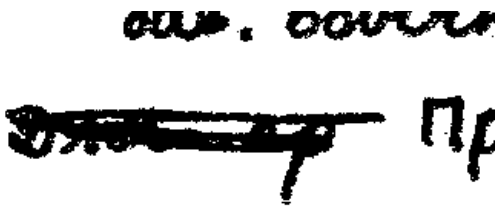


**Uncertain reading.** A separate upright mark stands between the source abbreviations предст. and алг. Its meaning is uncertain.

Original source details: GELFAND-P000033

$\begin{pmatrix} \cdot & \cdot \\ 0 & 1 \end{pmatrix}$ Unknown field.

[At right: So that for char. of fields it gives the usual expr.] A way of writing char. of $\forall$ alg. gr. [A short entry is cancelled.]



**Uncertain reading.** After алг. гр. [alg. gr.] a short entry is heavily cancelled; its beginning resembles Это... [This...]. The remaining strokes do not establish the text.

Original source details: GELFAND-P000033

Repr. of types 1 and 2 are recovered from char. $G$ is specified by a scheme of semigr. alg., assigned to $\forall$ field. Char. also.

[Cancelled; provisional reading: Function / Func. [Функция / Функц.; uncertain reading]]



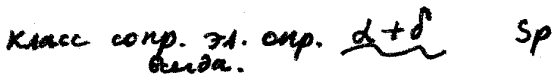
**Uncertain reading.** The cancelled source word resembles Функция / Функц. [Function / Func.]; the ending is obscured by horizontal strokes.

Original source details: GELFAND-P000033

Character — a functor of the field $K$. [At right: $\mathcal{F}(\cdot)$.] (A formula corresponds to the field $K$.) If there is an equality for several fields, then also always. (Similarly, e.g. only for finite ones.)

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \quad \text{with } \det = 1.$$

The conj. class of elements is determined by $\alpha + \delta$. [Beneath the source words сопр. эл.: вида. / велич. [of the form / quant.; provisional]. At right: Sp.]



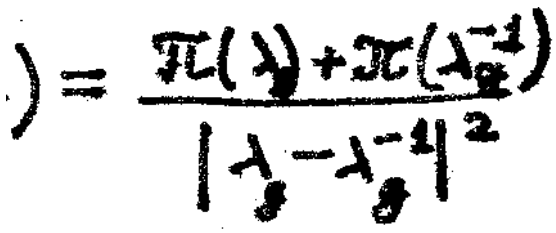
**Uncertain reading.** The word written beneath сопр. эл. is provisionally read as вида. or велич.

Original source details: GELFAND-P000033

GELFAND-P000034

What do we know? Char. of the gr. of real matr. with $\det = 1$. Complex —

$$\sigma(g) = \frac{\pi(\lambda_g) + \pi(\lambda_g^{-1})}{|\lambda_g - \lambda_g^{-1}|^2}.$$



**Uncertain reading.** The lower stroke by the first $\lambda$ in the numerator merges into the right parenthesis. The subscript $g$ is provisional.

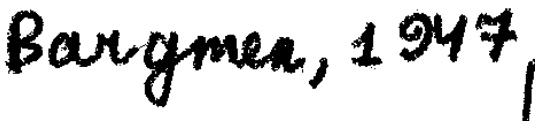
Original source details: GELFAND-P000034

[At right: Gelfand, Naimark, 1947.] $\pi(\lambda_g)$ — a mult. character of the field of complex numbers. Real.

$$\sigma(g) = \begin{cases} \dfrac{\pi(\lambda_g) + \pi(\lambda_g^{-1})}{|\lambda - \lambda^{-1}|} & \text{if } \lambda \text{ is real.,} \\ 0 & \text{if } \lambda \text{ is complex.} \end{cases}$$

$$\sigma'(g) = \begin{cases} \dfrac{\pi(\lambda_{\min})}{|\lambda - \lambda^{-1}|} & \text{if } \lambda \text{ is real.,} \\ \dfrac{\pi(\lambda_g) + \pi(\lambda_g^{-1})}{|\lambda - \lambda^{-1}|} - \dfrac{\pi(\lambda_g) - \pi(\lambda_g^{-1})}{\lambda_g - \lambda_g^{-1}} & \end{cases}$$

[No condition is written at right in the second row of the last brace. In the left margin, rotated:] Vanguten / Vantsuten, 1947.



**Uncertain reading.** The marginal surname resembles Вангутен / Ванцутен [Vanguten / Vantsuten]. Its joined middle strokes remain uncertain; the written year is 1947.

Original source details: GELFAND-P000034

[Cancelled; provisional reading: repr. [предст.; uncertain reading]]



**Uncertain reading.** The crossed-out source abbreviation resembles предст. [repr.]; a horizontal cancellation and two long strokes below it obscure parts of the letters.

Original source details: GELFAND-P000034

$p$-adic numbers.

$$\sigma(g) = \begin{cases} \dfrac{\pi(\lambda_g) + \pi(\lambda_g^{-1})}{|\lambda_g - \lambda_g^{-1}|} & \text{if } \lambda_g \in K \text{ is real.}, \\ 0 & \lambda_g \overline{\in} K. \end{cases}$$

[Nonmembership is written in the source with a bar over $\in$. At right, beside a long vertical stroke:] Gelfand, Graev, 1963.

Finite field.

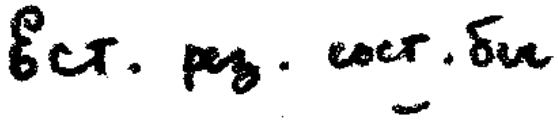
$$\sigma(g) = \begin{cases} \pi(\lambda_g) + \pi(\lambda_g^{-1}) & \lambda \in K, \\ 0 & \lambda \overline{\in} K, \end{cases}$$

[At right: Frobenius, 1895.]

$$\sigma'(g) = \begin{cases} 0 & \lambda \in K, \\ -\pi(\lambda_g) - \pi(\lambda_g^{-1}) & \lambda \overline{\in} K. \end{cases}$$

GELFAND-P000035

Nat. res. [рез.; provisional] [сост.; abbreviated] was that one must pass to the group of adeles.



**Uncertain reading.** The first and third source abbreviations read Ест. and сост. The middle is provisionally read as рез. [res.]; the middle strokes are joined.

Original source details: GELFAND-P000035

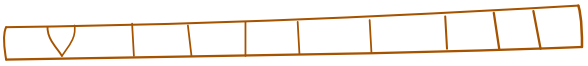
Gr. of adeles — the direct $\sum$ of the gr. of all $p$-adic numbers. In it there are rational numbers — the diagonal.

Vishik.



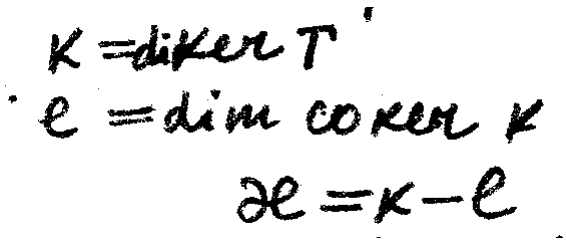
**Diagram and typed description.** Beneath the surname is a narrow strip with irregular long edges and transverse strokes. Two strokes near the left meet at the lower edge; the following dividers have unequal spacing. No labels are written inside.

Original source details: GELFAND-P000035



What one should work on in diff. eqns. For a lin. oper. with finite kernel and cokernel an index is introduced.

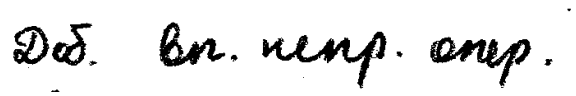
$$k = \dim \ker T, \qquad \ell = \dim \operatorname{coker} K, \qquad æ = k - \ell.$$




Original source details: GELFAND-P000035

**Uncertain reading.** The letter after coker resembles $K$. The looped index sign is represented by the provisional candidate æ; its exact letter has not been established.

1) If replaced by $\|T - T_1\| < \varepsilon$, then æ does not change. 2) Adding a compl. cont. [вп. непр.; abbreviated reading] oper. does not change the index.

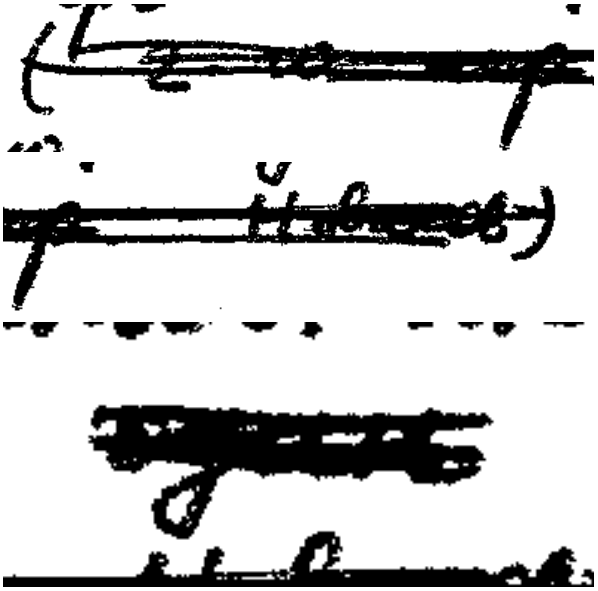



Original source details: GELFAND-P000035

**Uncertain reading.** The source writes Доб. вп. непр. опер. [Adding a compl. cont. oper.]. The joined abbreviations are shown in their written form; the full words are not written out.

æ allows classification of elliptic eqns. on manifolds. (Atiyah and Singer) Homotopy classification. Vishik considers that the index is the only homotopy inv.

A syst. with const. coeffs. in a ball has [a short entry is cancelled] as one more homot. inv. [Beneath this is a long cancelled parenthetical entry.]



Original source details: GELFAND-P000035

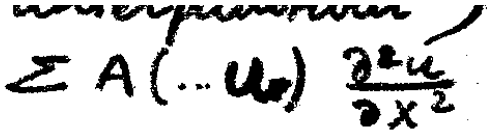
**Uncertain reading.** A short entry after имеет [has] is cancelled; its descending loop remains visible. Beneath it a long parenthetical entry is crossed out with two horizontal lines; its ending resembles инв. [inv.]. The text remains unreadable.

Can any one be transformed into any other (a large number of eqns. — stable). The question has an analogous meaning for a different class of nonlinear eqns. What is æ? What are the kernel and cokernel? $\sim$ a finite kernel. Quasilin. eqn.

$$Tu = v.$$

[At right: $u$ — a manifold. It is required to be finite dimensional.] Principal part of a nonlinear eqn. Cokernel $\sim$ [a short wavy line]. [At right: In each point, linearize the eqn., and everywhere it is the same.] Problem: study the index of a nonlinear elliptic eqn. (This is an integral question.) Quasilin.

$$\sum A(\dots u_0)\frac{\partial^2 u}{\partial x^2}.$$



Original source details: GELFAND-P000035

**Uncertain reading.** The lower sign on $u$ in the argument of $A$ resembles $0$ / $\omega$; the joined dark strokes do not establish which. The dots and absence of sum indices are retained.

16th. Linnik — “How to use funct. anal. in probability theory”.

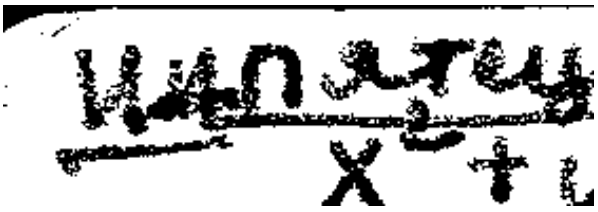


Original source details: GELFAND-P000035

**Uncertain reading.** The small marks after 16 are provisionally read as the Russian ordinal suffix ro. They could also be digits.

GELFAND-P000036

I. I. / I. M. Pyatetsky-Shapiro.



Original source details: GELFAND-P000036

**Uncertain reading.** I. I. / I. M. Pyatetsky-Shapiro. The second initial is merged; its exact reading is not established.

(Ilya Iosifovich). Linnik’s work.

$$x^2 + y^2 + z^2 = m.$$

[At left:] $\phi$.



Original source details: GELFAND-P000036

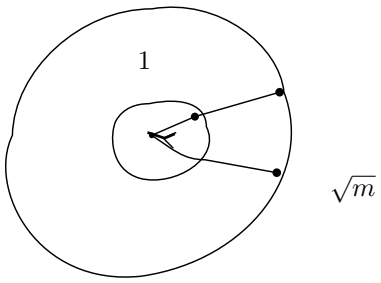
**Uncertain reading.** A separate looped sign with a right vertical stroke in the left margin resembles $\phi$. No meaning is written beside it.

For which $m$ are there integral solutions? It is known that solutions do not always exist. (When $m = 8a + 7$, there are no solutions.) [Beneath $8a + 7$: mod 8.] How the solutions are distributed on the surface of the 3-dimensional sphere. Rad. $m^{1/2}$.



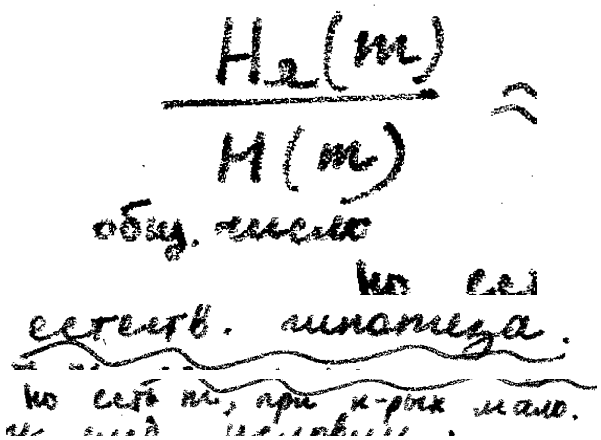

Original source details: GELFAND-P000036

**Diagram and typed description.** Two irregular rounded contours, labels $1$ and $\sqrt{m}$, diverging lines, three heavy points and a thick inner junction with a small leftward arrow. The junction of the inner strokes is not wholly clear.




Select a region on the unit sphere and see how many points project into it.

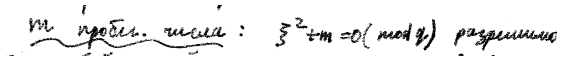
$$\frac{H_2(m)}{H(m)} \approx \text{mes}\,\Omega.$$



Original source details: GELFAND-P000036

**Uncertain reading.** The numerator subscript resembles 2; $\Omega$ remains an alternative. Beneath $H(m)$: total number. At right: nat. hypothesis; below: but there are $m$ for which few. The last abbreviated entry is not wholly clear.

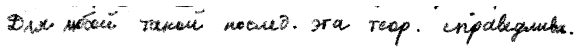
This holds under the following condition: if $q$ is a prime $m$ ranges over / пробл. [пробег. / пробл.; uncertain abbreviation] numbers: $\xi^2 + m = 0 \pmod q$ is solvable.



Original source details: GELFAND-P000036

**Uncertain reading.** Underlined: $m$ пробег. / пробл. числа [$m$ ranges over / пробл. numbers; uncertain abbreviated readings]. The middle and final strokes are joined; the second candidate is not expanded.

For any / small [любой / небол.; uncertain readings] such seq. this theorem holds.



Original source details: GELFAND-P000036

**Uncertain reading.** The joined word after For resembles любой [any]; it is followed by такой послед. [such seq.]. небол. [small; abbreviated reading] remains an alternative for the dark initial strokes.

Using $q$, construct a dyn. system on the sphere, whose behaviour obeys prob. laws: Fix $q$, consider [a short entry is cancelled] an integral point on the sphere and all orthog. matrices with entries of the form $a/q$.



Original source details: GELFAND-P000036

**Uncertain reading.** Before integral point, a short entry is cancelled with several dense strokes; the text has not been read.

($a$ is an integer.) There are finitely many of them, but many. They carry an integral point to a non-integral one. But among them there are finitely many carrying an integral point to an integral one. Construct such a flow (each point its own 48 orthog. matrices). $e$ is a characteristic fn.

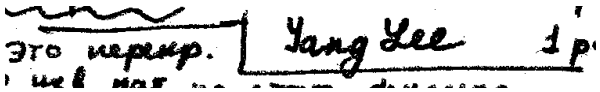
$$e(Tx) + e(T^2x).$$

[Rightward arrow: to the area.] This is a Markov chain.

GELFAND-P000037

Povzner. Naum Yakovlevich. Plan of stat. physics. Int. works.

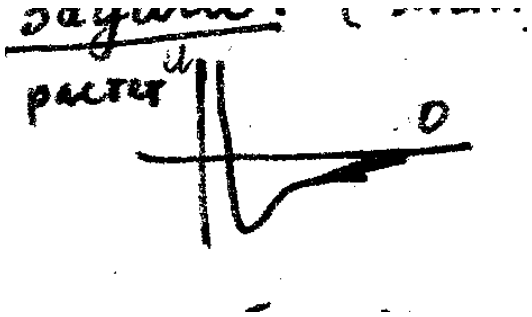
| | |
|---|---|
| Ruelle | 2–3 works |
| Yang Lee | 1 work |




Original source details: GELFAND-P000037

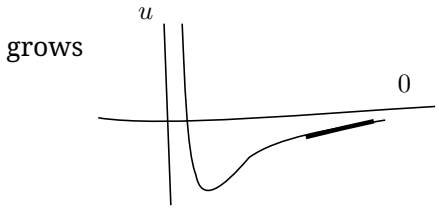
**Diagram and typed description.** The line steps down beside the list of works; its right segment runs beneath “Yang Lee”.

This is a revolution. Everything known mathematically about stat. physics. Kac — models. Problem. (Elem.)



Original source details: GELFAND-P000037

**Diagram and typed description.** A vertical with $u$, a horizontal with $0$ at right; the curve falls steeply below the horizontal and then rises towards it at right. The right part is thickened. At left: grows. The precise junction of the thick strokes is not wholly clear.




$$\min \sum_{i,j=1}^{N} u(x_i - x_j).$$



Original source details: GELFAND-P000037

**Diagram and typed description.** The line beneath the sum steps upward toward the note at right.

$x$ ranges in 2-dimensional space.

Set of $x$. There is a $C$ such that the min. config. has everything in a sphere of rad. $C$. (Easy to show $N^2$, but in the case of a sphere $N^3$.) [Separate thick lines beneath $N^2$ and case.]

16/XI A lin. operator in Hilbert space has an inv. subspace. Solved!!! Bull. Amer. Math. Soc. Lui de Brange and ...

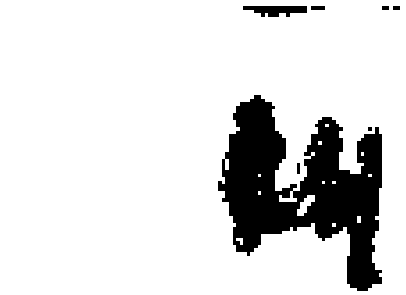



Original source details: GELFAND-P000037

**Uncertain reading.** The entry reads Lui de Brange and ...; the surname ending is looped, with a possible final s. Year: 1964.

1964. Connected with Livshits’s theory.

[Below at left:] $\phi$ / $\psi$.

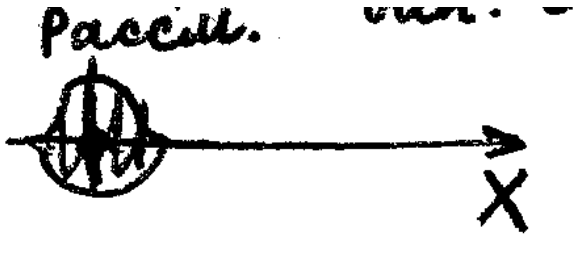


**Uncertain reading.** A separate sign in the left margin has two upper strokes and a right lower loop; graphic candidates are $\phi$ / $\psi$. There is no caption.

Original source details: GELFAND-P000037

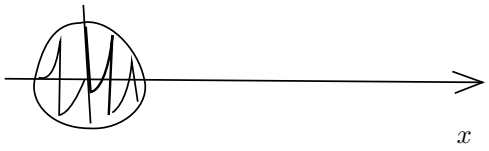
det by Schur. One must take the substitution group, and instead of all — repr. det of a + definite matrix is a + definite matrix.

Arnold. Consider a dyn. system and the corresponding quantum system.




Original source details: GELFAND-P000037

**Diagram and typed description.** A nearly round contour around the intersection of a vertical and a horizontal rightward arrow labelled $x$. Inside are irregular vertical and wavy strokes; some intersections merge.

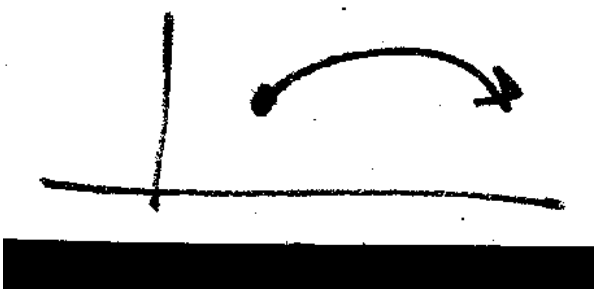



$$\psi(x)e^{\frac{i}{h}S(x)}.$$

Quantum particle.

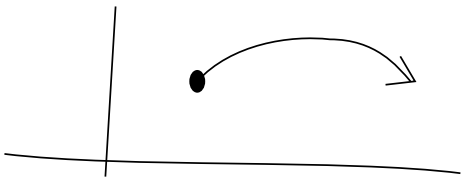
$$\psi \in \mathcal{D}(\mathbf{R}^n), \qquad S(x) = \int p\,dx.$$

$h$ is Planck's const. Seek the solution in the same form. A semiclass. asymptotic is obtained.

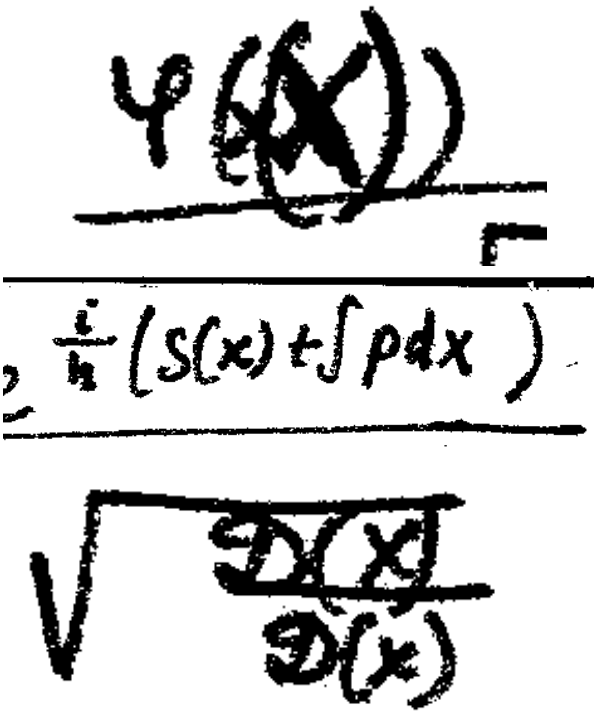


Original source details: GELFAND-P000037

**Diagram and typed description.** Unlabelled axes. A heavy point lies at right above the horizontal; an arch goes up and right from it, then down to a right arrowhead. No continuation beyond the frame is shown.



$$\frac{\psi(\xi(x))e^{\frac{i}{h}(S(x)+\int p\,dx)}}{\sqrt{\frac{D(X)}{D(x)}}}.$$



**Uncertain reading.** The argument of $\psi$ is overwritten: $\xi(x)$ / $x$. The sign after $S(x)$ resembles $+$ / $t$. Under the radical the upper argument letter resembles $X$ / $x$; no separate index is visible.

Original source details: GELFAND-P000038

[At right: norm. factor.] Holds only as $h \longrightarrow 0$, but when $D \neq 0$. (Through $t$, map. $D$ is the det of the new pts. in terms of the old ones.) The solution of the question reduces to top. inv. of a manifold. (At a turning point.) Index. [At right: During mov. one obtains a cohomology class.]

Linnik. Ideals of rings of fns. from the point of view of appl. to statistics. A. Cartan — on ideals of rings of holom. fns. (47 and 50). Analyt. sheaves appear (first with Oka). Appl. of the apparatus of analyt. sheaves. What is statistics?

$$(X, A, \mu) \xrightarrow{T} (Y, B).$$

[Beneath the first parentheses: space with measure. Beneath the second: space with a $\sigma$-system.] $T$ is a measurable mapping (observation fn.). $X$ is what we observe. The measure on $Y$ is directly induced [a short entry is cancelled] from $X$.

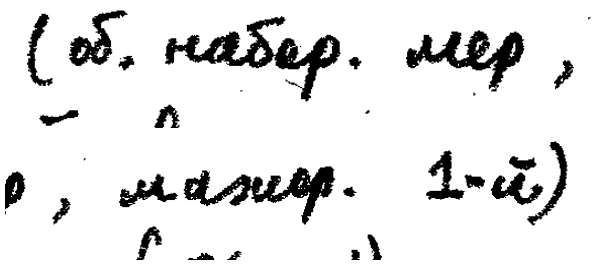


**Uncertain reading.** Between directly induced and from $X$, a short entry is heavily crossed out; the exact text is not established.

Original source details: GELFAND-P000038

$X$ is Euclidean space. $A$ — Borel sets.

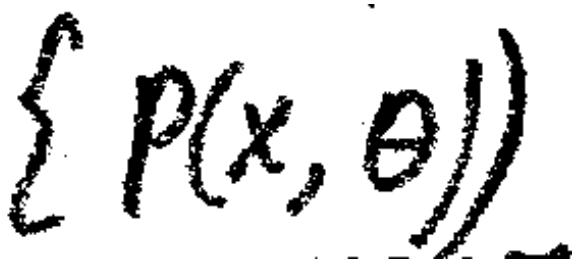
$\mu$ — measure. (об. [abbrev.] set of measures, major. [мажор.; provisional] 1st).



**Uncertain reading.** In parentheses: об. набор. мер, мажор. / массы 1-й / 1-ой [об.; set of measures; major. / mass; 1st, uncertain abbreviated readings]. The joined strokes before 1-й are closer to мажор.; the exact abbreviation and ending remain uncertain.

Original source details: GELFAND-P000038

There is a family of measures $\{P(x,\theta)\}$.



**Manuscript notation.** After $P(x,\theta)$ a second round parenthesis is written.

Original source details: GELFAND-P000038

[Beneath $P(x,\theta)$: given by a density.] We want to obtain info. about the parameter by observing $X$.

$$x_1, \ldots, x_n \in N(a,1).$$

[Beneath $N(a,1)$: Gaussian family. At right: 1 is precision, $a$ is the parameter.]

$$x_i = a + \Delta_i, \qquad \overline{X} \text{ arithm. mean}$$

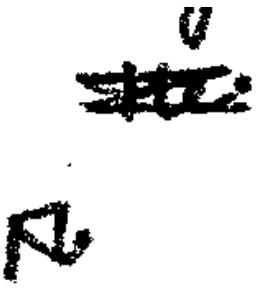
$\Delta_i$ — normal.

GELFAND-P000039

Stat. — finding estimates. Testing hypotheses $a = a_0$. $\theta \in \Omega$. ($\Omega$ — a comp. region in $\mathbf{R}^n$.) Null hypothesis $H_0 = \theta \in \omega$ (subspace). Testing with a fixed sample with randomization. A selected test is a fn. of the observations

$$0 \leq \varphi(x_1, \ldots, x_n) \leq 1.$$

[Beneath the formula: prob. meaning.] Observe with a random fn. and randomize with probabilities $\varphi$, $1 - \varphi$. [Cancellation at left and a separate sign below.]

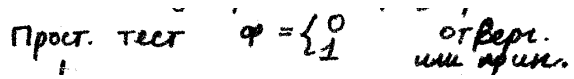


**Uncertain reading.** A short entry is heavily cancelled. Beneath it is a separate sign resembling $\pi$ / $\Pi$; no meaning is written.

Original source details: GELFAND-P000039

Simple test

$$\varphi = \begin{cases} 0 & \text{reject} \\ 1 & \text{or accept} \end{cases}$$



Original source details: GELFAND-P000039

**Manuscript notation.** Two short entries stand at right of 0 and 1. There are no visible lines connecting the entries to the values.

[Downward arrow to the example:]

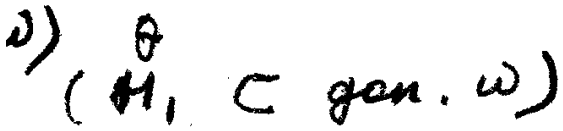
$$|\overline{x} - a_0| > \varepsilon \qquad \text{reject or accept}$$

Test quality is assessed as follows.

$$E(\varphi \mid H_0).$$

Expectation under hypothesis $H_0$ (param. $\in \omega$). [At right: meaningful only if 1 point.] $H_1$ — alternative.

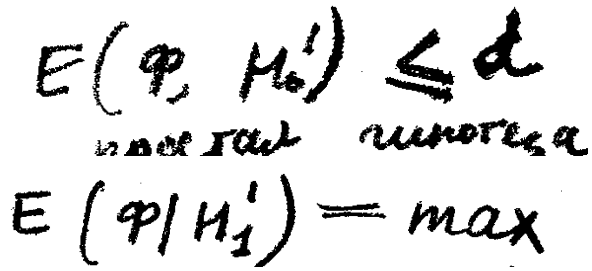
$(\overset{\theta}{H_1} \subset$ compl. $\omega)$.



Original source details: GELFAND-P000039

**Uncertain reading.** A separate $\theta$ stands over $H_1$. The following C-like sign may be $\subset$; it is followed by compl. $\omega$.

$$E(\varphi, H_a') \leq \alpha, \qquad E(\varphi \mid H_1') = \max.$$



Original source details: GELFAND-P000039

**Uncertain reading.** At left the lower sign at $H$ resembles $a$ / 0, and the upper stroke resembles / / 1. There is a comma between $\varphi$ and $H$. At right the upper stroke at $H_1$ is separately visible.

[Beneath the left expectation: simple hypothesis (fixed point). Beneath the right: (desirable).] Reject the true one not too often, but also reject [a short entry is cancelled] the false one often.



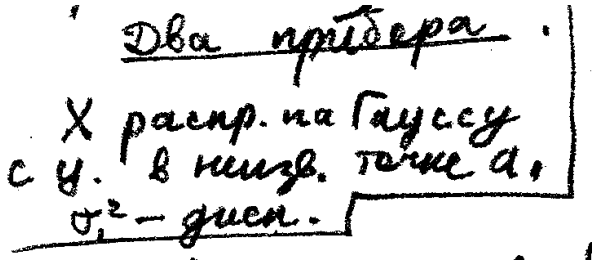


Original source details: GELFAND-P000039

**Uncertain reading.** Before false, a tiny sign is heavily crossed out; the text beneath the cancellation has not been read.

Similar tests. (At each point of the null hypothesis the prob. is $\alpha$.)

Two examples.




Original source details: GELFAND-P000039

**Diagram and typed description.** A stepped lower line and a tall right edge surround the entry about the Gaussian sample.

$X$ is Gaussian distributed with centre at an unknown point $a_1$.
$\sigma_1^2$ is the variance.

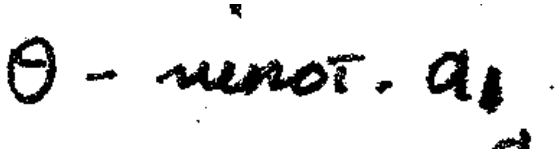
Testing by means of inv. tests.



Original source details: GELFAND-P000039

**Uncertain reading.** In the continuation: null, below it a short cancellation and гип. / ил. The joined strokes after the cancellation do not give an exact reading.

$\theta$ — mult. [мног.; provisional] $a_i$.



Original source details: GELFAND-P000039

**Uncertain reading.** $\theta$ — мног. / мнот. $a_i$. The final stroke of the abbreviation resembles г / т; the reading is not established.

Question. $a_1 = a_2$?

$$N(a_1, \sigma_1^2), \qquad x_1, x_2, \ldots, x_n,$$
$$N(a_2, \sigma_2^2), \qquad y_1, \ldots, y_n.$$

[At right: $\sigma_i^2$ are unknown param.]

$$H_0: \quad a_1 = a_2,$$

$$E(\varphi \mid H_0') = \alpha, \qquad E(\varphi \mid H_1) = \max.$$

$\varphi$ — ?

GELFAND-P000040

$\varphi$ is inv. or similar if for every $\theta$ the expectation of $\varphi$ is the same. (For every $\theta$ there is a prob. distrib.)

$$E(\varphi \mid H_0) = \alpha, \qquad E(\varphi \mid H_1') = \max.$$

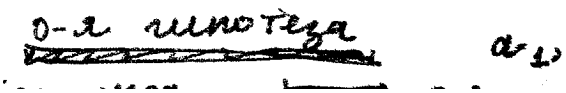
[At right: [mark]].

**Uncertain reading.** There is a separate dark mark to the right of the expectations. Its reading is uncertain.



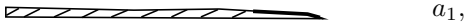
Original source details: GELFAND-P000040

$H_1'$ is one point in the complement. Bayesian measure is a measure on parameter space. (Given beforehand — a priori distribution.) [At left a rightward arrow to the lines:] The measure depends on a 4-tuple of points $(a_1, a_2, \sigma_1, \sigma_2)$. Null hypothesis.



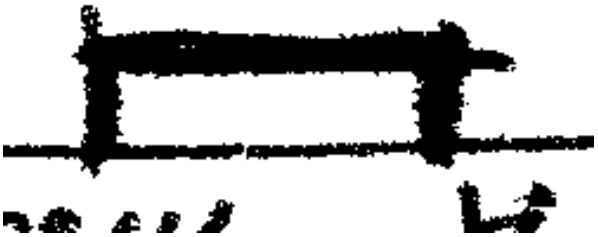
Original source details: GELFAND-P000040

**Diagram and typed description.** Beneath null hypothesis is an irregular narrow strip with diagonal strokes; $a_1$ and a comma stand separately at right. The right strokes merge with the upper edge.



$a_1,$

Bayes measure.

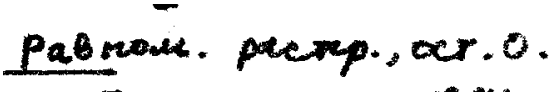


Original source details: GELFAND-P000040

**Diagram and typed description.** Beside Bayes measure is an empty rectangle: its thick top extends rightwards, and the lower line continues beyond both vertical sides. There is no label inside.



[At right: Unif. distrib., from [от.; provisional] 0.]

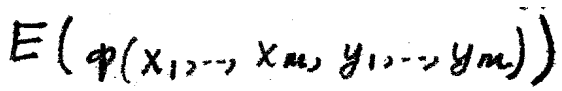


Original source details: GELFAND-P000040

**Uncertain reading.** The short entry before 0 is provisionally read as от. [from].

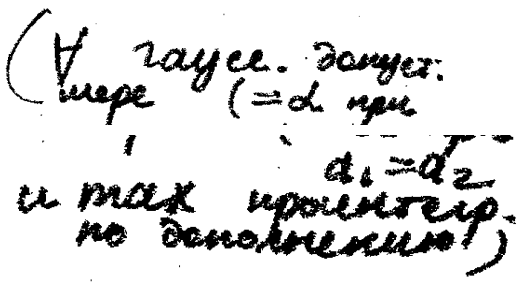
One must be able to solve for every Bayesian distrib.-measure. Result: A prob. distrib. in $X \times Y$ with a given Gaussian measure (4-param.). Find $\varphi(x, y)$ such that

$$E\big(\varphi(x_1,\dots,x_n,y_1,\dots,y_m)\big).$$



Original source details: GELFAND-P000040

**Uncertain reading.** The final subscript on $x$ resembles $n$ or $m$ and has two joined strokes. The subscript $m$ on $y$ is retained.



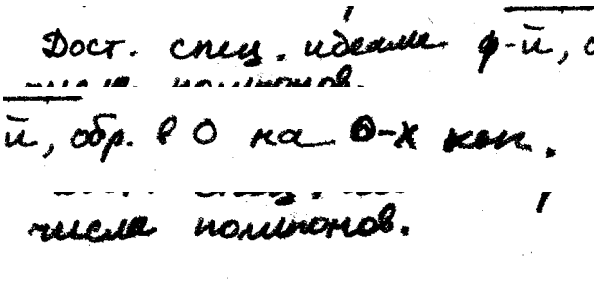
Original source details: GELFAND-P000040

**Uncertain reading.** At right: for every admiss. Gaussian measure ($=\alpha$ when $a_1=a_2$ and max integr. / param. [проинтегр. / параметр.; uncertain abbreviated readings] on the complement). The word after max is graphically closer to проинтегр.; neither candidate is asserted as certain.

Sufficient statistic — contains all info. about the parameters.

Problem not done. $\theta_1$, $\theta_2$ are complex variables. $f(\theta_1,\theta_2)$ in a ball, ideals are not known (All prim. ideals corresponding to the ideal equal to 0.) Glisson [Глиссон] — ball and centre holom.

Suff. spec. ideals of fns., van. at $0$ on a $\theta$-th fin. number of polynomials.



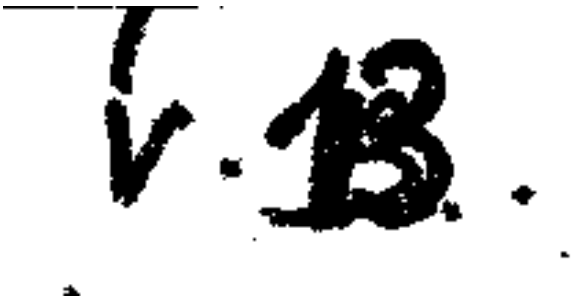
Original source details: GELFAND-P000040

**Uncertain reading.** The first sign after the source word на is provisionally read as $\theta$; it may be $0$.

GELFAND-P000041

Gleason — Gorin.

J. Math. Mech. v. 13. N 1 (1964).



Original source details: GELFAND-P000041

**Uncertain reading.** The number $13$ is written over earlier strokes. The underlying numeral is uncertain.

Consider a ball $z=(z_1,z_2)$,

$$|z_1|^2+|z_2|^2\le 1$$

and fns., an. inside and cont. on the bdry., with $f(0,0)=0$. Do there exist $g_1,g_2$ of the same type such that

$$f=z_1g_1+z_2g_2$$

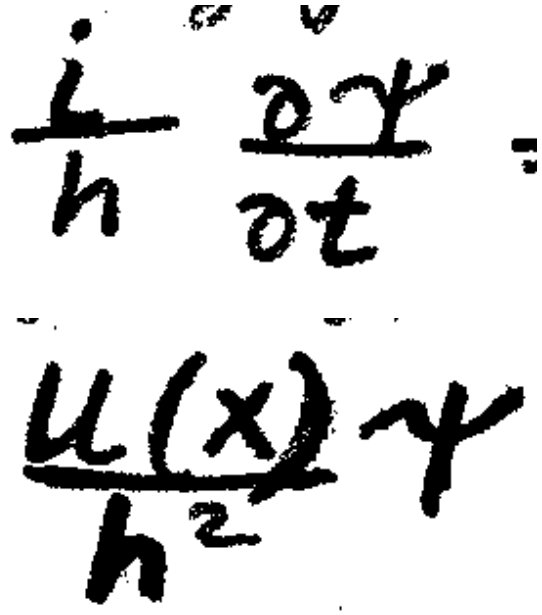
(max ideal; will $z_1,z_2$ be a basis? This can be done inside, but it is unknown whether it can be done on the boundary? For one var. it is trivial. For a bicylinder it is done (since the

Taylor series conv. on the bicylinder and one can take ...). Without bdry. — then the ball can be replaced by any domain of holomorphy.

23/XI. Arnold. "On Maslov's work".

Consider, in partic., the Cauchy prob. for the Schrödinger eqn. Config. space.

$$\frac{i}{h}\frac{\partial \Psi}{\partial t} = \Delta\Psi - \frac{U(x)}{h^2}\Psi$$



**Uncertain reading.** The factor before the derivative is written $i/h$; $h^2$ stands beneath $U(x)$ at right.

Original source details: GELFAND-P000041

$U(x)$ — potent. energy.

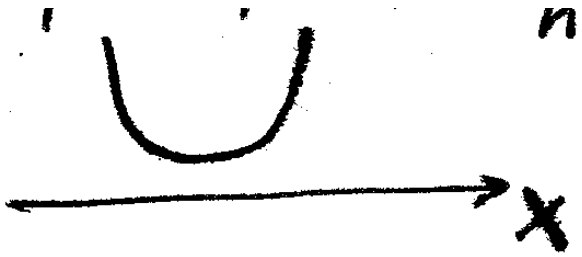
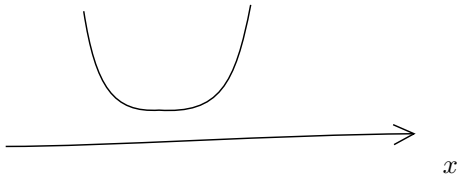


**Diagram and typed description.** An open-topped U-shaped curve lies above an almost horizontal line with a rightward tip labelled $x$; its bottom does not touch the line. No vertical axis is drawn.

Original source details: GELFAND-P000041



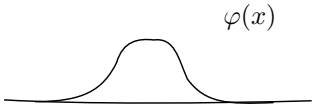

$$\Psi|_{t=0} = \varphi(x)e^{\frac{i}{h}S(x)}$$

[Beneath $\varphi(x)$: fin.]

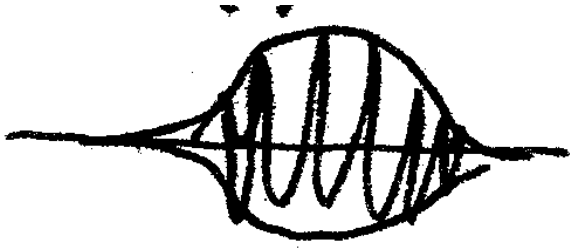


**Diagram and typed description.** A small rounded rise with steep sides stands over a horizontal stroke; $\varphi(x)$ is written above at right. Separate arrowheads cannot be distinguished.

Original source details: GELFAND-P000041

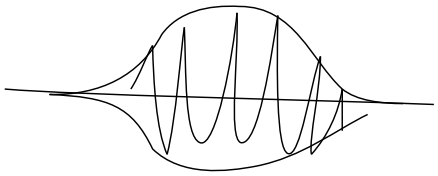



$S(x)$ can be $S(x) = px$.



Original source details: GELFAND-P000041

**Diagram and typed description.** Wavy and almost vertical strokes lie between rounded upper and lower contours, crossed by an almost horizontal stroke. Some individual connections at the ends and thickened parts cannot be distinguished securely.



Oscill. (rapid), [пропод.] $\varphi$.



**Uncertain reading.** After “Oscill. (rapid)” the manuscript writes пропод. Its expansion is uncertain. It is followed by $\varphi$.

Original source details: GELFAND-P000041

GELFAND-P000042

Seek the asympt. as $h \to 0$ in the form

$$\Psi = e^{\frac{i}{h}a(x,t)}.$$

$a$ — a complex fn. Substitute and obtain a seq. of asympt. formulas for the solution of the prob. The first 2 terms have the form

$$\Psi(x,t) = \frac{\varphi(X(x,t)}{\sqrt{\frac{\partial x(X,t)}{\partial X}}} e^{\frac{i}{h}(S(X(x,t))+\int p\,dx)+O(h)}$$

**Manuscript notation.** There is one closing parenthesis after $t$ in the numerator.



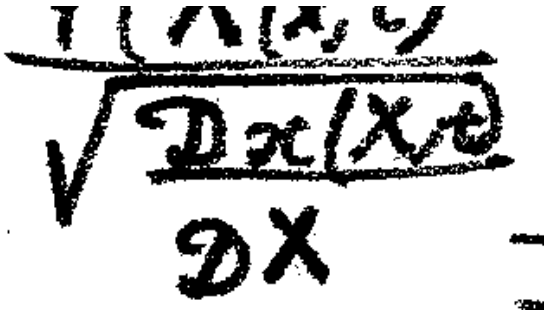
Original source details: GELFAND-P000042

**Uncertain reading.** The entry $+O(h)$ is on the raised line after the phase parenthesis; there is no separate lower line for it.



Original source details: GELFAND-P000042

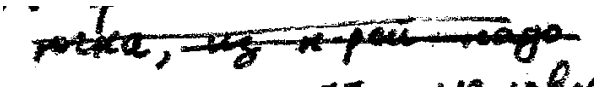
**Uncertain reading.** The first rounded sign in the numerator beneath the radical resembles $D$ / $\partial$; the denominator shows $\partial X$. The upper argument is written $(X,t)$.



Original source details: GELFAND-P000042

[From the denominator: — from unit.; see Titchmarsh and Landau.] $\varphi$ — given init. amplitude. $X$ at $t = 0 = x$.

$\underset{(x,t)}{X}$ — [cancelled: a point from which one must].



Original source details: GELFAND-P000042

**Uncertain reading.** Beneath the cancellation: точка, из к-рой надо / наше (a point from which one must / ours); the final letters are crossed by the stroke.

Classical init. condition.

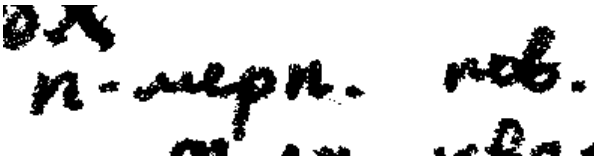
$$p = \dot{x}, \qquad \dot{x} = \frac{\partial S}{\partial x}$$



Original source details: GELFAND-P000042

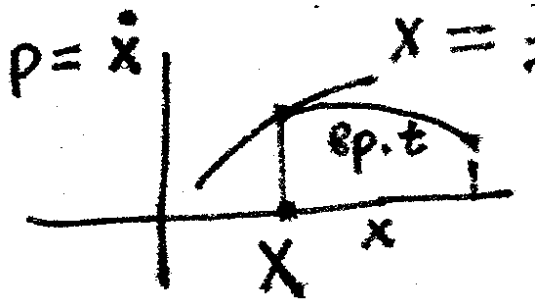
**Uncertain reading.** A dot is visible over each $x$ / $X$. The lower sign after $\partial S$ resembles $X$ / $x$; its case is not secure. No separate subscript is visible on the first sign.

$S$ is given at the init. instant. $n$-dim. пв. / пб.



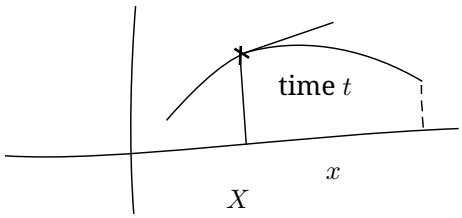
Original source details: GELFAND-P000042

**Uncertain reading.** After $n$-dim. is a short joined abbreviation resembling пв. / пб.; it is not expanded.




Original source details: GELFAND-P000042

**Diagram and typed description.** An almost horizontal line crosses a vertical one; an arch lies above them, with two vertical strokes from it. $X$ is below the left stroke; $x$ stands between the strokes. Inside: time $t$. An inclined stroke rises rightwards from the left junction, with a thickened start; its arrow direction is not established.



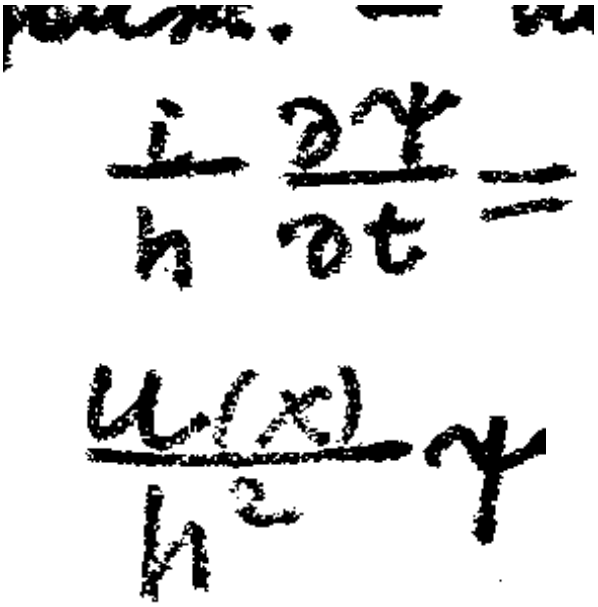

The formula is quasiclassical, valid when the Jacob. $\neq 0$. Quasicl. — when this formula holds.

$$\ddot{x} = -\frac{\partial U}{\partial x}$$

classical mechanics ($x$ — vector). In quantum mech., $\forall t$ there is a wave fn. $\Psi$. $|\Psi|^2$ — prob.

Eqn. of motion — Schröd.

$$\frac{i}{h}\frac{\partial \Psi}{\partial t} = \Delta\Psi - \frac{U(x)}{h^2}\Psi$$



**Uncertain reading.** The repeated equation has $i/h$ at left and $U(x)/h^2$ at right; the two fractions are separately visible.

Original source details: GELFAND-P000042

(if there is no $i$, then it is the heat eqn.; if initially at a point, it later spreads out). Connected with the asympt. of eigenvalues. $\Psi|_{t=0}$ is given. $\Psi(x)$



**Diagram and typed description.** A long almost horizontal stroke with a thickened right end follows $\Psi(x)$. There is no caption over it or secure arrowhead.

Original source details: GELFAND-P000042

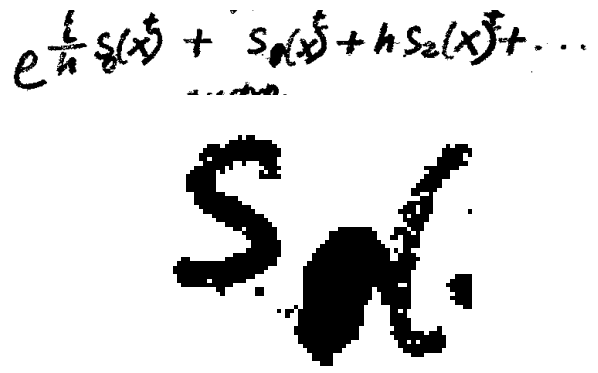
GELFAND-P000043

$S(x) = px$, then the de Broglie formula.

$$\frac{\partial S}{\partial x} = p$$

Usual method of finding the asympt. — seek the soln. in the form

$$e^{\frac{i}{h}S_0(x^t)+S_1(x^t)+hS_2(x^t)+\cdots}$$



**Uncertain reading.** A raised sign with a crossbar resembling $t$ stands over $x$ in all three arguments; readings $x^t$ / $\dot{x}$. The subscript of the second $S$ is blotted: $S_1$ / $S_0$.

Original source details: GELFAND-P000043

With respect to time [inserted above: diff.] $\frac{1}{h^2}$ ...The leading term must be sought as $e^{\frac{i}{h}S_0(x)}$. If $S$ is real, then oscill.; if complex, then there is decay. Obtain a 1st-order eqn., and the 1st fn. is real, and obtain oscill. Must take the amplit., hence 2 terms. Now must choose init. data for this asympt.

$\Omega$ / $\delta$ is not allowed, since there are many extraneous frequencies (strongly localized).



**Uncertain reading.** $\Omega$ or $\delta$: a loop with a horizontal lower stroke and a long left tail.

Original source details: GELFAND-P000043

$$\varphi(x,t)e^{\frac{i}{h}S(x,t)}$$

if substituted there, then for $S$ a 1st-order eqn.

$$\frac{\partial S}{\partial t} = H\left(x, \frac{\partial S}{\partial x}\right), \qquad H = U(x) + \frac{\dot{x}^2}{2}$$

to which an ordinary eqn. corresponds. $\varphi$ — density of the dyn. syst. (how they move according to class. mech.). Must take $x$ and find $X$ for it, so that our eqn. comes from $X$ to $x$ after $t$. The points carry $\varphi$ with them.

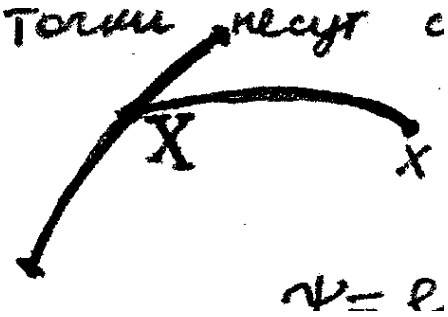
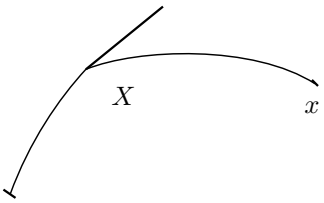



**Diagram and typed description.** A bending curve has $X$ below the left junction and $x$ at its right end. A thick inclined stroke passes through the left junction. Its arrowhead or direction is not established.




Original source details: GELFAND-P000043

$\dot{x}$ is given from $x$ (the dyn. syst. is given). For small $t$, single-valued.

$$\Psi = e^{iH}$$

unit. $\mathcal{D}$ — the Jacob. is added for unitarity. $e$ — the phase factor rotates with varying amplitude. Maslov's work, when $t$ is arbitrary rather than small.

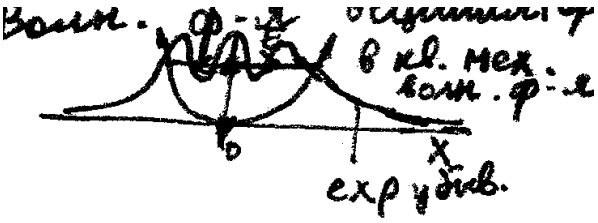
$t = 0$, Jacob. [an entry is cancelled] $= 1$ (identity transf.).



**Uncertain reading.** A heavily cancelled expression after Jacob. and before $= 1$; the individual signs are not established.

Original source details: GELFAND-P000043

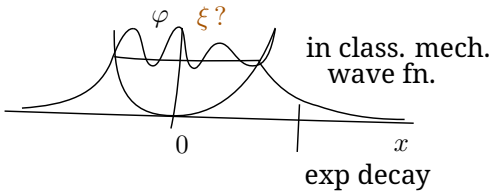
The wave fn. of the oscillator. Pendulum.






Original source details: GELFAND-P000043

**Diagram and typed description.** A parabolic arch, oscillating contour, almost horizontal stroke and central stroke intersect. Visible labels: $\varphi$, $0$, $x$; a small upper sign resembles $\xi$, but its lower part joins the curve. Captions: in class. mech. wave fn.; exp decay. Exact connections of dense strokes and the number of oscillations are not established.




GELFAND-P000044

If $t$ is large, it may happen that the curve of init. data ($n$-dim. пв.) moves.



Original source details: GELFAND-P000044

**Diagram and typed description.** Upper-left drawing: several bending arches cross inclined strokes, an almost horizontal line and a dense almost vertical bundle. Small tips appear at the upper-left end and inside the bundle; some individual connections overlap.



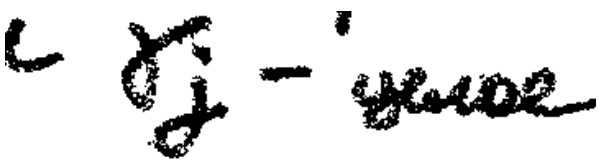
If there is a vertical tangent, then $\mathcal{D} = 0$, but afterwards there are 2 points and the formula is retained [(!) / (?)], but must add terms over the points $\gamma_j$ / $n_j$ — an integer.

**Uncertain reading.** After retained, a small parenthesized sign with a separate lower dot: (!) / (?). Its upper part is blotted.



Original source details: GELFAND-P000044

**Uncertain reading.** $\gamma_j$ or $n_j$: the upper sign has a loop and two raised strokes; $j$ lies below. The following entries resemble $n_j$.



Original source details: GELFAND-P000044

$$\int p\,dx) + i\,\frac{\pi}{2} n_j$$

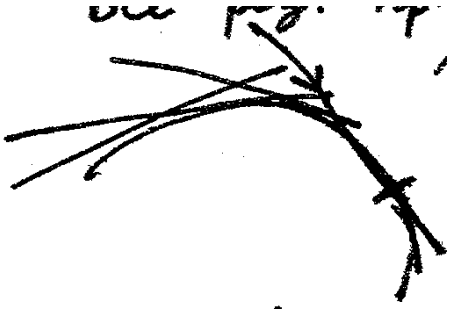
[A sign between $i$ and $\pi/2$ is cancelled.]



Original source details: GELFAND-P000044

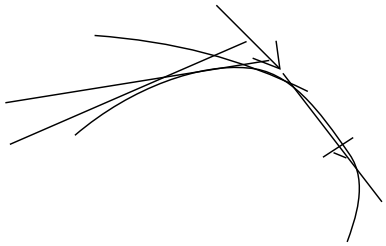
**Uncertain reading.** A sign between $i$ and $\pi/2$ is heavily cancelled; the readings $+$ / $-$ are not securely established.

Maslov's result — the $n_j$ have been found. $n_j = \pm 1$ from general considerations. Go around the branch point in the complex plane; this can determine $\oplus$ or $\ominus$. All results also apply in optics.



Original source details: GELFAND-P000044

**Diagram and typed description.** Several inclined rays cross a bending envelope. A thick arrow points downwards to the right at the upper tangency. Short crossing strokes lie on the lower-right branch; their connection to the envelope is uncertain.



The rays envelop a curve. We shall follow the phase along a ray. For small $\lambda$ — a phase jump of $\pi/2$ at the point of tangency. Answer $n_j$ — the Morse index.

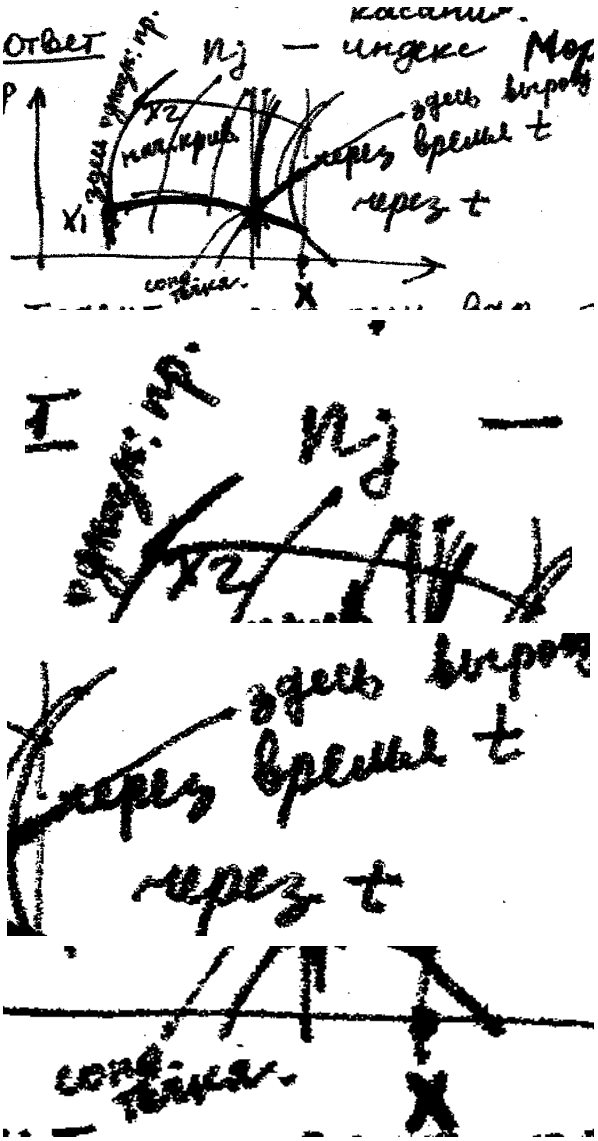
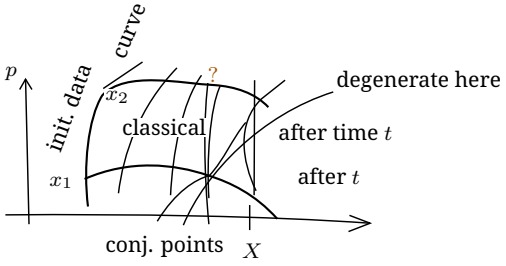



**Diagram and typed description.** The $p$ axis points upwards and the horizontal axis rightwards. A family of bending and almost vertical strokes; $x_1$, $x_2$, $X$. Captions: curve of init. data; classical; degenerate here; after time $t$; after $t$; conj. points. A small dense sign is above the central vertical stroke; its reading and some individual intersections are not established.




Original source details: GELFAND-P000044

A trajectory is a soln. of a variational problem by the principle of least action.

$$\delta u = 0$$

$$\delta^2 u$$

a quadratic form. It has an index [a word is cancelled] (number of neg. squares).



**Uncertain reading.** A short word left of (number of neg. squares) is cancelled by dense strokes; its reading is not established.

Original source details: GELFAND-P000044

Or [a word is cancelled] the number of conjugate points.

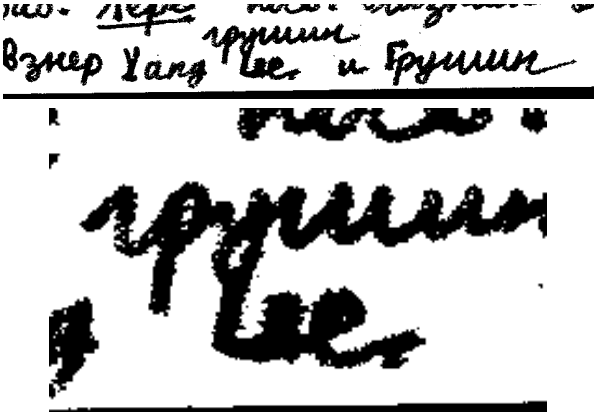


**Uncertain reading.** A word above number of conjugate points is covered by several horizontal strokes; its reading is not established.

Original source details: GELFAND-P000044

The 1st part of Leray's work is devoted to related questions.

Next time Povzner Yang Lee and Brushlin [Брушлин].



**Uncertain reading.** The line reads "Yang Lee". Above "Lee" is a short Russian addition resembling группам [to groups]. The final surname is provisionally read as Брушлин [Brushlin].

Original source details: GELFAND-P000044

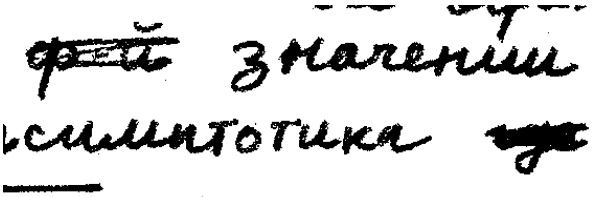
GELFAND-P000045

Berezin.

$$H = \frac{p^2}{2} + V(x)$$

(e.g. $V$ grows). Let the surfaces $p^2 + V(x) =$ const all be bounded.

Then the asympt. of eigenvalues [an abbreviation is cancelled]:



**Uncertain reading.** The cancelled abbreviation before values resembles ф-й / ф-и (fns.); horizontal strokes cross the upper letters.

Original source details: GELFAND-P000045

the volume of the region where $H \leq n\pi h$ — the asymptotic [a word is cancelled] of the eigenvalues.



**Uncertain reading.** A short word between asymptotic and eigenvalues is covered by dense ink; its reading is not established.

Original source details: GELFAND-P000045

———— · ————

Gelfand I.

$$-y'' + p(x)y = \lambda y$$

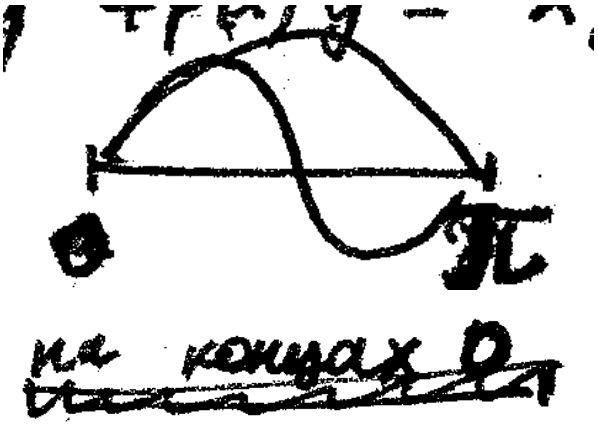
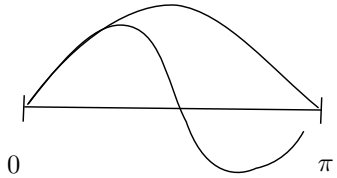


**Diagram and typed description.** A horizontal segment with short end strokes and labels $0$, $\pi$. One arch lies above the segment; another curve begins above it, crosses it, and falls below. A dense junction lies near the right end; the nearby underline under zero at the endpoints is crossed out.




Original source details: GELFAND-P000045

Zero at the endpoints [the underline is crossed out].

$p = 0$, eigenfns., $\lambda = -n^2$.



**Uncertain reading.** In $\lambda = -n^2$, a separate horizontal stroke is visible between the equality sign and $n$; the minus is retained.

Original source details: GELFAND-P000045

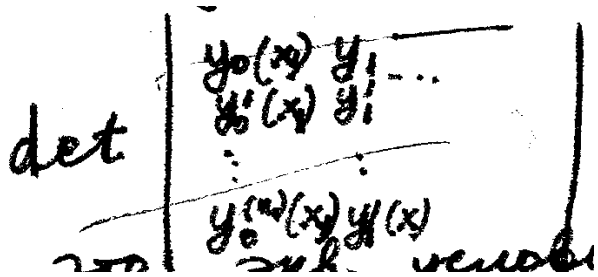
Claim. The 1st eigenfn. has no zeros (Morse index 1). The 2nd — ditto — has one zero. This holds for $\forall$ eqn. The zeros interlace. A more precise theorem. If $y_0, \dots, y_n$ are eigenfns., then

$$\sum_{i=1}^{n} a_i y_i(x)$$

will have $\leq n$ zeros!! (resembles polynomials).

[A cancelled determinant:]

$$\det \begin{vmatrix} y_0(x) & y_1 & \cdots \\ y_0'(x) & y_1' & \\ \vdots & \vdots & \\ y_0^{(n)}(x) & y'(x) & \end{vmatrix}$$

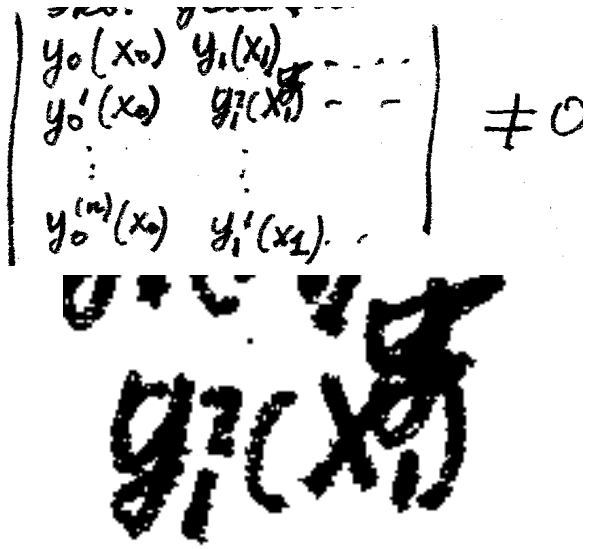


**Uncertain reading.** The lower second entry of the cancelled determinant resembles $y'(x)$ / $y_1^{(n)}(x)$; its subscript and derivative order are not established.

Original source details: GELFAND-P000045

This is equiv. to the condition

$$\begin{vmatrix} y_0(x_0) & y_1(x_1) & \cdots \\ y_0'(x_0) & y_1'(x_1) & \cdots \\ \vdots & \vdots & \\ y_0^{(n)}(x_0) & y_1'(x_1) & \end{vmatrix} \neq 0$$

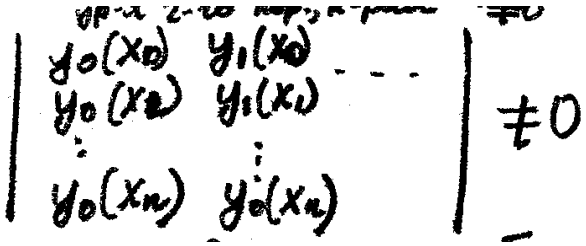


**Uncertain reading.** The upper sign of $y_1\prime(x_1)$ in the second entry of the second row is overwritten. A single raised stroke is visible in the lower second entry: $y_1\prime(x_1)$; a possible order $(n)$ is not established.

Original source details: GELFAND-P000045

Or [Above the following determinant:] this is the 1st eigenfn. of the 2nd-order eqn., which is $\neq 0$.

$$\begin{vmatrix} y_0(x_0) & y_1(x_0) & \cdots \\ y_0(x_1) & y_1(x_1) & \cdots \\ \vdots & \vdots & \\ y_0(x_n) & y_0(x_n) & \end{vmatrix} \neq 0$$



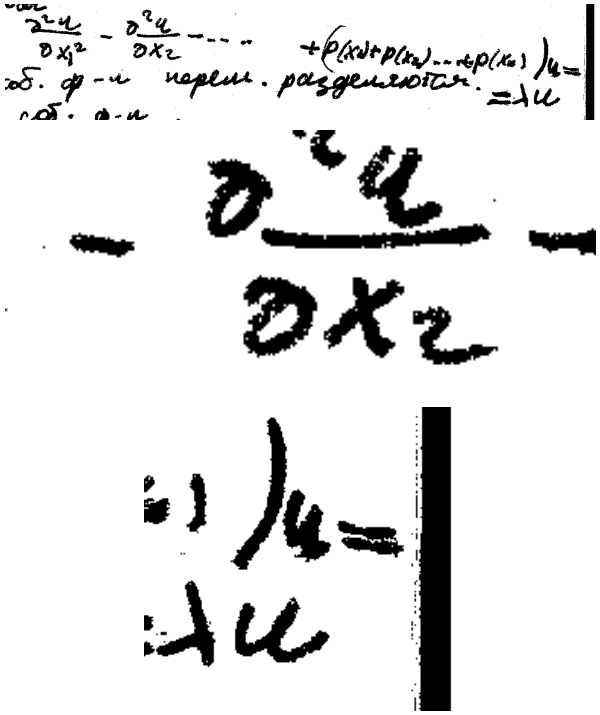
**Uncertain reading.** In the second entry of the bottom row the subscript is rounded, like $y_0$ in the first column. The dot above it aligns with the vertical ellipsis; a separate derivative stroke is uncertain.

Original source details: GELFAND-P000045

Hence a linear comb. has $\leq n$ zeros if $x_0 < x_1 < \cdots < x_n$.

Petrov. So that

$$\frac{\partial^2 u}{\partial x_1^2} - \frac{\partial^2 u}{\partial x_2} - \dots + (p(x_1) + p(x_2) \dots + p(x_n))\, u = \\ = \lambda u$$



**Uncertain reading.** The first denominator is $x_1^2$; the second shows $x_2$ without a square. At the right edge: $u =$; below, another equality sign before $\lambda u$. Both signs are retained.

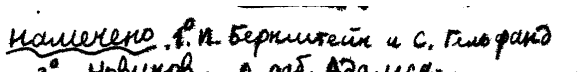
Original source details: GELFAND-P000045

Eigenfns., the variables separate. Eigenfns. $u_1(x_1)u_2(x_2) \dots u_n(x_n)$. The 1st eigenfn. $\neq 0$.

GELFAND-P000046

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Planned. [Number and initials unclear] Bernstein and S. Gilfand.

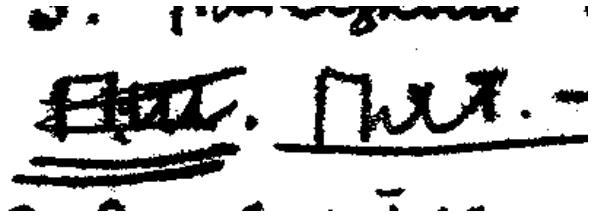


**Uncertain reading.** The first row begins with a long looped sign, then И / Н before Бернштейн. Readings: 1° I. Bernstein and S. Gilfand / F. N. Bernstein and S. Gilfand / 1° I. Bernstein and S. Gelfand. Full names are not established.

Original source details: GELFAND-P000046

[Separately at the right:] Dyson. 2°. Novikov, on the work of Adams. 3°. Dynkin. 4°. Vinberg. 5°. Pyatetskii-Shapiro.

[Cancelled.]

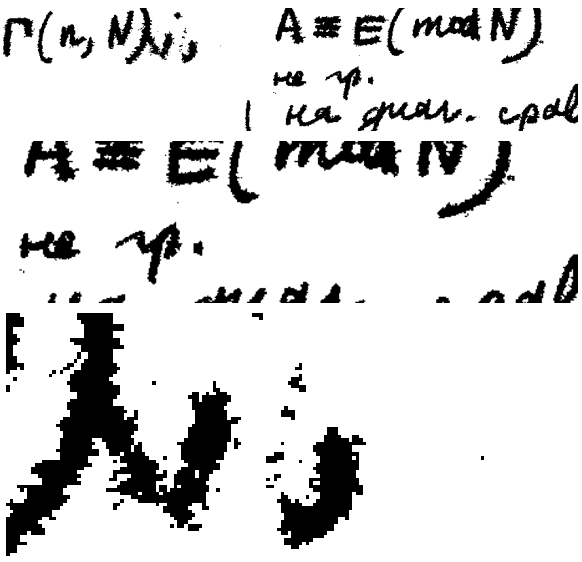


**Uncertain reading.** The short cancelled and doubly underlined heading before Pyat.-Shapiro resembles Пят. (Pyat.); dense strokes cover the earlier writing.

Original source details: GELFAND-P000046

Pyat.-Shapiro. Problem of Klein and Serre. Consider the gr. of integer matrices of order $n$, $\Gamma(n)$, with $\det = 1$. [Cancelled: How] describe all subgrs. $\Gamma'$ of finite index? Ex.

$$\Gamma(n, N), \qquad A \equiv E \pmod{N}$$



**Uncertain reading.** A small inscription below $\Gamma(n, N)$ resembles $\lambda ij$ / $i, j$; below $A \equiv E$: probably не гр. (not gr.), possibly на гр. (on gr.). The letters and the attachment of the small inscription to the formula are not established.

Original source details: GELFAND-P000046

(on the diag., congruent to $1$; off the diag., divisible by $N$).

GELFAND-P000047

Are these all the subgroups? Does $\Gamma'$ always contain $\Gamma(n, N)$ (a congruence subgr.). For $n = 2$ this is false (Klein). $n \geq 3$, this is true (Serre). Connection with Lie groups. Consider the matrices

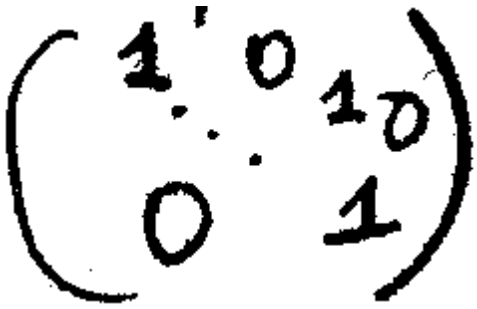


**Diagram and typed description.** Matrix schematic.

$$\begin{pmatrix} 1 & & 0 & & \\ & \ddots & & 1 & 0 \\ & & \ddots & & \\ 0 & & & 1 & \end{pmatrix}$$

Original source details: GELFAND-P000047

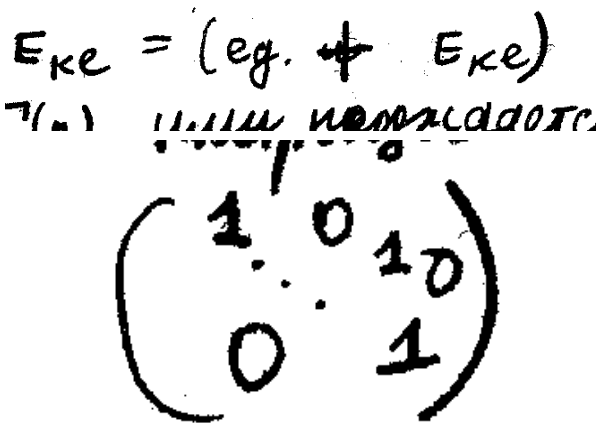
[Cancelled symbol.]



**Uncertain reading.** A cancelled and overwritten sign before the elementary-matrix notation resembles $E$; the underlying strokes are not established.

Original source details: GELFAND-P000047

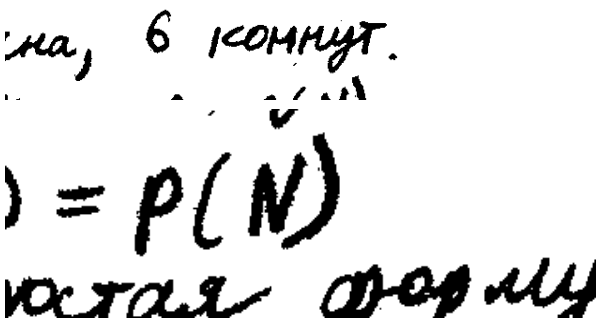
$E_{ke} = (\mathrm{id.} + E_{ke})$.



**Uncertain reading.** The same $E$-shaped sign occurs on both sides of the equality; its second subscript resembles $e$ / $\ell$. Readings $E_{ke} = (\mathrm{id.} + E_{ke})$ / $e_{k\ell} = (\mathrm{id.} + E_{k\ell})$. The schematic matrix retains the visible ones, zeros and dots; unwritten entries have not been supplied.

Original source details: GELFAND-P000047

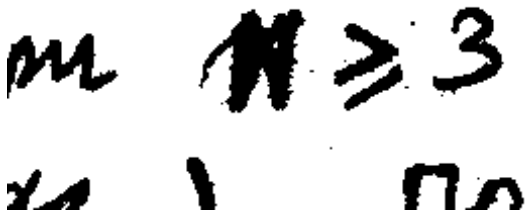
The gr. $\Gamma(n)$ is generated by them. Take these matrices and their conjugates and raise them to the power $N$: $E_{ke}^N$, $\Gamma^N$. For $n = 2$ it has infinite index; $n \geq 3$, finite index. $\Gamma/\Gamma^N$. $\boxed{n = 2}$ $N = 2, 3, 4, 5$ is finite, 6 commut. / коннут. [коммут. / коннут.; uncertain abbreviation]



**Uncertain reading.** After 6, the abbreviation коммут. / коннут. (commut. / коннут.; uncertain reading). Its initial letter has separate vertical and angled strokes. Below: $p = p(N)$; the letter resembles $p$. The abbreviation is not expanded.

Original source details: GELFAND-P000047

$\Gamma/\Gamma^N$ = fund. gr. of a surface, $p = p(N)$, a simple formula. Serre, Lazard, Bass (1964). Consider $a \equiv b\ (N)$, then for $n \geq 3$ / $N \geq 3$ there is a transf. $a \to b$ in $\Gamma/\Gamma^N$ (elem. transf.).



**Uncertain reading.** After then for, the sign before $\geq 3$ is heavily overwritten: $n \geq 3$ / $N \geq 3$. Vertical strokes resembling capital $N$ are visible.

Original source details: GELFAND-P000047

Then cohomol. technique.

Gelfand. Tomorrow at the conf. on elem. particles, there will be a talk about

[Cancelled.]



**Uncertain reading.** A group designation before boxed $SU(3)$ is densely cancelled. Readings $SU(3)$ / $SU(2)$; the digit beneath the ink is not established.

Original source details: GELFAND-P000047

$\boxed{SU(3)}$. Known particles are combined into groups, transforming into one another (multiplets). Another repr. of $SU(3)$. The idea $SU(2) \underset{\text{almost}}{\approx} SO(3)$. [Above the line: fns., lin. comb.] Combine by means of $\boxed{SU(2) \times SU(3)}$. The hypothesis is known also for $SU(6)$. Remark — actually the Lorentz group (with par. translations) (the Poincaré group). [Cancelled: $SU(2)$] $\longrightarrow$ Poincaré gr. Work of Todorov. Kadyshevsky, Muradyan, Tavkhelidze. Instead of $SU(6)$ — a 108-param. group.

GELFAND-P000048

Salam proposed a 144-param. group. Fulton, Wess. A 2-step gr. with a comm. normal divisor.
$SU(3) \otimes SU(2)$ in some sense.



**Uncertain reading.** The abbreviation before смысле (sense) resembles как. or нек.

Original source details: GELFAND-P000048

Lorentz transf. — complex matrices.

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \qquad \det = 1$$

(spinor repr. of the Lorentz gr.). Poincaré gr.

$$A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

Vector — Hermitian forms of order 2.

$$H = \begin{pmatrix} a & b \\ \bar{b} & c \end{pmatrix}$$

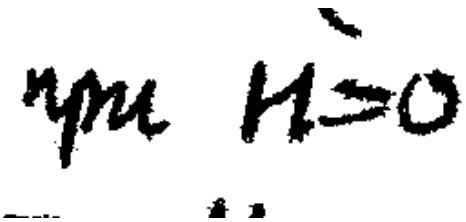
A vector in 4-dim. space; $a, c$ real, $b$ complex.

$$H \to AHA^*$$

Lorentz, det is preserved. Poincaré gr.

$$\begin{pmatrix} A & H \\ 0 & A^{*-1} \end{pmatrix}$$

This is a group for $H \geq 0$ / $H = 0$ rot. ($H$ adds). $H = 1$ — par. transl.



**Uncertain reading.** The condition on $H$ has an upper angled stroke and a lower horizontal stroke before $0$; readings $H \geq 0$ / $H = 0$.

Original source details: GELFAND-P000048

[Cancelled.]



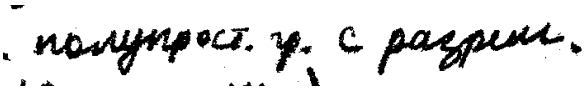
**Uncertain reading.** Before Matrix of order 4, a cancelled phrase begins gr.; the next word resembles Вращ. / Лоренца (rot. / Lorentz).

Original source details: GELFAND-P000048

Matrix of order 4.

$$\begin{pmatrix} A & H \\ 0 & A^{*-1} \end{pmatrix} \otimes SU(3)$$

Obtain a semisimple gr. with solv. (108-param.).



**Uncertain reading.** The abbreviation after with resembles разреш. (solv.); the ending is joined.

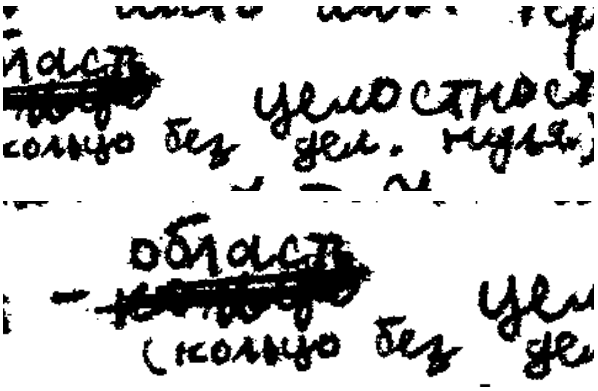
Original source details: GELFAND-P000048

One can describe what is needed to solve the theor. of elem. particles).

15/II

Kazhdan. Palam.'s result in purely alg. terms.

Let $A$ be an integral domain [domain inserted above a cancelled word] (ring without zero divisors), $I$ a prim. ideal.

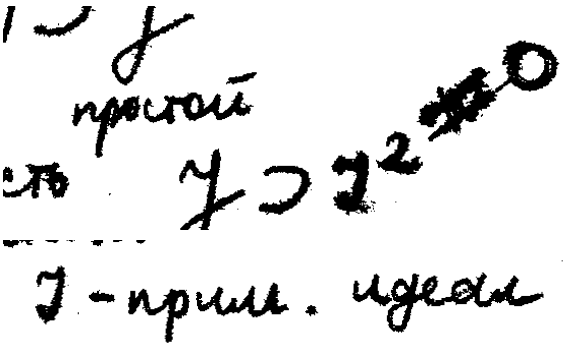


**Uncertain reading.** Область [domain] is inserted above a cancelled word; the candidate is кольцо [ring]. The dense strokes do not establish the whole word. The following целостности [integral, in integral domain] (ring without zero divisors) is legible.

Original source details: GELFAND-P000048

$$I \supset J$$

[Beneath $J$:] prime. And let $J \supset I^2$ [cancelled: $\neq$] $0$.



**Uncertain reading.** $I$ is introduced as a prim. ideal; prime is written beneath $J$ in $I \supset J$. Next: $J \supset I^2$, with a cancelled sign resembling $\neq$ before $0$. The letters and cancellation are retained.

Original source details: GELFAND-P000048

Every such ideal is given by diff. opers. of order 1.

GELFAND-P000049

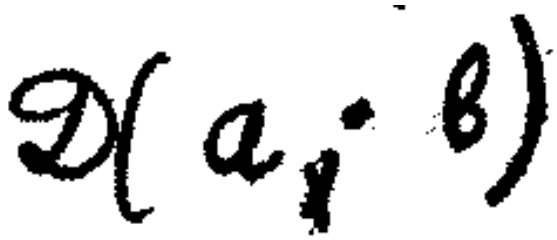
A diff. oper. is a $A$ / $K$ / $k$-lin. oper.



**Uncertain reading.** The heavily overwritten sign before -lin. resembles $A$ with a projecting upper stroke. $K$ and $k$ remain possible.

Original source details: GELFAND-P000049

$$D(a \cdot b) = aD(b) + D(a)b$$

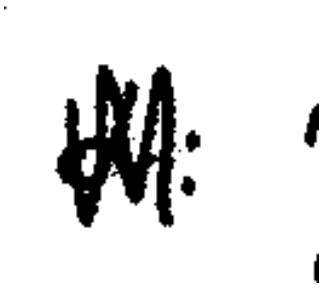


**Uncertain reading.** Below the multiplication dot in the argument of $D$ is a separate comma-like mark.

Original source details: GELFAND-P000049

An ideal is the zero of some collection of diff. opers. Let $A/J^2$ (of order 1 because of $J^2$). An operator with values in the ring $A/J$ instead of in the original ring. Diff. everywhere, and the ans. on a subset. $aD(b)$ — multiplication by $a$. On $J$ the oper. is $A$-linear [below: diff.]. Conversely, every $A$-lin. oper. on $J$

[Cancelled sign before $J$]:



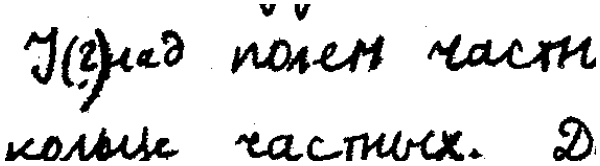
**Uncertain reading.** To the left of $J$, three dense, nearly vertical strokes with a loop are cancelled, followed by a colon. An individual underlying letter is not established.

Original source details: GELFAND-P000049

$$J \xrightarrow{f} A/J$$

Then it can be multiplied by an elem. $b$, again lin. [Above: $bf$.] $\exists b$ such that $bf$ extends to a diff. of the whole ring. Everything follows from this.

We shall consider $J(2)$ / $J(\xi)$ / $J(\ell)$ over the field of fractions.

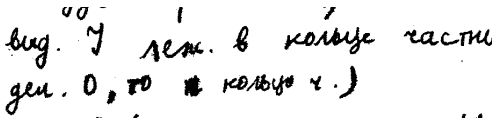


**Uncertain reading.** The parenthesized sign after $J$ resembles $2$ / $\xi$ / $\ell$, followed by over the field of fractions. Readings $J(2)$ / $J(\xi)$ / $J(\ell)$.

Original source details: GELFAND-P000049

Functionals exist, appar. $J$ lies in the ring of fractions.

Multiply by $b$ (since there are no zero divisors, then [a short sign is cancelled] ring of fr.).



Original source details: GELFAND-P000049

**Uncertain reading.** A short densely cancelled sign between then and ring of fr.; its reading is not established.

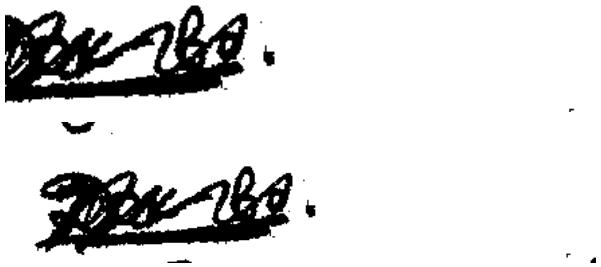
If $A/J$ is a field, then $\forall$ functional extends to a diff. (Need to take the localization of $A/J$). There is a map of the field

$$A/J \longrightarrow A$$

[Beneath the arrow: hom.] such that the reverse hom. is the identity.

$$A/J \longrightarrow A \longrightarrow A/J$$

[Cancelled.]



Original source details: GELFAND-P000049

**Uncertain reading.** The cancelled heading before For now, arb. resembles Док-во. (Proof.); its reading remains uncertain beneath the strike.

For now, arb. $A/J \subset A$, $J \subset A$. Any elem. decomposes into a $\Sigma$ ($\Sigma$ as a lin. space).

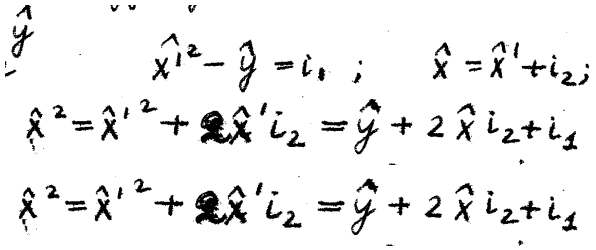
GELFAND-P000050

On $J$ it is already given. A lin. map $J \to A/J$ extends.

On the whole [of it], $0$.

A transcend. elem. anywhere.

$$\begin{aligned} y \to \hat{y}, \qquad & x^2 = y \\ \hat{x}'^2 - \hat{y} = i_1; \qquad & \hat{x} = \hat{x}' + i_2; \\ \hat{x}^2 = \hat{x}'^2 + 2\hat{x}' i_2 = \hat{y} + 2\hat{x} i_2 + i_1 & \end{aligned}$$



Original source details: GELFAND-P000050

**Uncertain reading.** The expansion has no $i_2^2$. The multiplier is $\hat{x}\prime$ in the first equality and $\hat{x}$ without a prime in the second. Both readings are retained.

Hence $i_2$ is found (since $A/J$ is a field). Instead of $I$, $J$ was written, and conversely.

---

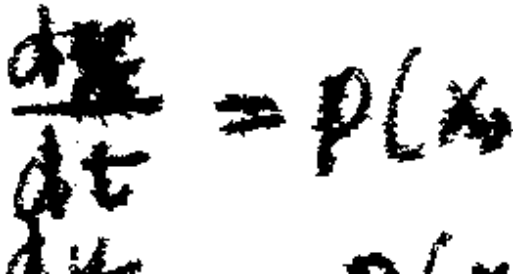
General form of a diff. oper. in the polynomial ring.

$$\sum P_i(x_1, \ldots, x_n) \frac{\partial}{\partial x_i}$$

$P_i$ are polynomials. Alg. theory of diff. eqns. over an arb. field. Dieudonné, Grothendieck: in alg. problems, passed to an arb. field.

Ordinary eqn.

$$\frac{dx}{dt} = P(x,y)$$
$$\frac{dy}{dt} = Q(x,y)$$



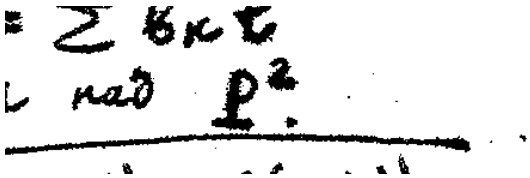
**Uncertain reading.** The letter in the numerator of the first equation is heavily overwritten; the visible candidate is $dx$, and the earlier letter is not established.

Original source details: GELFAND-P000050

$t$ is replaced by a local ring at $\forall$ point ($t$ a local parameter).

$$x = \sum a_k t^k$$
$$y = \sum b_k t^k$$

Diff. eqns. over $P^2$.



**Uncertain reading.** An underlined $P$-like letter with a raised 2: $P^2$ / $p^2$. No double stroke is visible.

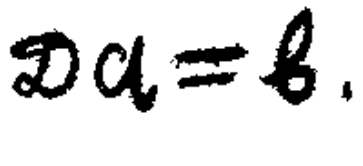
Original source details: GELFAND-P000050

$$D(x(t)) = f(x(t))$$

$D$ is a diff. operator in the ring of elems. $x$. Ringed space. Points $(x,y)$ and fibres. A ring at $\forall$ point. The fibre over each point is a power series in $t$ (a power series in $t$ at $\forall$ point).

GELFAND-P000051

$$Da = b.$$



**Uncertain reading.** In the left entry, $a$ / $\alpha$; no separate definition on this frame.

Original source details: GELFAND-P000051

Init. cond. — value in a max. ideal. Then localize. $D$ carries over to any localization. The soln. must be an ideal. (?) The whole ring need not be given. It is known locally. $x$ and $y$ need not be specified.

Palamodov.

$\mathcal{P}$ — ring of polyn. $P(z_1, \ldots, z_n)$

[Before $P$: a sign is cancelled.]



**Uncertain reading.** a short densely cancelled symbol

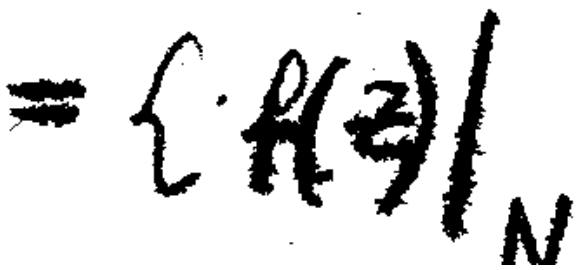
Original source details: GELFAND-P000051

$\mathcal{H}$ — space of all fns. in a domain of holom. (e.g. a ball)

$$\mathcal{J} \subset \mathcal{P}$$

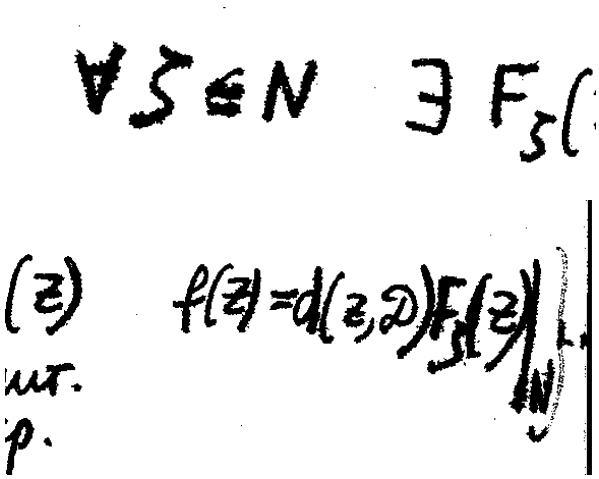
A prim. ideal with basis $P_1, \ldots, P_t$ Consider $\mathcal{J}\mathcal{H}$ Quotient space $\mathcal{H}/\mathcal{J}\mathcal{H}$ Construct $\mathcal{H}\{\mathcal{J}\}$, isomorphic to it by means of the operator $d(z, D)$

$$\mathcal{H}\{\mathcal{J}\} = \{h(z)\big|_N \quad \forall \zeta \in N \quad \exists F_\zeta(z) \quad f(z) = d(z, D)F_\zeta(z)\big|_N\}$$



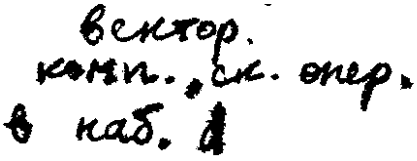
**Uncertain reading.** The first function symbol inside the brace resembles $h$ with a stroke across its lower part; the separate entry on the right reads $f(z)$.

Original source details: GELFAND-P000051



**Uncertain reading.** The point sign after $\forall$ and the subscript of $F$ resemble $\zeta$ / $\xi$; the operator letter in $d(z, D)$ resembles $d$ / $\delta$. The right formula reaches the frame edge; the lower $N$ and closing brace are visible. The mark below the right-hand $F$ matches the subscript on the first $F$; both are typeset as $\zeta$.

Original source details: GELFAND-P000051



**Uncertain reading.** Below the first $h(z)$: vector. comp. / конх. / конк.; see / an. / lin. oper. in coll. $d$ / $\lambda$ [комп. / конх. / конк., см. / ан. / лин. опер. в наб.; uncertain abbreviations].

Original source details: GELFAND-P000051

[Below $F_\zeta(z)$: analytic in a nbd.] Local theorem. There is a theorem with growth estimates. Res.

$$\mathcal{H}\{\mathcal{J}\} \cong \mathcal{H}/\mathcal{J}\mathcal{H}$$

The isom. is natural

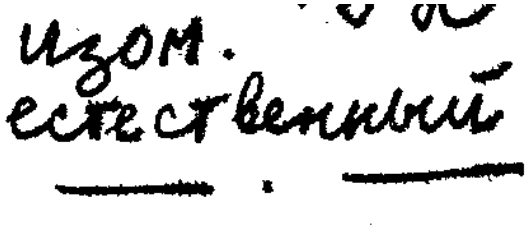


**Diagram and typed description.** Below the annotation are two nearly horizontal strokes with a separate dot between them.



Original source details: GELFAND-P000051

GELFAND-P000052

Dynin. $X$ — comp. manif. Over it, vect. bundles $E_0$ $E_1$

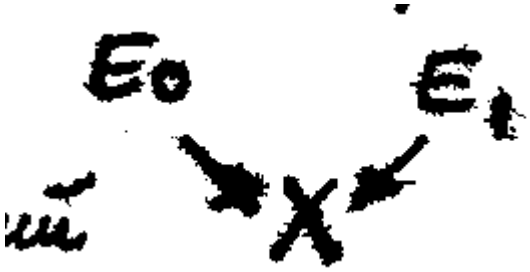
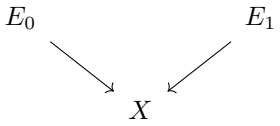


**Diagram and typed description.** Over $X$, two bundles $E_0$, $E_1$, with arrows to $X$.




Original source details: GELFAND-P000052

Sheaf of germs of $C^\infty$-fns. of the 1st and 2nd bundles:

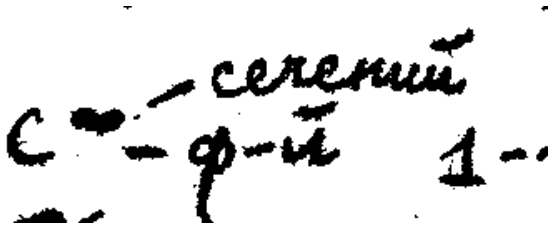
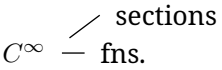


**Diagram and typed description.** Above fns., the word sections is inserted, with a short inclined stroke before it.



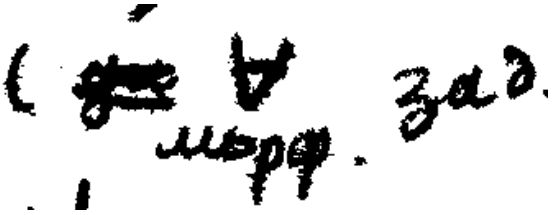

Original source details: GELFAND-P000052

$$\mathcal{E}(E_0) \qquad \mathcal{E}(E_1)$$

and morphisms

$$\mathcal{E}(E_0) \xrightarrow{A} \mathcal{E}(E_1)$$

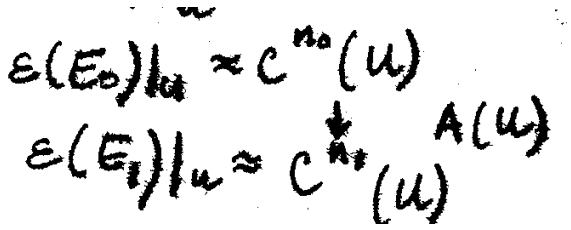
([An entry is cancelled; below it, morph.] $\forall$ given by diff. operators)



**Uncertain reading.** a short densely overwritten entry

Original source details: GELFAND-P000052

Locally, $\mathcal{E}(E_0)|_U$ — smooth vector fns.



Original source details: GELFAND-P000052

**Diagram and typed description.** Two local entries; in the first, the lower restriction sign is densely overwritten: $U$ / $a$. Downward arrow from $C^{n_0}(U)$ to $C^{n_1}(U)$, with $A(U)$ at right.

$$\begin{array}{lcl} \mathcal{E}(E_0)|_U \approx & & C^{n_0}(U) \\ & & \Big\downarrow {\scriptstyle A(U)} \\ \mathcal{E}(E_1)|_U \approx & & C^{n_1}(U) \end{array}$$

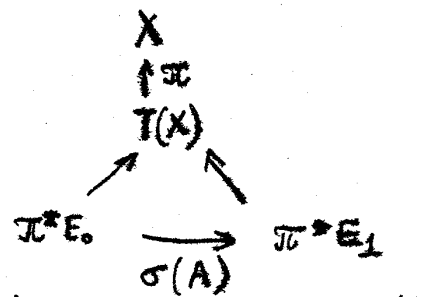
diff. oper. — a matrix. Symbol — princ. part of the matrix. Locally $\sigma(A)(x,\xi)$ — princ. part of the char. matrix.



Original source details: GELFAND-P000052

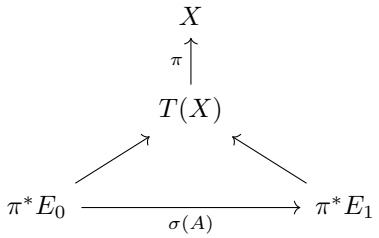
**Uncertain reading.** The second argument of $\sigma(A)$ is densely overwritten; forms $\xi$ / $\zeta$ / $D$ are visible.

(Select terms of highest degree Formalize. Tang. bundle — vect. $F$

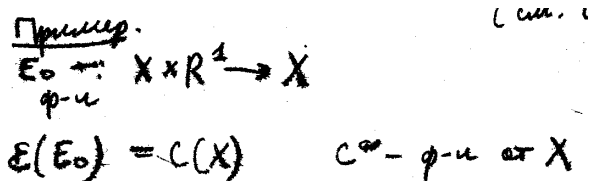


Original source details: GELFAND-P000052

**Diagram and typed description.** Diagram with $X$, $T(X)$, $\pi$, $\pi^*E_0$, $\pi^*E_1$, $\sigma(A)$.



$A$ ellipt. if $\sigma(A)$ is a monomorphism (see Schwartz) Example.



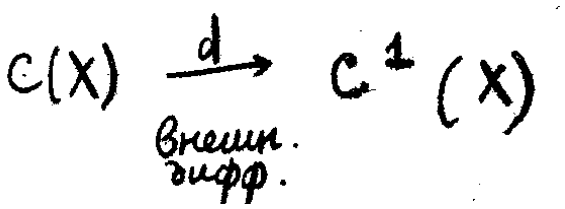
Original source details: GELFAND-P000052

**Diagram and typed description.** At $E_0$, a leftward arrow; below $E_0$, fns.

$$E_0 \longleftarrow X \times \mathbf{R}^1 \longrightarrow X$$
fns.

$$\mathcal{E}(E_0) = C(X) \qquad C^\infty \text{— fns. on } X$$

GELFAND-P000053



Original source details: GELFAND-P000053

**Diagram and typed description.** Below $d$, ext. diff.

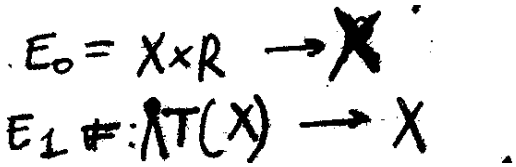
$$C(X) \xrightarrow[\text{ext. diff.}]{d} C^1(X)$$

Loc. $f(x) \longrightarrow$ grad $f(x)$ symbol — mult. by the vector $\xi \in T(X)$ This is an example of an overdet. elliptic system.

$$f(x) \xrightarrow{d} \sum \frac{\partial f}{\partial x_i}\, dx_i$$

$$f(x) \longrightarrow \begin{pmatrix} \frac{\partial}{\partial x_1} \\ \vdots \\ \frac{\partial}{\partial x_n} \end{pmatrix} f(x)$$

$$E_0 = X \times \mathbf{R} \longrightarrow X \qquad E_1 : \Lambda T(X) \longrightarrow X$$



Original source details: GELFAND-P000053

**Uncertain reading.** At the end of the first line, a heavily overwritten $X$. After $E_1$, a dense cancellation before the colon; the sign before $T(X)$ resembles $\Lambda$ / $\wedge$. No clear star at $T$ is visible.

[At right:] diff. forms — sections of such a bundle. Result.

$$\mathcal{E}(E_0) \xrightarrow{A} \mathcal{E}(E_1)$$

Take

$$0 \longrightarrow \ker A \longrightarrow \mathcal{E}(E_0) \xrightarrow{A} \mathcal{E}(E_1)$$

If $A$ is ellipt., then

$$\dim H^i(X, \ker A) < \infty$$

special cases. 1°. Let $A$ be ellipt. in Petrovskii's sense.

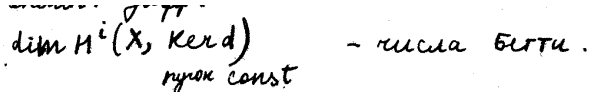
$$\mathcal{E}(\mathbf{R}^n) \xrightarrow{A} \mathcal{E}(\mathbf{R}^n)$$

Then the space of glob. solutions is finite-dimensional

$$H^0 \quad \text{— finite:} \qquad H^1 \quad \text{— defect. (fin. defect)}$$

$$H^1(X, \ker A)$$

2°. Ext. diff. oper.



Original source details: GELFAND-P000053

**Diagram and typed description.** Below $\ker d$, const. sheaf.

$\dim H^i(X, \underset{\text{const. sheaf}}{\ker d})$ — Betti numbers.

3°. Compl. an. Cartan–Serre–Kodaira theorem

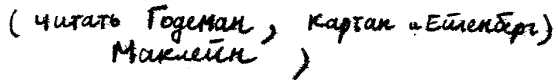
GELFAND-P000054



Original source details: GELFAND-P000054

**Diagram and typed description.** Several letters in the heading are densely overwritten.

Proof.



Original source details: GELFAND-P000054

**Diagram and typed description.** Mac Lane inserted below within the parenthesis; two closing parentheses at right.

(read Godement, Cartan and Eilenberg)

Mac Lane)

$\mathcal{P} \longrightarrow$ a sheaf. How to compute homology.

$$0 \longrightarrow \mathcal{P} \longrightarrow \mathcal{P}_0 \longrightarrow \mathcal{P}_1 \longrightarrow \cdots .$$

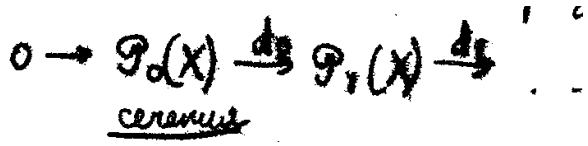
a fine resolution.

$$0 \longrightarrow \underset{\text{sections}}{\mathcal{P}_0(X)} \xrightarrow{d_0} \mathcal{P}_1(X) \xrightarrow{d_1} \cdots$$



Original source details: GELFAND-P000054

**Uncertain reading.** The arrow subscripts are heavily inked. The visible candidates are $0$ and $1$.



Original source details: GELFAND-P000054

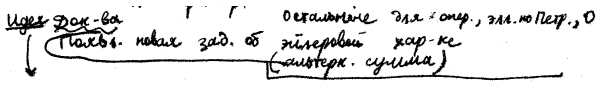
**Diagram and typed description.** Below $\mathcal{P}_0(X)$, sections underlined.

$\underset{\text{sections}}{\mathcal{P}_0(X)}$

$$H^1(X, \mathcal{P}) = \ker d_1 / \operatorname{Im} d_0$$

$$0 \longrightarrow \ker A \longrightarrow \mathcal{E}(\mathbf{R}^n) \xrightarrow{A} \mathcal{E}(\mathbf{R}^n)$$

this is a fine resolution. $H^1$ — quotient. The others, for an oper. ellipt. in Petrov.'s sense, 0.



Original source details: GELFAND-P000054

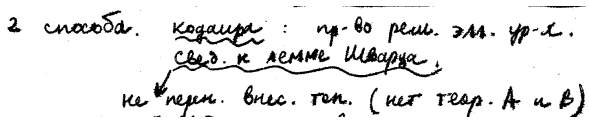
**Diagram and typed description.** At left, Idea underlined, beside Of proof with a wavy underline; downward arrow and a long oval/stepped line.

Idea Of proof

A new prob. arises on the Euler char.

(altern. sum)

2 methods.



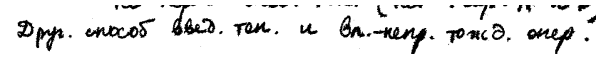
Original source details: GELFAND-P000054

**Diagram and typed description.** Kodaira and the line Reduc. to the Schwarz lemma have wavy lines; arrow to the following line.

Kodaira: space of solns. of an ellipt. eqn.

Reduc. to the Schwarz lemma,

does not carry over to higher hom. (no theorems $A$ and $B$)



Original source details: GELFAND-P000054

**Uncertain reading.** Another method: introd. top. / thin. and вп.-непр. ident. oper. [введ. / ввод. топ. / тон.; the initial abbreviations are not established.]

Resolution from Ban. spaces. Banach sheaf. Sheaf $\mathcal{P}$: $U \longrightarrow \mathcal{P}(U)$ (presheaf) of Fréchet type $\mathcal{P}(U)$ — $F$-spaces, and the restr. maps are cont. A series of new sheaves

$$C^{p,r}(\mathcal{P})$$



Original source details: GELFAND-P000054

**Diagram and typed description.** The arrow from $U$ has a wavy beginning and an arrowhead at right.

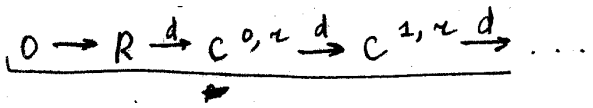
$U \rightsquigarrow C_U^{p,r}(\mathcal{P}(U))$

Vect.-valued diff. forms of degree $p$, smoothness $r$ The sheaf extends, but with no add. assumpt., it does not extend.

GELFAND-P000055

For $\mathcal{P} = \mathcal{E}(E)$ Sheaf — a Fréchet space. de Rham resolution.

$$0 \longrightarrow \mathbf{R} \xrightarrow{d} C^{0,r} \xrightarrow{d} C^{1,r} \xrightarrow{d} \cdots .$$



Original source details: GELFAND-P000055

**Diagram and typed description.** Below the sequence, a long bracket.

[Below it, a sign is cancelled.]



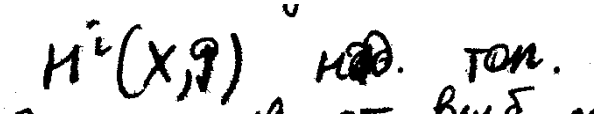
Original source details: GELFAND-P000055

**Uncertain reading.** a short densely cancelled sign

Coeffs. — maps of $U$ into diff. forms. coeffs. — smooth maps $U \longrightarrow \mathcal{P}(U)$ $r$ — smoothness exponent. Passing to the sheaf adds no new sections (but not in the gen. case, under add. assump.) Of the type of the de Rham resolution

$$0 \longrightarrow \mathcal{P} \longrightarrow \widehat{C}^{p,r}(\mathcal{P}) \xrightarrow{d} \widehat{C}^{1,r}(\mathcal{P}) \longrightarrow \cdots .$$

adjusted because of the order of smoothness Banach spaces result. $H^i(X, \mathcal{P})$ new [нов.; provisional] topol.



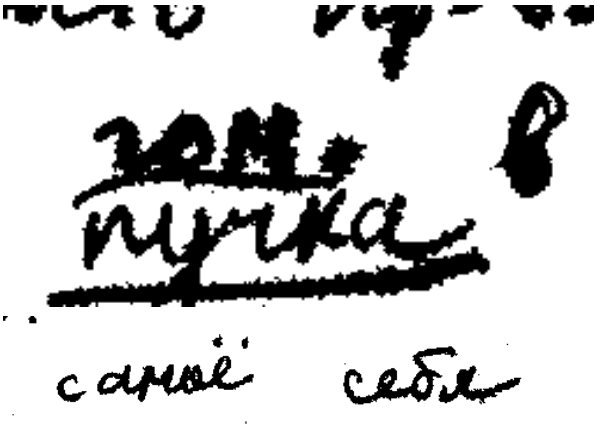
Original source details: GELFAND-P000055

**Uncertain reading.** The heavily overwritten abbreviation before топ. [topol.] resembles нов. [new]; над. is another possibility.

It does not depend on the choice of resolutions.

$$H^i(X, \mathcal{P})^{(r)} \longrightarrow H^i(X, \mathcal{P})^{(r-1)}$$

take the same Fréchet spaces (isomorphism) It is compl. cont. by the embedding theorem Gelfand. Problem: $X$ — comp. with a sheaf of lin. top. spaces on it. Ex. $X$ — a point. Just a space. Diff. oper. — endom. into itself.



Original source details: GELFAND-P000055

**Uncertain reading.** The abbreviation before “into itself” is heavily overwritten and resembles endom.; below it “of the sheaf” is inserted and underlined.

endom.
of the sheaf

[At right:] (fibres preserved.) When it is of finite index. $X$ — 1 point. $E+$ compl. cont. Riesz theorem. Then one can consider operators ellipt. of order $\infty$. Dynin. Hypoellipt. of the operator is needed. in the resolution, always the same sheaf ker $A$ Merely cont. coeffs., will it be of finite index? Next time. Adams (Novikov), Berezin ($P$ and $Q$)



Original source details: GELFAND-P000055

**Uncertain reading.** The last letter in the parenthesis resembles $Q$ with a lower right stroke; $O$ remains possible.

GELFAND-P000056



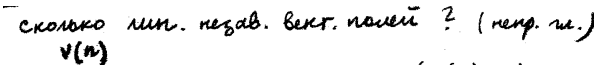
Original source details: GELFAND-P000056

**Diagram and typed description.** Date in a frame open at the top; below it, Adams; Novikov underlined twice.

22/II. Novikov S.P.

Adams.

$S^n$ how many lin. indep. vect. fields? (cont. f.)



Original source details: GELFAND-P000056

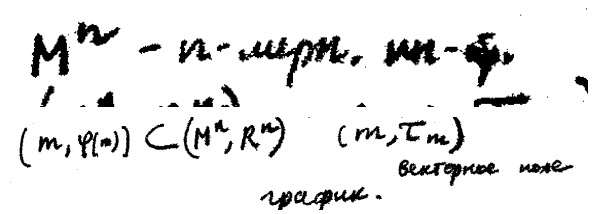
**Diagram and typed description.** Below how many, $\nu(n)$.

how many
$\nu(n)$

Triv. facts. Exist on odd-dim. sph. ($\nu(n) \geq 1$) on even-dim. sph., none ($\nu(n) = 0$)

$$\sum \text{ind. of sing.} = \text{Euler char.}$$

$M^n$ — $n$-dim. м-о. / м-ф.



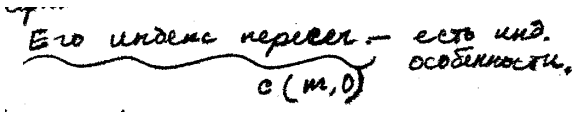
Original source details: GELFAND-P000056

**Uncertain reading.** м-о. / м-ф. [the last sign of the abbreviation is dense].

$(m, \varphi(m)) \subset (M^n, \mathbf{R}^n)$
graph.

$(m, T_m)$
vector field

$(m, 0)$ — canon. Its intersection index is the index of the singularity.



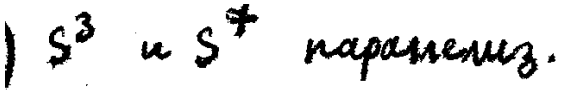

Original source details: GELFAND-P000056

**Uncertain reading.** Below the line, a wavy bracket to $c(m, 0)$ / $\varepsilon(m, 0)$.

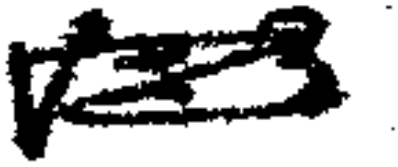


$c(m, 0)$

(Here only 1 and $-1$, but degree $n$). $E = 2$, so there are no fields. 2) $S^3$ and $S^7$ paralleliz.




**Uncertain reading.** Before the following equalities, a short cancelled entry; the upper sign of the second sphere resembles 7 / an asterisk.



Original source details: GELFAND-P000056

$\nu(3) = 3$, $\nu(7) = 7$ 3) $\nu(2n-1) \geq 1$ ($S^{2n-1}$)

4) If on $S^{2m-1}$ there are $k$ indep. fields ($\nu(2m-1) \geq k$), then on $S^{2pn-1}$, $\nu_{2pn-1} \geq k$

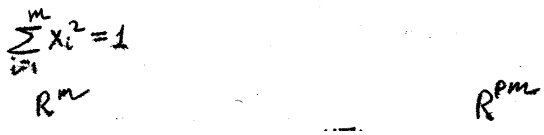



Original source details: GELFAND-P000056

**Uncertain reading.** After $p$ the one-humped sign resembles $n$ more than the $m$ on the preceding line. Readings $2pn-1$ / $2pm-1$.

In short

$$S^{m-1}, \quad \nu_m \geq k \quad \Longrightarrow \quad S^{pm-1}, \quad \nu_{pm-1} \geq k$$



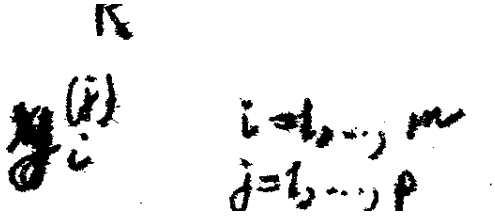

Original source details: GELFAND-P000056

**Diagram and typed description.** Below the sum, $\mathbf{R}^m$; at right, $\mathbf{R}^{pm}$.

$\sum_{i=1}^{m} x_i^2 = 1$ $\quad$ $\mathbf{R}^{pm}$

$\mathbf{R}^m$

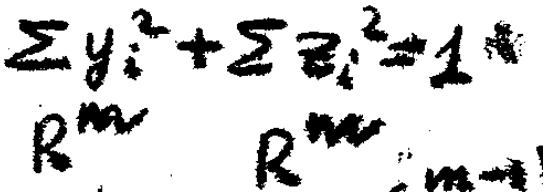
Coords. $p$ groups of $m$ each




Original source details: GELFAND-P000056

**Uncertain reading.** $x_i^{(j)}$ / $y_i^{(j)}$ [first symbol overwritten], $i = 1, \ldots, m$, $j = 1, \ldots, p$.

let $p = 2$.




Original source details: GELFAND-P000056

**Diagram and typed description.** Below each sum, $\mathbf{R}^m$. Asterisk after 1; no footnote on the frame.

$\sum_{\mathbf{R}^m} y_i^2 + \sum_{\mathbf{R}^m} z_i^2 = 1^*$

$$v(x) \text{ in } \mathbf{R}^m, \qquad v(x) \perp x \qquad (\text{field on } S^{m-1})$$



Original source details: GELFAND-P000056

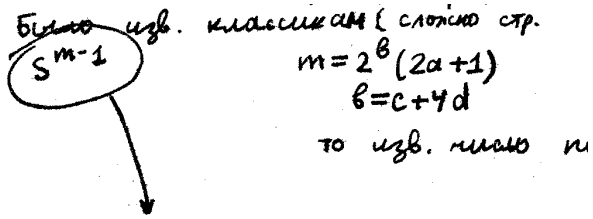
**Uncertain reading.** The initial letter in $v(x)$ resembles Latin $v$ / $\nu$; no separate quantifier sign is established.

GELFAND-P000057

Put

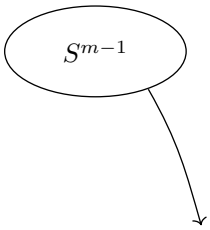
$$\widetilde{v}(y,z) = \left( \|y\| v\left(\frac{y}{\|y\|}\right),\ \|z\| v\left(\frac{z}{\|z\|}\right) \right)$$

$\|y\|$ to cope with $y = 0$, all in the 2nd comp., of mod $= 1$, and $\perp$ $(y, z)$ Indep. fields are obtained from indep. fields. Thus Hopf fields are obt. (On the right one can write other fields) can stabilize the problem: incr. dim. Was known to the classical authors (hard to construct.



Original source details: GELFAND-P000057

**Diagram and typed description.** At left, $S^{m-1}$ in an oval. The upper stroke overlaps Was. Arrow down.



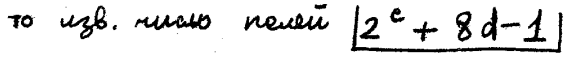

$$m = 2^b(2a+1) \qquad b = c + 4d \qquad 0 \le e \le 3,$$



Original source details: GELFAND-P000057

**Uncertain reading.** In the bounds $0 \le e \le 3$, the middle letter resembles $e$ / $c$.

then the known number of fields



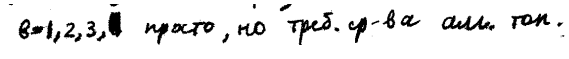
Original source details: GELFAND-P000057

**Diagram and typed description.** Frame open at the top; the exponent resembles $e$ / $c$.

$$2^e + 8d - 1$$

$$\nu_{m-1} \ge 2^c + 8d - 1$$

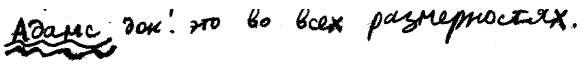
In cases 3, 7 gives max. $b = 0$ (even — triv.). $b = 1, 2, 3$, [a sign is overwritten], simple, but req. methods of alg. top.



Original source details: GELFAND-P000057

**Uncertain reading.** densely overwritten sign / downward arrow

$b \le 12$ Toda, but $m$ of order 3000 Adams proved this in all dimensions.



Original source details: GELFAND-P000057

**Diagram and typed description.** Two wavy underlines.

Adams



$*$ The reduct. part belongs to Atiyah and Toda. This leads to a problem on homot. gr. of spheres. finally

$$\nu_{m-1} = 2^c + 8d - 1$$

The odd factor plays no role.

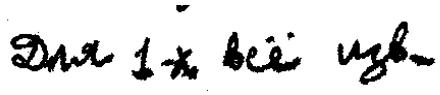
Here one assertion. $m \uparrow$ by a factor of 6, then the number of fields incr. by 8



Original source details: GELFAND-P000057

**Manuscript notation.** The factor is written as 6.

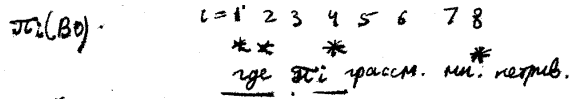
$(d+1)$ this is seen from the constr. in Bott periodicity. For the first ones all is known.



Original source details: GELFAND-P000057

**Uncertain reading.** After 1 are a short stroke and an x-shaped sign. The proposed reading is 1-x [the first ones].

2 parall. sph., then Bott periodicity



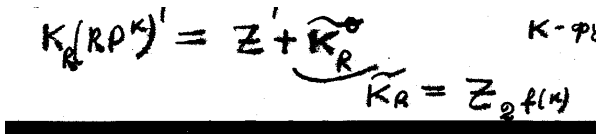
Original source details: GELFAND-P000057

**Diagram and typed description.** Asterisks below 1, 2, 4, 8.

$\pi_i(BO)$.

| $i=$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| | * | * | | * | | | | * |

where $\pi_i$ of Grassm. mfld. is nontriv. $\mathbf{R}P^k$ — project. space Consid.



Original source details: GELFAND-P000057

**Uncertain reading.** In the lower entry $f(k)$ is raised relative to 2. Readings $2^{f(k)}$ / $2f(k)$. The small upper sign on $\widetilde{K}$ resembles 0. From the second summand a bracket to the lower entry.

$K_{\mathbf{R}}(\mathbf{R}P^k) = \mathbf{Z} + \widetilde{K}^0_{\mathbf{R}}$

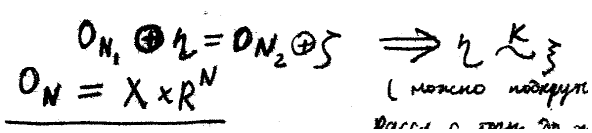
$\widetilde{K}_{\mathbf{R}} = \mathbf{Z}_{2^{f(k)}}$

$K$-functor, see “Матем.” [Math.] Atiyah and Hirzebruch

GELFAND-P000058

Elem. of $K$ — real vect. bundle, but with an equiv. relation. Add. — direct $\Sigma$. Mult. — tensor prod. of 2 vect. bund. 1 elem. in $K$, if

$$O_{N_1} \oplus \eta = O_{N_2} \oplus \zeta \quad \Longrightarrow \quad \eta \overset{K}{\sim} \xi$$



Original source details: GELFAND-P000058

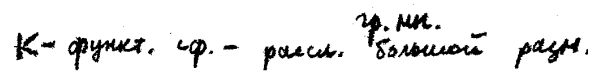
**Uncertain reading.** After the second sum the sign resembles $\zeta$; after the equivalence it resembles $\xi$ / $\zeta$. The initial oval permits $O$ / 0.

$$O_N = X \times \mathbf{R}^N$$

[At right:] (can twist it) Bund. up to equiv. — element of the $K$-funct. Coll. — $K$-functor. Can add, mult.! Triv. bund. ($\Sigma$ of tang. and norm. — triv., there is an inverse elem.) To each space assigns a ring) Over the sphere $K$-functor

$$K_{\mathbf{R}}(S^i) = \pi_i(BO)$$

(how many homot. maps to Grassm. mfld.)

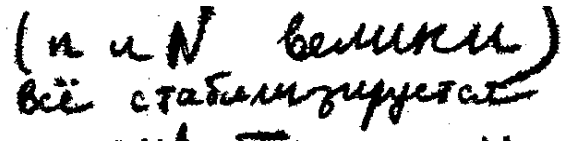


Original source details: GELFAND-P000058

**Diagram and typed description.** Above large, the insertion Grassm. mfld.

$K$-funct. of sph. — bund. of $\overset{\text{Grassm. mfld.}}{\text{large}}$ dim.

$BO$ — Grassm. mfld. — $n$-dim. lin. in $N$-dim. sp. ($n$ and $N$ large)



Original source details: GELFAND-P000058

**Uncertain reading.** Below, everything stabilizes; the long tail of the parenthesis overlaps the end of the word.

everything stabilizes

Vect. bund. with given base ⟵ evident $\pi_i$.

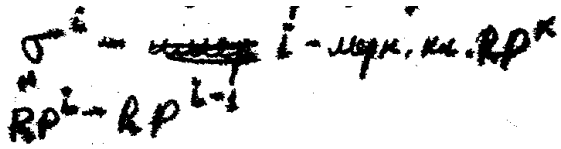


Original source details: GELFAND-P000058

**Manuscript notation.** The abbreviation before $\pi_i$ reads очев. [evident].

To the mfld. corresp. tang. spher. maps. To the normal sheaf — norm. spher. map. Next an artificial device. $\mathbf{R}P^k$ subdiv. into cells (1 in each dim.) All vect. bund. over $\mathbf{R}P^k$ are constr. thus.

$\sigma^i$ — $i$-dim. cell of $\mathbf{R}P^k$

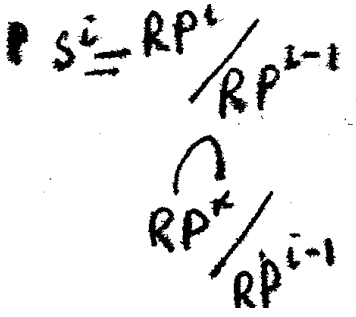


Original source details: GELFAND-P000058

**Uncertain reading.** After $\sigma^i$ a short entry densely cancelled; below it, $\mathbf{R}P^i \longrightarrow \mathbf{R}P^{i-1}$. Above the first $\mathbf{R}P^i$ a small unidentified sign.

$\mathbf{R}P^i \longrightarrow \mathbf{R}P^{i-1}$

consid.



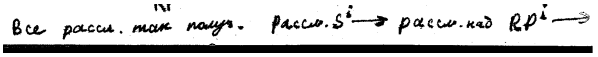
Original source details: GELFAND-P000058

**Uncertain reading.** Before $S^i$ a dense short sign, its reading not established. Below, $\mathbf{R}P^k/\mathbf{R}P^{i-1}$ with a curved arrow.

$S^i = \mathbf{R}P^i/\mathbf{R}P^{i-1}$

$\mathbf{R}P^k/\mathbf{R}P^{i-1}$

All bund. are obtained thus. Bund. $S^i \longrightarrow$ bund. over $\mathbf{R}P^i \longrightarrow$

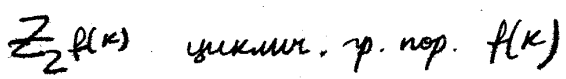


Original source details: GELFAND-P000058

**Undeciphered passage.** The arrow in the bottom line reaches the right edge; the next object is not visible in this frame.

$\longrightarrow$ bund. on $\mathbf{R}P^k$. $f(k)$ — number of asterisks to the left of $k$ **Z** — invar. — dimension Crit. points appear on cycles

$$\mathbf{Z}_{2f(k)} \qquad \text{cyclic gr. of order } f(k)$$



Original source details: GELFAND-P000059

**Uncertain reading.** The index on **Z** permits $2^{f(k)}$ / $2f(k)$; the order caption visibly has $f(k)$, without a separate 2.

[At left an entry is cancelled.]



Original source details: GELFAND-P000059

**Uncertain reading.** short dense cancellation

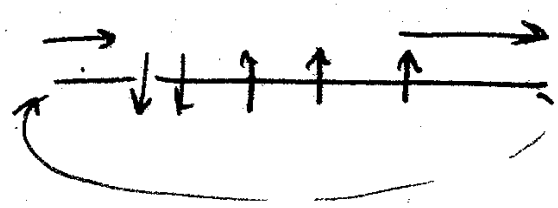
$$\mathbf{R}P^k \in \mathbf{R}P^{k+1}$$



Original source details: GELFAND-P000059

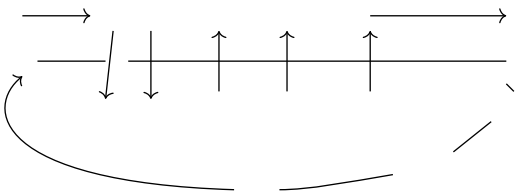
**Manuscript sign.** The sign has the middle stroke of $\in$. A separate $\subset$ appears lower on the page.

there is a norm. sheaf — Möbius strip (this is a generator) of the group it must be added to itself $2^{f(k)}$ times

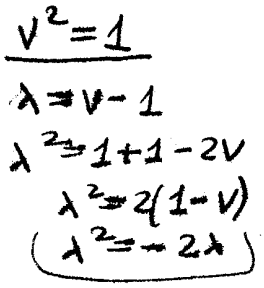


Original source details: GELFAND-P000059

**Diagram and typed description.** At left an almost horizontal line; five transverse arrows: two on the left down, the next three up. Two rightward arrows above the line, short on the left, long on the right. Below, an open oval arc with a gap at the lower right; its left end rises to the line and has an arrow, the right end is faint. No letter labels.



$$\underline{\nu^2 = 1} \qquad \lambda = \nu - 1$$
$$\lambda^2 = 1 + 1 - 2\nu$$
$$\lambda^2 = 2(1 - \nu)$$

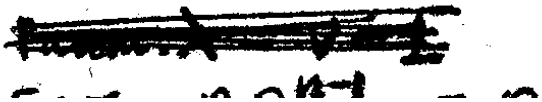


Original source details: GELFAND-P000059

**Uncertain reading.** The narrow sign in $\nu$ resembles $\nu$ / $v$ / $V$; some equality signs are overwritten. The last line is in a rounded frame open at the top.

$$\lambda^2 = -2\lambda$$

mult. structure. [A line is cancelled.]



Original source details: GELFAND-P000059

**Uncertain reading.** Long, densely crossed-out line on the left below the formulas

There is

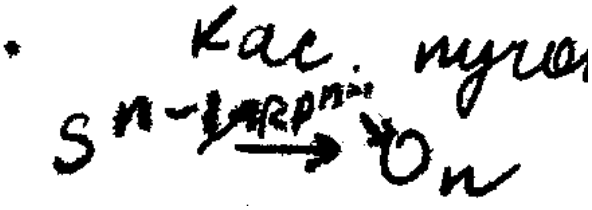


Original source details: GELFAND-P000059

**Diagram and typed description.** Above the right side are two horizontal strokes of different lengths with a dot between them.

$$\mathbf{R}P^{n-1} \subset O_n$$

$S^n$ is assigned to the hyperpl. of the equator $\longrightarrow$ into the orthog. group. Tang. sheaf — map.

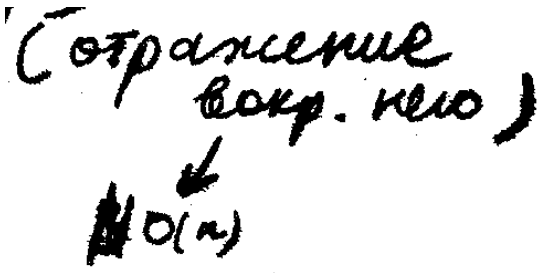


Original source details: GELFAND-P000059

**Uncertain reading.** Above the arrow, a tight insertion $\mathbf{R}P^{n-1}$ / $\mathbf{R}P^n$ with an additional slanted arrow pointing to $O_n$; some signs are dense.

$$\begin{array}{ccc} & & \mathbf{R}P^{n-1} \\ & \nearrow & \\ S^{n-1} & \longrightarrow & O_n \end{array}$$

At symmetric points, the same. it is 2-sheeted (Gluing on the equator) rotation about it



Original source details: GELFAND-P000059

**Uncertain reading.** From the parenthesis about reflection an arrow down to $O(n)$; before $O(n)$ a black overwritten stroke, reading not established.

(reflection about it)

$$\downarrow$$

$$O(n)$$

axes of proj. space.

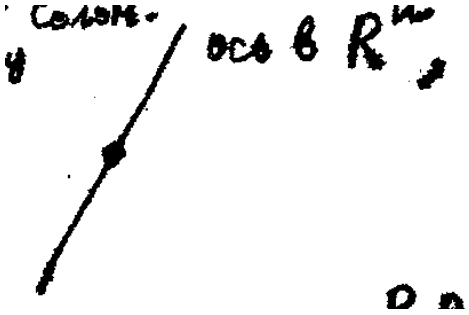
**Uncertain reading.** Albert Solom. / Soln.

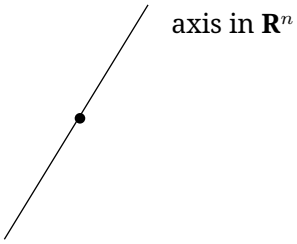


Original source details: GELFAND-P000059

Schwarz.

**Diagram and typed description.** Slanted line with a point and the label axis in $\mathbf{R}^n$.

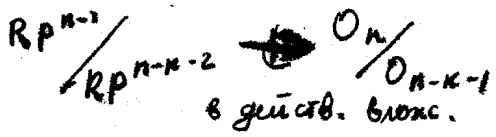





Original source details: GELFAND-P000059

$$\mathbf{R}P^{n-1}/\mathbf{R}P^{n-k-2} \longrightarrow O_n/O_{n-k-1}$$

**Uncertain reading.** In the lower left exponent, $n-k-2$ / $n-n-2$; the arrow between the quotients is densely overwritten.



Original source details: GELFAND-P000059

in fact an embedding.

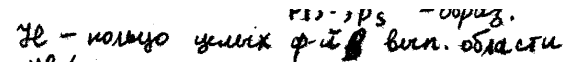
GELFAND-P000060

1/III

V.P. Palamodov. $\mathcal{P}$ — ring of polyn. in $\mathbf{C}^n$, $\mathcal{J} \subset \mathcal{P}$ — ideal, primary, $P_1, \dots, P_s$ — gener.

$\mathcal{H}$ — ring of entire funct. [a sign is cancelled] of a conv. region

**Uncertain reading.** after funct. a short sign is densely crossed out

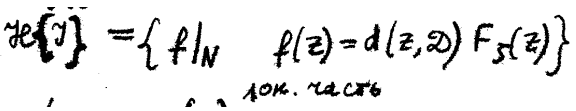


Original source details: GELFAND-P000060

$\mathcal{H}/\mathcal{J}\mathcal{H}$ is described thus

$$\mathcal{H}\{\mathcal{J}\} = \{f|_N \quad f(z) = d(z,\mathcal{D})F_\zeta(z)\}$$

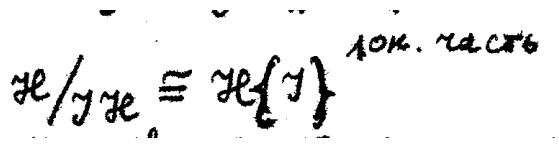
**Uncertain reading.** The subscript on $F$ resembles $\zeta$ / $s$.



Original source details: GELFAND-P000060

$$\mathcal{H}/\mathcal{J}\mathcal{H} \cong \mathcal{H}\{\mathcal{J}\}$$ local part

**Position of the note.** The note “local part” is written above and to the right of the second formula.

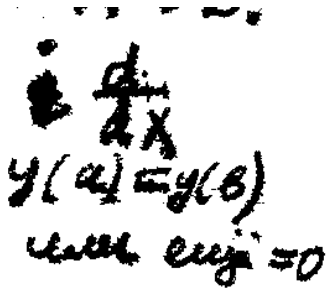


Original source details: GELFAND-P000060

Kazhdan says that this is obt. algebraically.

• Gelfand. Nelson's example. 2 self-adj. operators $A$ and $B$. what changes a self-adj. oper? (not symm. closed) [In the right margin:]

$$i\frac{d}{dx}, \qquad y(a) = y(b)$$

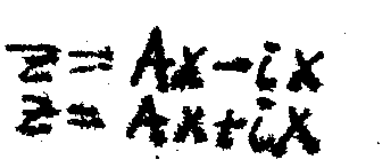


**Uncertain reading.** Below is written "or also $= 0$"; no argument precedes the equality sign.

Original source details: GELFAND-P000060

Essentially self-adj. $A$ (symm.), if there are no vect. orthog. to all vect. of the form

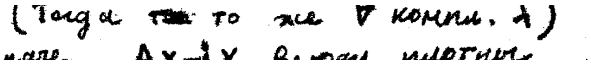
$$\overline{z} = Ax - ix, \qquad z = Ax + ix$$



**Uncertain reading.** The first $z$ has a long upper bar. The upper stroke of the second may be part of the letter or a separate mark.

Original source details: GELFAND-P000060

(Then [an entry is cancelled] the same $\forall$ complex $\lambda$)

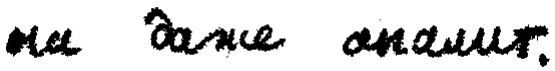


**Uncertain reading.** short dense cancellation

Original source details: GELFAND-P000060

otherwise $Ax - ix$ are everywhere dense $A - iE$ has inv. (in general?: if closed)

$$\left(i\frac{d}{dx} \quad \text{[on / not] even analyt.} \quad \begin{matrix} f(a) = f(b) & \text{essentially self-adj.} \\ f(a) = f(b) = 0 & \text{only symm.} \end{matrix}\right)$$

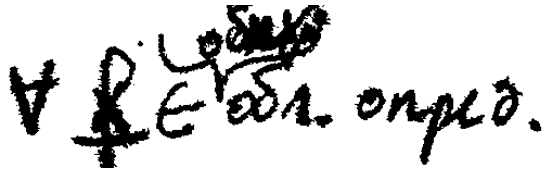


**Uncertain reading.** Before the conditions is written "[on / not] even analyt."; the last letter of the first word is joined.

Original source details: GELFAND-P000060

$A$ and $B$ essentially self-adj., $\alpha A + \beta B$ essentially self-adj., $\alpha, \beta$ — real.

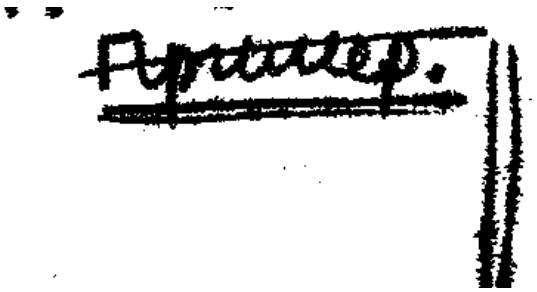
$$\forall f \in \text{domain of def.} \qquad ABf = BAf.$$



**Uncertain reading.** Above the domain of definition a short entry is densely overwritten; below it a curved stroke.

Original source details: GELFAND-P000060

Nevertheless $A$ and $B$ do not commute.



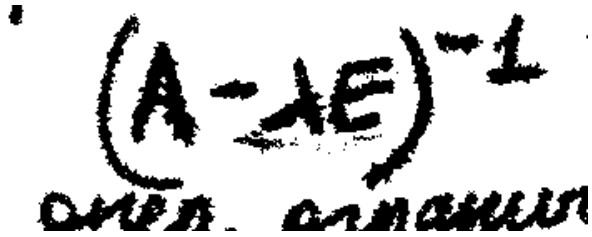
Original source details: GELFAND-P000060

**Diagram and typed description.** At left Example. is crossed out and doubly underlined; to its right a double vertical stroke.

~~Example.~~ ||

$A$ and $B$ comm., if reasonabl. funct. of them commute.

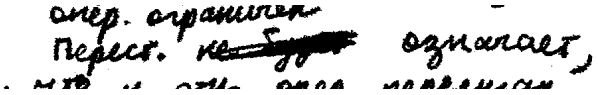
$$(A-\lambda E)^{-1}$$
oper. bounded



Original source details: GELFAND-P000060

**Uncertain reading.** Below $\lambda E$ lower strokes; whether a separate underline or parts of letters is not established.

Comm. [Cancelled: not...] also with $B$ means that these oper. too comm.



Original source details: GELFAND-P000060

**Uncertain reading.** The cancelled entry begins with a mark resembling не [not]; the remaining letters are obscured.

$e^{iA}$ — unit. $e^{iB}$ do not commute

[An entry is cancelled; beside it (∀ bounded oper.).]

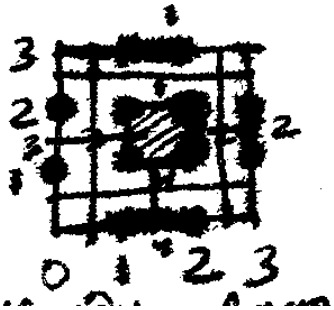


Original source details: GELFAND-P000060

**Uncertain reading.** Short densely crossed-out entry

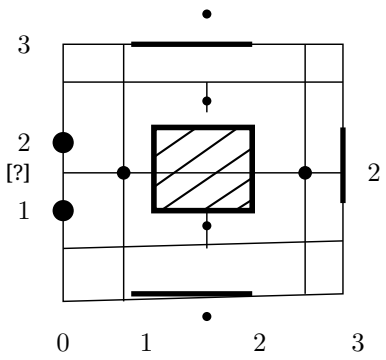
GELFAND-P000061

Nelson's example.



Original source details: GELFAND-P000061

**Diagram and typed description.** Square grid, heavy points and central hatching; labels $0, 1, 2, 3$ below, $1, 2, 3$ at left and $2$ at right. Between $1$ and $2$ at left, an overwritten sign; the central overlapping strokes and exact gluing are not established.



From the region between the squares, remove the heavy points The segments are glued each point is glued to the nearest boundary point along the horiz. or vert. The corners are glued together. Obtain a surface of genus $\underline{4}$ / $\underline{1}$ with one removed point



**Uncertain reading.** $\underline{4}$ / $\underline{1}$

Original source details: GELFAND-P000061

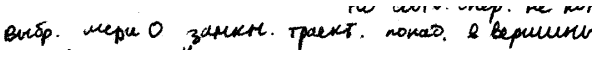
$$X = \frac{d}{dx}, \qquad y = \frac{d}{dy}$$



**Uncertain reading.** The second name to the left of the equality is $y$ / $Y$; subsequently $iY$ is written.

Original source details: GELFAND-P000061

$iX$, $iY$ — formally self-adj. They commute But the corresp. oper. do not commute. Remove a measure-$0$ signif. / that is [значен. / значит.] traject. hit the vertices.



**Uncertain reading.** signif. / that is [значен. / значит.]

Original source details: GELFAND-P000061

$e^X$ and $e^Y$ — unit. oper. which do not commute.

$$e^{\frac{d}{dx}} f(x) = f(x+1)$$

(Taylor formula) Prove first for analytic funct., then for the whole Hilb. space by a limiting passage (the oper. is unitary) And it has to be moved.

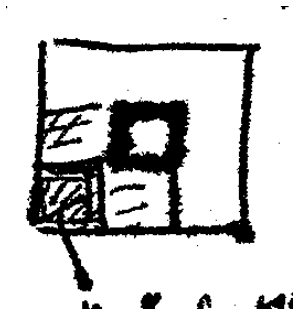
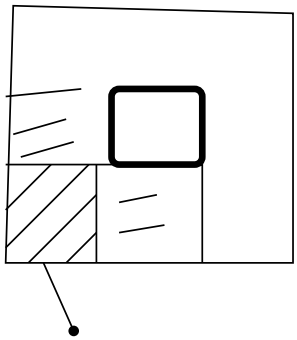


**Diagram and typed description.** Square with a hole; hatching to the left of the hole and in the lower left region, two short strokes to its right. A slanted line runs from the lower left region down towards the caption. The boundaries of the overlapping strokes are not established.

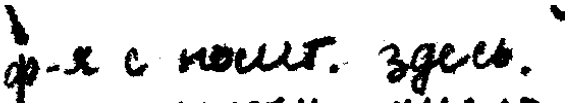
Original source details: GELFAND-P000061



$e^Y$ — region by $1$ upwards, $e^X$ will turn round

$$e^X$$

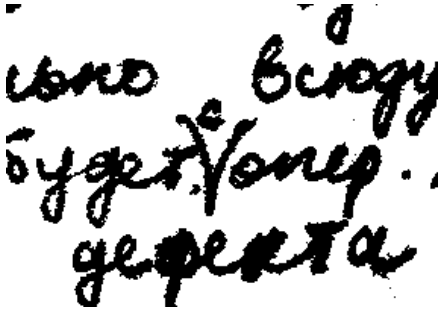
funct. with compact. / comp. [компт. / комп.] here.



Original source details: GELFAND-P000061

**Uncertain reading.** compact. / comp. [компт. / комп.]

Exclude a finite number of traject. $e^{tX}$ — shift along the traject. of oper. $X$ locally everywhere they commute what will happen to [an overwritten sign] the Laplace oper.?



Original source details: GELFAND-P000061

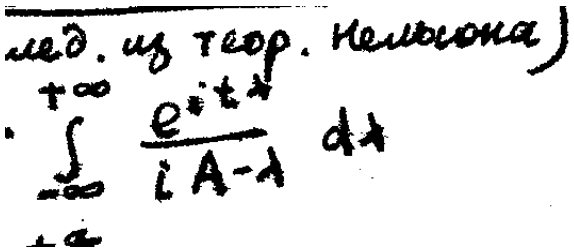
**Uncertain reading.** before oper., a dark overwritten stroke

deficiency index

$$\Delta f = if$$
$$\underline{\Delta = X^2 + Y^2}$$

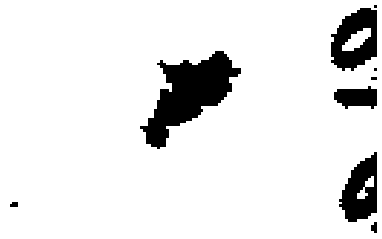
(this follows from Nelson's theorem) It is not self-adj. (has a $C^\infty$ solution, bound. cond., and at the point only $\int$ square.) The shift oper. themselves are def. on $L^2$, and not on periodic funct. 3 def $e^{itA}$:

$$1^\circ = \int_{-\infty}^{+\infty} \frac{e^{it\lambda}}{iA - \lambda}\, d\lambda$$
$$2^\circ = \int_{-\infty}^{+\infty} e^{it\lambda}\, dE_\lambda$$
$$3^\circ : \quad \frac{df}{dt} = iAf$$



Original source details: GELFAND-P000061

**Uncertain reading.** In the first entry the denominator is $iA - \lambda$; no separate factor before the integral is visible.



Original source details: GELFAND-P000061

**Uncertain reading.** Before the fraction in the third line, a short dark stroke; the exact reading is not established.

$\underset{\text{sol.}_{t,0}}{U} f_0 = f$ — unit. oper.

GELFAND-P000062

Sinai. Homol. / Posit. [Гомол. / Полож.] in dyn. systems.



**Uncertain reading.** Homol. / Posit. [Гомол. / Полож.]

Original source details: GELFAND-P000062

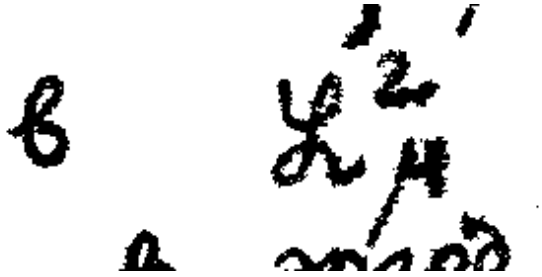
For an ergod. dyn. syst. [Crossed out: a short densely crossed-out entry; inserted above: with discr. spec.] the spectrum is always simple



**Uncertain reading.** a short densely crossed-out entry

Original source details: GELFAND-P000062

$M$, $\mu$, $\{S_t\}$ — one-param. gr. in $M$ In $L^2_M$ / $L^2_H$ a one-param. gr. of unit. oper.



**Uncertain reading.** $L^2_M$ / $L^2_H$

Original source details: GELFAND-P000062

$$\underbrace{U_t}_{\text{simple}} f(x) = f(S_t x)$$

In the ergod. case every eigenvalue is simple Lebesgue spectrum: $\forall f$

$$(U_t f, f) = \int e^{it\lambda} p(\lambda)\, d\lambda$$

($p(\lambda)$ — integr.) (In general always $\int e^{it\lambda}\, d\mu$, here $\mu$ abs. cont.) Countably-mult. Leb. spec.: $\exists$ orthog. $f_1, f_2, \ldots$

$$(U_t f_i, f_j) = 0 \qquad (i \neq j)$$

and generate everything. In all known ex. the spectrum is countably-mult. Ex. Transf. of the torus.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \det = 1, \quad \text{integer}$$

$$x \longrightarrow ax + by$$
$$y \longrightarrow cx + dy$$

Need to find autocorr. / autog. [авток. / автог.] Expand in a Fourier series and act.



**Uncertain reading.** autocorr. / autog. [авток. / автог.]

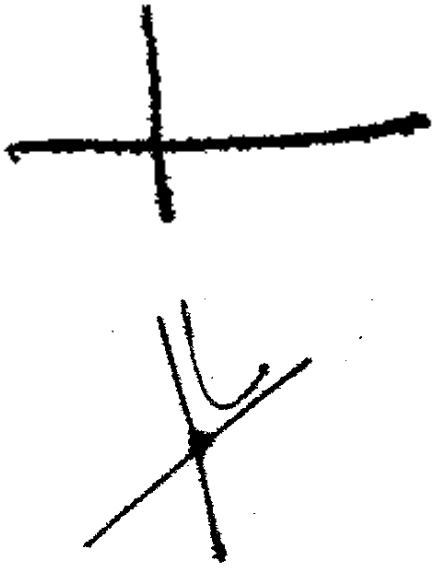
Original source details: GELFAND-P000062

$$f(x,y) \longrightarrow f(ax+by, cx+dy)$$
$$e^{2\pi i(mx+ny)} \longrightarrow e^{2\pi i((am+cn)x+(bm+dn)y)}$$

A matrix acts on the integer latt. Need to take trajectories

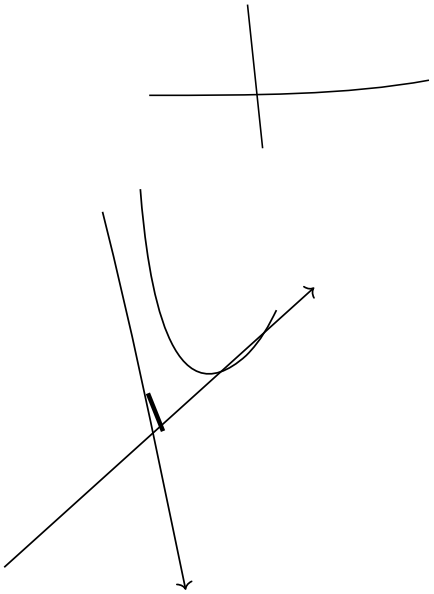
$$(m,n) \longrightarrow S(m,n) = (m',n')$$

The traject. is a hyperbola.



Original source details: GELFAND-P000062

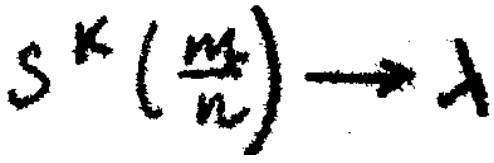
**Diagram and typed description.** Above, two crossing strokes without explicit arrows. Below, an oblique line and two curved strokes; thickening at the crossing. The upper right end of the oblique line and the bottom end of the long curved stroke resemble arrows; the membership of the overlapping strokes is not established.



Real eigenvalues — contraction along one axis and expansion along the other (det $= 1$)

$$S^k \begin{pmatrix} m \\ n \end{pmatrix} \longrightarrow \lambda$$

($\lambda$ — eigenvalue)



Original source details: GELFAND-P000062

**Uncertain reading.** After the column $S^k(m,n)$, an arrow to $\lambda$; no additional factor at the column is visible.

The waves become ever denser and direct. Waves $\Longrightarrow$ towards an eigendir. The space decomposes into spaces of funct. with frequencies on hyperbolas. On the 1st hyperbola. A sequence of orthog. vect., each passes to the next



Original source details: GELFAND-P000063

**Diagram and typed description.** Above, a row of nine separate points.

(shift by 1 at the integer pts.) cont. Lebesgue spectrum. With the Bernoulli scheme — charact. of the group. Theorem. Measure $d\sigma(\lambda)$

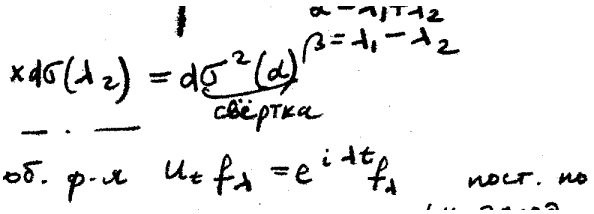


Original source details: GELFAND-P000063

**Diagram and typed description.** Two crossing strokes; arrows to the right and upwards. No labels.

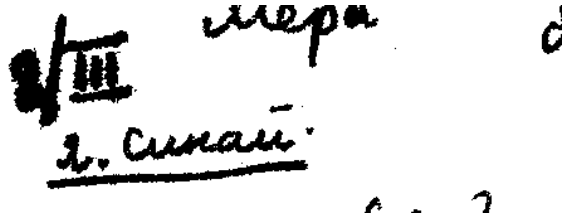
$\sigma \times \sigma$

$$\alpha = \lambda_1 + \lambda_2$$
$$\beta = \lambda_1 - \lambda_2$$
$$\text{measure} \quad d\sigma(\lambda_1) \times d\sigma(\lambda_2) = \underbrace{d\sigma^2(\alpha)}_{\text{convolution}}$$



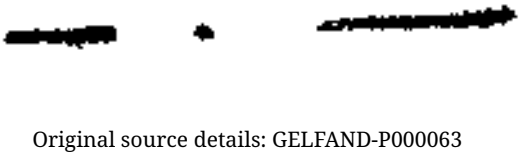

Original source details: GELFAND-P000063

**Uncertain reading.** After the product of measures, $d\sigma^2(\alpha)$ is written, with convolution below; no separate factor in $\beta$ is visible.



Original source details: GELFAND-P000063

**Uncertain reading.** Before /III, a densely overwritten sign: 8 / 4 / 1; below, Ya. Sinai / 2. Sinai.



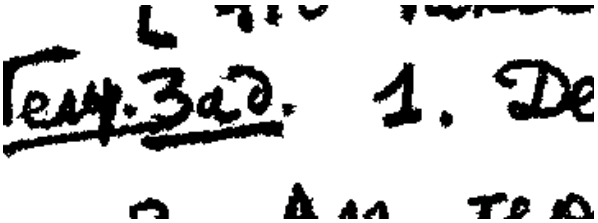
Original source details: GELFAND-P000063

**Diagram and typed description.** A short horizontal stroke, a separate point, a longer almost horizontal stroke.

$M, \mu, \{S_t\}$. An eigenfunct.

$$U_t f_\lambda = e^{i\lambda t} f_\lambda$$

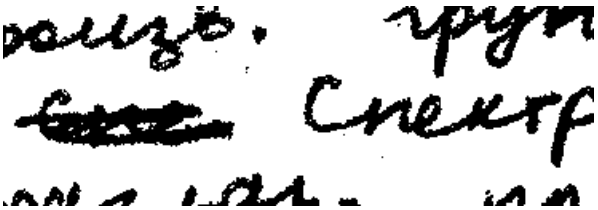
const. in mod (for an ergod. system) 15/III Todorov $SU$ 22/III-65 In a year there will be a math. congress in Moscow. (March — April) [what is a “simple root”? connection with matrices] Gelf. [Гелф. / Гельф.] Probl.



**Uncertain reading.** Gelf. [Гелф. / Гельф.]

Original source details: GELFAND-P000063

1. Do alg. geom. over noncomm. rings!! 2. Alg. theory of diff. eqs. Povzner. Small parameter. Berezin. On funct. of many variables. Need to study not all funct., but specific ones — occurring in repr. theory and in stat. physics. Povzner a “wish” to Berezin. Kirillov. Ordinary dyn. syst. — time $R$ or $Z$. Can have an arbitrary group preserving measure. Classific., invar.? [Before Spectrum, crossed out: a short entry under dense strokes.]



**Uncertain reading.** a short entry under dense strokes

Original source details: GELFAND-P000063

Spectrum — the set of irred. repr. into which the space of funct. on $M$ decomposes

$$\begin{array}{ll} M & G \\ \underset{\text{funct.}}{H} & \underset{\text{unit. oper.}}{U_g : f(x) \longrightarrow f(g^{-1}x)} \end{array}$$

$H$ decomposes into irred. compon., which are called the spectrum. $G$ comm. — repr. [прдпр. / прдпрд.] are one-dimensional.

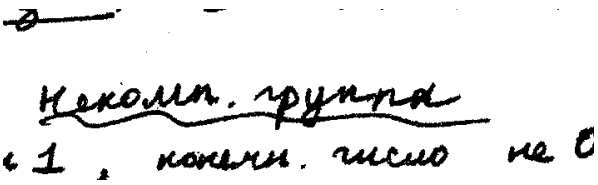


**Uncertain reading.** repr. [прдпр. / прдпрд.]

Original source details: GELFAND-P000063

$G$ acts ergodically (no inv. funct. except const) For a compact gr. All transitive and reduce to describing subgroups. Ex. 1) Gordin and Whitehead.

$$G = \prod Z_2$$



**Uncertain reading.** Noncomm. group / Noncomp. group [Некомм. / Некомп.].

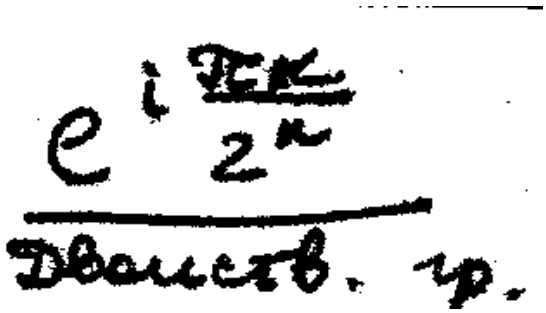
Original source details: GELFAND-P000063

Sequences of $0$ and $1$, a finite number not $0$. $\sum$ usual. For it, the problem of describing repr. of virtual subgroups is solved

GELFAND-P000064

2) Dyadic-rat. numbers on the circle.

$$e^{i\frac{2\pi k}{2^n}}$$



**Uncertain reading.** Overwritten strokes in the numerator: $2\pi k$ / $\pi k$.

Original source details: GELFAND-P000064

Countable. Dual gr. — 2-adic numbers. On it, already noncomm. facts. Point spectrum and cont. sp. Possibly, [crossed out: ϕ / illegible] prod. of funct., decomposed by sections.



**Uncertain reading.** ϕ / illegible

Original source details: GELFAND-P000064

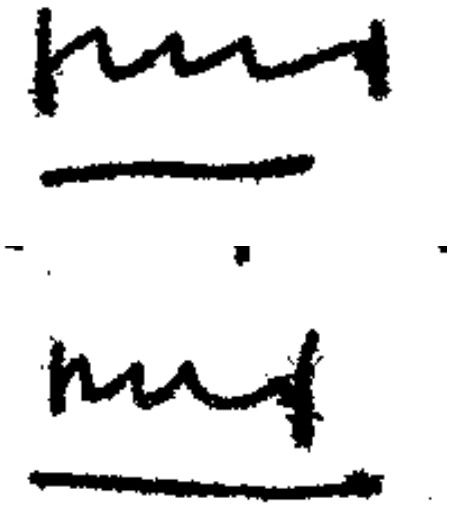
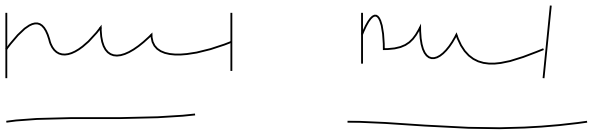


**Diagram and typed description.** Two short sawtooth-like sections over horizontal underlines; the terminal strokes and some bends are uncertain.



Original source details: GELFAND-P000064

Cont. sp. gives a funct. decomposed over a discrete set. Oseledets: also held for an ordinary system. 2-dim. time.

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}$$

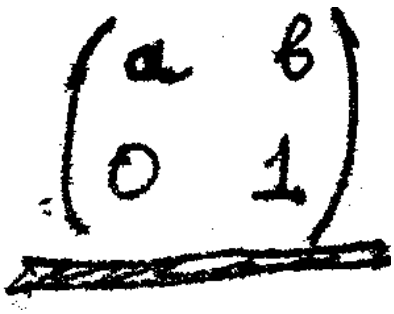


**Diagram and typed description.** Below the matrix, a long double underline with oblique strokes.



Original source details: GELFAND-P000064

Sinai's flows (equipped). Irred. repr. are few: one-dim. $|a|^{i\lambda}(\operatorname{sign} a)^{\varepsilon}$! And one infinite-dim. Connection with the Lorentz group.



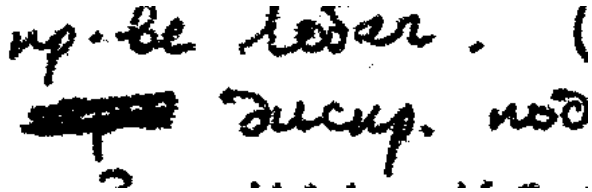
**Diagram and typed description.** Wavy underline; below with [c], a short descending stroke with a bend.



Original source details: GELFAND-P000064

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

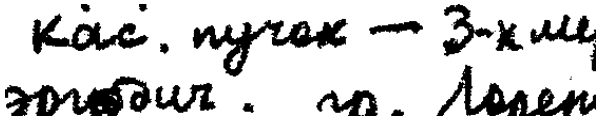
Acts on Lobach. space (upper half-plane). About quot. by [crossed out: illegible] a discrete subgr. Will be compact, of negative curv.



**Uncertain reading.** illegible

Original source details: GELFAND-P000064

Tang. / oblique [Kac. / Koc.] bundle — a 3-dim. compact manif., on which acts an ergod. Lorentz gr.



**Uncertain reading.** Tang. / oblique [Kac. / Koc.]

Original source details: GELFAND-P000064

Discrete spectrum because of compactness. Homologies of dyn. systems. Operator

$$U_g = \alpha(x, g) f(xg).$$

A homomorphism then. $\alpha(x, g)$ — a cocycle. $|\alpha| = 1$.

$$\alpha \in Z^1(G, \underset{\text{gr. of functions}}{F}).$$

$\alpha$ equiv. if a continuous transf. exists. This happens if

$$\frac{\alpha}{\alpha_1} \in B^1(G, F)$$

General form of a boundary

$$\alpha_0(x, g) = \frac{\beta(xg)}{\beta(x)}$$

— homolog. to zero. $Z^1/B^1$ — homologies. Computed only if the gr. is transitive (then this is the set of repr. of the stabilizer subgr.). Homol. 0, not all const, but $e^{2\pi i t\lambda}$.



**Diagram and typed description.** At the right, a caption with a long curved underline.

$\lambda$ from the spectrum.

Original source details: GELFAND-P000064

If a subgr., then the homol. gr. determines the dyn. system.

GELFAND-P000065

The collection of homol. is called a repr. of a virtual subgroup (although there is no subgr.). 2-dim. homol. of the spectrum of a dyn. system. $f_\lambda$ — eigenfunct. Choose 1 fixed eigenfunct.

$$f_\lambda f_\mu = C_{\lambda,\mu} f_{\lambda\mu}$$

$C_{\lambda,\mu}$ — 2-dim. cocycles on the spectrum of a dyn. system.

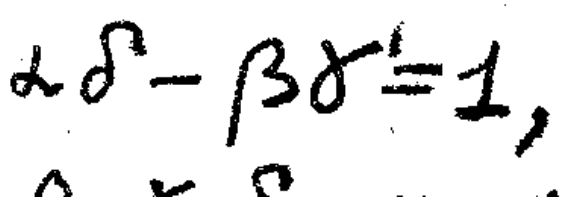
$$C_{\lambda,\mu} C_{\lambda\mu,\nu} = C_{\lambda,\mu\nu} C_{\mu,\nu}$$

Equiv. to 1

$$C_{\lambda,\mu} = \frac{a_\lambda a_\mu}{a_{\lambda\mu}}$$

In the classical case these homol. are trivial ("the syst. is determined by the spectrum"). It is not known even whether a dyn. syst. is determined by its spectrum for finite groups (Gelfand thinks not). Rotation group of the icosahedron.

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \qquad \alpha\delta - \beta\gamma' = 1$$



**Uncertain reading.** By $\gamma$ in the determinant condition, an upper stroke is visible; $\gamma'$ / $\gamma$.

Original source details: GELFAND-P000065

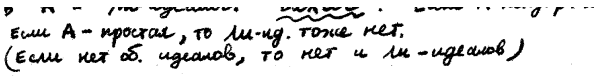
$\alpha, \beta, \gamma, \delta$ mod 5. (At a vertex, 5 triangles.) Klein. Lectures on the Icosahedron.

$$T_g f(x) = f(xg)$$

Need to recover $X$. There are shift oper., a group. But there are also oper. of mult. by a funct. (but if there is a ring, then the repr. in the ring determines the dyn. system). 29/III Viktor Nikolaevich. Connection between Lie algebras and assoc. algebras.

$$\begin{array}{ccc} L & \longrightarrow & U \\ \text{Lie alg.} & & \text{univ. envelop. alg.} \end{array}$$

Harish-Chandra — if $L$ is simple, what can be said about $U$? Jacobson's problem: what is needed for $U$ to be primitive (a faithful irred. module over it exists). Example. $e_1, e_2$, $[e_1, e_2] = e_1$ — $U$ primitive. $[e_1, e_2] = 0$ — $U$ non-primitive. (This is a polynomial ring, and a commut. primitive alg. is a field.) $A^{(-)}$ — a Lie alg. obtained from an assoc. alg. $A$ (ab-ba). Which properties of $A$ carry over to $A^{(-)}$? In $A$ — Lie ideals. Baxter. If $A$ is Jacobson semisimple.

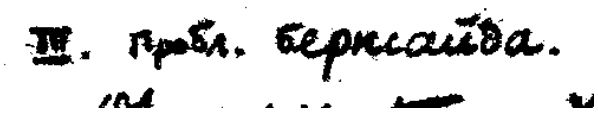



Original source details: GELFAND-P000065

**Diagram and typed description.** If $A$ is simple, then there are no Lie ideals either. (If there are no ordinary ideals, then there are no Lie ideals either.)

GELFAND-P000066

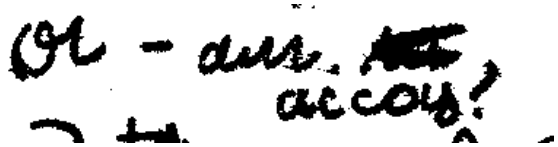
II / III. Burnside's probl. Kostrikin's result.




Original source details: GELFAND-P000066

**Uncertain reading.** II / III

($A$ — an alg. [Crossed out: illegible; below it: assoc.?]




Original source details: GELFAND-P000066

**Uncertain reading.** illegible

$X$ — a finite set, [crossed out: illegible]

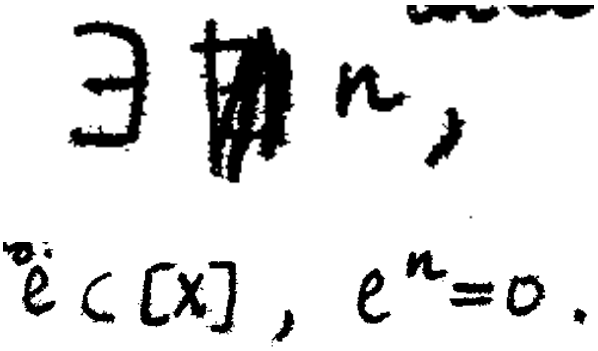


Original source details: GELFAND-P000066

**Uncertain reading.** illegible

generating $A$, $[X]$ ($\supset X$.
generated Lie alg.

$\exists$ [crossed out: $\forall$] $n$, $\quad e \subset [X], \quad e^n = 0.$




Original source details: GELFAND-P000066

**Uncertain reading.** $\exists\, n$, an overwritten sign before $n$; relation $e \subset [X]$ / $e \in [X]$.

Then $A$ nilpot. (i.e. $\forall$ elem. is nilpotent). Gelfand. Dixmier — unsolved problem: 2 rings $P(x, \frac{d}{dx})$ and $P(x, \frac{d}{dx}, y, \frac{d}{dy})$ (polynomials). Are these rings isomorphic? S. Gelfand. Dyson. 1. Orthog. ensemble.

$U^T = U$ symm.
$UU^* = 1$ and unit.

Inv. for the unit. gr.

$$U \longrightarrow VUV^T$$

(a homogeneous space of the unit. group). Density of the distribution of eigenval. $P(\theta_1, \dots, \theta_N) d\theta_1 \dots d\theta_N$ — measure of matrices, eigenval. (their args.) in $(\theta_1, \theta_1 + d\theta_1); \dots$. Stabilizer subgr.: $E \longrightarrow VV^T = E$ (real orthog. matrices). An inv. measure is introduced. Turns out

$$p_1(\theta_1, \dots, \theta_N) = C_1 \prod_{i<j} |e^{i\theta_i} - e^{i\theta_j}|$$

2. All unit. matrices. (Probab. of eigenvalues!) Group — unit. [Inner autom.] $U \longrightarrow VU$ [then a crossed-out sign: illegible, resembling $V^*$.]



Original source details: GELFAND-P000066

**Uncertain reading.** illegible, resembling $V^*$

$$p_2(\theta_1, \dots, \theta_n) = C_2 \prod_{i<j} |e^{i\theta_i} - e^{i\theta_j}|^2$$



Original source details: GELFAND-P000066

**Uncertain reading.** In the last argument of $p_2$, the index is $n$ / $N$.

(Inv. measure of the unit. gr. $\int$ over conjugacy classes.) Distinct eigenval. are more probable. [Above $\theta$: a circled letter: $\Delta$ / illegible; a triangular sign inside an oval. .]



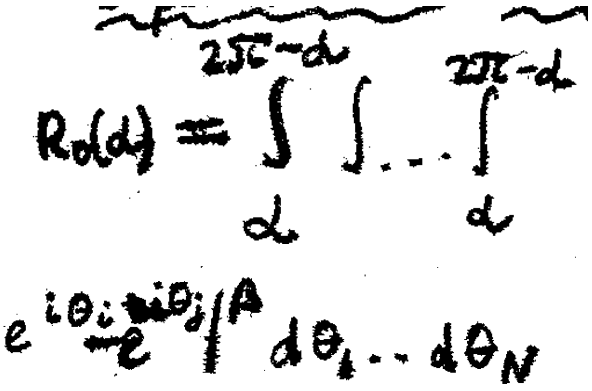
Original source details: GELFAND-P000066

**Uncertain reading.** $\Delta$ / illegible; a triangular sign inside an oval.



Consider all matr., $-\alpha < \theta < \alpha$ contains no eigenvalues. The probability of this is calculated.

$$R_\alpha(\alpha) = \int_\alpha^{2\pi-\alpha} \int \dots \int_\alpha^{2\pi-\alpha} |e^{i\theta_i} - e^{i\theta_j}|^\beta \, d\theta_1 \dots d\theta_N$$



Original source details: GELFAND-P000066

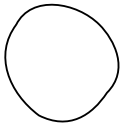
**Uncertain reading.** In this line, no definite $\prod$ is visible before the absolute value; the index of $R$ resembles $\alpha$, argument $\alpha$ / $d$.

$\beta = 1, 2$. For Dyson, $\forall\beta$. Physical assumptions are made. Phys. analogy.

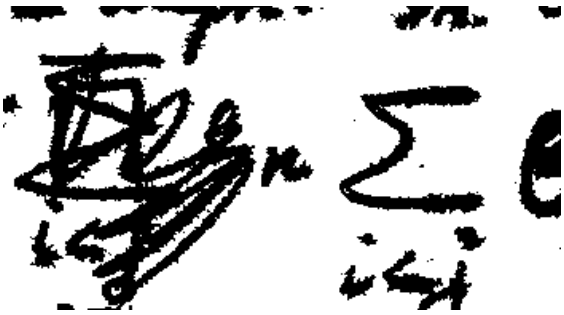


Original source details: GELFAND-P000066

**Diagram and typed description.** A circle without labels.



Conducting wire in the form of a circle, on it $N$ charges, repelling by 2-dim. electrostat. Then the energy (potent.) = $W$. [Crossed out before the sum: $\prod_{i<j}$ / illegible.]



Original source details: GELFAND-P000066

**Uncertain reading.** $\prod_{i<j}$ / illegible

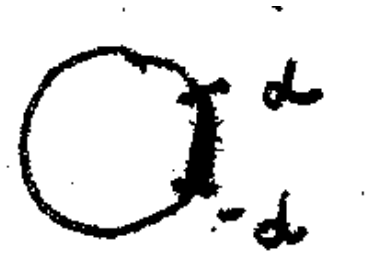
$$W = \sum_{i<j} \ln |e^{i\theta_i} - e^{i\theta_j}|$$

(Free charges.) With this, the gr. can. stat. functions.

$$\int_0^{2\pi} \int_0^{2\pi} \dots \int_0^{2\pi} e^{[-\beta W]}\, d\theta_1 \dots d\theta_N$$

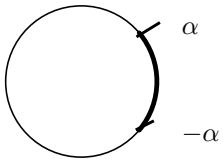
GELFAND-P000067

$\beta = \frac{1}{T}$. $T = 1$ and $1/2$ correspond to the orthog. and unit. ensembles. The segment $(-\alpha, \alpha)$ is forbidden.



Original source details: GELFAND-P000067

**Diagram and typed description.** A circle with a short dark right-hand section between the labels $\alpha$ and $-\alpha$. The thick terminal strokes do not establish arrows.

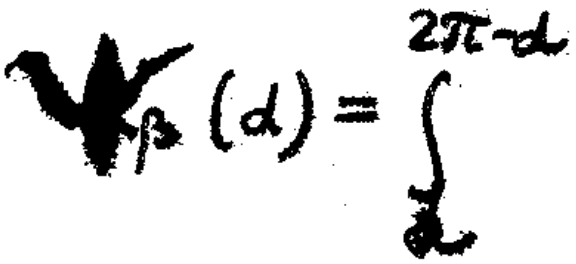



Free energy.

$$R_\beta(\alpha) = e^{\beta[F(\alpha) - F(0)]}$$

$F$ — free energy. (This equality can be taken as its def.) Consider matrices with no eigenval. in the interval $-\alpha < \theta < \alpha$. Their probability

$$\Psi_\beta(\alpha) = \int_\alpha^{2\pi-\alpha} \dots \int_\alpha^{2\pi-\alpha} \prod_{i<j} |e^{i\theta_i} - e^{i\theta_j}|^\beta \, d\theta_1 \dots d\theta_N$$



**Uncertain reading.** The initial letter $\Psi$ in the integral is overwritten; the upper reading is $\Psi$, the lower one is not established.

Original source details: GELFAND-P000067

$$R_\beta(\alpha) = \frac{\Psi_\beta(\alpha)}{\Psi_\beta(0)}$$

(Need to calculate.) Phys. assumptions. Let $N$ be large, introduce a density of eigenval. (instead of eigenval.). $\sigma(\theta)d\theta$. $\sigma(\theta)$ — $\delta$-funct. at the points $\theta_1, \dots, \theta_n$. $\sigma(\theta)d\theta \longrightarrow$ number of eigenval. in the interval $d\theta$.

$$R_2(\alpha) = e^{(N^2 \ln \cos \frac{\alpha}{2})}$$

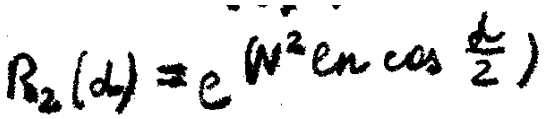
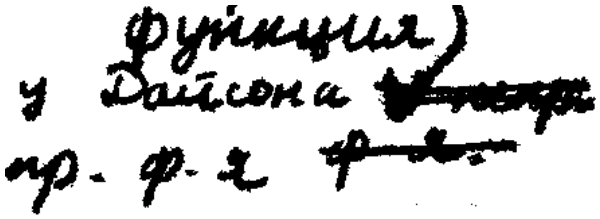


**Diagram and typed description.** The exponent is enclosed in round parentheses.

Original source details: GELFAND-P000067

Assumptions. 1) The distribution of eigenvalues is given by $\overbrace{\sigma_\alpha(\theta)}$. Under Dyson's assumpt., between $(-\alpha, \alpha)$ they are 0. (For $\forall$ matr., its own function.) Generalization — for Dyson [crossed out: an illegible word and a short fragment] any cont. funct. ($N \to \infty$).



**Uncertain reading.** an illegible word and a short fragment

Original source details: GELFAND-P000067

2) $F = V + F_1$. def:

$$V = \underbrace{-\frac{1}{2} \int_\alpha^{2\pi-\alpha} \int \sigma_\alpha(\theta)\sigma_\alpha(\varphi) \ln|e^{i\theta} - e^{i\varphi}| \, d\theta \, d\varphi}_{\text{Coul. energy}}$$

def:

$$F_1 = \int_\alpha^{2\pi-\alpha} \sigma_\alpha(\theta) f_\beta(\sigma_\alpha(\theta)) \, d\theta$$

$f_\beta$ — a certain funct., written out explicitly.

GELFAND-P000068

The assumpt. consists in

$$R_\beta(\alpha) = e^{\beta(F(\alpha)-F(0))}$$

where $\sigma$ is a funct. giving the minimum of the functional $F$ (with $\int_\alpha^{2\pi-\alpha} \sigma(\theta)d\theta = N$).

$$f_\beta = U_\beta(\sigma) - \beta^{-1} S_\beta(\sigma)$$

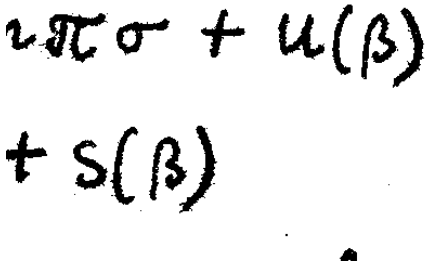
where [in $U_\beta$, the first term is crossed out: illegible]



**Uncertain reading.** illegible

Original source details: GELFAND-P000068

$$U_\beta(\sigma) = -\frac{1}{2}\ln \pi\sigma + u(\beta)$$
$$S_\beta(\sigma) = \ln \frac{N}{2\pi\sigma} + S(\beta)$$

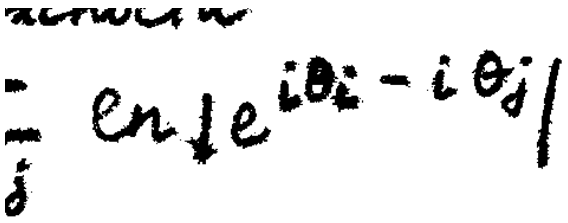


**Uncertain reading.** In the terms independent of $\sigma$, $u(\beta)$ / $U(\beta)$ and $S(\beta)$ / $s(\beta)$.

Original source details: GELFAND-P000068

$U$ — potent. energy, $S$ — entropy in the system. $N$ charged points on a circle.

$$W = \sum_{i<j} \ln |e^{i\theta_i - i\theta_j}|$$



**Uncertain reading.** On the right-hand side of $W$, after the first $e$, $i\theta_i - i\theta_j$ is visible; a second $e$ is not established.

Original source details: GELFAND-P000068

$\beta = \frac{1}{T}$. $T$ — temperature. The distribution of these points will be the same as the distribution of eigenval.

$$\int\int = \int\int e^{\beta \sum \ln |e^{i\theta_i} - e^{i\theta_j}|} \prod d\theta_j$$

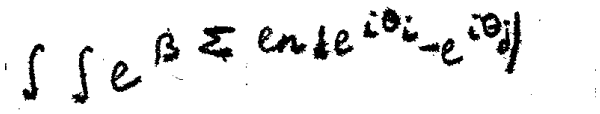


**Diagram and typed description.** The exponent of the last integral reads $\beta \sum \ln |e^{i\theta_i} - e^{i\theta_j}|$; no minus is visible before $\beta$.

Original source details: GELFAND-P000068

The canonical $\int$ of statistical physics in the plane. Next time: Minlos and S. Gelfand and Bernstein — monoidal transformations.

GELFAND-P000069

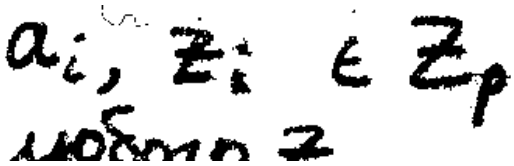
[Above the heading, a line is crossed out: illegible.]

**Uncertain reading.** illegible



Original source details: GELFAND-P000069

5/IV. I. Bernstein. Grothendieck top. and monoidal transformations. Dioph. eq. $F(x) = 0$. $F$ — polyn. with integer coeff., $x_1, \ldots, x_n$. Usually reduce mod $p$. Everything in $Z_p$. $a_i, z_i \in Z_p$.



**Uncertain reading.** A small raised stroke is visible at $z_i$; $z_i$ / $z_i^1$ / $z_i'$.

Original source details: GELFAND-P000069

If the sol. is known $\forall Z_p$, then there is inform. for any $Z$. Weil. Take $F$ over the alg. closure of $Z_p$. Polyn. — surf. How many points over $Z_p$? $\mathcal{L}$ / $Y$ — closure of $Z_p$.

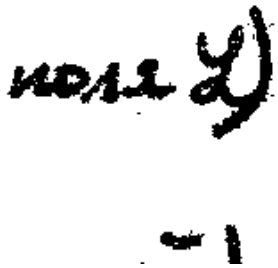
**Uncertain reading.** $\mathcal{L}$ / $Y$



Original source details: GELFAND-P000069

$x \longrightarrow x^p$ — homomorphism (since the field has char. $p$) (automorphism of the ring $\mathcal{L}$ / $Y$).

**Uncertain reading.** $\mathcal{L}$ / $Y$



Original source details: GELFAND-P000069

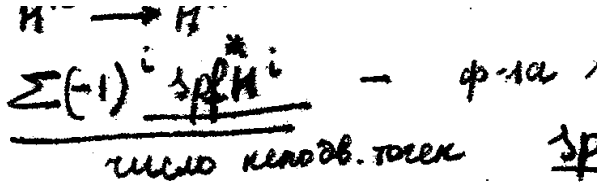
The fixed elem. of the base field: $x^p = x$ in the base field and only these (since $\leq p$ solutions).

$$(x_1, \ldots, x_n) \longrightarrow (x_1^p, \ldots, x_n^p)$$

How many fixed points? These will be in $Z_p$. The coeff. are fixed under the automorphism. The number of fixed points is found using the Lefschetz formula. $f : X \longrightarrow X$. $X$ — top. space. If $f$ is in gen. position, then one can find $H^n \longrightarrow H^n$.

$$\frac{\sum(-1)^i \operatorname{Sp} f^* H^i}{\text{number of fixed points}}$$

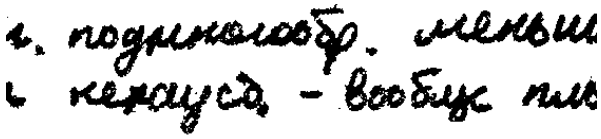
— Lefschetz formula!



**Uncertain reading.** The exponent over $f$ is overwritten; the upper layer resembles *, the lower one $n$.

Original source details: GELFAND-P000069

Sp $f^* H^i$ ($H^i$ — cohomol.). Closed sets — $\bigcup$ of a finite number of alg. submanifolds of smaller dimensions. (Zariski top.) It is негусса / негуща — generally bad.

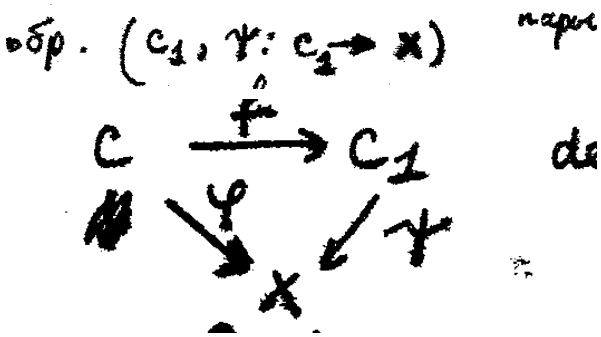


**Uncertain reading.** негусса / негуща

Original source details: GELFAND-P000069

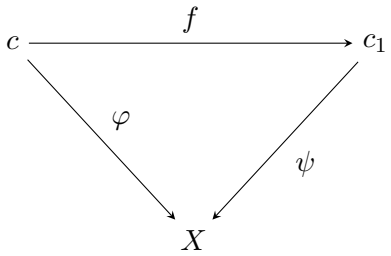
Grothendieck generalizes and improves. Definition. Grothendieck top. $C$ — category, $X$ — distinguished object (one). New categ. $C_X$ — objects — $(c, \varphi : c \longrightarrow X)$, $c \in C$. Mapping
pair

$(c_1, \psi : c_1 \longrightarrow X)$.



**Uncertain reading.** The letter in the second pair and on the right arrow: $\psi$ / $\gamma$.

Original source details: GELFAND-P000069




def commutative. [To the left of the diagonal arrow, a stroke is crossed out: illegible.]



**Uncertain reading.** illegible

Original source details: GELFAND-P000069

The elements of $C_X$ are called open sets [inserted: over] $X$. A topology is introduced in $X$. This is no special case of ordinary top. — the reverse.

GELFAND-P000070

$U$ — elem. of $C_X$ — open sets. Fix a fam. of mappings
object

$$\{U_\alpha \longrightarrow U\}_{\alpha \in I}$$

There are terribly many such families. Some of these families are called coverings of $U$, if $1^\circ$. $\{U \xrightarrow{\text{id}} U\}$ [Crossed out and inserted above: is; the crossed-out word is illegible] a covering.



**Uncertain reading.** is; the crossed-out word is illegible

Original source details: GELFAND-P000070

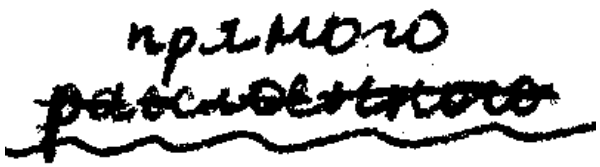
Example. $X$ — space (top.). $C$ — open sets, mapping — inclusion (if no incl., then none). $C_X$ is almost $C$ (incl. in $X$). Covering $U_\alpha$ — completely covers $U$. $2^\circ$. $\{U_\alpha \to U\}$ — cov. $\forall \alpha$ there is a covering $\{U_{\alpha\beta} \to U_\alpha\}$, $\alpha \in I$. Then $\{U_{\alpha\beta} \to U\}$ — also a covering. $3^\circ$. [Before "def." a symbol is crossed out: illegible.]

**Uncertain reading.** illegible



Original source details: GELFAND-P000070

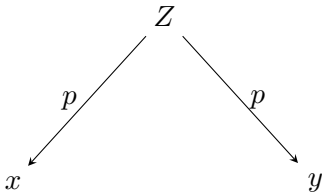
Def. [crossed out: fibred; above: direct] product.

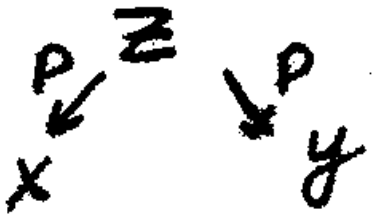


**Uncertain reading.** fibred

Original source details: GELFAND-P000070

[On the right:] $X \times Y$. $A$ — categ., def. of product: $x, y \in A$. Product — object $Z$ and

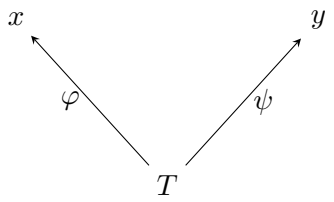



**Uncertain reading.** Both projections carry the same $p$-like sign; $p$ / $\rho$, indices not established.

Original source details: GELFAND-P000070

If $T \in A$, and $\varphi, \psi$ are given




then there exists a unique $f$

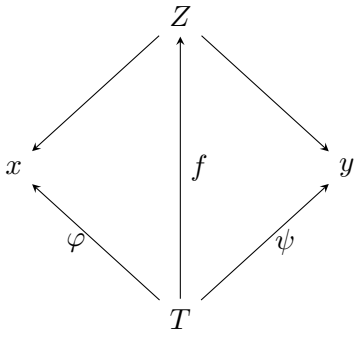
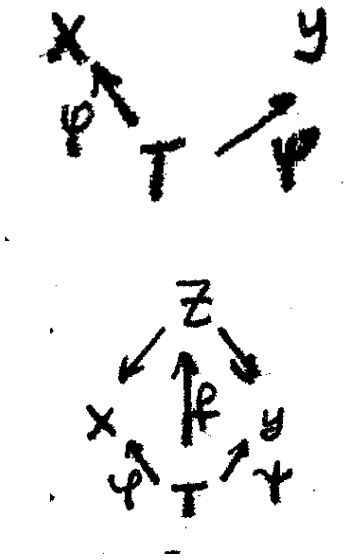



**Diagram and typed description.** The two arrows from $T$ point to $x, y$; in the next diagram $f$ points from $T$ to $Z$.



Original source details: GELFAND-P000070

(If it exists, then it is called a product.) [Crossed out: illegible.]

**Uncertain reading.** illegible



Original source details: GELFAND-P000070

The product is defined uniquely up to isomorphism $\underset{1}{\overset{\longrightarrow}{\longleftarrow}}$.



**Diagram and typed description.** Two opposing arrows, upper rightwards, lower leftwards; 1 below them.

Original source details: GELFAND-P000070

Example. $X$ — top. space, elem. — open sets.

$$U \times V = U \cap V = Z$$

Mappings to $U, V$ — incl. Ax. 3°. In categ. $C_X$ there exists the product of any 2 [above: — two.] elements.

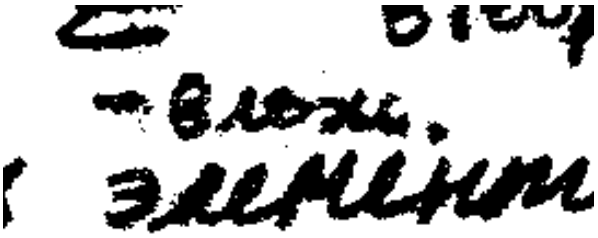


**Diagram and typed description.** Above "2", "— two." is inserted.

Original source details: GELFAND-P000070

(Here this is the intersect. of $\forall$ 2 sets.) Reason in $C$. Look at the categ. of sets with all mappings (product rel. to $X$

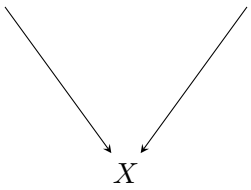



**Diagram and typed description.** Two arrows with unlabelled upper objects meet at $X$.



Original source details: GELFAND-P000070

pairs $(x, y)$ map to 1 element. (If $X$ is a point, then the direct product.)

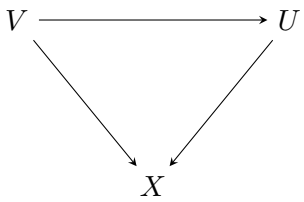
GELFAND-P000071

Ax. 4. (intersection is defined)

Given $f : V \to U$ (open sets) and a covering $\{U_\alpha \to U\}$.

$$\{V \times_U U_\alpha \to V\} \quad \text{will also be a covering.}$$

Product in categ. $C_U$.



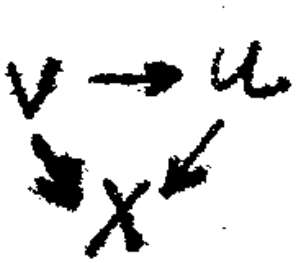
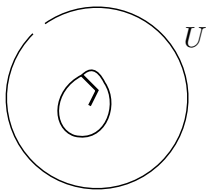



**Diagram and typed description.** An arrow $V \to U$; two oblique arrows from $V$ and $U$ meet at $X$. The arrows have no labels.

Original source details: GELFAND-P000071




$U_\alpha \cap V$

also coverings

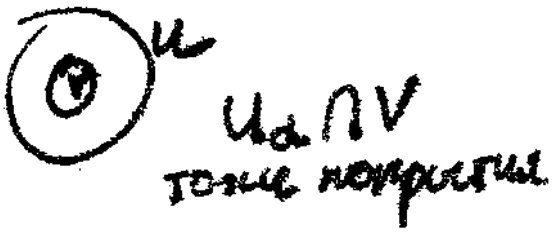


**Diagram and typed description.** A circle with an external letter $U$; the inner sign resembles $O$ / $0$, with a heavy upper stroke and a stroke inside.

Original source details: GELFAND-P000071



**Uncertain reading.** At the extreme right edge, a separate vertical stroke resembling $1$; a complete leaf number is not readable.

Original source details: GELFAND-P000071

Presheaf. $U \to P(U)$ (ab. group)

$$U \to V, \qquad P(U) \leftarrow P(V).$$

And ax. $1 \to 1$ and functor.

Not every presheaf is a sheaf.



**Uncertain reading.** “This is a functor” is densely crossed out / overwritten.

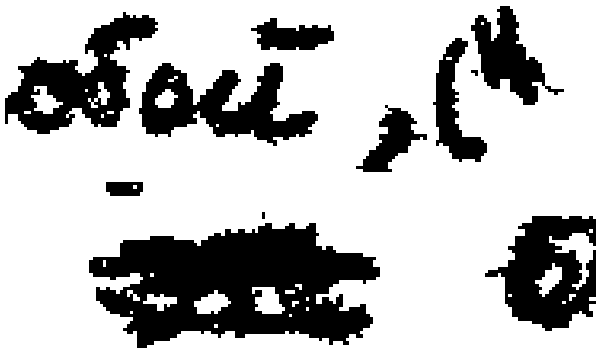
Original source details: GELFAND-P000071

Let there be a sect. $P(U)$ and a covering $\{U_\alpha \to U\}$. Then a mapping is def. $P(U) \to P(U_\alpha)$, hence $P(U) \to \prod P(U_\alpha)$.

Sheaf ax. — 1°. This is an embedding.

$$0 \to P(U) \to \prod P(U_\alpha).$$

2°. If $\xi_\alpha \in P(U_\alpha)$, $\xi_\beta \in P(U_\beta)$ are compat. with each other, [?],



**Uncertain reading.** after the comma, short upper strokes; before "images", a crossed-out sign is illegible

Original source details: GELFAND-P000071

$$\begin{array}{ccc} P(U_\alpha) & & \\ & \searrow & \\ & & P(U_\alpha \times_U U_\beta) \\ & \nearrow & \\ P(U_\beta) & & \end{array}$$

the images coincide,



**Uncertain reading.** At the left, a note "$U_\alpha$-mor. $U$" / "$U_\alpha$-kor. $U$".

Original source details: GELFAND-P000071

then there exists [Inserted above: sect.] $\xi \in P(U)$ such that they are obtained from it.

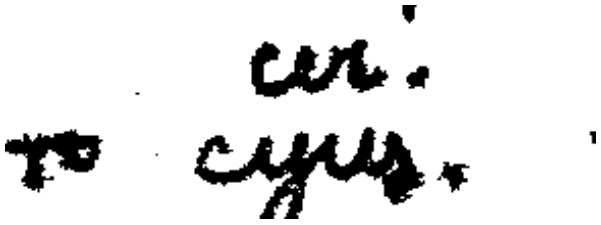


**Diagram and typed description.** Above "exists", "sect." is inserted.

Original source details: GELFAND-P000071

Top. over a point — unusual sheaves. To a point — fields, this is Galois theory, and

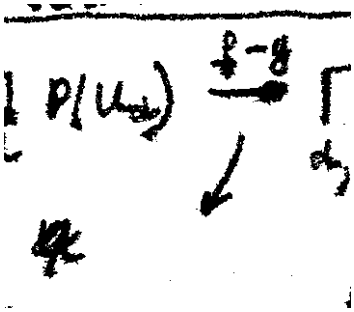


Original source details: GELFAND-P000071

**Uncertain reading.** “gon.” / “gom.” [гон. / гом.] — topol. of Galois groups.

Sheaf ax. in the form of an exact seq.

$$0 \to P(U) \to \prod_{\alpha} P(U_\alpha) \overset{f-g}{\underset{\swarrow}{\rightrightarrows}} \prod_{\alpha,\beta} P(U_\alpha \times_U U_\beta).$$

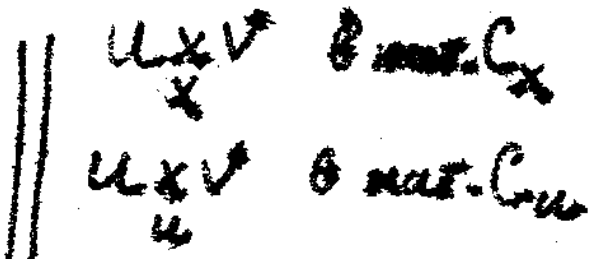


Original source details: GELFAND-P000071

**Uncertain reading.** Below, at the end of the arrow pointing down and left, a densely crossed-out sign; the exact reading is not established.

[In the right margin:]

$\| \; U \times_X V$ in [crossed out: illegible] $C_X$;
$\| \; U \times_U V$ in categ. $C_U$.



Original source details: GELFAND-P000071

**Uncertain reading.** In the upper marginal note, between “in” and $C_X$, a densely overwritten fragment.

2 mappings.

$$U_\alpha \times_U U_\beta \to U_\alpha,$$
$$U_\alpha \times_U U_\beta \to U_\beta.$$

Whence

$$P(U_\alpha) \to P(U_\alpha \times_U U_\beta),$$
$$P(U_\beta) \to P(U_\alpha \times_U U_\beta).$$

Therefore def. [?],

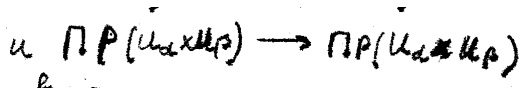


Original source details: GELFAND-P000071

**Uncertain reading.** before $P(U_\alpha)$, densely crossed-out signs, probably a product sign

$$\prod P(U_\alpha) \to \prod P(U_\alpha \times U_\beta),$$

and [?]:



**Uncertain reading.** in the next line, both on the left and on the right, $\prod P(U_\alpha \times U_\beta)$ is visible

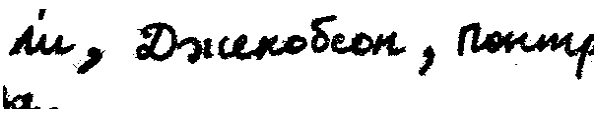
Original source details: GELFAND-P000071

$$\prod P(U_\alpha \times U_\beta) \to \prod P(U_\alpha \times U_\beta).$$

$$\prod_\alpha P(U_\alpha) \xrightarrow{f} \prod P(U_\alpha \times U_\beta),$$
$$\prod_\beta P(U_\beta) \xrightarrow{g} \prod_{\alpha,\beta} P(U_\alpha \times U_\beta).$$

GELFAND-P000072

Gindikin. Eigenval. on a symm. space. Need to know roots. Sem. Sophus Lie, [?], Pontryagin.

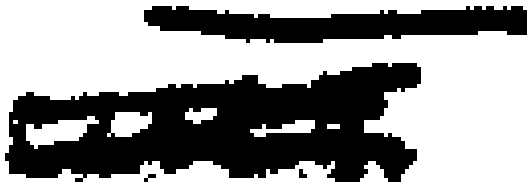


**Uncertain reading.** “Dzheloben” / “Dzhelobe” / “Dzhekobson” [Джелобен / Джелобе / Джекобсон]

Original source details: GELFAND-P000072

G. Topic: study of Dyson, $\forall$ symm. spaces.

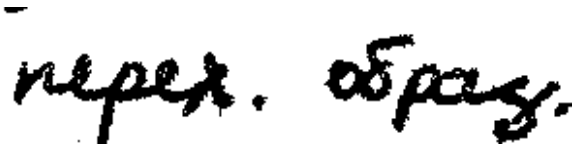
$12/\overline{\text{IV}}$. Stepin. Kirillov's probl. Construct an example [crossed out: an illegible word]



**Uncertain reading.** an illegible word

Original source details: GELFAND-P000072

of a dyn. syst. with cont. spectrum, dyadic-rat. numbers on the circle $G$. Every elem. — of finite order. (not with discr.) $g \to T_g$. Suff. to specify



**Uncertain reading.** “perekh.” / “pred.” [перех. / пред.]

Original source details: GELFAND-P000072

images of $1/2^k$.

$$T^2_{1/2^k} = T_{1/2^{k-1}}.$$

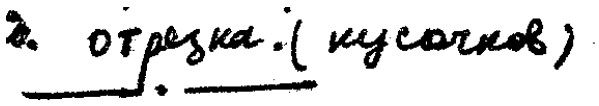
$T_0 = E$, $T_1$ — [?], refl.



**Uncertain reading.** a crossed-out word resembling “rotation” / “refl.”

Original source details: GELFAND-P000072

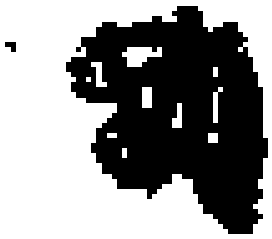
Next, break up the segments. (pieces) preserving orientation.



Original source details: GELFAND-P000072

**Uncertain reading.** The underlined word's ending: "segments" / "segment" [отрезки / отрезка]; a full stop follows it.

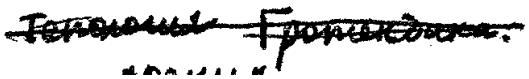
[On the right:] $1/2^k \to T_k$.



Original source details: GELFAND-P000072

**Uncertain reading.** At the left before the heading, a densely overwritten sign; the exact reading is not established.

What is it desirable for participants in the seminar to know?

1. Representations. Lie groups and Lie algebras. Symmetric spaces. Simple Lie groups.
2. 

Original source details: GELFAND-P000072

**Uncertain reading.** "Homologies" / "Homolies" and "Representations" are crossed out.

   Sheaves. Homol. algebra (fundamentals). Vector bundles.
3. Elements of algebraic geometry.
4. Quantum mechanics (nonrelativ.). Cauchy $\int$ of several variables according to Gindikin will be incl.

an illegible sign.



Original source details: GELFAND-P000072

**Uncertain reading.** an illegible sign

   Nuclear spaces and measures (Wiener measure).

S. Gelfand. $R_2(\alpha)$ calculated exactly. [?]:



Original source details: GELFAND-P000072

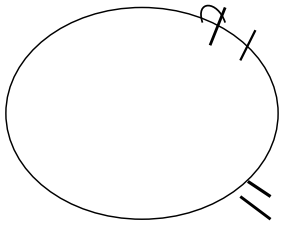
**Uncertain reading.** In the exponent after the parenthesis, a densely overwritten sign before 2, probably $n^2$

$$R_2(\alpha) = \left(\cos\frac{\alpha}{2}\right)^{n^2}.$$

(This is asserted, but not proved.) This probl. is replaced by another for which this is the exact answer.

Dyson's 3rd paper. $S(\theta)$.

$$(\theta_1, \theta_1 + d\theta_1), \qquad (\theta_2, \theta_2 + d\theta_2).$$



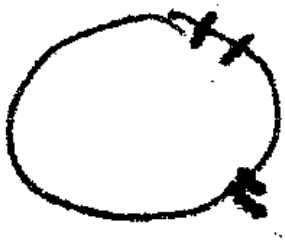


**Diagram and typed description.** A circle; at the upper right, two short transverse strokes and a short curved stroke; at the lower right, a heavy mark pointing outward. The marks have no labels.

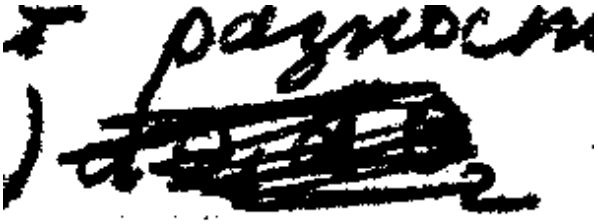
Original source details: GELFAND-P000072

Prob. that there is an eigenval. in the 1st interval, one in the 2nd, and none between. This is

$$F(\theta_1, \theta_2)d\theta_1 d\theta_2.$$

It depends on the difference.

$$F(\theta_1; \theta_2) = S(\theta_1 - \theta_2).$$



**Uncertain reading.** Between $F(\theta_1; \theta_2)$ and the equals sign, densely crossed-out signs; the original entry is illegible.

Original source details: GELFAND-P000072

This ~~$S(\theta)$~~ —



**Diagram and typed description.** At the bottom, $S(\theta)$ is crossed out.

Original source details: GELFAND-P000072

[On the right:] easy to prove.

$$S(\theta) = \frac{d^2R}{d\theta^2}.$$

GELFAND-P000073

$R(2\alpha)$ — prob. that there are no eigenval. in $(-\alpha, \alpha)$.

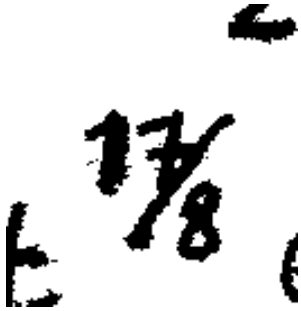
$$R_1(2\alpha) = \det \left| \delta_{pq} - \frac{1}{\pi} \int_{-\alpha}^{\alpha} \cos p\theta \cos q\theta \, d\theta \right|, \qquad p, q = \frac{1}{2}, \ldots, m - \frac{1}{2}, \quad 2m \geq n.$$



**Uncertain reading.** The relation sign: $2m \geq n$ / $2m = n$; its upper stroke slopes and the sign is heavily written.

Original source details: GELFAND-P000073

Asymptotics [?]:



**Uncertain reading.** In the exponent on $t$, $17/8$ is visible; the first sign is heavily written.

Original source details: GELFAND-P000073

$$R_1(t) \sim A t^{17/8} \exp\left[-\frac{1}{4}t^2 - \frac{1}{2}t\right], \qquad \underline{\underline{t = \alpha n}}.$$

And other results (analogous).

Minlos. Elems. of the matrix (Hermit.) — random numbers with Gauss. distr.

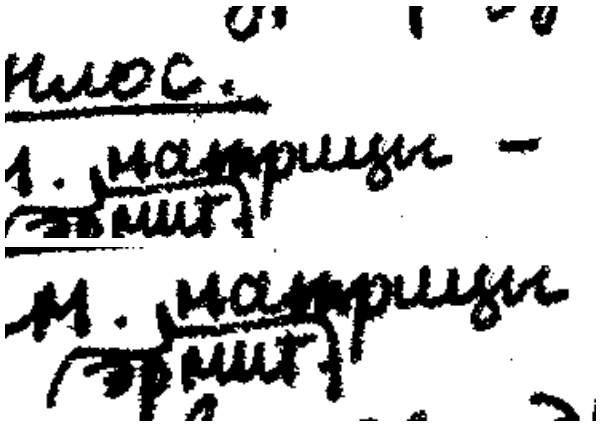


**Diagram and typed description.** Below the word “matrix”, “(Hermit.)” is inserted.

Original source details: GELFAND-P000073

Brownian motion.

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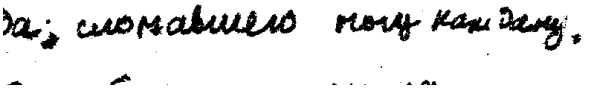


**Diagram and typed description.** Two horizontal strokes with a point between them.

Original source details: GELFAND-P000073

What is the angle of $k$-dim. subspaces in an $n$-dim. space?

Kazhdan. Res. of Gromov from Leningrad; [?].



Original source details: GELFAND-P000073

**Uncertain reading.** “who caught a bird on the move” [схватившего птицу на ходу / схватившему птицу на ходу]; the participle’s ending is uncertain

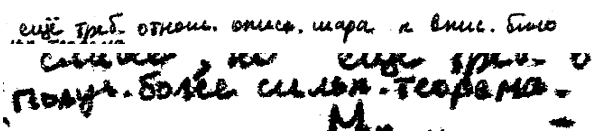
Dvoretzky’s res. $\forall \varepsilon, k\ \exists$ a space of a suff. large number of dims., such that $\forall$ [crossed out: an illegible word beneath several heavy strokes]



Original source details: GELFAND-P000073

**Uncertain reading.** an illegible word beneath several heavy strokes

convex centrally symm. body $\exists$ a section almost a ball ($k$-dimensional). Dvoretzky did it head-on (70 pp.). (In a space of a large number of dims., there is almost a circle in a section.) Gromov is weaker, but [?].



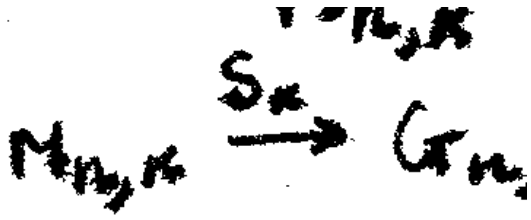
Original source details: GELFAND-P000073

**Uncertain reading.** “also req. relat. size of the ball and conv. was” [ещё треб. относит. велич. шара и вып. было]; the short phrase is not fully legible

[Inserted below:] a stronger theorem is obt.

Def. $G_{n,k}$. $M_{n,k}$ — a $k$-dim. plane and a point of its unit sphere.

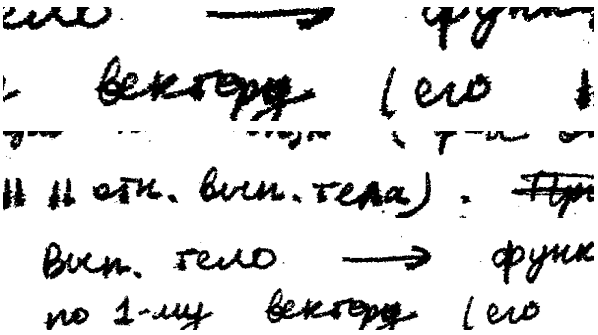
$$M_{n,k} \xrightarrow{S_k} G_{n,k}.$$



Original source details: GELFAND-P000073

**Uncertain reading.** Above the arrow, $S_k$; the small lower index is heavily written.

A bundle of spheres over $G_{n,k}$ (Grassm.). Convex body $\to$ a function on $M_{n,k}$ [?].



Original source details: GELFAND-P000073

**Uncertain reading.** (func. determ. by the 1st vector (its H и отк. convex body)); the two middle signs and the following abbreviation are unclear

[On the right, crossed out: a short word beginning with “П”; the remainder is illegible.]



**Uncertain reading.** a short word beginning with “П”; the remainder is illegible

Original source details: GELFAND-P000073

Let us expand it in sph. funcs. If $n$ is large, then $\exists$ a fibre such that the first $\ell$ funcs. vanish (except $1$). $\forall$ repr. of the unit. group $\exists n$ such that $\forall$ funcs. on $M_{n,k}$ $\exists$ a fibre such that the proj. $= 0$. Conv. — compactness on funcs.

**Uncertain reading.** a densely overwritten sign; the exact reading is not established.

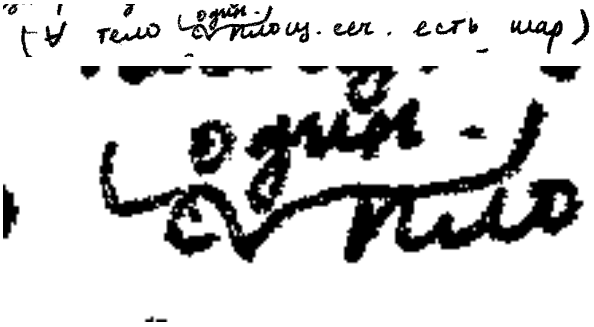


Original source details: GELFAND-P000073

Fourier coeffs. — many $0$.

GELFAND-P000074

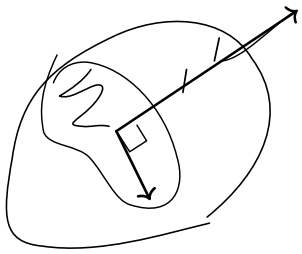
Kirillov. A convex body (centrally symm.) in $R^3$ ($R^n$). Draw plane sections through the centre. The areas of all its sects. are known. The body is recovered from the areas of the sections.



**Uncertain reading.** ($\forall$ body [inserted “odin.” / “odinak.” (equal)] sect. areas is a ball).

Original source details: GELFAND-P000074

Consider $S^2$ and a func. on it. Can it be recovered if the $\int$ over great circles is known (even funcs.). This probl. is solved triv. using expansion in sph. funcs.



sect. $\longrightarrow$ norm.

vect.



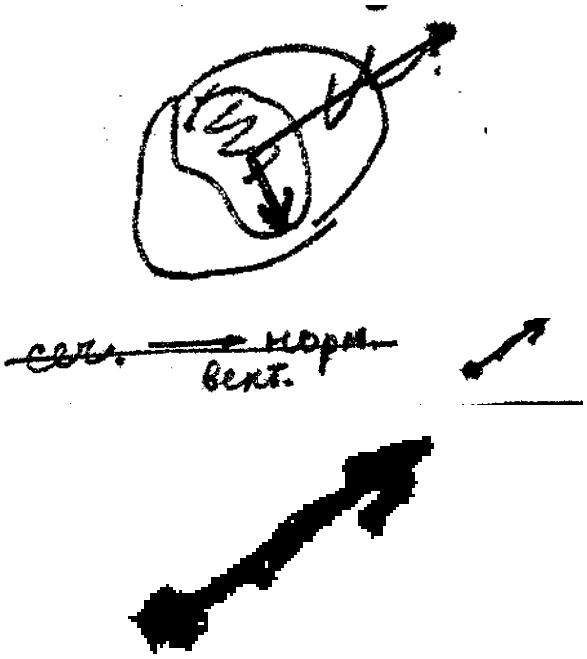



**Diagram and typed description.** An outer oval and an irregular inner contour; an oblique vector pointing upward to the right, a short downward arrow, an angle mark and short transverse strokes. Beside them, “sect.” and “norm.” are crossed out; “vect.” is inserted below the latter. Farther right, a separate oblique arrow has a heavy mark at its lower end.

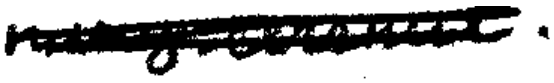
Original source details: GELFAND-P000074



**Uncertain reading.** Inside the inner contour, a winding sign; its exact reading as a letter is not established.

Original source details: GELFAND-P000074

On the sphere, a vect. of the same direction. A func. on the sphere $\to$ [crossed out: an illegible entry beneath several heavy strokes]



**Uncertain reading.** an illegible entry beneath several heavy strokes

Original source details: GELFAND-P000074

$r^2/2$ (so that the $\int$ equals the area). A convex body $K \longrightarrow$ a func. on the sphere. We obtain a probl. of $\int$ geometry. If all the shadows of the convex body (areas) are known, is the body known? Reduces.

19/IV. Faddeev. Ludwig (the younger). What is the characteristic determ. of the Sturm–Liouville oper.? On the half-line. (For a matrix — zeros — eigenval., Fredholm det.) $A$ — a matrix, $w(\lambda) = \det(A - \lambda E)$.



**Diagram and typed description.** Notation for the characteristic determinant: $w(\lambda)$.

Original source details: GELFAND-P000074

Then

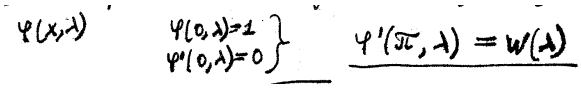
$$\frac{d}{d\lambda} \ln w(\lambda) = -\operatorname{Tr}(A - \lambda E)^{-1}.$$

(Check using eigenval.) Determ. $w(\lambda)$ up to mult const. The same in $\int$ eqns.

$$Ly = -y'' + q(x)y, \qquad y'(0) = y'(\pi) = 0.$$

The resolvent — an $\int$ oper. with a smooth kernel and the trace exists. The determ. — an entire func. of $\lambda$ and its zeros — eigenval. of the St.–Liouv. probl.

$$\psi(x,\lambda), \qquad \left.\begin{matrix}\psi(0,\lambda) = 1\\ \psi'(0,\lambda) = 0\end{matrix}\right\} \quad \underline{\psi'(\pi,\lambda) = w(\lambda).}$$



Original source details: GELFAND-P000074

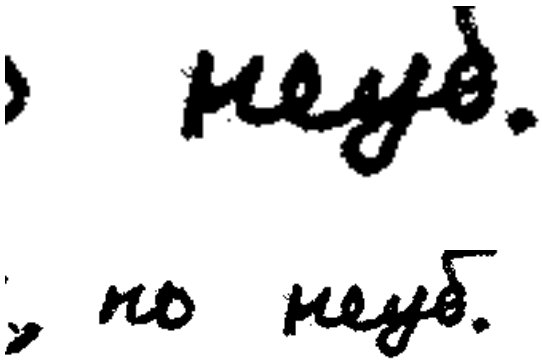
**Uncertain reading.** The normalized solution's letter: $\psi$ / $\varphi$; an open left branch, a right loop and a descending stroke.



Original source details: GELFAND-P000074

**Diagram and typed description.** Two short horizontal strokes with a point between them.

$y(0) = 0$, $q(x) \to 0$, $x \to \infty$ suff. rapidly. The res. has no trace; the resolvent does not decrease, by [?],

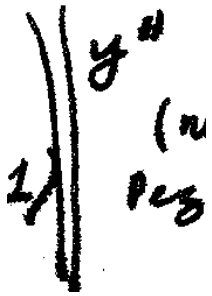


Original source details: GELFAND-P000074

**Uncertain reading.** "nevb." / "neub." [невб. / неуб.]

the part has a triv. form. Regularize it.

$$r(x,x,\lambda) = \frac{1}{2i\sqrt{\lambda}} - O(1).$$



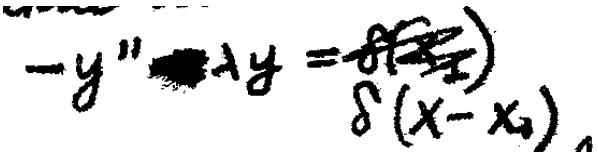
Original source details: GELFAND-P000074

**Diagram and typed description.** In the right margin, a double almost vertical stroke and $y''$.

$y''$ The resolvent is not an entire func. (Poles, we obtain — the spectrum.) The res. — the kernel $K(x,y,\lambda)$.

GELFAND-P000075

What singularities? [?]:

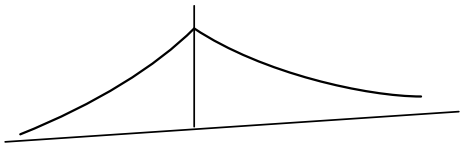


Original source details: GELFAND-P000075

**Uncertain reading.** After $-y''$, the sign before $\lambda y$ is overwritten; the initial right-hand side is crossed out and $\delta(x - x_0)$ is written below

$$-y'' - \lambda y = \delta(x - x_0).$$

It will be the resolvent. There are 2 lin. indep. sols. So as not to grow. The 1st deriv. must have a jump.

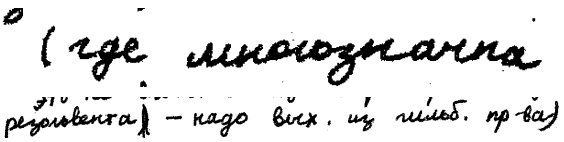




Original source details: GELFAND-P000075

**Diagram and typed description.** The horizontal axis rises slightly to the right. Two branches meet at a peak above it; the left rises, the right decreases. A vertical segment goes from the axis through the peak and a little above it. There are no labels.

$\lambda > 0$ does not work (sin and cos), $e^{\pm\sqrt{-\lambda}\,x}$. When crossing the axis, $\sqrt{\ }$ changes sign. This does not fit into the general theor. of operators.



Original source details: GELFAND-P000075

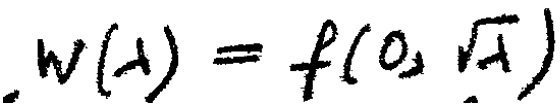
**Uncertain reading.** (where the resolvent is multivalued); the beginning of “multivalued” [многозначна] is heavily written.

— one must go outside the Hilbert space. 2-dim. case: ln (many-sheeted).

For the cont. spectrum, let

$$f(x, \sqrt{\lambda}) \sim e^{i\sqrt{\lambda}x}, \qquad x \to \infty.$$

Then $w(\lambda) = f(0, \sqrt{\lambda})$ (in the phys. lit., the Jost func.).



Original source details: GELFAND-P000075

**Diagram and typed description.** Notation in the line with the Jost function: $w(\lambda)$.

On the whole axis, it is no longer expressed through 1 solution.

$$f_1(x, \sqrt{\lambda}) \sim e^{i\sqrt{\lambda}x}, \quad x \to +\infty,$$
$$f_2(x, \sqrt{\lambda}) \sim e^{-i\sqrt{\lambda}x}, \quad x \to -\infty.$$

Their Wronskian — is the determ.

$$f_1' f_2 - f_1 f_2' = W(\lambda).$$

(Does not depend on the point — $\to$ the 2nd coeff. $= 0$.) How, knowing the determ., to write an identity for eigenval.

$$\frac{1}{2\pi i}\oint h(\lambda)\frac{W'(\lambda)}{W(\lambda)}\,d\lambda =$$

det — an entire func. of $\lambda$, discr. $0$ on the $+$ axis. For good $h(\lambda)$

$$= \sum h(\lambda_i) \quad \text{over zeros } \lambda_i \text{ of } W(\lambda).$$

$h(\lambda) = \lambda^s$.



**Uncertain reading.** Below $h(\lambda) = \lambda^s$, a short expression is crossed out: the first sign is $1$ / $\lambda$, followed by poorly distinguishable strokes.

Original source details: GELFAND-P000075

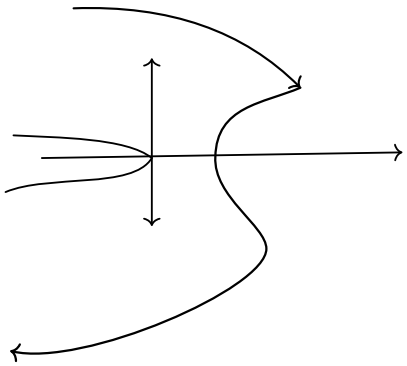
[Before the next line, densely crossed out: a short entry; its exact reading is not established.]



**Uncertain reading.** a short entry; its exact reading is not established

Original source details: GELFAND-P000075

Deform the contour. We obtain.



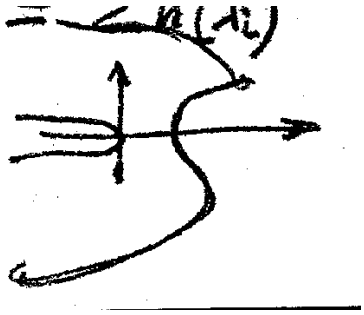


**Diagram and typed description.** Axes with arrows upward, downward and to the right; on the left, two short lines merge at the origin. The upper curve goes downward to the right to an arrow; from its end, an S-shaped line bends to the right of the origin and goes downward to the left to another arrow. There are no labels.

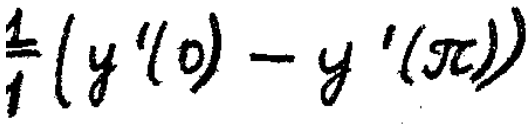
Original source details: GELFAND-P000075

$$\int_{-\infty}^{0} |\lambda|^s \frac{W'(\lambda)}{W(\lambda)}\, d\lambda = \sum_i \lambda_i^s.$$

(Re $s$, so that the $\int$ converges.) And its analyt. cont. automatically regularizes it. ($\int$ — the jump.) $W$ has known asymptotics (therefore its decrease is known).

$$\frac{W'(\lambda)}{W(\lambda)} = d_1 - \frac{d_2}{\sqrt{\lambda}} - \frac{d_3}{\lambda} - \dots$$

$s$ is taken negative and then it is proved that the left and right sides — are analyt. funcs. of $s$. [?]:



**Uncertain reading.** On the right, the letter $y$ with a prime: $y'(0)$ and $y'(\pi)$.

Original source details: GELFAND-P000076

$$\sum_n (\lambda_n - n^2 - c_0) = \frac{1}{4}\big(y'(0) - y'(\pi)\big).$$



**Diagram and typed description.** Two short separating strokes with a point between them.

Original source details: GELFAND-P000076

How to calculate the trace?

$$\sum \lambda_n = \int K(x,x)\, dx.$$

Diverges. (Sum of diag. elems.) Diag. elems.

$$-\delta''(x_1 - x_2) + q(x)\delta(x_1 - x_2).$$

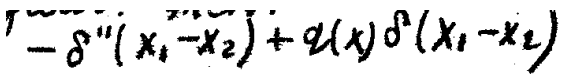
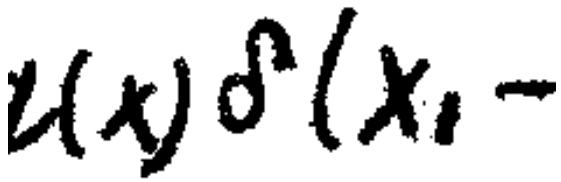


**Diagram and typed description.** In the diagonal element, the first $\delta$ has two primes; the second $\delta$ has none.



Original source details: GELFAND-P000076

$x_1 = x_2$: nothing is calculated. $-y''$ has eigenval. $n^2$!! Subtract

$$\sum (\lambda_n - n^2) = \mathrm{Sp}\, q(x)\delta(x_1 - x_2) = \delta(0) \int_0^\pi q(x)\, dx.$$

$\to c_0$, so that $\int q(x)$ [crossed out: a sign resembling $d$ and densely overwritten signs] $= 0$.



**Uncertain reading.** a sign resembling $d$ and densely overwritten signs

Original source details: GELFAND-P000076

$$\sum(\lambda_n - n^2 - c) = 0, \qquad \delta(0) = 0.$$

$c$ — asymptotics. [Separately on the right:] $\int_0^\pi q(x)\,dx$. [On the left, crossed out: “this is a func.” [это ф-я] / a short illegible entry.]



**Uncertain reading.** “this is a func.” [это ф-я] / a short illegible entry

Original source details: GELFAND-P000076

True if there is a $\delta$-func. everywhere (periodic conds.). At the end, incomplete $\delta$-funcs.; there one obtains $1/4$. G. $\mathrm{Sp}(AB - BA) = 0$. Problem.



**Diagram and typed description.** Before Gindikin’s name, two short separating strokes with a point between them.

Original source details: GELFAND-P000076

Gindikin. Integration of funcs. of several complex vars. $(z, w)$: a bicyl. $D_z \times D_w$. $\int$ of a func.

$$f(z,w) = \frac{1}{(2\pi i)^2} \iint_\Omega \frac{f(\zeta,\eta)\,d\zeta\,d\eta}{(\zeta - z)(\eta - w)}.$$

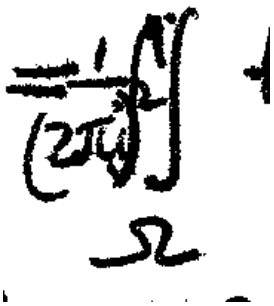


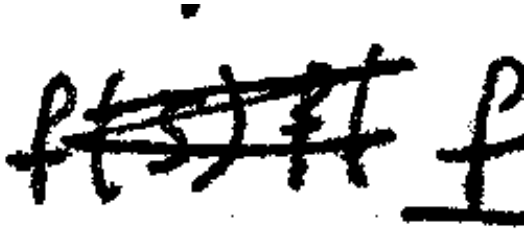
**Diagram and typed description.** A double integral with lower notation $\Omega$ and factor $1/(2\pi i)^2$.

Original source details: GELFAND-P000076

[In the numerator before $f(\zeta,\eta)$, crossed out: $f$ with a parenthesis; the arguments and following signs are densely overwritten.]



**Uncertain reading.** $f$ with a parenthesis; the arguments and following signs are densely overwritten



Original source details: GELFAND-P000076

The next interesting example — a Weil domain (generaliz. bicyl.). There is a set of funcs. $\chi_i(z,w)$, $i=1,\dots,\ell$ — holom. And $D_i$ are specified in the complex plane (1-dim.). A Weil domain: $(z,w)\colon \chi_i(z,w)\in D_i$. 2 funcs., $z$ and $w$, $z\in D_z$, $w\in D_w$ — a bicylinder!

GELFAND-P000077

Let there be two of them, $\chi_1,\chi_2$ — a skew bicylinder. This is simply a bicyl. in other coords.! ($\chi_1,\chi_2$ — new vars.). The int. can be written from a change of vars. Formula: Weil.

$$f(z,w)-f(\zeta,\eta)=(z-\zeta)P_1(z,w,\zeta,\eta)+(w-\eta)P_2(\ ).$$

Lem. There is a decomposition. But this is wrong; [crossed out: a short entry beneath dense horizontal strokes; its reading is undetermined]



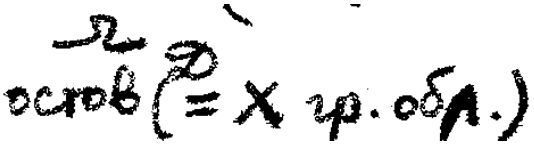
**Uncertain reading.** a short entry beneath dense horizontal strokes; its reading is undetermined

Original source details: GELFAND-P000077

— possible in any domain of holomorphy. (Gelfer's theor.) $\forall D$, $(z,w)\in D$, $(\zeta,\eta)\in D$; for a domain of holom. (Cartan's theor. B). Apply to $\chi_i$:

$$\chi_i(z,w)-\chi_i(\zeta,\eta)=(z-\zeta)P_1^i(\ )+(w-\eta)P_2^i(\ ).$$

$$f(z,w)=\frac{1}{(2\pi i)^2}\int_\Omega \frac{(P_1^1P_2^2-P_1^2P_2^1)f(\zeta,\eta)\,d\zeta\,d\eta}{(\chi_1(\zeta,\eta)-\chi_1(z,w))(\chi_2(\zeta,\eta)-\chi_2(z,w))}.$$



**Uncertain reading.** Below $\Omega$: "skeleton ($\partial=\times$ bdy. of dom.)"; the sign above the beginning of the parenthesis: $\partial$ / $D$ / an illegible sign.

Original source details: GELFAND-P000077

If there are $\ell$ funcs., then the formula has the form $\chi_1, \dots, \chi_\ell$:

$$f(z,w) = \frac{1}{(2\pi i)^2} \sum_{i,j} \int_{\Omega_{ij}} \frac{(P_i^1 P_j^2 - P_i^2 P_j^1) f(\zeta,\eta)\, d\zeta\, d\eta}{(\chi_i(\zeta,\eta) - \chi_i(z,w))(\chi_j(\zeta,\eta) - \chi_j(z,w))}.$$

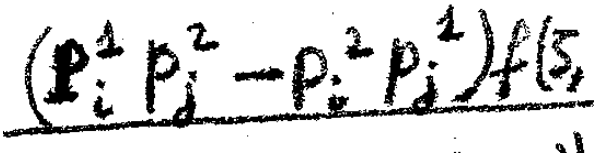
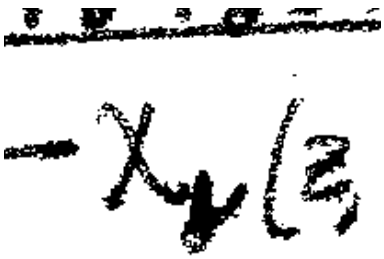


**Diagram and typed description.** In the numerator: $P_i^1 P_j^2 - P_i^2 P_j^1$; $1, 2$ above, $i, j$ below.

Original source details: GELFAND-P000077



**Uncertain reading.** The lower index of the right-hand $\chi$ in the first denominator factor: $i$ / $j$ / a densely overwritten illegible sign.

Original source details: GELFAND-P000077

$\Omega_{ij}$

$\downarrow$

preimage of the boundary
of the corresp. dom.

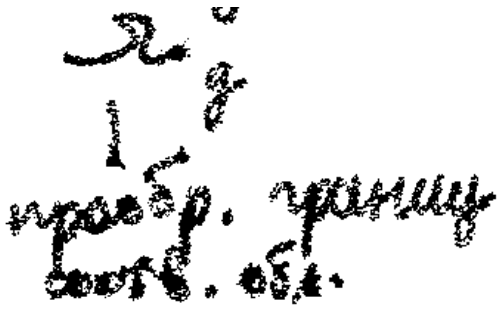


**Diagram and typed description.** A downward arrow from $\Omega_{ij}$ to “preimage of the boundary of the corresp. dom.”.

Original source details: GELFAND-P000077

Domain $\chi_i(z,w) \in D_i$.

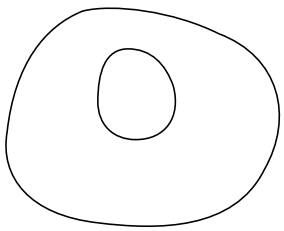
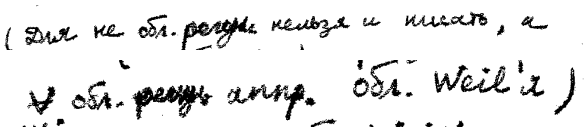




**Diagram and typed description.** Two closed rounded contours, the small one inside the large one; without labels or arrows.

Original source details: GELFAND-P000077

Decomp. into a $\sum$ of funcs. analyt. in larger doms.



**Uncertain reading.** (For non-dom. [“result.” / illegible], it is impossible even to write, whereas $\forall$ dom. [“result.” / illegible] is approx. by Weil doms.).

Original source details: GELFAND-P000077

Gindikin: A contin. analogue of a Weil dom. Dom. $\chi(z, w, \alpha) \in D_\alpha$.

$$\chi(\alpha, z, w) - \chi(\alpha, \zeta, \eta) = (z - \zeta)P_1(\alpha, z, w, \zeta, \eta) + (w - \eta)P_2(\alpha, \ldots).$$

Smooth in $\alpha$.

GELFAND-P000078

$$f(z, w) = \int \frac{d\alpha}{(2\pi i)^2} \int \frac{P_1(\alpha, \ldots)P_2'(\,) - P_2P_1'}{(\chi(\alpha, \zeta. \, - \chi(\alpha, z, \ldots)} f(\zeta, \eta)\, d\zeta\, d\eta.$$

[Above the numerator:] Wronskian det. (Derivs. in $\alpha$

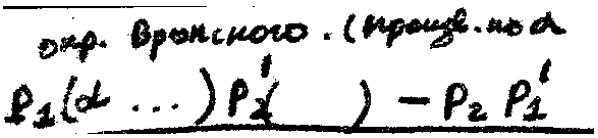
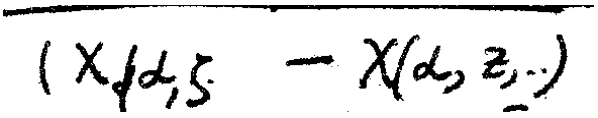


**Diagram and typed description.** Above the numerator,“Wronskian det. (Derivs. in $\alpha$”; primes at $P_2$ and the last $P_1$.

Original source details: GELFAND-P000078



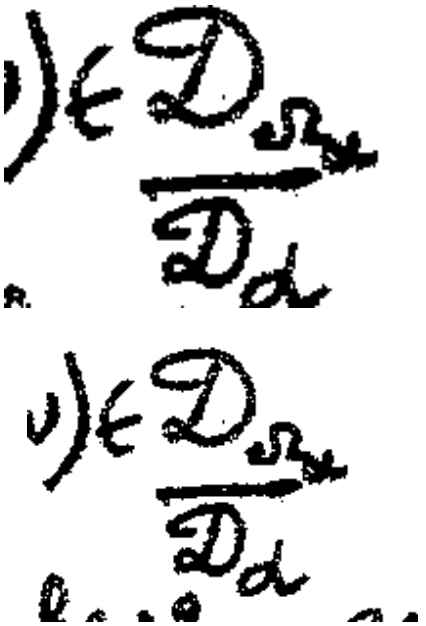
**Uncertain reading.** In the denominator, one $\chi$ difference; opening parentheses before the first $\chi$ and its $\alpha$; after the first $\zeta$, a short sign: point / comma / illegible; on the right, one closing parenthesis after the ellipsis. A small upper sign is undetermined.

Original source details: GELFAND-P000078

Some new conds. on the dom. (general position, etc.).

$$\chi(\alpha, z, w) \in D_\alpha$$

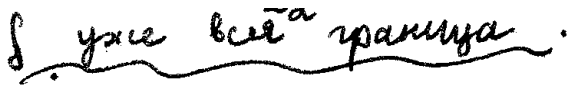
$$\overline{D_\alpha}$$



Original source details: GELFAND-P000078

**Uncertain reading.** At the upper $D$, the lower sign is densely overwritten: $\alpha$ / an illegible corrected sign; below $D$, a nearly horizontal stroke thickened at the right (underline / arrow?); below that a separate $D_\alpha$.

$\int$ already over the entire boundary.



Original source details: GELFAND-P000078

**Diagram and typed description.** A wavy underline beneath "$\int$ already over the entire boundary.".

In a partic. case [crossed out: a short entry beneath two dense horizontal strokes, with a lower slanting stroke; its reading is undetermined]



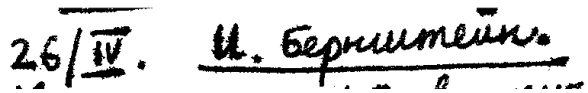
Original source details: GELFAND-P000078

**Uncertain reading.** a short entry beneath two dense horizontal strokes, with a lower slanting stroke; its reading is undetermined

the formula for the inverse Radon transform. $\chi_\alpha$ — linear, then tube doms.

GELFAND-P000079

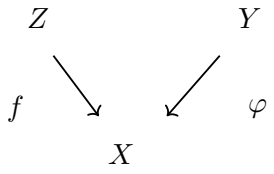
[The upper part of the sheet is without text.] $26/\overline{\overline{\text{IV}}}$. I. Bernstein.



Original source details: GELFAND-P000079

**Diagram and typed description.** The date $26/\text{IV}$ with two strokes above IV and one below; the name is underlined separately.

Cohom. with coeffs. in a sheaf in Grothendieck top. All coverings are fixed. In categ. $C$, $\forall$3objects $X, Y, Z$, $\exists T = Z \times_X Y$. Categ. of sets.



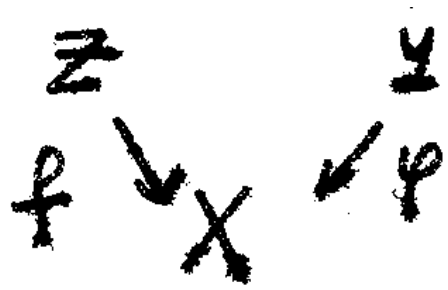
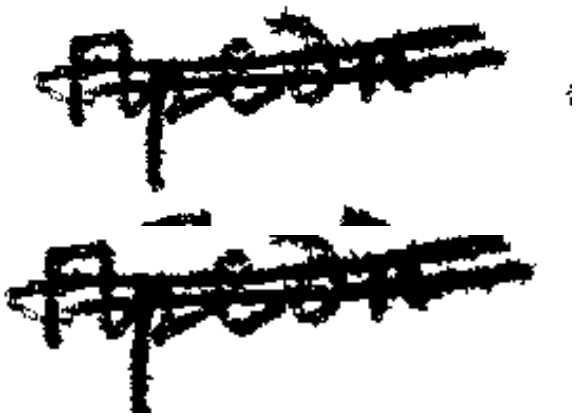


**Diagram and typed description.** $Z$ and $Y$ above $X$; two slanting arrows toward $X$, $f$ on the left, $\varphi$ on the right.

Original source details: GELFAND-P000079

$T$: take $Z \times Y$ and in it those pairs $(z, y)$ such that $f(z) = \varphi(y)$. They form $T$ and there are maps. If $f$ and $\varphi$ are to a point, this is the ordinary direct prod. If $f$ and $\varphi$ are embeddings in $X$, then $Z \times_X Y = Z \cap Y$ in $X$. [Crossed out: "Pres..." [Предп...] / a short entry beneath dense horizontal strokes; the ending is undetermined.]



**Uncertain reading.** "Pres..." [Предп...] / a short entry beneath dense horizontal strokes; the ending is undetermined

Original source details: GELFAND-P000079

Open sets — a category. A presheaf $S$ — a contravar. functor from this categ. to the categ. of ab. groups.

$$U \to X$$
$$\Gamma(U)$$

A substantive example — the topology of a point (in fact of a field).

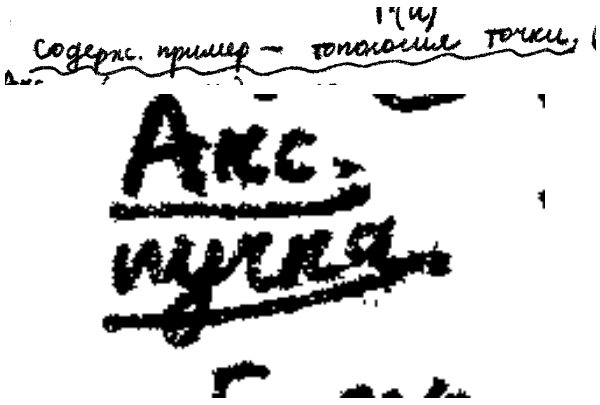


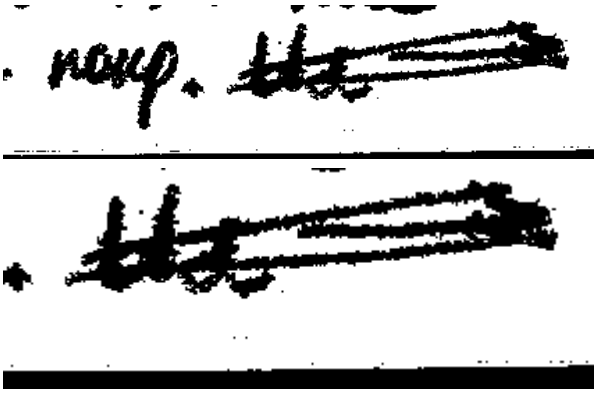
**Diagram and typed description.** A wavy underline beneath "A substantive example — the topology of a point"; "Ax." and "of a sheaf." are underlined separately.

Original source details: GELFAND-P000079

Ax. of a sheaf. $\{U_\alpha \to U\}$ a covering. There is $\Gamma(U) \to \Gamma(U_\alpha)$; hence there is $\Gamma(U) \to \prod_\alpha \Gamma(U_\alpha)$.

$$0 \to \Gamma(U) \to \prod_\alpha \Gamma(U_\alpha).$$

This is an embedding. $U_\beta$ — this too is a covering. [Crossed out: $U$ with overwritten lower signs: $\alpha$ / $\beta$ / illegible; long horizontal strokes converge on the right.]



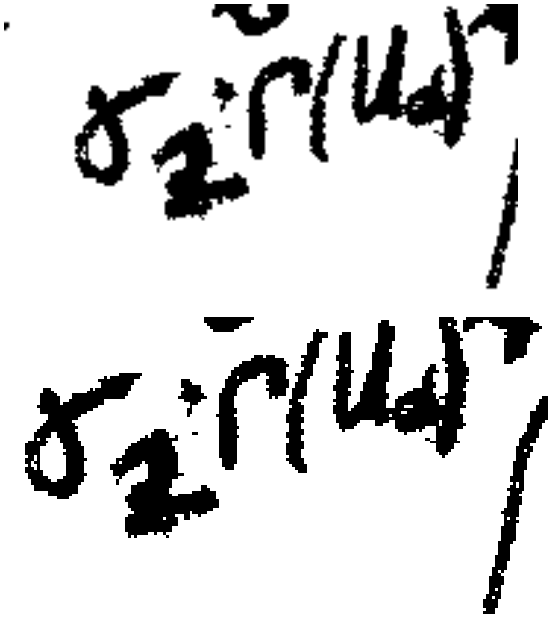
**Uncertain reading.** $U$ with overwritten lower signs: $\alpha$ / $\beta$ / illegible; long horizontal strokes converge on the right

Original source details: GELFAND-P000079

$U_\alpha \times_U U_\beta \to U_\alpha$.

GELFAND-P000080

There is $\Gamma(U_\alpha) \to \Gamma(U_\alpha \times_U U_\beta)$ and into $\prod_\beta \Gamma(U_\alpha \times_U U_\beta)$. [Below the top line, to the left of the product:] $\delta_2\Gamma(U_\alpha)$.

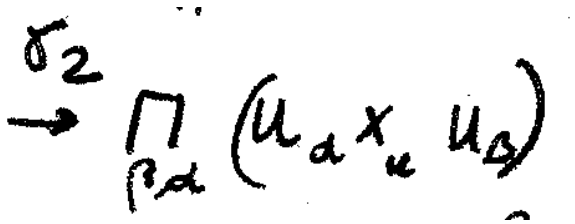


**Uncertain reading.** The lower digit at $\delta$: 2 / 1; a dense lower stroke.

Original source details: GELFAND-P000080

$$0 \to \Gamma(U) \to \prod_\alpha \Gamma(U_\alpha) \xrightarrow{\delta_1 - \delta_2} \prod_{\alpha,\beta} \Gamma(U_\alpha \times_U U_\beta).$$

$$\prod_\beta \Gamma(U_\beta) \xrightarrow{\delta_2} \prod_{\beta\alpha} (U_\alpha \times_U U_\beta).$$



**Uncertain reading.** On the right, a parenthesis immediately after $\prod_{\beta\alpha}$, without a visible $\Gamma$.

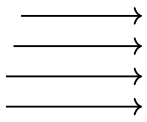
Original source details: GELFAND-P000080

This seq. must be exact at the 2nd factor. (If the sect. over the pieces is $0$, it is obtained from the $0$th sect. over all of $U$.) Those agreeing on intersections can be glued.

Continuation of the seq.

$$0 \to \Gamma(U) \to \prod_{\alpha} \Gamma(U_\alpha) \overset{\delta_1}{\underset{\delta_2}{\rightrightarrows}} \prod_{\alpha,\beta} \Gamma(U_\alpha \times U_\beta) \overset{\longrightarrow}{\underset{\longrightarrow}{\rightrightarrows}} \prod_{\alpha,\beta,\gamma} \Gamma(U_\alpha \times U_\beta \times U_\gamma).$$

[On the right:] 3 maps by pairs.



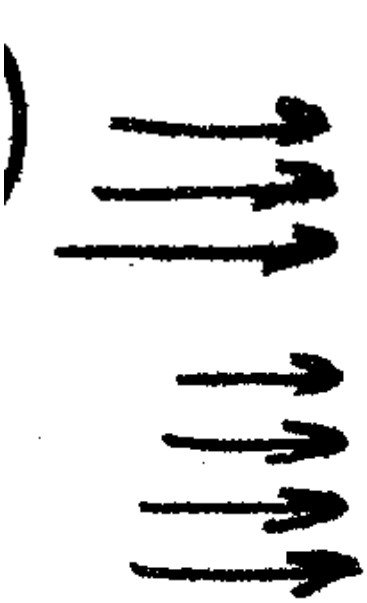


**Diagram and typed description.** Three parallel rightward arrows between the products and, below, four separate parallel rightward arrows.

Original source details: GELFAND-P000080

$U_\alpha \times U_\beta \times U_\gamma \to U_\beta \times U_\gamma$. Assoc. from uniqueness of the prod.

$$\prod_{\beta,\gamma} \Gamma(U_\beta \times U_\gamma) \to \prod_{\alpha} \Gamma(U_\alpha \times U_\beta \times U_\gamma).$$

Next 4 maps.

$$0 \to \Gamma(U) \to \prod \Gamma(U_\alpha) \overset{\delta_0-\delta_1}{\longrightarrow} \prod_{\alpha,\beta} \Gamma(U_\alpha \times U_\beta) \overset{\delta_0-\delta_1+\delta_2}{\longrightarrow} \prod \cdots \overset{\delta_0-\delta_1+\delta_2-\delta_3}{\longrightarrow} .$$

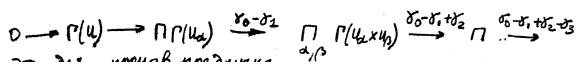
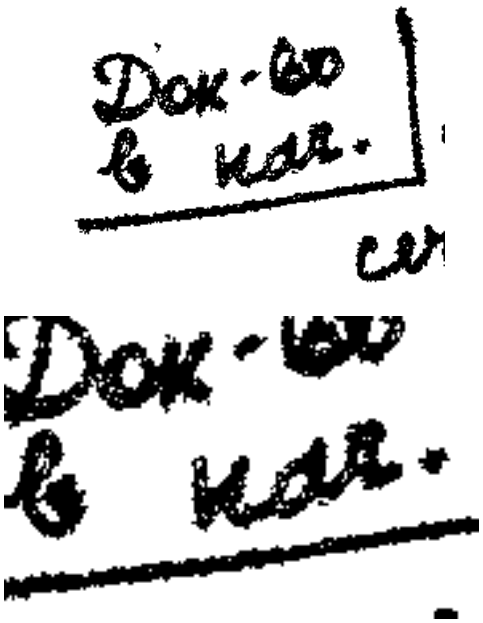


**Diagram and typed description.** Upper arrow labels: $\delta_0 - \delta_1$, $\delta_0 - \delta_1 + \delta_2$, $\delta_0 - \delta_1 + \delta_2 - \delta_3$; after the last product, an ellipsis.

Original source details: GELFAND-P000080

This is for an arbitrary presheaf. The compos. of 2 adjacent maps $= 0$. (Easy to check.) [On the right:] ker / Im.

Proof
in cat.



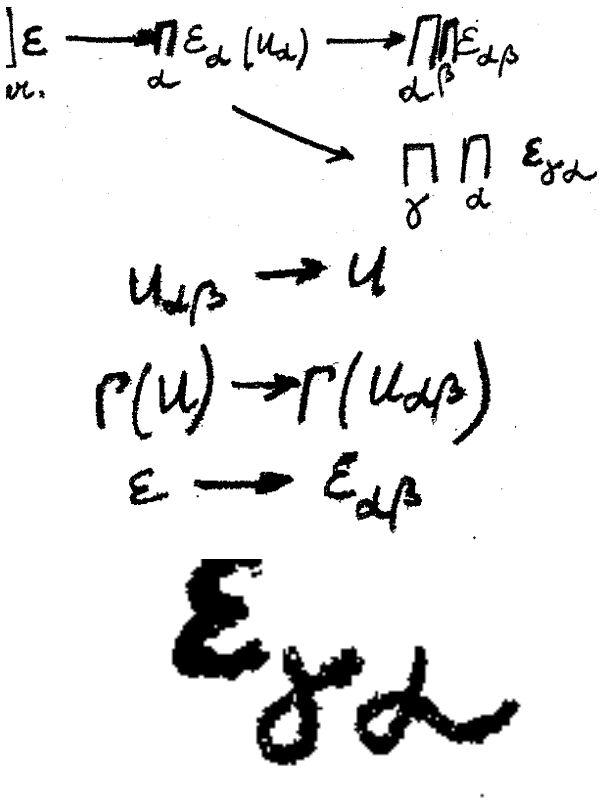
**Uncertain reading.** On the left, “Proof in cat.” / “Proof in id.”; the last letters are undetermined, with an angular stroke to the right and below.

Original source details: GELFAND-P000080

$$\varepsilon \longrightarrow \prod_{\alpha} \varepsilon_\alpha(U_\alpha) \longrightarrow \prod_{\alpha}\prod_{\beta} \varepsilon_{\alpha\beta}$$

sect.

$$\prod_{\gamma}\prod_{\alpha} \varepsilon_{\gamma\alpha}$$



**Uncertain reading.** In the lower term $\prod_\gamma \prod_\alpha \varepsilon_{\gamma\alpha}$: the order and last letter of the indices are uncertain ($\gamma\alpha$ / $\gamma\delta$); a slanting arrow downward to the right. Below the first $\varepsilon$, “sect.”.

Original source details: GELFAND-P000080

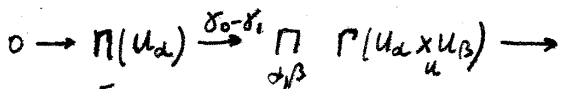
[In the right margin:]

$$U_{\alpha\beta} \to U,$$
$$\Gamma(U) \to \Gamma(U_{\alpha\beta}),$$
$$\varepsilon \to \varepsilon_{\alpha\beta}.$$

Def. the same. ker / Im. Čech cohom. covering.

GELFAND-P000081

$$0 \longrightarrow \prod(U_\alpha) \overset{\delta_0-\delta_1}{\longrightarrow} \prod_{\alpha\beta} \Gamma(U_\alpha \times_U U_\beta) \longrightarrow$$



**Uncertain reading.** Before $(U_\alpha)$, $\prod$ is visible; a separate $\Gamma$ is not established.

Original source details: GELFAND-P000081

In this seq. sect.



**Uncertain reading.** The last abbreviation: “сеч.” / “cer.” (“sect.” / “seg.”); upper joined strokes.

Original source details: GELFAND-P000081

$$H_0(S) = \Gamma(U)$$

Čech cohom. of the whole space. Embedded $\{U_\gamma \to U\}$ in $\{U_\alpha \to U\}$, if $\forall\gamma$ a map is given.

$$\begin{array}{ccc} U_\gamma & \longrightarrow & U_\alpha \\ & \searrow \quad \swarrow & \\ & U & \end{array}$$

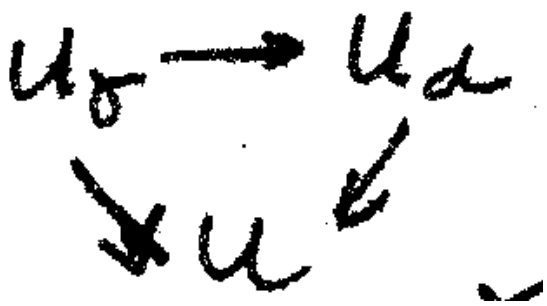


**Diagram and typed description.** Above, $U_\gamma \to U_\alpha$; both slanting arrows point to the lower $U$.

Original source details: GELFAND-P000081

Obtain a map of the chain / of chains (“цепочки” / “цепочек”), commut. with $\underset{\text{(functor!)}}{\delta}$ and therefore a map of cohomology.

**Uncertain reading.** of the chain / of chains (“цепочки” / “цепочек”)



Original source details: GELFAND-P000081



**Diagram and typed description.** Below $\delta$, “(functor!)” is inserted.

Original source details: GELFAND-P000081

This map does not depend on the manner of embedding. (Easy to understand.) Ind. limit $\hat{H}(U, P)$. [On the right:] $\underline{P}$ — a sheaf.

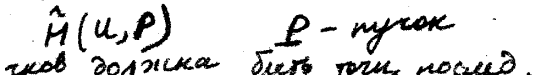
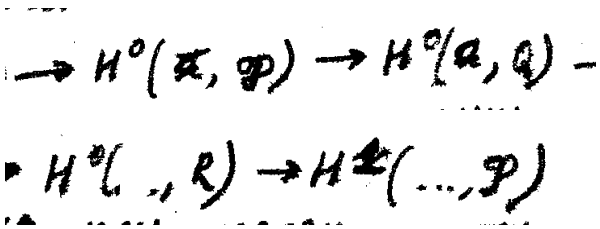


**Diagram and typed description.** The separately written $P$ on the right is underlined; the letter in $\hat{H}(U, P)$ has no underline.

Original source details: GELFAND-P000081

For an exact seq. of sheaves there must be an exact seq. of cohom.

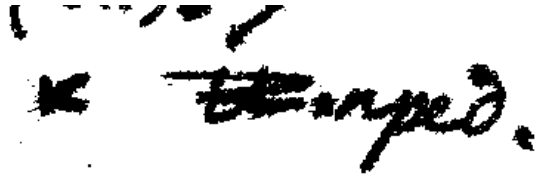
$$0 \to H^0(x, \Phi) \to H^0(a, Q) \to H^0(\dots, R) \to H^1(\dots, \Phi)$$



**Uncertain reading.** The first argument $x$ / $X$; the next resembles $a$; the sheaf letter $\Phi$ / $P$. The upper digit of the last $H$: 1, with dense overlaid strokes.

Original source details: GELFAND-P000081

For coverings they exist, but on passing to [crossed out before “limit”: illegible]
the limit, exactness is not preserved.



**Uncertain reading.** illegible

Original source details: GELFAND-P000081

1. Paracomp. spaces in top. it is done. New def. (ax.) now only for sheaves. Exact seq. of presheaves

$$0 \to P \to Q \to R \to 0$$

[Before “If”, a mark is crossed out: illegible.]



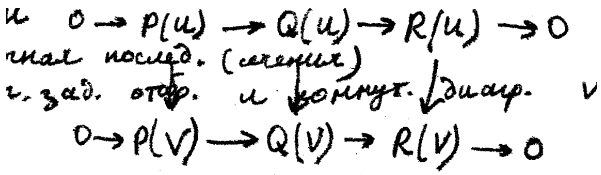
**Uncertain reading.** illegible

Original source details: GELFAND-P000081

If $\forall U$

$$\begin{array}{ccccccccc} 0 & \to & P(U) & \to & Q(U) & \to & R(U) & \to & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \to & P(V) & \to & Q(V) & \to & R(V) & \to & 0 \end{array}$$

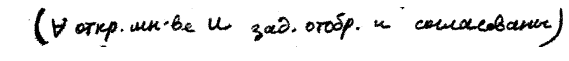
exact seq. (sections).
($\forall$ sect. given maps and commut. diagr. $V \to U$



**Uncertain reading.** Two rows with $P(U), Q(U), R(U)$ and $P(V), Q(V), R(V)$; three downward arrows cross the intervening handwritten annotation. The abbreviation before “given”: “сеч.” / “сек.” (“sect.”).

Original source details: GELFAND-P000081

($\forall$ open set $U$ given maps and gluing.)



**Undeciphered passage.** After “gluing.”, one closing parenthesis is visible; the parenthesis of the preceding annotation is not closed separately.

Original source details: GELFAND-P000081

Exact seq. of sheaves.

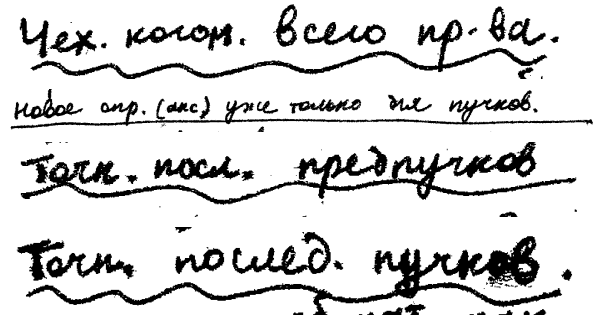


**Diagram and typed description.** Wavy underlines at “Čech cohom. of the whole space.”, “Exact seq. of presheaves” and “Exact seq. of sheaves.”; a straight line at “New def. (ax.) now only for sheaves.”.

Original source details: GELFAND-P000081

$S$ — an ab. category.

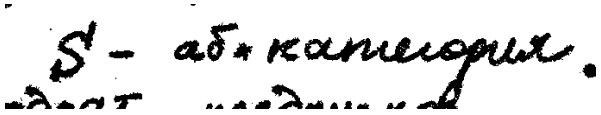


**Diagram and typed description.** Before “ab. category”, the letter $S$ with upper and lower terminal strokes.

Original source details: GELFAND-P000081

Sheaves — an ab. cat. as a subcat. of presheaves. Ker and Coker are defined. Ax.: isom. of the image and coimage.

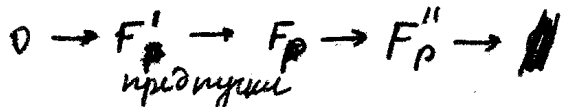
GELFAND-P000082

$$0 \to F' \to F \to F'' \to 0$$

Exact seq. (as sheaves).

$$0 \to F'_p \to F_p \to F''_p \to [\text{crossed out: } 0]$$

[Below $F'_p$: presheaves.]



**Uncertain reading.** Lower indices $p$ / $\rho$; the last 0 is heavily crossed out.

Original source details: GELFAND-P000082

Not necess. 0(!) — the categ. is wider!. In another category.

$$0 \to iF' \to iF \to iF'' \longrightarrow$$

The def. of cohomology is based on this being [crossed out: illegible]

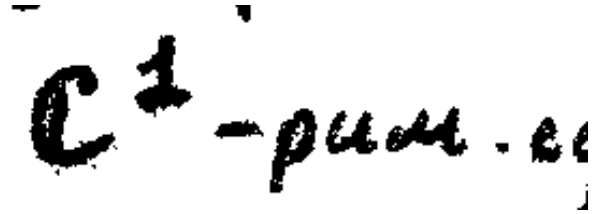
a nonexact seq.

**Uncertain reading.** illegible



Original source details: GELFAND-P000082

(In the difference between sheaves and presheaves — the whole theor. of cohom.) $C^1$ — the Riem. sph. and 2 points.



**Uncertain reading.** $C^1$ / $P^1$; the initial sign is C-like.

Original source details: GELFAND-P000082

Exact seq. of kogf. — exact seq. of sheaves.



Original source details: GELFAND-P000082

**Uncertain reading.** “когф.” / “коэф.” / “когр.” (kogf. / koef., coeff. / kogr.); the middle letter has dense strokes.

Sheaf of germs [an arrow is crossed out: arrow / illegible]
of hol. fns. $= 0$ at the 2 points. $\mathcal{J}$.

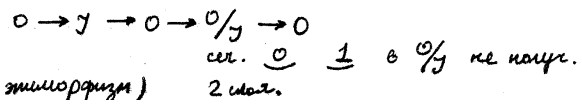


Original source details: GELFAND-P000082

**Uncertain reading.** arrow / illegible

$$0 \to \mathcal{J} \to O \to O/\mathcal{J} \to 0$$

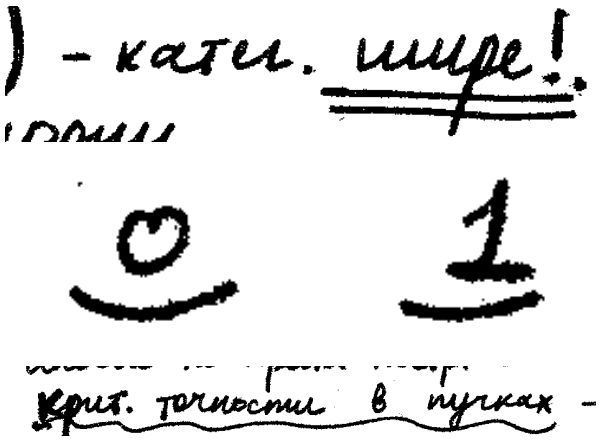
Sect. $\underset{\smile}{0}$, $\underset{\smile}{1}$ in $O/\mathcal{J}$ are not obtained.



Original source details: GELFAND-P000082

**Uncertain reading.** Before $0, 1$, the abbreviation “сек.” / “сеч.” (“sect.”); both numbers are underlined with arcs.

2 stalks. (Sect. is not an epimorphism.) Locally this map is onto. Sometimes the 0th sheaf is constr. from a presh. Criterion of exactness in sheaves — if locally exact.



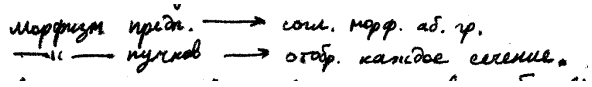
Original source details: GELFAND-P000082

**Diagram and typed description.** At “wider!”, a double straight line; below $0$ and $1$, arcs; at “Criterion of exactness in sheaves”, a wavy line.

Morphism ...

$$\text{exact seq. of sheaves} \iff \text{exact seq. of “stalks”}.$$

Morphism of presh. $\longrightarrow$ a fam. of morph. of ab. gr. ___"___ of sheaves $\longrightarrow$ maps of each section.

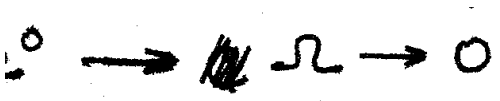


Original source details: GELFAND-P000082

**Diagram and typed description.** At the start of the second line, a repetition sign: two short vertical strokes between horizontal strokes.

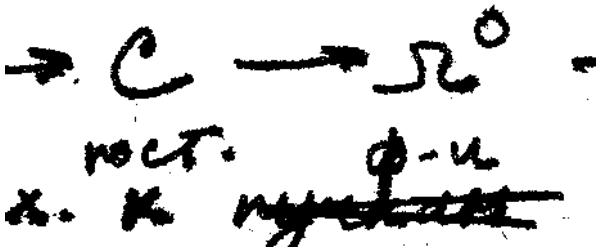
The addit. cat. of presheaves and sheaves are abelian. If there is an exact seq. of sheaves, then the seq. is exact locally. (Also in top. spaces.) (Here stalks cannot be def. in principle.) Presh. exact on fine coverings. Presh. compared on all sets, and sheaves on fine ones.

$$0 \to \underset{\text{const.}}{C} \to \underset{\text{fns.}}{\Omega^0} \to [\text{a sign is crossed out}]\Omega \to 0$$



Original source details: GELFAND-P000082

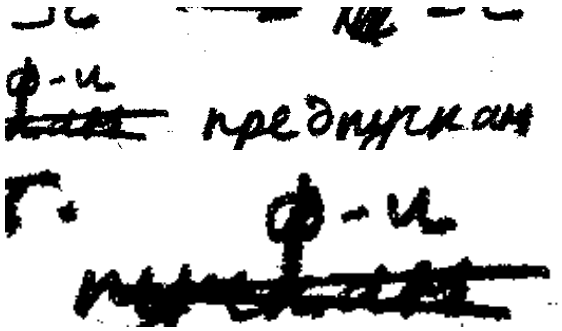
**Uncertain reading.** The sign before $\Omega$ is heavily crossed out; illegible.



Original source details: GELFAND-P000082

**Diagram and typed description.** Below $C$, “const.”; below $\Omega^0$, “fns.”.

Pass to [crossed out: sheaves / illegible] presheaves.

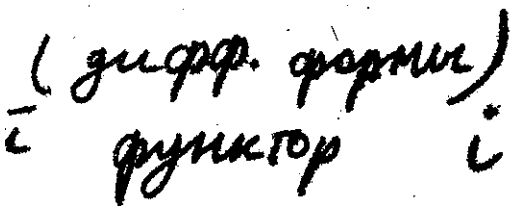


Original source details: GELFAND-P000082

**Uncertain reading.** sheaves / illegible

$$0 \to C \to f$$

(diff. forms!)



Original source details: GELFAND-P000082

**Uncertain reading.** On the right, the annotation “(diff. forms!)” / “(diff. forms)”; a dot near the lower end of the parenthesis. Below, separately, “functor $i$”.



Original source details: GELFAND-P000082

**Uncertain reading.** $f$ / $\mathcal{F}$.

[A short fragment is crossed out: illegible.]



**Uncertain reading.** illegible

Original source details: GELFAND-P000082

Cohom. $\longrightarrow$ derived functor $i$.

GELFAND-P000083

$$0 \to F' \to F \to F'' \to 0$$

Req. that there be an exact seq. [On the right:] $H$ — presheaves. 1.

$$0 \to iF' \to iF \to iF'' \to H^1(F') \to H^1(F) \\ \to H^1(F'') \to H^2(F') \longrightarrow \dots$$

(Ax. of derived functors.)

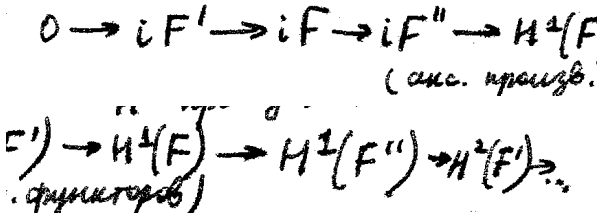


**Diagram and typed description.** After $iF''$, the terms $H^1(F')$, $H^1(F)$, $H^1(F'')$, $H^2(F')$; then an arrow and an ellipsis.

Original source details: GELFAND-P000083

$$i : S \longrightarrow \underline{P}$$

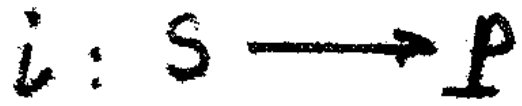


**Diagram and typed description.** In $i : S \longrightarrow P$, a straight line below the final $P$.

Original source details: GELFAND-P000083

[Crossed out: illegible]
a functor (left exact).

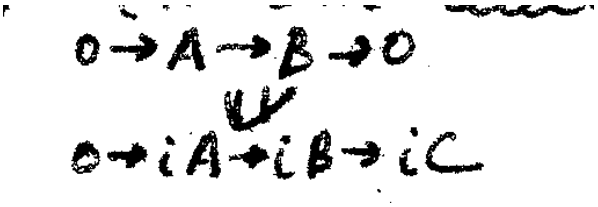


**Uncertain reading.** illegible

Original source details: GELFAND-P000083

[On the right:]

$$0 \to A \to B \to 0$$
$$\Downarrow$$
$$0 \to iA \to iB \to iC$$

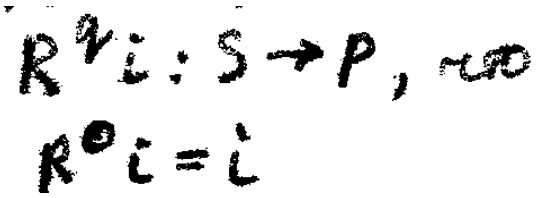


Original source details: GELFAND-P000083

**Uncertain reading.** In the upper row, 0 is visible after $B$; in the lower row the last term is $iC$, without a corresponding $C$ above.

Derived functor

$$R^q i : S \to P, \quad \text{but}$$
$$R^0 i = i$$



Original source details: GELFAND-P000083

**Uncertain reading.** The upper index in $R^q i$: $q$ / $\nu$, with an upper loop and a slanting lower stroke; in the next row, $R^0 i = i$.

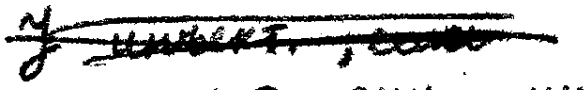
1:

$$0 \to R^0(A) \longrightarrow \dots$$

Exact seq. 2. For inject. objects $S$

$$R^q(S) = 0, \qquad q > 0.$$

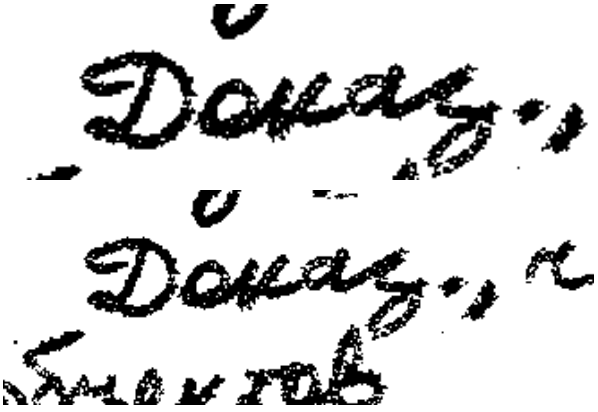
[A row is crossed out: “For inject.” / illegible.]



Original source details: GELFAND-P000083

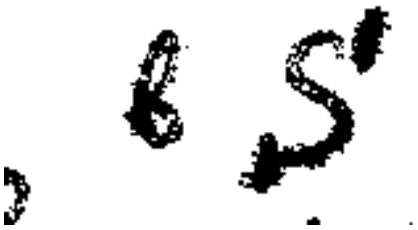
**Uncertain reading.** “For inject.” / illegible

Prov., that they exist if in $S'$ there are enough inject. objects (an embedding into an inj. obj. exists for any).



Original source details: GELFAND-P000083

**Diagram and typed description.** At the start of the row, the abbreviation “Доказ.” (“Prov.”); a comma after the full stop.



Original source details: GELFAND-P000083

**Diagram and typed description.** In the words “if in $S'$ there are enough”, a separate upper stroke after $S$.

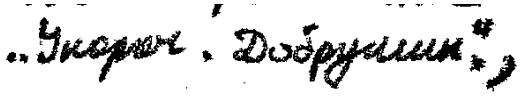
See Grothendieck, “Lectures on hom. algebra”. Val.: on the whole space, of the derived functor, is called cohom. of the whole space. Next time. ~~Berezin~~



Original source details: GELFAND-P000083

**Diagram and typed description.** “Next time.” is underlined with a straight line; the following “Berezin” is crossed out.

“Another form. of Plancherel for diff. oper.” Berezin and Sinai “Ukoren. Dobrushin” / “Igor. Dobrushin” (“Укорен. Добрушин” / “Игорь. Добрушин”), possibly, Halmos.

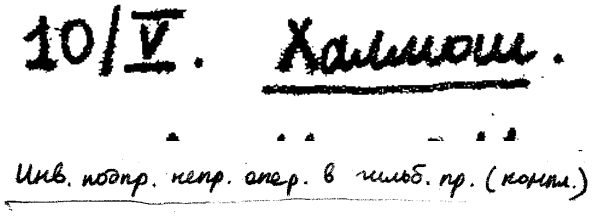


Original source details: GELFAND-P000083

**Uncertain reading.** “Ukoren. Dobrushin” / “Igor. Dobrushin” (“Укорен. Добрушин” / “Игорь. Добрушин”)

GELFAND-P000084

$10/\overline{\text{V}}$. Halmos. Inv. subsp. of a cont. oper. in Hilb. sp. (compl.)

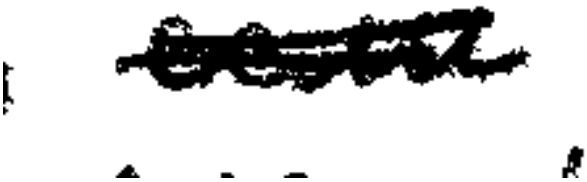


Original source details: GELFAND-P000084

**Diagram and typed description.** The $V$ in the date has an upper and a lower bar; “Halmos.” and the heading are underlined.

$$A : H \to H$$

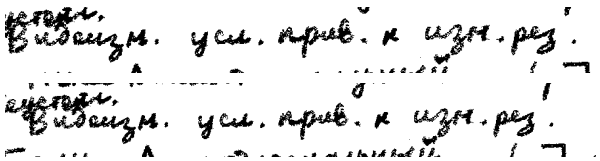
Inv. sp. — closed and inv. $M$. $\exists M$ — ? ($M \neq 0, H$) The gen. probl. is not solved, [crossed out: illegible]



Original source details: GELFAND-P000084

**Uncertain reading.** illegible

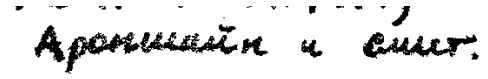
(known if $A$ is Herm. or norm.). Several class. — for completely cont. (von Neumann, ...). [In the margin: counterex.] In finite / In compact (“В конечн.” / “В компактн.”) cond. reduce to known res.



Original source details: GELFAND-P000084

**Uncertain reading.** In finite / In compact (“В конечн.” / “В компактн.”)

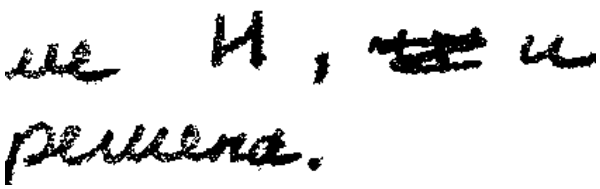
Aronshain [?] and Smith.



Original source details: GELFAND-P000084

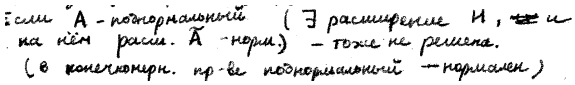
**Uncertain reading.** Aronshain (“Ароншайн”); interior strokes of the name are uncertain.

If $A$ is subnormal (∃ an extension of $H$, crossed out: illegible; and on it an ext. $\tilde{A}$ is norm.), — also not solved.



Original source details: GELFAND-P000084

**Uncertain reading.** illegible



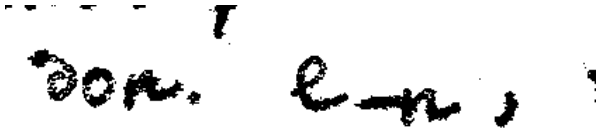
Original source details: GELFAND-P000084

**Diagram and typed description.** The definition is enclosed in round parentheses; a wavy stroke above $A$ in “ext. $\tilde{A}$”.

(In a finite-dim. sp., subnormal $\longrightarrow$ normal.) In infinite dim. — the shift

$$e_1 \to e_2,\ e_2 \to e_3, \dots$$

If add / prove (“доп.” / “док.”) $e_{-n}$, then it will be normal.



Original source details: GELFAND-P000084

**Uncertain reading.** add / prove (“доп.” / “док.”)

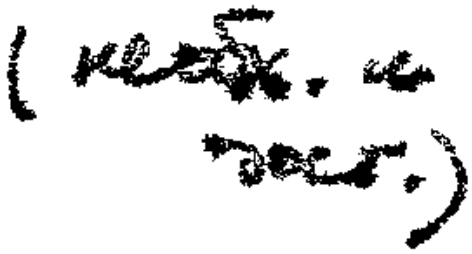
Internal conditions of subnorm.

$$A^*A \longrightarrow AA^* \geq 0$$



Original source details: GELFAND-P000084

**Uncertain reading.** Between $A^*A$ and $AA^*$: $\longrightarrow$ / $-$; the right end of the stroke is pointed.



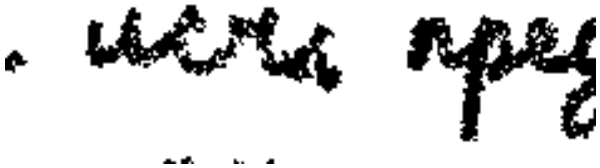
Original source details: GELFAND-P000084

**Uncertain reading.** (necess. and suff.); the words have overwritten strokes.

(The adj. of a subnorm. need not be subnorm.) See Halmos, Bram. Generalizations. $A^2$ completely cont. — positive ans. (∃ an inv. subsp.) A. Robinson, Bernstein in terms of pred. calc. / inn. pred. / inf. pred. (“исч. предик.” / “инн. предик.” / “инф. предик.”)



**Uncertain reading.** pred. calc. / inn. pred. / inf. pred. (“исч. предик.” / “инн. предик.” / “инф. предик.”)



Original source details: GELFAND-P000084

Proof of the schem. method.

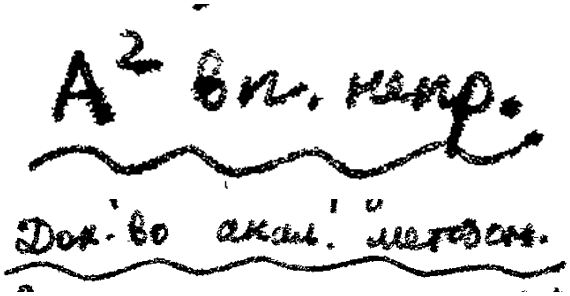
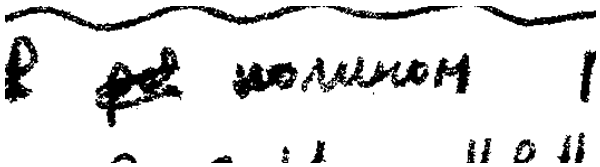


**Diagram and typed description.** Wavy lines below “$A^2$ completely cont.” and “Proof of the schem. method.”

Original source details: GELFAND-P000084

$p$ [A sign is crossed out: ∀ / illegible]

a polynomial $p \neq 0$, $p(A)$ completely cont. Then ∃ an inv. subsp.



**Uncertain reading.** ∀ / illegible

Original source details: GELFAND-P000084

$e \in H$, $\|e\| = 1$.

Rasst. $\{e, Ae, A^2e, \ldots\} = H$



**Uncertain reading.** The abbreviation before the brace: “Расст.” / “Рассм.” (“Rasst.” / “Rassm.”); the final joined strokes are uncertain.

Original source details: GELFAND-P000084

(Otherwise there is a subsp.) ∴ therefore.

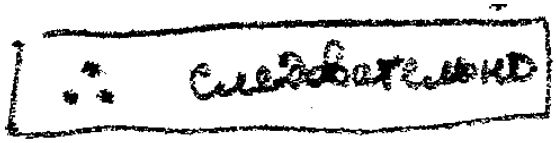
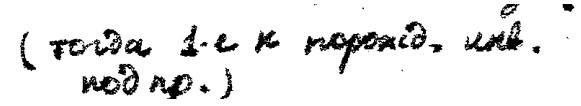


**Diagram and typed description.** Three dots in a triangle and the word “therefore” are enclosed in a box.

Original source details: GELFAND-P000084

$e, Ae, \ldots$ may be considered lin. indep. Otherwise there is a finite-dim. subsp. (then the 1st to the generated / the 1st to each (“1-е к порожд.” / “1-е к каждой”) inv. subsp.)



**Uncertain reading.** the 1st to the generated / the 1st to each (“1-е к порожд.” / “1-е к каждой”)

Original source details: GELFAND-P000084

Orthogonalize

$$\{e, Ae, \ldots, A^{n-1}e\} = \{e_1, e_2, \ldots, e_n\}$$

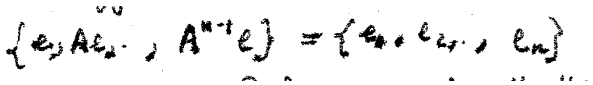


**Diagram and typed description.** The last exponent is $n-1$; on the right, the indices $1$, $2$, $n$.

Original source details: GELFAND-P000084

$e_1, e_2, \ldots$ — $\perp$, $\| \| = 1$. The matrix has the form

$$a_{mn} = (Ae_n, e_m)$$

$$\begin{pmatrix} * & * & * \\ * & * & * \\ 0 & * & * \\ 0 & 0 & * \\ 0 & 0 & 0 \end{pmatrix}$$

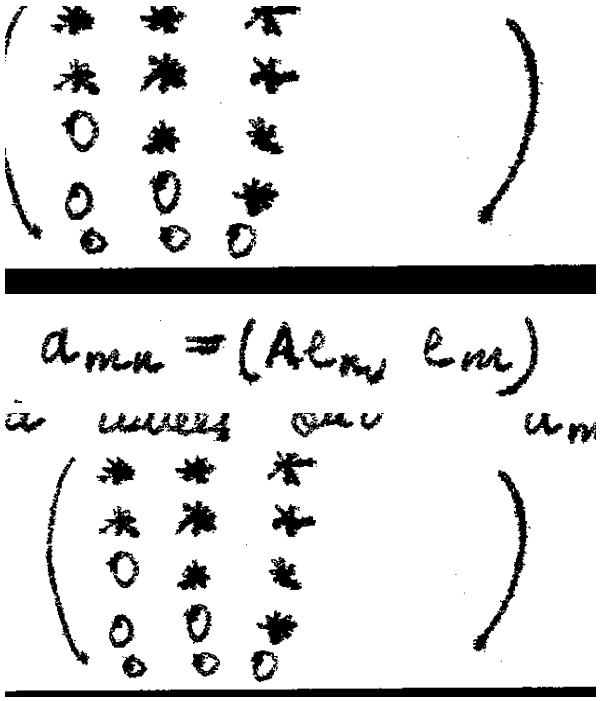


**Diagram and typed description.** The three lower zeros continue below the ends of the round parentheses; above, $a_{mn} = (Ae_n, e_m)$.

Original source details: GELFAND-P000084

GELFAND-P000085

If 1 beyond the diag., then 0. Image of $e$:

$$Ae = \lambda_1 e_1 + \lambda_2 e_2$$

and so on.

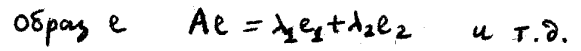


**Diagram and typed description.** In “Image of $e$” and $Ae$, no lower index; on the right, $\lambda_1 e_1 + \lambda_2 e_2$.

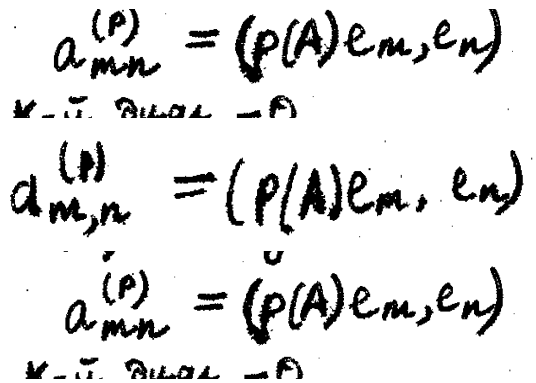
Original source details: GELFAND-P000085

$A^k$ has the matrix $a^{(k)}_{mn}$ (add $k$ diagonals below). (Farther than $k$ units from the main diag., 0.) On the $k$th diag.

$$a^{(k)}_{n+k,n} = \prod_{j=1}^{k} a_{n+j,n+j-1}$$

Matrix $p(A)$, deg $k$.

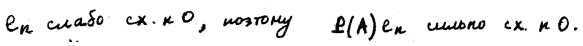
$$a^{(p)}_{mn} = (p(A)e_m, e_n)$$



**Uncertain reading.** For $p(A)$, the notation $(p(A)e_m, e_n)$; on P000084 it was $(Ae_n, e_m)$.

Original source details: GELFAND-P000085

Below the $k$th diag. — 0. On the $k$th diag. — the same as for the $k$th power of $A$ (up to a factor). $e_n$ converges weakly to 0, therefore $p(A)e_n$ converges strongly to 0.



Original source details: GELFAND-P000085

**Diagram and typed description.** First "converges weakly to $0$", then "converges strongly to $0$".

$$\prod_{j=1}^{k} d_{n+j,n+j-1} \longrightarrow 0, \qquad n \to \infty.$$

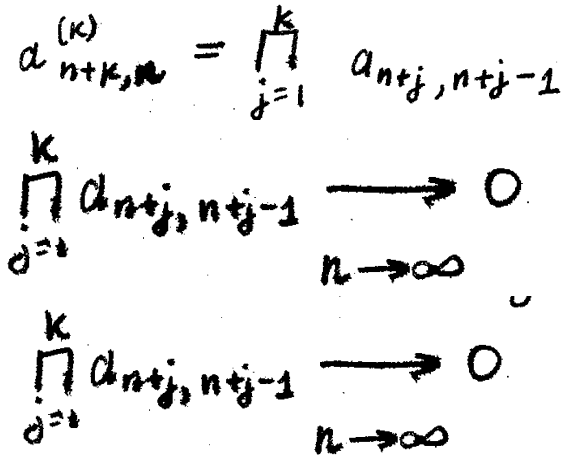


**Diagram and typed description.** In the first product, the letter $a$; in the product with an arrow to $0$, the letter $d$. In both, the limits $j = 1, k$ and the indices $n + j$, $n + j - 1$.

Original source details: GELFAND-P000085

Product of numbers below the diagonal. Change both indices.

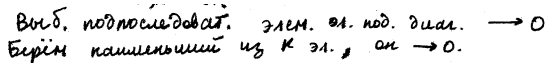
$$d^{(p)}_{m,n} = (p(A)e_m, e_n)$$



**Uncertain reading.** The upper sign on $d$: $(p)$ / $(1)$; a short joined stroke inside the parentheses.

Original source details: GELFAND-P000085

Choose a subseq. of elem. from / elem. (“от.” / “эл.”) below the diag. $\to 0$.




**Uncertain reading.** from / elem. (“от.” / “эл.”)

Original source details: GELFAND-P000085

Take the smallest of the $k$ elem., it $\to 0$.

$$a_{k(n)+1,k(n)} \longrightarrow 0$$

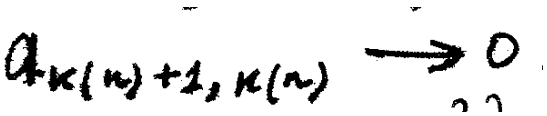



**Diagram and typed description.** In the row with an arrow to $0$: $a_{k(n)+1,k(n)}$.

Original source details: GELFAND-P000085

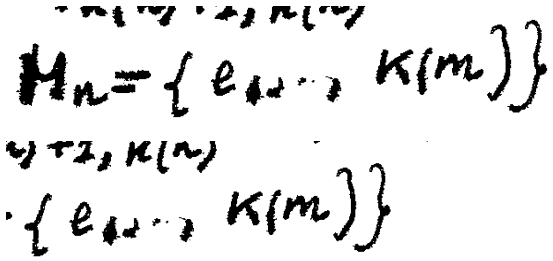
Consider. [A symbol is crossed out: illegible.]

**Uncertain reading.** illegible



Original source details: GELFAND-P000085

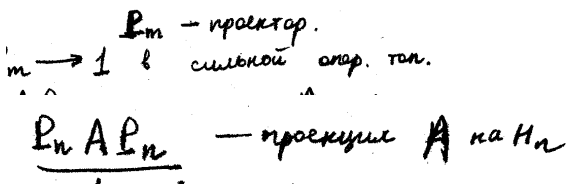
$$H_n = \{e_1, \dots, k(m)\}$$




**Uncertain reading.** The last term $k(m)$ / $k(n)$, without a confidently visible $e$ before it; the lower index on $H$: $n$ / $m$.

Original source details: GELFAND-P000085

$P_m$ — a projector. $P_m \to 1$ in the strong oper. top. $\underline{P_nAP_n}$ — the projection of $A$ onto $H_n$.



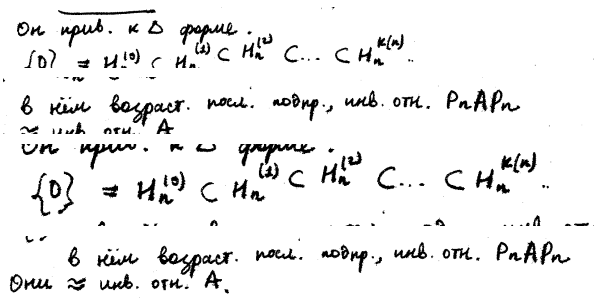

**Uncertain reading.** First the index $m$, then $n$ in $P_nAP_n$ and $H_n$.

Original source details: GELFAND-P000085

It is reduced to $\Delta$ form.

$$\{0\} = H_n^{(0)} \subset H_n^{(1)} \subset H_n^{(2)} \subset \dots \subset H_n^{k(n)}.$$

In it, an increasing seq. of subsp., inv. with respect to $P_nAP_n$. They are $\approx$ inv. with respect to $A$.



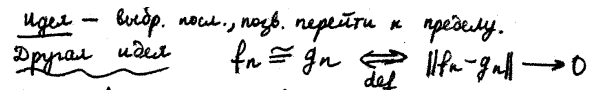
Original source details: GELFAND-P000085

**Diagram and typed description.** The sign $\Delta$ before “form”; in the chain, the upper $(0)$, $(1)$, $(2)$ and $k(n)$; in the last row, $\approx$.

GELFAND-P000086

Idea — choose a seq., allowing passage to the limit. Another idea

$$f_n \cong g_n \quad \underset{\text{def}}{\iff} \quad \|f_n - g_n\| \to 0$$



Original source details: GELFAND-P000086

**Diagram and typed description.** A straight underline below “Idea”, a wavy underline below “Another idea”; def below the two-way arrow.

If $f_n$ is bounded $\Rightarrow$

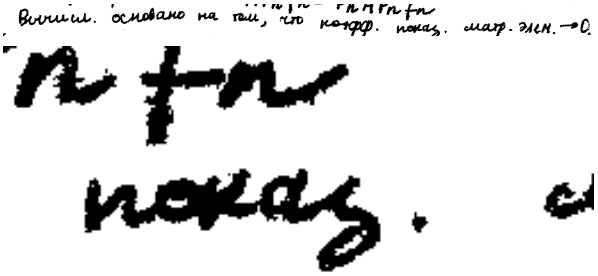
$$AP_nf_n \cong P_nAP_nf_n.$$



Original source details: GELFAND-P000086

**Diagram and typed description.** On the left $AP_nf_n$, on the right $P_nAP_nf_n$.

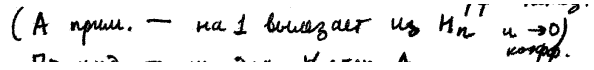
The calc. is based on the fact that the coeff. of the unknown / off-diag. (“неиз.” / “недиаг.”) matrix elem. $\to 0$.



Original source details: GELFAND-P000086

**Uncertain reading.** unknown / off-diag. (“неиз.” / “недиаг.”)

($A$ appl. — goes 1 outside $H_n$ and $\to 0$.) [Added below the end: coeff.]

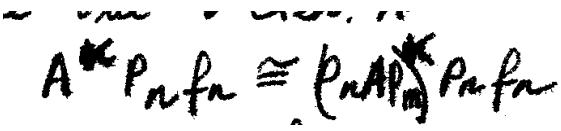


Original source details: GELFAND-P000086

**Diagram and typed description.** In parentheses “$A$ appl. — goes $1$ outside $H_n$ and $\to 0$”; below the end, the addition “coeff.”.

By ind., the same for $\forall$ powers of $A$:

$$A^kP_nf_n \cong (P_nAP_n)^kP_nf_n.$$

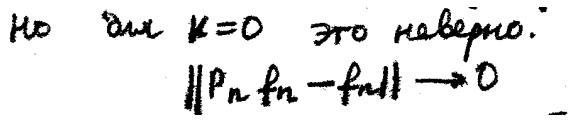


Original source details: GELFAND-P000086

**Uncertain reading.** Inside the parentheses the third $P_n$ is overwritten; the exponent $k$ has dense strokes.

But for $k = 0$ this is false.

$$\|P_nf_n - f_n\| \to 0.$$



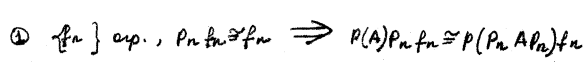

Original source details: GELFAND-P000086

**Diagram and typed description.** In the row "But for $k = 0$ this is false", a visible $0$; below, $\|P_n f_n - f_n\| \to 0$.

But if this req. is imposed on $f_n$, then this will be true for all powers (and for polyn.). 1) $\{f_n\}$ bounded,

$\Rightarrow$

$$p(A)P_n f_n \cong p(P_n A P_n) f_n.$$

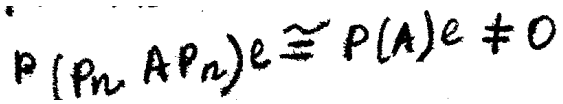



Original source details: GELFAND-P000086

**Uncertain reading.** $P_n f_n \cong f_n$ / $P_n f_n \neq f_n$

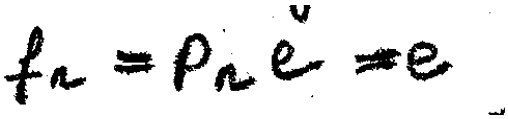
Spec. particular case

$$f_n = P_n e \cong e$$
$$p(P_n A P_n)e \cong p(A)e \neq 0.$$




Original source details: GELFAND-P000086

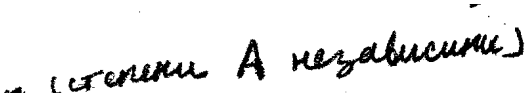
**Diagram and typed description.** An $e$ follows both $p(P_n A P_n)$ and $p(A)$; at the end $\neq 0$.




Original source details: GELFAND-P000086

**Uncertain reading.** The second relation in $f_n = P_n e \cong e$: $\cong$ / $\neq$, with a crossing diagonal stroke.

[On the right: powers of $A$ are independent.]

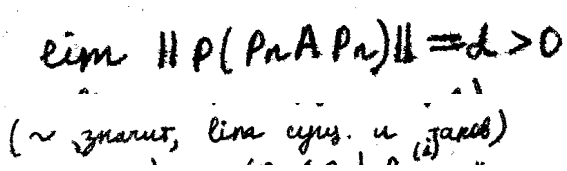



Original source details: GELFAND-P000086

**Uncertain reading.** Overwritten strokes near "powers of $A$"; the lower layer is illegible.

Hence

$$\lim \|p(P_n A P_n)\| = d > 0.$$



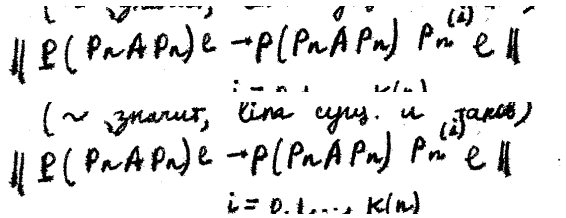

Original source details: GELFAND-P000086

**Uncertain reading.** After the closing parenthesis inside the norm, no confidently visible $e$; in the following parentheses, a single wavy sign $\sim$.

($\sim$ means, the lim exists and is such.)

$$\|p(P_n A P_n)e \longrightarrow p(P_n A P_n)P_n^{(i)} e\|$$

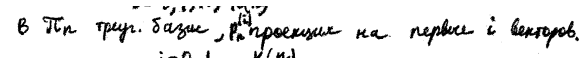
$i = 0, 1, \ldots, k(n)$.




Original source details: GELFAND-P000086

**Uncertain reading.** Between the terms inside the norm, a visible arrow rather than an ordinary minus sign.

In $\Pi_n$, triang. basis, $P_n^{(i)}$ — projection onto the first $i$ vectors.



**Uncertain reading.** $\Pi_n$ / $H_n$.

Original source details: GELFAND-P000086

$i = 0, 1, \ldots, k(n)$. $i = 0 \Rightarrow$ obtain the zero subsp. $i = k(n) \equiv 0$.

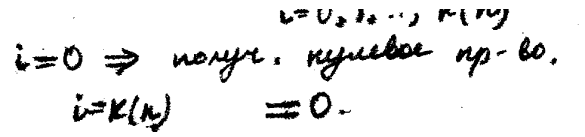


**Diagram and typed description.** The first row $i = 0 \Rightarrow$ "obtain the zero subsp."; the next $i = k(n) \equiv 0$.

Original source details: GELFAND-P000086

$$\exists n: \quad \|p(P_nAP_n)e\| > \frac{3}{4}d.$$



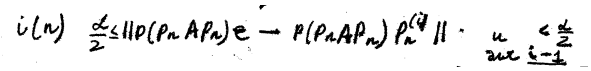
**Diagram and typed description.** Before the norm $\exists n$ :; inside it a final $e$; after the norm $> \frac{3}{4}d$.

Original source details: GELFAND-P000086

$i(n)$:

$$\frac{d}{2} \leq \|p(P_nAP_n)e \longrightarrow p(P_nAP_n)P_n^{(i)}\|$$

and $< \frac{d}{2}$ for $\underline{i-1}$.

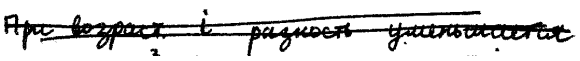


**Uncertain reading.** The parameter $d$ / $\alpha$; in this last norm after $P_n^{(i)}$, no confidently visible $e$.

Original source details: GELFAND-P000086

GELFAND-P000087

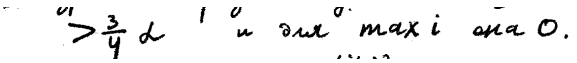
[Crossed out: As $i$ incr., the diff. decreases.]



**Uncertain reading.** Abbreviations under two lines: "возрастан." / "возрастан", "разност." / "разность" ("incr.", "diff." / "difference").

Original source details: GELFAND-P000087

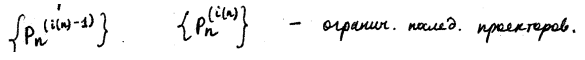
$> \frac{3}{4}d$ and for max $i$ it is $0$.



**Uncertain reading.** The parameter in the upper row: $d$ / $\alpha$.

Original source details: GELFAND-P000087

$$\{P_n^{(i(n)-1)}\}, \qquad \{P_n^{(i(n)}\}$$



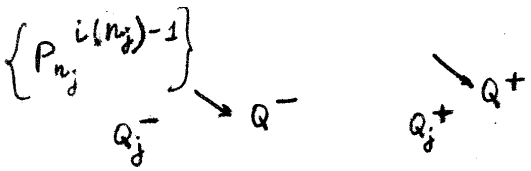
**Uncertain reading.** In the first exponent $(i(n) - 1)$; in the second a visible opening parenthesis before $i(n)$, without a separate second closing parenthesis.

Original source details: GELFAND-P000087

— bounded seq. of projectors. Choose a weakly conv. subseq.

$$\{P_{n_j}^{i(n_j)-1}\} \searrow Q^- \qquad \searrow Q^+$$

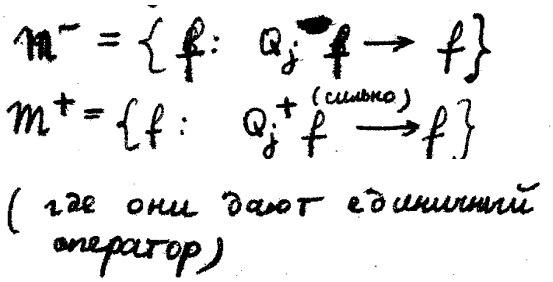
$$Q_j^- \qquad\qquad Q_j^+$$



Original source details: GELFAND-P000087

**Diagram and typed description.** On the left $\{P_{n_j}^{i(n_j)-1}\}$, an inclined arrow to $Q^-$; below the left part $Q_j^-$. On the right an inclined arrow to $Q^+$, below it $Q_j^+$. The upper starting point of the right arrow has no separate label.

$$m^- = \{f : Q_j^- f \to f\}$$
$$m^+ = \{f : Q_j^+ f \overset{\text{(strongly)}}{\longrightarrow} f\}.$$



Original source details: GELFAND-P000087

**Uncertain reading.** The minus above the first $Q_j$ is overwritten; $m$ / $M$ in the left-hand notation.

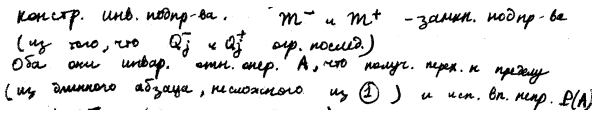
[On the right: where they give the identity operator.] (On $M^-$ them $Q^-$ is $1$.)



Original source details: GELFAND-P000087

**Uncertain reading.** $M^-$ is inserted before "them"; the original phrase is not amended.

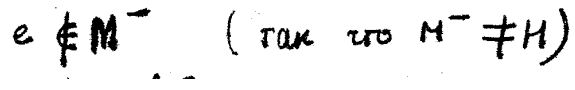
Constr. of inv. subsp. $m^-$ and $m^+$ — closed subsp. (from the fact that $Q_j^-$ and $Q_j^+$ are bounded seq.). Both are inv. with respect to the oper. $A$, which is obtained by passage to the limit (from the long paragraph, uncomplicated from ①) and using comp. $p(A)$.



Original source details: GELFAND-P000087

**Diagram and typed description.** Here lower-case $m^-$, $m^+$; the reference has a circled $1$; at the end $p(A)$.

$$e \notin M^- \qquad (\text{so that } M^- \neq H).$$



Original source details: GELFAND-P000087

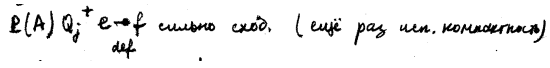
**Diagram and typed description.** Capital $M^-$ in both places; before it $e \notin$, in parentheses $M^- \neq H$.

$M^+ \neq \{0\}$ — this is already less simple.

$$\{Q_j^+ e\} \quad \text{converges weakly.}$$

$$p(A)Q_j^+ e \longrightarrow \underset{\text{def}}{f}$$

Strongly conv. (compactness used once more).



Original source details: GELFAND-P000087

**Diagram and typed description.** Below $f$ the label def; before it an arrow, on the left $p(A)Q_j^+ e$.

$$f \neq 0, \qquad f \in M^+.$$

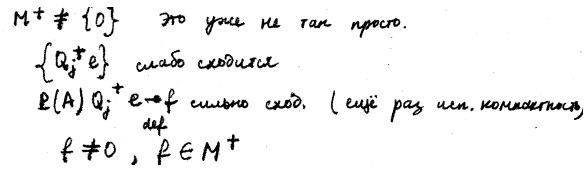
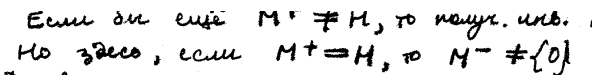



**Diagram and typed description.** The sequence has $Q_j^+ e$ with a $+$; below, $f \neq 0$, $f \in M^+$.

Original source details: GELFAND-P000087

If also $M^+ \neq H$, then we obtain an inv. subsp. But here, if $M^+ = H$, then $M^- / N^- \neq \{0\}$.




**Uncertain reading.** $M^- / N^-$

Original source details: GELFAND-P000087

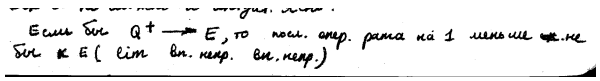
The proof is not difficult and is intuitively clear. If $Q^+ \to E$, then the seq. of oper. of rank $1$ less [A short fragment is crossed out: illegible.]

not would / cl. (“бы” / “бл.”) to $E$ (lim comp. comp.).

**Uncertain reading.** illegible



Original source details: GELFAND-P000087




**Uncertain reading.** would / cl. (“бы” / “бл.”)

Original source details: GELFAND-P000087

GELFAND-P000088

Unsolved problems. (and solved ones)

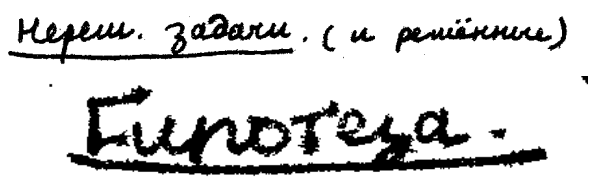



**Diagram and typed description.** “Unsolved problems.” is underlined; the parentheses are outside the underline. Below, “Conjecture.” is separately underlined.

Original source details: GELFAND-P000088

$p$ — entire fn., $p(A)$ — comp., then triv. $\exists M$ (use for polynomials). de Branges — Rovnyak announced $\forall$ cont. oper., but with an error in the proof. From their reasoning follows:

$$\|A\| < 1, \quad 1 - A^*A \text{ comp.} \quad \Rightarrow \quad \exists \text{ inv. subsp.}$$

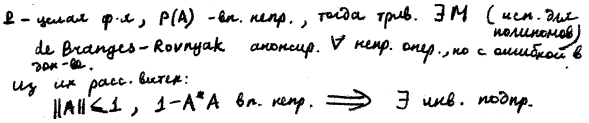
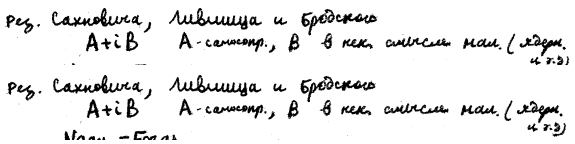



**Diagram and typed description.** The first sign is $<$; then $1 - A^*A$ and a double arrow to $\exists$.

Original source details: GELFAND-P000088

Res. of Sakhnovich, Livshits and brodsky. $A + iB$, $A$ — self-adj., $B$ in some sense small (nuclear, etc.).




Original source details: GELFAND-P000088

**Diagram and typed description.** At the start $A + iB$; “nuclear, etc.” continues on the second line.

Nagy — Foias / Foas

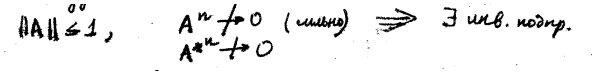



Original source details: GELFAND-P000088

**Uncertain reading.** Foias / Foas

$$\|A\| \leq 1, \qquad \begin{matrix} A^n \not\longrightarrow 0 \\ A^{*n} \not\longrightarrow 0 \end{matrix} \quad \text{(strongly)},$$

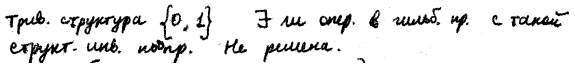
$\Rightarrow \exists$ inv. subsp.




Original source details: GELFAND-P000088

**Uncertain reading.** Both arrows are crossed by an inclined stroke.

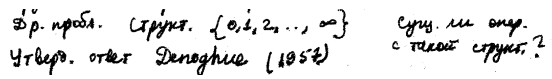
Triv. structure $\{0, 1\}$.




Original source details: GELFAND-P000088

**Diagram and typed description.** Two elements $\{0, 1\}$; then the question about an operator in Hilb. sp. and “Not solved.”

$\exists$ an oper. in Hilb. sp. with such a struct. of inv. subsp.? Not solved. Another probl. Struct. $\{0, 1, 2, \ldots, \infty\}$. Does an oper. with such a struct. exist? Affirm. answer Donoghue (1957).




Original source details: GELFAND-P000088

**Diagram and typed description.** Written $\{0, 1, 2, \ldots, \infty\}$; the date in parentheses (1957).

$$\begin{pmatrix} 0 & \frac{1}{2} & & & \\ & 0 & \frac{1}{4} & & \\ & & \cdot & \frac{1}{8}\,[?] & \\ 0 & & & \cdot & \\ & & & & \cdot \end{pmatrix}$$

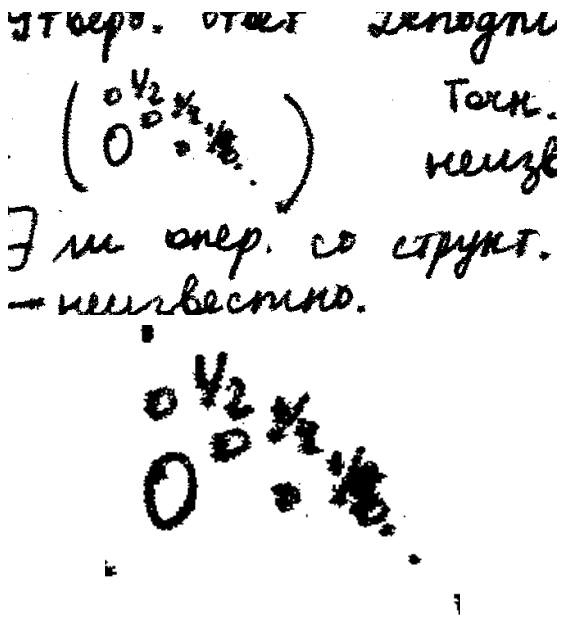


**Diagram and typed description.** Within round parentheses two small $0$s, $1/2$, $1/4$; the third densely overwritten coefficient $1/8$ / $1/6$; a large $0$ below and separate dots down to the right.

Original source details: GELFAND-P000088

Exact cond. on the numbers above the diag. are unknown (so that precisely these occur). $\exists$ an oper. with the struct. of all integers $\to$ unknown.

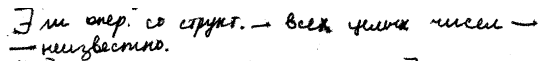


**Diagram and typed description.** “all integers”; then an arrow and “unknown”.

Original source details: GELFAND-P000088

$$[0,1] \longmapsto$$

— such a structure $\exists$ operator.

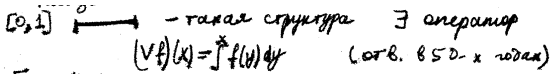


**Diagram and typed description.** On the left the label $[0,1]$, then a segment with two end strokes. Below $(Vf)(x) = \int_0^x f(y)\,dy$.

Original source details: GELFAND-P000088

$$(Vf)(x) = \int_0^x f(y)\,dy$$

(answ. in the 1950s). Conjecture. A measure on the line $\mu$.

$$-\infty \qquad\qquad +\infty$$

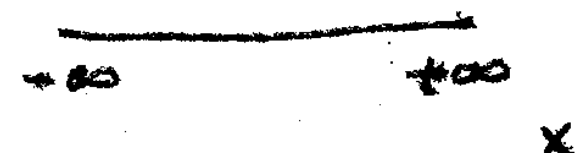
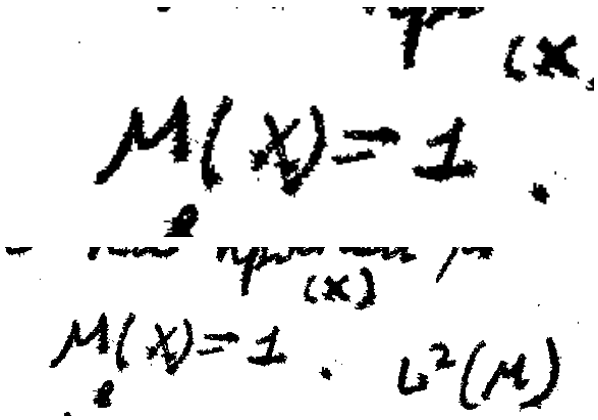


**Diagram and typed description.** Below the line $-\infty$ on the left, $+\infty$ on the right.

Original source details: GELFAND-P000088

$$\mu(x) = 1, \qquad L^2(\mu).$$

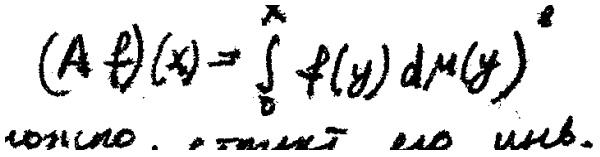
[Above the first argument: $(x)$.]



**Uncertain reading.** Visible $\mu(x) = 1$ and another $(x)$ above; $\mu$ / $M$. Below the first letter a separate short mark: $2$ / illegible sign.

Original source details: GELFAND-P000088

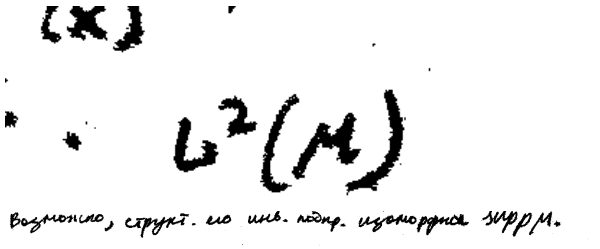
$$(Af)(x) = \int_0^x f(y)\, d\mu(y).$$



**Uncertain reading.** The lower limit is visible as $0$.

Original source details: GELFAND-P000088

Possibly, the struct. of inv. subsp. is isomorphic to supp $\mu$.



**Uncertain reading.** In $L^2(\mu)$ and after supp the letter $\mu$ / $M$.

Original source details: GELFAND-P000088

GELFAND-P000089

Possible counterexample. $X$ — sp. with measure $\mu$. $T$ — transf., preserving measure (in Russian a dyn. system). $\varphi$ — bounded fn. [Crossed out before the formula: Oper. / illegible.]



**Uncertain reading.** Oper. / illegible

Original source details: GELFAND-P000089

[Above the formula: in $L^2$.]

$$(Af)(x) = \varphi(x) f(Tx).$$

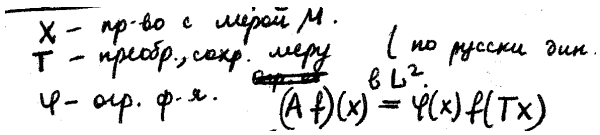
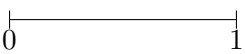


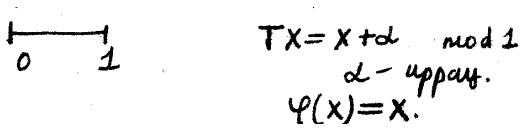
**Diagram and typed description.** Above $X, T, \varphi$; above the formula "in $L^2$"; on the right $f(Tx)$.

Original source details: GELFAND-P000089

$\exists$ examples of transf. with simple cont. spectrum.



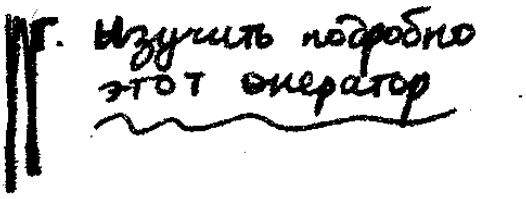
$$Tx = x + \alpha \pmod 1, \qquad \alpha \text{ — irr.}$$
$$\varphi(x) = x.$$




Original source details: GELFAND-P000089

**Diagram and typed description.** A segment with two end strokes, labels $0$, $1$ below. On the right $Tx = x + \alpha \pmod 1$, "$\alpha$ — irr.", then $\varphi(x) = x$.

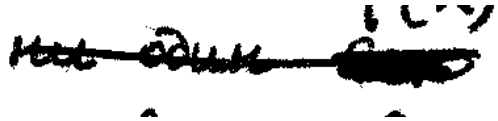
Study in detail
this operator




Original source details: GELFAND-P000089

**Diagram and typed description.** In the right margin two vertical strokes; a wavy line below "this operator".

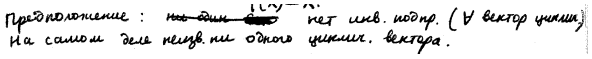
Conjecture: [Crossed out: on one… / on a giv… ("на один…" / "на дан…") / illegible.] no inv. subsp. (∀ vector cyclic).




Original source details: GELFAND-P000089

**Uncertain reading.** on one… / on a giv… ("на один…" / "на дан…") / illegible

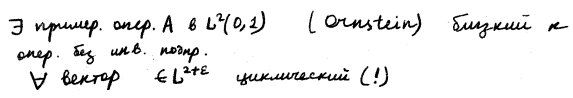
In fact, not a single cyclic vector is known.




Original source details: GELFAND-P000089

**Diagram and typed description.** In parentheses ∀ vector cyclic; below "not a single cyclic vector is known".

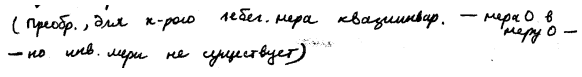
∃ an example of an oper. $A$ in $L^2(0,1)$ (Ornstein), close to an oper. without inv. subsp. ∀ vector $\in L^{2+\varepsilon}$ cyclic (!).




Original source details: GELFAND-P000089

**Diagram and typed description.** In the first line $L^2(0,1)$ and the name Ornstein; below ∀, $\in L^{2+\varepsilon}$ and (!).

(A transf. for which Lebesgue measure is quasi-inv. — measure $0$ to measure $0$ — but an inv. measure does not exist.)

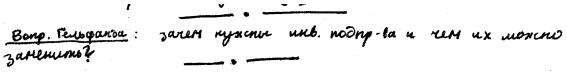



Original source details: GELFAND-P000089

**Diagram and typed description.** "measure $0$ to measure $0$"; after the line break "but an inv. measure does not exist".

— • —

Gelfand's question: why are inv. subsp. needed, and what can replace them?

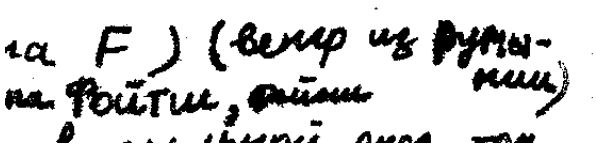



Original source details: GELFAND-P000089

**Diagram and typed description.** "Gelfand's question" is underlined; before and after the question, separators with a dot between two lines.

— • —

Another professor, speaking English (beginning with F) (a Hungarian from Romania), something like Foytsh / Foyshi (“Фойтш” / “Фойши”)



Original source details: GELFAND-P000089

**Uncertain reading.** Foytsh / Foyshi (“Фойтш” / “Фойши”)

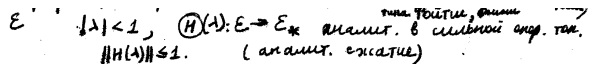
[A repeated entry is densely overwritten: oyshi / Foyshi (“ойши” / “Фойши”) / illegible.]



Original source details: GELFAND-P000089

**Uncertain reading.** oyshi / Foyshi (“ойши” / “Фойши”) / illegible

$$E, \qquad |\lambda| < 1, \qquad \textcircled{H}(\lambda) : E \longrightarrow E_*,$$
$$\|H(\lambda)\| \leq 1.$$



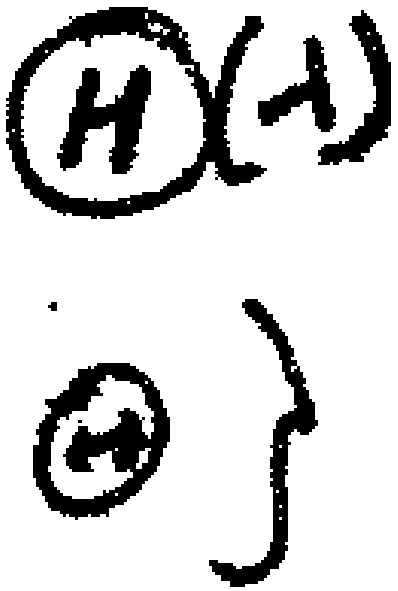
Original source details: GELFAND-P000089

**Diagram and typed description.** Separately $E$; $|\lambda| < 1$; the first letter $H$ is circled, the right $E$ has a lower star; below $\|H(\lambda)\| \leq 1$.

[The letter $H$ in the first entry is circled.] Analyt. in the strong oper. top. (analyt. contraction).

$$\{E, E_*, \textcircled{H}\}$$

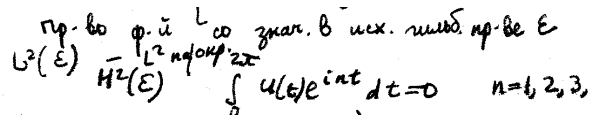
[The last letter in the triple is circled.]



Original source details: GELFAND-P000089

**Uncertain reading.** The circled sign in the triple: $H$ / $\Theta$ / illegible; above, the circled $H$ has two clear upright strokes.

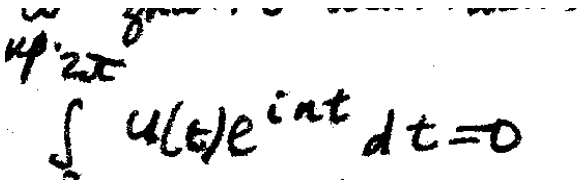
Sp. of fns. with values in the initial Hilb. sp. $E$. $L^2(E)$ — $L^2$ on the circ. $2\pi$.



Original source details: GELFAND-P000089

**Diagram and typed description.** In the upper line Hilb. sp. $E$; then $L^2(E)$, $2\pi$ and $H^2(E)$; the exponent has $+int$.

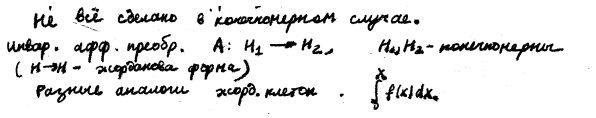
$$H^2(E) \qquad \int_0 u(t)e^{int}\,dt = 0, \qquad n = 1, 2, 3, \ldots$$



Original source details: GELFAND-P000089

**Uncertain reading.** The integral clearly has a lower $0$; no separate upper limit is visible.

(boundary values of analyt. fns.). Not everything has been done in the non-finite-dimensional case. Inv. aff. transf. $A : H_1 \longrightarrow H_2$,



Original source details: GELFAND-P000089

**Uncertain reading.** The abbreviation aff. / diff. (“афф.” / “дифф.”); below $H \to H$ and “Jordan form”.

$H_1, H_2$ — finite-dimensional. ($H \to H$ — Jordan form.) Various analogues of Jordan blocks.

$$\int_0^x f(x)\,dx.$$



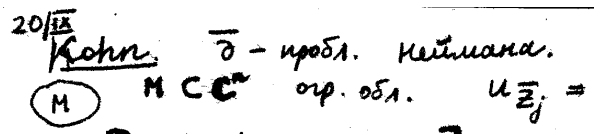
Original source details: GELFAND-P000089

**Diagram and typed description.** Upper limit $x$, lower $0$; the integrand and the differential also have $x$.

GELFAND-P000090

$20/\overline{\text{IX}}$. Kohn. $\bar\partial$ — Neumann probl.

$M$



Original source details: GELFAND-P000090

**Diagram and typed description.** Date $20/\text{IX}$: strokes above and below IX; Kohn underlined; on the left $M$ in an oval.

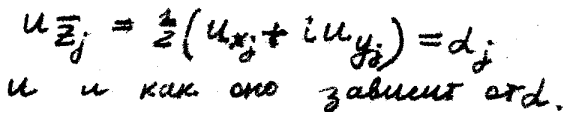
$M \subset C^n$ bounded dom.



Original source details: GELFAND-P000090

**Uncertain reading.** The exponent above $C$ is dense: $n$ / $m$.

$$u_{\bar z_j} = \frac{1}{2}(u_{x_j} + iu_{y_j}) = \alpha_j.$$

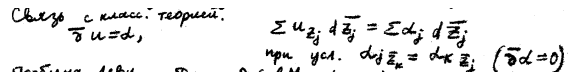


Original source details: GELFAND-P000090

**Diagram and typed description.** In the first definition a bar above $z_j$; on the right $\frac{1}{2}$, the sign $+$ and $i$, then $\alpha_j$.

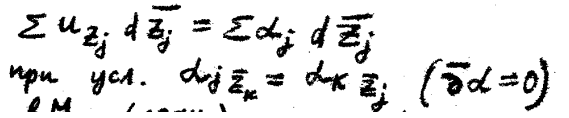
Given $\alpha_j$, when does $\exists u$ and how does it depend on $\alpha$. Connection with class. theorems.

$$\bar\partial u = \alpha,$$
$$\sum u_{z_j}\, d\bar z_j = \sum \alpha_j\, d\bar z_j$$
under cond. $\alpha_{j\bar z_k} = \alpha_{k\bar z_j} \qquad (\bar\partial\alpha = 0).$



Original source details: GELFAND-P000090

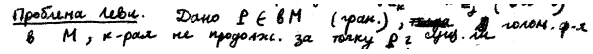
**Diagram and typed description.** Bars above $z_j$ in the differentials; condition $\alpha_{j\bar z_k} = \alpha_{k\bar z_j}$, on the right $(\bar\partial\alpha = 0)$.



Original source details: GELFAND-P000090

**Uncertain reading.** In the sum the subscript $z_j$ has no confidently visible bar; above, in the definition, $\bar z_j$ is clear.

Levi problem.



Original source details: GELFAND-P000090

**Diagram and typed description.** The heading has a straight underline; in the question $p$, $M$, the sign $\exists$ and abbreviations.

Given $p \in \partial M$ (bound.), [Crossed out: which (“какая”) / illegible.]

$\exists$ a holom. fn. in $M$, which does not continue beyond the point $p$? Exists.



Original source details: GELFAND-P000090

**Uncertain reading.** which (“какая”) / illegible

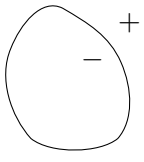
The class. Levi theor. solves this probl. locally:



Original source details: GELFAND-P000090

**Diagram and typed description.** A wavy line under the whole sentence from “The class.” to “locally:”.

$\frac{r}{x}$ — equ. of the boundary.
0



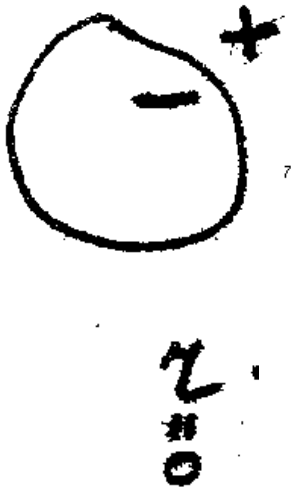



**Diagram and typed description.** An irregular closed curve, $-$ inside and $+$ outside above right. Below $r$ a small crossed sign resembling $x$, then $0$; their exact reading and role are not established.

Original source details: GELFAND-P000090

$$\underline{dr \neq 0}$$

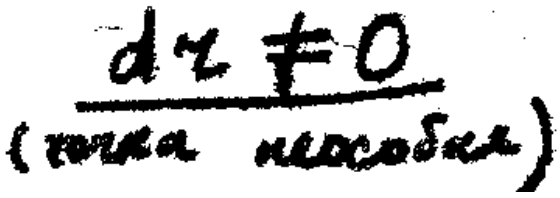



**Diagram and typed description.** The entry $dr \neq 0$ has a straight underline; below "(point nonsingular)".

Original source details: GELFAND-P000090

(point nonsingular). At pt. $p \in \partial M$

$$\sum r_{z_i}(p)a^i = 0 \qquad \text{hyperpl.}$$

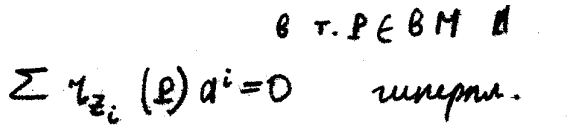
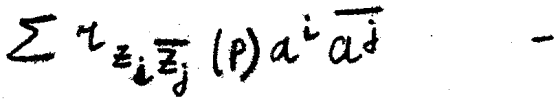



**Diagram and typed description.** Above the formula "At pt. $p \in \partial M$"; in the formula $a^i$ and $= 0$; on the right "hyperpl.".

Original source details: GELFAND-P000090

$$\sum r_{z_i \bar{z}_j}(p)a^i \bar{a}^j$$

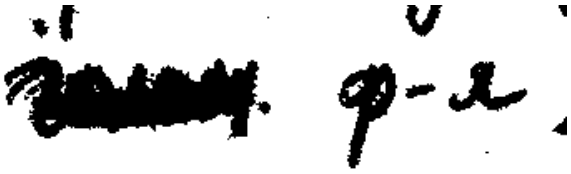
— Levi form.




**Uncertain reading.** The boundary function's letter resembles $r$ / $\tau$, the boundary sign $\partial$ / $b$.

Original source details: GELFAND-P000090

If it is pos. def. on the indicated hyperplane,
then there exists in a neighb. of pt. $p$ [Crossed out: illegible.]
a fn., not cont. beyond the point $p$.



**Uncertain reading.** illegible

Original source details: GELFAND-P000090

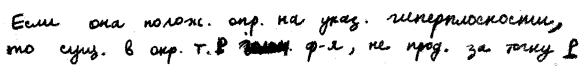
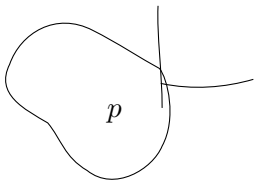


**Diagram and typed description.** Abbreviations “pos. def.”, “on the indicated hyperplane”, then “exists in a neighb. of pt. $p$”.

Original source details: GELFAND-P000090

($\exists$ a polynomial $W$, such that the zeros are arranged thus:




)

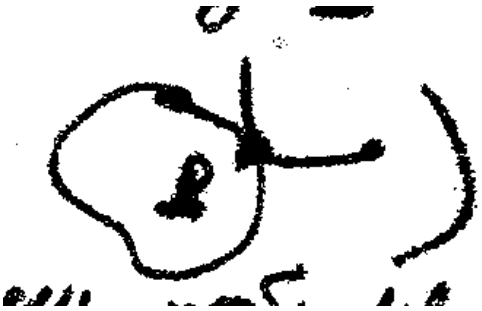


**Diagram and typed description.** An irregular closed curve; at the right edge a nearly vertical stroke and a stroke running out to the right. Label $p$ inside, left and below their junction. The exact position of the dense overlapping strokes is not established.

Original source details: GELFAND-P000090

A good answer to that probl. ($\bar\partial$) leads to a glob. sol. of the Levi probl. Let the $\bar\partial$-probl. be solvable for the dom. $M$ ($\forall\alpha$, $\bar\partial\alpha = 0$, $\exists u$, $\bar\partial u = \alpha$). Let

$$g = \begin{cases} 1 & \text{in a neighb. of } p, \\ 0 & \text{outside } U \end{cases}$$

— cut-off fn.

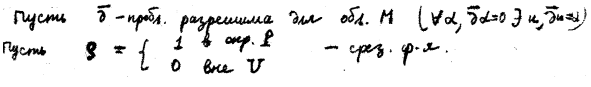


**Diagram and typed description.** The entry $g$ has a brace, $1$ “in a neighb. of $p$”, $0$ “outside $U$”; on the right “cut-off fn.”.

Original source details: GELFAND-P000090

Denote $\alpha = \bar\partial\left(\frac{g}{W}\right)$. $W$ — the previously constr. polynomial. $\alpha$ — smooth up to the boundary.

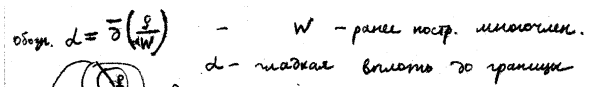
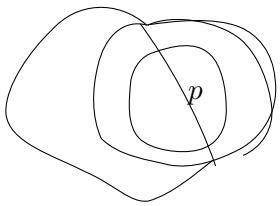


**Diagram and typed description.** On the left $\alpha = \bar\partial(g/W)$; on the right $W$ — the previously constr. polynomial; below “smooth up to the boundary”.

Original source details: GELFAND-P000090

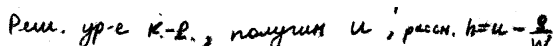




Original source details: GELFAND-P000090

**Diagram and typed description.** Three irregular nested contours and an inclined stroke; a gap at the lower right of the outer contour. $p$ inside the small contour. No labels $U$ or boundary are visible at the contours; dense intersections are not fully distinguishable.

Solve the equ. someh., obtain $u$, consider $h = u - \frac{g}{W}$.



Original source details: GELFAND-P000090

**Uncertain reading.** After “equ.” the abbreviation “к-л.” (someh.); the second letter is dense, the exact reading is not fully established. Then $u$, $h = u - \frac{g}{W}$.

$h$ holom. (since $\bar{\partial} = 0$) [A stroke crossed out: arrow (“стрелка”) / illegible.]



Original source details: GELFAND-P000090

**Uncertain reading.** arrow (“стрелка”) / illegible

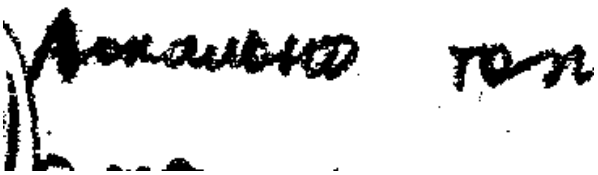
$u$ does not continue.



Original source details: GELFAND-P000090

**Uncertain reading.** Before “does not continue” the letter $u$ is visible.

Analogo / Analogously (“Аналого” / “Аналогично”) is also possible, but there it continues beyond the point.



Original source details: GELFAND-P000090

**Uncertain reading.** Analogo / Analogously (“Аналого” / “Аналогично”)

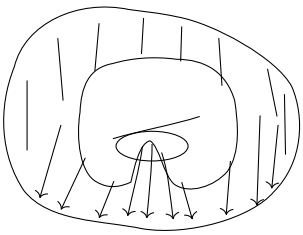




**Diagram and typed description.** An outer contour; an inner contour with a lower notch; hatching between them. At the notch a small oval and an oblique stroke above it; several long strokes with downward arrows. Dense intersections and some stroke ends are not fully distinguishable.

Original source details: GELFAND-P000090

[Crossed out to the right of the drawing: illegible.]



**Uncertain reading.** illegible

Original source details: GELFAND-P000090

Must take into acc. the geom. structure of the entire domain as a whole.

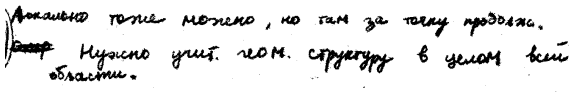


**Diagram and typed description.** The last sentence breaks to “domain.”; above “but there it continues beyond the point”.

Original source details: GELFAND-P000090

GELFAND-P000091

$a^{p,q}$ — space of forms of type $p, q$.

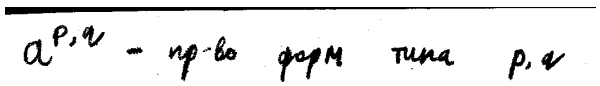
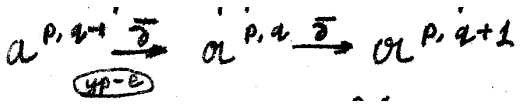


**Diagram and typed description.** $a^{p,q}$; space of forms of type $p, q$.

Original source details: GELFAND-P000091

$$a^{p,q-1} \xrightarrow[\text{eqn}]{\bar\partial} a^{p,q} \xrightarrow{\bar\partial} \alpha^{p,q+1}.$$

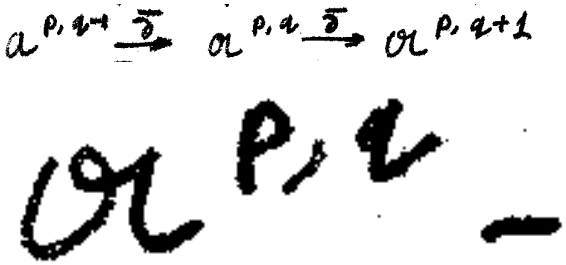


Original source details: GELFAND-P000091

**Diagram and typed description.** Two arrows with $\bar\partial$; under the first, eqn is enclosed in an oval.



**Uncertain reading.** The initial and middle letters have the form $a$; the last in the chain and the following notation have the form $\alpha$. The local $a/\alpha$ variation is retained.



Original source details: GELFAND-P000091

General theorems. $\alpha^{p,q}$ — Hilb. scalar product. $\varphi, \psi \in D_{\bar\partial^*}$.

$$Q(\varphi,\psi) = (\bar\partial\varphi, \bar\partial\psi) + (\bar\partial^*\varphi, \bar\partial^*\psi).$$

[At the right:] ($\bar\partial$ on smooth forms of 1st degree without boundary conditions).



Original source details: GELFAND-P000091

**Diagram and typed description.** $D_{\bar\partial^*}$, two terms in $Q(\varphi,\psi)$; at the right, parentheses about smooth forms of 1st degree without boundary conditions.

$$\boxed{\bar\partial^*\bar\partial\varphi + \bar\partial\bar\partial^*\varphi = \alpha}$$

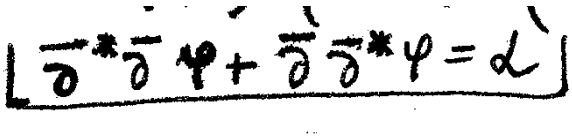


**Diagram and typed description.** Equation with two terms and right side $\alpha$; frame at the left, bottom and right, with no top side.

Original source details: GELFAND-P000091

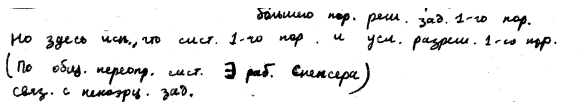
With the aid of a pos. def. prob. [Crossed out: illegible] of higher order solve a prob. of 1st order.



Original source details: GELFAND-P000091

**Uncertain reading.** After prob., a former short expression is densely crossed out; an initial 1-/4-like stroke, the rest illegible.

But here it is used that the syst. is of 1st order and the solvability cond. is of 1st order. (On gen. overdet. syst. ∃ work of Spencer.) Connection with the Neumann prob.

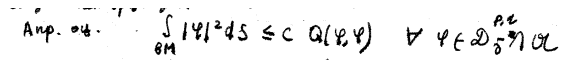


Original source details: GELFAND-P000091

**Diagram and typed description.** Syst. of 1st order, solvability cond. of 1st order; in parentheses ∃ work of Spencer; then connection with the Neumann prob.

A priori est.

$$\int_{\partial M} |\varphi|^2 \, dS \leq CQ(\varphi, \varphi) \qquad \forall \varphi \in D_{\bar{\partial}^*} \cap \alpha^{p,q}.$$



Original source details: GELFAND-P000091

**Diagram and typed description.** The integral has $\partial M$, integrand $|\varphi|^2 dS$; $\leq CQ(\varphi, \varphi)$ and $\forall \varphi \in D_{\bar{\partial}^*} \cap \alpha^{p,q}$.

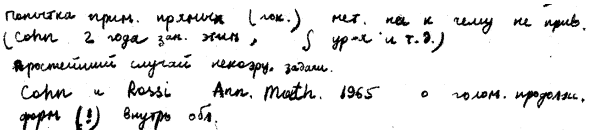
(but not min. $\sum$ of squares of 1st derivs., $n = 1 \sim \int$ Dirichlet).



Original source details: GELFAND-P000091

**Diagram and typed description.** Before $\sum$, min.; then squares of 1st derivs., $n = 1 \sim \int$ Dirichlet.

Attempt to apply projection (loc.) no, led nowhere. (Cohn spent 2 years on this, $\int$ eqns and so on.) Simplest case of the Neumann problem. Cohn and Rossi Ann. Math. 1965 on holom. continuation of forms (!) into the domain.



Original source details: GELFAND-P000091

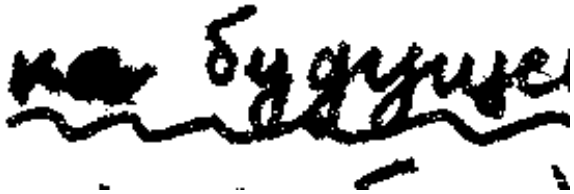
**Diagram and typed description.** Attempt to apply projection (loc.); Cohn in two places; 2 years, $\int$ eqns; Rossi, Ann. Math. 1965; (!) after forms.

Plans for the future.



Original source details: GELFAND-P000091

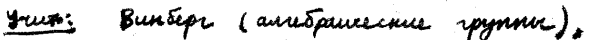
**Diagram and typed description.** Wavy line beneath the heading Plans for the future.



Original source details: GELFAND-P000091

**Uncertain reading.** The word for in the heading is densely overwritten; the earlier strokes are illegible.

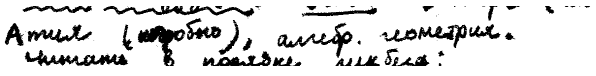
Study: Vinberg (algebraic groups),



Original source details: GELFAND-P000091

**Diagram and typed description.** Study: has a straight underline; Vinberg (algebraic groups).

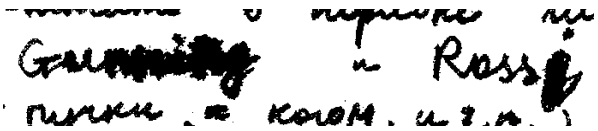
Atiy[?] (...detail[?]; the beginning in parentheses is densely overwritten), alg. geometry.



Original source details: GELFAND-P000091

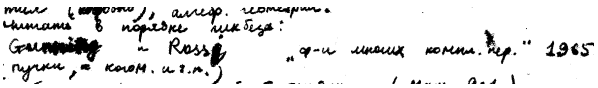
**Uncertain reading.** Atiy[?] (...detail[?]; the beginning in parentheses is densely overwritten), alg. geometry.

read for basic instruction: [Crossed out: Latin $G$... / Cyrillic Г...; the rest of the name is illegible. ] and Rossi “functions of several complex variables” 1965 (sheaves, cohom. and so on).



Original source details: GELFAND-P000091

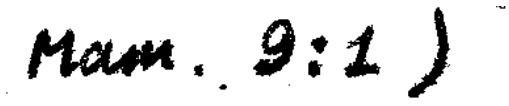
**Uncertain reading.** Latin $G$... / Cyrillic Г...; the rest of the name is illegible.



Original source details: GELFAND-P000091

**Diagram and typed description.** After the cancelled name, and Rossi; functions of several complex variables, 1965; in parentheses, sheaves, cohom. and so on.

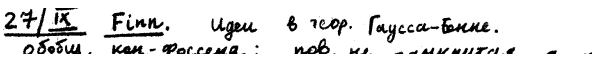
Review by Dieudonné of the work of Grothendieck (Man. / Mat.[?] 9:1); the abbreviated first word is not entirely certain.



Original source details: GELFAND-P000091

**Uncertain reading.** (Man. / Mat.[?] 9:1); the abbreviated first word is not entirely certain.

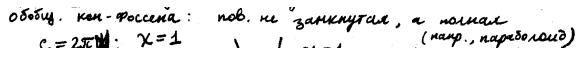
27/IX Finn. Ideas in the Gauss–Bonnet theor.



Original source details: GELFAND-P000091

**Diagram and typed description.** Separate straight lines under the date and Finn; short bars above and below IX.

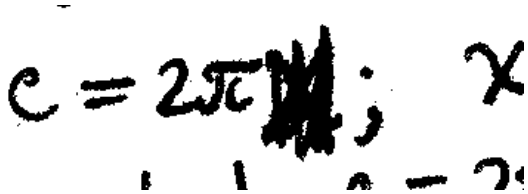
Generalization of Cohn–Vossen: the surf. is not closed, but complete (e.g., a paraboloid).



Original source details: GELFAND-P000091

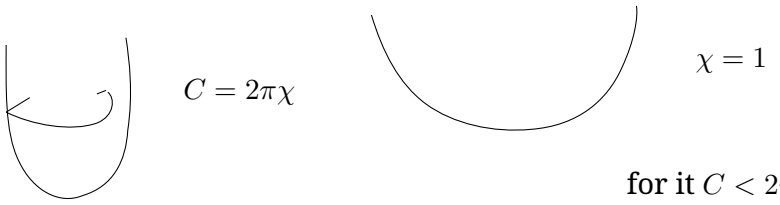
**Diagram and typed description.** The surf. is not closed, but complete; in parentheses, e.g., a paraboloid.

$C = 2\pi$ [Crossed-out sign: illegible]; $\chi = 1$.



Original source details: GELFAND-P000091

**Uncertain reading.** After the first $C = 2\pi$ a sign is densely crossed out; its exact reading is not established.



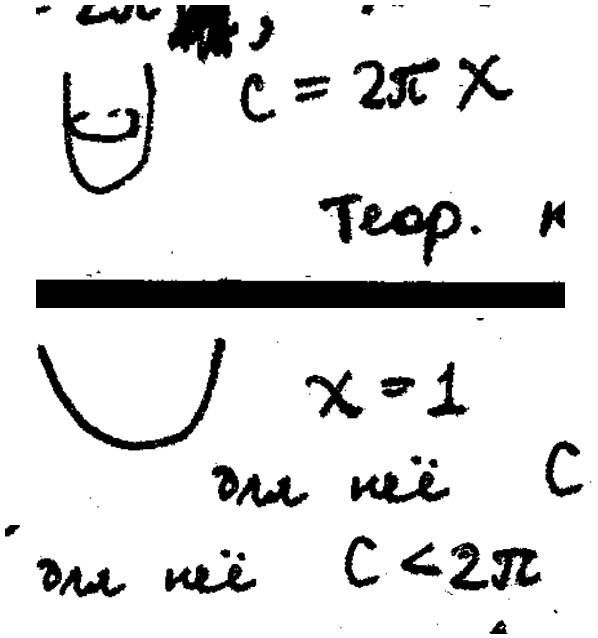




**Diagram and typed description.** Two lines opening upward; the small one at the left has a short curved transverse stroke with thickened ends. Its exact role and direction are not established. To its right $C = 2\pi\chi$; by the large one $\chi = 1$, below, for it $C < 2\pi$.

Original source details: GELFAND-P000091

($C = \int K\, dS$).

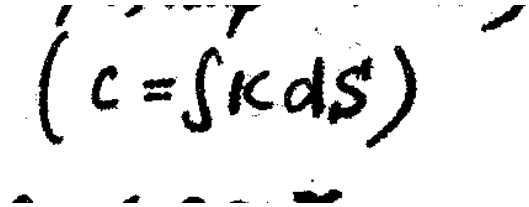



**Diagram and typed description.** $C = \int K dS$ is enclosed in round parentheses.

Original source details: GELFAND-P000091

Cohn–Vossen theor.: $\forall$ complete surf. $C \leq 2\pi\chi$.

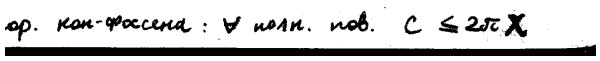



**Diagram and typed description.** The last line has $\forall$ complete surf. and $C \leq 2\pi\chi$.

Original source details: GELFAND-P000091

GELFAND-P000092

Aim of the work — to replace the inequality by an equality.

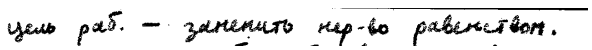



**Diagram and typed description.** Aim of the work — to replace the inequality by an equality.

Original source details: GELFAND-P000092

Next time a review by Volevich[?]

on var., anal. bundles.



Original source details: GELFAND-P000092

**Uncertain reading.** Volevich[?]

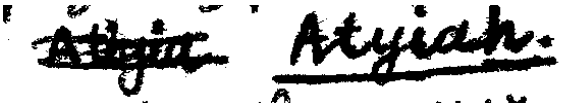
$Kohn$'om[?]



Original source details: GELFAND-P000092

**Uncertain reading.** $Kohn$'om[?]; dense beginning of the name and ending after the apostrophe.

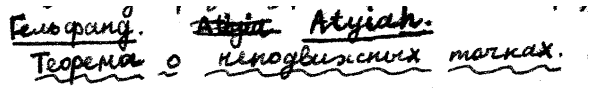
Gelfand. [Crossed out: Atiy...; ending illegible.] Atiyah.



Original source details: GELFAND-P000092

**Uncertain reading.** [Crossed out: Atiy...; ending illegible.] Atiyah.

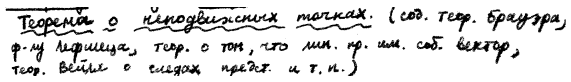
Fixed-point theorems.



Original source details: GELFAND-P000092

**Diagram and typed description.** Straight lines under Gelfand. and Atiyah.; wavy line under Fixed-point theorems.

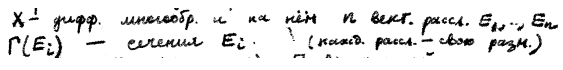
(contains Brouwer's theor., the Lefschetz formula, the theor. that a lin. space has an eigenvector, Weyl's theor. on traces of reps. and so on.)



Original source details: GELFAND-P000092

**Diagram and typed description.** At the beginning of the parentheses, contains; then Brouwer's theor., the Lefschetz formula, eigenvector, Weyl's theor. on traces of reps.

$X$ — diff. manifold and on it $n$ vector bundles $E_1, \dots, E_n$. $\Gamma(E_i)$ — sections of $E_i$. (each bundle — its own dim.)



Original source details: GELFAND-P000092

**Diagram and typed description.** $X$, $n$, $E_1, \dots, E_n$ and $\Gamma(E_i)$; in parentheses, each bundle — its own dim.; the last ending of the Russian word for its own is tight.

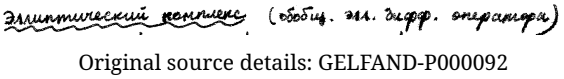
Example: $X \times I$ ($I$ — a straight line). Space of functions.



Original source details: GELFAND-P000092

**Diagram and typed description.** Straight underline under Example:; $X \times I$, $I$ — a straight line, space of functions.

Elliptic complex (generalizes ell. diff. operators).

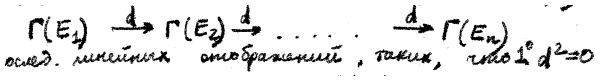


Original source details: GELFAND-P000092

**Diagram and typed description.** Wavy line under Elliptic complex; then generalizes ell. diff. operators.

$$\Gamma(E_1) \xrightarrow{d} \Gamma(E_2) \xrightarrow{d} \cdots \xrightarrow{d} \Gamma(E_n).$$

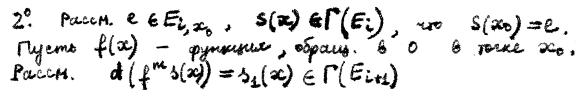
a seq. of linear maps such that $1^\circ$ $d^2 = 0$.



Original source details: GELFAND-P000092

**Diagram and typed description.** Three arrows with $d$; $\Gamma(E_1)$, $\Gamma(E_2)$, ellipsis, $\Gamma(E_n)$; $1^\circ$, $d^2 = 0$.

$2^\circ$. Consider $e \in E_{i,x_0}$, $s(x) \in \Gamma(E_i)$, such that $s(x_0) = e$. Let $f(x)$ be a function taking the value $0$ at the point $x_0$. Consider $d(f^m s(x)) = s_1(x) \in \Gamma(E_{i+1})$.

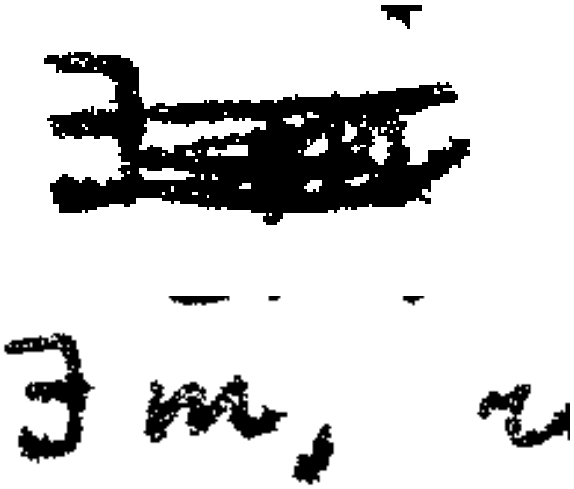


Original source details: GELFAND-P000092

**Diagram and typed description.** $e \in E_{i,x_0}$; $s(x_0) = e$; function $f(x)$ vanishes at $x_0$; $d(f^m s(x)) = s_1(x) \in \Gamma(E_{i+1})$.

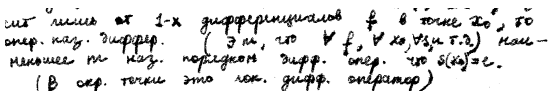
Then, if [Crossed out: $\exists$ with an illegible continuation.] $\exists m$,

such that $\forall x_0$ $(f(x_0) = 0)$ $s_1(x_0)$ depends only on the 1st differentials of $f$ at the point $x_0$, the operator is called differential. ($\exists m$ such that $\forall f$, $\forall x_0$, $\forall s$ and so on.) The least $m$ is called the order of the diff. operator such that $s(x_0) = e$. (In a neighbourhood of the point this is a local diff. operator.)



Original source details: GELFAND-P000092

**Uncertain reading.** [Crossed out: $\exists$ with an illegible continuation.] $\exists m$,



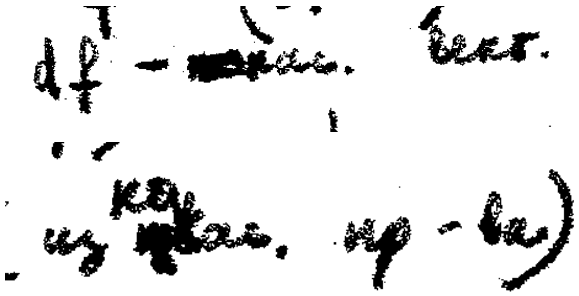
Original source details: GELFAND-P000092

**Diagram and typed description.** Three $\forall$ signs in parentheses; the least $m$, then such that $s(x_0) = e$; in the final parentheses, in a neighbourhood of the point.

$$df = (\eta, dx).$$

$df$ — [Short word crossed out: illegible] vector; $\eta$ — gradient (from [tang. inserted above densely overwritten strokes] space).

$\eta \in T_x^*$.



**Uncertain reading.** $df$ — [Short word crossed out: illegible] vector; $\eta$ — gradient (from [tang. inserted above densely overwritten strokes] space).

Original source details: GELFAND-P000092

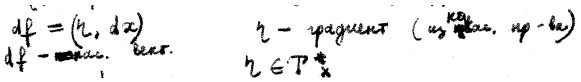


**Diagram and typed description.** $df = (\eta, dx)$; alongside, $\eta$ — gradient; below, $\eta \in T_x^*$.

Original source details: GELFAND-P000092

$$\sigma(d, \eta)e$$

— a lin. function of $e$, called the symbol of the diff. operator.



**Uncertain reading.** First argument: $d$ added above a densely cancelled sign; the former sign is illegible.

Original source details: GELFAND-P000092

Example. Laplace operator. $m = 2$.

$$\frac{\partial^2(f^2(x)s(x))}{\partial x^2} = 2\left(\frac{\partial f}{\partial x}\right)^2 s(x).$$

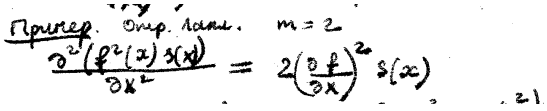


**Diagram and typed description.** Straight underline under Example.; $m = 2$; $f^2(x)s(x)$ in the second derivative; on the right $2(\partial f/\partial x)^2 s(x)$.

Original source details: GELFAND-P000092

Symbol $2(\eta_1^2 + \eta_2^2 + \cdots + \eta_n^2)$.

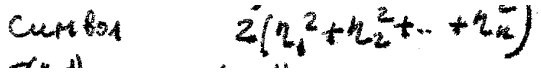


**Diagram and typed description.** Factor $2$; $\eta_1^2 + \eta_2^2 + \cdots + \eta_n^2$.

Original source details: GELFAND-P000092

Consider.

$$0 \longrightarrow E_{1,x_0} \overset{\sigma(\eta,d)}{\longrightarrow} E_{2,x_0} \overset{\sigma(\eta,d)}{\longrightarrow} \cdots \longrightarrow E_{n,x_0} \longrightarrow 0.$$

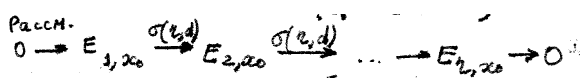
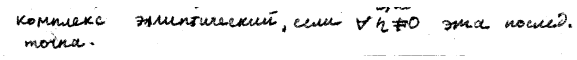


**Diagram and typed description.** Zero endpoints; $E_{1,x_0}$, $E_{2,x_0}$, ellipsis, $E_{n,x_0}$; $\sigma(\eta, d)$ over the first two arrows.

Original source details: GELFAND-P000092

The complex is elliptic if $\forall \eta \neq 0$ this seq. is exact.

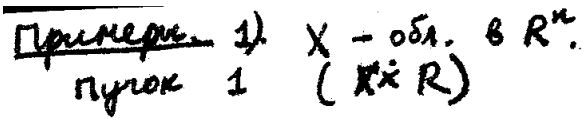


Original source details: GELFAND-P000092

**Diagram and typed description.** The last line has $\forall \eta \neq 0$ and this seq. is exact.

GELFAND-P000093

Examples. 1) $X$ — a domain in $R^n$. sheaf 1 ($X \times R$).

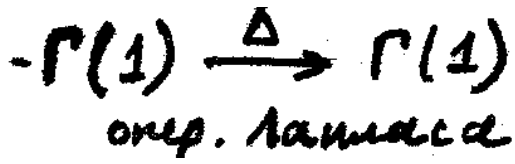


Original source details: GELFAND-P000093

**Diagram and typed description.** Straight underline under Examples.; 1), $X$, $R^n$, sheaf 1, $X \times R$.

$$-\Gamma(1) \xrightarrow{\Delta} \Gamma(1).$$

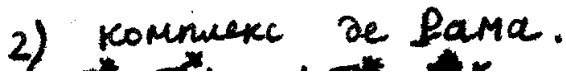
Laplace operator.



Original source details: GELFAND-P000093

**Uncertain reading.** A short horizontal stroke before the first $\Gamma(1)$: minus / dash.

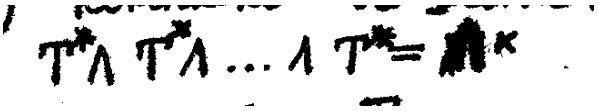
2) de Rham complex.



Original source details: GELFAND-P000093

**Diagram and typed description.** 2) de Rham complex.

$$T^* \wedge T^* \wedge \cdots \wedge T^* = \Lambda^k\,[?].$$



Original source details: GELFAND-P000093

**Uncertain reading.** The $\Lambda$ itself is densely overwritten; small exponent $k$[?], also $n$- / $x$- / $\kappa$-like.

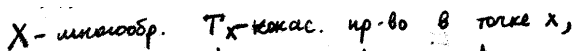
$T$ — [Beginning of the word densely overwritten; former letters illegible] tang. space.



Original source details: GELFAND-P000093

**Uncertain reading.** [Beginning of the word densely overwritten; former letters illegible] tang. space.

$X$ — manifold. $T_x$ — tang. space at the point $x$,



Original source details: GELFAND-P000093

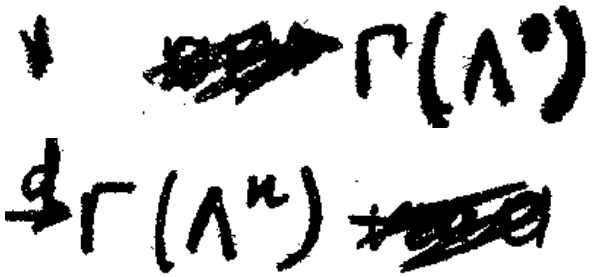
**Diagram and typed description.** $X$ — manifold; $T_x$ — tang. space at the point $x$.



Original source details: GELFAND-P000093

**Uncertain reading.** [At the right: compact entry crossed out; T-/Phi-like strokes, exact reading undetermined.]

$$\Gamma(\Lambda^0) \xrightarrow{d} \Gamma(\Lambda^1) \xrightarrow{d} \cdots \xrightarrow{d} \Gamma(\Lambda^n).$$



Original source details: GELFAND-P000093

**Uncertain reading.** [Signs densely cancelled before the first $\Gamma(\Lambda^0)$, a separate short stroke at the left; after $\Gamma(\Lambda^n)$, an arrow and a $0$-like sign are crossed out.]

[At the right:] sections — diff. forms.

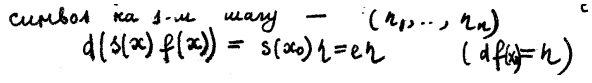


Original source details: GELFAND-P000093

**Diagram and typed description.** At the right, sections — diff. forms.; wavy line under diff. forms.

Symbol at the 1st step — $(\eta_1, \ldots, \eta_n)$.

$$d(s(x)f(x)) = s(x_0)\eta = e\eta \qquad (df(x_0) = \eta).$$

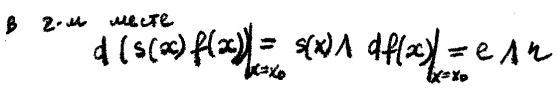


Original source details: GELFAND-P000093

**Diagram and typed description.** At the 1st step $(\eta_1, \ldots, \eta_n)$; $s(x_0)\eta = e\eta$; in parentheses $df(x_0) = \eta$.

In the 2nd place

$$d(s(x)f(x))\big|_{x=x_0} = s(x) \wedge df(x)\big|_{x=x_0} = e \wedge \eta.$$



Original source details: GELFAND-P000093

**Diagram and typed description.** Two restrictions $x = x_0$; $s(x) \wedge df(x)$ and $e \wedge \eta$.

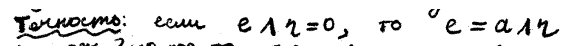
(each subsequent symbol — multiplication by $\eta$).



Original source details: GELFAND-P000093

**Diagram and typed description.** Each subsequent symbol — multiplication by $\eta$, in parentheses.

Exactness: if $e \wedge \eta = 0$, then $e = a \wedge \eta$.

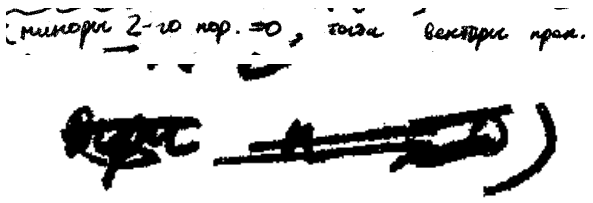


Original source details: GELFAND-P000093

**Diagram and typed description.** Wavy line under Exactness:; $e \wedge \eta = 0$, $e = a \wedge \eta$.

(minors of the 2nd ord. $= 0$, then the vectors [pryam.: meaning unclear] [Final signs crossed out; dense entry, arrow and 0-like ending; exact reading undetermined.]

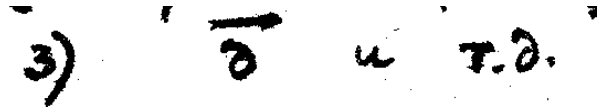
)



Original source details: GELFAND-P000093

**Uncertain reading.** [Final signs crossed out; dense entry, arrow and 0-like ending; exact reading undetermined.]

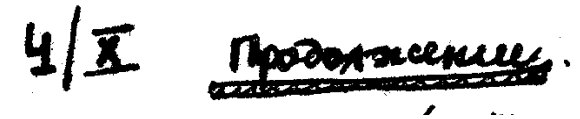
3) $\bar{\partial}$ and so on.



Original source details: GELFAND-P000093

**Diagram and typed description.** 3); horizontal bar above $\partial$; and so on.

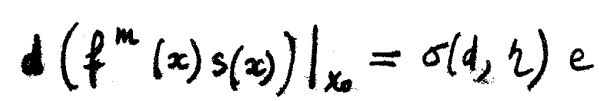
$4/\overline{\text{X}}$ Continuation. $\Gamma(E)$ — seq.



Original source details: GELFAND-P000093

**Diagram and typed description.** Date $4/\overline{\text{X}}$; Continuation. underlined with two straight lines.

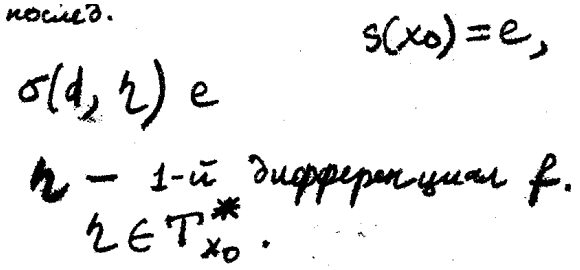
$$d(f^m(x)s(x))\big|_{x_0} = \sigma(d,\eta)e, \qquad s(x_0) = e.$$



Original source details: GELFAND-P000093

**Diagram and typed description.** $d(f^m(x)s(x))\big|_{x_0} = \sigma(d,\eta)e$; at the right $s(x_0) = e$.

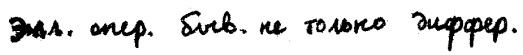
$\eta$ — 1st differential of $f$. $\eta \in T^*_{x_0}$.



Original source details: GELFAND-P000093

**Diagram and typed description.** At the right $s(x_0) = e$; $\eta$ — 1st differential of $f$; $\eta \in T^*_{x_0}$.

Ell. operators can be other than differential.



Original source details: GELFAND-P000093

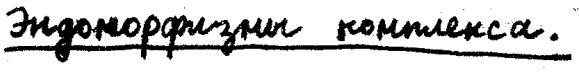
**Diagram and typed description.** Ell. operators can be other than differential.



Original source details: GELFAND-P000093

**Uncertain reading.** [Short word crossed out before the heading; dense horizontal strokes and a descending stroke; former letters illegible.]

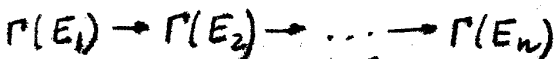
Endomorphism of a complex.



Original source details: GELFAND-P000093

**Diagram and typed description.** Straight underline under Endomorphism of a complex.

$$\Gamma(E_1) \longrightarrow \Gamma(E_2) \longrightarrow \cdots \longrightarrow \Gamma(E_n).$$



Original source details: GELFAND-P000093

**Diagram and typed description.** $\Gamma(E_1) \longrightarrow \Gamma(E_2) \longrightarrow \cdots \longrightarrow \Gamma(E_n).$

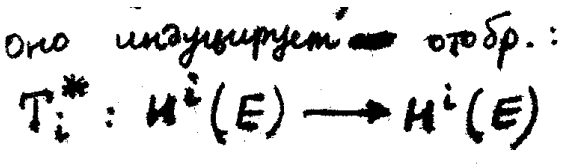
It induces [Short sign / word densely cancelled before map; illegible.]
a map:

$$T_i^* : H^i(E) \longrightarrow H^i(E).$$



Original source details: GELFAND-P000093

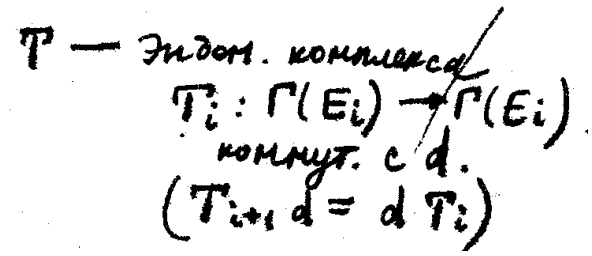
**Uncertain reading.** [Short sign / word densely cancelled before map; illegible.]



Original source details: GELFAND-P000093

**Diagram and typed description.** It induces ... a map:; $T_i^* : H^i(E) \longrightarrow H^i(E)$.

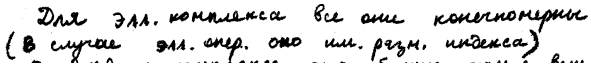
[At the right:] $T$ — endom. of the complex, $T_i : \Gamma(E_i) \to \Gamma(E_i)$, commutes with $d$. $(T_{i+1}d = dT_i)$. ↙




Original source details: GELFAND-P000093

**Uncertain reading.** At the right a slanting arrow crosses the end of the word complex and passes near the arrow between $\Gamma(E_i)$ and $\Gamma(E_i)$; its exact connection is unclear. Formulas: $T_i : \Gamma(E_i) \to \Gamma(E_i)$, $(T_{i+1}d = dT_i)$.

For an ell. complex they are all finite-dimensional. (In the case of an ell. operator it has an index difference.)




Original source details: GELFAND-P000093

**Diagram and typed description.** They are all finite-dimensional; in parentheses, In the case of an ell. operator it has an index difference.

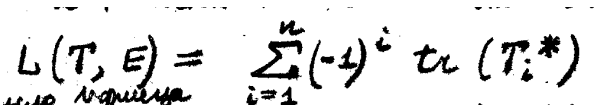
In the de Rham complex these are the usual cohom. with real coeffs.




Original source details: GELFAND-P000093

**Diagram and typed description.** In the de Rham complex these are the usual cohom. with real coeffs.

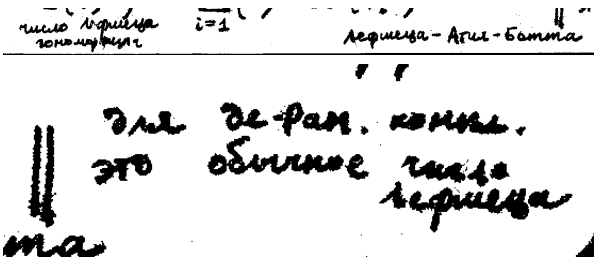
$$L(T, E) = \sum_{i=1}^{n} (-1)^i \operatorname{tr}(T_i^*).$$




Original source details: GELFAND-P000093

**Diagram and typed description.** $L(T, E)$; sum from $i = 1$ to $n$; $(-1)^i \operatorname{tr}(T_i^*)$.

[Under $L$:] Lefschetz number, homological. [At the right:] ⇓ for the de Rham complex this is the usual Lefschetz number. [At the bottom:] Lefschetz-Atiyah-Bott.

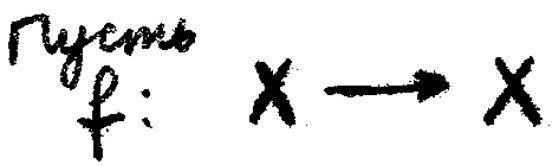



Original source details: GELFAND-P000093

**Diagram and typed description.** Under $L$: Lefschetz number, homological; at the right a downward arrow with two strokes and for the de Rham complex this is the usual Lefschetz number; at the bottom Lefschetz-Atiyah-Bott.

GELFAND-P000094

Let $f : X \longrightarrow X$.

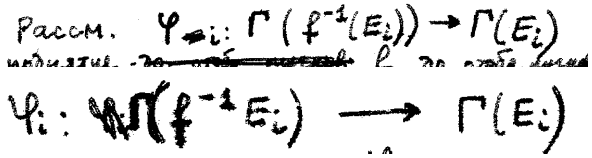



Original source details: GELFAND-P000094

**Diagram and typed description.** Let; $f : X \longrightarrow X$.

Consider.

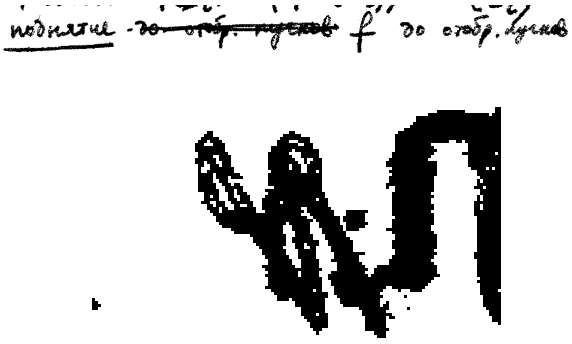
$$\psi_i : \Gamma(f^{-1}(E_i)) \longrightarrow \Gamma(E_i).$$



**Uncertain reading.** At the first $\psi_i$, a dense additional stroke precedes the clear $i$; former index / sign illegible. The first entry has $\Gamma(f^{-1}(E_i))$, the repeated entry $\Gamma(f^{-1}E_i)$.

Original source details: GELFAND-P000094

lifting — ~~by a map of sheaves~~ $f$ by a map of sheaves.



**Uncertain reading.** By a map of sheaves is crossed out, partly obscured by strokes. A short sign is densely cancelled before the second $\Gamma$; exact reading undetermined.

Original source details: GELFAND-P000094

$$\psi_i : \Gamma(f^{-1}E_i) \longrightarrow \Gamma(E_i),$$

$$E_{i,f^{-1}x} \xrightarrow{\psi_{i,x}} E_{i,x}.$$



**Uncertain reading.** The left fibre index visibly has $f^{-1}x$.

Original source details: GELFAND-P000094

It is not uniquely determ. by $f$.

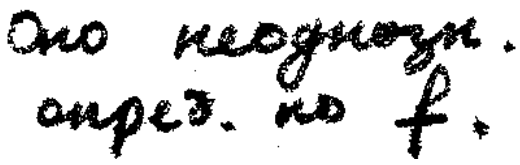


**Diagram and typed description.** It is not uniquely determ. by $f$.

Original source details: GELFAND-P000094

de Rham sheaf: $f$ uniquely induces a map of de Rham sheaves. (natural lifting).

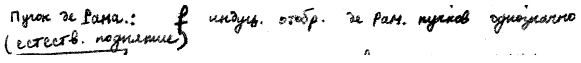


**Diagram and typed description.** $f$ uniquely induces a map of de Rham sheaves; natural lifting in parentheses.

Original source details: GELFAND-P000094

$$\boxed{f : X \longrightarrow X}.$$

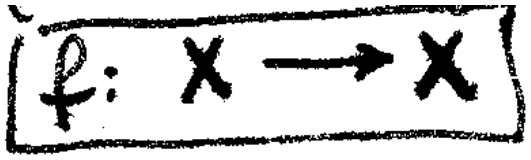



**Diagram and typed description.** Rectangular frame around $f : X \longrightarrow X$; the natural-lifting parenthesis is above its upper border.

Original source details: GELFAND-P000094

Suppose that $f$ has only a finite number of isol. fixed points of the simplest structure.

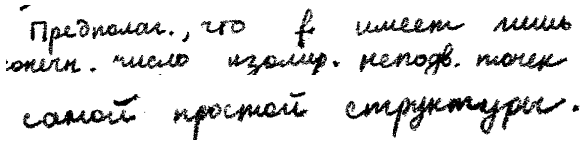



**Diagram and typed description.** Suppose that $f$ has only a finite number of isol. fixed points of the simplest structure.

Original source details: GELFAND-P000094

Brouwer theorem for the interval $[0, 1]$. $f : x \to f(x)$.

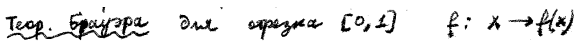
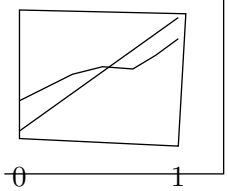



**Diagram and typed description.** Wavy line under Brouwer theorem; $[0, 1]$; $f : x \to f(x)$.

Original source details: GELFAND-P000094




$\rho$ — fixed point of this map.
At it we have $\psi_i(\rho)$ : [An $E$-like fibre entry crossed out; indices illegible.] $E_{i,\rho} \to E_{i,\rho}$.
In the tang. space$^{(x)[?]}$ there is a linear transformation.

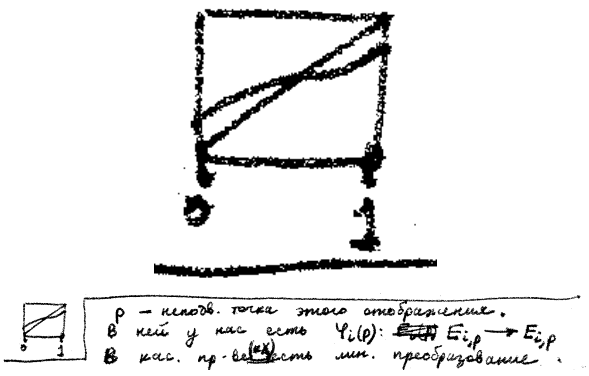



**Diagram and typed description.** Rectangle labelled $0$ and $1$, two ascending lines, dense crossing. The crossing geometry is unclear. An angular line runs above and to the left of the adjoining lines about $\rho$, the fibre and the tangent space, then extends left below the drawing.

Original source details: GELFAND-P000094

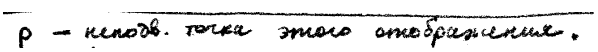
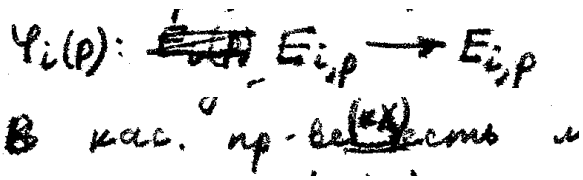



**Diagram and typed description.** $\rho$ — fixed point of this map.

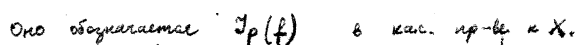
Original source details: GELFAND-P000094




**Uncertain reading.** Before the new $E_{i,\rho}$ an $E$-like entry with unread indices is cancelled. An insertion $(x)$[?] is above the underlined space; the precise letter is uncertain.

Original source details: GELFAND-P000094

It is denoted $J_\rho(f)$ in the tang. space to $X$.



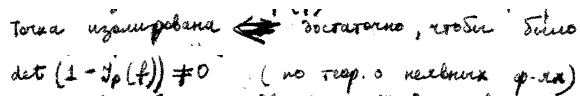
Original source details: GELFAND-P000094

**Diagram and typed description.** $J_\rho(f)$; in the tang. space to $X$.

The point is isolated $\Leftarrow$ it is sufficient that

$$\det(1 - J_\rho(f)) \neq 0.$$

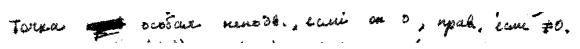
(by the implicit-function theorem).



Original source details: GELFAND-P000094

**Diagram and typed description.** Heavy leftward arrow $\Leftarrow$; $\det(1 - J_\rho(f)) \neq 0$; by the implicit-function theorem.

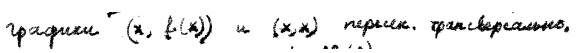
The point [short crossed-out word illegible] singular fixed, if it is $0$, regular, if $\neq 0$.



Original source details: GELFAND-P000094

**Uncertain reading.** After point, a short word is densely cancelled; former letters cannot be recovered. Then singular fixed, if it is $0$, regular, if $\neq 0$.

The graphs $(x, f(x))$ and $(x, x)$ intersect transversally.

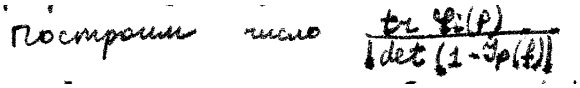


Original source details: GELFAND-P000094

**Diagram and typed description.** $(x, f(x))$ and $(x, x)$; intersect transversally.

Consider the number

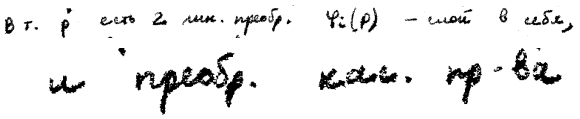
$$\frac{\operatorname{tr} \psi_i(\rho)}{|\det(1 - J_\rho(f))|}.$$



Original source details: GELFAND-P000094

**Diagram and typed description.** $\operatorname{tr} \psi_i(\rho)$ above $|\det(1 - J_\rho(f))|$.

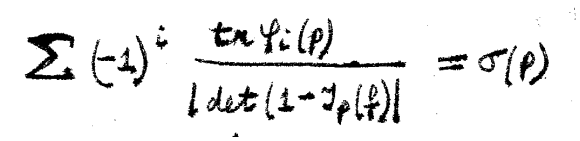
At the point $\rho$ there are $2$ lin. transf. $\psi_i(\rho)$ — the fibre into itself, and a transf. of the tang. space.



Original source details: GELFAND-P000094

**Diagram and typed description.** At the point $\rho$ — $2$ lin. transf.; $\psi_i(\rho)$ — the fibre into itself; transf. of the tang. space.

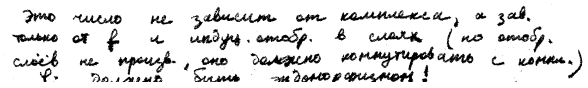
$$\sum (-1)^i \frac{\operatorname{tr} \psi_i(\rho)}{|\det(1 - J_\rho(f))|} = \sigma(\rho).$$



Original source details: GELFAND-P000094

**Diagram and typed description.** $\sum(-1)^i$ with no visible limits; $\operatorname{tr} \psi_i(\rho)$; $|\det(1 - J_\rho(f))|$; $\sigma(\rho)$.

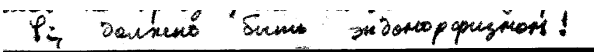
This number does not depend on the complex, but depends only on $f$ and the induced maps in the fibres (but the map of fibres is not arbitrary, it must commute with the complex).



Original source details: GELFAND-P000094

**Diagram and typed description.** Does not depend on the complex; $f$ and induced maps in the fibres; in parentheses, but the map of fibres is not arbitrary, it must commute with the complex.

$\psi_i$ must be an endomorphism!



Original source details: GELFAND-P000094

**Diagram and typed description.** $\psi_i$ must be an endomorphism!

GELFAND-P000095

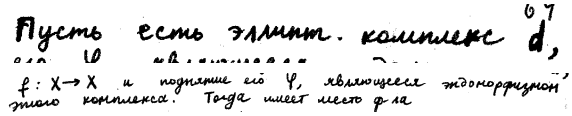
Theorem of Atiyah–Bott.



Original source details: GELFAND-P000095

**Diagram and typed description.** Only Theorem is underlined; Atiyah–Bott; number 64.

Let there be an ellipt. complex $d$, $f : X \to X$ and its lifting $\psi$, which is an endomorphism of this complex. Then the formula holds



Original source details: GELFAND-P000095

**Diagram and typed description.** Ellipt. complex $d$; $f : X \to X$; its lifting $\psi$, which is an endomorphism of this complex.

$$L(T, E) = \sum_{\rho} \sigma(\rho)$$

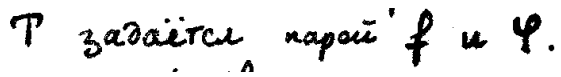
over all fixed points.



Original source details: GELFAND-P000095

**Diagram and typed description.** $L(T, E) = \sum_{\rho} \sigma(\rho)$; over all fixed points.

$T$ is given by the pair $f$ and $\psi$.



Original source details: GELFAND-P000095

**Diagram and typed description.** $T$ is given by the pair $f$ and $\psi$.

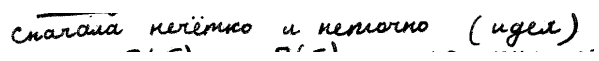
Proof.



Original source details: GELFAND-P000095

**Diagram and typed description.** Proof is underlined.

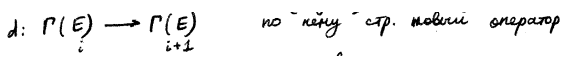
First unclearly[?] and incompletely (ideas)



Original source details: GELFAND-P000095

**Uncertain reading.** First word after First: unclearly[?] / inexactly[?]; crowded middle strokes.

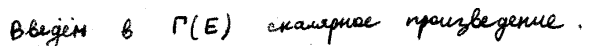
$d : \Gamma(E)_i \longrightarrow \Gamma(E)_{i+1}$, construct a new operator from it.




Original source details: GELFAND-P000095

**Diagram and typed description.** $d : \Gamma(E)_i \longrightarrow \Gamma(E)_{i+1}$; indices $i$, $i+1$ are outside the parentheses; construct a new operator from it.

Introduce a scalar product on $\Gamma(E)$.

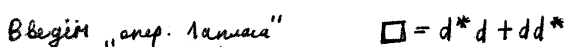



Original source details: GELFAND-P000095

**Diagram and typed description.** Introduce a scalar product on $\Gamma(E)$.

Introduce the "Laplacian oper."

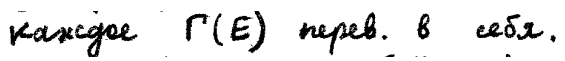
$$\Box = d^*d + dd^*.$$




Original source details: GELFAND-P000095

**Diagram and typed description.** Laplacian oper.; $\Box = d^*d + dd^*$.

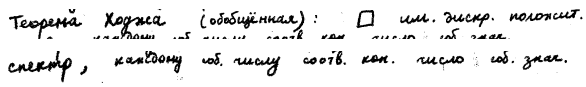
Maps each $\Gamma(E)$ into itself.




Original source details: GELFAND-P000095

**Diagram and typed description.** Maps each $\Gamma(E)$ into itself.

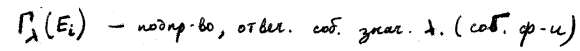
Hodge theorem (generalized): $\Box$ has a discr. positive spectrum, to each eig. number corresponds a finite number of eig. values.




Original source details: GELFAND-P000095

**Diagram and typed description.** Discr. positive spectrum; to each eig. number corresponds a finite number of eig. values.

$\Gamma_\lambda(E_i)$ — subspace corresp. to the eig. value $\lambda$. (eig. functions)




Original source details: GELFAND-P000095

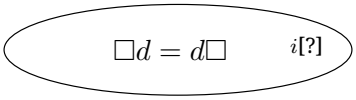
**Diagram and typed description.** $\Gamma_\lambda(E_i)$; subspace corresp. to the eig. value $\lambda$; eig. functions.

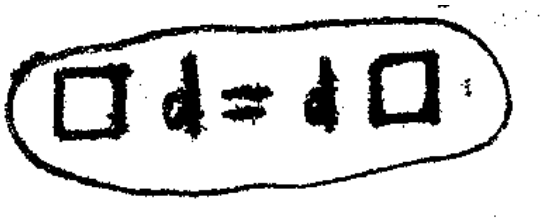
$$d : \Gamma_\lambda(E_i) \longrightarrow \Gamma_\lambda(E_{i+1}).$$




Original source details: GELFAND-P000095

**Diagram and typed description.** $d : \Gamma_\lambda(E_i) \longrightarrow \Gamma_\lambda(E_{i+1})$.

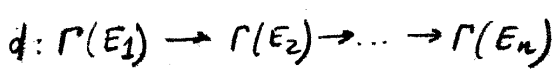



Original source details: GELFAND-P000095

**Uncertain reading.** $\Box d = d\Box$ in an oval; a separate small $i$-like stroke[?] is to the right of the last square.

$$d : \Gamma(E_1) \longrightarrow \Gamma(E_2) \longrightarrow \cdots \longrightarrow \Gamma(E_n).$$

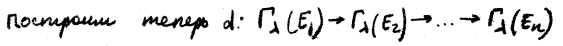


Original source details: GELFAND-P000095

**Diagram and typed description.** $d : \Gamma(E_1) \to \Gamma(E_2) \to \cdots \to \Gamma(E_n)$.

Now consider

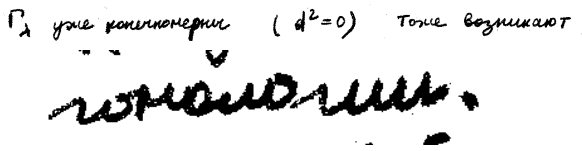
$$d : \Gamma_\lambda(E_1) \longrightarrow \Gamma_\lambda(E_2) \longrightarrow \cdots \longrightarrow \Gamma_\lambda(E_n).$$



Original source details: GELFAND-P000095

**Diagram and typed description.** Now consider; $d : \Gamma_\lambda(E_1) \to \Gamma_\lambda(E_2) \to \cdots \to \Gamma_\lambda(E_n)$.

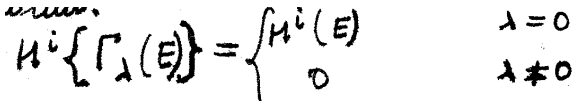
$\Gamma_\lambda$ are already finite-dimensional ($d^2 = 0$), homologies arise here too.



Original source details: GELFAND-P000095

**Diagram and typed description.** $\Gamma_\lambda$ are already finite-dimensional; $(d^2 = 0)$; homologies arise here too.

$$H^i\{\Gamma_\lambda(E)\} = \begin{cases} H^i(E), & \lambda = 0, \\ 0, & \lambda \neq 0. \end{cases}$$



Original source details: GELFAND-P000095

**Diagram and typed description.** $H^i\{\Gamma_\lambda(E)\}$; $H^i(E)$ at $\lambda = 0$, $0$ at $\lambda \neq 0$.

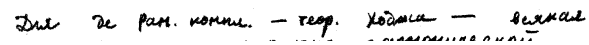
(indices exist only for defin.[?] oper.)



Original source details: GELFAND-P000095

**Uncertain reading.** Abbreviation after for: defin.[?] / diffin.[?]; unexpanded.

For the de Rham compl. — Hodge theorem — every [crossed-out expression illegible]



Original source details: GELFAND-P000095

**Diagram and typed description.** For the de Rham compl. — Hodge theorem — every.

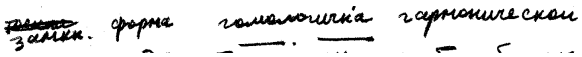
[Crossed-out expression illegible] closed form homologous to a harmonic one.



**Uncertain reading.** Two densely crossed-out expressions: after every and before closed; former letters undetermined.



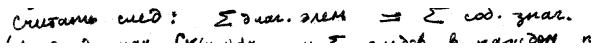
Original source details: GELFAND-P000095



Original source details: GELFAND-P000095

**Diagram and typed description.** Closed form homologous to a harmonic one; two short rightward arrows under homologous.

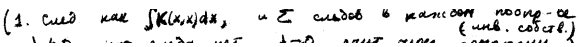
Calculate the trace: $\sum$ diag. elem. $\equiv \sum$ eig. values.



Original source details: GELFAND-P000095

**Diagram and typed description.** Calculate the trace; $\sum$ diag. elem. $\equiv \sum$ eig. values; three horizontal bars.

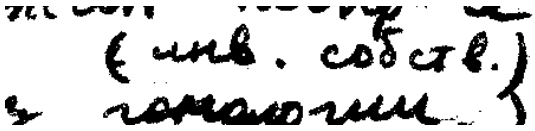
(1. Trace as $\int K(x,x)\,dx$, and $\sum$ of traces in each subspace



Original source details: GELFAND-P000095

**Diagram and typed description.** (1.; trace as $\int K(x,x)\,dx$; and $\sum$ of traces in each subspace.

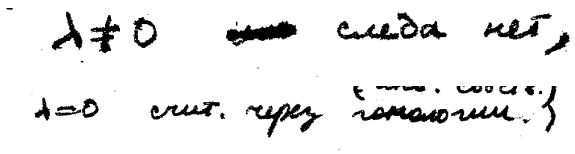
(inv. eig.)



Original source details: GELFAND-P000095

**Diagram and typed description.** Inv. eig. in parentheses.

$\lambda \neq 0$ [short sign densely crossed out, illegible] no trace, $\lambda = 0$ calc. through homologies.
}[?]

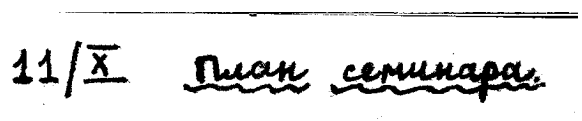


Original source details: GELFAND-P000095

**Uncertain reading.** After $\lambda \neq 0$, a short sign / arrow[?] is densely cancelled. No trace is legible, then $\lambda = 0$ calc. through homologies. A large closing bracket-like sign at right: brace / parenthesis[?].

GELFAND-P000096

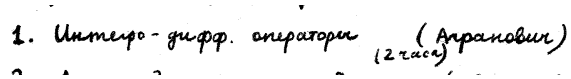
$11/\overline{\underline{X}}$ Seminar plan.



Original source details: GELFAND-P000096

**Diagram and typed description.** $11/\overline{\underline{X}}$; upper and lower bars at $X$; Seminar and plan have wavy underlining.

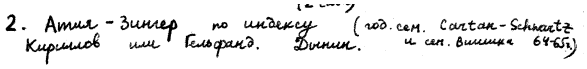
1. Integro-diff. operators (Agranovich). (2 hours)



Original source details: GELFAND-P000096

**Diagram and typed description.** Integro-diff. operators; Agranovich; beneath the parenthesis, (2 hours).

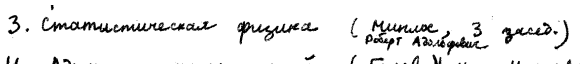
2. Atiyah–Singer on the index. Kirillov or Gelfand. Dynin. (yearly sem. Cartan-Schwartz and Vishik's sem. 64–65.)



Original source details: GELFAND-P000096

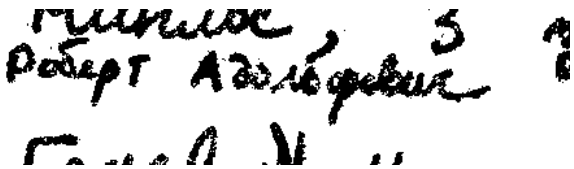
**Diagram and typed description.** Atiyah–Singer on the index; Kirillov or Gelfand; Dynin; yearly sem. Cartan-Schwartz and Vishik's sem. 64–65.

3. Statistical physics ( Minlos , 3 meetings).
Robert Adl'fovich[?]



Original source details: GELFAND-P000096

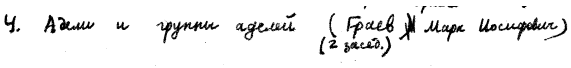
**Diagram and typed description.** Statistical physics; Minlos; 3 meetings; first name and patronymic below the surname.



Original source details: GELFAND-P000096

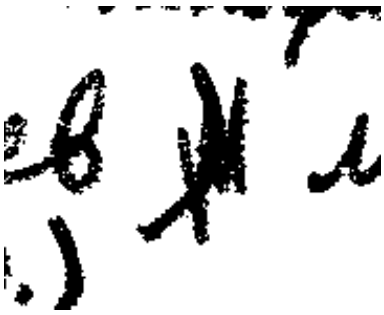
**Uncertain reading.** Below Minlos: Robert Adl'fovich[?] / Robert Adol'fovich[?]; the small loop after d is uncertain.

4. Adeles and adele groups (Graev [crossed-out M-like sign, illegible] Mark Iosifovich). (2 meetings)



Original source details: GELFAND-P000096

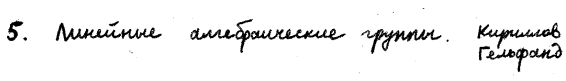
**Diagram and typed description.** Adeles and adele groups; Graev Mark Iosifovich; (2 meetings).



Original source details: GELFAND-P000096

**Uncertain reading.** Between Graev and Mark, an M-like sign is densely cancelled; former reading undetermined.

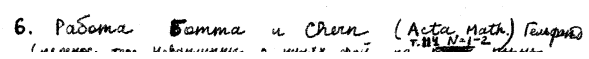
5. Linear algebraic groups. Kirillov Gelfand



Original source details: GELFAND-P000096

**Diagram and typed description.** Linear algebraic groups; Kirillov / Gelfand in a column at right.

6. Work of Bott and Chern (Acta Math.) Gelfand Vol. 114[?] $N = 1–2$



Original source details: GELFAND-P000096

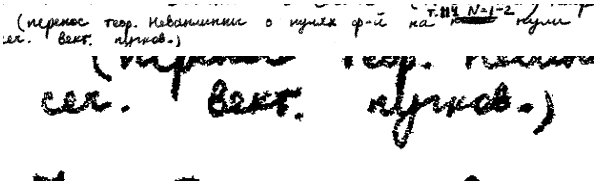
**Diagram and typed description.** Work of Bott and Chern; Acta Math.; Gelfand; volume notation below the journal name.



Original source details: GELFAND-P000096

**Uncertain reading.** Dense volume signs: Vol. 114[?] / Vol. III[?]; $N = 1–2$. A short sign / word is crossed out below.

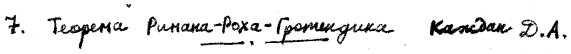
(transfer[?] of Nevanlinna's theorem on zeros of funct. to [densely crossed-out fragment illegible] curves sect. of vect. sheaves.)



Original source details: GELFAND-P000096

**Uncertain reading.** First word in parentheses transfer[?] / transferred[?]. A fragment is densely cancelled before curves; then sect. of vect. sheaves.

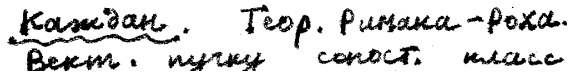
7. Riemann–Roch–Grothendieck theorem. Kazdan[?] D. A.



Original source details: GELFAND-P000096

**Uncertain reading.** Two straight underlines below parts of the title: the ann–Roch area and the start of Grothendieck; precise boundaries uncertain. Surname Kazdan[?] / Kazhdan[?].

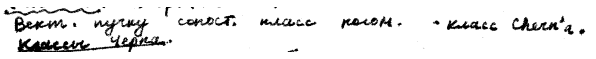
Kazdan[?]. Riemann–Roch theorem.



Original source details: GELFAND-P000096

**Uncertain reading.** The surname with a wavy underline: Kazdan[?] / Kazhdan[?]; dense sign between a and d.

To a vect. sheaf corresp. a cohom. class — Chern class.



Original source details: GELFAND-P000096

**Diagram and typed description.** To a vect. sheaf corresp. a cohom. class; Chern class, the Latin surname with a Russian ending in the source.

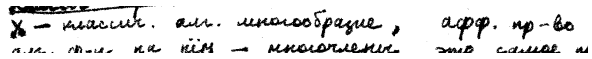
Chern classes.



Original source details: GELFAND-P000096

**Diagram and typed description.** Chern classes is underlined; the start of classes is written densely.

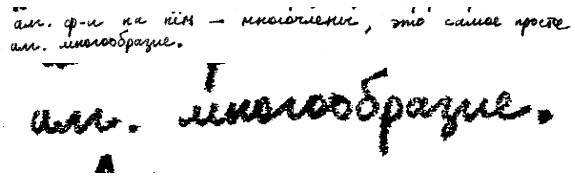
$X$ — classical alg. variety, aff. space



Original source details: GELFAND-P000096

**Diagram and typed description.** $X$ — classical alg. variety, aff. space.

alg. funct. on it — polynomials, this is the simplest[?] alg. variety.



**Uncertain reading.** Alg. funct. on it — polynomials; this is the simplest[?] / this is the most simple[?]; ending of the last word dense; then alg. variety.

Original source details: GELFAND-P000096

Affine alg. varieties. Let a collection of polynomials be given

$$p_1(x_1, \ldots, x_n), \ldots, p_k(x_1, \ldots, x_n)$$

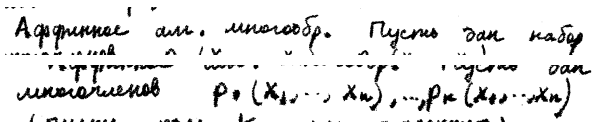


**Diagram and typed description.** Affine alg. varieties; collection of polynomials $p_1(x_1, \ldots, x_n), \ldots, p_k(x_1, \ldots, x_n)$.

Original source details: GELFAND-P000096

(let the field $K$ be alg. closed)

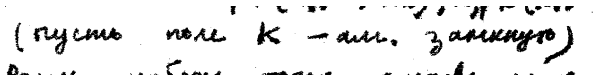


**Diagram and typed description.** Let the field $K$ be alg. closed; uppercase $K$.

Original source details: GELFAND-P000096

Consider collections of points — satisfy an eqn., aff. alg. var.-

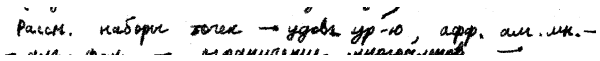


**Diagram and typed description.** Consider collections of points; satisfy an eqn.; aff. alg. var.- with a final dash.

Original source details: GELFAND-P000096

— alg. funct. — restrictions of polynomials —

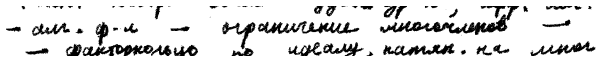


**Diagram and typed description.** — Alg. funct. — restrictions of polynomials —.

Original source details: GELFAND-P000096

— quotient ring by the ideal spanned by polyn.

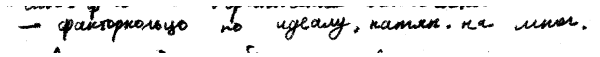


**Diagram and typed description.** Quotient ring by the ideal spanned by polyn.

Original source details: GELFAND-P000096

Alg. prevariety — space loc. isomorphic to an aff. variety.

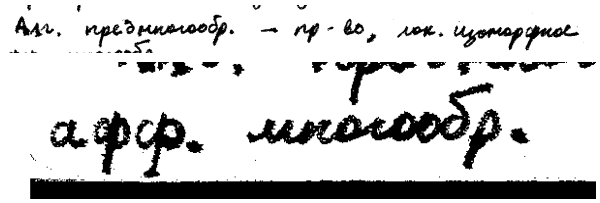


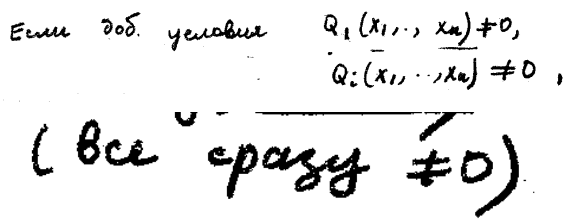
**Diagram and typed description.** Alg. prevariety; space loc. isomorphic to an aff. variety.

Original source details: GELFAND-P000096

GELFAND-P000097

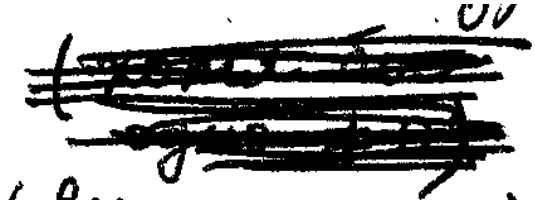
If add. conditions

$$Q_1(x_1, \ldots, x_n) \neq 0,$$
$$Q_i(x_1, \ldots, x_n) \neq 0,$$




**Uncertain reading.** $Q_1$, $Q_i$; $x_n$) underlined in the first line. Between the lines below $Q_1$, a dot ·, and a short dash – farther right; their role is undetermined. At right, (all at once $\neq 0$).

Original source details: GELFAND-P000097



**Uncertain reading.** Two lines at right are densely crossed out; an initial parenthesis, loops, horizontal and slanting strokes remain visible; no reliable reading established.

Original source details: GELFAND-P000097

(all at once $\neq 0$) then also an alg. var.




**Diagram and typed description.** Then also an alg. var.

Original source details: GELFAND-P000097

funct. — from rat. funct. with poles outside these points.

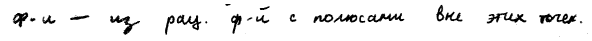



**Diagram and typed description.** Funct. — from rat. funct. with poles outside these points.

Original source details: GELFAND-P000097

Example: line, alg. funct. — polynomials.

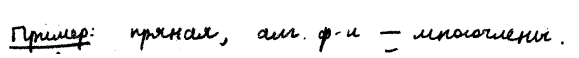



**Diagram and typed description.** Example is underlined; line, alg. funct. — polynomials.

Original source details: GELFAND-P000097

Space given with a stock of funct.

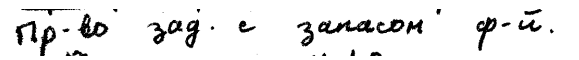
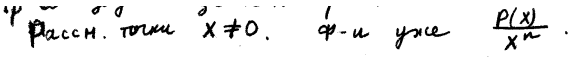



**Diagram and typed description.** Space given with a stock of funct.

Original source details: GELFAND-P000097

Consider points $x \neq 0$. Funct. now $\frac{P(x)}{x^m}$[?].




**Uncertain reading.** Consider points $x \neq 0$; fraction $\frac{P(x)}{x^m}$[?]. The denominator exponent is $m$- / $n$-like and written densely.

Original source details: GELFAND-P000097

Aff. var. — zeros of some funct. and others do not turn into $0$.

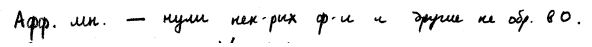
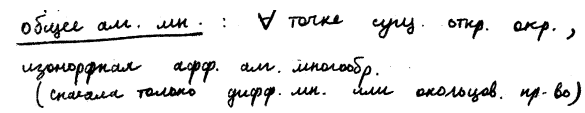



**Diagram and typed description.** Aff. var. — zeros of some funct. and others do not turn into $0$.

Original source details: GELFAND-P000097

General alg. var.: ∀ point there exists an open nbhd., isomorphic to an aff. alg. variety. (at first only a diff. var. or ringed space)



Original source details: GELFAND-P000097

**Diagram and typed description.** General alg. var. underlined; ∀ point, open nbhd., isomorphic to an aff. alg. variety; below, parentheses concerning a diff. var. or ringed space.

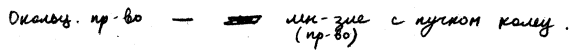
Ringed space — [short crossed-out fragment illegible]



Original source details: GELFAND-P000097

**Uncertain reading.** Between the dash and var., a short sign / word is densely cancelled; reading undetermined.

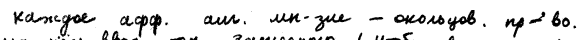
var. with a sheaf of rings.
(space)



Original source details: GELFAND-P000097

**Diagram and typed description.** Var. with a sheaf of rings; (space) inserted below var.

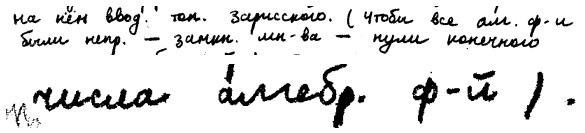
Every aff. alg. var. — ringed space.



Original source details: GELFAND-P000097

**Diagram and typed description.** Every aff. alg. var. — ringed space.

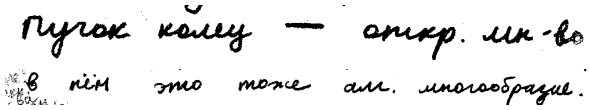
On it one introduces the Zariski top. (So that all alg. funct. are cont. — closed sets — zeros of a finite number of alg. funct. /[?].



Original source details: GELFAND-P000097

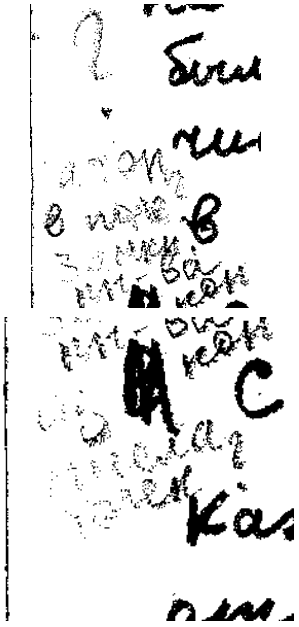
**Uncertain reading.** Zariski top.; opening parenthesis before So that; after alg. funct., a slash /[?] / closing parenthesis )[?], then a period.

Sheaf of rings — an open set in it this is also an alg. variety.



Original source details: GELFAND-P000097

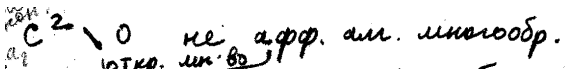
**Diagram and typed description.** Sheaf of rings — an open set in it this is also an alg. variety.



Original source details: GELFAND-P000097

**Uncertain reading.** In the left margin, faint, dotted and loop-like traces; before the following formula, a dense A-like sign. No reliable complete words read; origin of the traces undetermined.

$C^2 \setminus 0$ is not an aff. alg. variety.

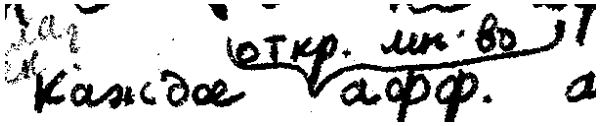


Original source details: GELFAND-P000097

**Diagram and typed description.** $C^2 \setminus 0$ is not an aff. alg. variety; written $C$, without an additional vertical bar.

open set

Every aff. alg. variety[?] cont... [ending crossed out]



Original source details: GELFAND-P000097

**Diagram and typed description.** Above aff., open set is inserted with a lower brace and a downward point; insertion position retained.



Original source details: GELFAND-P000097

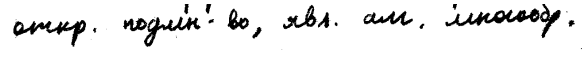
**Uncertain reading.** Ending of the word variety: handwritten -iya[?] / -iye[?]; last sign ya-/e-like.



Original source details: GELFAND-P000097

**Uncertain reading.** Cont...: the ending is densely crossed out; zhit-like letter tops remain above the strokes[?]. Possible whole word contains[?]; cancelled letters not securely determined.

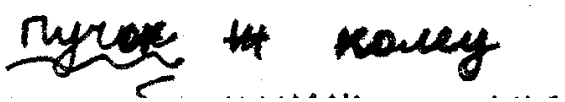
open subset, being an alg. variety.



Original source details: GELFAND-P000097

**Diagram and typed description.** Open subset, being an alg. variety.

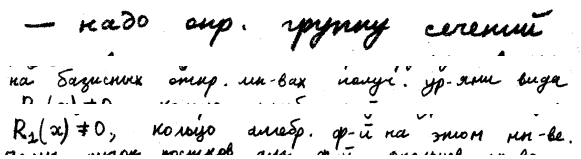
Sheaf [n-/i-like sign, illegible] of rings — must def. the group of sections



Original source details: GELFAND-P000097

**Uncertain reading.** Sheaf has a wavy underline. Before of rings, a separate dense n-/i-like sign; whether cancellation or insertion is undetermined.

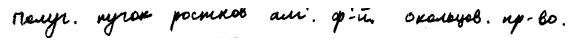
on basic open sets obtained by eqns. of the form $R_2(x) \neq 0$[?], the ring of alg. funct. on this set.



Original source details: GELFAND-P000097

**Uncertain reading.** The group of sections on basic open sets; eqns. of the form $R_2(x) \neq 0$[?] / $R_1(x) \neq 0$[?]; subscript with a curved top and a long base, 2-like. Then the ring of alg. funct. on this set.

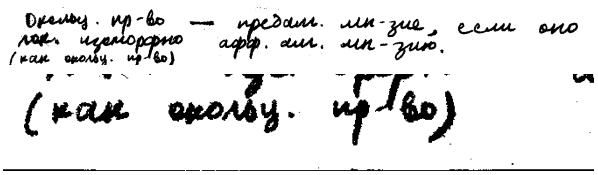
Obtain a sheaf of germs of alg. funct. ringed space.



Original source details: GELFAND-P000097

**Diagram and typed description.** Obtain a sheaf of germs of alg. funct. ringed space; no separate comma visible after funct.

Ringed space — prealg.[?] var., if it is loc. isomorphic to an aff. alg. var. (as a ringed space)



Original source details: GELFAND-P000097

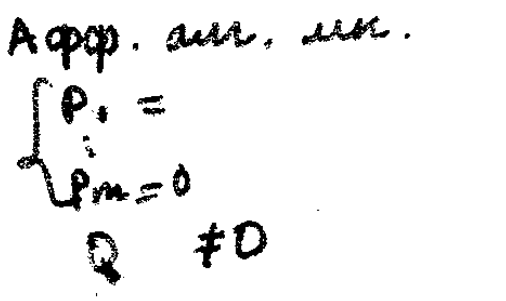
**Uncertain reading.** Abbreviation after the dash prealg.[?] / prev.[?]; dense letters after pred. In the next line loc. is written densely; then isomorphic to an aff. alg. var.; below, (as a ringed space).

GELFAND-P000098

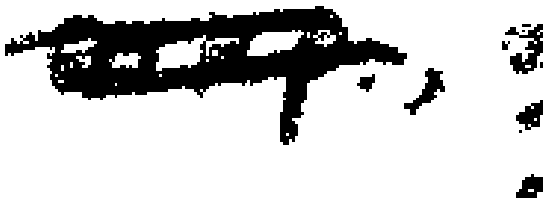
Aff. alg. var.

$$\begin{cases} p_1 = \\ \vdots \\ p_m = 0 \end{cases}$$
$$Q \neq 0$$

[Crossed out: open[?], remainder illegible] closed set.



**Uncertain reading.** Under the left brace, $p_1 =$ with no visible right-hand side, dots, $p_m = 0$; below, $Q \neq 0$. At right, dense cancellation before closed set; otkr.-like remnants (open)[?], full reading undetermined.



Original source details: GELFAND-P000098

$R_i(x_1, \ldots, x_n) = 0$. $R_i$ — polynomials.

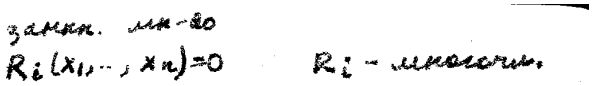


**Diagram and typed description.** Closed set; $R_i(x_1, \ldots, x_n) = 0$, $R_i$ — polynomials.

Original source details: GELFAND-P000098

Basis of open sets — one condition $U_S : S \neq 0$. (such is cont. in every open — — leave $1$ condition).

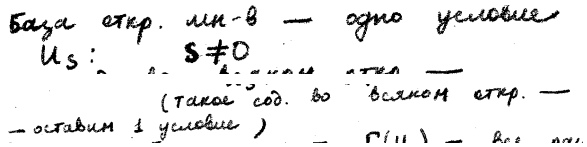


**Diagram and typed description.** $U_S : S \neq 0$; in parentheses two separate dashes: after open and before leave $1$ condition.

Original source details: GELFAND-P000098

The group of sections is given thus —

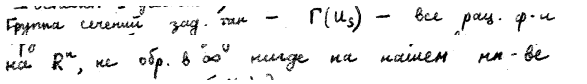


**Diagram and typed description.** The group of sections is given thus —; then $\Gamma(U_S)$ and rat. funct. on $R^n$. Written $R$, without an additional vertical bar.

Original source details: GELFAND-P000098

$\Gamma(U_S)$ — all rat. funct. on $R^n$ which do not turn into $\infty$ anywhere on our set.

$$\Gamma(U_S) = [?] \left\{ \frac{P(x)}{Q(x)} \right\}$$



**Uncertain reading.** Between $\Gamma(U_S)$ and the brace, two horizontal bars with dense overwriting between them: $=$ with alteration / rewritten sign; $\neq$ remains an unestablished alternative. No definite diagonal stroke read.

Original source details: GELFAND-P000098

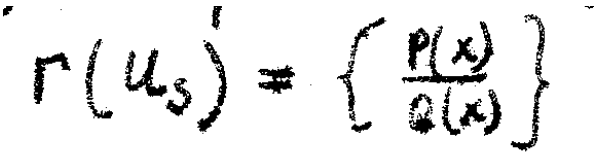


**Diagram and typed description.** $\Gamma(U_S)$, overwritten sign, $\left\{ \frac{P(x)}{Q(x)} \right\}$; no separate period visible after the right brace.

Original source details: GELFAND-P000098

Identify, if on $U_S$ one value. This is a ring.

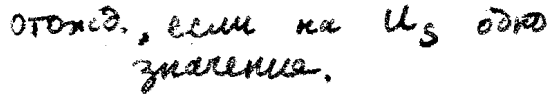
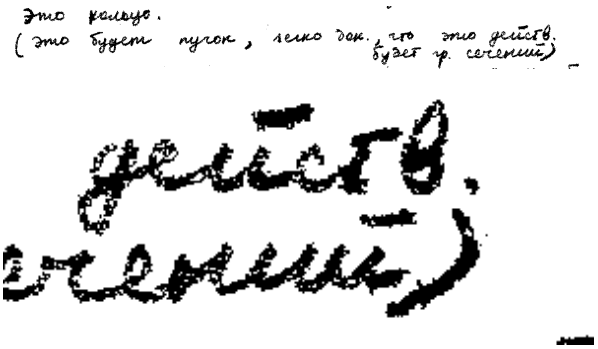


**Diagram and typed description.** Identify, if on $U_S$ one value.

Original source details: GELFAND-P000098

(This will be a sheaf, easy to prove that this will really be the gr. of sections.)



**Uncertain reading.** This is a ring. In parentheses: This will be a sheaf, easy to prove that this will really be the gr. of sections. A short horizontal stroke lies below the final letter of the translated really; its role is undetermined. The upper breve belongs to the source letter y-short.

Original source details: GELFAND-P000098

Isom. of sheaves. obtained by the definition of an isomorphism

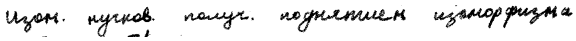


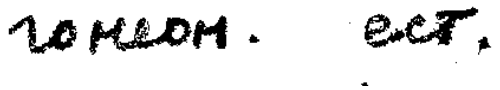
**Diagram and typed description.** Isom. of sheaves. obtained by the definition of an isomorphism; period after sheaves.

Original source details: GELFAND-P000098

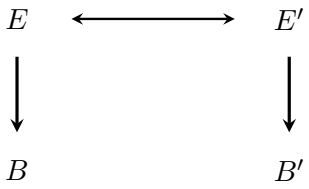
of bases[?] homeom. nat.

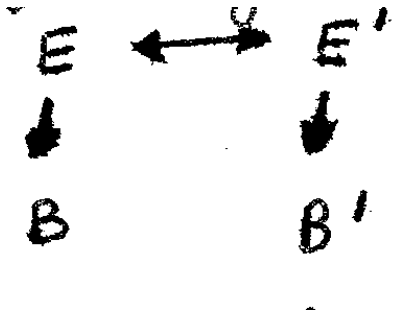


**Uncertain reading.** At the line end, baz.[?] / bez.[?]; the middle letter has a loop, the last a descending stroke. Next line: homeom. nat.; the abbreviation remains uncertain.



Original source details: GELFAND-P000098





**Uncertain reading.** Top row $E \leftrightarrow E'$, two heavy downward arrows to $B$, $B'$. The upper ends and arrowheads are dense; their exact outlines are undetermined. No horizontal arrow visible between $B$ and $B'$.

Original source details: GELFAND-P000098

If $E$ is a subsheaf of funct., then $B \longrightarrow B'$ induces $E' \longrightarrow E$.

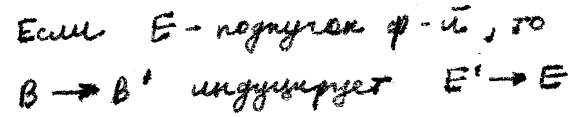


**Diagram and typed description.** In the line, $B \longrightarrow B'$ induces $E' \longrightarrow E$; the prime strokes are raised at the second letters.

Original source details: GELFAND-P000098

(A priori only commut.) Require that the isom. be an isom. arising from considerations of the base. (natural!) simply coincidence!

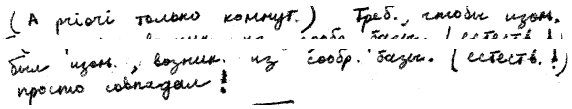
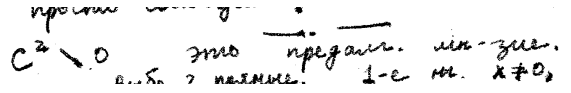


**Diagram and typed description.** (A priori only commut.) Require that the isom. be an isom. arising from considerations of the base. (natural!) simply coincidence!

Original source details: GELFAND-P000098

$C^2 \setminus 0$ is a prealg. var.



**Uncertain reading.** $C^2 \setminus 0$ is a prealg. var. Above prealg., two separated horizontal strokes, the right higher; a dot beside the lower. Their role and association with the words are undetermined. Written $C$, without an additional vertical bar.

Original source details: GELFAND-P000098

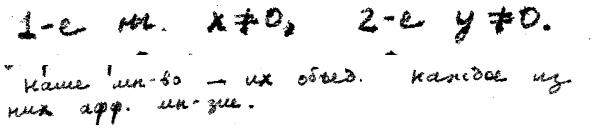
Choose 2 straight lines[?].



Original source details: GELFAND-P000098

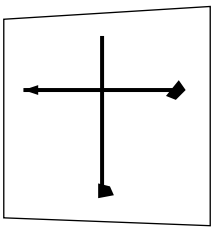
**Uncertain reading.** After Choose 2, cramped pryamye (straight lines)[?] / pryamki[?]; the middle and final strokes are dense.

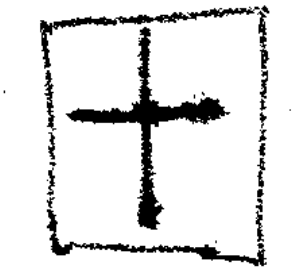
1st on $x \neq 0$, 2nd $y \neq 0$. Our set — their union. Each of them an aff. var.



Original source details: GELFAND-P000098

**Diagram and typed description.** 1st on $x \neq 0$, 2nd $y \neq 0$. Our set — their union; each of them an aff. var.

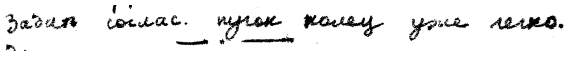




Original source details: GELFAND-P000098

**Uncertain reading.** Within a rectangular frame, heavy vertical and horizontal lines intersect. The left end is pointed, the right and lower ends thickened; the role of the thickenings and exact form of possible arrowheads are undetermined. No letter labels or separate circle visible at the intersection.

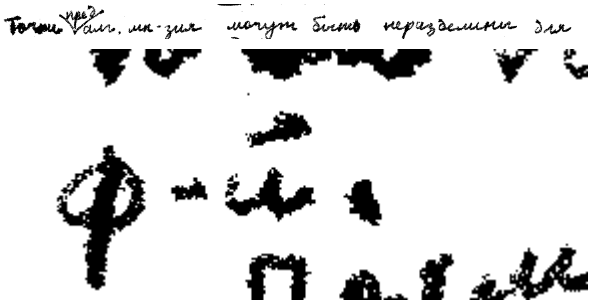
Then agree. sheaves of rings already clear[?].



Original source details: GELFAND-P000098

**Uncertain reading.** Then agree. sheaves of rings already clear[?]; final word clear[?] / easy[?]. Below the line are two separate strokes, under the end of agree. and part of sheaves; exact association of the underlining remains uncertain.

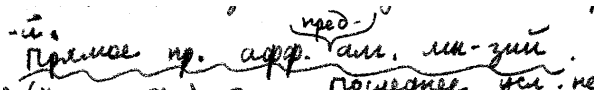
Points of $\overset{\text{pre}}{\vee}$ alg. var. may be inseparable for funct.



Original source details: GELFAND-P000098

**Diagram and typed description.** Above the gap before alg., pre is inserted with a V-shaped insertion mark. Points is written densely; next-line ending funct.

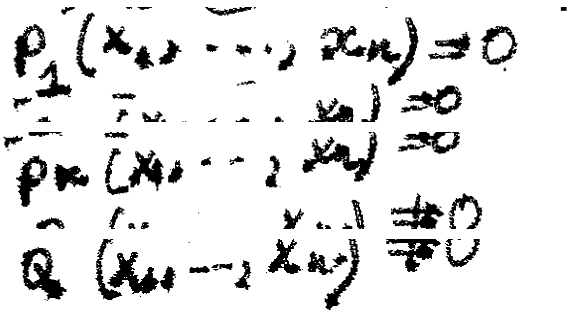
pre-
Example of pr. aff. alg. var.



Original source details: GELFAND-P000098

**Diagram and typed description.** The heading Example of pr. aff. alg. var. has a wavy underline. Above alg., pre- is inserted with a lower curly brace. The abbreviation pr. is not expanded.

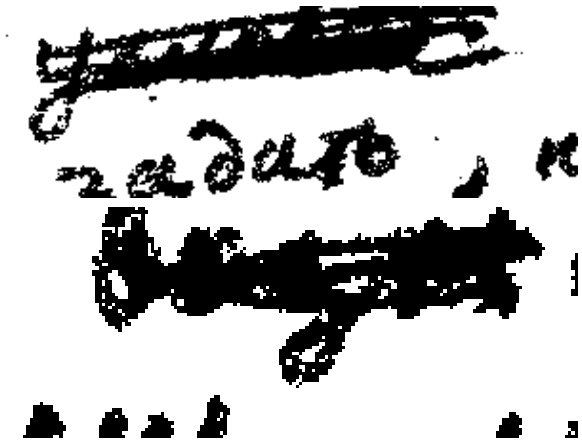
$$p_1(x_1, \ldots, x_n[?]) = 0,$$
$$, \quad - \qquad - [?]$$
$$p_k(x_1, \ldots, x_n[?]) = 0,$$
$$Q(x_1, \ldots, x_n[?]) \neq 0.$$



Original source details: GELFAND-P000098

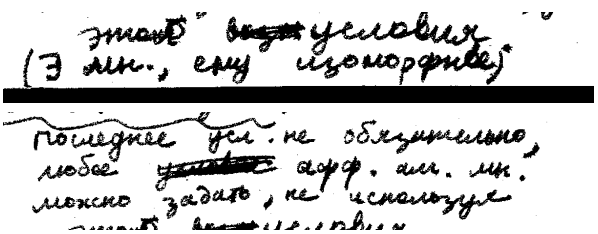
**Uncertain reading.** In the three lines, the final indices $x_n$[?] / $x_m$[?] are dense; in the second line there is an additional upper stroke. The subscript $1$ on the first $p$ has a dense horizontal base. Between the first and second lines, three separate short marks: the left comma-/dot-like, and two dashes; their role is undetermined.

The last cond. is not obligatory, any [crossed-out word illegible] aff. alg. var. can be given without using these [crossed-out word illegible] conditions (∃ var., isomorphic to it).



Original source details: GELFAND-P000098

**Uncertain reading.** After any, a word densely cancelled with long horizontal strokes and a loop at left; below, after these, a second densely crossed-out word. No secure letter-by-letter reading of either established.



Original source details: GELFAND-P000098

**Diagram and typed description.** The last cond. is not obligatory; then without using these ... conditions; below, (∃ var., isomorphic to it). The source ending usloviyakh is retained.

GELFAND-P000099

Take $(x, y)$ with the condition

$$\begin{cases} P_i(x) = 0, \\ Q(x)y - 1 = 0 \end{cases}$$

$$(x_1, \ldots, x_n, y)$$

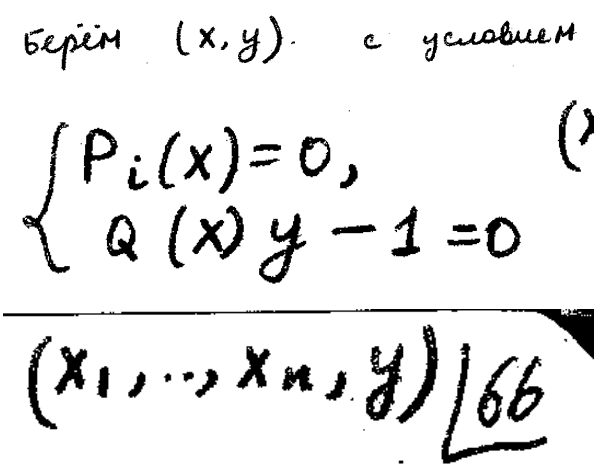


**Diagram and typed description.** System with a left brace: $P_i(x) = 0$, $Q(x)y - 1 = 0$. At the upper right, $(x_1, \ldots, x_n, y)$; the number 66 with an angular stroke to its left and below.

Original source details: GELFAND-P000099

They are isomorphic.



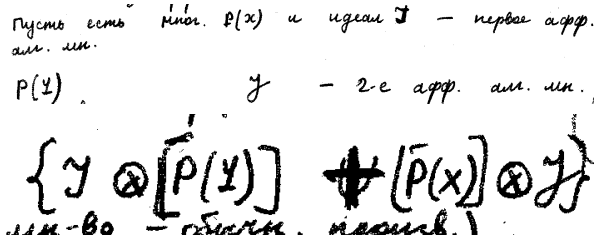
**Diagram and typed description.** They are isomorphic.

Original source details: GELFAND-P000099

Let there be a rng.[?] $P(x)$ and an ideal $I$[?] — the first aff. alg. var. $P(y)$ $J$ — the 2nd aff. alg. var.

Their product — — the ring $P(x, y)$ with the ideal

$$\{I[?] \otimes [P(y)] + [?]\,[P(x)] \otimes J\}$$



**Uncertain reading.** The first ideal $I$[?] / $J$[?]: upper bar and lower left hook, with a lower loop in the formula. Between the summands, a densely overwritten $+$[?] / $\oplus$[?]. The abbreviation before the first ring is kol.[?] / kol'.[?].

Original source details: GELFAND-P000099

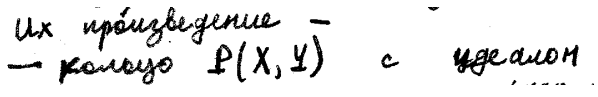


**Diagram and typed description.** Their product — and the following line — the ring $P(x, y)$ with the ideal: two separate dashes.

Original source details: GELFAND-P000099

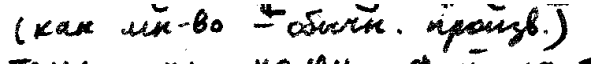
? [?]



Original source details: GELFAND-P000099

**Uncertain reading.** At right, a circled ?-/2-like sign; a separate lower dot below the main curved stroke. Reading undetermined.

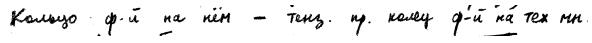
(as a set — the usual prod.)



Original source details: GELFAND-P000099

**Diagram and typed description.** (as a set — the usual prod.).

The ring of funct. on it — the tensor pr. of the rings of funct. on those var.

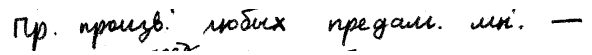


Original source details: GELFAND-P000099

**Diagram and typed description.** The ring of funct. on it — the tensor pr. of the rings of funct. on those var.

Pr. prod. of any prealg. var. — [crossed-out text illegible]

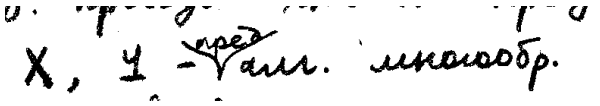




Original source details: GELFAND-P000099

**Diagram and typed description.** Pr. prod. of any prealg. var. —. The abbreviation Pr. is retained.

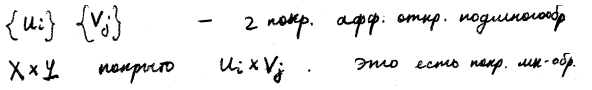
$X, Y$ — $\overset{\text{pre}}{\vee}$ alg. varieties.



Original source details: GELFAND-P000099

**Diagram and typed description.** Above the gap before alg., pre is inserted, with a V-shaped insertion mark below; in the line, $X, Y$.

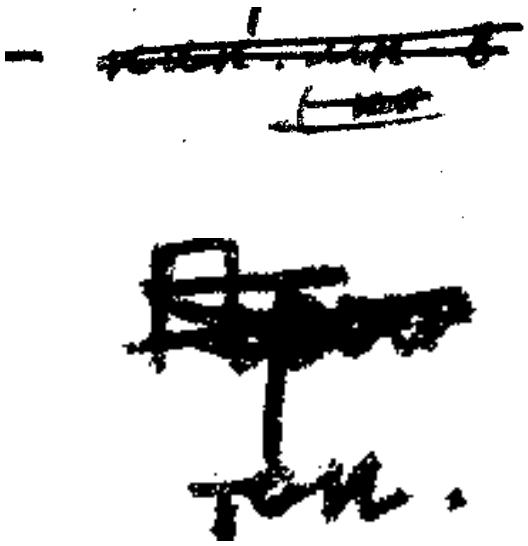
$\{U_i\}$ $\{V_j\}$ — 2 covers by aff. open subvarieties. $X \times Y$ is covered by $U_i \times V_j$. This is a napr.[?] var.



Original source details: GELFAND-P000099

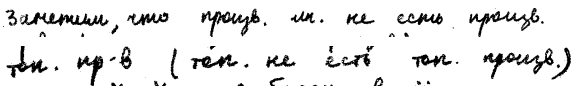
**Uncertain reading.** Upper line: $\{U_i\}$, $\{V_j\}$ — 2 covers by aff. open subvarieties; below, $X \times Y$ is covered by $U_i \times V_j$. After This is, a napr.[?]-like abbreviation, read uncertainly.

[Densely crossed-out initial text illegible] Note that the prod. of var. is not the prod. of top. spaces (the top. is not the top. of the prod.).



Original source details: GELFAND-P000099

**Uncertain reading.** After the dash in the product line, dense cancellation; below, a separate angular stroke and a black mark. Before Note, initial text is densely crossed out. Letter-by-letter readings of the cancellations undetermined.

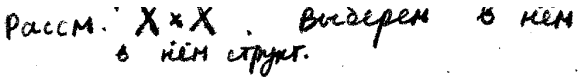


Original source details: GELFAND-P000099

**Diagram and typed description.** Note that the prod. of var. is not the prod. of top. spaces (the top. is not the top. of the prod.).

Consider $X \times X$. Choose in it

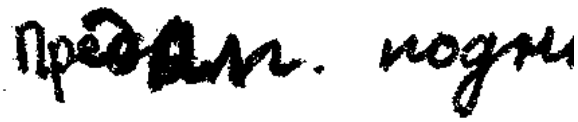
in it a struct.



Original source details: GELFAND-P000099

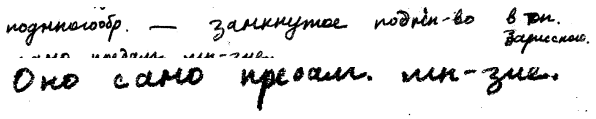
**Diagram and typed description.** In the line: Consider $X \times X$. Choose in it; below the line, separately, in it a struct.

Prealg.[?] subvar. — a closed subset in the top.[?] of Zariski. It is itself a prealg. var.



Original source details: GELFAND-P000099

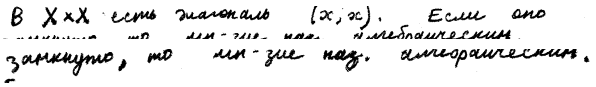
**Uncertain reading.** At the beginning, Prealg.[?]; the middle letters are densely overwritten; earlier strokes unread.



Original source details: GELFAND-P000099

**Uncertain reading.** Subvar. — a closed subset in the top.[?] of Zariski. The source abbreviation tp.[?] / top.[?] is cramped. Next line: It is itself a prealg. var.

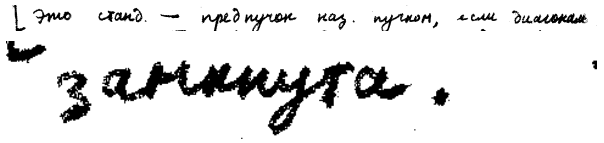
In $X \times X$ there is the diagonal $(x, x)$. If it is closed, the var. is called algebraic.



Original source details: GELFAND-P000099

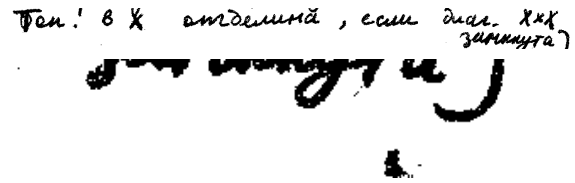
**Diagram and typed description.** In $X \times X$ there is the diagonal $(x, x)$. If it is closed, the var. is called algebraic.

[This is stand. — a presheaf is called a sheaf if the diagonal is closed.
The top. in $X$ is separated if the diag. $X \times X$ is closed.)



Original source details: GELFAND-P000099

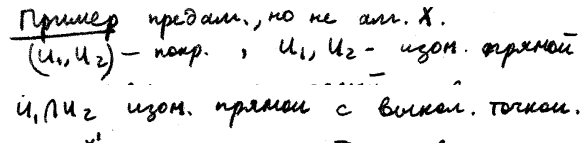
**Uncertain reading.** A presheaf is called a sheaf if the diagonal is closed. The first word is written predpuchok (presheaf); its cramped joined strokes are retained in the adjacent original.



Original source details: GELFAND-P000099

**Uncertain reading.** The top. in $X$ is separated if the diag. $X \times X$ is closed.); the aside begins with a square bracket and ends with a round parenthesis. Association of the unequal brackets undetermined.

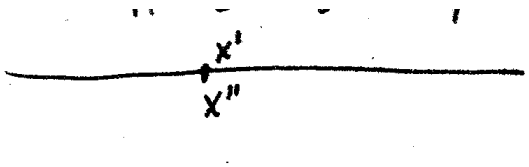
Example of a prealg. but non-alg. $X$. $(U_1, U_2)$ — a cover, $U_1, U_2$ — isom. to a line. $U_1 \cap U_2$ isom. to a line with a point removed.



Original source details: GELFAND-P000099

**Diagram and typed description.** Example has a straight underline; $(U_1, U_2)$, $U_1, U_2$, $U_1 \cap U_2$ and a removed point.

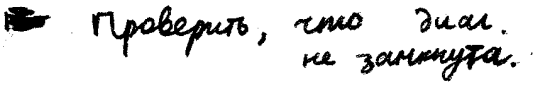


Original source details: GELFAND-P000099

**Uncertain reading.** One nearly horizontal line with a thickened point; above it $x'$, below it $x''$. Exact outline of the thickening undetermined.

Check that the diag. is not closed.

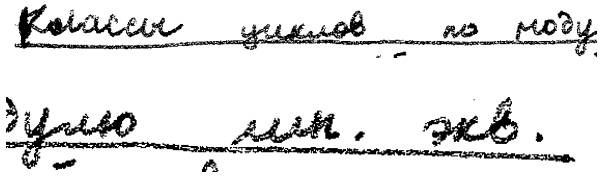


Original source details: GELFAND-P000099

**Uncertain reading.** Before Check, a separate dense black sign with two rightward projections; its role and exact outline undetermined. Check that the diag. is not closed.

GELFAND-P000100

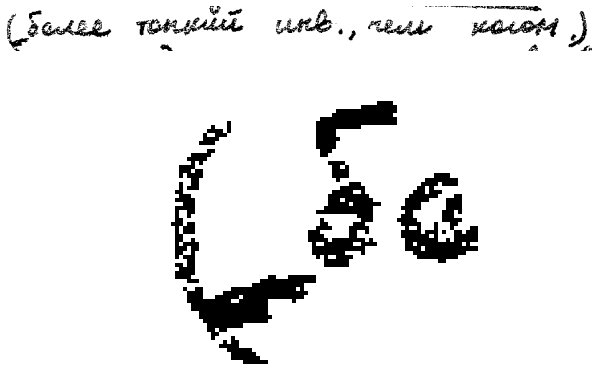
Classes of cycles modulo lin. equiv.



Original source details: GELFAND-P000100

**Diagram and typed description.** The entire heading Classes of cycles modulo lin. equiv. has a straight underline.

[a more precise inv. than cohom.)



**Uncertain reading.** A more precise inv. than cohom.; the opening bracket is angular, with a lower rightward foot, and its reading as [ / ( is not fully established; the closing parenthesis is round.

Original source details: GELFAND-P000100

$X$ — an alg. variety, subvar. — a closed set in the top. of Zariski.

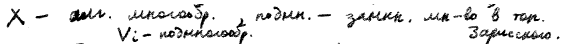


**Diagram and typed description.** $X$ — an alg. variety; subvar. — a closed set in the top. of Zariski.

Original source details: GELFAND-P000100

$V_i$ — subvarieties.

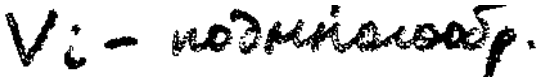
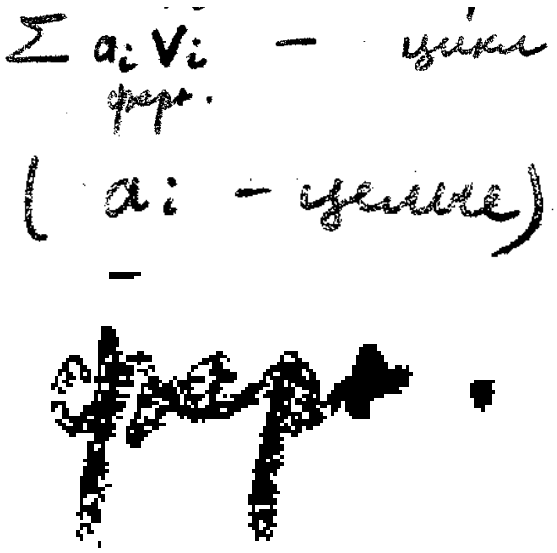


**Diagram and typed description.** $V_i$ — subvarieties; lower index $i$.

Original source details: GELFAND-P000100

$\sum_{\text{for.[?]}} a_i V_i$ — cycles $\quad (a_i$ — integers$)$

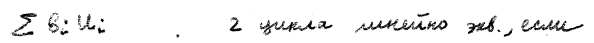


**Uncertain reading.** Below the summand, the short source annotation for.[?] / form.[?]; after the distinct f-, o- and r-shaped letters, a cramped upper exit stroke. Formula $\sum a_i V_i$ and, at right, $(a_i$ — integers$)$.

Original source details: GELFAND-P000100

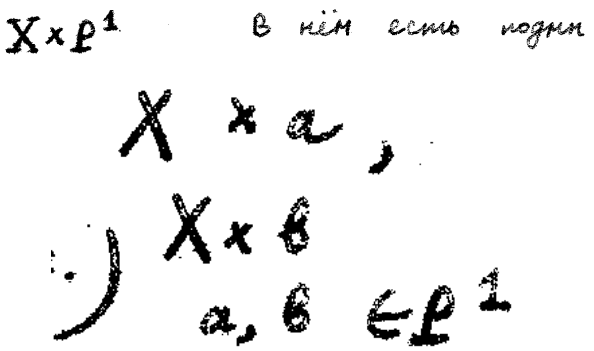
$\sum b_i U_i$. 2 cycles are linearly equiv. if

$X \times P^1$ in it there are subvar. $X \times a$, $X \times b$ $a, b \in P^1$



Original source details: GELFAND-P000100

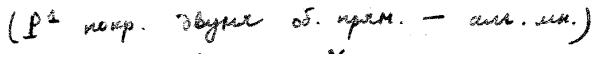
**Diagram and typed description.** $\sum b_i U_i$. 2 cycles are linearly equiv. if; a period after $U_i$.



Original source details: GELFAND-P000100

**Diagram and typed description.** $X \times P^1$; the words in it there are subvar.; at right $X \times a$, below it $X \times b$ and $a, b \in P^1$.

($P^1$ cov. by two ord. lines — an alg. var.)



Original source details: GELFAND-P000100

**Diagram and typed description.** ($P^1$ cov. by two ord. lines — an alg. var.); the source abbreviation ob. and both parentheses.

They are can. isom. to $X$.



Original source details: GELFAND-P000100

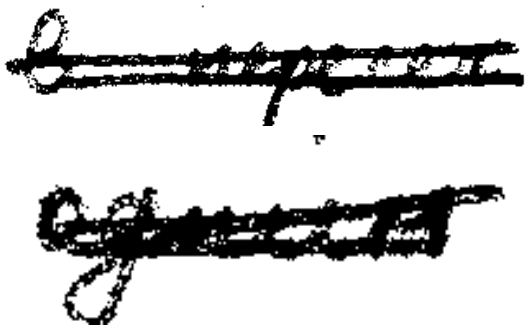
**Uncertain reading.** At left, a black vertical star-/letter-like mark with projections. Its approximate form is reproduced; letter reading and purpose undetermined.



Original source details: GELFAND-P000100

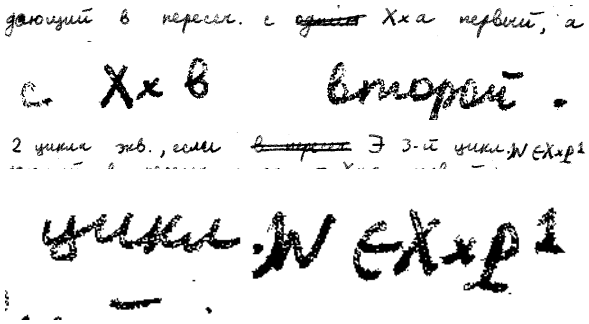
**Diagram and typed description.** They are can. isom. to $X$.

2 cycles are equiv. if [crossed out: illegible] ∃ a 3rd cycle, $W[?] \in X \times P^1$, giving in intersect. with [crossed out: illegible] $X \times a$ the first, and with $X \times b$ the second.



Original source details: GELFAND-P000100

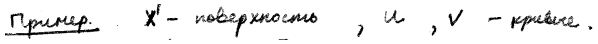
**Uncertain reading.** After if, an expression with an initial v-like stroke is cancelled; after with, another word with a lower loop is cancelled. Full readings of the cancellations undetermined.



Original source details: GELFAND-P000100

**Uncertain reading.** ∃ a 3rd cycle, $W[?] \in X \times P^1$; the letter $W[?]$ has cramped ascending strokes and may look N-like. A short lower comma follows cycle. The continuation with $X \times a$ and $X \times b$ occupies three lines.

Example. $X$ — a surface, $U, V$ — curves.



Original source details: GELFAND-P000100

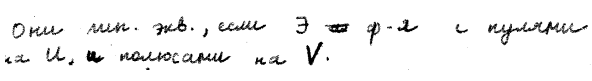
**Diagram and typed description.** Example. is underlined including the period; $X$ — a surface, $U, V$ — curves.

They are lin. equiv. if ∃ [crossed-out sign illegible] a funct. with zeros on $U$ and poles on $V$.



Original source details: GELFAND-P000100

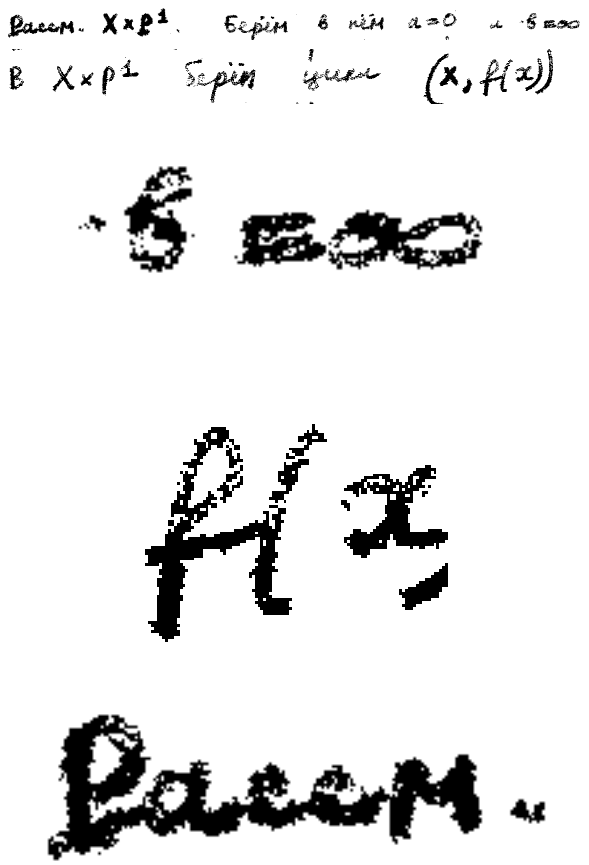
**Uncertain reading.** Between the existence sign and funct., a small dense sign: a closed horizontal loop with a transverse stroke; letter reading undetermined.



Original source details: GELFAND-P000100

**Diagram and typed description.** They are lin. equiv. if ∃ ... a funct. with zeros on $U$ and poles on $V$.

Consider $X \times P^1$. Take in it $a = 0$ and $b =$[?] $\infty$. In $X \times P^1$ take the cycle $(x, f(x))$



**Uncertain reading.** Before $\infty$, after $b$, a dense short horizontal mark, $=$[?]; two separate bars cannot be distinguished confidently. On the next line, $(x, f(x))$: a tall $f$ stroke with a crossbar. The initial source Rassm. (Consider) has dense lower strokes; a separate underline is not established.

Original source details: GELFAND-P000100

Def. of Chern[?] classes (alg. varieties)



**Uncertain reading.** In the current word Chern[?], inner and final strokes are densely overwritten; earlier letters are not separable.

Original source details: GELFAND-P000100

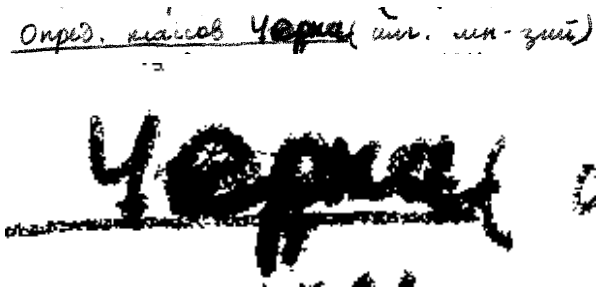


**Diagram and typed description.** A straight line runs under Def. of classes and the name Chern; at right, (alg. varieties).

Original source details: GELFAND-P000100

They are giv. with fin.[?] axioms.



Original source details: GELFAND-P000100

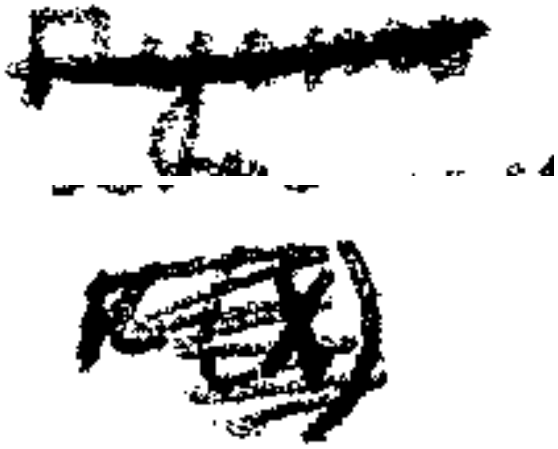
**Uncertain reading.** Before axioms, a short source kon.[?] / nek.[?], corresponding to fin.[?] / some[?]; cramped middle joins do not establish letter order.



Original source details: GELFAND-P000100

**Diagram and typed description.** They are giv. with ... axioms.; a period at the end of the line.

[Crossed-out initial sign illegible] Functor from the category [crossed-out circled sign illegible]



Original source details: GELFAND-P000100

**Uncertain reading.** At left before Functor, an earlier sign with a tall left stroke is densely cancelled; after category, another sign is cancelled and enclosed by a loop. Letter readings undetermined.



Original source details: GELFAND-P000100

**Diagram and typed description.** Functor from the category; separate cancellations at right and left.

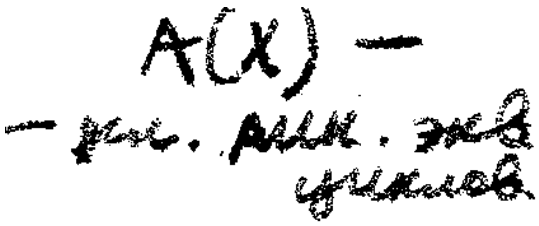
Algebr. sheaf, algebr. $K$-functor.



Original source details: GELFAND-P000100

**Diagram and typed description.** Algebr. sheaf, algebr. $K$-functor.

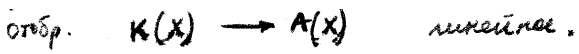
$A(X)$ —
— gr. of lin. equiv.
cycles



Original source details: GELFAND-P000100

**Diagram and typed description.** At right $A(X)$ —; on the next line another dash before gr. of lin. equiv., with cycles below.

The map $K(X) \longrightarrow A(X)$ is linear.



Original source details: GELFAND-P000100

**Diagram and typed description.** The map $K(X) \longrightarrow A(X)$ is linear.; an arrow directed to the right.

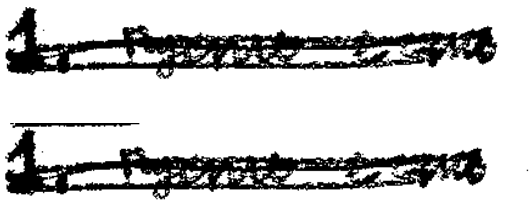
Axioms.



Original source details: GELFAND-P000100

**Diagram and typed description.** Axioms. has a straight underline.

1. [Crossed-out continuation illegible]



Original source details: GELFAND-P000100

**Uncertain reading.** Final entry: 1. and a densely crossed-out line with two long transverse strokes. Below it, down to the lower edge, there are no further legible axioms.

GELFAND-P000101

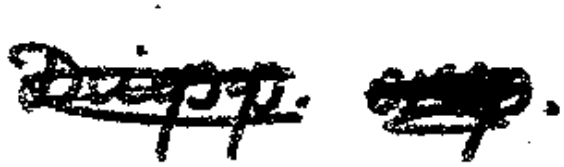
$18/\overline{X}$. I.M. Gelfand.

$X$

$E_1, \ldots, E_n$

$$\Gamma(E_1) \xrightarrow{d} \Gamma(E_2) \xrightarrow{d} \cdots \xrightarrow{d} \Gamma(E_n)$$

[Crossed out at right: Diff.[?] oper.[?]] Ell. complex.



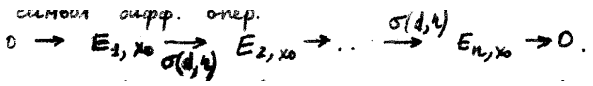
Original source details: GELFAND-P000101

**Uncertain reading.** Two densely crossed-out entries: Diff.[?] / Differ.[?] and oper.[?]. The distinct remnants do not establish all cancelled letters.

$$d\big(f^m(x)s(x)\big)\big|_{x_0} = \sigma(d,\eta)s(x_0), \qquad df\big|_{x_0} = \eta$$

$\sigma$ — symbol of the diff. oper.

$$0 \longrightarrow E_{1,x_0} \underset{\sigma(d,\eta)}{\longrightarrow} E_{2,x_0} \longrightarrow \cdots \overset{\sigma(d,\eta)}{\longrightarrow} E_{n,x_0} \longrightarrow 0.$$



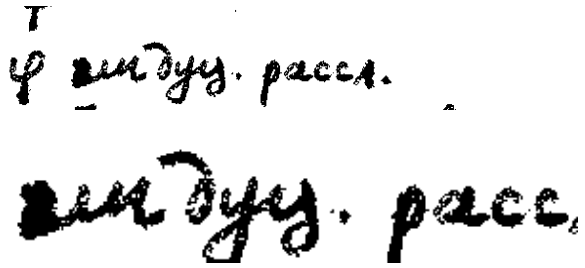
Original source details: GELFAND-P000101

**Diagram and typed description.** At the first arrow between fibres, $\sigma(d,\eta)$ is below; at the last, above; zeros at both ends.

Map. $T : \Gamma(E) \longrightarrow \Gamma(E)$. $\qquad \Gamma(E_i) \longrightarrow \Gamma(E_i)$

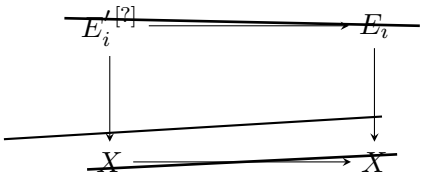
$f : X \longrightarrow X$

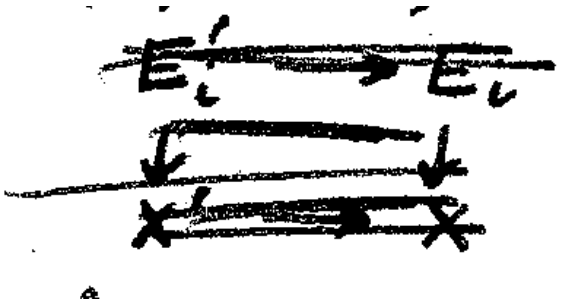
$\varphi$ induc.[?] bundle.



Original source details: GELFAND-P000101

**Uncertain reading.** Between $\varphi$ and bundle, the connected source abbreviation induc.[?] / endom.[?]; the first letter is densely overwritten, with a tall middle stroke and an upper arch.



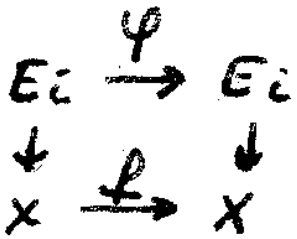



Original source details: GELFAND-P000101

**Uncertain reading.** Crossed-out square: upper $E_i$ labels, with a possible added upper stroke at left; rightward and downward arrows, $X \to X$ below. Several long cancellation strokes; exact hidden indices and outlines undetermined.

$$\begin{array}{ccc} E_i & \xrightarrow{\varphi} & E_i \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & X \end{array}$$

**Diagram and typed description.** Upper arrow $E_i \xrightarrow{\varphi} E_i$, two downward arrows; lower arrow $X \xrightarrow{f} X$.



Original source details: GELFAND-P000101

(In the simplest cases $\varphi$ determines $f$ — — forms $f$ determ. the map of tang. vectors)

Endom., if it commut. with $d$.

$T$ is an oper. in a $C^\infty$-space. The theorem reduces to the study of the trace (it is calc. in $2$ ways)

Theorem 1. The complex has finite [crossed-out word illegible] homologies.

**Uncertain reading.** Between finite and homologies, a word is densely crossed out; an upper arch and a lower stroke remain. Full reading undetermined.

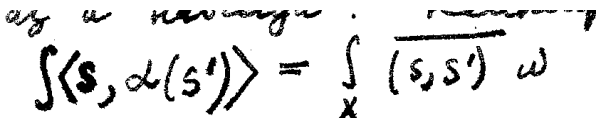


Original source details: GELFAND-P000101

Let $\omega^n$ be a form of ord. $n$ on $X$, not taking the value $0$. Use it to introduce a Hermitian structure. (Let it exist once and for all. Noninvariant, but simpler.)

$$\int \langle s, \alpha(s') \rangle = \int_X \overline{(s, s')}\, \omega.$$

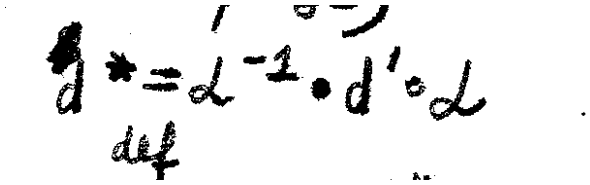
**Diagram and typed description.** At left, angle brackets containing $s, \alpha(s')$; at right, integral over $X$, a bar over $(s, s')$, then $\omega$.



Original source details: GELFAND-P000101

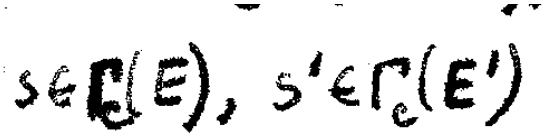
$$d^* \underset{\text{def}}{=} \alpha^{-1} \cdot d' \circ \alpha.$$

**Uncertain reading.** The top of the initial $d$ is dense, with a star beside it. Below the equality, def; at right $\alpha^{-1} \cdot d' \circ \alpha$. No separate sign is securely readable inside the dense part.



Original source details: GELFAND-P000101

$s \in \Gamma(E)$, $s' \in \Gamma_c(E')$.



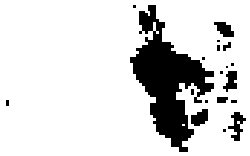
**Uncertain reading.** The lower part of the first $\Gamma$ is overwritten; the second $\Gamma$ has an open lower loop, read as $c$, possibly o-like.

Original source details: GELFAND-P000101

$E'$ — the dual sheaf.

$$\Box \underset{\text{def}}{=} d^*d + dd^*$$

[A separate black mark below and to the left of the definition.]



**Uncertain reading.** A separate dense blot with small dots and slanting strokes; letter reading and purpose undetermined.

Original source details: GELFAND-P000101

($d'$ — simply the adjoint operator)

$\Box$ — an ellipt. operator, taking each $\Gamma(E_i)$ into itself.

$\Gamma_\lambda(E)$ — sections satisfying the eqn. $\Box s = \lambda s$. Since $\Box$ is elliptic, for $\forall \lambda$ a fin. number of distinct lin. indep. eig. functs. [dense separate sign illegible].



**Uncertain reading.** After eig. functs., a separate dense closed black entry with inner vertical strokes; an H-like reading is not established.

Original source details: GELFAND-P000101

$$\square d = d\square$$

$$\Gamma_\lambda(E_1) \xrightarrow{d} \Gamma_\lambda(E_2) \xrightarrow{d} \cdots \longrightarrow \Gamma_\lambda(E_n)$$

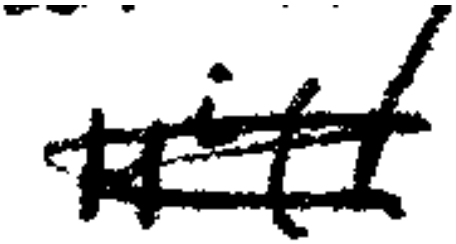




**Diagram and typed description.** The second closing parenthesis has a low outgoing stroke touching the next line; no separate comma established.

Original source details: GELFAND-P000101

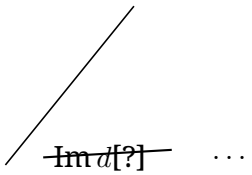
Introduce the homology groups [below, crossed-out formula illegible].

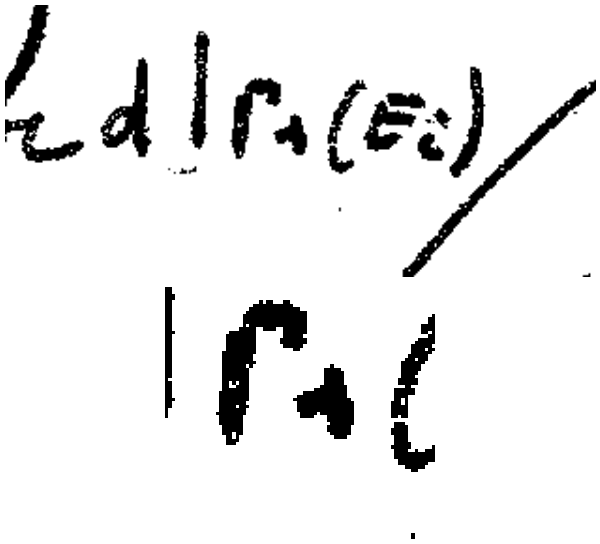


**Uncertain reading.** Below homology groups, two long transverse strokes cross a short formula; upright bases and an upper slanting stroke remain. Full reading undetermined.

Original source details: GELFAND-P000101

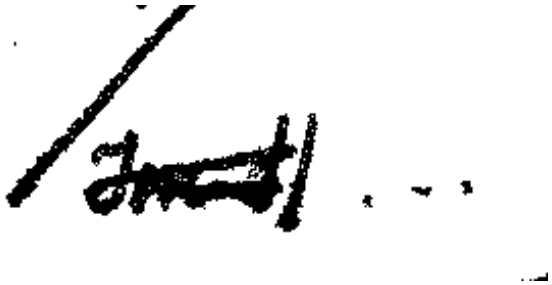
$$H^i(\Gamma_\lambda(E)) = \operatorname{Ker} d\big|_{\Gamma_{\lambda[?]}(E_i)}$$



**Uncertain reading.** After the vertical bar, $\Gamma$ with a dense lower sign $\lambda$[?], then $(E_i)$; the initial Ker has a capital K.

Original source details: GELFAND-P000101



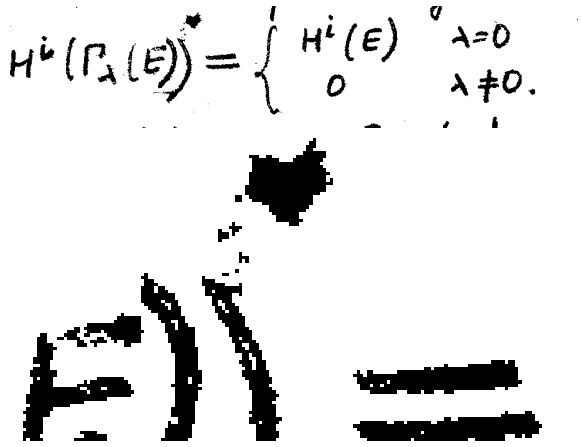
**Uncertain reading.** At right a slanting quotient stroke; below, densely crossed-out Im $d$[?], then three separate dots. The cancelled middle strokes remain unread.

Original source details: GELFAND-P000101

GELFAND-P000102

Without proof this work asserts Hodge's theorem

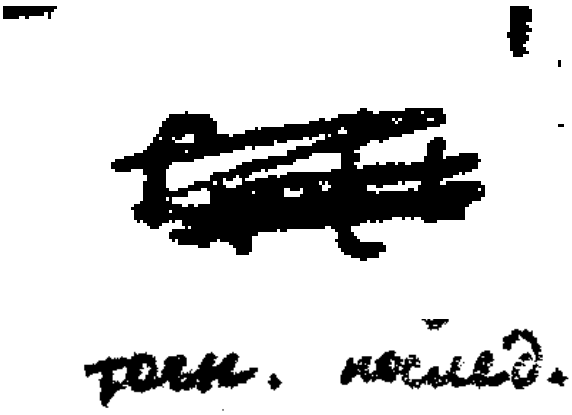
$$H^i(\Gamma_\lambda(E))^{*[?]} = \begin{cases} H^i(E) & \lambda = 0 \\ 0 & \lambda \neq 0. \end{cases}$$



Original source details: GELFAND-P000102

**Uncertain reading.** Above the right closing parentheses, a small dense star-like mark; its exact attachment to the parentheses and notation is undetermined.

Dynin. For $\lambda \neq 0$ — here $0$. ($d$ for $\lambda \neq 0$ maps eig. functs. [crossed-out word illegible] exact compl.

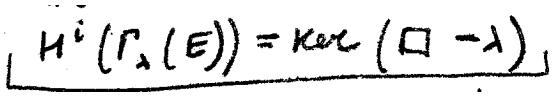


Original source details: GELFAND-P000102

**Uncertain reading.** Before exact compl., a short entry is densely crossed out. After it, exact compl.; the final source letters are cramped, with a separate arch above them. The cancelled letters remain unread.

$$H^i(\Gamma_\lambda(E)) = \mathrm{Ker}(\Box - \lambda)$$

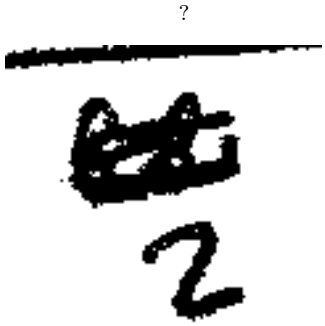
this is Hodge's theor. $\forall \lambda$



Original source details: GELFAND-P000102

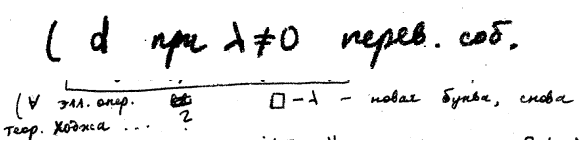
**Diagram and typed description.** Three sides of a frame below the formula and at both edges, open at the top. At right Ker$(\Box - \lambda)$, with an initial capital K.

(∀ ell. oper. [crossed out: illegible] $\square - \lambda$ — a new letter, again Hodge's theor....



Original source details: GELFAND-P000102

**Uncertain reading.** After ell. oper., a short abbreviation is densely cancelled; a question mark lies immediately below it. Full reading undetermined.



Original source details: GELFAND-P000102

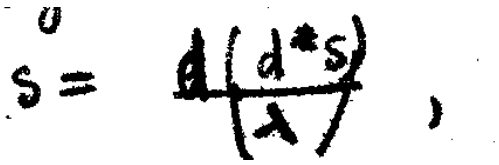
**Diagram and typed description.** Visible opening round parentheses before $d$ and before ∀ ell. oper.; the last line has three separate dots after Hodge. No closing parentheses read in this fragment.

Bernstein. Same. $H^i(\Gamma_\lambda(E)) = 0$, $\lambda \neq 0$. Let $s \in \Gamma_\lambda(E_i)$ and let $ds = 0$. [?] $\square s = dd^*s$. Then $\lambda s = d(d^*s)$ and $s = d\left(\frac{d^*s}{\lambda}\right)$, $d^*s$ is an eig. funct. ($d^*$ commut. with $\square$).



Original source details: GELFAND-P000102

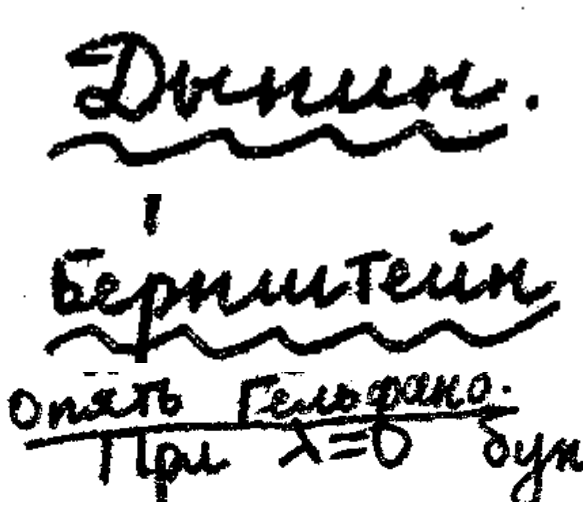
**Uncertain reading.** Before $\square$, a tall dense stroke with a closed upper loop and a lower slanting exit: $d$[?] / opening parenthesis[?]. Reading undetermined.



Original source details: GELFAND-P000102

**Diagram and typed description.** The entry $s = d\left(\frac{d^*s}{\lambda}\right)$: a long horizontal bar passes near the base of the outer $d$ and across the parentheses; a separate comma at right.

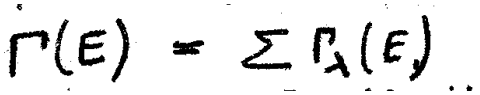
$d^*s \in C^\infty$ follows from the theor. on ell. eqns. Gelfand again. For $\lambda = 0$ transfer Hodge's proof literally.




Original source details: GELFAND-P000102

**Diagram and typed description.** Wavy lines below Dynin. and Bernstein.; a straight line below Gelfand again. In the line before the last heading, theor. and ell. are separately underlined.

$$\Gamma(E) = \sum \Gamma_\lambda(E) \qquad \text{(discr. spectrum)}$$



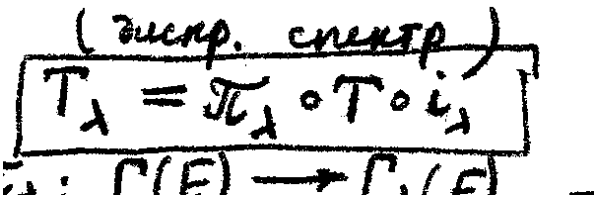
Original source details: GELFAND-P000102

**Diagram and typed description.** $\Gamma(E) = \sum \Gamma_\lambda(E)$; no separate index visible below the summation sign.

Besides the oper. $T$, introduce

$$\boxed{T_\lambda = \pi_\lambda \circ T \circ i_\lambda,}$$

where $i_\lambda : \Gamma_\lambda(E) \longrightarrow \Gamma(E)$, $\pi_\lambda : \Gamma(E) \longrightarrow \Gamma_\lambda(E)$ — orthog. projection.



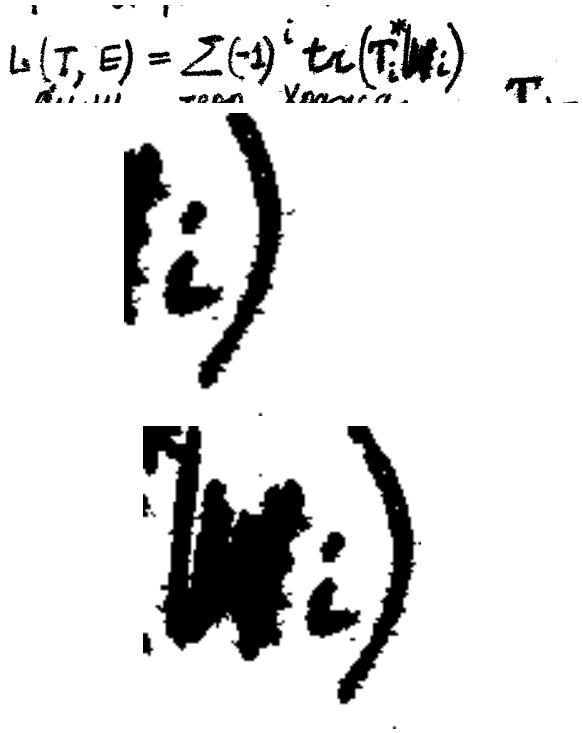

Original source details: GELFAND-P000102

**Diagram and typed description.** Rectangular frame around $T_\lambda = \pi_\lambda \circ T \circ i_\lambda$; a final comma inside the frame. Below, the line with $\pi_\lambda$ and orthog. projection; above the frame (discr. spectrum).

(oper. $T$ in this cell) Lefschetz number for $T$.

$T$ — a map of sections, thus a map of sections $\Gamma(E) \to \Gamma(E)$

$$L(T, E) = \sum (-1)^i \operatorname{tr}(T_i^* | H_i[?]).$$

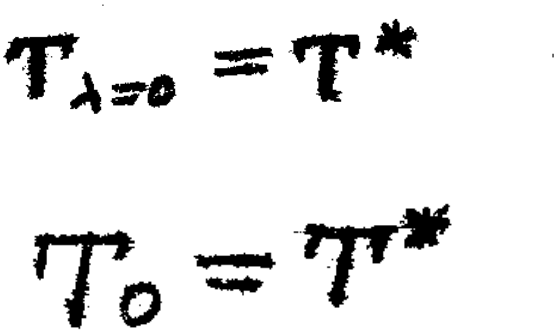


**Uncertain reading.** After the vertical bar, an H-like letter is densely overwritten; a separate $i$ lies at the lower right, with a dot above it. The main letter and index placement remain uncertain: $H_i[?]$ / $H^i[?]$.

Original source details: GELFAND-P000102

By Hodge's theor. $T_{\lambda=0} = T^*$.

$\Pi_0[?] = T^{*[?]}$



**Uncertain reading.** At left $T_{\lambda=0} = T^*$. At right, the first letter has a long top bar and a lower $0$: $\Pi_0[?]$ / $\pi_0[?]$ / $T_0[?]$. Above the right $T$, a dense star-like entry; one or two overlapping stars cannot be distinguished securely.

Original source details: GELFAND-P000102

$$\sum(-1)^i \operatorname{tr}(T_0|E_i) = L(T,E)$$
$$\sum(-1)^i \operatorname{tr}(T_\lambda|E_i) = 0 \qquad \lambda \neq 0$$

[At right, two densely crossed-out entries illegible.]



**Uncertain reading.** Two small dense cancellations in the right margin: the upper one has a lower loop, the lower a long rising slanting stroke. Letter readings undetermined.

Original source details: GELFAND-P000102

(the trace in hom. is the same as the trace in the spaces themselves)
(By Hodge's theor. $H^i(E) = \Gamma_0(E_i)$)

$$\Box s = 0 \Longrightarrow \underline{ds = 0}.$$

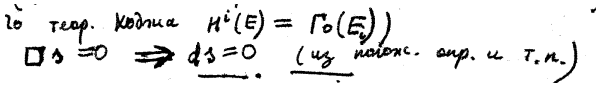
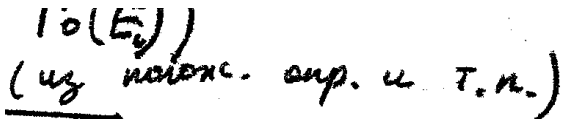


**Diagram and typed description.** In $\Box s = 0 \Longrightarrow ds = 0$, a thick overwritten arrow; below $ds = 0$, a horizontal line, with a separate dot at right.

Original source details: GELFAND-P000102

(from immed.[?] def. and so on)



**Uncertain reading.** After from, a dense connected source abbreviation neposr. (immed.)[?] / polozh. (positive)[?]; at right def. and so on, then a closing round parenthesis. Dense overwriting above the early middle strokes.

Original source details: GELFAND-P000102

Must give meaning to $\sum(-1)^i \operatorname{tr}(T|E_i)$. Regulariz. it.
In 2 ways.

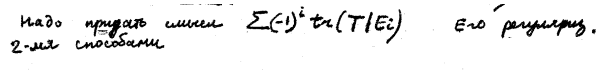


**Diagram and typed description.** First line: Must give meaning, the sum, and Regulariz. it at right. Below at left: In 2 ways.

Original source details: GELFAND-P000102

GELFAND-P000103

In finite $\Gamma_\lambda(E)$

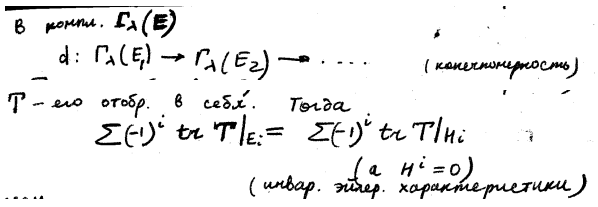
$$d : \Gamma_\lambda(E_1) \longrightarrow \Gamma_\lambda(E_2) \longrightarrow \cdots$$

(finite-dimensionality) $T$ — its map into itself. Then

$$\sum (-1)^i \operatorname{tr} T|_{E_i} = \sum (-1)^i \operatorname{tr} T|_{H_i}$$

(and $H^i = 0$)

(invar. of the Euler characteristic)




Original source details: GELFAND-P000103

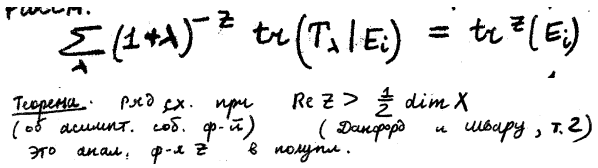
**Diagram and typed description.** The first chain has two arrows, then an ellipsis; finite-dimensionality in parentheses. In the trace equality, $E_i$ and $H_i$ have subscripts after the restriction bars; the next parenthesis has $H^i = 0$ with superscript i.

Consider.

$$\sum_\lambda (1+\lambda)^{-z} \operatorname{tr}(T_\lambda | E_i) = \operatorname{tr}^z(E_i)$$

Theorem. The series conv. for $\operatorname{Re} z > \frac{1}{2} \dim X$ (on the asympt. of eig. functions) (Dunford and Schwartz, vol.2)

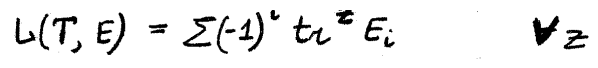
This is an anal. funct. of $z$ in the half-plane.




Original source details: GELFAND-P000103

**Diagram and typed description.** The exponent of $(1+\lambda)$ is $-z$, with a separate short minus stroke. At right, $\operatorname{tr}^z(E_i)$. Beneath, Theorem. is underlined; $\operatorname{Re} z > \frac{1}{2} \dim X$ and two parentheses.

$$L(T, E) = \sum (-1)^i \operatorname{tr}^z E_i \qquad \forall z$$

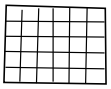



Original source details: GELFAND-P000103

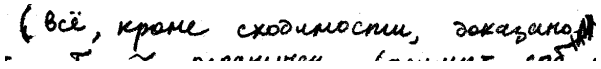
**Diagram and typed description.** At left, $L(T, E)$; at right the trace has superscript z, then $E_i$ without parentheses. Farther right, a separate $\forall z$ with a dense first sign.

(everything except convergence is proved [densely overwritten sign illegible]

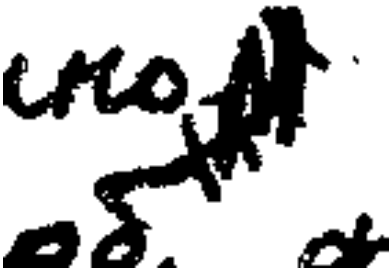
—



)



**Uncertain reading.** After proved, a dense vertical entry with a lower oblique looplike exit. Former letters remain inseparable even under enlargement; exact reading undetermined.



Original source details: GELFAND-P000103



**Diagram and typed description.** After the dash, a small rectangular grid of horizontal and vertical strokes. The typographic recreation shows the visible lines; dense intersections and borders are not fully distinguishable. No letter label or definite role of the grid is read.

Original source details: GELFAND-P000103

Since $T \sim$ is bounded (asympt. of eig. functions)

(on anal. cont., a pole at $z = 0$)

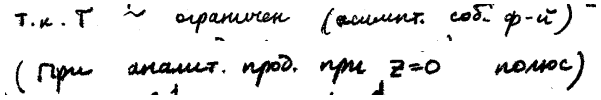


**Diagram and typed description.** After T, a separate wavy $\sim$ sign, then bounded; asympt. of eig. functions in parentheses. Beneath, a separate parenthesis: on anal. cont., a pole at $z = 0$.

Original source details: GELFAND-P000103

Example. $S^1$




$d = \frac{d}{dt}$, ell. oper., $\Box = \frac{d^2}{dt^2}$.

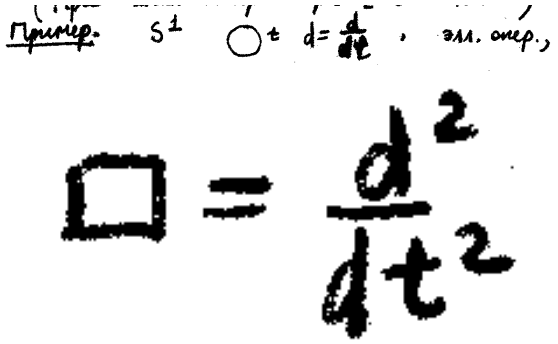



**Diagram and typed description.** A circle with t at right. The initial d in the first derivative's denominator is densely overwritten, followed by t; former strokes are inseparable. The entry for □ is $d^2/dt^2$ with a positive sign.

Original source details: GELFAND-P000103

$T = (f, \varphi)$,

$$dS(x) = s(x)$$
$$Ts(x) = \varphi(x)s(f(x))$$

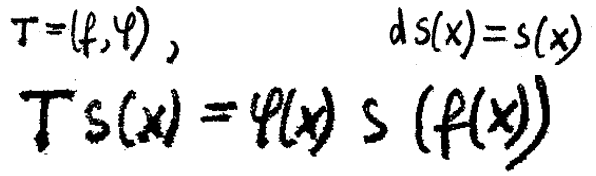



**Diagram and typed description.** At left, $T = (f, \varphi)$ and a comma. At upper right, $dS(x) = s(x)$: the first S is uppercase, the second s lowercase. Beneath, $Ts(x) = \varphi(x)s(f(x))$.

Original source details: GELFAND-P000103





**Diagram and typed description.** At left beneath $T = (f, \varphi)$, a separate wavy line; below it, an almost horizontal stroke with a thickened right end. The typographic recreation preserves the general course of the lines; direction of the thickening and purpose of the sketch are undetermined. There are no axis or function labels.

Original source details: GELFAND-P000103

Definition [?] of a pseudodiff. oper.:



**Uncertain reading.** Between the heading and pseudodiff., a dense overwritten sign with a right vertical and several transverse strokes: P[?] / illegible. No secure letter reading established.

Original source details: GELFAND-P000103

$X$ — manifold. $\Gamma(E)$, $\Gamma(F)$ — 2 bundles on $X$

$$P : \Gamma_c(E) \longrightarrow \Gamma(F)$$

$\Gamma_c(E)$ — sections with compact support. [Crossed out: Called[?]]



**Uncertain reading.** Before Lin. oper. is called, a short abbreviation with a long lower oblique cancellation stroke. Source Naz.[?] (Called[?]); former letters are densely joined and not securely read. A separate period follows.

Original source details: GELFAND-P000103

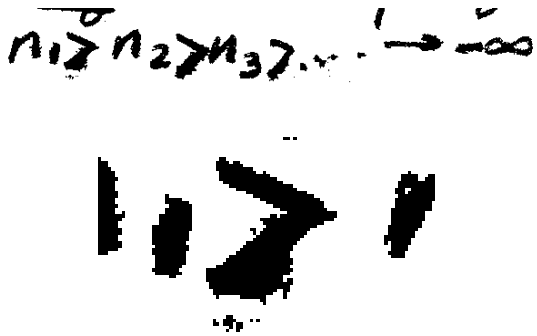
A lin. oper. is called pseudodiff. if $\exists$ a seq. of numbers $n_1 > n_2 > n_3 > \cdots \to -\infty$
and satisfies the following conditions

$$e^{-i\lambda g} \underline{P}(e^{i\lambda g} s)$$

$g \in \Gamma(1)$ ($C^\infty$ funct.), 1 — trivial sheaf, $s(x) \in \Gamma_c(E)$, $dg \neq 0$ on supp $s$, if there is an asympt. expansion.

GELFAND-P000104

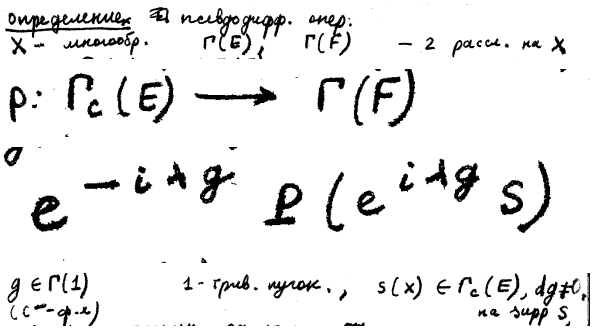
$$\sim \sum_0^\infty P_j(s,g)\lambda^{n_j}$$



**Uncertain reading.** Three right-pointing signs. The first and third have dense lower ends and separate tiny traces; the middle one is overwritten and joins the following n. Separate lower equality strokes are not established: $>$[?] / $\geq$[?].



Original source details: GELFAND-P000103



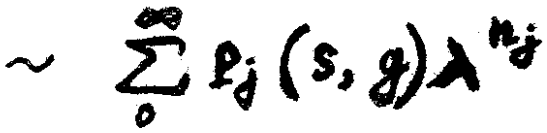

Original source details: GELFAND-P000103

**Diagram and typed description.** The heading Definition has a straight underline. In the exponential expression, P has a short underline; exponents $-i\lambda g$ and $i\lambda g$. In parentheses, $C^\infty$ funct. At right, $dg \neq 0$; beneath, on supp $s$ and a comma.



Original source details: GELFAND-P000103

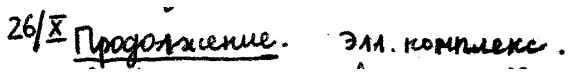
**Uncertain reading.** To the right of asympt. expansion., a short dense isolated mark: an upper loop, a horizontal exit to the right and a lower projection. Letter reading and association undetermined.



Original source details: GELFAND-P000104

**Diagram and typed description.** Continuation of the formula: a separate $\sim$, lower limit $0$ without $j =$, upper $\infty$, then $P_j(s,g)\lambda^{n_j}$.

$26/\overline{X}$ Continuation. Ell. complex.



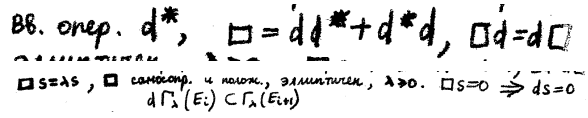

Original source details: GELFAND-P000104

**Diagram and typed description.** X in the date has separate over- and underlines. Continuation. has its own underline; then Ell. complex.

In $E_i$ — a Hermitian metric, and in $X$ — integration. Introduce oper. $d^*$, $\Box = dd^* + d^*d$, $\Box d = d\Box$

$\Box s = \lambda s$, $\Box$ self-adj. and positive, elliptic, $\lambda > 0$. $\Box s = 0 \Longrightarrow ds = 0$

$$d\Gamma_\lambda(E_i) \subset \Gamma_\lambda(E_{i+1})$$

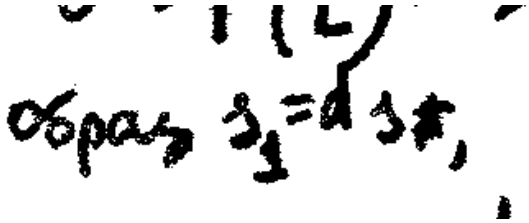



Original source details: GELFAND-P000104

**Uncertain reading.** At right in $\Box d = d\Box$, the last square's right edge is broken and faint. Its other sides are distinguishable; the exact right boundary of the sign is damaged in the held scan. Beneath, separate entries $\lambda > 0$ and $\Box s = 0 \Longrightarrow ds = 0$.

$$0 \xrightarrow{d} \Gamma(E) \xrightarrow{d} \Gamma(F) \xrightarrow{d} 0$$

$d$ — simply an elliptic operator, $ds = 0$ — the kernel. Image $s_1 = ds_*$,



**Uncertain reading.** The first s has a lower sign resembling 1, possibly $i$. The star-like sign after the second s lies close to the baseline; its height is ambiguous: $s_1 = ds_*$ / $s_i = ds^*$ / illegible index.

Original source details: GELFAND-P000104

Homology: Ker $d|_E$, Coker $d|_F$. The Lefschetz number — the index

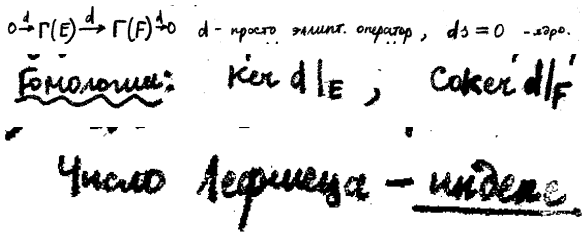
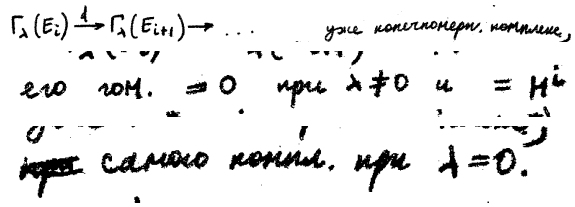


**Diagram and typed description.** All three arrows of the first chain are labelled d. Homology: has a wavy underline; Ker and Coker begin with capitals. Index has a separate straight underline.

Original source details: GELFAND-P000104

$$\Gamma_\lambda(E_i) \xrightarrow{d} \Gamma_\lambda(E_{i+1}) \longrightarrow \cdots$$

already a finite-dimensional complex, its hom. $= 0$ for $\lambda \neq 0$ and $= H^i$ [crossed out: for] [illegible phrase] for $\lambda = 0$.



**Uncertain reading.** After $H^i$, the source word при (for) is densely crossed out. Then joined loops: самое конеч.[?] (most finite.[?]) / само конеч.[?] (itself finite.[?]); the first word is not securely read. Finally, for $\lambda = 0$.

Original source details: GELFAND-P000104

$T_i^*$ $H^i$

$$\sum(-1)^i \operatorname{tr}(T_i^*) = L(T, E)$$

Compute through fixed points.

$T$ — a map of $X$, commutes with $d$.

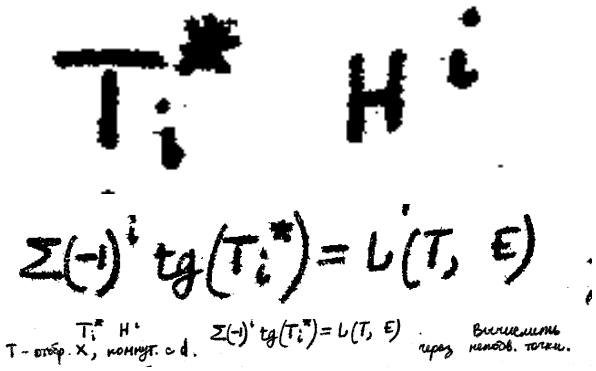


**Diagram and typed description.** At left, separate $T_i^*$ and $H^i$, without an arrow between them. At right, an equality with $L(T, E)$. Beneath, T — a map of X, commutes with d.; at right, Compute through fixed points.

Original source details: GELFAND-P000104

$T$ is a map of $\Gamma_\lambda(E_i)$ Introd. operators $\pi_\lambda$ and $i_\lambda$. inclusion of $\Gamma_\lambda$ into all of $\Gamma$, and projection onto [crossed out: illegible entry]

$\Gamma_\lambda(E_i)$.



**Uncertain reading.** After onto, a short densely cancelled entry with an upper loop and transverse strokes. Former letters remain inseparable; followed by a separate active $\Gamma_\lambda(E_i)$.

Original source details: GELFAND-P000104

$$T_\lambda = \pi_\lambda \cdot T i_\lambda$$

(select the blocks corresp. to $\Gamma_\lambda$)

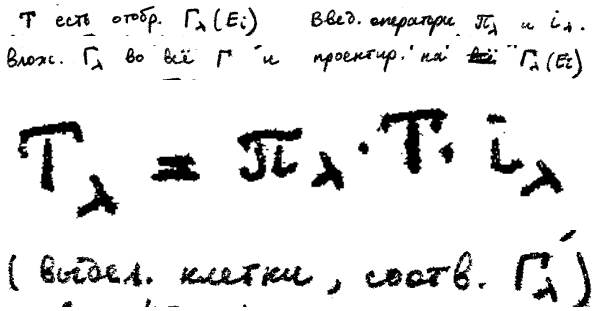
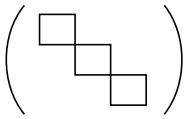


**Diagram and typed description.** Entries for inclusion of $\Gamma_\lambda$ into all of $\Gamma$ and projection onto $\Gamma_\lambda(E_i)$. The formula has one multiplication dot before T; no separate dot between T and $i_\lambda$. Beneath, a parenthesis about selecting blocks.

Original source details: GELFAND-P000104

Commut. unnecessary, suff. [dense overwritten sign]



in the blocks.



**Uncertain reading.** Between suff. and the sketch, a dense vertical entry with two dark clusters and a lower oblique exit. No letter reading established; the sign is retained separately from the blocks.

Original source details: GELFAND-P000104

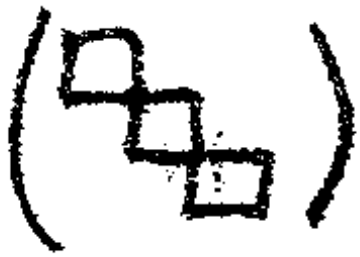


**Diagram and typed description.** Three empty squares descend from upper left to lower right within parentheses; neighbouring corners touch. The typographic recreation preserves the three outlines and parentheses; exact sizes and dense joints remain undetermined.

Original source details: GELFAND-P000104

Then $\boxed{T^* = T_0}$

$$0 \longrightarrow \Gamma_0(E_1) \xrightarrow{d} \Gamma_0(E_2) \xrightarrow{d} \cdots$$

all $d = 0$ since $\Box s = 0$.

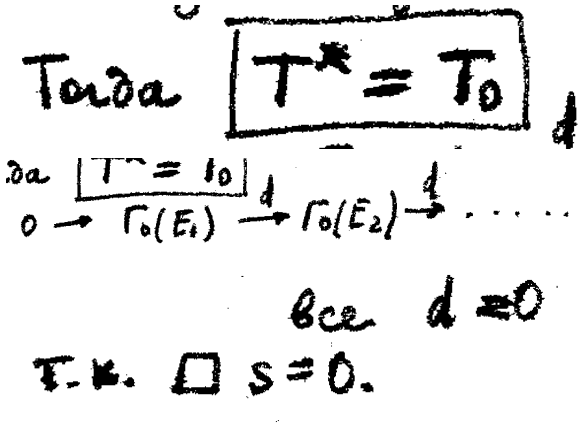



**Diagram and typed description.** The equality $T^* = T_0$ has a full rectangular frame. The first arrow from 0 is unlabelled, the next two have d; then an ellipsis. At right, all $d = 0$; beneath, since $\Box s = 0$.

Original source details: GELFAND-P000104

All homology in $\Gamma_0$

$$\Gamma(E_i) = \Gamma_0(E_i) + \quad \Gamma_{\lambda_1}(E_i) \quad + \quad \cdots$$
$$\downarrow \qquad\qquad |$$
$$\text{homol. } 0$$

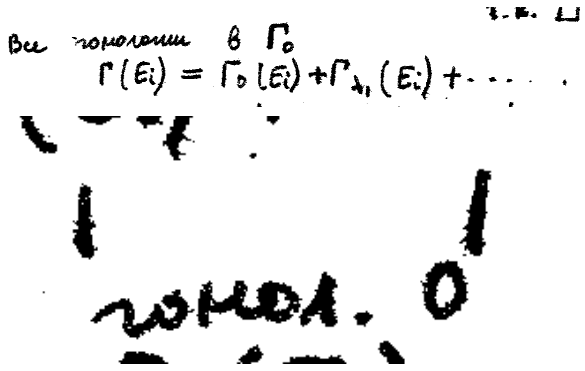
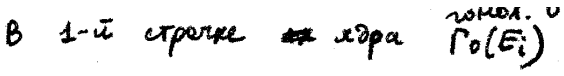



**Diagram and typed description.** A short downward arrow beneath $\Gamma_{\lambda_1}(E_i)$; a separate vertical stroke with a dense lower end beneath the ellipsis. Beneath, homol. 0. No arrow beneath $\Gamma_0(E_i)$. A separate arrowhead on the right stroke is not established.

Original source details: GELFAND-P000104

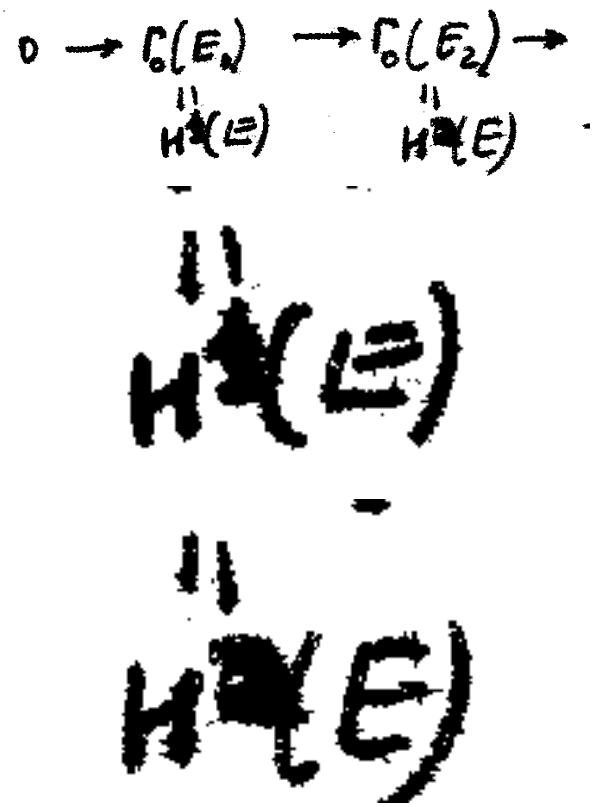
In the 1st row [crossed out: illegible word] kernel $\Gamma_0(E_i)$




**Uncertain reading.** Before kernel, a short horizontal cancelled entry with dense joined strokes. No secure letter reading.

Original source details: GELFAND-P000104

$$0 \longrightarrow \Gamma_0(E_1) \longrightarrow \Gamma_0(E_2) \longrightarrow$$
$$\| \qquad\qquad \|$$
$$H^{[?]}(E) \qquad H^{[?]}(E)$$



Original source details: GELFAND-P000104

**Uncertain reading.** Beneath two terms of the chain, vertical double strokes and entries $H^{[?]}(E)$. Both superscripts are densely overwritten: $H^{*}(E)$ / $H^{i}(E)$ / illegible degree. Former strokes remain inseparable even under enlargement. The chain's last arrow has no visible continuation or ellipsis.

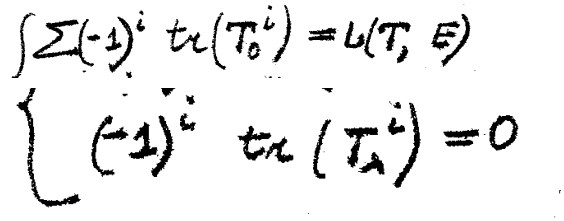
Remark. $\pi_\lambda$ commutes with $d$.



Original source details: GELFAND-P000104

**Diagram and typed description.** Remark. has a straight underline; then $\pi_\lambda$ commutes with d.

$$\begin{cases} \sum (-1)^i \operatorname{tr}(T_0^i) = L(T, E) \\ (-1)^i \operatorname{tr}(T_\lambda^i) = 0 \end{cases}$$



Original source details: GELFAND-P000104

**Uncertain reading.** A shared left brace. Only the first row starts with $\sum$; no separate sum is visible before $(-1)^i$ in the second row. T has superscript i in both rows; lower signs 0 and $\lambda$.

GELFAND-P000105

If there is a seq. of spaces

$$0 \longrightarrow A_1 \xrightarrow{d} A_2 \xrightarrow{d} \cdots \xrightarrow{d} A_n \longrightarrow 0, \qquad d^2 = 0.$$

and maps $T_i$, comm. with $d$. Then

$$\sum_{\text{in chains}} (-1)^i \operatorname{tr} T_i = \sum_{\text{in homol.}} (-1)^i \operatorname{tr}(T_i^*) \qquad \text{(Euler's formula)}$$

(Euler's formula is obtained by taking $T = E$)

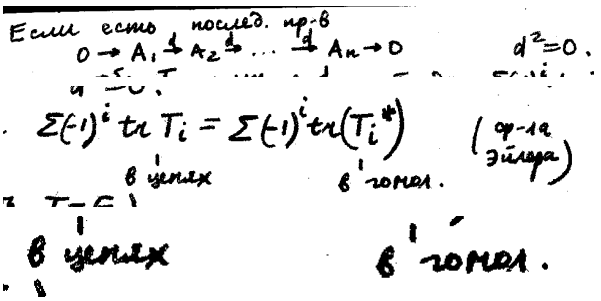



**Diagram and typed description.** The chain's end arrows are unlabelled, its internal arrows have d. Beneath the left sum, a separate stroke and in chains; beneath the right, a stroke and in homol. At right, a parenthesis Euler's formula.

Original source details: GELFAND-P000105

Proof: represent.

$$A_i = \underset{Z \text{ 3 summands.}}{Z} + C_Z, \qquad Z = B + \text{compl.}$$

and then consider $T$ there.

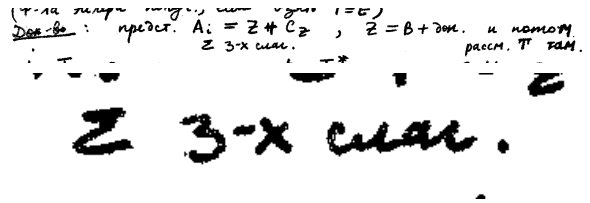
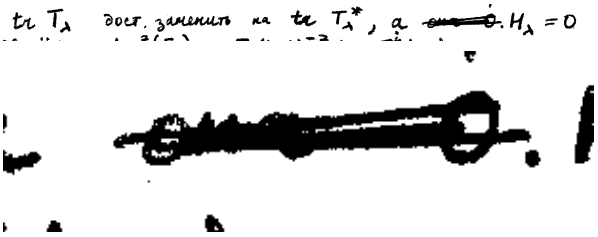



**Diagram and typed description.** Proof has a straight underline; the colon is to the right of the line. Beneath the first Z, a separate entry Z 3 summands. The second entry is $Z = B +$ compl.; then and then consider T there.

Original source details: GELFAND-P000105

It suffices to replace $\operatorname{tr} T_\lambda$ by $\operatorname{tr} T_\lambda^*$, and [crossed out: illegible entry] $H_\lambda = 0$.




**Uncertain reading.** Before $H_\lambda = 0$, a long densely cancelled entry with loops at both ends and a horizontal stroke. Former letters remain inseparable.

Original source details: GELFAND-P000105

Introduce

$$\operatorname{tr}^z(E_i) = \sum_\lambda (1+\lambda)^{-z} \operatorname{tr}(T_\lambda^* | E_i)$$

for $\operatorname{Re} z > \frac{1}{2} \dim X$,

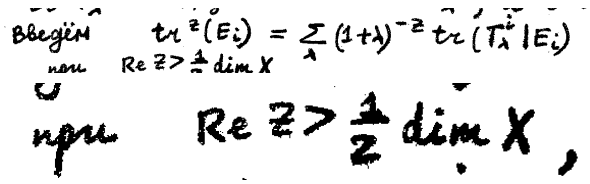
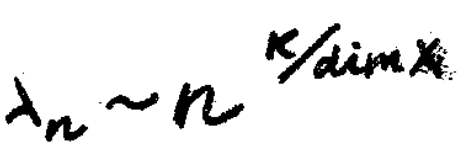



**Diagram and typed description.** The sum has subscript $\lambda$, the factor exponent $-z$, and $T_\lambda$ a superscript star. The next line is for $\operatorname{Re} z > \frac{1}{2} \dim X$, followed by a comma.

Original source details: GELFAND-P000105

[At right:]

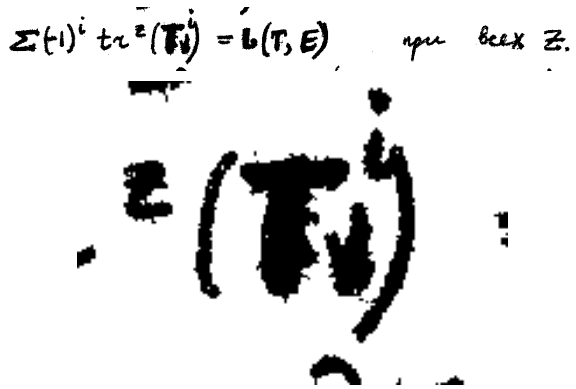
$$\lambda_n \sim n^{K/\dim X}$$



**Uncertain reading.** The exponent's numerator is a K-like sign with a straight left stroke and two right branches: $K$ / $\kappa$.

Original source details: GELFAND-P000105

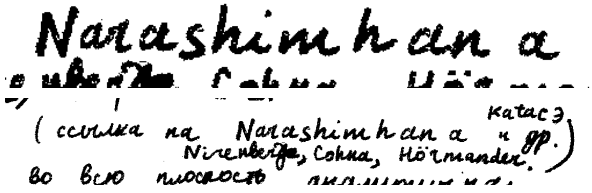
$$\sum(-1)^i \operatorname{tr}^z(T_i^i) = L(T,E) \qquad \text{for all } z.$$



**Uncertain reading.** In parentheses after tr, a dense T with a lower i-like sign and a separate upper i-like sign: $T_i^i$ / $T_i$ / illegible superscript.

Original source details: GELFAND-P000105

Continuation theorem (reference to Narashimhan a[?] and others.)



**Uncertain reading.** A Latin name with joined m/h/n-like loops and a separate final a: Narashimhan a[?] / Narasimhcan a / Narasimhana. Beneath, Nirenberg with a densely overwritten ending; then two more names.

Original source details: GELFAND-P000105

[Above and others.: katase.[?]]



**Uncertain reading.** A separate small entry above and others.: source катасэ. / катасэ. (katase.[?]) / illegible. The last sign resembles э; some forms also permit a Latin c.

Original source details: GELFAND-P000105

[Beneath: Nirenberg, Cohna[?], Hörmander.]



**Uncertain reading.** The middle Latin name between commas: Cohna / Cohn a. Between h and final a, dense joined strokes; exact separation of letters is undetermined.

Original source details: GELFAND-P000105

$\mathrm{tr}^z T^i$ extends to the whole plane, is analytic throughout the plane (!) and at $z = 0$

$$\mathrm{tr}^0(T^i) = \mathrm{tr}^{[?]}(T^i)$$

where $\mathrm{tr}^b$ — the trace, calc. in the usual way through generalized functions.

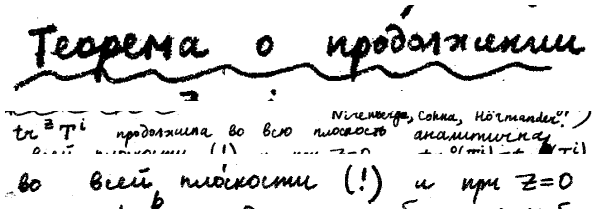
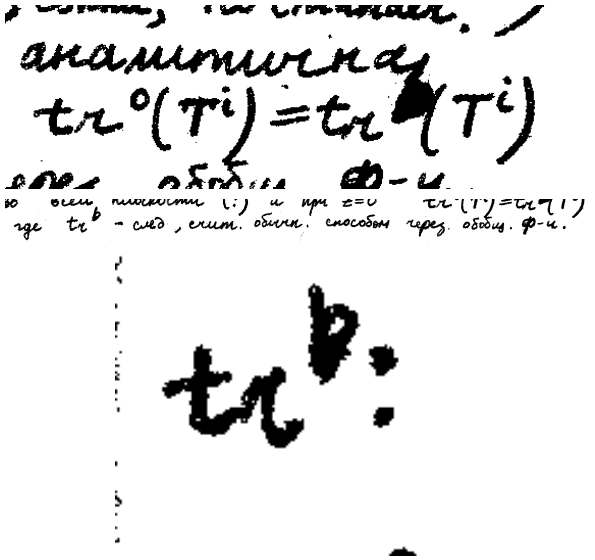


**Diagram and typed description.** The heading Continuation theorem has a wavy underline. The text twice mentions the whole plane, with a separate (!) and at $z = 0$. T has superscript i.

Original source details: GELFAND-P000105

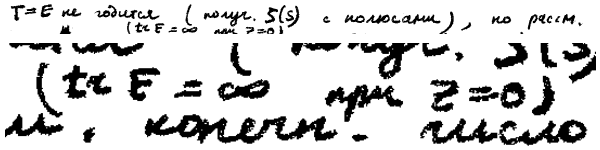


**Uncertain reading.** The second tr's superscript is densely overwritten and joins surrounding strokes: b[?] / illegible. Beneath and later at left, $\mathrm{tr}^b$ is clearly written.

Original source details: GELFAND-P000105

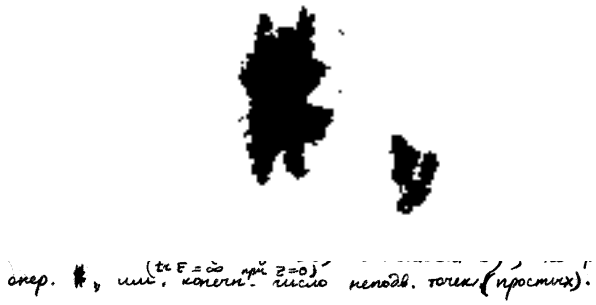
$T = E$ does not work (giving $\zeta(s)$[?] with poles), but consider [Insertion beneath $T = E$:] ($\mathrm{tr}\, E = \infty$ at $z = 0$).

an oper. [illegible sign], having a finite no. of fixed points (simple).



**Uncertain reading.** Before (s), a large zigzag sign with a rounded lower exit: ζ[?] / Σ[?]. Beneath the line, a separate parenthesis $\mathrm{tr}\, E = \infty$ at $z = 0$.

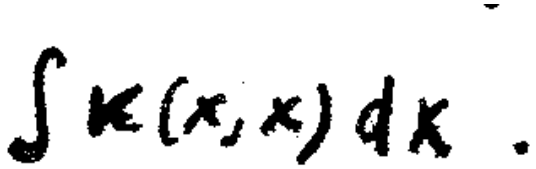
Original source details: GELFAND-P000105



Original source details: GELFAND-P000105

**Uncertain reading.** After oper., a dense vertical overwritten entry with several transverse projections and lower legs; no secure letter reading. Then a comma, having a finite no. of fixed points (simple).

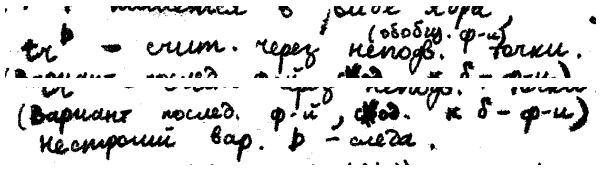
tr$^b$: $T$ is given as a kernel, $\int \kappa(x,x)\,dx$.



Original source details: GELFAND-P000105

**Uncertain reading.** The kernel letter has a K-/к-like form: $\kappa$ / $K$. A short dense sign in the integral, a taller left stroke in the two-argument formula. The handwritten height does not establish alphabet or case.

tr$^b$ — calc. through fixed points. [Above fixed: (generalized functions).] (A version using a seq. of functions, conv.[?] to a $\delta$-function) Some variant of the b-trace.



Original source details: GELFAND-P000105

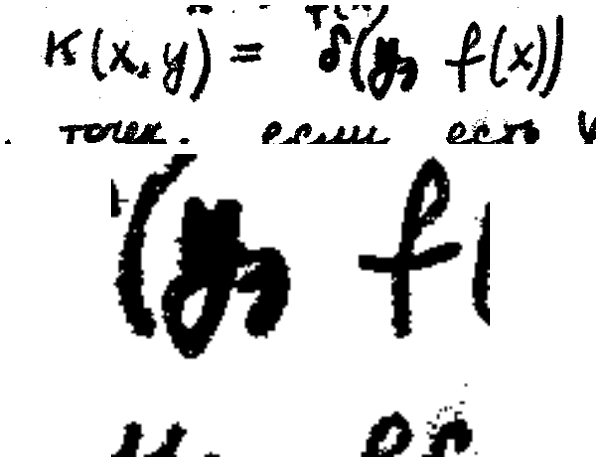
**Uncertain reading.** In the parenthesis, the word before to a $\delta$-function is densely overwritten: source сход.[?] (conv.[?]) / illegible beginning. Former strokes remain inseparable. Above the preceding fixed, a separate insertion (generalized functions).

[At right:] $\delta(x,a)$ — $\delta$ at the point $a$.

$$s(x) \longrightarrow \varphi s(f(x))$$

(transf.[?] of traces) $x \longrightarrow f(x)$

$$\kappa(x,y) = \delta(y\,[?]\,f(x))$$



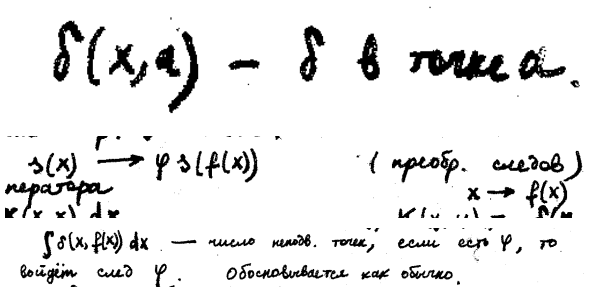
Original source details: GELFAND-P000105

**Uncertain reading.** The first argument and following sign to the right of $\delta$ are densely overwritten. Readings $y - f(x)$ / $y, f(x)$ / illegible. No separate untouched minus stroke can be distinguished between the overwritten y and f.

The operator's trace $\int \kappa(x,x)\,dx$.

$$\int \delta(x, f(x))\,dx$$

— the number of fixed points; if there is $\varphi$, the trace of $\varphi$ enters. Justified as usual.

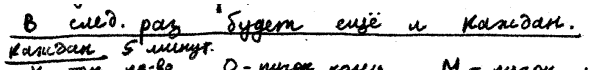



Original source details: GELFAND-P000105

**Diagram and typed description.** At right, $\delta(x,a)$ with a comma and an entry about the point a. The lower integral $\delta(x, f(x))$ also has a comma. At left, the map with $\varphi$; at right, a parenthesis source пробр. следов (transf.[?] of traces) and a separate $x \longrightarrow f(x)$.

Next time Kazhdan will also speak.

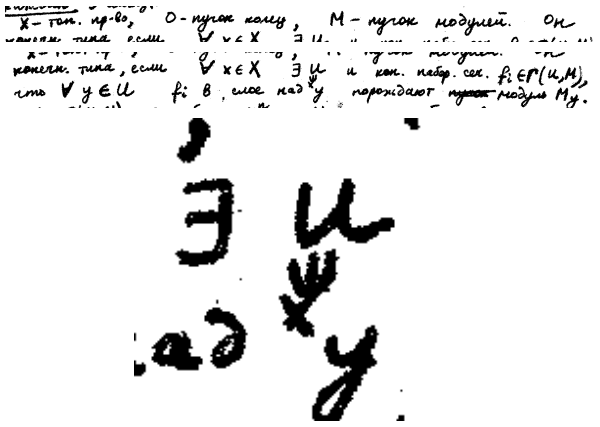
Kazhdan 5 minutes.




Original source details: GELFAND-P000105

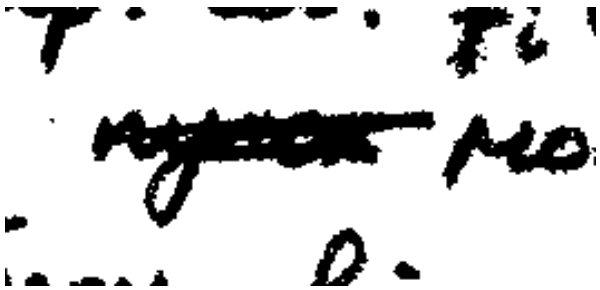
**Diagram and typed description.** The entire first phrase has a straight underline. Beneath, Kazhdan is separately underlined, followed by 5 minutes.

$X$ — top. space, $O$ — sheaf of rings, $M$ — sheaf of modules. It is of finite type if $\forall x \in X\ \exists U \underset{x}{\Downarrow}$ and a fin. set of sections $f_i \in \Gamma(U, M)$, such that $\forall y \in U$, the $f_i$ in the fibre over $y$ generate [crossed out: illegible word] the module $M_y$.



Original source details: GELFAND-P000105

**Diagram and typed description.** Beneath U after $\exists$, a double downward arrow to x. This x is separate from y in the next line over y. The text first has $\forall x \in X$, then $\forall y \in U$.



Original source details: GELFAND-P000105

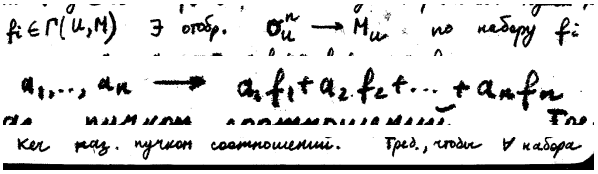
**Uncertain reading.** Between generate and module, a long densely cancelled word with a lower loop at left and a horizontal stroke to the right. Former letters remain inseparable.

$$f_i \in \Gamma(U, M) \qquad \exists \text{ map } O_U^n \longrightarrow M_U \qquad \text{from the set } f_i$$
$$a_1, \ldots, a_n \longrightarrow a_1 f_1 + a_2 f_2 + \cdots + a_n f_n.$$

Ker is called the sheaf of relations. Req., that $\forall$ set

GELFAND-P000106

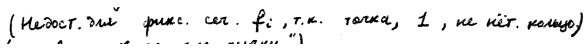
of sections the sheaf of rel. should be of finite type.



Original source details: GELFAND-P000105

**Diagram and typed description.** O has superscript n and subscript U; M has subscript U. The coefficient arrow points right; at right a linear combination through $a_n f_n$ and a period. On the last line the abbreviation source Треб. (Req.) has a separate following comma; the line ends $\forall$ set.

(Insuff. for fixed sections $f_i$, since a point, 1, a non-Noeth. ring)

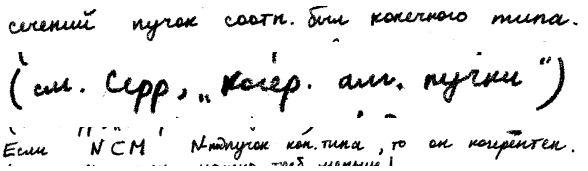


Original source details: GELFAND-P000106

**Uncertain reading.** Between a point, and a non-Noeth. ring, a separate 1 with a following comma. The sign has no label; its role is undetermined.

(see Serre, “Coher. alg. sheaves”)

If $N \subset M$, N — a subsheaf of fin. type, then it is coherent. (For Noeth. rings one can require less)



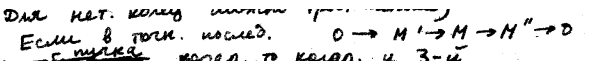
Original source details: GELFAND-P000106

**Diagram and typed description.** The page begins with the continuation of sections the sheaf of rel. should be of finite type. The reference has a comma after Serre, then quotation marks. The next assertion has a separate $N \subset M$, then N — a subsheaf of fin. type.

If in an exact seq.

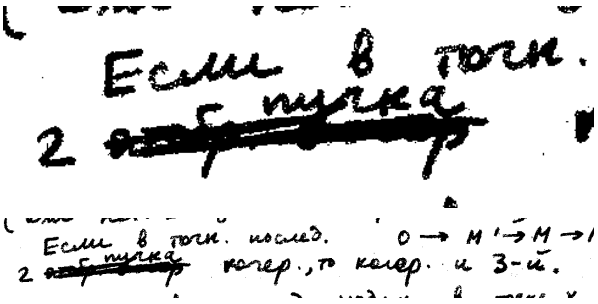
$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

2 [densely crossed out: illegible word; written above: sheaves[?]] are coher., the 3rd is coher. too.



Original source details: GELFAND-P000106

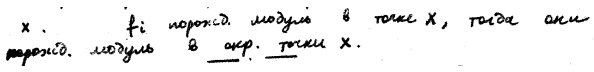
**Diagram and typed description.** The chain has M with one prime, M without primes, and M with two primes; all four arrows are unlabelled.



Original source details: GELFAND-P000106

**Uncertain reading.** Above the densely cancelled entry, source пучка (sheaves)[?] / illegible. Beneath the addition, long horizontal strokes and lower loops; the former word remains inseparable.

$x$. The $f_i$ gen. the module at the point $x$; then they gen. the module in a neighb. of the point $x$.



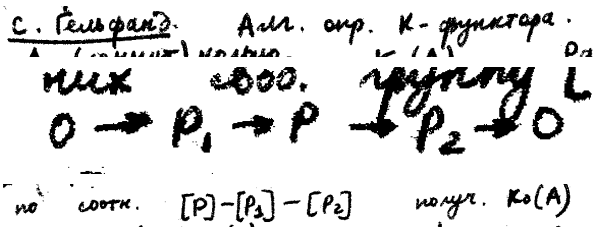
Original source details: GELFAND-P000106

**Diagram and typed description.** An isolated x followed by a period at the beginning. In the lower line, two separate straight strokes at neighb. and point; none beneath the last x.

S. Gelfand. Alg. def. of the $K$-functor. $A$ — a (commut.) ring. $K_0(A)$. Consider left proj. modules over $A$ and form the free group $[P]$ on them, abelian. Quotient by exact seq.

$$0 \longrightarrow P_1 \longrightarrow P \longrightarrow P_2 \longrightarrow 0$$

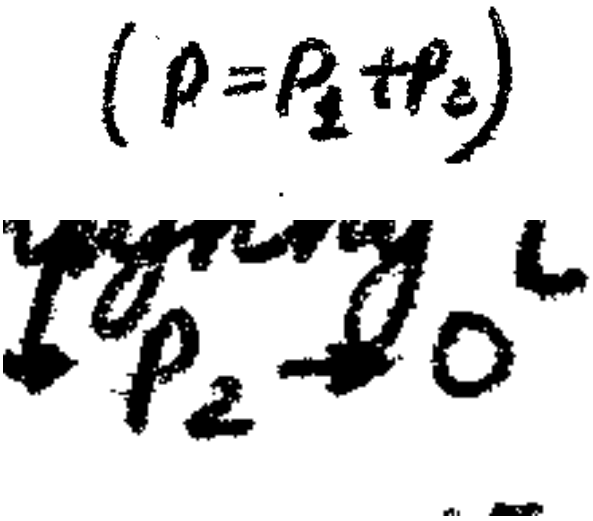
by the rel. $[P] - [P_1] - [P_2]$, obtaining $K_0(A)$



Original source details: GELFAND-P000106

**Diagram and typed description.** S. Gelfand. is separately underlined. The chain has $P_1$, P, $P_2$; the next line has $[P] - [P_1] - [P_2]$, with no $= 0$ written beside it, then obtaining $K_0(A)$.

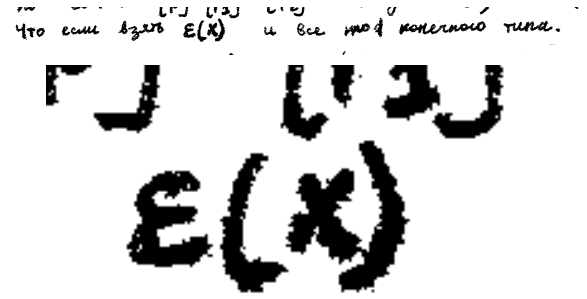
$(P = P_1 + P_{[?]})$.



Original source details: GELFAND-P000106

**Uncertain reading.** The last lower sign in the parenthesis has two rounded parts and a dense lower curve: 2[?] / 3[?]. The neighbouring chain and relation have lower 2 signs with flatter ends. The final index remains undetermined.

What if one takes $E(X)$ and all modules of finite type.



Original source details: GELFAND-P000106

**Uncertain reading.** Before (X), a rounded E-like letter with three exits to the right: E / calligraphic E / an $\varepsilon$-like form. Alphabet and exact case are not established by this sign.

$P$ — in topology, a loc. triv. bundle (sheaf of rings of cont. functions)

A proj. mod over $A$ is loc. free in the localization at a prime ideal (all if $A$ is commut.). A proj. mod over a loc. [crossed out: ring] is free.

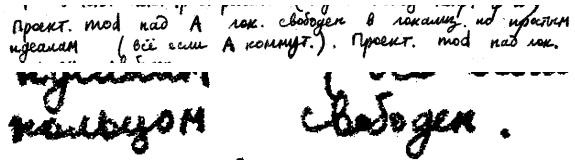


**Diagram and typed description.** After loc., the next line has ring crossed out, then a separate is free. The parenthesis says all if A is commut.

Original source details: GELFAND-P000106

$1/\overline{XI}$ Smirnov. Positive-def. functions on a group. Describe the extreme points. (cont. representations) for groups $G$ obtained from a nilp. alg. by introducing mult. by the Campbell — Hausdorff formula.

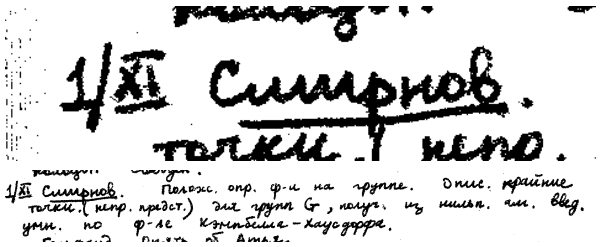


**Diagram and typed description.** XI in the date has separate over- and underlines. Smirnov. is separately underlined. Then Positive-def. functions on a group and the entry about the Campbell — Hausdorff formula.

Original source details: GELFAND-P000106

Gelfand. Again on Atiyah. $S(x) \in \Gamma(E_i)$, $\varphi(x)s(f^{-1}(x))$. This is $\int \kappa(x,y)s(y)\,dy$. For fixed $y$, $\kappa(x,y) \in \mathrm{Hom}(E_y, E_x)$,

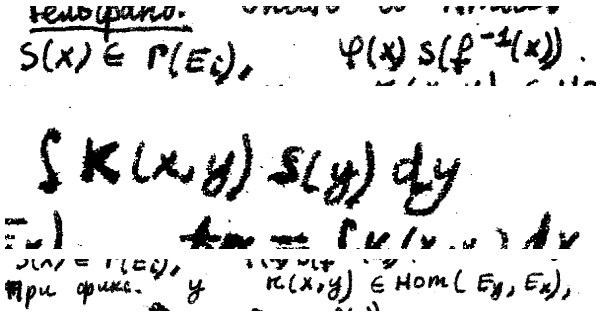
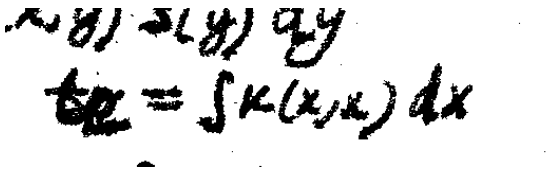


**Diagram and typed description.** An uppercase S(x) at the beginning; lowercase s after $\varphi(x)$ and in the integral. The first f has an explicit superscript $-1$. Hom has lower signs y and x. The kernel letter has K-/κ-like strokes.

Original source details: GELFAND-P000106

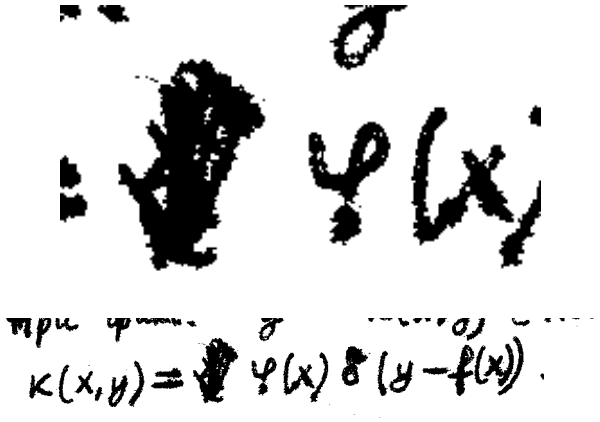
[Densely crossed out: tr[?]] $= \int \kappa(x,x)\,dx$.



**Uncertain reading.** Before the equality, a dense overwritten entry: tr[?] / illegible. The upper transverse stroke and lower loops merge; former lines remain inseparable.

Original source details: GELFAND-P000106

$$\kappa(x,y) = \text{[crossed out: ?]}\,\varphi(x)\delta(y-f(x)).$$



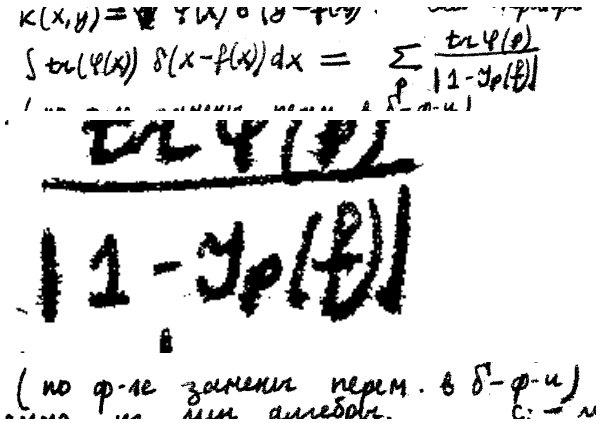
**Uncertain reading.** Between the equality and $\varphi(x)$, a dense cancelled vertical sign with upper and lower loops. Exact reading undetermined. A minus stroke is visible in the following $\delta$ argument: $y - f(x)$.

Original source details: GELFAND-P000106

The operator's trace $\int \operatorname{tr}(\kappa(x,x))\,dx$.

$$\int \operatorname{tr}(\varphi(x))\delta(x-f(x))\,dx = \sum_{\rho} \frac{\operatorname{tr}\varphi(\rho)}{|1-J_\rho(f)|}.$$

(by the change-of-variables formula for the $\delta$-function)



**Uncertain reading.** The sign before subscript $\rho$ is curved, with a short upper exit and a hooked lower part: $J_\rho(f)$[?] / $D_\rho(f)$ / $Д_\rho(f)$. Two vertical outer bars. No separate det entry is visible in the denominator.

Original source details: GELFAND-P000106

Lemma from lin. algebra. $C_i$ — lin. spaces. $d^2 = 0$. $T$ commutes with $d$.

$$\begin{array}{ccccccccccc} 0 & \longrightarrow & C_1 & \xrightarrow{d} & C_2 & \xrightarrow{d} & \cdots & \xrightarrow{d} & C_n & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle T_1} & & \downarrow{\scriptstyle T_2} & & & & \downarrow{\scriptstyle T_n} & & \\ 0 & \longrightarrow & C_1 & \xrightarrow{d} & C_2 & \longrightarrow & \cdots & \longrightarrow & C_n & \longrightarrow & 0 \end{array}$$

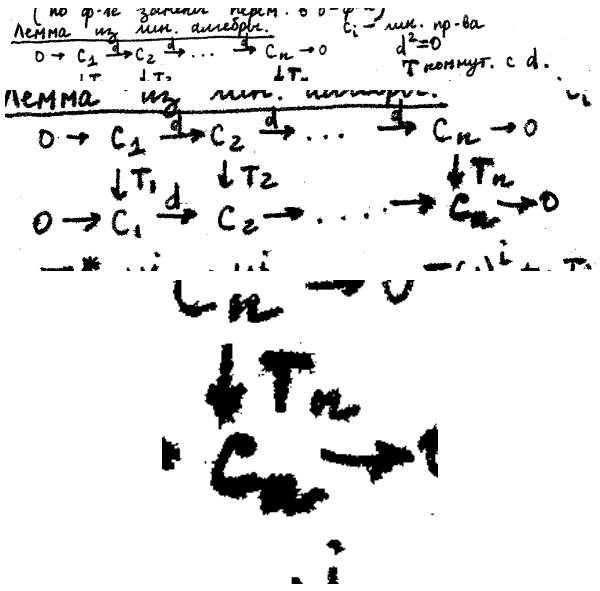


**Diagram and typed description.** Two horizontal chains, with 0 at their ends. The upper row has three internal d labels; the lower row has d only between $C_1$ and $C_2$. Three downward vertical marks labelled $T_1$, $T_2$, $T_n$. The right mark has a dense diamond-like lower end; beneath it the lower $C_n$ is densely overwritten. The typographic recreation preserves direction and labels; dense contours remain inseparable.

Original source details: GELFAND-P000106

$$\frac{\operatorname{Ker} d|C_i}{\operatorname{Im} C_{i-1}} = H^i.$$

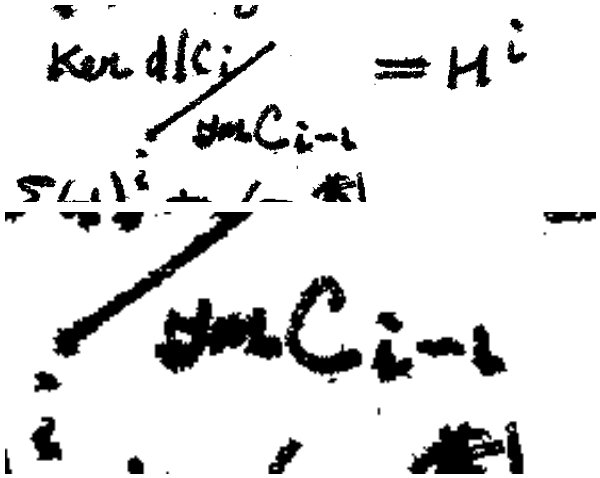


**Diagram and typed description.** The manuscript fraction bar is oblique. Above it, Ker $d|C_i$; beneath, a dense Im $C_{i-1}$, with no separate d or restriction bar visible after Im. At right, $= H^i$.

Original source details: GELFAND-P000106

$T_i^* : H^i \longrightarrow H^i$.

$$\sum (-1)^i \operatorname{tr} T_i = \sum (-1)^i \operatorname{tr}(T_i^*).$$

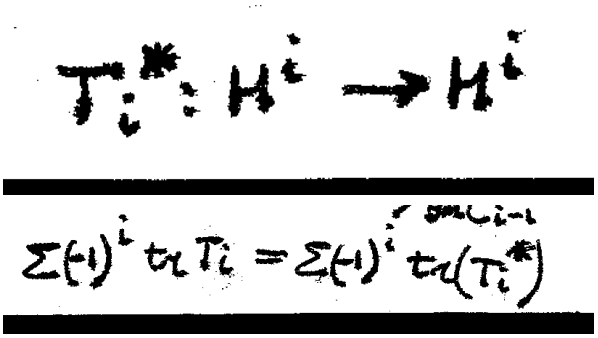
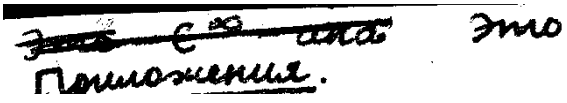


**Diagram and typed description.** At left, $T_i^* : H^i \longrightarrow H^i$. In the last equality both sums have superscript i on $(-1)$; tr $T_i$ at left, tr($T_i^*$) at right.

Original source details: GELFAND-P000106

GELFAND-P000107

[Crossed out at the top: This[?] $C^\infty$ that[?]; the rest illegible.]



**Uncertain reading.** A long stroke crosses the top line; dense writing beneath it on the left, $C^\infty$ in the middle, that[?] on the right. The word boundaries and exact reading are not established.

Original source details: GELFAND-P000107

This is a finite analogue.

Applications.

1. Usual Lefschetz formula. $X \xrightarrow{f} X$ isol. sing. points. De Rham complex. [On the right:]

$$\left(\sum \text{mult. of fixed points } = \right.$$
$$= \sum \text{traces on cohom.}$$

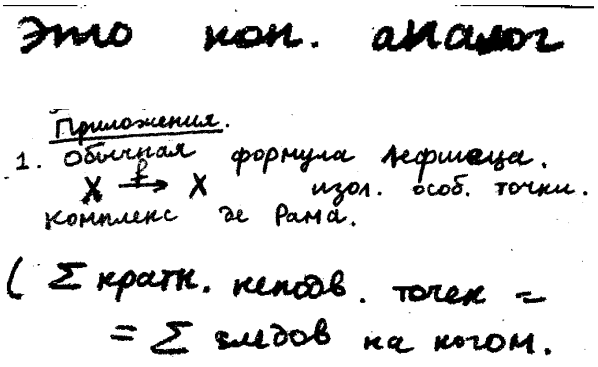
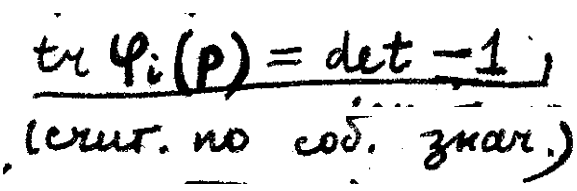


**Diagram and typed description.** This is a finite analogue. is separate from the crossed line. Applications. is underlined. On the right an opening parenthesis precedes the sum; an equals sign ends the upper line and another begins the lower one. No closing parenthesis is visible.

Original source details: GELFAND-P000107

$$\operatorname{tr} \varphi_i(\rho) = \det -1\,;$$

(computed by eigenvalues)



**Uncertain reading.** After det a horizontal minus and a 1-like sign; the readings $-1$ / $-I$ and an illegible argument remain undistinguished. A long line below the entire expression turns upward at the right.

Original source details: GELFAND-P000107

[On the right:]

$$\sum \pm 1 \quad \text{over fixed points.}$$

$$\sum (\pm 1)\lambda_{i_1} \cdots \lambda_{i_k} \qquad J_\rho = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

[A diagonal ellipsis lies between $\lambda_1$ and $\lambda_n$ in the matrix.]

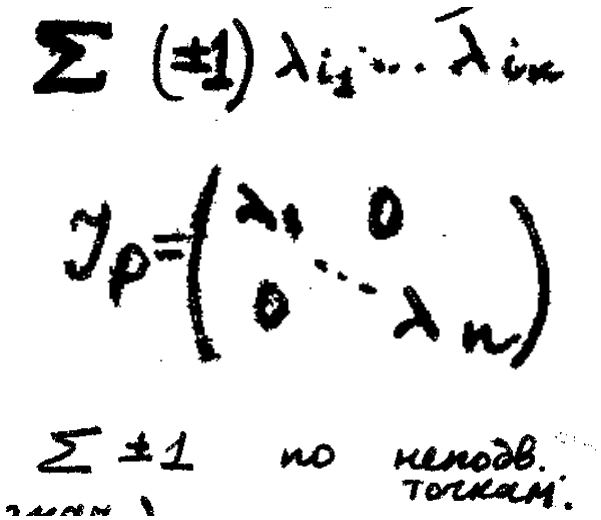


**Diagram and typed description.** The sum has a separate $(\pm 1)$, then $\lambda_{i_1} \cdots \lambda_{i_k}$. On the right a J-like letter with subscript $\rho$; inside parentheses $\lambda_1$ at upper left, $\lambda_n$ at lower right, zeros at the other corners and a diagonal ellipsis between them.

Original source details: GELFAND-P000107

2. Complex case. $f : X \longrightarrow X$ — holom.

$$\sum f_{i_1 \cdots i_k}\, dz_{i_1} \wedge \cdots \wedge dz_{i_k}$$

$f_{i_1 \cdots i_k}$ holom. and holom. diff. $H^{i,0}($

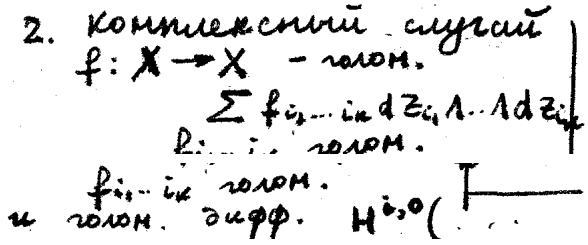
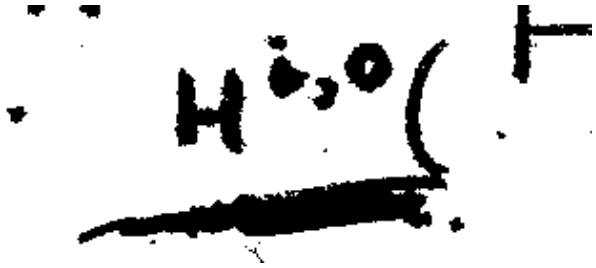


**Diagram and typed description.** To the right of the lines of forms a long vertical stroke downward and a horizontal stroke to the right from it; the corner has no arrows or labels. The parenthesis at $H^{i,0}($ remains open.

Original source details: GELFAND-P000107

[Densely crossed out below $H^{i,0}($: illegible.]



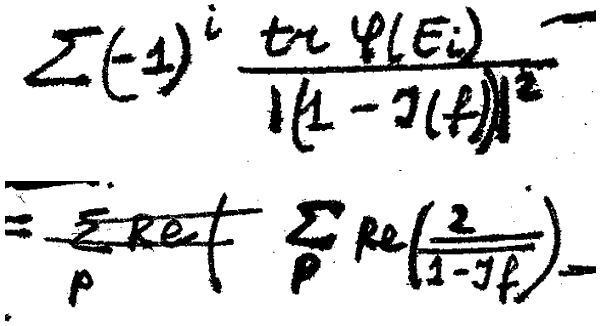
**Uncertain reading.** Below the open parenthesis dense horizontal strokes slope upward to the right; individual letters are not distinguished.

Original source details: GELFAND-P000107

$$\sum (-1)^i \frac{\operatorname{tr} \varphi(E_i)}{|(1 - J(f))|^2} =$$

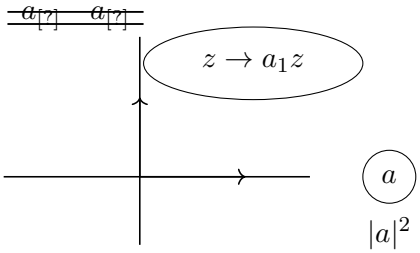
[Crossed out: $\sum_\rho \operatorname{Re}($; followed by:]

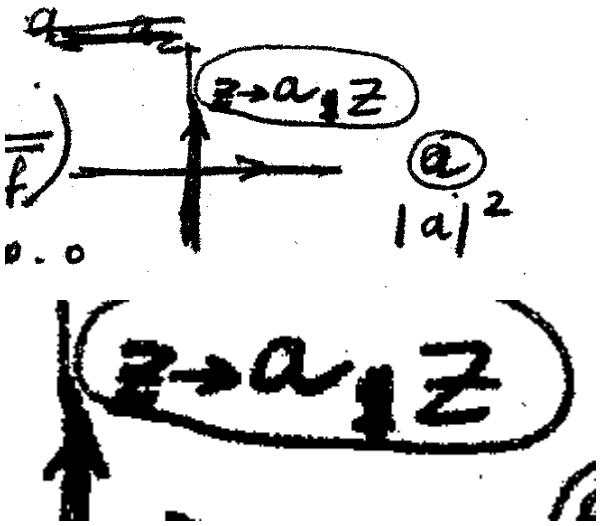
$$\sum_\rho \operatorname{Re}\left(\frac{2}{1 - J_f}\right)$$



**Uncertain reading.** The vertical strokes at the denominator are thickened: absolute value / double bars. Inside, a J-like letter / handwritten Cyrillic D; the numerator of the active sum on the right is 2, with two close horizontal lines below it.

Original source details: GELFAND-P000107

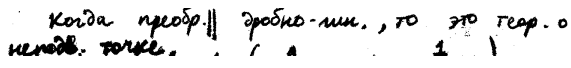


**Uncertain reading.** Two intersecting lines, arrows upward and to the right; the horizontal line continues beyond the arrowhead. In an oval $z \to a_1 z$, separately a circled a, below it $|a|^2$. Above the vertical line crossed a-like signs with illegible lower strokes; the exact reading of this writing is not established.

Original source details: GELFAND-P000107

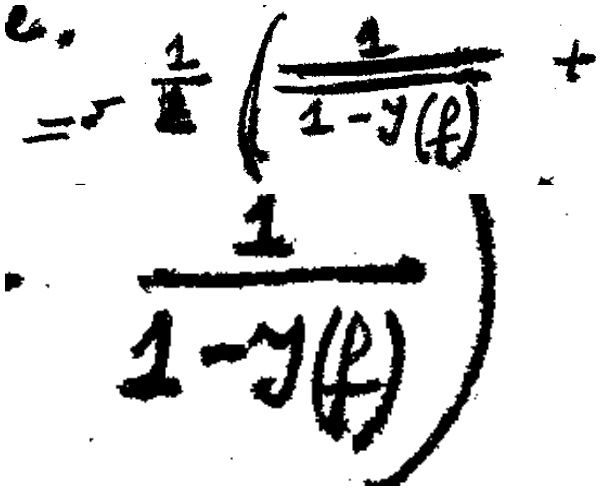
When the transf. ‖ is fractional-linear, this is the fixed-point theorem.



Original source details: GELFAND-P000107

**Diagram and typed description.** Two vertical lines stand between transf. and fractional-linear. The final words fixed point. are on the next line.

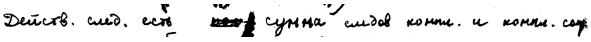
$$= \frac{1}{[?]} \left( \frac{1}{\overline{\overline{1 - J(f)}}} + \frac{1}{1 - J(f)} \right)$$



**Uncertain reading.** Below the first separate 1 a dense sign: 2[?] / 1[?] / illegible. The first inner fraction has two long horizontal lines, the second one; the first extra line may indicate conjugation of the fraction or denominator.

Original source details: GELFAND-P000107

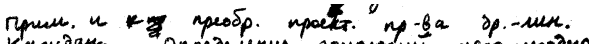
The real trace, is [crossed out: illegible] the sum of the complex and complex-conjugate traces.



Original source details: GELFAND-P000107

**Uncertain reading.** Between is and the sum a densely crossed horizontal record with a vertical stroke at the right; individual letters are not distinguished.

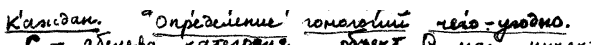
Appl. also to [heavily overwritten: illegible] fractional-linear transf. of projective space.



Original source details: GELFAND-P000107

**Uncertain reading.** After also to two dense superposed signs. Under the abbreviation for space two separate short lines; the preceding overwritten writing is illegible.

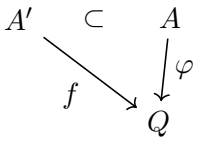
Kazhdan. Definition of homology of anything.

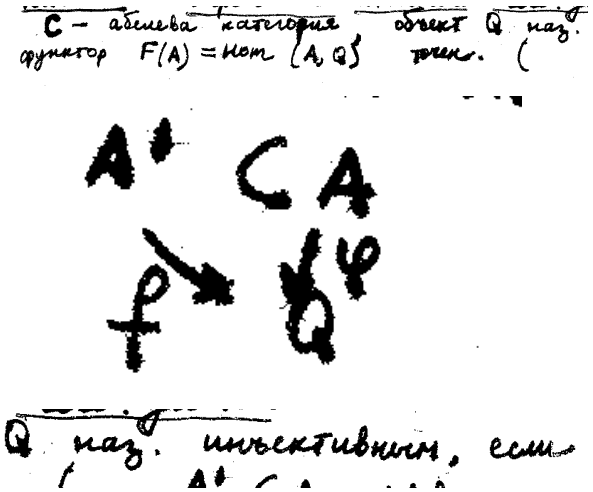


Original source details: GELFAND-P000107

**Diagram and typed description.** The surname and three parts of the heading have separate underlines. The Russian words for of anything are joined by a hyphen.

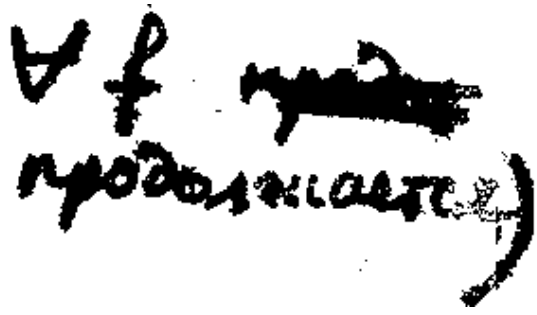
$C$ — an abelian category. an object $Q$ is called injective if the functor $F(A) = \mathrm{Hom}(A, Q)$ is exact. ($A' \subset A \; \forall f$ [crossed out: an illegible word] extends)



Original source details: GELFAND-P000107

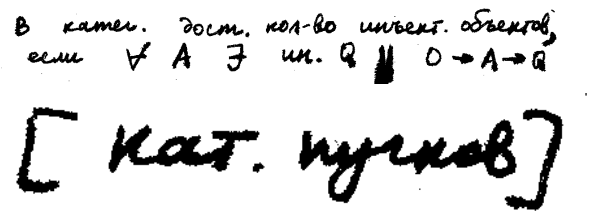
**Diagram and typed description.** Within the parentheses $A' \subset A$; from $A'$ an arrow downward to the right labelled f, from A downward to Q an arrow with $\varphi$ at the right. There is no superscript at $F(A)$.



Original source details: GELFAND-P000107

**Uncertain reading.** After $\forall f$ above extends a densely crossed word; its letters are not separated.

The categ. has enough injective objects if $\forall A\ \exists$ inj. $Q \parallel 0 \longrightarrow A \longrightarrow Q$. [cat. of sheaves]

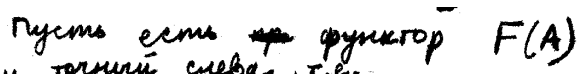



Original source details: GELFAND-P000107

**Diagram and typed description.** Before $0 \to A \to Q$ two vertical lines, with the lower part thickened. On the right a separate record in square brackets cat. of sheaves.

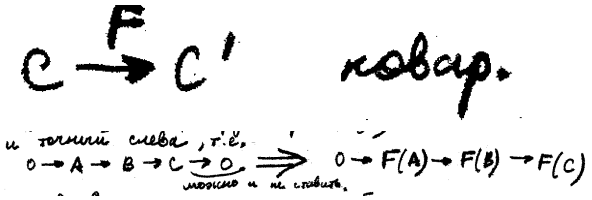
Let there be [crossed out: illegible] a functor $F(A)\ C \xrightarrow{F} C'$ covar. and left exact, i.e.

$$0 \longrightarrow A \longrightarrow B \longrightarrow \underbrace{C \longrightarrow 0}_{\text{may also be left out.}} \Longrightarrow 0 \longrightarrow F(A) \longrightarrow F(B) \longrightarrow F(C)$$



Original source details: GELFAND-P000107

**Uncertain reading.** Between there be and a functor a small densely crossed record, with a downward projecting stroke; the exact reading is not established.



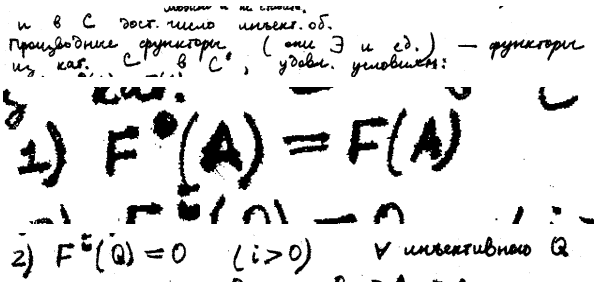
Original source details: GELFAND-P000107

**Diagram and typed description.** F above the arrow at $C \to C'$. Beneath $C \to 0$ in the original sequence an arc with may also be left out. In the sequence after the large arrow the last visible term is $F(C)$, followed by no dot or zero.

and $C$ has enough inj. obj. Derived functors (they $\exists$ and are unique) — functors from cat. $C$ to $C'$, satisfying the conditions:

1) $F^0(A) = F(A)$

2) $F^{[?]}(Q) = 0\ (i > 0)\ \forall$ injective $Q$

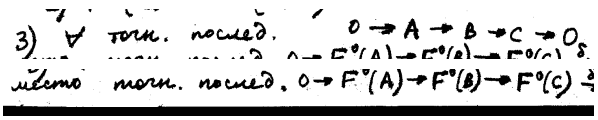


Original source details: GELFAND-P000107

**Uncertain reading.** In condition 1 the superscript is 0. In condition 2 the superscript above F is filled in, with a small upward projection: i[?] / 0[?] / illegible; separately $(i > 0)$ at the right.

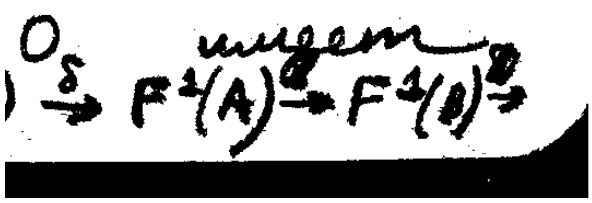
3) $\forall$ exact seq. $0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$ there is an exact seq.

$$0 \longrightarrow F^0(A) \longrightarrow F^0(B) \longrightarrow F^0(C) \xrightarrow{\delta} F^1(A) \longrightarrow F^1(B) \xrightarrow{\delta}$$



Original source details: GELFAND-P000107

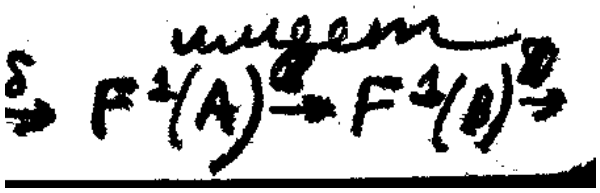
**Diagram and typed description.** At the bottom the sequence begins 0, then three terms with superscript 0, then an arrow with $\delta$, two terms with superscript 1 and another arrow with $\delta$.



**Undeciphered passage.** The last arrow is at the right edge; the following term is not visible in this frame.

Original source details: GELFAND-P000107

[Above the final arrows: induced[?] with a stroke continuing to the right.]



**Uncertain reading.** The readings are induc.[?] / induc...[?]; the ending merges into a long rightward stroke ending beside the last $\delta$. The precise attachment of the stroke is not established.

Original source details: GELFAND-P000107

GELFAND-P000108

Moreover 3') Let there be another exact seq. and a map.

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & 0 \end{array}$$

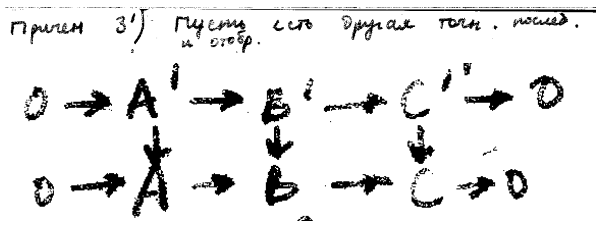
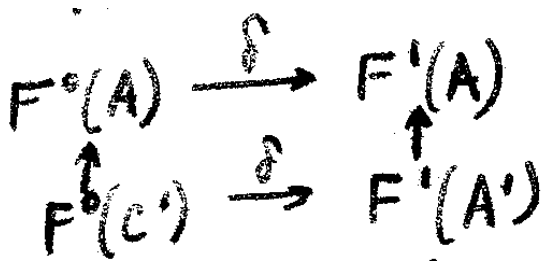


**Diagram and typed description.** Two rows $0 \to A' \to B' \to C' \to 0$ and $0 \to A \to B \to C \to 0$; three vertical arrows downward from the primed letters. The arrows have no labels.

Original source details: GELFAND-P000108

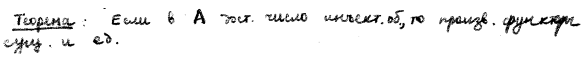
$$\begin{array}{ccc} F^0(A) & \xrightarrow{\delta} & F^1(A) \\ \uparrow & & \uparrow \\ F^0(C') & \xrightarrow{\delta} & F^1(A') \end{array}$$



**Uncertain reading.** At upper left $F^0(A)$: the two slanting sides and crossbar of A are visible. At upper right $F^1(A)$, below $F^0(C')$ and $F^1(A')$; both vertical arrows point upward.

Original source details: GELFAND-P000108

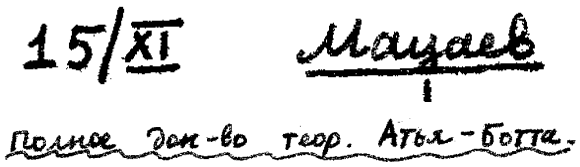
Theorem: If $A$ has enough inj. obj., the derived functors exist and are unique.



Original source details: GELFAND-P000108

**Diagram and typed description.** A separate large A in the hypothesis. Theorem is underlined, with the colon to the right of the line. Exist and are unique continues the writing on the next line.

15/XI Matsaev Complete proof of the Atiyah-Bott theorem.

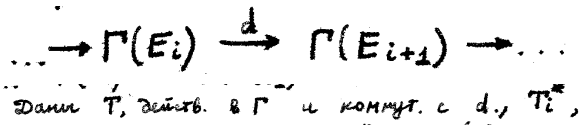


Original source details: GELFAND-P000108

**Diagram and typed description.** XI has lines above and below. A straight underline below the surname; a wavy underline below Complete proof of the Atiyah-Bott theorem.

$$\cdots \longrightarrow \Gamma(E_i) \xrightarrow{d} \Gamma(E_{i+1}) \longrightarrow \cdots$$

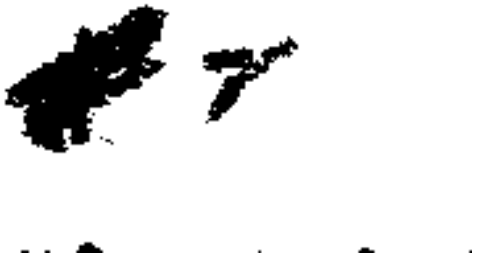
Given $T$, acting on $\Gamma$ and commuting with $d$, $T_i^*$,



Original source details: GELFAND-P000108

**Diagram and typed description.** d above the middle arrow of the complex, subscript $i+1$ on the right. On the next line $T_i^*$, followed by a comma.

[Densely crossed out: an illegible sign; to the right a T-like sign[?].]

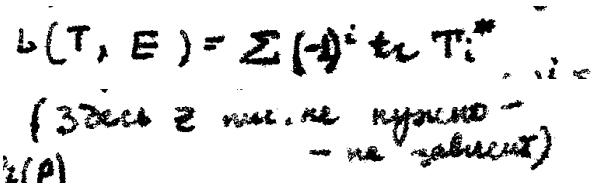


Original source details: GELFAND-P000108

**Uncertain reading.** Dense crossed writing slopes upward to the right; farther right a separate small sign with a horizontal top and slanting stem: T[?] / illegible.

$$L(T, E) = \sum (-1)^i \operatorname{tr} T_i^*$$

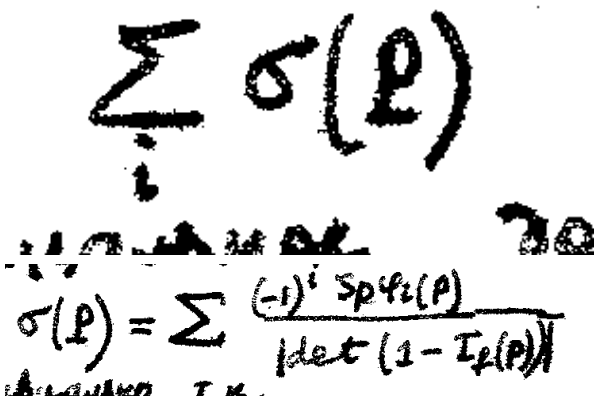
[On the right:] (Here we do not need $z$ — does not depend)



Original source details: GELFAND-P000108

**Diagram and typed description.** T in the sum has a superscript star and subscript i. In the right parenthesis z, then we do not need, below — does not depend.

$$\sum_i \sigma(\rho) \qquad \sigma(\rho) = \sum \frac{(-1)^i \operatorname{Sp} \varphi_i(\rho)}{|\det(1 - I_f(\rho))|}$$



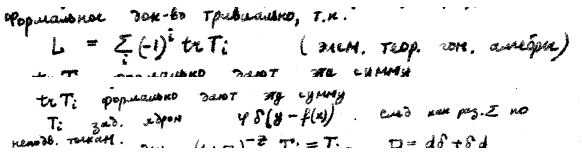
Original source details: GELFAND-P000108

**Uncertain reading.** Beneath the first sum a small sign with a separate dot above: i[?] / illegible index. In the denominator an I-like letter with a wide base / a J-like letter; followed by subscript f and $(\rho)$.

The formal proof is trivial, since

$$L = \sum_i (-1)^i \operatorname{tr} T_i$$

(elem. theorem of lin. algebra) $\operatorname{tr} T_i$ formally give this sum. $T_i$ is given by the kernel $\varphi\delta(y - f(x))$. The trace is precisely $\sum$ over fixed points.



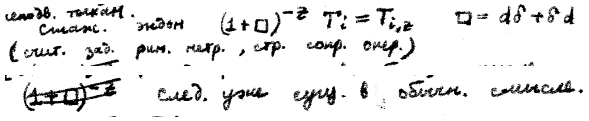
Original source details: GELFAND-P000108

**Diagram and typed description.** In the formal sum T has subscript i without a star. In the kernel a horizontal minus follows y, then $f(x)$. Fixed points. is carried to the next line.

Smooth by the kernel

$$(1 + \Box)^{-z} T_i = T_{i,z}, \qquad \Box = d\delta + \delta d$$

(assume a Riem. metric is given, Herm. adjoint oper.) [Crossed out: $(1 + \Box)^{-z}$.] The trace already exists in the usual sense.



Original source details: GELFAND-P000108

**Diagram and typed description.** Exponent $-z$; a comma in the right-hand $T_{i,z}$. A separate formula with a square, d and $\delta$. Below, a crossed copy of $(1 + \Box)^{-z}$.

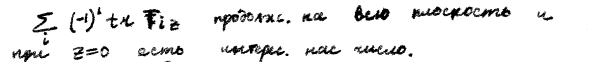
$$\operatorname{tr} T_{iz} = \sum_\lambda \pi_\lambda T i_\lambda (1 + \lambda)^{-z}$$



Original source details: GELFAND-P000108

**Uncertain reading.** The indices of the left-hand T are densely overwritten: iz / i,z. On the right $\pi_\lambda$, T, $i_\lambda$ and $(1 + \lambda)^{-z}$; no separate tr is visible after the sum.

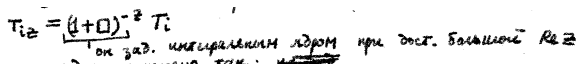
$\sum_i (-1)^i \operatorname{tr} T_{iz}$ continues to the whole plane and at $z = 0$ is an integral as a number.



Original source details: GELFAND-P000108

**Diagram and typed description.** Subscript i in the sum; after the whole plane and, the continuation at $z = 0$ is an integral as a number.

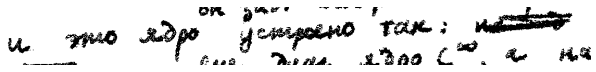
$$T_{iz} = \underbrace{(1+\square)^{-z}}_{\text{it is given by an integral kernel for sufficiently large } \operatorname{Re} z} T_i$$



Original source details: GELFAND-P000108

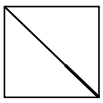
**Diagram and typed description.** Beneath $(1+\square)^{-z}$ a straight bracket with a short downward stroke; the explanation below it ends in Re $z$.

and this kernel is arranged as follows: [crossed out: an illegible word]



Original source details: GELFAND-P000108

**Uncertain reading.** After arranged as follows: two dense, almost horizontal crossing strokes; the original letters are not separated.

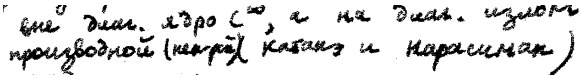




Original source details: GELFAND-P000108

**Uncertain reading.** An unlabelled square; the diagonal goes from upper left to lower right, thickening at the latter corner. Thickening / arrowhead[?]; it has no separate label.

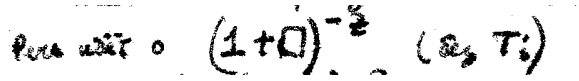
off the diag. the kernel is $C^\infty$, and on the diag. breaks in the derivative (contr.[?] to Katz and Narasimhan)



Original source details: GELFAND-P000108

**Uncertain reading.** In parentheses the abbreviation contr.[?] with a dense ending, then to Katz and Narasimhan. Before the parenthesis derivative.

The discussion concerns $(1+\square)^{-z}$ (without $T_i$).

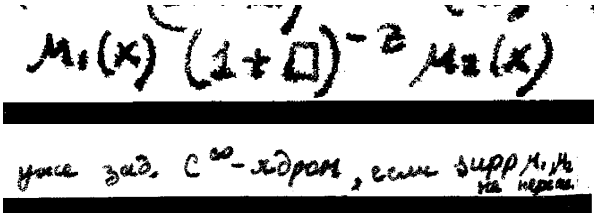


Original source details: GELFAND-P000108

**Diagram and typed description.** The final parenthesis reads without $T_i$. A separate $(1+\square)^{-z}$ precedes it.

$$\mu_1(x)(1+\square)^{-z}\mu_2(x)$$

is already given by a $C^\infty$ kernel if supp[?] $\mu_1$ $\mu_2$ at the edges.



**Uncertain reading.** The outer letters $\mu_1, \mu_2$ / $M_1, M_2$; the first sign is more angular. After if, supp[?] / diff.[?] / illegible; below, at the edges. The precise separator between the two letters after the abbreviation is not distinguished.



Original source details: GELFAND-P000108

GELFAND-P000109





**Diagram and typed description.** An empty quadrilateral in the upper-left margin; no diagonal or labels. The side ends do not join at lower right; a small gap also at upper left.

Original source details: GELFAND-P000109

When can one correctly multiply two generalized fn.s with singularities on two transversal manifolds.

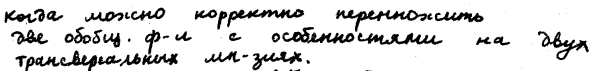


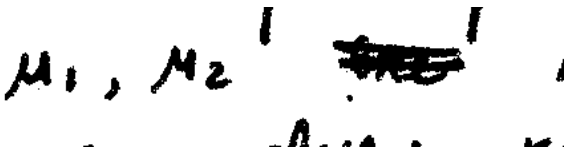
**Diagram and typed description.** The writing starts When and ends manifolds. There is no question mark at the end.

Original source details: GELFAND-P000109

$\mu_1, \mu_2$ [crossed out: illegible]

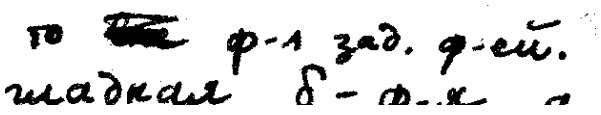
$\cap$ supp. off the curve[?], then [crossed out: illegible]

the fn. is given by a fn.



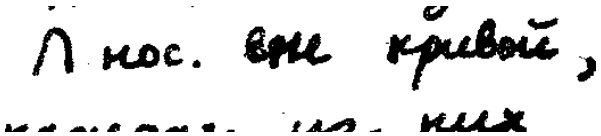
**Uncertain reading.** Between $\mu_1, \mu_2$ and $\cap$ a short densely crossed record. The first $\mu$ has an angular top and a long lower stroke.

Original source details: GELFAND-P000109



Original source details: GELFAND-P000109

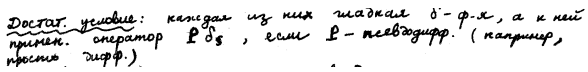
**Uncertain reading.** After then and before fn., dense crossed writing; the original signs are not separated.



Original source details: GELFAND-P000109

**Uncertain reading.** Off the curve[?] / off the straight line: the initial letter of the final word is dense, the next has a long lower stroke.

Suff. condition: each of them is a smooth $\delta$-fn., and to it is applied an operator $P\delta_S$, if $P$ is pseudodiff. (for example, simply diff.)

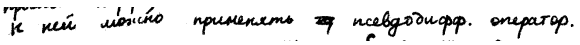


Original source details: GELFAND-P000109

**Diagram and typed description.** A wavy underline below Suff. condition. After operator, a separate $P\delta_S$; then if $P$ is pseudodiff. The parenthesis ends simply diff.)

To it one can apply [densely crossed out: an illegible sign]

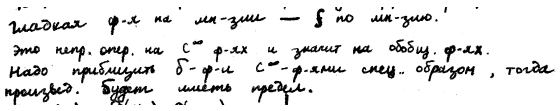
a pseudodiff. operator.



Original source details: GELFAND-P000109

**Uncertain reading.** Between apply and pseudodiff., a short dense sign with a lower slanting stroke.

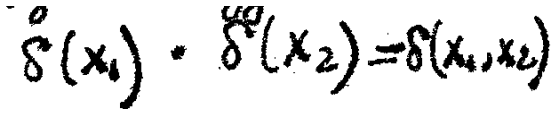
A smooth fn. on a manifold — $\int$ over the manifold. This is a cont. oper. on $C^\infty$ fn.s and therefore on generalized fn.s. One must approximate the $\delta$-fn.s by $C^\infty$ fn.s in a special way, then the product will have a limit.



Original source details: GELFAND-P000109

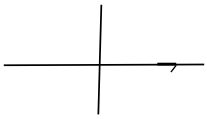
**Diagram and typed description.** After the dash a separate integral sign without limits. Then two sentences on the cont. oper. and approximation of $\delta$-fn.s; ending in limit.

$$\delta(x_1) \cdot \delta(x_2) = \delta(x_1, x_2)$$



Original source details: GELFAND-P000109

**Diagram and typed description.** A dot between the first two factors; on the right $\delta(x_1, x_2)$ without a separate final period.

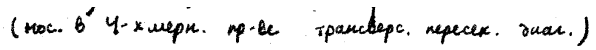




Original source details: GELFAND-P000109

**Uncertain reading.** Intersecting vertical and horizontal lines without labels. To the right of the intersection a short thickening and slanted stroke: arrowhead[?] / unevenness of the line. No separately marked point.

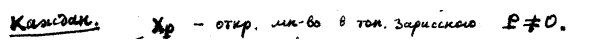
(supp. in 4-dim. space transvers. intersect. of diag.)



Original source details: GELFAND-P000109

**Diagram and typed description.** The parenthesis below the formula: supp. in 4-dim. space transvers. intersect. of diag.

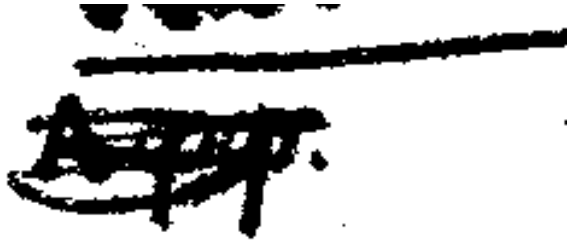
Kazhdan. $X_P$ — an open set in the Zariski top. $P \neq 0$.



Original source details: GELFAND-P000109

**Diagram and typed description.** The surname Kazhdan. is underlined. Then $X_P$, on the right $P \neq 0$.

[On the left, crossed out: an illegible word.]



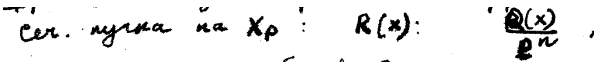
Original source details: GELFAND-P000109

**Uncertain reading.** Below the underlined surname and to the left of This is a basis, several dense crossing strokes.

This is a basis of open sets on an aff. alg. variety. Sect. of the sheaf on $X_P$: $R(x)$: $\frac{Q(x)}{P^n}$,

[A short sign crossed out: =[?] / illegible.]

they are identified if the diff. becomes 0 on the variety.



Original source details: GELFAND-P000109

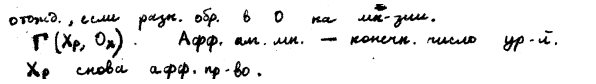
**Diagram and typed description.** Colons after $X_P$ and after $R(x)$. Numerator $Q(x)$, denominator $P^n$; a comma on the right.



Original source details: GELFAND-P000109

**Uncertain reading.** Before they, a short dark sign with two almost horizontal strokes; original sign =[?].

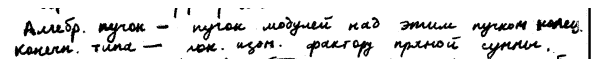
$\Gamma(X_P, O_X)$. Aff. alg. variety — a finite number of eq.s. $X_P$ is again an aff. space.



Original source details: GELFAND-P000109

**Diagram and typed description.** A period after $\Gamma(X_P, O_X)$. The final line reads precisely again an aff. space.

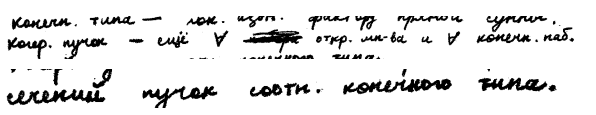
An algebraic sheaf — a sheaf of modules over this sheaf of rings. Of finite type — loc. isom. to a quotient of a direct sum.



Original source details: GELFAND-P000109

**Diagram and typed description.** Two lines: the definition of an algebraic sheaf and Of finite type — loc. isom. to a quotient of a direct sum.

A coher. sheaf — further ∀ [crossed out: illegible] open set and ∀ finite coll. of sections, the sheaf of rel.s is of finite type.



Original source details: GELFAND-P000109

**Uncertain reading.** After the first ∀ a densely crossed word. Then open set and ∀ finite coll.; continuation on the next line of sections, the sheaf of rel.s is of finite type.

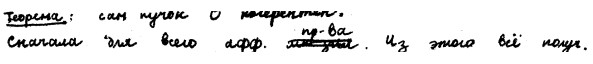
Theorem: the sheaf $O$ itself is coherent.



Original source details: GELFAND-P000109

**Diagram and typed description.** Theorem is underlined, with the colon to the right. After the sheaf a separate O, then itself is coherent.

First for the whole aff. [crossed out: variety[?] / illegible.] [Inserted above: space.] From this everything follows autom.



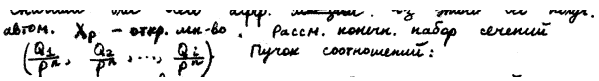
Original source details: GELFAND-P000109

**Uncertain reading.** Beneath the insertion space, a crossed record variety[?]. After it a period; From this everything follows continues on the same line, autom. below.

$X_P$ — an open set. Consider a finite collection of sections

$$\left(\frac{Q_1}{P^n}, \frac{Q_2}{P^n}, \ldots, \frac{Q_i}{P^n}\right)$$

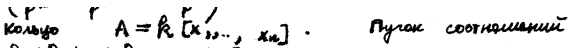
Sheaf of relations:



Original source details: GELFAND-P000109

**Diagram and typed description.** The tuple has subscripts $1, 2, \ldots, i$, denominators $P^n$; no separate period after the parenthesis. On the right Sheaf of relations:

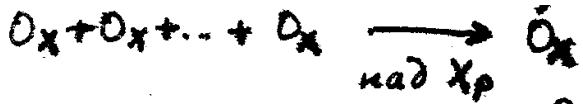
Ring $A = k[x_1, \ldots, x_n]$. [On the right again: Sheaf of relations]



Original source details: GELFAND-P000109

**Diagram and typed description.** Ring $A = k[x_1, \ldots, x_n]$ with a final period. On the right Sheaf of relations is repeated without a final sign.

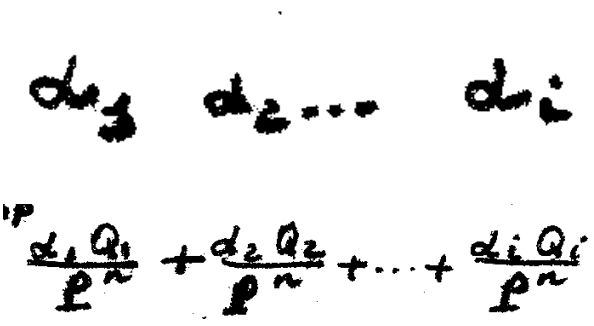
$$O_X + O_X + \cdots + O_X \underset{\text{over } X_P}{\longrightarrow} O_X$$



Original source details: GELFAND-P000109

**Diagram and typed description.** Everywhere $O_X$; plus signs between the initial summands. Beneath the rightward arrow over $X_P$.

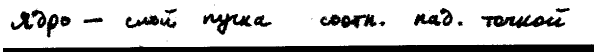
$$d_{3[?]}\, d_2 \cdots d_i \longrightarrow \frac{d_1 Q_1}{P^n} + \frac{d_2 Q_2}{P^n} + \cdots + \frac{d_i Q_i}{P^n}$$



Original source details: GELFAND-P000109

**Uncertain reading.** The first lower index on the left resembles 3 / 1; its lower end is very dense. The first $d$ on the right has index 1. Then on the left $d_2$, ellipsis and $d_i$; no period at the end of the formula.

Kernel — the fibre of the sheaf of rel.s over a point



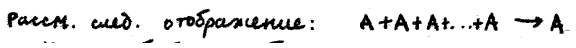
Original source details: GELFAND-P000109

**Diagram and typed description.** The final line reads fibre, then of the sheaf of rel.s over a point. No separate final sign after the last word.

GELFAND-P000110

Consider the following map:

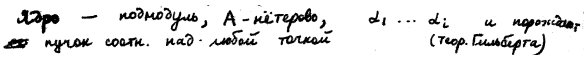
$$A + A + A + \cdots + A \longrightarrow A$$



Original source details: GELFAND-P000110

**Diagram and typed description.** A colon after map; no labels on the rightward arrow. No separate period after the final A.

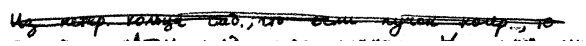
One can dispense with $P$, the rel.s are the same, Kernel — a submodule, $A$ is Noetherian, $d_1 \cdots d_i$ and generate [crossed out: an illegible sign] the sheaf of rel.s over any point (Hilbert's theorem)



Original source details: GELFAND-P000110

**Uncertain reading.** On the left before the sheaf of rel.s, a short crossed sign. On the preceding line $d_1$, ellipsis, $d_i$ and generate; the right end of the last word reaches the image edge.

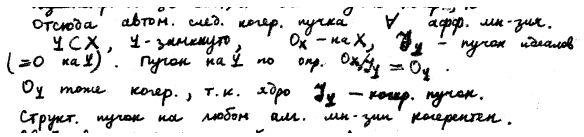
[The whole line crossed out: From coher. of the ring it follows that every sheaf is coher., then[?]; individual portions illegible.]



Original source details: GELFAND-P000110

**Uncertain reading.** Long strokes cross the whole line. The reading From coher. of the ring it follows that every sheaf is coher., then is partial; exact letters beneath the dense portions are not established.

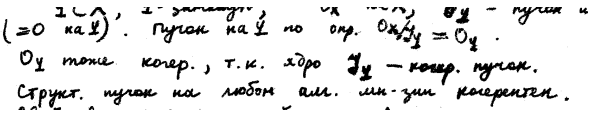
From this autom. follows coher. of the sheaf $\forall$ aff. variety. $Y \subset X$, $Y$ is closed, $O_X$ is on $X$, $J_Y$[?] — the sheaf of ideals ($= 0$ on $Y$).



Original source details: GELFAND-P000110

**Diagram and typed description.** A separate $\forall$ after coher. of the sheaf. On the next line $Y \subset X$, then $Y$ is closed.

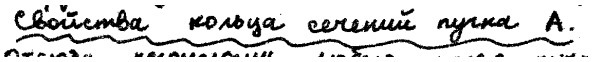
The sheaf on $Y$ by def. $O_X/J_Y = O_Y$. $O_Y$ is also coher., since the kernel $J_Y$ is a coher. sheaf. The struct. sheaf on any alg. variety is coherent.



Original source details: GELFAND-P000110

**Uncertain reading.** The ideal letter J[?] / I-like: a curved stroke at the top, a dense loop below. In the formula a slash between $O_X$ and this letter with subscript Y; on the right $= O_Y$. After kernel on the next line the same letter.

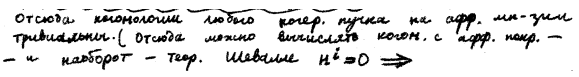
Properties of the ring of sections of the sheaf $A$.



Original source details: GELFAND-P000110

**Diagram and typed description.** The whole line Properties of the ring of sections of the sheaf A. has a wavy underline.

From this the cohomology of any coher. sheaf on an aff. variety is trivial. (From this one can compute cohom. using an aff. cover — — and conversely — Chevalley's theorem $H^i = 0 \Longrightarrow$

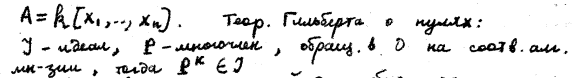


Original source details: GELFAND-P000110

**Undeciphered passage.** After $H^i = 0$ a double arrow pointing right; there is no continuation to its right. The parenthesis opened before the second From this is not closed.

$$A = k[x_1, \ldots, x_n].$$

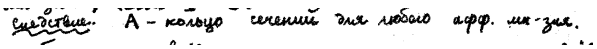
Hilbert's theorem on zeros: $J$ is an ideal, $P$ is a polynomial vanishing on the corresp. alg. variety, then $P^k \in J$



Original source details: GELFAND-P000110

**Diagram and typed description.** A final period after the ring $A = k[x_1, \ldots, x_n]$. A colon after Hilbert's theorem on zeros. P has exponent k; no separate period after $J$.

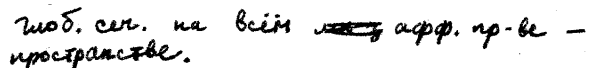
Corollary. $A$ is the ring of sections for any aff. variety.



Original source details: GELFAND-P000110

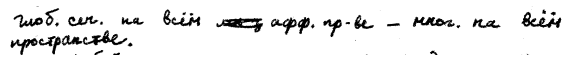
**Diagram and typed description.** A wavy underline beneath Corollary. Then A is the ring of sections for any aff. variety.

Glob. sect. on the whole [crossed out: illegible] aff. space — polyn.[?] on the whole space.



Original source details: GELFAND-P000110

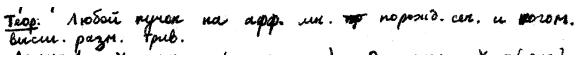
**Uncertain reading.** Between whole and aff., a short crossed record; its lower-right end forms a loop.



Original source details: GELFAND-P000110

**Uncertain reading.** After the dash the abbreviation mnog.[?] / mnozh.; the final sign joins the preceding one closely. Polyn.[?] is a tentative reading; the abbreviation is not expanded.

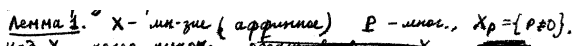
Theor. Any sheaf on an aff. variety [dense sign: illegible] is generated by sect. and its cohom. in higher dim. is triv.



Original source details: GELFAND-P000110

**Uncertain reading.** Between variety and is generated, a separate dense sign with a lower stroke. No word coher. is visible before sheaf in this line.

Lemma 1. $X$ — a variety (affine), $P$ — a polyn., $X_P = \{P \neq 0\}$.

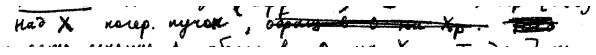


Original source details: GELFAND-P000110

**Diagram and typed description.** Lemma 1. has a straight underline. Affine is in parentheses. On the right $X_P = \{P \neq 0\}$.

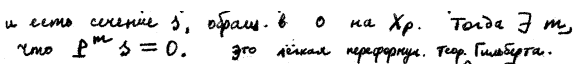
A coher. sheaf over $X$, [crossed out: vanishing on $X_P$[?].] [Separately crossed out on the right: illegible.]

and there is a section $s$ vanishing on $X_P$. Then $\exists m$ such that $P^m s = 0$. This is an easy reformulation of Hilbert's theorem.



Original source details: GELFAND-P000110

**Uncertain reading.** After sheaf, long crossing strokes; portions of vanishing, on $X_P$[?] are distinguished. Farther right a separate densely crossed record.



Original source details: GELFAND-P000110

**Diagram and typed description.** The uncanceled line and there is a section s vanishing on $X_P$. Then $\exists m$ such that $P^m s = 0$.

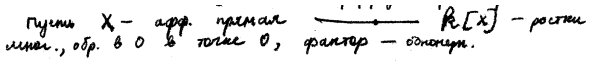
Let $X$ be the aff. line



Original source details: GELFAND-P000110

**Uncertain reading.** A horizontal line with a marked point; no label at the point. At the left end a short downward slant: left arrowhead[?] / bend in the line. No arrowhead is visible on the right.

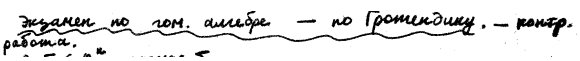
$k[x]$ — germs of polyn.s vanishing at the point 0, quotient — one-dim.



Original source details: GELFAND-P000110

**Diagram and typed description.** To the right of the drawing $k[x]$ — germs; on the next line of polyn.s vanishing at the point 0, quotient — one-dim.

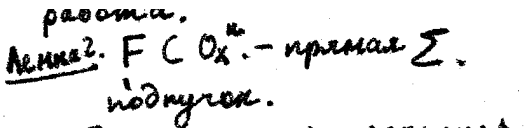
Examination in hom. algebra — following Grothendieck. — class test.



Original source details: GELFAND-P000110

**Diagram and typed description.** The wavy underline runs from Examination to Grothendieck. After it class, and test. is carried below without an underline.

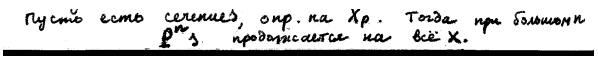
Lemma 2. $F \subset O_X^{n[?]}$ — direct $\sum$. [Below, with an insertion sign: subsheaf.]



Original source details: GELFAND-P000110

**Uncertain reading.** After direct, a separate $\sum$ / $\Sigma$ with a period to the right. The superscript of $O_X$ is n[?] / m-like; the small upper strokes merge. Subsheaf is inserted beneath F.

Let there be a section $s$ defined on $X_P$. Then for large $m$, $P^m s$ extends to the whole $X$.



Original source details: GELFAND-P000110

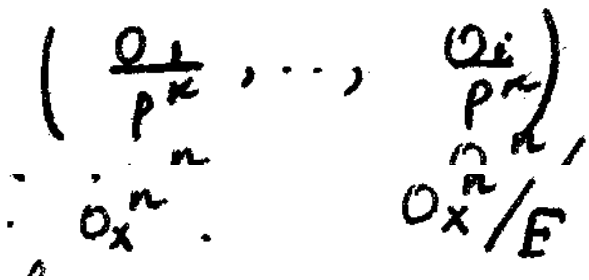
**Diagram and typed description.** The section s is defined on $X_P$. On the last line $P^m s$ extends to the whole X. A final period is visible.

GELFAND-P000111

(also Hilbert's theorem).

$$\left(\frac{O_1[?]}{p^k}, \ldots, \frac{O_i[?]}{p^k}\right)$$

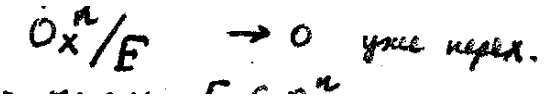
mult. by $p^k$, but this may not be a sect. of $O_X^n$.



Original source details: GELFAND-P000111

**Uncertain reading.** Rounded O-like signs in the numerators, with lower $1$ and $i$; their lower portions join the fraction bars. Readings $O$ / $Q$ are not fully distinguished. Beneath, $O_X^n$ and $O_X^n/F$.

$O_X^n/F \longrightarrow 0$ already nepr.[?]



Original source details: GELFAND-P000111

**Uncertain reading.** After already, a short joined entry source непр.[?] (nepr.[?]); its middle and final strokes merge densely. The abbreviation is not expanded. Before already, an arrow to a separate 0.

Now it already suffices for the proof that the cohom. of $F \subset O_X^n$ are zeros. $X$ — an aff. variety, $(X_{p_1}, X_{p_2}, \ldots, X_{p_i})$ a cover. The cohomology of $X$ with coefficients in $F$ in this cover are zeros.

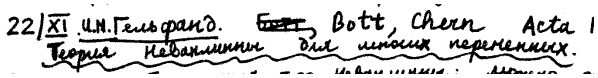
$22/\overline{XI}$. I.M. Gelfand. [Crossed out: an illegible entry.] Bott, Chern Acta Math., 1965, 114 $N = 1$–2.



Original source details: GELFAND-P000111

**Uncertain reading.** Before Bott, a densely canceled entry with a Cyrillic Б-like beginning; its former letters are not established. A separate comma follows the cancellation.

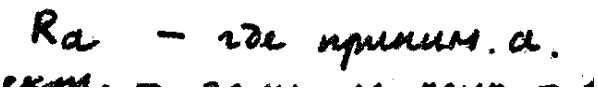
Theories of Nevanlinna for several variables.



Original source details: GELFAND-P000111

**Diagram and typed description.** XI has separate over- and underlines; the surname with initials has a straight underline. The heading Theories of Nevanlinna for several variables. has a long wavy underline.

Picard's theorem, the more precise theorem of Nevanlinna: one can estimate the number of zeros in a circle of radius $R$ (may be less). $R_a$ — where it takes $a$.



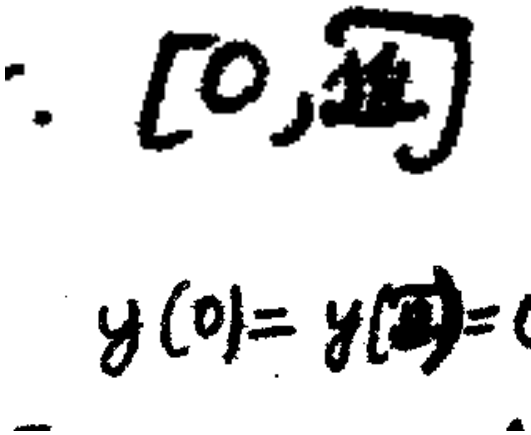
Original source details: GELFAND-P000111

**Diagram and typed description.** Before the lower $a$, a large R-like letter with a slanting right leg. At right, a dash and where it takes $a$.

Most values are taken equ. often. Defect — if not taken, then $1$; if less often, then a numer. char. The $\sum$ of defects does not exceed two. $\frac{a(z)}{b(z)}$ omits at most $2$ values. $(a(z), b(z))$ a merom. fn. — a curve in $C^2$, a value — a line (only the ratio matters). This curve intersects all lines except, perhaps, two. Generalization. In $3$-dim. space an entire curve $(a(z), b(z), c(z))$ intersects all planes passing through $0$, except, perhaps, three. (In $n$-dim. a finite number.)

For us this? $-y'' + p(x)y = \lambda y$ — an eqn. on the int. $[0, \pi]$ with conditions [At right:] $y(0) = 0$, $y'(0) = 0$.

[Crossed out: an illegible short sign.] $\lambda$ [ne sushch.; abbreviation unexpanded], $y(0) = y(\pi) = 0 \Rightarrow$ a countable number of eig. values.



**Uncertain reading.** The right endpoint of the interval and the argument in the second entry $y(0) = y(\pi) = 0$ are densely overwritten. The current sign has $\pi$-like strokes; the former entry is not established.

Original source details: GELFAND-P000111

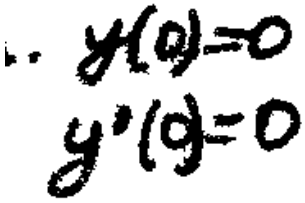


**Diagram and typed description.** Two separate lines in the upper right field: $y(0) = 0$ and $y'(0) = 0$. Both final signs are oval, with a white centre.

Original source details: GELFAND-P000111



**Uncertain reading.** Written source не сущ. (ne sushch.), with a period and following comma. The expansion of this abbreviation is not established on this line.

Original source details: GELFAND-P000111



**Uncertain reading.** Immediately before $\lambda$, a short dense cancellation with a loop on the left and horizontal strokes. The former sign is unread.

Original source details: GELFAND-P000111

Possibly, for most boundary conditions there is an eigenvalue $y(0) = 0$, $ay'(\pi) + by(\pi) = 0$. Consider the solution

$$\begin{cases} y(0,\lambda) = 0 \\ y'(0,\lambda) = 1. \end{cases}$$

It is analytic in $\lambda$.

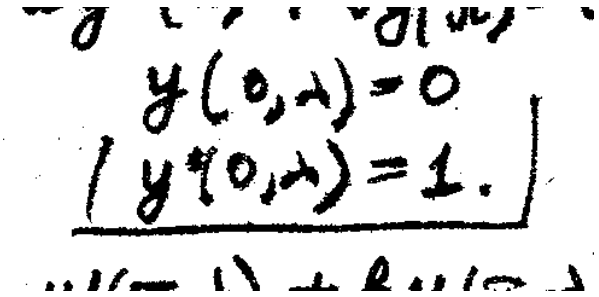
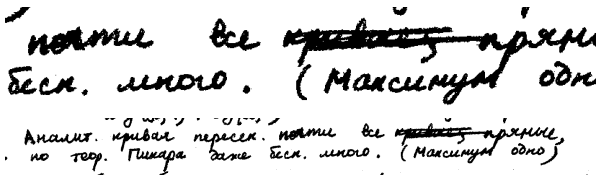


**Diagram and typed description.** Two lines $y(0,\lambda) = 0$, $y'(0,\lambda) = 1$.. Below them a horizontal line turns upward vertically at right. To the left of the second line a separate short curved stroke, unconnected to the lower line.

Original source details: GELFAND-P000111

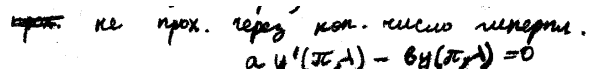
Eig. values — when $ay'(\pi,\lambda) + by(\pi,\lambda) = 0$ has a nonzero solution. An analytic curve intersects almost[?] all [crossed out: an illegible word], lines, and by Picard's theorem even infinitely many. (At most one)



**Uncertain reading.** Before all, a densely joined word: source почти (almost) / наши (our); its middle strokes are not fully distinguished. After all a long crossed-out entry; active lines follows at right. The next line ends (At most one).

Original source details: GELFAND-P000111

How from general considerations do we obtain completeness (a curve in a function space) — this has not been done. One must prove a theorem in a Banach space — an analytic curve in a Banach space (possibly, false) [Crossed out: an illegible entry.] does not pass through a finite number of hyperplanes.



**Uncertain reading.** Before does not pass, a separate dense cancellation with a long downward stroke. The former letters are not separated.

Original source details: GELFAND-P000111

$$ay'(\pi,\lambda) - by(\pi,\lambda) = 0.$$

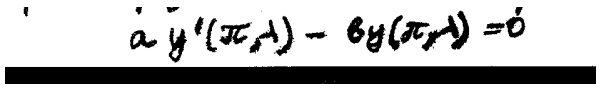
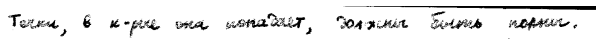


**Diagram and typed description.** In the last formula a separate horizontal minus stroke between $ay'$ and $by$. The derivative stroke at $y'$ is visible; both arguments are $(\pi,\lambda)$.

Original source details: GELFAND-P000111

GELFAND-P000112

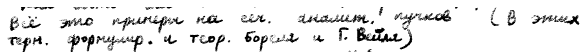
The points into which it falls must be poln.[?]



**Uncertain reading.** The line ends poln.[?] / polny; the middle strokes are densely joined, with a separate dot at the right. The ending is not established.

Original source details: GELFAND-P000112

What is to be done for several complex variables? All these are examples of sl.[?] analytic sheaves (in these terms both Borel's theorem and G. Weyl's theorem are formulated).

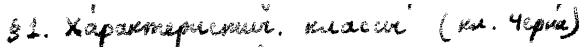


Original source details: GELFAND-P000112

**Uncertain reading.** Before analytic there is a short sl.-like entry with a dense right stroke and a dot. The abbreviation is unexpanded.

There is no satisfactory theory analogous to Nevanlinna's theorems. Distribution of the zeros of $n$ functions of $n$ variables — ?

§ 2. Characteristic classes (Chern classes)



Original source details: GELFAND-P000112

**Diagram and typed description.** At the left are § and a separate 2 with a dot. No underline is visible below the heading. At the right, (Chern classes).

$X$ — manifold, $E$ — vector bundle, $T = T(X)$ — [Crossed out: unreadable entry.] tangent bundle.

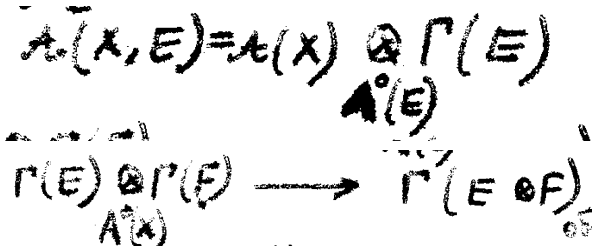


Original source details: GELFAND-P000112

**Uncertain reading.** Before the abbreviation for tangent a short prefix is densely crossed out; its individual earlier letters are not established.

$A^i(X)$ — sheaf of differential forms of order $i$. $\sum A^i(X) = A(X)$ — a ring. $d$ — differential in $A(X)$, $A(X, E)$ — sheaf of differential forms on $X$ with values in $E$.

$$A(X, E) = A(X) \otimes_{A^{0[?]}(E)} \Gamma(E).$$



Original source details: GELFAND-P000112

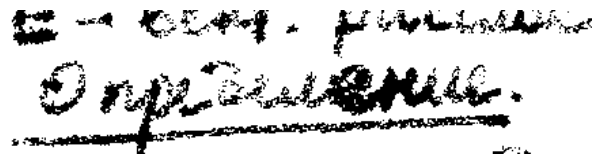
**Uncertain reading.** Below the first tensor sign, $A^{0[?]}(E)$: above $A$ is a round mark with a small opening, $0$ / $i$; in parentheses an E-like letter. Below the next tensor sign, a separate $A^0(X)$.

Notation.

$$\Gamma(E) \otimes_{A^0(X)} \Gamma(F) \longrightarrow \Gamma(E \otimes F)$$

— this multiplication is denoted simply by a dot. $E$ — vector bundle over $X$. Definition. A connection on $E$ is a differential operator $D$ on $\Gamma(E)$ satisfying the requirement:

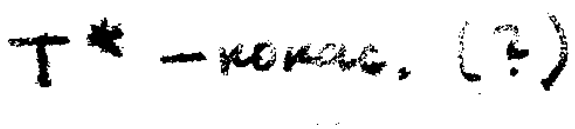
$$D : \Gamma(E) \longrightarrow \Gamma(T^* \otimes E), \qquad D(fs) = df \cdot s + f \cdot Ds, \quad s \in \Gamma(E),\ f \in A^0(X).$$



Original source details: GELFAND-P000112

**Uncertain reading.** In the underlined Definition. the middle letters are densely overwritten; the earlier strokes have not been read separately.

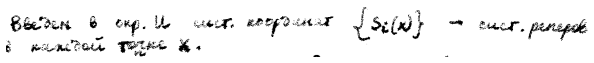
$T^*$ — kokas.[?] (!)



Original source details: GELFAND-P000112

**Uncertain reading.** After $T^*$, a short entry kokas. / konas.; the second consonant is densely joined to neighbouring strokes. A separate (!) is at the right.

Introduce local coordinates $\{s_i(x)\}$ in a neighbourhood $U$ — linear frames at each point $x$.



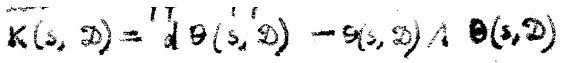
Original source details: GELFAND-P000112

**Diagram and typed description.** Before the brace the words local coordinates are written; after the brace, linear frames. The handwritten word koordinat has a long upper arc at d and a separate final stroke.

$$Ds_i = \sum_{j=1}^{n} \theta_{ij} s_j.$$

$\theta_{ij}$ — connection matrix $\theta(s, D)$. Its values — differential forms.

$$K(s, D) = d\theta(s, D) - \theta(s, D) \wedge \theta(s, D).$$

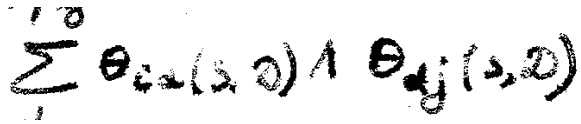


Original source details: GELFAND-P000112

**Diagram and typed description.** After the first $\theta(s, D)$ is a separate horizontal minus stroke; between the following matrices is $\wedge$.

Not $0$, since this is a product of matrices

$$\sum_{l} \theta_{il}(s, D) \wedge \theta_{lj}(s, D).$$

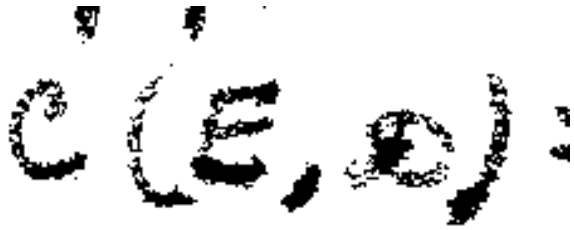


Original source details: GELFAND-P000112

**Diagram and typed description.** Below the sum is a separate $l$; the first $\theta$ has lower $i, l$, the second lower $l, j$. Both factors have arguments $(s, D)$.

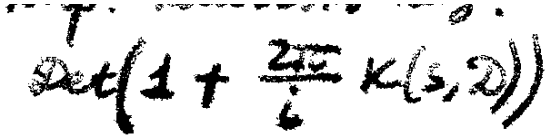
This exterior $2$-form is called the curvature. $2$-forms commute with one another, so a det of forms makes sense. The characteristic classes are called

$$C(E, D[?]) = \mathrm{Det}\left(1 + \frac{2\pi}{i} K(s, D)\right).$$



Original source details: GELFAND-P000112

**Uncertain reading.** In $C(E, D[?])$ the second argument is pale and round, with a dense left stroke; $D$ / a C-like letter.



Original source details: GELFAND-P000112

**Diagram and typed description.** In the fraction $2\pi$ is above the bar and $i$ below; before the fraction is a plus, and at the right $K(s, D)$.

These are closed 2-forms. [At the right:] Forms of different orders — and a collection of characteristic classes.

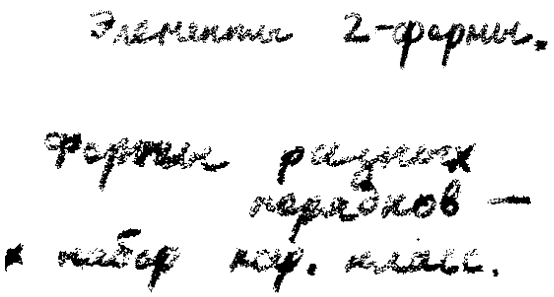


**Diagram and typed description.** The bottom line literally has 2-forms. A separate entry in the right margin: Forms of different orders — and a collection of characteristic classes.

Original source details: GELFAND-P000112

GELFAND-P000113

73

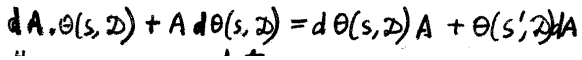
Prove that $C$ does not depend on the choice of basis and is a closed form; in fact it does not depend on the choice of connection either. Independence of the choice of basis.

$$\begin{aligned} s'_i = \sum A_{ij} s_j, \qquad s' &= As; \\ Ds' &= dA \cdot s + A\theta(s, D)s, \\ Ds' &= \theta(s', D)s', \\ dA + A\theta(s, D) &= \theta(s', D)A. \end{aligned}$$

Substitute this into the definition of $K$.

$$\boxed{AK(s, D) = K(s', D)A}$$

$$dA \cdot \theta(s, D) + Ad\theta(s, D) = d\theta(s, D)A + \theta(s', D)dA.$$



**Uncertain reading.** In the first $d\theta$ on the right the argument is $s$ without a separate prime. In the next $\theta$ the argument is $s'$ with a prime.

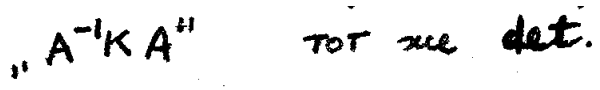
Original source details: GELFAND-P000113



**Diagram and typed description.** On the right between $d\theta(s, D)A$ and $\theta(s', D)dA$ is a separate plus with intersecting horizontal and vertical strokes.

Original source details: GELFAND-P000113

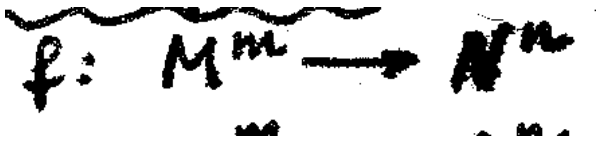
"$A^{-1}KA^{+1}$[?]" the same det.



**Uncertain reading.** Above the final $A$ are two dense raised strokes, currently $+1$[?]; the right vertical stroke is close to the closing quotation mark. No separate minus is visible.

Original source details: GELFAND-P000113

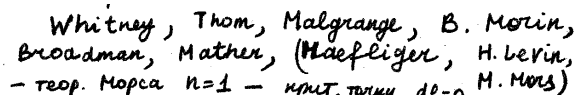
V. I. Arnold. Progress in the theory of singularities of differentiable mappings.
$f : M^m \longrightarrow N^n$ — 2 manifolds.



**Uncertain reading.** In the target $N^n$ the capital letter is densely overwritten; the individual earlier strokes are unread.

Original source details: GELFAND-P000113

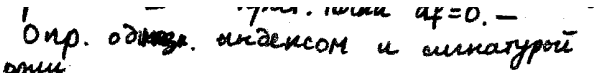
$R^m \longrightarrow R^n$, $0 \longrightarrow 0$. Whitney, Thom, Malgrange, B. Morin, Broadman, Mather, (Haefliger, H. Levin, M. Mors).



Original source details: GELFAND-P000113

**Diagram and typed description.** The Latin list has Broadman, H. Levin and M. Mors; before Morin is a B-like capital.

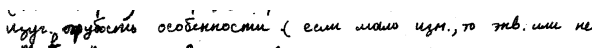
The simplest case — Morse's theorem $n = 1$ — critical points $df = 0$. ← isolated for general position. Determined by [Crossed out: an od...-like entry, continuation unreadable.] index and signature. (Morse lemma) — a model for the theory.



Original source details: GELFAND-P000113

**Uncertain reading.** After the abbreviation for determined a short token with an od...-like beginning is cancelled. The middle and ending are densely covered by strokes.

Study the roughness[?] of singularities (if changed slightly, equivalent or not).

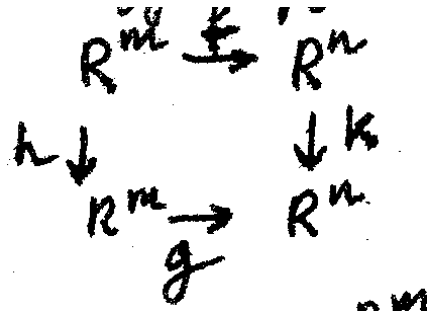


Original source details: GELFAND-P000113

**Uncertain reading.** The beginning of the word after study is densely overwritten. The current reading is grubost'[?] (roughness), with a b-like upper arc and ending ost'; the earlier strokes are not established.

$$\begin{array}{ccc} R^m & \xrightarrow{f} & R^n \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle k} \\ R^m & \xrightarrow[g]{} & R^n \end{array}$$

$f, g$ are equivalent if obtained by a change of variables.



Original source details: GELFAND-P000113

**Diagram and typed description.** The upper and lower rows are $R^m \to R^n$, with $f$ above and $g$ below. Vertical arrows point down: $h$ on the left, $k$ on the right.

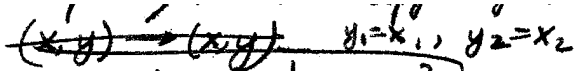
$d^2 f$ without zeros — reducible to a quadratic form. For the komp.[?] source manifold they form an everywhere pl.[?] open set.



Original source details: GELFAND-P000113

**Uncertain reading.** After for is a short komp.[?] / norm.; after everywhere a short pl.[?] / na. with a dot. The abbreviations are unexpanded. At the right, open set.

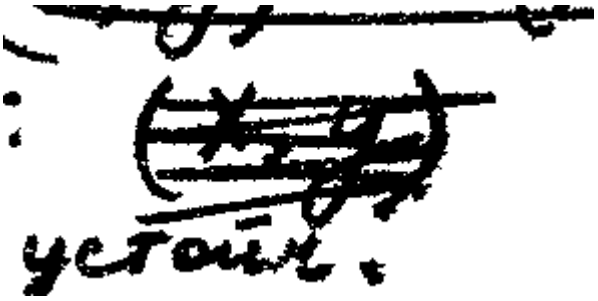
$R^m \longrightarrow R^1$ is solved. Whitney's first result (55) $R^2 \longrightarrow R^2$. An example of singularities. Singular points — rank of the differential $= 1$ and they are of 2 classes (folds and cusps). They are structurally stable (0 can be avoided). Ordinary point ~~$(x,y) \Longrightarrow (x,y)$~~ $y_1 = x_1$, $y_2 = x_2$.



Original source details: GELFAND-P000113

**Uncertain reading.** The entry $(x,y) \longrightarrow (x,y)$ is crossed out by two long horizontal lines. At the right, separately $y_1 = x_1$, $y_2 = x_2$.

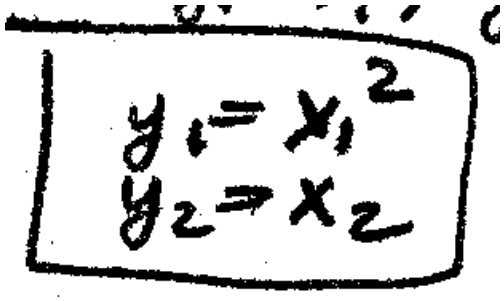
Fold point: [Crossed out: $(x,y)$[?], other strokes unread.] structurally stable.



Original source details: GELFAND-P000113

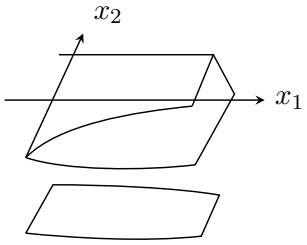
**Uncertain reading.** Below the heading an $(x,y)$-like entry is densely crossed out; below the lower slanting strokes the individual letters are not separated.

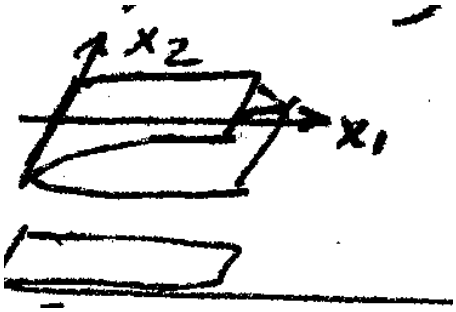
$$\boxed{\begin{aligned} y_1 &= x_1^2, \\ y_2 &\to x_2. \end{aligned}}$$



Original source details: GELFAND-P000113

**Diagram and typed description.** Two lines in one frame. The first is $y_1 = x_1^2$. In the second, between $y_2$ and $x_2$ is an arrow-like sign, a horizontal stroke with a point at the right.



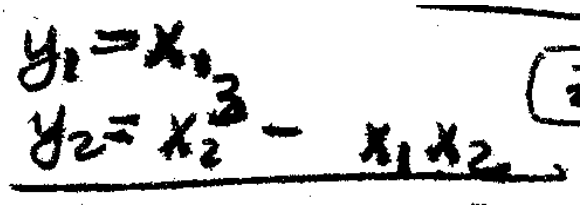



Original source details: GELFAND-P000113

**Diagram and typed description.** Two separate oblique strips. The upper one has a curved front edge and an internal line; the $x_1$ axis points right, $x_2$ obliquely upward. The lower strip has no labels.

Cusp point:

$$y_1 \overset{[?]}{\to} x_1,$$
$$y_2 = x_2^3 - x_1 x_2.$$



**Uncertain reading.** At $y_1$ is a sign with two closely spaced strokes, $\to$ / $=$; at the right $x_1$. The second line is $y_2 = x_2^3 - x_1 x_2$, with a separate minus. A long straight line is below both lines.

Original source details: GELFAND-P000113

$z^2$ singularity unstable; under deformation splits into 3 cusp points.

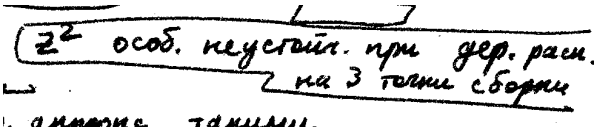
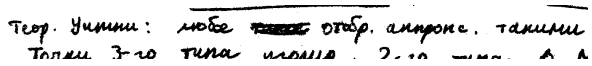


**Diagram and typed description.** At the right an entry is framed, open at the right edge, with a step in the lower outline. The text begins with $z^2$, and the next line has into 3 cusp points.

Original source details: GELFAND-P000113

Whitney's theorem: any [Crossed out: unreadable entry.] mapping is approximated by such ones.



**Uncertain reading.** Between any and the abbreviation for mapping is a short dense cancellation. The earlier letters are not established.

Original source details: GELFAND-P000113

Points of the 3rd type are isolated, of the 2nd type in $M^m$ a nonsingular 1-dimensional manifold; in $N$ it has singularities.

GELFAND-P000114

Cusp. Jacobian

$$\begin{vmatrix} 1 & 0 \\ -x_2 & 3x_2^2 - x_1 \end{vmatrix}.$$

$3x_2^2 - x_1 = 0$ a parabola. At 0 a fold, but more complicated as a whole.

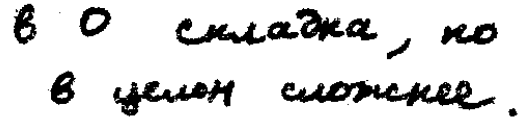
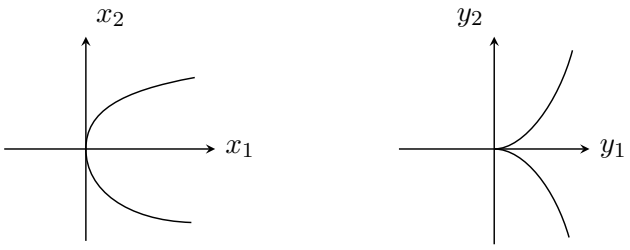


**Diagram and typed description.** At the right are separate lines At 0 a fold, but / more complicated as a whole.

Original source details: GELFAND-P000114



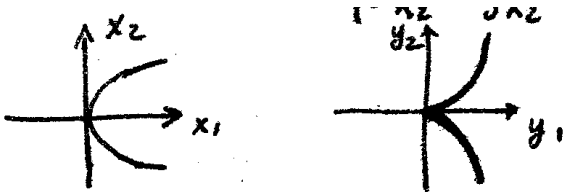



**Diagram and typed description.** At the left a parabola with vertex at the origin, $x_1$ right,$x_2$ up. At the right two branches with a common point,directed right; $y_1$ right,$y_2$ up.

Original source details: GELFAND-P000114

Where does the parabola go? [Crossed out: $y_2 = [?]y_1^{3/2}$.]



**Uncertain reading.** In the cancelled entry $y_2$,an equals sign and $y_1^{3/2}$ are visible. The coefficient and intervening strokes are densely covered.

Original source details: GELFAND-P000114

$y_2 = -Cy_1^{3/2}$.

$$\boxed{y_2^2 = -y_1^3}$$

$y_1 > 0$.

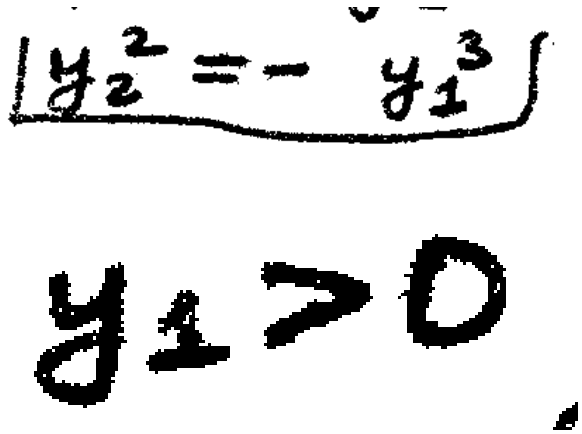


**Diagram and typed description.** The corner frame has $y_2^2 = -y_1^3$,with a separate minus. Below and to the left is a separate $y_1 > 0$.

Original source details: GELFAND-P000114

The mapping is not a fold on the whole parabola. Away from the point $0$ a fold. Consider lines parallel to the $x_2$ axis. $x_1 = \text{const} \Rightarrow y_1 = \text{const}$. $y_2 = x_2^3 - Cx_2$.

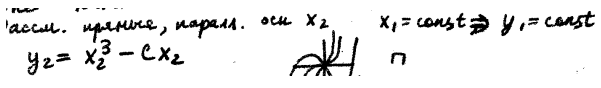
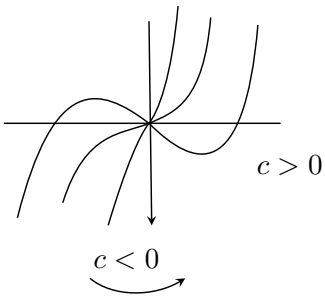
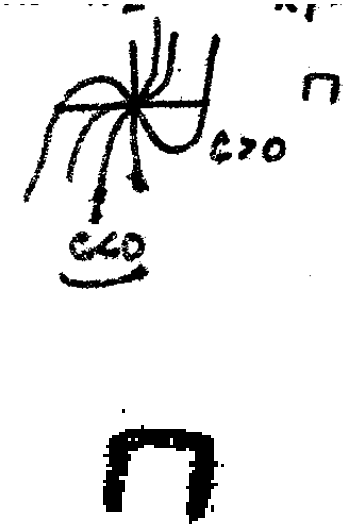


**Diagram and typed description.** In the line $x_1 = \text{const}$ an arrow points to $y_1 = \text{const}$. Below is $y_2 = x_2^3 - Cx_2$,with a large C-like letter before $x_2$.

Original source details: GELFAND-P000114




**Uncertain reading.** Several curves pass through the common intersection of the axes. At the right,next to the curve with turns,is $c > 0$; below,$c < 0$ and a curved arrow. At the right is a separate Cyrillic-P-/ П-like mark,meaning unestablished.

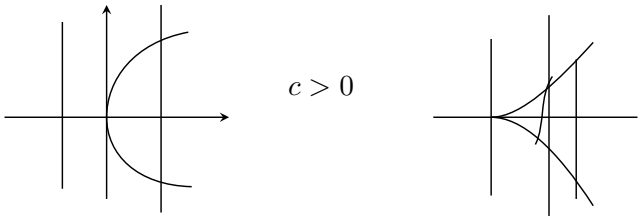
Original source details: GELFAND-P000114

$c < 0$ the lines are mapped izom.[?] $c > 0$.




Original source details: GELFAND-P000114

**Uncertain reading.** After the abbreviation for mapped,a short izom.[?] / monot.-like word has densely joined middle strokes. The abbreviation is unexpanded.



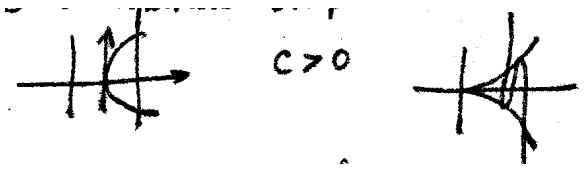
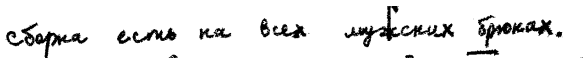




**Diagram and typed description.** At the left a parabola,axes and two separate verticals. At the right a pointed two-branch curve with several vertical strokes and a short curved internal line. Between the sketches is $c > 0$.

Original source details: GELFAND-P000114

A cusp occurs on all luchesnykh[?] branakh[?].



Original source details: GELFAND-P000114

**Uncertain reading.** In the first disputed word a vertical diagram stroke crosses the middle: luchevykh / luchesnykh. In the second,the initial strokes are b-/g-like: branakh / granyakh. The full reading is not established.

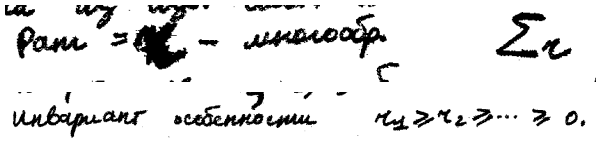
Problem. [Crossed out: $z^2$[?] and unreadable strokes.] $z \longrightarrow z^2 + \varepsilon\overline{z}$ — general position. $\varepsilon = 0$ singularity at $0$; for $\varepsilon \neq 0$ it moves, there will be folds and cusps. And only this is stable.



Original source details: GELFAND-P000114

**Uncertain reading.** After Problem. a z-like sign with an upper $2$ is cancelled; the right part of the entry is covered by long strokes.

Scientific definition of a cusp. One of the known systems of invariants. Rank $= r$[?] — manifold $\Sigma_r$.



Original source details: GELFAND-P000114

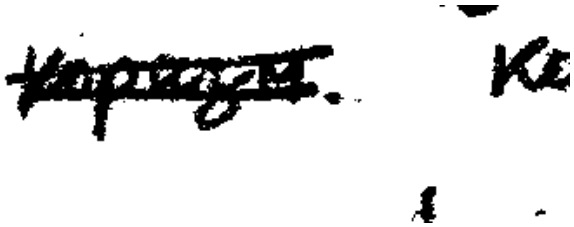
**Uncertain reading.** After the equals sign the letter is densely overwritten; currently $r$[?],with earlier round strokes unread separately. Below,similar signs occur at $\Sigma_r$ and in the sequence $r_1, r_2$.

$M^m$ is partitioned into manifolds. ~~Codim.~~

Codim. sing. = product of coranks $(m-r)(n-r)$

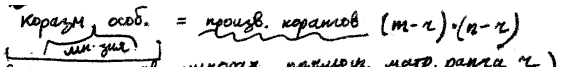
manifold

(This is a question about independent minors of a rectangular matrix of rank $r$.)



Original source details: GELFAND-P000114

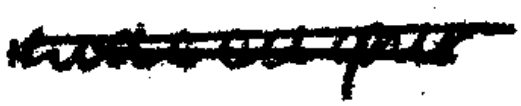
**Uncertain reading.** The first Codim. is crossed out by long horizontal lines. Below,a separate repetition Codim. sing.



Original source details: GELFAND-P000114

**Diagram and typed description.** Below sing. is a brace pointing up with the label manifold; below that,a long corner frame. Product of coranks has a wavy underline; at the right,$(m-r)(n-r)$.

$\Sigma_r$ — smooth manifolds in $R^m$; consider our mapping on the manifolds themselves (parabola). Rank on these manifolds. $\Sigma_{r_1,r_2}$, $r_2 < r_1$ — submanifold — rank levels. $\Sigma_{r_1,r_2,r_3}$. To each [Crossed out: unreadable word.] singularity corresponds a set of numbers $r_1, r_2, r_3$.



**Uncertain reading.** After to each a long word,resembling osobennosti,is cancelled. Upper and lower horizontal strokes cover most letters.

Original source details: GELFAND-P000114

Invariant of a singularity $r_1 \geq r_2 \geq \cdots \geq 0$.

GELFAND-P000115

Whitney sing.: $\Sigma_0$ ordinary. The first index is the corank. Cusp: $\Sigma_{10}$. It is proved that this is a manifold.

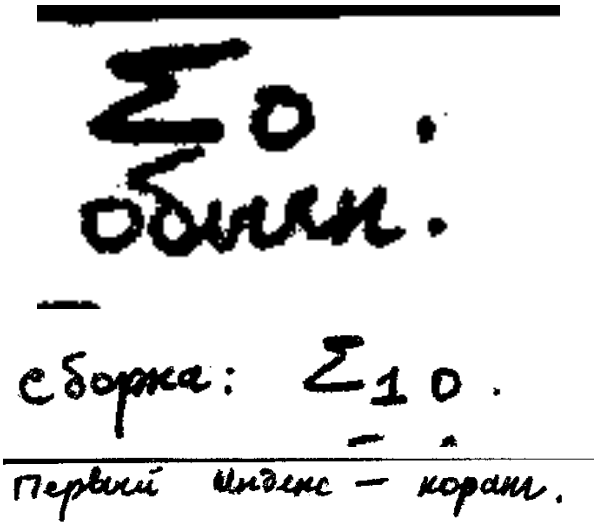



**Diagram and typed description.** For Whitney,$\Sigma_0$,with ordinary below it; for the cusp,$\Sigma_{10}$. At the right,The first index is the corank.

Original source details: GELFAND-P000115

[Crossed out: D...]

For the partition in the space of Taylor-series segments. (Brodman) — the old naive approach.

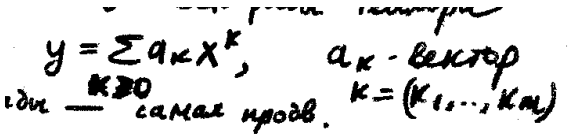


**Uncertain reading.** Before for the partition,a short cancelled entry; a Cyrillic-D-like beginning is visible,the other letters covered by strokes.

Original source details: GELFAND-P000115

It is known that $\Sigma$ is not a complete classification. (There is an example $R^9 \longrightarrow R^9$.) General theory. Consider the space of Taylor series. (3 classes — $C^\infty$, analytic, formal power series.) [At the right:] $R^m \longrightarrow R^n$; $C^\infty$ has Taylor series.

$$y = \sum_{\substack{[?]\\ k \geq 0}} a_k x^k, \qquad a_k \text{ — a vector}, \quad k = (k_1, \ldots, k_m).$$



**Uncertain reading.** Below the sum,$k \geq 0$[?] / $k > 0$; the small strokes of the sign are joined. At the right,$a_k$ is a vector,$k = (k_1, \ldots, k_m)$.

Original source details: GELFAND-P000115

Local theories. $C^\infty$ — the most difficult, analytic — the most interesting, formal series — the simplest. Mappings are equivalent if there is a change. A partition into pieces arises — a stratification. Study of this partition.

$$\begin{array}{ccc} R^m & \xrightarrow{\quad f \quad} & R^n \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle k} \\ R^m & \xrightarrow{\quad g \quad} & R^n \end{array}$$

$h, k$ — formal diffeomorphisms (1st terms with Jacobian $\neq 0$).

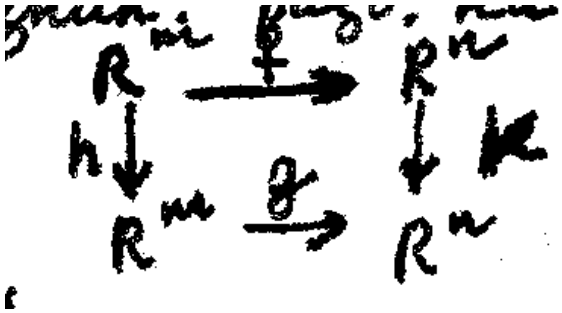
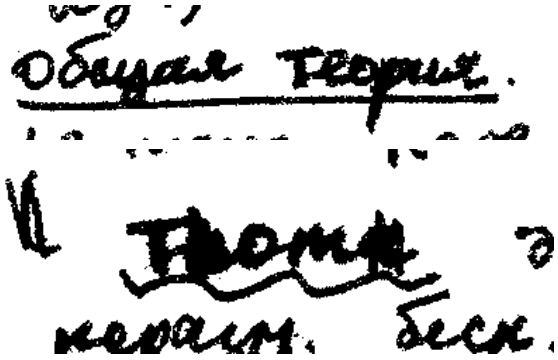


**Diagram and typed description.** Two rows $R^m \to R^n$: above,$f$; below,$g$; vertical arrows down,$h$ on the left,$k$ on the right.

Original source details: GELFAND-P000115

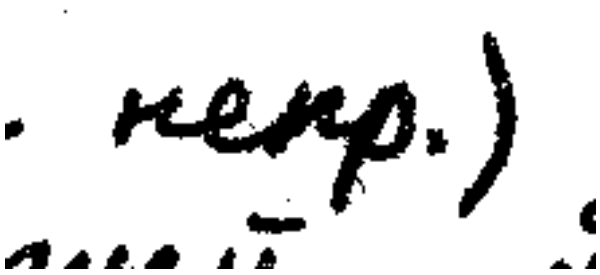
There is an example ($> 9$) with infinitely many invariants. ↓ Thom proved that in the space of power series a part of infinite codimension is singled out, that the rest are stable in the topological sense.



**Uncertain reading.** General theory. has a straight underline,with middle letters overwritten. Thom has a wavy underline; at the left,a short arrow down. Earlier strokes of the name cannot be read separately.

Original source details: GELFAND-P000115

(Thom's invariants in $R^9$ are invariants of smooth,but not nepr.[?])



**Uncertain reading.** The final abbreviation is nepr. / nerp.; the joined middle strokes do not give an unambiguous letter order.

Original source details: GELFAND-P000115

(Thom does not want to publish the proof,he is working on biology. The proof was recounted,for a whole year,but nobody understood it: Malgrange, Cartan, Arnold and others.) $\bigcup X_\gamma + X_\infty$, $X_\gamma$ singularities stable in the topological sense.

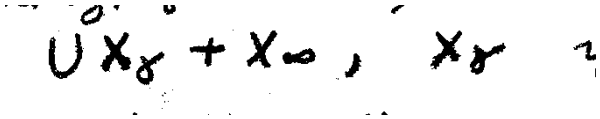


**Diagram and typed description.** Before $X_\gamma$ is a large union sign without a lower index; between $X_\gamma$ and $X_\infty$ is a separate plus.

Original source details: GELFAND-P000115

For some dimensions Thom's analytic difficulties do not arise. (Singularities differentially stable.) Apparently the problem is solved where there is [Crossed out: unreadable word.] a classification. (In small dimensions.)



**Uncertain reading.** Between there is and classification,a long cancelled entry covered by several horizontal strokes.

Original source details: GELFAND-P000115

There are $\Sigma$ that are differentially stable $\Sigma_{1,\ldots,1,0}$. Not all $\Sigma$ are differentially stable. Another result of Mother. (Milnor's student.)

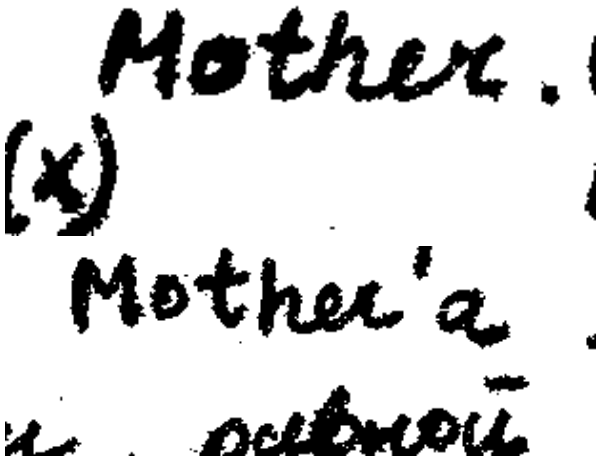
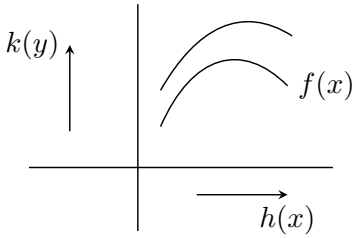


**Diagram and typed description.** The name has a round $o$: Mother. In the lower repetition,Mother'a.

Original source details: GELFAND-P000115

$\delta y = \varphi(x)$. $h(x)$ — variation in $X$ — a vector field.



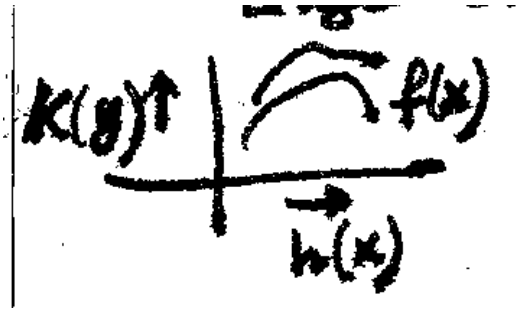
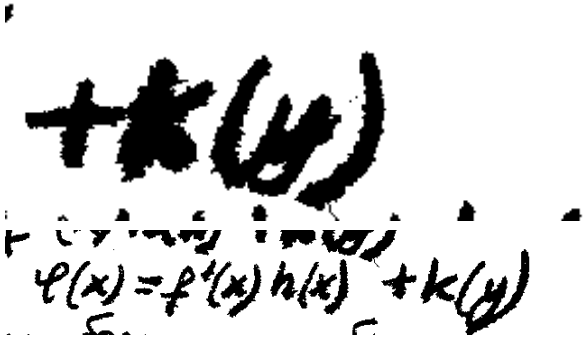



**Diagram and typed description.** Crossing axes and two nearby arcs above the horizontal,with $f(x)$ at the lower arc. A separate horizontal arrow right below the axis is labelled $h(x)$,and a vertical arrow up at the left,$k(y)$.

Original source details: GELFAND-P000115

Change of variables — the function changes by $f'(x)h(x) + k(y)$. In the 1st approximation one must solve the equation

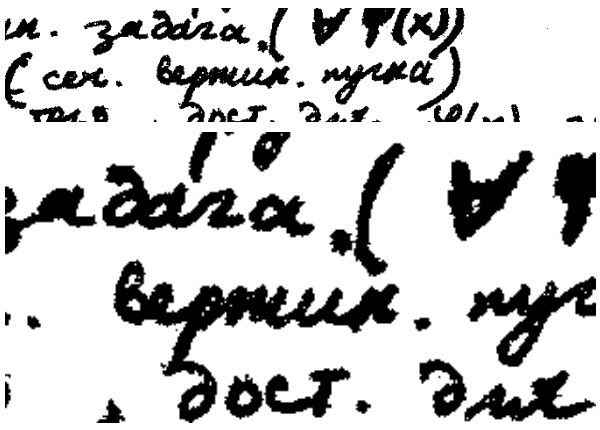
$$\varphi(x) = f'(x)h(x) + k(y).$$



**Uncertain reading.** In the first formula,$k(y)$ is densely overwritten; below,in a separate equation,is a clean $k(y)$. The earlier strokes of the first letter are unread.

Original source details: GELFAND-P000115

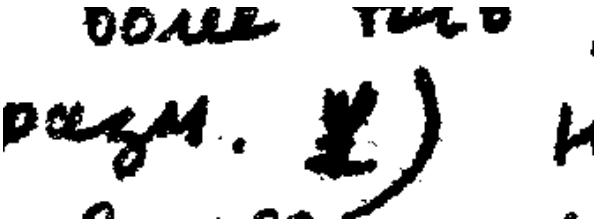
The corresponding theorem says that for stability it is necessary and sufficient that the corresponding linear problem be solvable. ($\forall\varphi(x)$) (sek. verkhnikh.[?] sheaves[?])



**Uncertain reading.** The beginning is sek.; the next word is verkhnikh.[?] / ver...nikh.-like,the full reading unestablished. The last word is puchki / puchka.

Original source details: GELFAND-P000115

(This is Mother's 1st theorem. Moreover,it suffices for $\varphi(x)$ given by a polynomial of degree equal to dim. $\ell$[?].)



Original source details: GELFAND-P000115

**Uncertain reading.** After dim.,a densely overwritten sign,$\ell$- / $N$-like; earlier strokes are covered.

But the proof is incorrect. Modern technique (nonlinear functions in $C^\infty$ spaces). All dimensions for which everything is good have been found. $R^2 \longrightarrow R^2, R^3$ (Whitney); in $R^4$ unknown, in $R^5$ already known (there are no singularities at all any more). Immersion (immersion). For $m = n$ up to $8$ all mappings are stable (approx.).

GELFAND-P000116

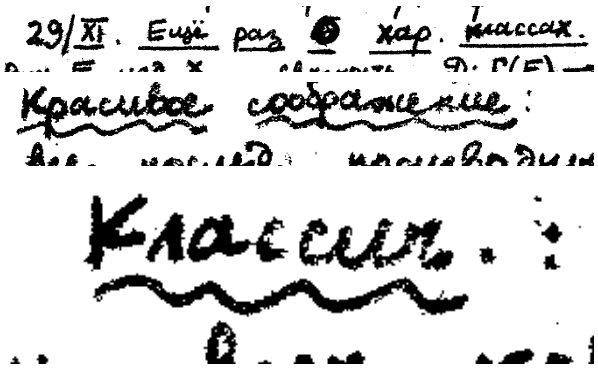
Thom — a continuous invariant (stable). It is not topological — it is an invariant of smooth mappings. Under continuous deformation one can deform them. 29/XI. Once again about[?] characteristic classes. [Crossed out: unreadable word.]

(I. M. Gelfand.)



Original source details: GELFAND-P000116

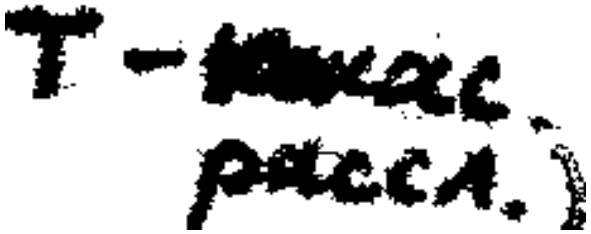
**Uncertain reading.** After classes.,a long densely cancelled entry; the letters are covered by horizontal strokes.



Original source details: GELFAND-P000116

**Uncertain reading.** Before characteristic,a round densely overwritten letter,o[?],with its own straight underline. Parts of the heading are underlined separately. Beautiful observation. and Classification: below have wavy underlines.

Consider $E$ over $X$,a connection $D : \Gamma(E) \to \Gamma(T^* \otimes E)$. $f$ (function) $\in A^0$, $s \in \Gamma(E) \Rightarrow$ $D(fs) = f \cdot Ds + df \cdot s$. [At the right:] (Here $T$ is the tang.[?] bundle.)



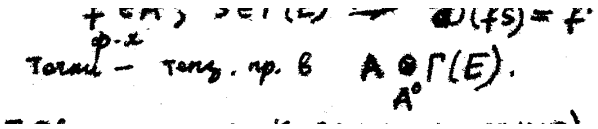
**Uncertain reading.** Before and at the beginning of kas.,a dense correction; the earlier strokes cannot be read separately.

Original source details: GELFAND-P000116

[In the margin:] ② More precisely — tensor product in $A \otimes_{A^0} \Gamma(E)$.

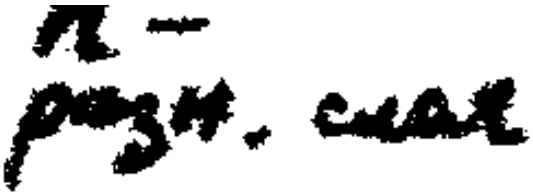


**Diagram and typed description.** Below $f$ is the label function,joined to the letter by a short stroke. In the tensor product,$A^0$ is below $\otimes$. In the right margin,2 is circled.



Original source details: GELFAND-P000116

Cartan's moving frame (locally): $s_i$ $(i = 1, \ldots, n)$, $n$ — dim.[?] of the fibre.
Linearly independent at $\forall$ point.



**Uncertain reading.** In $n$ — dim. of the fibre,the abbreviation for dimension is densely overwritten; earlier letters are covered.

Original source details: GELFAND-P000116

$D(s_i) = \sum \theta_{ij} s_j$,coefficients from $A^1$ (forms). $\|\theta_{ij}\|$ is called the connection matrix (in Cartan — $\omega$).

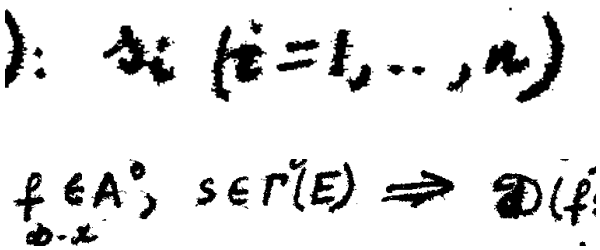


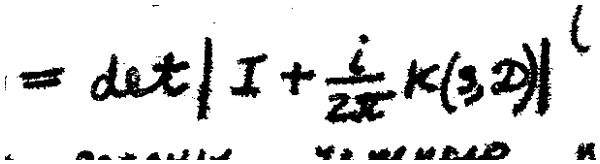
**Diagram and typed description.** For sections an s-like letter recurs: $s \in \Gamma(E)$,$s_i$,$s_j$; below,$s = (s_1, \ldots, s_n)$. The initial $D$ in $D(fs)$ is overwritten.

Original source details: GELFAND-P000116

$$\|K_{ij}\| = K(s,D) = d\theta(s,D) - \theta(s,D) \wedge \theta(s,D), \qquad s = (s_1, \dots, s_n).$$

(This is a matrix.) (Matrix product,but the product is exterior.) The form

$$C(E,D) = \det \left| I + \frac{i}{2\pi} K(s,D) \right|.$$



Original source details: GELFAND-P000116

**Diagram and typed description.** In the determinant,before $K(s,D)$,a fraction with $i$ in the numerator and $2\pi$ in the denominator.

The entries are forms of even order,they commute and one can take det. A $\sum$ of forms of different orders. These are the characteristic classes associated with the given bundle. This does not depend on the choice of connection. [Idea of the proof according to Gelfand: closedness and this are proved,$\theta$[?] changed.

The form no longer depends on the connection,$\int$ can be expressed through $\int$ over the boundary. A variation of the connection inside has no effect.]



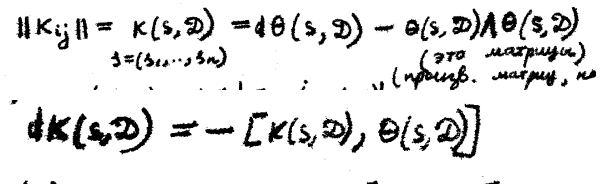
Original source details: GELFAND-P000116

**Uncertain reading.** Before changed,a dense $\theta$- / paragraph-like sign,with a round upper and dense lower stroke.

It does not depend on the choice of frame. Bianchi identity:

$$dK(s,D) = -[K(s,D), \theta(s,D)].$$

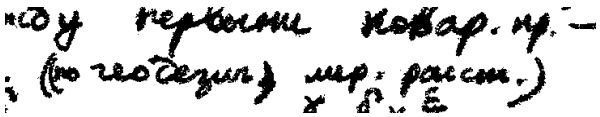
(Derivative of the curvature tensor.)



Original source details: GELFAND-P000116

**Diagram and typed description.** The curvature formula has a minus before $\theta \wedge \theta$. The Bianchi identity has a minus before the commutator; the order inside the brackets is $K(s,D), \theta(s,D)$.

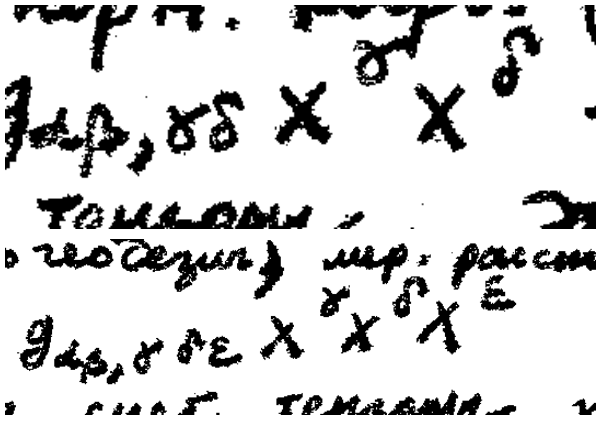
Beautiful observation. $g_{\alpha\beta}(x)$ — a Riemannian metric,we shall take all successive derivatives of $g_{\alpha\beta}(x)$ and form tensors from them. How does one find all tensors that can be formed from $g_{\alpha\beta}(x)$? Classification: they are functions of the curvature tensor and all covariant derivatives of the curvature tensor (of any order) (algebraic expressions in them). But what algebraic relations? (Between first covariant derivatives — the Bianchi identity.) Introduce normal coordinates (po[?] geodesics) mr.[?] bundle.



Original source details: GELFAND-P000116

**Uncertain reading.** At the beginning of the parentheses,a short po[?]-like word. After the parentheses,mr.[?] / metr.-like abbreviation before bundle; the full reading is unestablished.

$$g_{\alpha\beta}(x) = g_{\alpha\beta}(0) + g_{\alpha\beta,\delta\delta}x^\delta x^\delta + g_{\alpha\beta,\delta\delta\varepsilon}[?]x^\delta x^\delta x^\varepsilon.$$



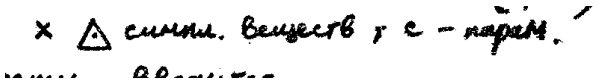
Original source details: GELFAND-P000116

**Uncertain reading.** In the second term,two $\delta$-like letters in the subscripts and at both $x$'s. In the third,the subscripts are $\delta\delta\varepsilon$[?] / $\gamma\delta\varepsilon$,with the middle letter overwritten; the superscripts are also $\delta$- / $\gamma$-like,the last $\varepsilon$.

In a normal system they are all tensors. This system of tensors,and here the algebraic relations are simpler.

GELFAND-P000117

V. I. Arnold. Stable flow. $\dot{x} = f(x, C)$; $g^t$ — a group of diffeomorphisms. There are chemical and biological models (Thom). $x$ △ sums of substances,$c$ — a parameter.

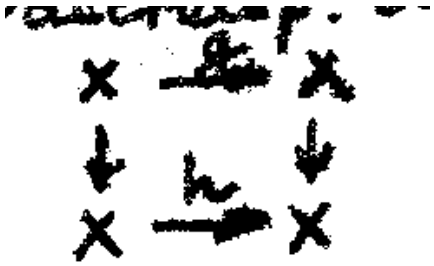


Original source details: GELFAND-P000117

**Diagram and typed description.** Between $x$ and sums of substances,a triangle with a dot inside. The right parameter label uses lowercase $c$,the equation uppercase $C$.

Consider one (stable). A number of kogrubost[?] is introduced.

$$\begin{array}{ccc} X & \xrightarrow{g^t[?]} & X \\ \downarrow & & \downarrow \\ X & \xrightarrow{h[?]} & X \end{array}$$



Original source details: GELFAND-P000117

**Uncertain reading.** Four $X$'s; above the upper horizontal arrow,an overwritten $g^t$[?],above the lower,an h-like letter. Horizontals right,verticals down,with no labels at the verticals.

Stable if for $\forall$ nearby $h$ there is a perturbation of $X$. One cannot consider only stable ones (when parameters change,one may pass through nonstable parameters). $(x, C)$ — a point of kogrubost[?] $1$,if for nearby parameter values stable,etc. As a mapping $X \times C \longrightarrow X \times C$,stable. (How many terms of the Taylor series must be taken.) A centre — an infinite degree of nonstability.

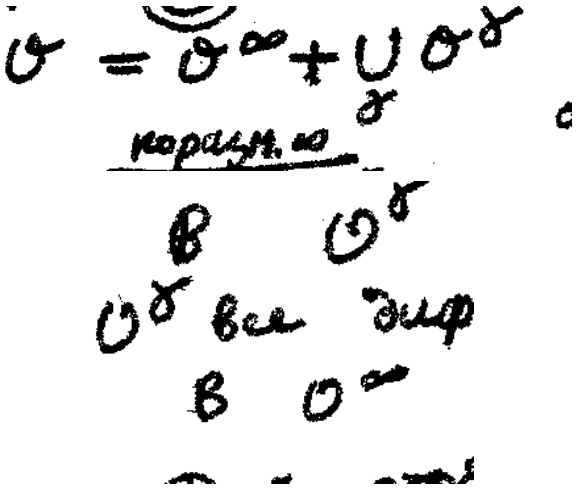


Original source details: GELFAND-P000117

**Diagram and typed description.** Next to the entry about a centre,three nested closed round lines. They have no labels.

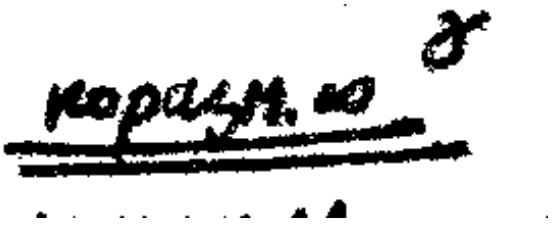
Thom's theorem on germs of Taylor series $\vartheta = \vartheta^{\infty} + \bigcup_{\gamma} \vartheta^{\gamma}$.

[Below the formula:] codim. $\infty$



Original source details: GELFAND-P000117

**Uncertain reading.** The recurring letter is $\vartheta$[?] / $\theta$ / O-like,with a large upper loop; the superscripts are $\infty$ and gamma-like.



Original source details: GELFAND-P000117

**Diagram and typed description.** Below the formula,a separate annotation codim. $\infty$,underlined twice.

In $\vartheta^\gamma$,an open subset of finite negrub.[?] In $\vartheta^\gamma$ singularities are described by finitely many parameters,inside [Crossed out: unreadable short token.]

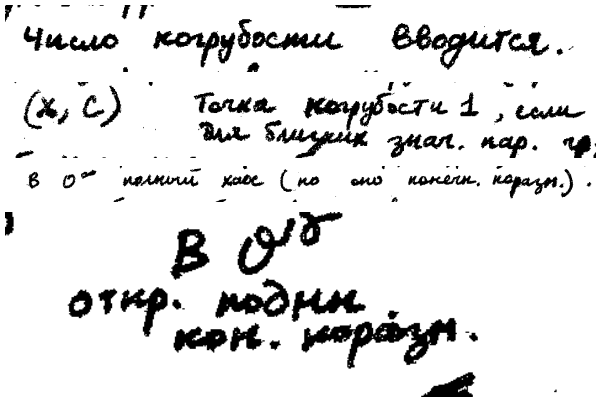
$\vartheta^\gamma$,all are diffeomorphic. (Tr-ie for differential equations.)



**Uncertain reading.** After inside,a short dense cancellation; separately at its left,another short horizontal stroke.

Original source details: GELFAND-P000117

In $\vartheta^\infty$,complete chaos (but they are of finite codim.[?]). All $1$-parameter ones are in $\vartheta^\infty$.




**Uncertain reading.** In the first lines after number and point,a ko-/negrubosti-like word; literally kogrubosti[?]. In the right annotation,finite negrub.[?]; in the lower parentheses,finite korazm.[?] / kogrub.-like abbreviation. Initial and middle strokes are joined.

Original source details: GELFAND-P000117

For self-mappings,the invariants are unknown. (F. Bendixson's theorem: $\frac{dy}{dx} = \frac{P}{Q}$ on $CP^2$,then by finitely many $\sigma$-processes one reaches Bendixson singular points $x^p \frac{dy}{dx} = \lambda y + P(x, y)$. Already an invariant relative to $\sigma$-processes. A $\sigma$-process — substitute $\frac{y[?]}{x}$.

For $3$ variables unknown. On a birationally equivalent manifold,simple singular points. Acta Math. $1901$.)



**Uncertain reading.** At the beginning of the parentheses,a separate F-like Latin letter with a dot. The beginning of the next word is overwritten; the current reading is theorem.

Original source details: GELFAND-P000117

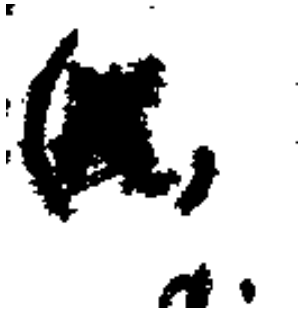


**Uncertain reading.** The numerator is $y$[?] / h-like,densely overwritten; the denominator $x$. Beside it,a short pointer bent down.

Original source details: GELFAND-P000117

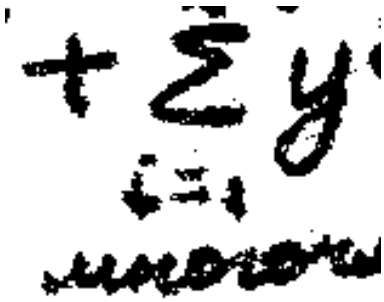
On the Weierstrass preparation theorem. (Malgrange.)

$$P(y) = y^n + \sum_{i=1}^{n-1} y^i a_i(x,z).$$



Original source details: GELFAND-P000117

**Uncertain reading.** In the first argument of $a_i$,a dense cancellation,currently $x$[?]. Below,a separate clean $a_i(x,z)$.



Original source details: GELFAND-P000117

**Diagram and typed description.** The first sum's lower limit is explicitly $i=1$,the upper $n-1$.

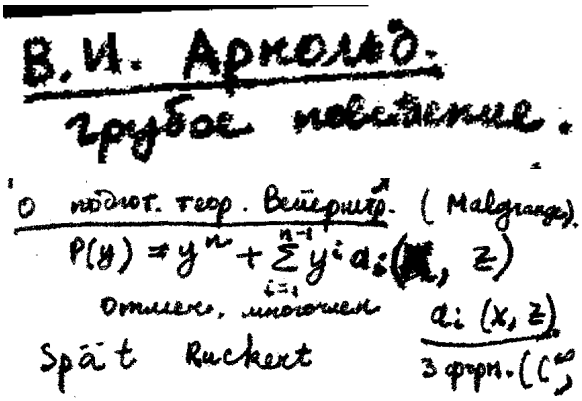
Distinguished polynomial $a_i(x,z)$,$a_i(0,0)=0$.



Original source details: GELFAND-P000117

**Uncertain reading.** The second $0$ in $a_i(0,0)=0$ is densely overwritten; earlier strokes are covered.

Spät Ruckert.



Original source details: GELFAND-P000117

**Diagram and typed description.** The name V. I. Arnold. is underlined,the next line Stable flow. separate. The heading On the Weierstrass preparation theorem. is underlined; below,$a_i(x,z)$ is also underlined. The Latin entry is Spät Ruckert.

$P(y,x,z)$; $y$ — one variable, $x=(x_1,\dots,x_m)$,$z=(z_1,\dots,z_n)$ — spare ones. 3 forms ($C^\infty$,analytic,formal series).

$$f(x,y,z)=P(y,x,z)\qquad Q(y,x,z)\qquad +\sum_{0}^{n-1} y^i b_i(x,z).$$

of the same class

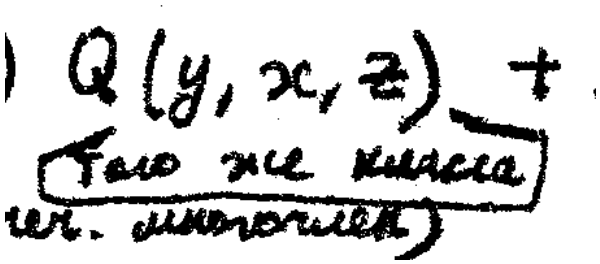


**Diagram and typed description.** Below $Q(y,x,z)$,a frame with the label of the same class,joined to the right of the entry by a short line. The remainder sum's lower limit is $0$.

Original source details: GELFAND-P000117

(Division is possible.) (One can do this on relative sets.) [[This works for Bendixson: the process must be finite.]] The quotient by the ideal generated by $P$ has finite dimension [Crossed out: unreadable token.] over

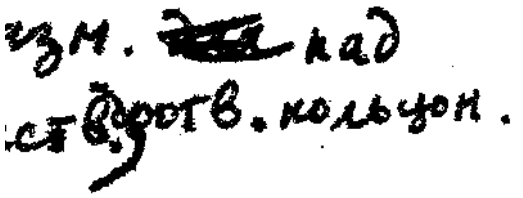
[Below at the right:] the corresponding ring.

**Uncertain reading.** Before over,a short entry is densely cancelled; the hidden letters are unread.



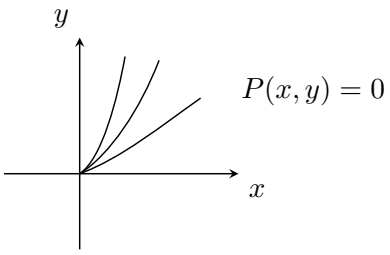
Original source details: GELFAND-P000117

**Diagram and typed description.** Over stands at the end of the upper line. The corresponding ring. is below at the right,at the end of the next line.



Original source details: GELFAND-P000117

(Malgrange: for $C^\infty$ this is also true,but there is no longer uniqueness.) 1) Let $P$ nearby have $n$ real roots — by interpolation it will be $C^\infty$,but (the approximating polynomial by Lagrange's theorem is uniquely determined by the description — this is difficult).



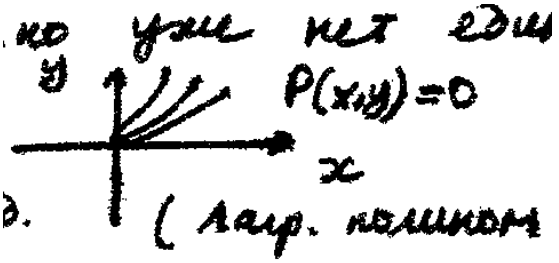
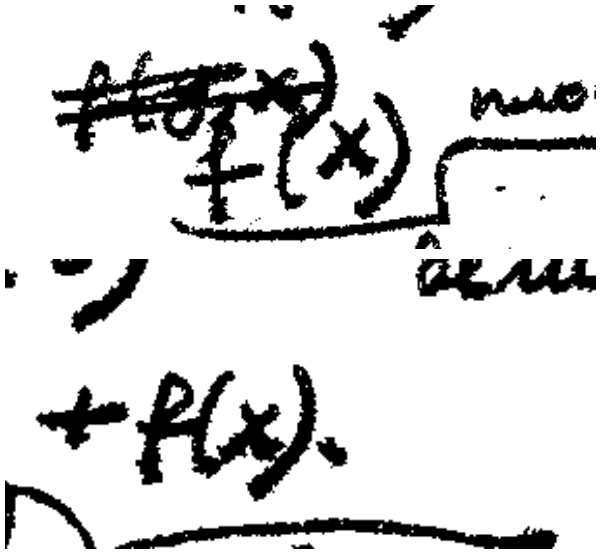



**Diagram and typed description.** Axes:$x$ right,$y$ up. Three separate branches leave the origin into the first quadrant. At the right,the label $P(x, y) = 0$.

Original source details: GELFAND-P000117

2) [Crossed out: $f(y, x)$-like entry.]

$f(x)$ flat at $0$ (all derivatives $0$).



**Uncertain reading.** In the cancelled entry,the arguments look like y,x. Below it,a clean $f(x)$. The first division formula ends with a separate $+f(x)$.

Original source details: GELFAND-P000117

Divide by $y^2 + x^2$. Trivial answer.

$$f(x) = (y^2 + x^2) \cdot 0 + f(x).$$

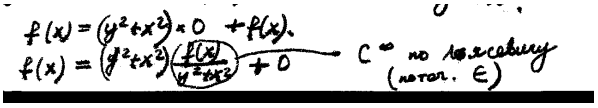
$$f(x) = (y^2 + x^2) \left( \frac{f(x)}{y^2 + x^2} \right) + 0$$

$C^\infty$ by Łojasiewicz (potom. $E$[?]).



Original source details: GELFAND-P000117

**Uncertain reading.** In the last parentheses,potom. and a separate $E$- / epsilon-like sign. The abbreviation's expansion is unestablished.

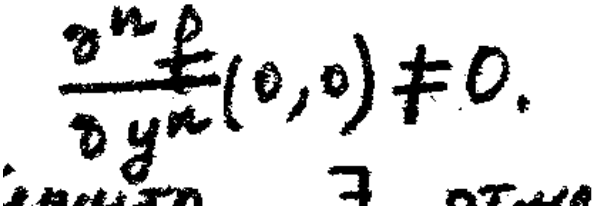


Original source details: GELFAND-P000117

**Diagram and typed description.** The fraction $f(x)/(y^2 + x^2)$ is circled. A curved line with a round endpoint runs right from it toward the entry $C^\infty$ by Łojasiewicz. At the right of the equality,a separate $+0$.

GELFAND-P000118

Another formulation. $f(y, x)$,$f(0, 0) = 0$. Let $\frac{\partial^n f}{\partial y^n}(0, 0) \neq 0$. This is possible if $f$ is not flat.

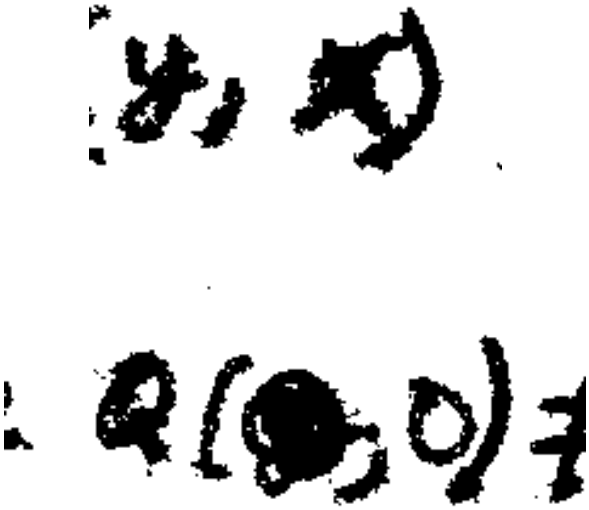


Original source details: GELFAND-P000118

**Diagram and typed description.** In the condition,$n$ is the derivative order in $y$,followed by $(0, 0) \neq 0$.

Weierstrass's theorem:∃ a distinguished polynomial $P$,

$$f(y,x) = P(y,x)Q(y,x), \qquad Q(0,0) \neq 0.$$



**Uncertain reading.** In $P(y,x)$ the second argument is densely overwritten,the current letter $x$[?]. In $Q(0,0)$ the first $0$ is overwritten; earlier strokes are covered.

Original source details: GELFAND-P000118

V. I. Arnold.

Diffeomorphisms of the annulus,preserving area,homotopic to the identity.

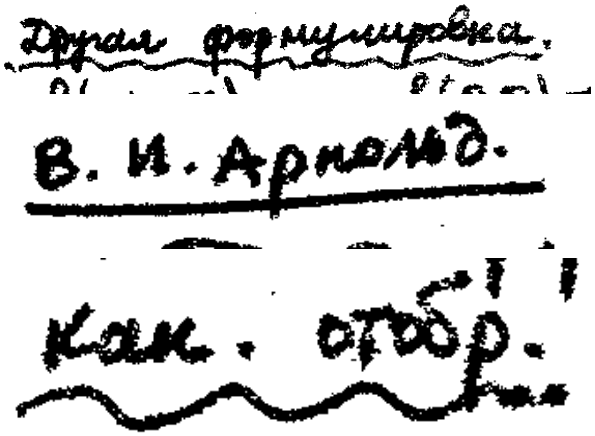



**Diagram and typed description.** Another formulation. has a wavy underline; Arnold's name a straight one. Canonical mappings. below also has a wavy underline.

Original source details: GELFAND-P000118

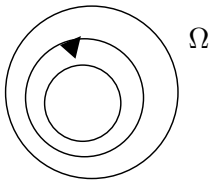





**Diagram and typed description.** Three nested round lines; at the top of the middle one,a dense triangular stroke. At the right,$\Omega$.

Original source details: GELFAND-P000118

$\forall\gamma$ — a closed curve,going around $0$, $\underset{\text{intersection points}}{A\gamma \cap \gamma} \geq 2$ (otherwise one does not obtain area preservation). Has significance for the $3$-body problem. For $n$ bodies [Crossed out: unreadable word.]

the original problem — tori,annulus $\Omega = T^n \times B^n$,$B^n \subset R^n$ — a domain. Diffeomorphism,homotopic to the identity,$A : \Omega \longrightarrow \Omega$. $\Omega = \underset{q, \bmod 2\pi}{T^n}$. In $R^n$,coordinates $p = (p_1, \ldots, p_n)$.



**Uncertain reading.** After for $n$ bodies,a dense cancellation before original problem; the hidden letters are unread.

Original source details: GELFAND-P000118

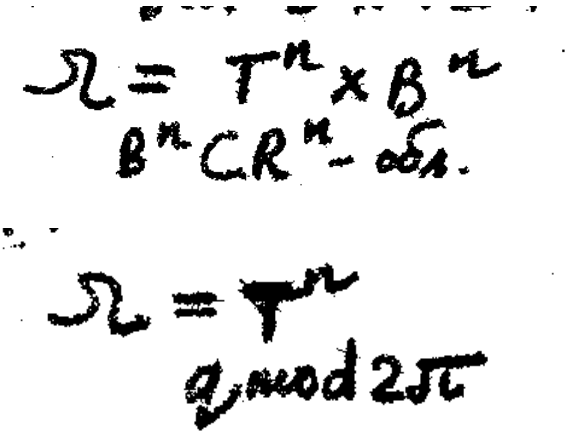


**Diagram and typed description.** The upper entry is $\Omega = T^n \times B^n$,below a separate $\Omega = T^n$; under $T^n$,the label $q, \bmod 2\pi$.

Original source details: GELFAND-P000118

Consider the form $p\,dq = p_1\,dq_1 + \cdots + p_n\,dq_n$. Canonical mappings. preserve the form:

$$\oint_{A\gamma} p\,dq = \oint_{\gamma} p\,dq.$$

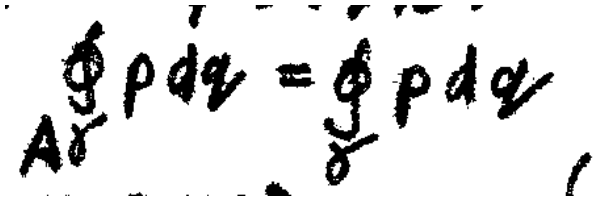


**Diagram and typed description.** Both integrals have closed contours; below the left sign,$A\gamma$,below the right,$\gamma$.

Original source details: GELFAND-P000118

(Only these are considered.) (This is globally canonical.) For $n = 1$ this is the enclosed area. Transfer this to the $n$-dimensional case; instead of $\gamma$,$T^n$ — but there it bounds nothing. Theorem. Consider the torus $T : p = 0$,then $AT \cap T \geq 2^n$ (at least $2^n$ points).

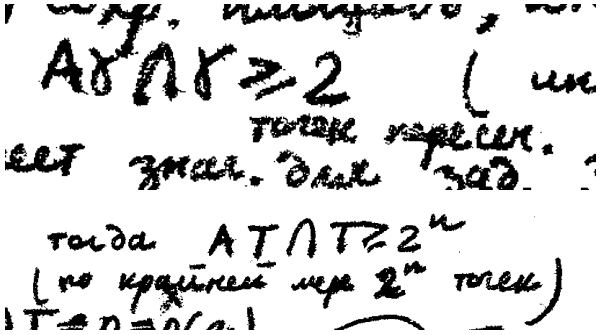
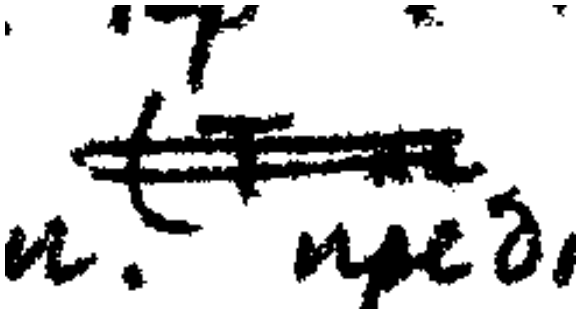


**Diagram and typed description.** The first entry is $A\gamma \cap \gamma \geq 2$,with intersection points below it. In the theorem,$AT \cap T \geq 2^n$; below,in words,at least $2^n$ points.

Original source details: GELFAND-P000118

Proof: under the additional [Crossed out: unreadable entry.]

assumption $AT \Rightarrow p = p(q)$.



**Uncertain reading.** Between under the additional and assumption,a short densely cancelled entry,with an opening-parenthesis-like stroke at the left.

Original source details: GELFAND-P000118

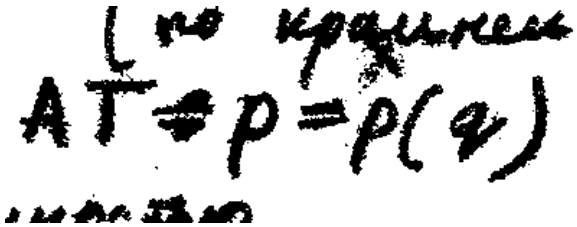
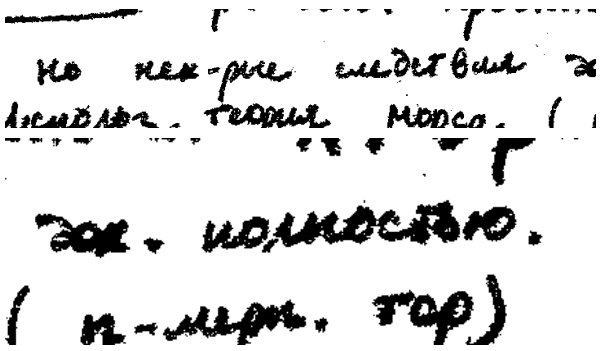


**Diagram and typed description.** After $AT$,a thick arrow-like sign right,then $p = p(q)$.

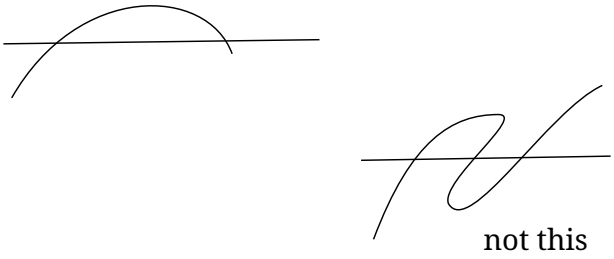
Original source details: GELFAND-P000118

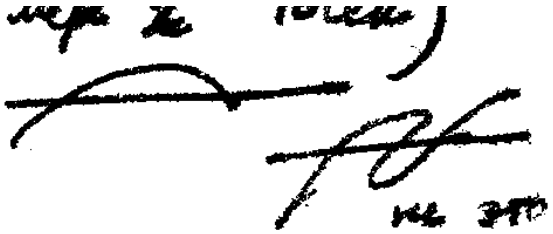
But some consequences dok.[?] completely.



Original source details: GELFAND-P000118

**Uncertain reading.** The word after some reads consequences; following it,a dense short abbreviation,dok.[?] / ekh.-like. Its full expansion is unestablished.



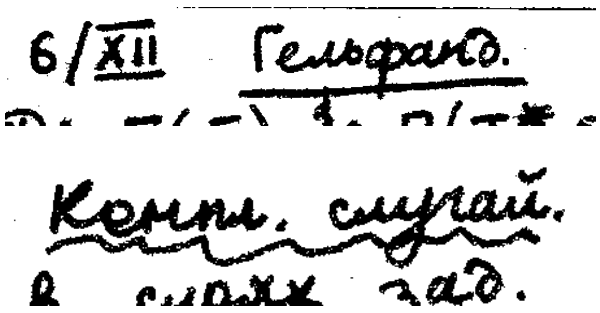
Original source details: GELFAND-P000118

**Diagram and typed description.** Above,a straight line and a single-valued arched curve with two intersections. Below,a different straight line and a looped self-crossing curve; below it,not this.

Use Morse's theorem ($n$-dimensional torus).

GELFAND-P000119

6/XII Gelfand.



Original source details: GELFAND-P000119

**Diagram and typed description.** Straight lines above and below XII; Gelfand's name has a straight underline. Complex case. below has a wavy underline.

On $X$, a vector bundle $E$. $T^*$ — cotangent. $D : \Gamma(E) \xrightarrow{\delta} \Gamma(T^* \otimes E)$.



Original source details: GELFAND-P000119

**Uncertain reading.** Above the arrow, a small $\delta$- / $s$-like sign; its exact reading is unestablished.

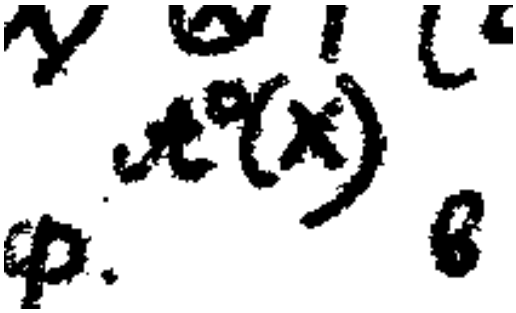
$D(fs) = df \cdot s + f \cdot Ds$.



Original source details: GELFAND-P000119

**Uncertain reading.** The $D$ in the last $Ds$ is densely overwritten; the earlier strokes are unread.

$A(X) = \sum A^p(X)$ — forms. $A(X, E) = A(X) \otimes_{A^0(X)} \Gamma(E)$ — forms with coefficients in the bundle.

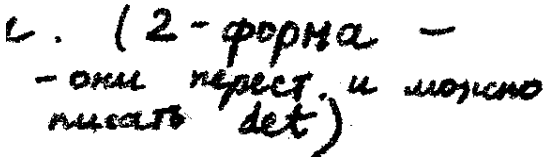


Original source details: GELFAND-P000119

**Diagram and typed description.** Below the tensor-product sign, $A^0(X)$.

$s_i$ — a set of sections, at $\forall$ point a frame in the fibre over $x$. $D(s_i) = \sum \theta_{ij} s_j$; $\theta$ — connection matrix. Curvature $K = d\theta - \theta \wedge \theta$ and characteristic classes (2-form — they commute and one can write det)

$$C(E, D) = \det\left(1 + \frac{i}{2\pi} K\right).$$



Original source details: GELFAND-P000119

**Diagram and typed description.** In parentheses, singular 2-form, then they commute and one can write det.

This does not depend on the choice of connection. Complex case. An analytic vector bundle is given. It is Hermitian if a Hermitian metric is given in the fibres.
$A(X) = \sum A^{p,q}(X)$; $A^{p,q}(X)$:

$$df_1 \wedge \cdots \wedge df_p \wedge d\overline{f}_{p+1} \wedge \cdots \wedge d\overline{f}_{p+q}.$$

($f_i$ — an analytic section.)



**Uncertain reading.** The last index $p+q$ / $p+g$ is densely corrected; its last letter remains disputed.

Original source details: GELFAND-P000119

$$\begin{aligned} d &= d' + d'', \\ d' &: A^{p,q} \longrightarrow A^{p+1,q}, \\ d'' &: A^{p,q} \longrightarrow A^{p,q+1}, \\ D &= D' + D'', \\ D' &: \Gamma(E) \longrightarrow A^{1,0}(X,E), \\ D'' &: \Gamma(E) \longrightarrow A^{0,1}(X,E), \\ \Gamma(T^* \otimes E) &= A^1(X,E). \end{aligned}$$

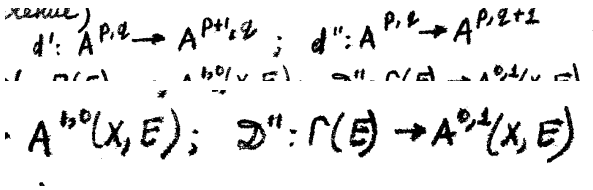


**Diagram and typed description.** For $d'$, at the right $p+1, q$; for $d''$, $p, q+1$. For $D'$, type $1,0$; for $D''$, $0,1$.

Original source details: GELFAND-P000119

The connection decomposes into a sum of 2 connections. $s_i$ given, consider $N_{ij} = \langle s_i, s_j \rangle_x$ — functions of $X$. (Hermitian matrix.) Theorem: if there is a Hermitian space, then there is a unique connection compatible with it. From the Hermitian bundle $N$, a connection is uniquely determined such that

$1^\circ$ $D$ preserves $N$; $2^\circ$ $s \in \Gamma_U(E)$ analytic $\Rightarrow D''s = 0$.

(Preserves the holomorphic structure.) (Parallel transport — a linear transformation.)

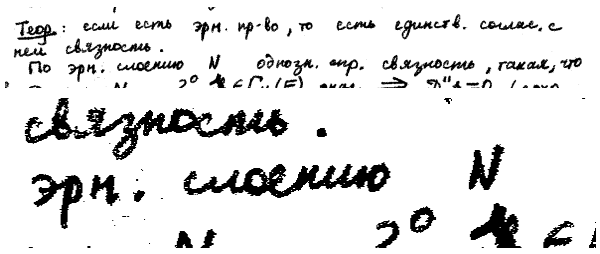
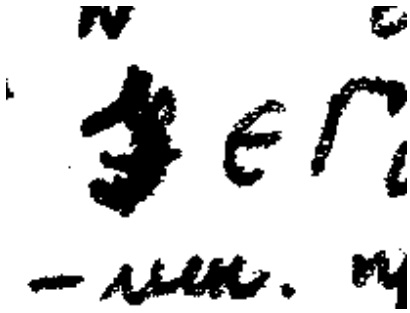


**Diagram and typed description.** At the start, Theorem is underlined. Below, From the Hermitian bundle $N$; the word is bundle.

Original source details: GELFAND-P000119



**Uncertain reading.** The $s$ in item 2 is densely overwritten; the earlier sign is unread. Below $\Gamma$, subscript $U$.

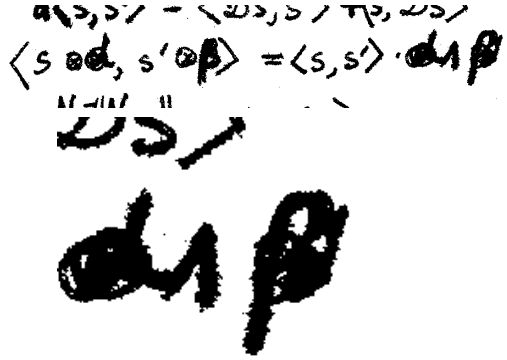
Original source details: GELFAND-P000119

$$d\langle s, s'\rangle = \langle Ds, s'\rangle + \langle s, Ds'\rangle.$$

(Differentiation formula.)

$$\langle s \otimes \alpha, s' \otimes \beta\rangle = \langle s, s'\rangle \cdot \alpha \wedge \beta.$$

(The scalar product of forms is a form.)



**Uncertain reading.** No separate bar over $\beta$ is visible in this product.

Original source details: GELFAND-P000119

$$N = \|N_{ik}\| = \langle s_i, s_k\rangle, \qquad dN = \theta N + N\bar{\theta}^t.$$

$\theta$ of type $1, 0$ — equivalent to condition 2. $d'N = \theta N$.

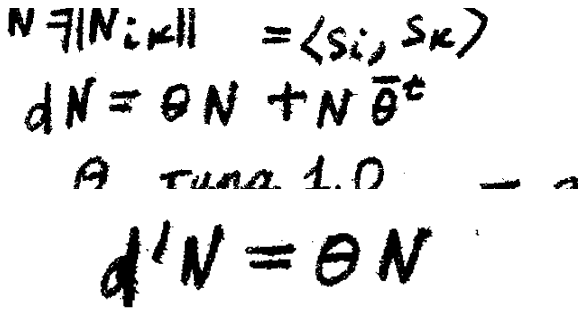


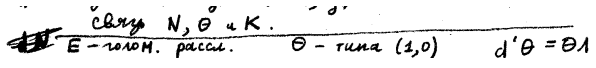
**Diagram and typed description.** In the matrix, indices $ik$; in the second term $N\bar{\theta}^t$, a bar over $\theta$ and a superscript $t$ at its right. At the bottom, $d'N = \theta N$.

Original source details: GELFAND-P000119

$\theta = d'N \cdot N^{-1}$. (The connection constructed from the Hermitian metric must have this form.) And conversely, from $N$ one easily obtains a connection.

$$d\langle s_i, s_j\rangle = \langle s_i, Ds_j\rangle + \langle Ds_i, s_j\rangle.$$

Relation between $N$, $\theta$ and $K$.



Original source details: GELFAND-P000120

**Diagram and typed description.** Below Relation between $N$, $\theta$ and $K$, a long straight line.

[Crossed out: a short unreadable entry.]



Original source details: GELFAND-P000120

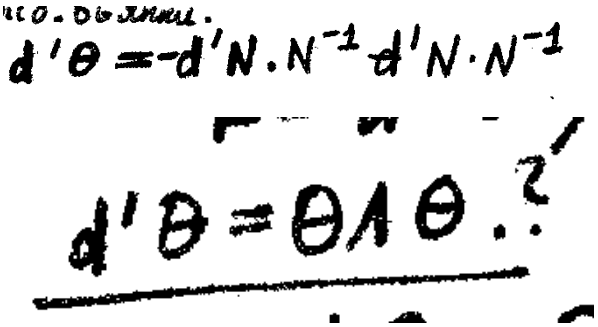
**Uncertain reading.** At the left below the heading, dense cancelled strokes; their exact letters are unestablished.

$E$ — a holomorphic bundle. $\theta$ — of type $(1,0)$.

$$K = d''\theta, \qquad d''K = 0,$$
$$d'K = -[K,\theta] \quad \text{Bianchi identity.}$$

$d'\theta = \theta \wedge \theta$. $\underline{d'\theta = \theta \wedge \theta}$ ? $\theta = d'N \cdot N^{-1}$, differentiate.

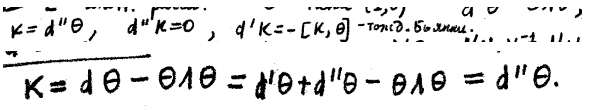
$$d'\theta = -d'N \cdot N^{-1} d'N \cdot N^{-1}.$$



Original source details: GELFAND-P000120

**Uncertain reading.** A minus in the differentiated product. Above at the right, a positive equality; at the left, its underlined repetition with a question mark.

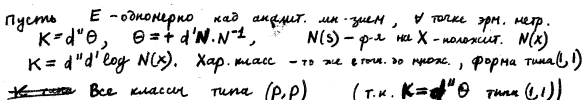
$$K = d\theta - \theta \wedge \theta$$
$$= d'\theta + d''\theta - \theta \wedge \theta = d''\theta.$$



Original source details: GELFAND-P000120

**Diagram and typed description.** In the identity $d'K = -[K,\theta]$, a minus before the bracket. Below, the equality chain ends with $d''\theta$.

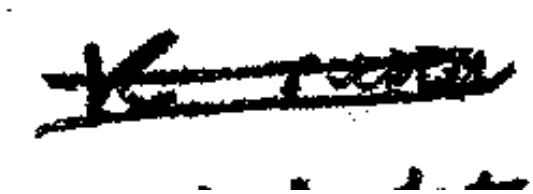
Let $E$ be one-dimensional over an analytic manifold, at $\forall$ point a Hermitian metric. $K = d''\theta$, $\theta = +d'N \cdot N^{-1}$. $N(s)$ — a function on $X$ — positive $N(x)$. $K = d''d' \log N(x)$. The characteristic class — the same up to a factor, a form of type $(1,1)$.



Original source details: GELFAND-P000120

**Diagram and typed description.** An explicit plus before $d'N$. On one line, $N(s)$, then $N(x)$. Below, $K = d''d' \log N(x)$.

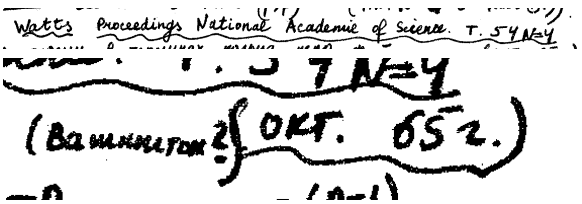
[Crossed out: a $K$-like beginning, the continuation unreadable.]



Original source details: GELFAND-P000120

**Uncertain reading.** Before All classes, a short $K$-like fragment is cancelled; the hidden strokes are unread.

All classes of type $(p,p)$ (since $K = d''\theta$ is of type $(1,1)$). Watts Proceedings National Academie of Science. vol. $54$, $N = 4$ homology in terms of the ring of continuous functions. (Washington? Oct. $65$)



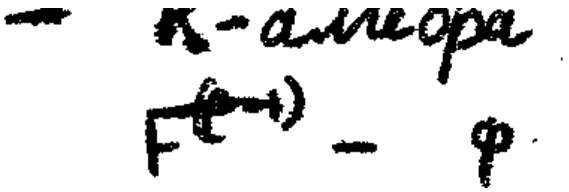
Original source details: GELFAND-P000120

**Diagram and typed description.** Watts and Proceedings National Academie of Science. vol. $54$, $N = 4$ have wavy underlines. In parentheses, Washington?; Oct. $65$ also has a wavy underline.

$\Lambda$ — a $K$-algebra ($K$ — a given ring). (In the application, $k = R$.)

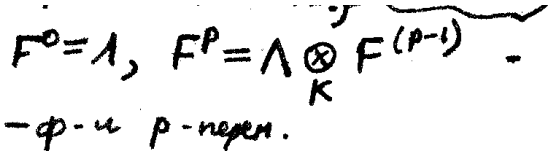
$$F^0 = \Lambda, \qquad F^p = \Lambda \otimes_K F^{(p-1)}.$$

functions of $p$ variables. $F^{(p-1)}$ — $\varphi$.



Original source details: GELFAND-P000120

**Uncertain reading.** In the left annotation, exponent $p-1$ / $p+1$, with a dense intervening stroke.



Original source details: GELFAND-P000120

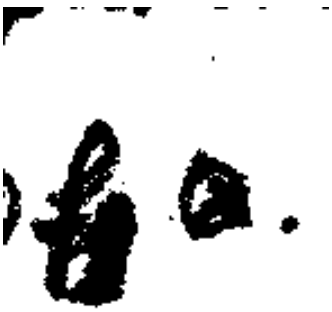
**Diagram and typed description.** At the right, $F^0 = \Lambda$, then $F^p = \Lambda \otimes_K F^{(p-1)}$; below them, functions of $p$ variables.

$$\mu_F(f_0 \otimes \cdots \otimes f_p) = f_0 f_1 \cdots f_p.$$

(Restriction map to the diagonal.)

$$d_i^p(f_0 \otimes f_1 \otimes \cdots \otimes f_p) = f_0 \otimes f_1 \otimes \cdots \otimes f_{i-1} \otimes 1 \otimes f_i \otimes \cdots \otimes f_p.$$

$d^p = \sum(-1)^i d_i^p$. $\{d^p, F^{(p)}\}$ — a complex.



**Uncertain reading.** The $f_1$ in the left parentheses is densely overwritten; the earlier index is unread.

Original source details: GELFAND-P000120

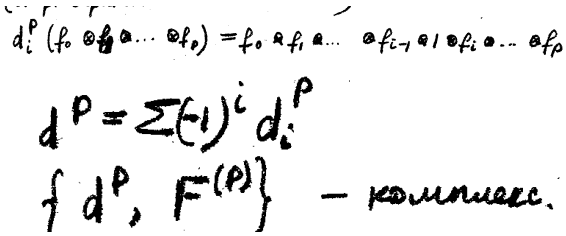
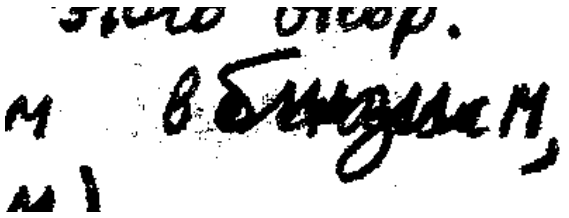


**Diagram and typed description.** Between $f_{i-1}$ and $f_i$, an inserted $1$. The sum has no written limits; below, $\{d^p, F^{(p)}\}$.

Original source details: GELFAND-P000120

Let $M$ be a maximal ideal. $\mu_F^{-1}(M) \Rightarrow$ a maximal ideal in $F^{(p)}$ — arises from this map. (The point on the diagonal corresponding to them.) $f$ is called vanishing near $M$, if $f$ lies in the kernel of the localization of $F^p$ at $\mu_p^{-1}(M)$.



**Uncertain reading.** Near and the final $M$ are densely corrected; $M$ is visible, the earlier letters are unread.

Original source details: GELFAND-P000120

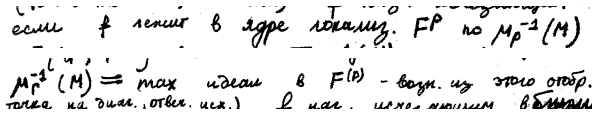
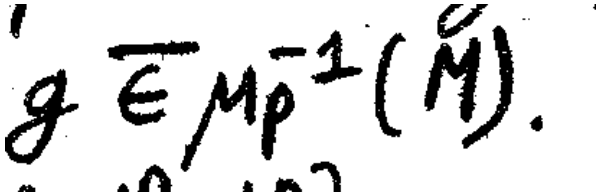


**Diagram and typed description.** In the first entry, $\mu_F^{-1}(M)$; in the localization line, $\mu_p^{-1}(M)$. Both $p$ and $(p)$ occur on $F$.

Original source details: GELFAND-P000120

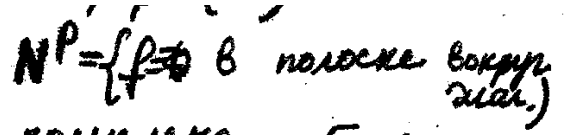
$\exists g$ such that $gf = 0$, $g \overline{\in} \mu_p^{-1}(M)$.



**Uncertain reading.** Above the sign $\in$ beside $g$, a separate horizontal bar.

Original source details: GELFAND-P000120

$N^p = \{f$ [Crossed out: an unreadable entry.] in a strip around the diagonal $\}$.

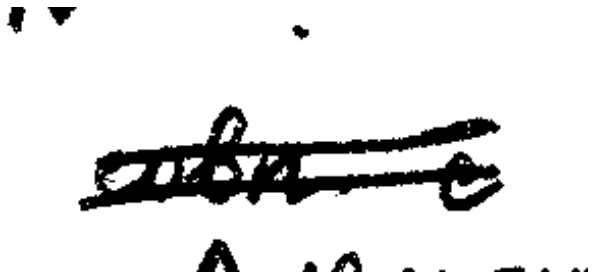


Original source details: GELFAND-P000120

**Uncertain reading.** After $f$, a densely cancelled $= 0$-like entry. It remains crossed out.

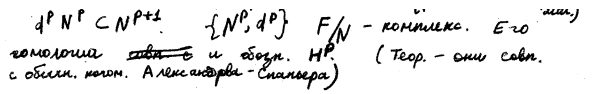
$d^p N^p \subset N^{p+1}$. $\{N^p, d^p\}$, $F/N$ — a complex. Its homology [Crossed out: several unreadable letters.] is denoted $H^p$.

(Theorem — they coincide with the usual Alexander–Spanier cohomology.)



Original source details: GELFAND-P000120

**Uncertain reading.** After homology, several letters are cancelled by two lines.



Original source details: GELFAND-P000120

**Diagram and typed description.** In the line, Its homology; in the final parentheses, Alexander–Spanier cohomology. Both $\{N^p, d^p\}$ and $F/N$ are written.

GELFAND-P000121

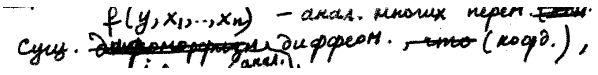
On the work of N. Levinson Duke, 1961 p. 345–353

$f(y, x_1, \ldots, x_n)$ — analytic in many variables. [Crossed out: a short unreadable entry.] There exists [Crossed out: a word resembling Diffeomorphism.] a diffeom.

[Inserted: anal.] [Crossed out: that[?].] (of coord.), such that

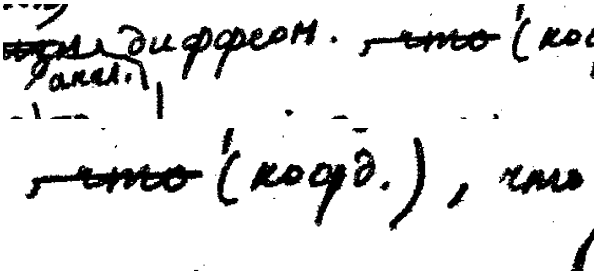
(local property)

Given $\left| \begin{array}{l} \dfrac{\partial^i f}{\partial y^i}(0,0) = 0 \\ \\ \dfrac{\partial^n f}{\partial y^n}(0,0) \neq 0 \end{array} \right|$ $\left| \begin{array}{l} i = 0, \ldots, n-1 \\ \\ d^n(x) y^n + \sum_0^{n-1} d^i(x) y^i \\ \\ d^n(0) \neq 0. \qquad d^i(0) = 0. \end{array} \right|$



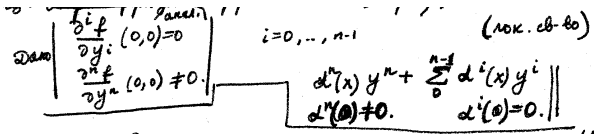
Original source details: GELFAND-P000121

**Uncertain reading.** After variables, a short dense cancellation. After There exists, a long cancelled word with a Diffeomorphism-like beginning; covered letters have not been supplied.



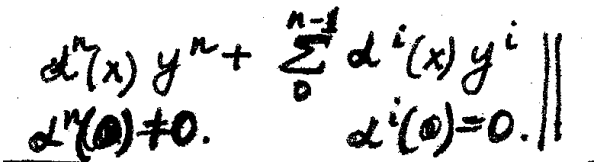
Original source details: GELFAND-P000121

**Uncertain reading.** A separate anal. stands below the main line near the cancelled word. Before (of coord.), a cancelled entry resembling that; after the parentheses, a separate clean that.



Original source details: GELFAND-P000121

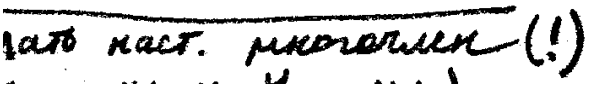
**Diagram and typed description.** Given at the left; the two conditions have a separate left vertical and a right vertical. A lower stepped line runs to the polynomial at the right; beyond it, two vertical lines. Above it, (local property).



Original source details: GELFAND-P000121

**Diagram and typed description.** The first coefficient has superscript $n$; the sum has lower limit $0$, upper $n-1$; then $d^i(x)y^i$. Beneath, separate $d^n(0) \neq 0$ and $d^i(0) = 0$.

Also, if there are $2$ variables, one can make an actual polynomial (!)



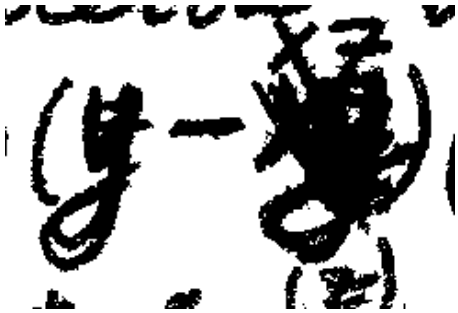
Original source details: GELFAND-P000121

**Diagram and typed description.** Before polynomial, the abbreviation nast., with a separate full stop. After polynomial, a separate (!).

(if $n = 3$, this is not always possible — Whitney's example)

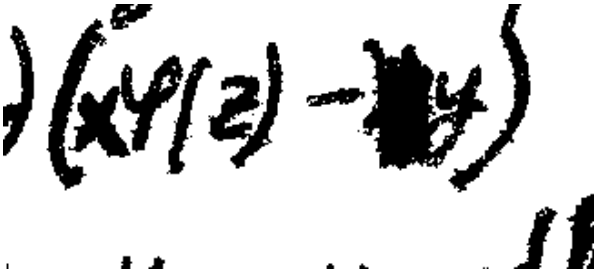
$$f(x, y, z) = xy(y - x)(y - xz[?])(x\varphi(z) - y[?]), \qquad \varphi(0) = 0,$$

$\varphi(z)$ — any transcendental function. Zeros — ($4$ surfaces)



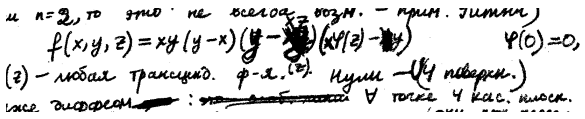
Original source details: GELFAND-P000121

**Uncertain reading.** In the fourth factor, after the minus, a densely corrected entry; $xz$ is inserted above it. The covered lower strokes have not been read.



Original source details: GELFAND-P000121

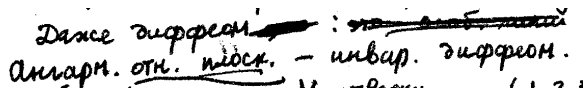
**Uncertain reading.** In the last factor, after the minus, a dense correction before a y-like letter; the preceding strokes have not been established separately.



Original source details: GELFAND-P000121

**Diagram and typed description.** In the example line, a separate $\varphi(0) = 0$. Below, after Zeros, the entry ($4$ surfaces); further down, at $\forall$ point $4$ tangent planes.

Even a diffeom. [The end of the abbreviation is crossed out]: [Crossed out: an unreadable phrase.]



Original source details: GELFAND-P000121

**Uncertain reading.** After Even, a diffeom.-like entry is visible, with its ending densely covered; after the colon, a long cancellation with several lines. In the next line, the words ratio of planes are underlined.

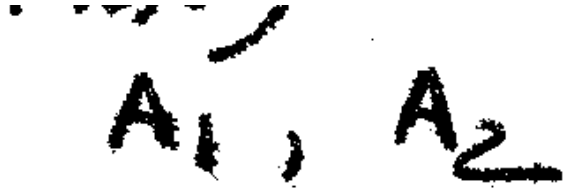
At $\forall$ point $4$ tangent planes. (They pass through one line.) Anharmonic ratio of planes — an invariant of diffeomorphisms. Singularities of curves. $\forall$ quadruple $(1, 2, 3, 4; \quad 1, 2, 3, 5)$ [Crossed out: $\forall$ and a short unreadable entry.]



Original source details: GELFAND-P000121

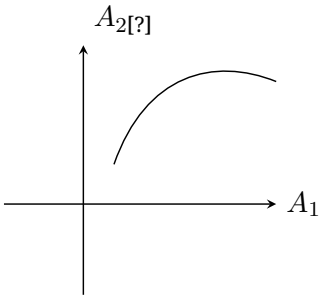
**Uncertain reading.** Before On this, a separately cancelled $\forall$ and a following short sign or word. These signs are crossed by one long line.

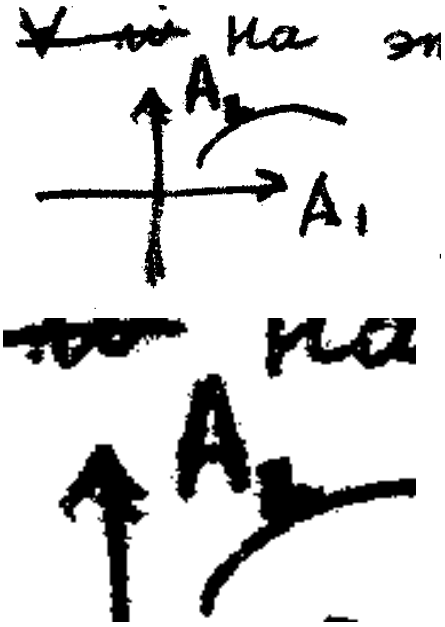
On this singular line, this is a function of $z$. $A_1,\ A_2$.



Original source details: GELFAND-P000121

**Diagram and typed description.** At the right, separate $A_1, A_2$; the lower horizontal stroke of $2$ is extended.

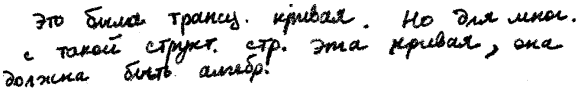


Original source details: GELFAND-P000121

**Uncertain reading.** The horizontal axis points right with $A_1$; the vertical points up. Beside its $A$, the lower right stroke merges with the beginning of the arc; subscript 2[?]. The arc in the first quadrant initially rises, then descends gently.

This was a transcendental curve. But for a polynomial with such a structure, str.[?] this curve, it must be algebraic.



Original source details: GELFAND-P000121

**Uncertain reading.** After structure, a short str.-like entry; its abbreviation has not been expanded. The final line reads must be algebraic.

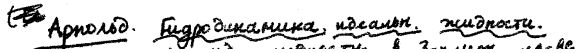
[Crossed out: a short unreadable sign.]



Original source details: GELFAND-P000121

**Uncertain reading.** At the left before Arnold., a separate densely cancelled sign with a rounded left stroke.

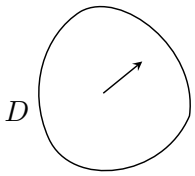
Arnold. Hydrodynamics, ideal fluid.



Original source details: GELFAND-P000121

**Diagram and typed description.** Arnold. has a straight underline. Hydrodynamics, ideal and fluid. have separate wavy underlines.

Stationary flows of ideal fluid in 3-dimensional space. $D$ — a bounded domain, and ideal fluid flows there (stationary flow).



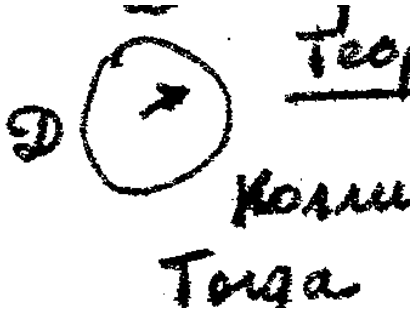
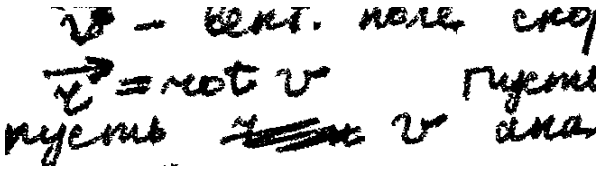



**Diagram and typed description.** A closed rounded contour; $D$ at the left. Inside, a short arrow points right and up.

Original source details: GELFAND-P000121

Theorem. $\vec{v}$ — the velocity vector field (tangent on the boundary).

$$\vec{\tau}[?] = \operatorname{rot} v.$$



**Uncertain reading.** The letter at the left has a rightward arrow above it. The letter is $\tau$- / $r$-like; its exact reading is unestablished. After rot, a separate $v$ with no visible overhead arrow.

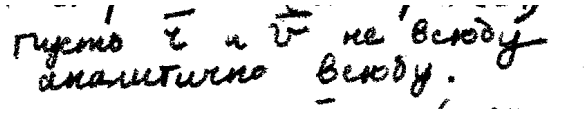
Original source details: GELFAND-P000121

Let $\vec{\tau}$[?] and $\vec{v}$ be not everywhere collinear; let [Crossed out: an unreadable word.] $v$ be analytic everywhere.



**Uncertain reading.** Before the last $v$, a separate dense cancellation; a short initial k-like stroke is visible, the other letters covered.

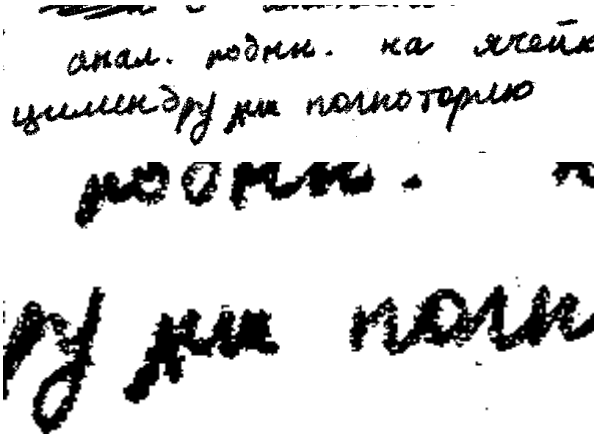
Original source details: GELFAND-P000121



**Uncertain reading.** In the condition not everywhere collinear, the first disputed letter has a short horizontal stroke overhead with no clearly separate head; $v$ has a rightward arrow.

Original source details: GELFAND-P000121

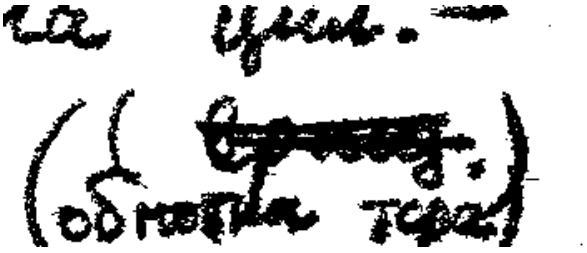
Then $D$ is divided by an analytic subset into cells (a finite number) — finite tsilindru[?] [na / ili?] polnotoriyu[?], and in them the flow is arranged as follows. On the cylinder — rotations.



Original source details: GELFAND-P000121

**Uncertain reading.** After finite, the ending is tsilindru[?] / cylinders; between it and polnotoriyu, a short na- / ili-like entry. Joined strokes do not give an unambiguous reading; no standard term has been substituted.

On the torus, $\dot{\varphi}_1 = \lambda_1, \quad \dot{\varphi}_2 = \lambda_2$ ([Crossed out: a word beginning vrashch...] [Below:] winding of the torus.)



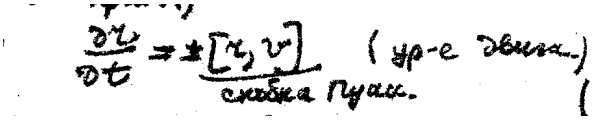
Original source details: GELFAND-P000121

**Uncertain reading.** In the parentheses, the upper entry is densely crossed out; vrashch...-like initial strokes are visible. The lower separate line is winding of the torus.

If everywhere collinear, there are examples when this is not so. Ideal fluid

$$\frac{\partial v}{\partial t} = \pm\underbrace{[\overline{\tau}, v]} \qquad \text{(eq. of mot.)}$$

Poisson bracket



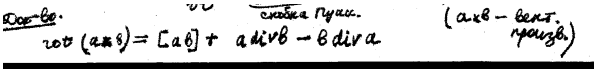
Original source details: GELFAND-P000121

**Diagram and typed description.** Before the bracket, a separate $\pm$. A short bar above its first letter. Below the bracket, a line with a short upturn at the right and the label Poisson bracket; at the right, (eq. of mot.).

Proof:

$$\mathrm{rot}(a * b) = [a\ b] + a\,\mathrm{div}\,b - b\,\mathrm{div}\,a$$

($a * b$ — vector product.)



Original source details: GELFAND-P000121

**Diagram and typed description.** Between $a$ and $b$, a star-shaped sign. At the right, $[a\ b]$ with no written comma, then $+a\,\mathrm{div}\,b - b\,\mathrm{div}\,a$. In parentheses, vector product.



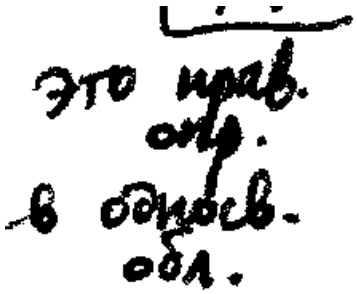
**Diagram and typed description.** At the upper right, 78; an angular line at its left and below it, with no upper or right side.

Original source details: GELFAND-P000122

$v$ is called the velocity field of ideal fluid if

1) $\operatorname{div} v = 0$
2) $[v, \operatorname{rot} v] = 0$
3) Tangent on the boundary.

this is a prav.[?] definition in a simply connected domain.



**Uncertain reading.** To the right of the three conditions, a separate note: this is a prav.[?] definition in a simply connected domain. The disputed abbreviation has prav.- / prib.-like strokes; the full word has not been expanded.

Original source details: GELFAND-P000122

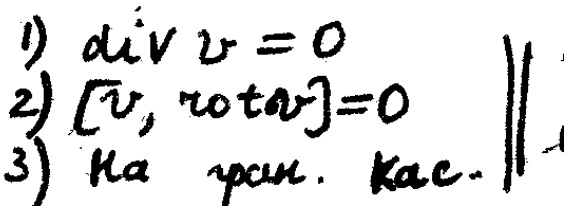


**Diagram and typed description.** The conditions are numbered $1), 2), 3)$. The second has $[v, \operatorname{rot} v] = 0$; the third, Tangent on the boundary. At the right, two separate parallel vertical lines.

Original source details: GELFAND-P000122

At $\forall$ point, 2 vector fields, $R^2$ — the orbits of the plane. 2 vector fields and the div of both 0, commuting, then the orbits are compact (everything in a finite domain) (If $b_1 = 0$ — the 1st Betti number $= 0$ One must preserve circulation also over large domains.

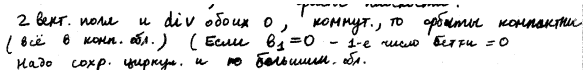


**Diagram and typed description.** In the condition, separate $b_1 = 0$ and the 1st Betti number $= 0$. An opening parenthesis before If; no closing parenthesis is visible after the last 0. The next line is One must preserve circulation also over large domains.

Original source details: GELFAND-P000122

Mathematical theorem. $b_1 = 0$, finite mn., $2$ commuting fields with div $= 0$, then the orbits are compact.

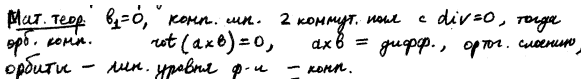
$$\text{rot}(a * b) = 0, \qquad a \times b = \text{diff.}, \quad \text{orthogonal sections},$$

orbits — level lines of a function — compact.



Original source details: GELFAND-P000122

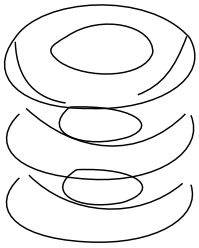
**Uncertain reading.** After $a \times b =$, the entry is diff.; its abbreviation has not been expanded. In the first product inside rot, the sign is star-shaped.



Original source details: GELFAND-P000122

**Diagram and typed description.** Mathematical theorem. is underlined. After $b_1 = 0$, finite mn.; below, the rot of the product equals $0$. At the end, a separate line: orbits — level lines of a function — compact.

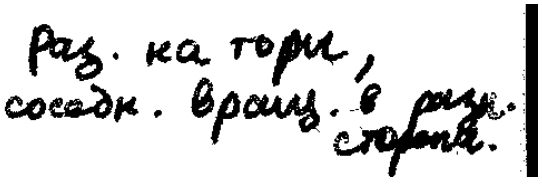
If there is no boundary, the compact orbits are tori; if noncompact, then cylinders. The action of the line — winding. (Experiments are visible in photographs.) There are cylinders too. $2$ rotating cylinders, fluid there.



Division into tori, neighbouring ones rotate in different directions.



**Diagram and typed description.** Three overlapping ring contours with inner loops and short side strokes. No separate labels or arrows are visible on the rings. At the right, Division into tori, neighbouring ones rotate in different directions.



Original source details: GELFAND-P000122

13/XII. Arnold. What is ideal fluid?

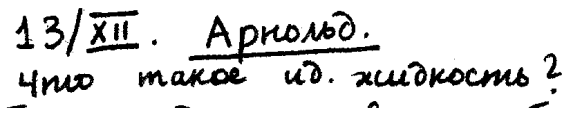



**Diagram and typed description.** Separate straight lines above and below XII; Arnold. is underlined separately. Below, the question What is ideal fluid?

Original source details: GELFAND-P000122

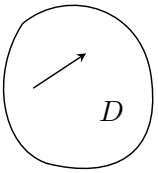
Fluid flow in a bounded domain, [Crossed out: a short unreadable sign.] Interesting to study instability.

**Uncertain reading.** Before Interesting, a separate densely crossed-out short sign; its letter or meaning is unestablished.



Original source details: GELFAND-P000122

$v$ — velocity field, $\operatorname{div} v = 0$ (incompressible). (Makes sense on a Riemannian manifold.)



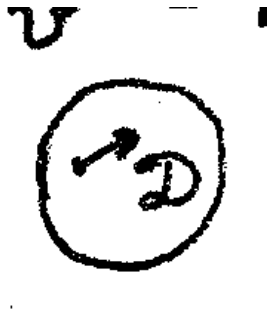




**Diagram and typed description.** A closed oval contour. Inside, a right-and-up arrow; $D$ is also inside, to the right and below the arrow.

Original source details: GELFAND-P000122

Newton's equation

$$\frac{DV}{Dt} = \mathcal{F}.$$

$\mathcal{F}$ — force. $\frac{DV}{Dt}$ — acceleration of a particle (not the partial derivative).

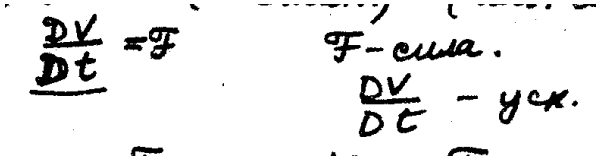



**Diagram and typed description.** In Newton's equation, numerator $DV$, denominator $Dt$; a separate straight line beneath the whole fraction. At the right, a calligraphic F-like letter, and its repetition beside force.

Original source details: GELFAND-P000122

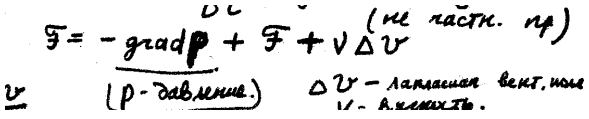
$v(x, y, z, t)$. If there are no external forces

$$\mathcal{F} = -\underline{\operatorname{grad} p[?]} + \mathcal{F} + \nu\Delta v$$

(They are potential.)

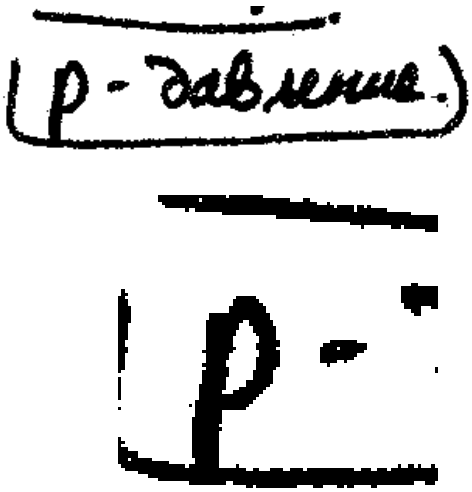
$$\boxed{p[?] \text{ — pressure.}}$$

$\Delta v$ — Laplacian of the vector field. $\nu$ — viscosity.



Original source details: GELFAND-P000122

**Diagram and typed description.** In the equality, after the underlined grad with a p-like letter, separate $+\mathcal{F}$ and $+\nu\Delta v$. Below, $\Delta v$ — Laplacian of the vector field and $\nu$ — viscosity.

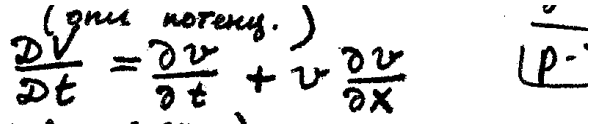


Original source details: GELFAND-P000122

**Uncertain reading.** In the pressure label, a p-/$\rho$-like letter with a long lower stroke. Around the label, left, lower and right lines; no upper side is drawn. The letter's shape differs from the disputed sign in the gradient of the system below.

$$\frac{DV}{Dt} = \frac{\partial v}{\partial t} + v\frac{\partial v}{\partial x}$$

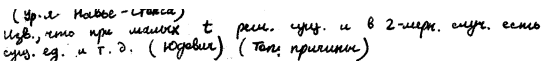
(Navier–Stokes equation)



Original source details: GELFAND-P000122

**Diagram and typed description.** In the left fraction, numerator $DV$ with a capital V-like letter; the right fractions have small $v$. The last denominator is $\partial x$.

It is known that for small $t$ a solution exists, and in the 2-dimensional case there is existence, uniqueness, etc. (Yudovich) (Applicable there)



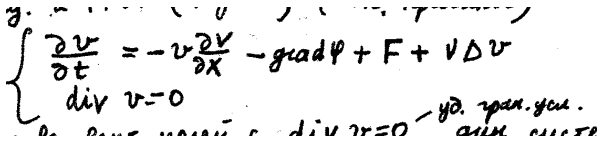
Original source details: GELFAND-P000122

**Diagram and typed description.** After existence, uniqueness, etc., two separate parentheses: (Yudovich) and (Applicable there). A closing parenthesis follows there; no continuation is written inside it.

$$\begin{cases} \dfrac{\partial v}{\partial t} = -v\dfrac{\partial v}{\partial x} - \operatorname{grad}\varphi[?] + F + \nu\Delta v \\ \operatorname{div} v = 0 \end{cases}$$

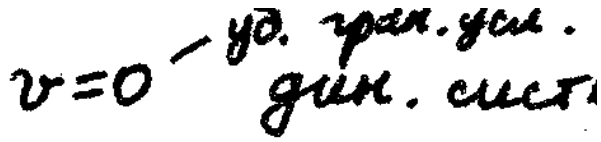
ud. bound. cond.

In the space of vector fields with $\operatorname{div} v = 0$ — a dynamical system.



Original source details: GELFAND-P000122

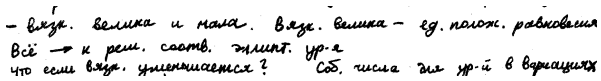
**Uncertain reading.** After grad, the letter is $\varphi$- / p-like, with a long stroke through the rounded part. A minus precedes it. Then a separate plain $F$, followed by $+\nu\Delta v$. A common brace at the left.



Original source details: GELFAND-P000122

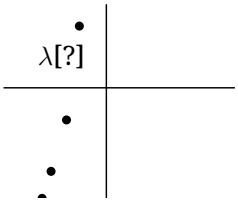
**Diagram and typed description.** A short oblique pointer leads from $\operatorname{div} v = 0$ to a note above the line: ud. bound. cond. The first abbreviation is ud.; it has not been expanded.

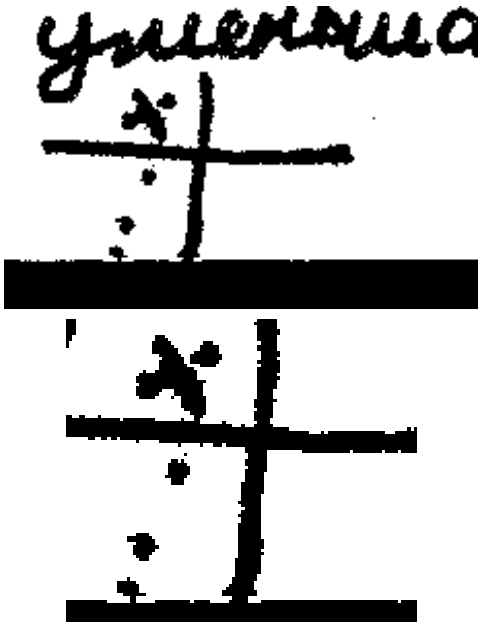
Limiting cases — — viscosity large and small. Large viscosity — a unique equilibrium position. Everything $\to$ the solution of the corresponding elliptic equation. What if viscosity decreases? Eigenvalues for the variational equations (stable solution) $\lambda$ moves right.



Original source details: GELFAND-P000122

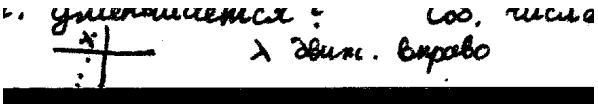
**Diagram and typed description.** In the question, decreases is underlined; below, (stable solution), then a separate $\lambda$ moves right.



Original source details: GELFAND-P000122

**Uncertain reading.** Two intersecting lines with no clearly separate arrowheads. Several points at the left; above the horizontal, a dense $\lambda$-like sign and a separate point. The sign's reading is unestablished.

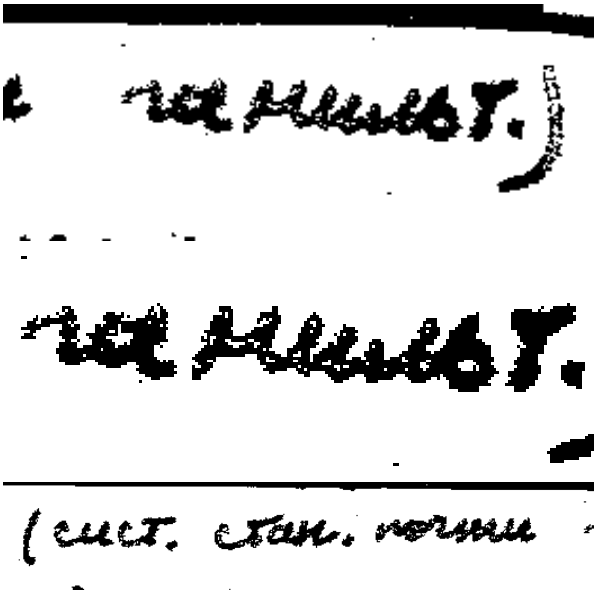


Original source details: GELFAND-P000122

**Undeciphered passage.** The lower edge of the scan passes through the vertical stroke and the lowest point of the diagram. Their continuation beyond the edge is not visible.

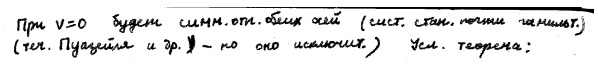
GELFAND-P000123

For $\nu = 0$, we shall be symm. about both axes (syst. stationary points Hamil.[?])
(Poiseuille flow and others) — but it is unstable.)



Original source details: GELFAND-P000123

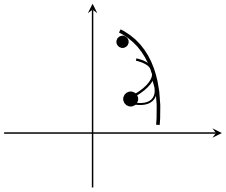
**Uncertain reading.** In the parentheses, syst. stationary points, then a Hamil.-like abbreviation; its full word is unestablished. Alternative reading: nachnut.

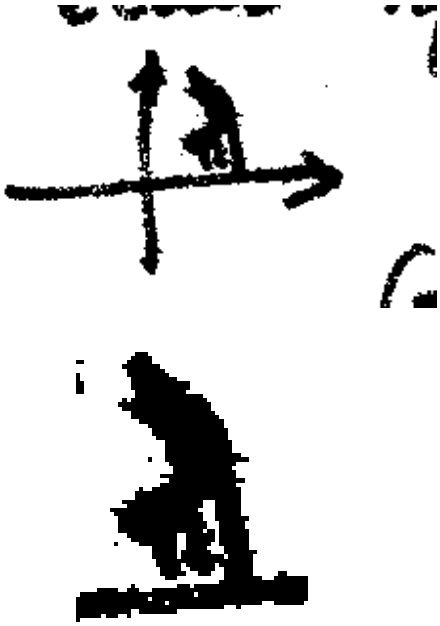


Original source details: GELFAND-P000123

**Diagram and typed description.** In the first line, $\nu = 0$, then we shall be symm. about both axes. After the following line, a separate closing parenthesis after unstable.

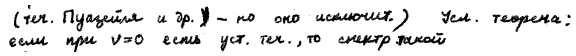
Conditional theorem: if at $\nu = 0$ there is stable flow, then the spectrum is like this





**Uncertain reading.** Axes with arrows up and right. To the right above the intersection, a dense curved trace with short adjoining strokes. Its reading as a letter or separate points is unestablished.

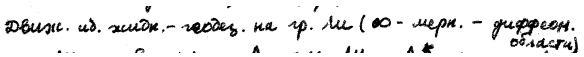
Original source details: GELFAND-P000123



Original source details: GELFAND-P000123

**Diagram and typed description.** Conditional theorem: above a separate line, if at $\nu = 0$ there is stable flow, then the spectrum is like this. The drawing has no readable axis labels.

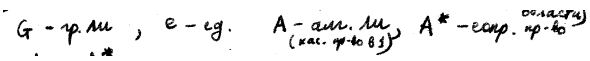
Motion of ideal fluid — geodesics on a Lie gr. ($\infty$-dimensional — diffeomorphisms of the domain)



Original source details: GELFAND-P000123

**Diagram and typed description.** In the heading line, geodesics on a Lie gr.; at the right in parentheses, $\infty$-dimensional — diffeomorphisms of the domain. The word domain continues below.

$G$ — Lie gr., $e$ — identity. $A$ — Lie alg. (tangent space at $1$), $A^*$ — dual space.



Original source details: GELFAND-P000123

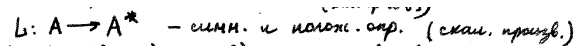
**Diagram and typed description.** Beside $A$, parentheses: tangent space at $1$; the numeral $1$ is written. Beside $A^*$, separately, dual space.

$L : A \to A^*$ — symm. and positive definite (scalar product)

$$\langle a, b\rangle = \langle La, b) = (a, Lb).$$

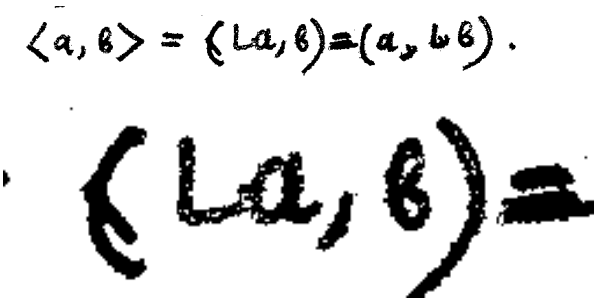
Defines a left-inv. metric on the gr.

$$\langle a, b\rangle_g = \langle \widetilde{\Lambda}_{g^{-1}} a, \widetilde{\Lambda}_{g^{-1}} b\rangle. \qquad L_g : TG_g \to T^*G_g$$



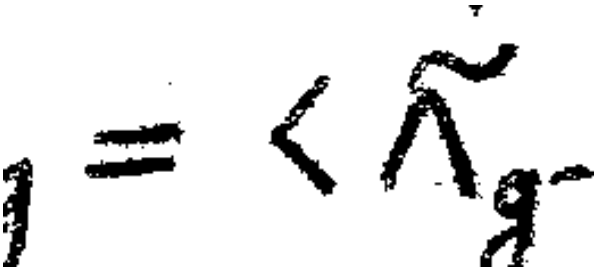
Original source details: GELFAND-P000123

**Diagram and typed description.** A single rightward arrow from $A$ to $A^*$; after it, symm. and positive definite, then scalar product in parentheses.



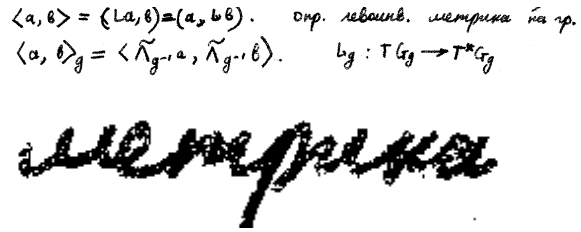
Original source details: GELFAND-P000123

**Uncertain reading.** The first opening bracket after the equality has an angular upper part and two lower strokes, one rounded. Its bracket type is unestablished. After $b$, a round closing bracket; the last pair is $(a, Lb)$.



Original source details: GELFAND-P000123

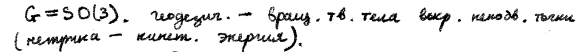
**Uncertain reading.** Wavy strokes above both $\Lambda$-like signs; the choice between capital $\Lambda$ and small $\lambda$ is not finally established. Subscripts $g^{-1}$.



Original source details: GELFAND-P000123

**Diagram and typed description.** In the line before the formula, the Russian word metrika ends in a. The left pair has subscript $g$; at the right, separately, $L_g : TG_g \to T^*G_g$.

$G = SO(3)$. geodesics — rotation of a rigid body about a fixed point (metric — kinetic energy).

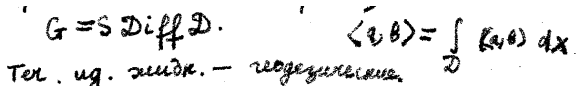


Original source details: GELFAND-P000123

**Diagram and typed description.** After $SO(3)$, a separate full stop; then geodesics — rotation of a rigid body about a fixed point. On the next line, parentheses: metric — kinetic energy.

$$G = \mathrm{SDiff}\, D. \qquad \langle a, b\rangle = \int_D (a, b)\, dx.$$

Ideal fluid flows — geodesics.



Original source details: GELFAND-P000123

**Diagram and typed description.** The integral has lower subscript $D$, and the integrand pair $(a, b)$ is in round brackets. A separate next line: Ideal fluid flows — geodesics.

$$G: \quad x \to ax + b \qquad a > 0.$$

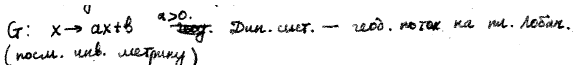
[Crossed out: a short unreadable sign or word beneath $a > 0$.]

Dynamical syst. — geodesic flow on the Lobachevsky plane. (Look at the inv. metric.)



Original source details: GELFAND-P000123

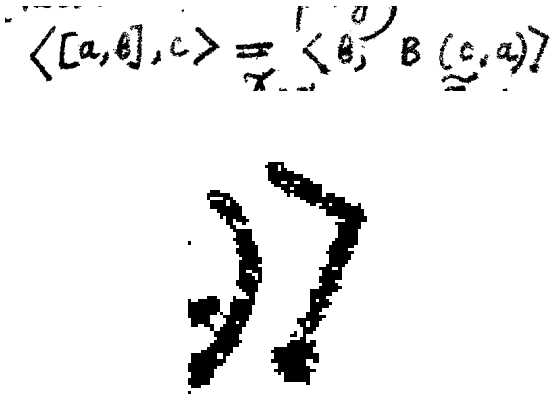
**Uncertain reading.** Below the active $a > 0$, a densely crossed-out entry with a lower loop; its letters and sign are unestablished.



Original source details: GELFAND-P000123

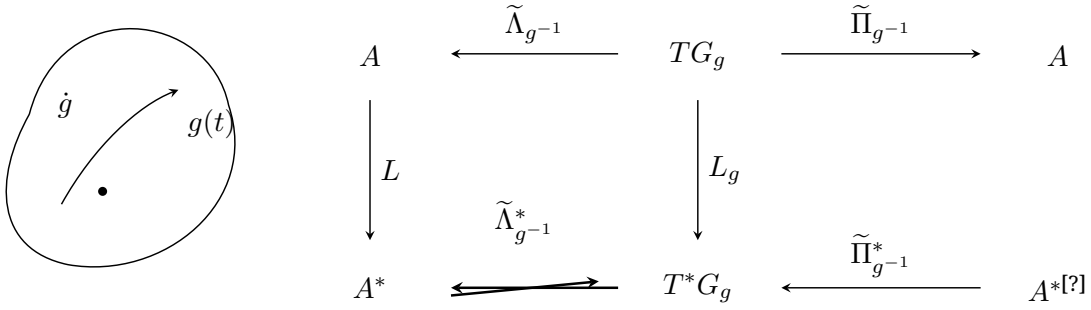
**Diagram and typed description.** Arrow $x \to ax + b$; above the crossed-out entry, $a > 0$. At the right, Dynamical syst. — geodesic flow on the Lobachevsky plane; below, separate parentheses: Look at the inv. metric.

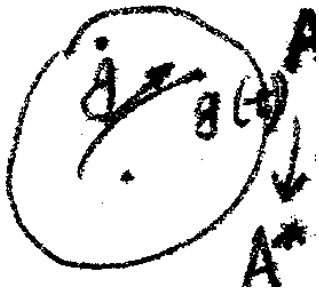
$$\langle [a, b], c\rangle = \langle b, B(c, a)\rangle$$



Original source details: GELFAND-P000123

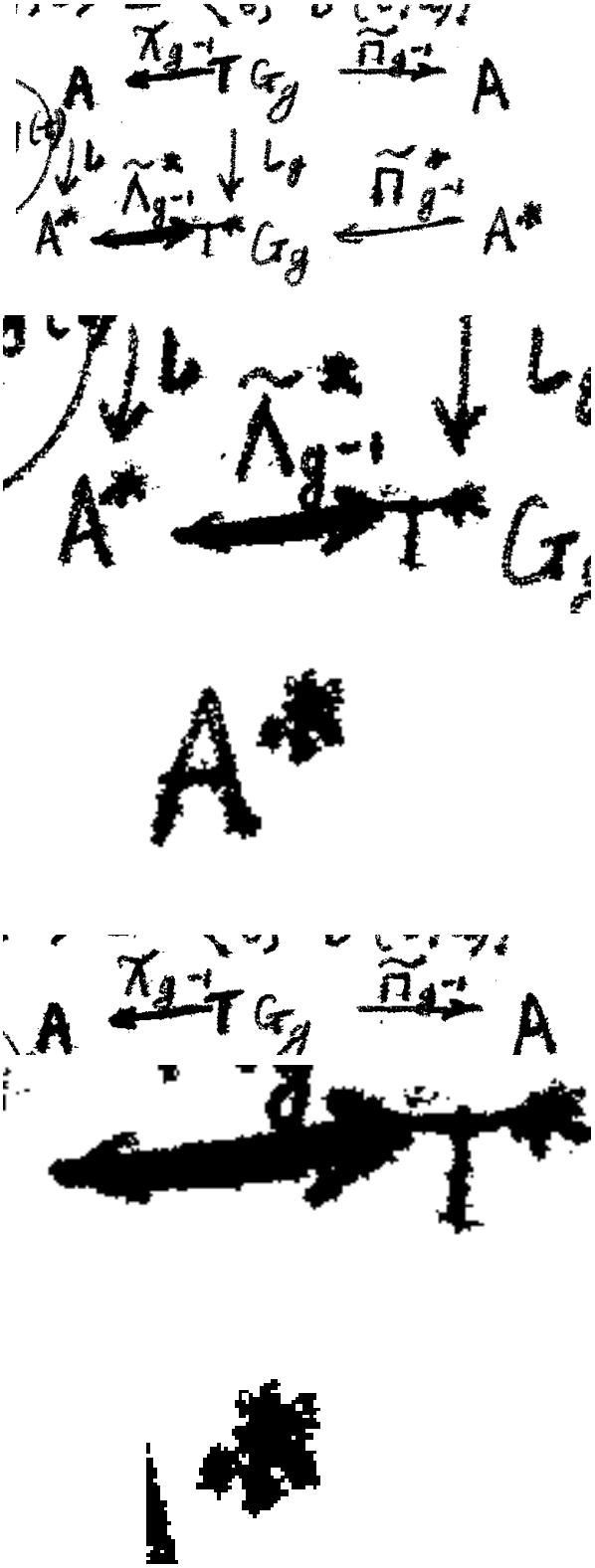
**Diagram and typed description.** In the first pair, $[a, b]$ and $c$; in the second, $b$ and $B(c, a)$. After the round bracket, the final angle bracket, with a short thickening at the bottom.



Original source details: GELFAND-P000123

**Diagram and typed description.** At the left, a closed oval contour; inside, a curve with an arrow, $\dot{g}$, $g(t)$ and a separate lower point. The label $g(t)$ touches the contour's right edge.



**Uncertain reading.** Upper row $A \leftarrow TG_g \rightarrow A$, with wavy $\Lambda$- and $\Pi$-like signs above the arrows, subscripts $g^{-1}$. Separate downward arrows $L$ and $L_g$. In the lower row, $A^*$ at the left, $T^*G_g$ in the middle, and at the right $A$ with a dense star or overlaid stars: $A^*$ / $A^{**}$. The lower right arrow points left. The lower left line has two overlaid tips in opposite directions; its active direction is unestablished. Wavy signs with stars and subscripts $g^{-1}$ above the lower arrows.

Original source details: GELFAND-P000123

1966 yr. 7/II Kushnirenko: “On invariant dynamical systems”

I. M. Gelfand. on the envelop. alg. of a Lie algebra

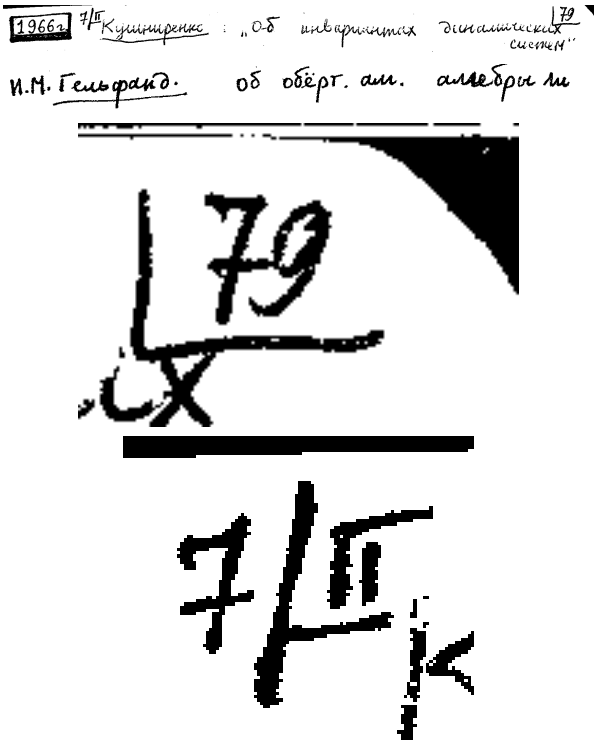



**Diagram and typed description.** The year is in a rectangular frame with a thickened lower stroke. At 79, left and lower frame sides. Lines above and below II in the date; Kushnirenko and Gelfand. are underlined separately from the date and initials. The Russian title ends with sistem.

Original source details: GELFAND-P000124

Every Noeth. ring (without zero divisors) — an Ore ring

$$\begin{pmatrix} \forall a, b \neq 0 & \exists x, y \neq 0, \\ & xa = yb \end{pmatrix}$$

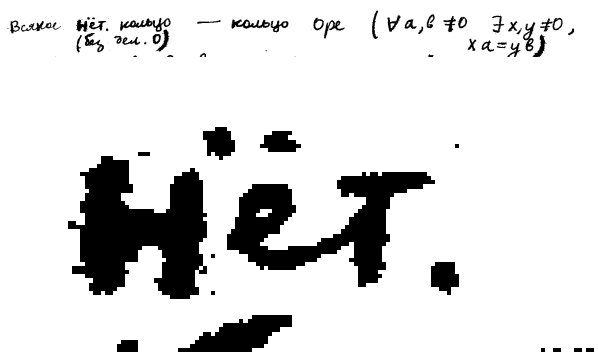



**Diagram and typed description.** Two dots over e in the Russian abbreviation nyot. At the right, parentheses with $\forall a, b \neq 0$, $\exists x, y \neq 0$; below, $xa = yb$.

Original source details: GELFAND-P000124

An Ore ring embeds in a skew field (these are est. classes) (elements of the form $ab^{-1}$)

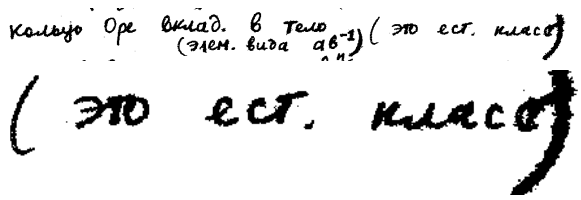


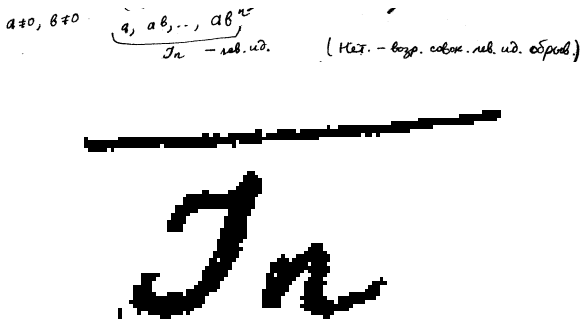
**Diagram and typed description.** In the right parentheses, these are est. classes; est. is an abbreviation. Below, separate parentheses: elements of the form $ab^{-1}$, with negative one on $b$.

Original source details: GELFAND-P000124

$a \neq 0, b \neq 0.$

$$\underbrace{a, ab, \ldots, ab^n}_{\mathcal{I}_n[?] \text{ — left ideal}}$$

(Noeth. — an increasing collection of left ideals stops.)



**Uncertain reading.** A long lower brace beneath the sequence. Below it, a capital I-/J-like letter with a lower loop and subscript $n$, then left ideal; its exact letter is unestablished. At the right, separate parentheses beginning Noeth.

Original source details: GELFAND-P000124

$$ab^{n+1} = x_0 a + x_1 ab + \cdots + x_n ab^n, \qquad x_0 a = (\quad) b$$

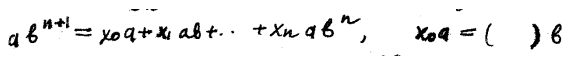
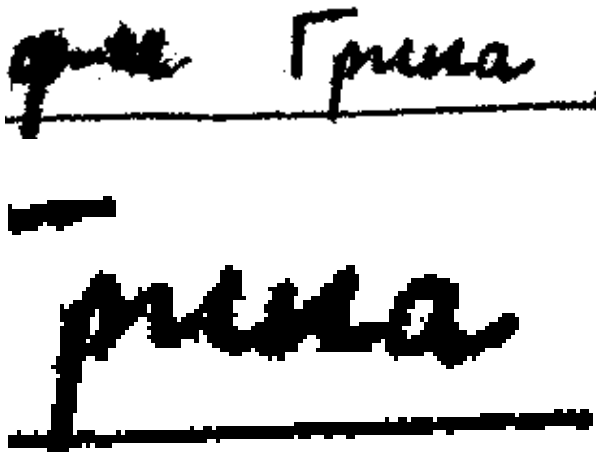


**Diagram and typed description.** In the first equality, powers $n+1$ and $n$, subscripts $0, 1, n$. In the second, after $x_0 a =$, round brackets with an empty interval, then $b$.

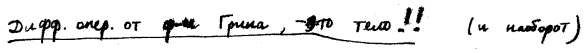
Original source details: GELFAND-P000124

Diff. oper. of [Crossed out: an f-n-like abbreviation.] Grin[?], this is a skew field!! (and conversely)



**Uncertain reading.** After of, a short densely cancelled entry with an f-like beginning. The following name reads Grin / Grim; its penultimate letter is unestablished.

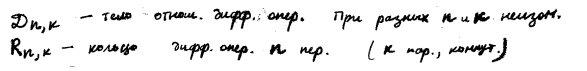
Original source details: GELFAND-P000124



Original source details: GELFAND-P000124

**Diagram and typed description.** The line from Diff. oper. of through skew field!! has a common underline. A comma after the disputed name; at the right, separate parentheses: and conversely. Two exclamation marks.

$D_{n,k}$ — skew field of ratios of diff. oper. For different $n$ and $k$, nonisom. $R_{n,k}$ — ring of diff. oper. in $n$ variables ($k$ param., commuting).



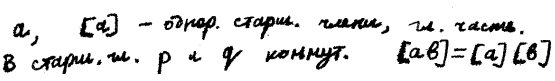
Original source details: GELFAND-P000124

**Diagram and typed description.** Two different capitals, $D$ and $R$, with the same subscripts $n, k$. Beside $R$, separately, $n$ variables and parentheses: $k$ param., commuting.

$a$, $[a]$ — oper. leading term, principal part. In the leading part, $p$ and $q$ commute. $[ab] = [a][b]$

$$b = a_1 a_2^{-1} \qquad \text{Consider} \quad \frac{[a_1]}{[a_2]}$$

it depends only on $b$, and not on the manner of representation.



Original source details: GELFAND-P000124

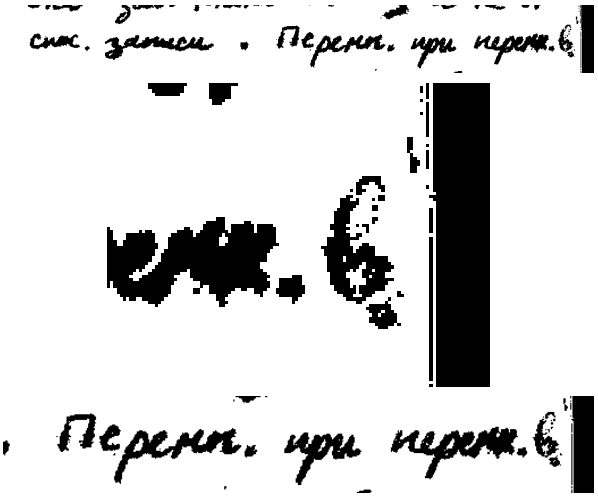
**Diagram and typed description.** A separate comma at $a$, $[a]$. Square brackets around the principal parts; below, $p$ and $q$, then $[ab] = [a][b]$.



Original source details: GELFAND-P000124

**Diagram and typed description.** In $b = a_1 a_2^{-1}$, negative one on $a_2$. At the right, a fraction with $[a_1]$ and $[a_2]$ in square brackets; below, it depends only on $b$, and not on the manner of representation.

Peremn.[?] under norm.[?] $b$[?]



Original source details: GELFAND-P000124

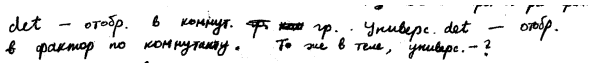
**Uncertain reading.** The next fragment: a Peremn.-like abbreviation, under, a norm.- / peremn.-like abbreviation, and a final v-/b-like letter at the line edge. The abbreviations and final letter are not unambiguously established; no continuation is visible after it.

det — a map to a commut. [Crossed out: two short signs or words.] gr. Universal det — a map to the quotient by the commutator subgroup. The same in a skew field, universal — ?



Original source details: GELFAND-P000124

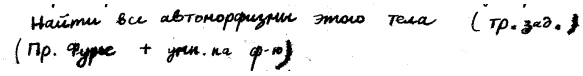
**Uncertain reading.** Between commut. and gr., two separate dense cancellations. The first has an f-/f-ya-like lower stroke; the second, covered kom-like strokes. The active gr. is at the right of both.



Original source details: GELFAND-P000124

**Diagram and typed description.** At the start, Latin det; then Universal det — a map to the quotient by the commutator subgroup. The final phrase, The same in a skew field, universal — ?, ends with a question mark.

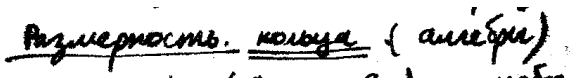
Find all automorphisms of this skew field (difficult problem). (Fourier tr. + multiplication by a function)



Original source details: GELFAND-P000124

**Diagram and typed description.** In the first line, all automorphisms of this skew field and parentheses: difficult problem. In the second, Fourier tr., a separate plus, then multiplication by a function.

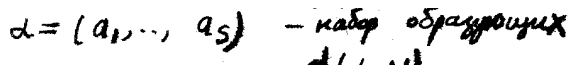
Dimension. of a ring (algebra)



Original source details: GELFAND-P000124

**Diagram and typed description.** Dimension. is underlined once, of a ring with two separate lower lines. Parentheses: algebra, after ring.

$\alpha = (a_1, \ldots, a_s)$ — a set of generators

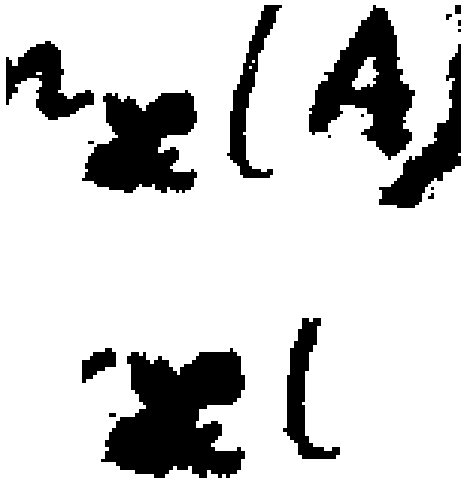


Original source details: GELFAND-P000124

**Diagram and typed description.** At the left, $\alpha$; a tuple in round brackets, with extreme subscripts $1$ and $s$. After the dash, a set of generators.

$$\sup_{\alpha} \overline{\lim}_{N\to\infty} \frac{d(\alpha, N)}{\ln N} = \mathrm{Dim}_{\chi[?]}(A)$$

$d(\alpha, N)$ — the number of linearly independent monomials of degree $\leq N$.



**Uncertain reading.** Below Dim, a dense $\chi$- / z- / $\alpha$-like sign; its exact reading is unestablished. At the right, round brackets with $A$.

Original source details: GELFAND-P000124

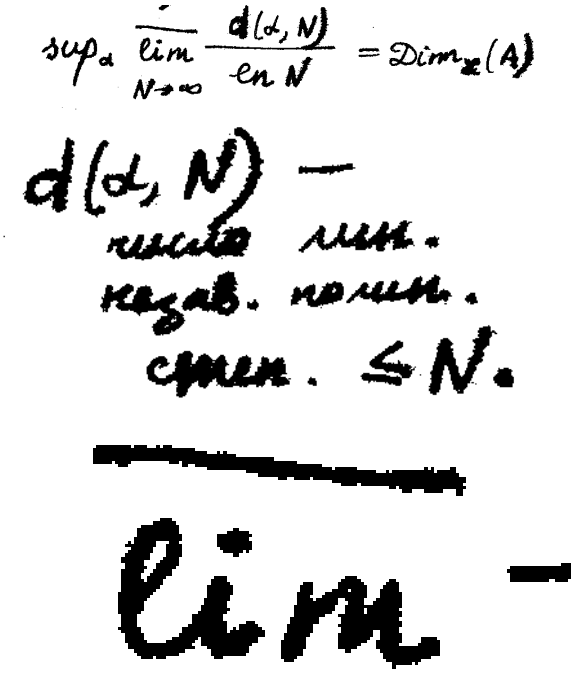


**Diagram and typed description.** A lower $\alpha$ at sup; a separate horizontal line above lim, with $N \to \infty$ below. Numerator $d(\alpha, N)$, denominator $\ln N$. At the right, separately, $d(\alpha, N)$ — the number of linearly independent monomials of degree $\leq N$.

Original source details: GELFAND-P000124

$$\dim R_{n,k} = 2n + k.$$

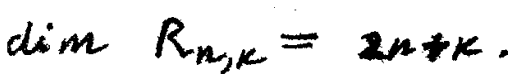
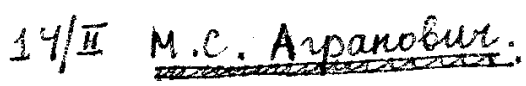


**Diagram and typed description.** In the final line, dim with small d, $R_{n,k}$ and $2n + k$, then a full stop.

Original source details: GELFAND-P000124

GELFAND-P000125

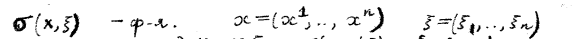
14/II M. S. Agranovich.



Original source details: GELFAND-P000125

**Diagram and typed description.** Lines above and below II in the date. Beneath the name M. S. Agranovich., two lines with oblique hatching between them.

$\sigma(x,\xi)$ — a fn. $x=(x^1,\dots,x^n)$, $\xi=(\xi_1,\dots,\xi_n)$.

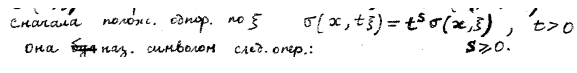


Original source details: GELFAND-P000125

**Diagram and typed description.** The first sign is $\sigma$; the coordinates of $x$ have superscripts $1, n$, and those of $\xi$ have subscripts $1, n$.

Initially pos. homog. in $\xi$

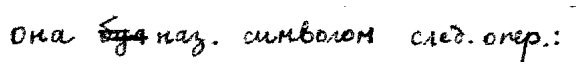
$$\sigma(x,t\xi)=t^s\sigma(x,\xi),\qquad t>0,\quad s\geq 0.$$



Original source details: GELFAND-P000125

**Diagram and typed description.** The exponent of $t$ is $s$. After $t>0$, a separate $s\geq 0$ followed by a period.

it [Crossed out: bud[?].] is called the symbol of the following oper.:

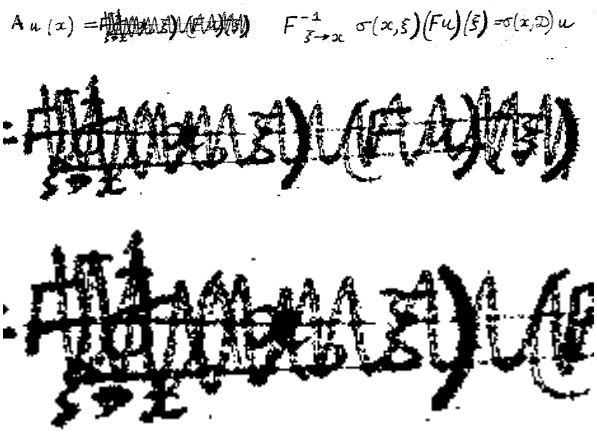


**Uncertain reading.** After it and before is called, a short crossed-out bud-like word; the exact reading of the covered strokes is undetermined.



Original source details: GELFAND-P000125

$$Au(x)=\text{[Crossed out: } F^{-1}_{\xi\to x}\sigma(x[?],\xi)(Fu)(\xi)]$$
$$F^{-1}_{\xi\to x}\sigma(x,\xi)(Fu)(\xi)=\sigma(x,D)u$$

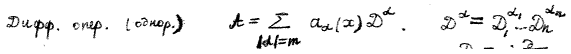


**Uncertain reading.** After $Au(x)=$, a crossed-out formula with $F^{-1}_{\xi\to x}$, $\sigma$, parentheses and $(Fu)(\xi)$. Interior strokes, including the first argument of $\sigma$, are covered; the reading of the entire crossed-out expression remains incomplete. A separate active formula is to the right.

Original source details: GELFAND-P000125

Differential oper. (homog.)

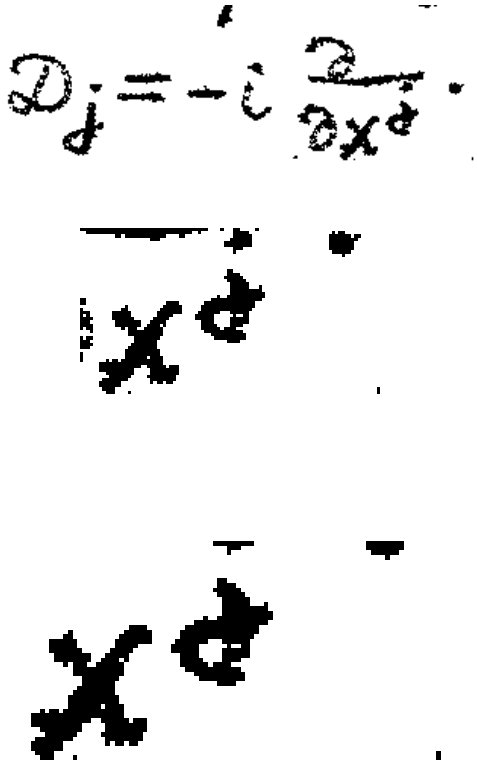
$$A = \sum_{|\alpha|=m} a_\alpha(x) D^\alpha. \qquad D^\alpha = D_1^{\alpha_1} \cdots D_n^{\alpha_n}$$



Original source details: GELFAND-P000125

**Diagram and typed description.** The summation condition is $|\alpha| = m$. To the right of $D^\alpha$ are factors with subscripts $1, n$ and superscripts $\alpha_1, \alpha_n$.

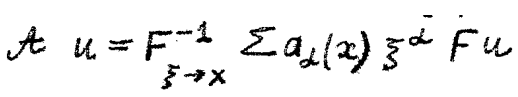
$$D_j = -i \frac{\partial}{\partial x^{\alpha[?]}}.$$



Original source details: GELFAND-P000125

**Uncertain reading.** The superscript of $x$ in the denominator has a rounded counter and a hooked upper stroke: $\alpha$- / a- / i- / j-like. The exact sign is undetermined.

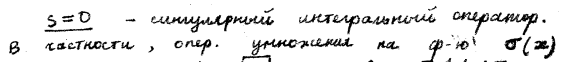
$$Au = F^{-1}_{\xi \to x} \sum a_\alpha(x) \xi^\alpha F u$$



Original source details: GELFAND-P000125

**Diagram and typed description.** Beneath $F^{-1}$, the direction $\xi \to x$. In the sum, $a_\alpha(x)$ and $\xi^\alpha$, followed by $Fu$.

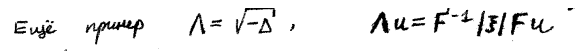
$\underline{s = 0}$ — a singular integral operator. In particular, the oper. of multiplication by the fn. $\sigma(x)$.



Original source details: GELFAND-P000125

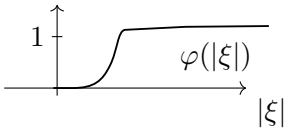
**Diagram and typed description.** Only $s = 0$ is underlined. The second line says multiplication by the fn. $\sigma(x)$.

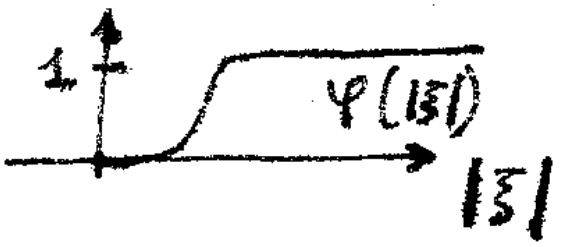
Another example $\Lambda = \sqrt{-\Delta}, \qquad \Lambda u = F^{-1}|\xi| F u.$



Original source details: GELFAND-P000125

**Diagram and typed description.** Capital $\Lambda$ appears twice. Under the root are a minus sign and $\Delta$; to the right, the modulus $|\xi|$.



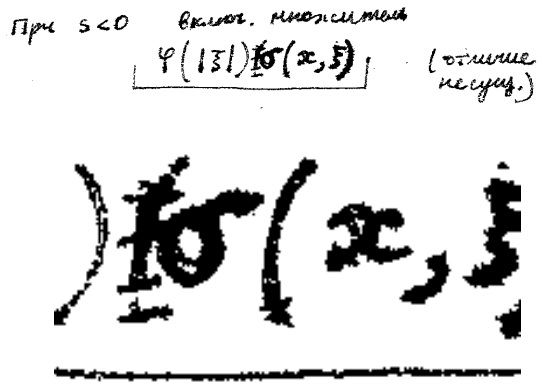
Original source details: GELFAND-P000125

**Diagram and typed description.** Two axes with arrows; the horizontal one is labelled $|\xi|$. A mark $1$ on the vertical axis. The curve runs along zero, rises and becomes almost horizontal; the label is $\varphi(|\xi|)$. The right end of the curve extends beyond the horizontal arrow.

For $s < 0$ include the factor [Before $\sigma$, a K-like sign is crossed out.]

$$\underline{\varphi(|\xi|)\sigma(x,\xi)}$$

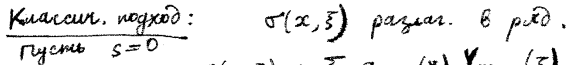
(the difference is inessential)



Original source details: GELFAND-P000125

**Uncertain reading.** Dense overlapping strokes at $\sigma$, including a K-like sign; the exact reading of the cancelled sign is undetermined. Beneath the product is a lower line with short vertical edges. To the right, a separate parenthesis: the difference is inessential.

Classical approach: expand $\sigma(x,\xi)$ in a series. Let $s = 0$.

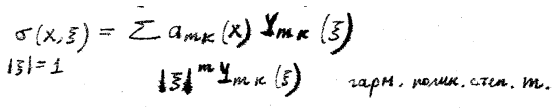


Original source details: GELFAND-P000125

**Diagram and typed description.** Classical approach: is underlined. To the right, expand $\sigma(x,\xi)$ in a series; beneath it, Let $s = 0$.

$$\sigma(x,\xi) = \sum a_{mk}(x) Y_{mk}(\xi), \qquad |\xi| = 1$$

$|\xi|^m Y_{mk}(\xi)$ — a harmonic polyn. of degree $m$.

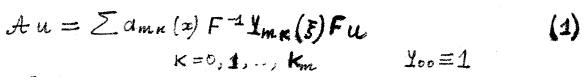


Original source details: GELFAND-P000125

**Diagram and typed description.** Both occurrences of $Y$ have subscript $mk$. Beneath the first formula is $|\xi| = 1$; the next line has $|\xi|^m Y_{mk}(\xi)$ and harmonic polyn. of degree $m$.

$$Au = \sum a_{mk}(x) F^{-1} Y_{mk}(\xi) Fu \qquad (1)$$

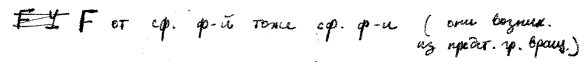
$k = 0, 1, \ldots, k_m$, $Y_{00} \equiv 1$.



Original source details: GELFAND-P000125

**Diagram and typed description.** The formula has number (1) at the right. Beneath it, the range $k = 0, 1, \ldots, k_m$ and a separate $Y_{00} \equiv 1$.

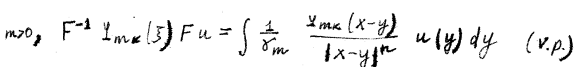
[Crossed out: $FY$[?].] $F$ of spherical fns. also gives spherical fns. (they arose from reps. of the rotation grp.)

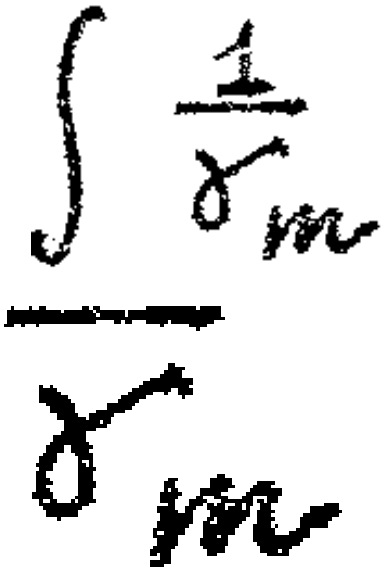


Original source details: GELFAND-P000125

**Uncertain reading.** Before the active $F$, F- and Y-like signs are crossed out. The parenthesis they arose from reps. of the rotation grp. belongs to the following phrase.

$$m > 0, \qquad F^{-1} Y_{mk}(\xi) Fu = \int \frac{1}{\gamma_m[?]} \frac{Y_{mk}(x-y)}{|x-y|^n} u(y)\, dy \qquad (\text{v.p.})$$





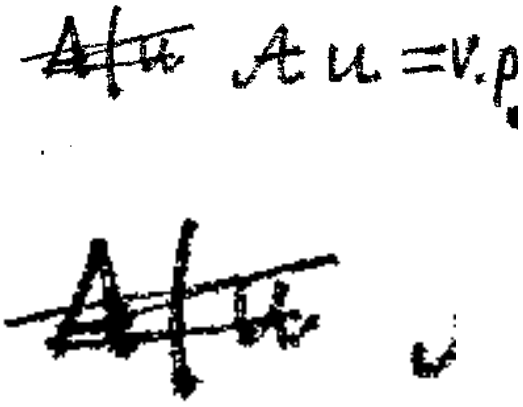
Original source details: GELFAND-P000125

**Uncertain reading.** In the denominator of the separate fraction $1/(\gamma_m[?])$, a looped Greek sign with an upper crossing stroke; candidates $\gamma_m$ and $\delta_m$. Subscript $m$ is visible.

[Crossed out: $Au$[?].]

$$Au = \text{v.p.} \int K(x, x-y)u(y)\,dy$$

(the calculations are formal)



**Uncertain reading.** To the left of the active $Au$, an earlier Au-like beginning is crossed out with several intersecting strokes, including a tall vertical one.

Original source details: GELFAND-P000125

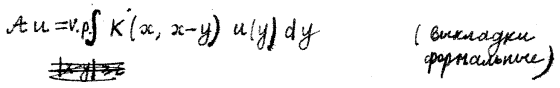


**Diagram and typed description.** The active $Au$ is to the right of the cancelled beginning. The integrand is $K(x, x-y)u(y)\,dy$, preceded by v.p.; to the right, the parenthesis the calculations are formal.

Original source details: GELFAND-P000125

3rd def. Consider the fn.

[Crossed out: $x - y$[?], parentheses and a z-like sign.] $K(x, z)$ pos. homog. in $z$ of degree $-n$.

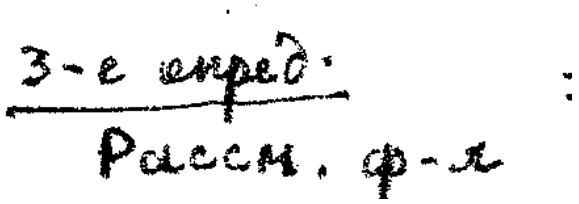
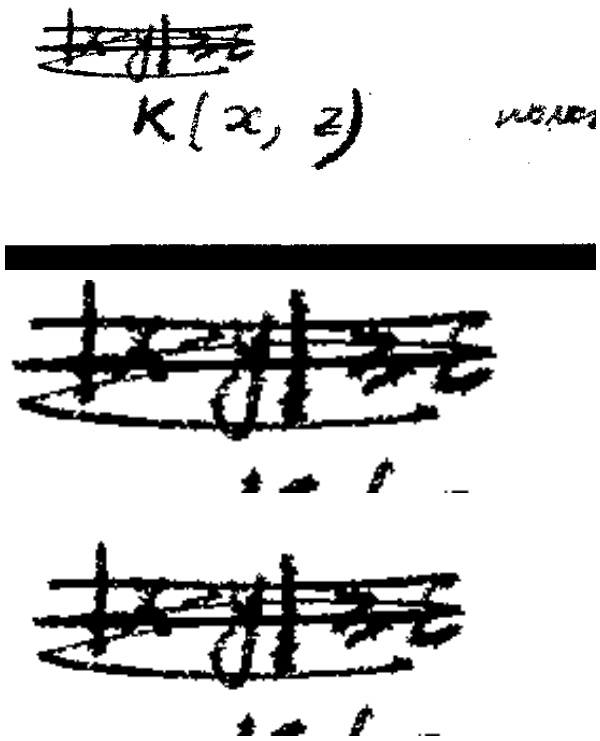


**Diagram and typed description.** A separate line beneath 3rd def. Then Consider the fn.

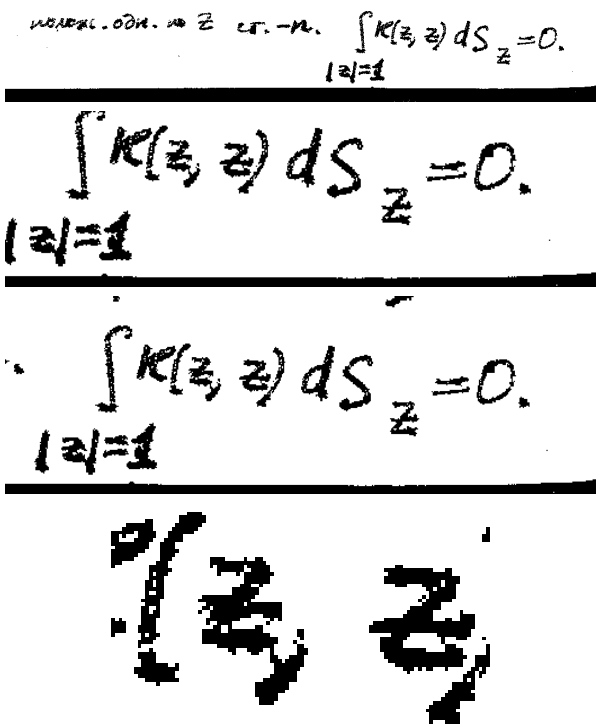
Original source details: GELFAND-P000125



**Uncertain reading.** Above the active $K(x, z)$, densely cancelled signs: an x-y-like portion, parentheses and a z-like sign at the right. The order and complete reading of the cancelled argument are undetermined.

Original source details: GELFAND-P000125

$$\int_{|z|=1} K(z[?], z)\, dS_z = 0.$$



**Uncertain reading.** The end says degree $-n$. Beneath the integral is $|z| = 1$, and $dS$ has subscript $z$. The first argument of $K$ is a z- / $\xi$- / x-like sign with dark horizontal strokes; its exact reading is undetermined. The second argument is z-like.

Original source details: GELFAND-P000125



**Diagram and typed description.** The frame around 80 has left and bottom sides.

Original source details: GELFAND-P000126

Let it stabilize as $x \to \infty$.

$$\sigma(x, \xi) = \sigma(\infty, \xi) + \sigma'(x, \xi)$$

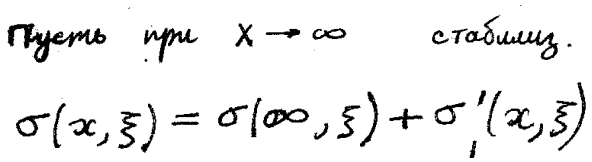
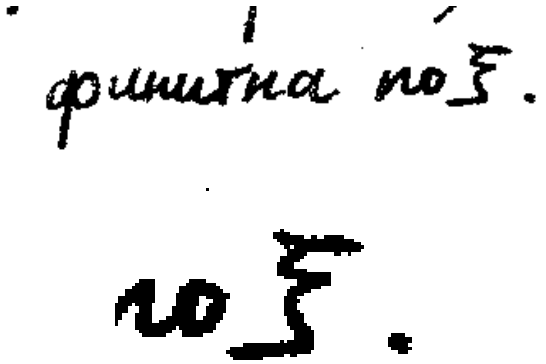


**Diagram and typed description.** The first line has $x \to \infty$. In the decomposition, the last $\sigma$ has a prime; the first summand has argument $\infty$.

Original source details: GELFAND-P000126

$\sigma'$ is compactly supported in $\xi$.



**Uncertain reading.** After compactly supported in, a visible $\xi$-shaped sign with upper and lower horizontal strokes.

Original source details: GELFAND-P000126

$$\sigma(x, D) = \sigma(\infty, D) + \sigma'(x, D)$$

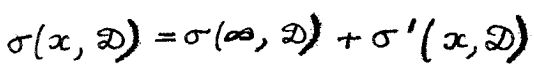
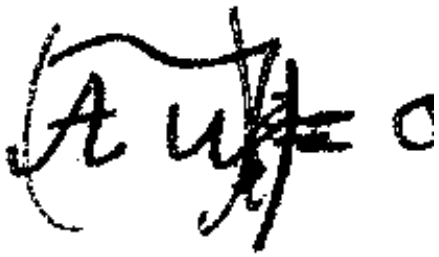
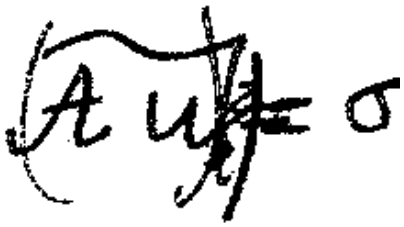


**Diagram and typed description.** The second argument is $D$ in all three places. The last $\sigma$ has a prime and is preceded by a plus sign.

Original source details: GELFAND-P000126

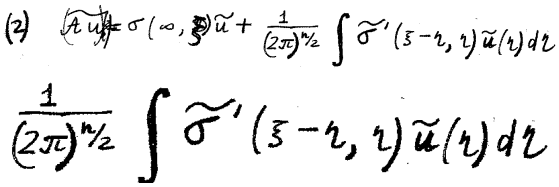
$$\widetilde{Au}[?] = \sigma(\infty, \xi[?])\widetilde{u} + \frac{1}{(2\pi)^{n/2}} \int \widetilde{\sigma}'(\xi - \eta, \eta)\widetilde{u}(\eta)\, d\eta \qquad (2)$$





Original source details: GELFAND-P000126

**Uncertain reading.** A wavy line above Au; surrounding arcs at the left and below. The right portion at u and the equality sign is densely overwritten; the exact reading of the corrected strokes is undetermined.



Original source details: GELFAND-P000126

**Uncertain reading.** The second argument in $\sigma(\infty, \xi[?])$ is overwritten and $\xi$-like. The denominator is $(2\pi)^{n/2}$; the integrand is $\widetilde{\sigma}'(\xi - \eta, \eta)\widetilde{u}(\eta)\, d\eta$. Number $(2)$ is at the left.

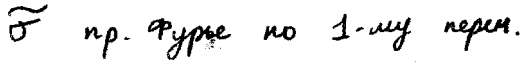
Cohn and Nirenberg



Original source details: GELFAND-P000126

**Diagram and typed description.** In the side bracket, the Latin names Cohn and Nirenberg, with and between them. The left line is almost straight and vertical, and the right line is curved.

$\widetilde{\sigma}$ is the Fourier transf. in the 1st variable.

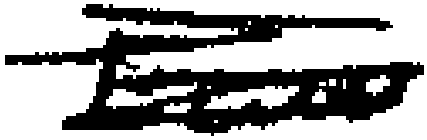


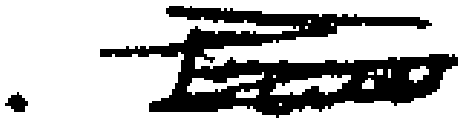
Original source details: GELFAND-P000126

**Diagram and typed description.** A wavy line above $\sigma$. The end says in the 1st variable, with digit $1$.

To what extent are they equiv.? If the symbols are very good. [Crossed out: Was[?].]

Насколько они эквиван.? Если симв. очень хорошие. Есть





**Uncertain reading.** After good., a short cancelled word resembling Was; its middle strokes are covered by intersecting lines.

Original source details: GELFAND-P000126

In the general case the defs. are inequivalent (lead to different classes of oper.)

В общем случае опр. неэквив. (прив. к разн. кл. опер.)

Original source details: GELFAND-P000126

**Diagram and typed description.** After inequivalent, a separate parenthesis: lead to different classes of oper.

In which spaces the oper. is bounded.

$$H^\ell = W_2^{(\ell)}, \qquad \|u\|_\ell^2 = \int |\widetilde{u}|^2 (1+|\xi|^2)^\ell \, d\xi$$

В каких пр-вах опер. огран.

$H^\ell = W_2^{(\ell)}$ $\|u\|_\ell^2 = \int |\widetilde{u}|^2 (1+|\xi|^2)^\ell d\xi$

Original source details: GELFAND-P000126

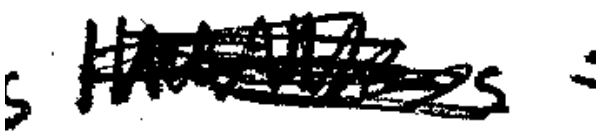
**Diagram and typed description.** The superscript of $H$ is $\ell$; $W$ has subscript $2$ and superscript $(\ell)$. Both the norm and weight have $\ell$; $|\widetilde{u}|$ and $|\xi|$ are squared.

$$\|Au\|_{\ell-s} \quad \text{[Crossed out: } \|[?]\|_s\text{]} \quad \leq C\|u\|_\ell$$

for a smooth fn. $\sigma$, expansion in a series is rapid.

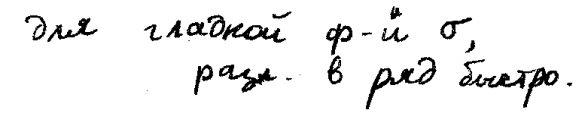
$\|Au\|_{\ell-s}$ [illegible] $\leq C\|u\|_\ell$





**Uncertain reading.** Between the active left norm and $\leq C\|u\|_\ell$, a densely crossed-out term. Norm-like vertical lines and an s-like sign at the lower right are visible; the interior reading is undetermined.

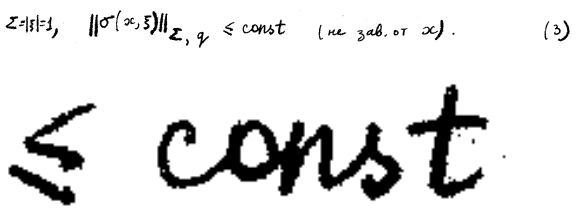
Original source details: GELFAND-P000126




Original source details: GELFAND-P000126

**Diagram and typed description.** Two lines at the right: for a smooth fn. $\sigma$, then expansion in a series is rapid.

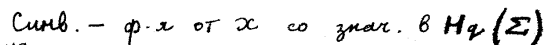
$$\Sigma : |\xi| = 1, \qquad \|\sigma(x,\xi)\|_{\Sigma,q} \leq \text{const} \quad (\text{indep. of } x). \qquad (3)$$




Original source details: GELFAND-P000126

**Diagram and typed description.** The norm has subscripts $\Sigma, q$. The sign before const has a lower line; at the right, the parenthesis indep. of $x$. Number (3) is at the right.

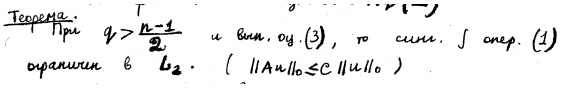
The symbol is a fn. of $x$ with values in $H_q(\Sigma)$.




Original source details: GELFAND-P000126

**Diagram and typed description.** The subscript of $H$ is $q$, with $\Sigma$ in parentheses.

Theorem. For $q > \dfrac{n-1}{2}$ and est. (3) holding, the sing. $\int$ oper. (1) is bounded in $L_2$. ($\|Au\|_0 \leq C\|u\|_0$)



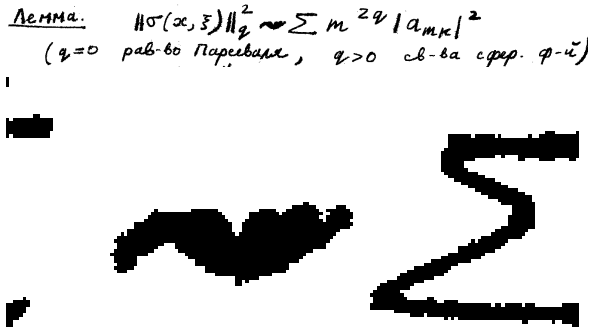

Original source details: GELFAND-P000126

**Diagram and typed description.** Theorem. is underlined. The condition is $q > (n-1)/2$, with references (3) and (1); the end has $L_2$ and a separate parenthesis with norms having subscript 0.

Lemma.

$$\|\sigma(x,\xi)\|_q^2 \sim \sum m^{2q} |a_{mk}|^2$$

($q = 0$ Parseval's equality, $q > 0$ properties of spherical fns.)

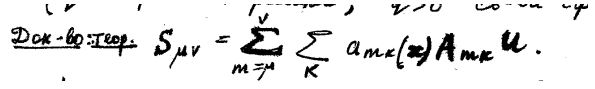



Original source details: GELFAND-P000126

**Diagram and typed description.** Lemma. is underlined. A dense wavy sign between the norm and sum; the exponent of $m$ is $2q$, and $a$ has subscripts $mk$. Beneath it, a separate parenthesis with $q = 0$ and $q > 0$.

Proof of the thm.

$$S_{\mu\nu} = \sum_{m=\mu}^{\nu} \sum_{k} a_{mk}(x) A_{mk} u.$$



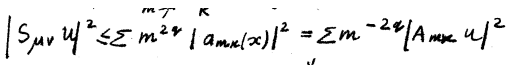

Original source details: GELFAND-P000126

**Diagram and typed description.** Proof of the thm. is underlined. At the left, $S_{\mu\nu}$; at the right, the first sum has limits $m = \mu$ and $\nu$, the second has subscript $k$, and the end is $A_{mk}u$.

$$|S_{\mu\nu}u|^2 \le \sum m^{2q}|a_{mk}(x)|^2 = \sum m^{-2q}|A_{mk}u|^2$$



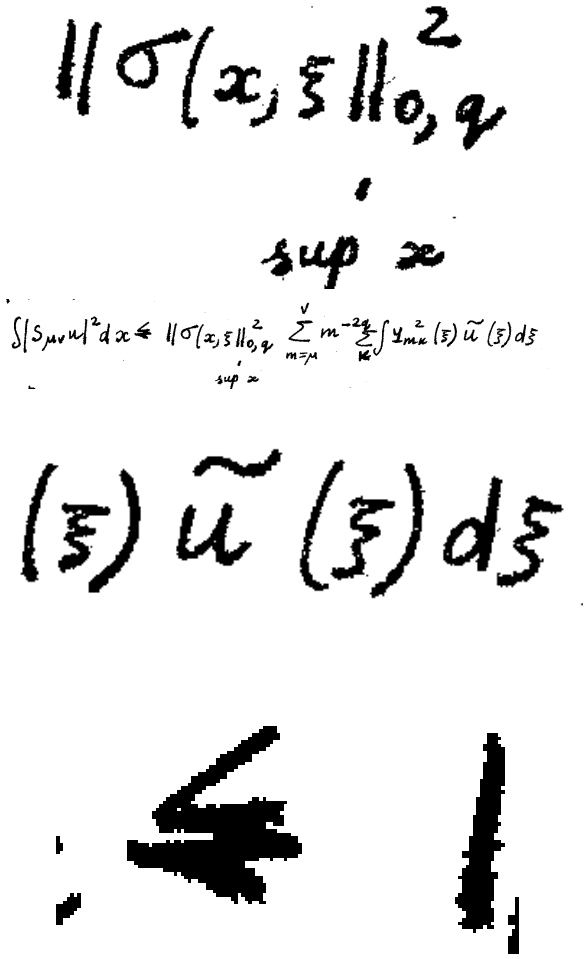
**Uncertain reading.** Two horizontal lines between the sums, an equality sign. The exponent of the first $m$ is $2q$, and that of the second is $-2q$.



Original source details: GELFAND-P000126

$$\int |S_{\mu\nu}u|^2\,dx \le [?]\|\sigma(x,\xi\|^2_{0,q}$$
$$\sup_x$$
$$\sum_{m=\mu}^{\nu} m^{-2q} \sum_k \int Y^2_{mk}(\xi)\widetilde{u}(\xi)\,d\xi$$

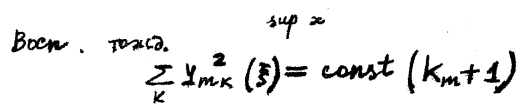
**Uncertain reading.** After $dx$, an overwritten $\le$- / $\Leftarrow$-like sign. No closing argument parenthesis is visible after $\xi$ in the norm; sup x is written separately beneath it. $Y_{mk}$ is squared; it is followed only by $\widetilde{u}(\xi)$, without a visible modulus or square.



Original source details: GELFAND-P000126

Use the identity

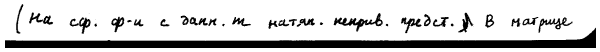
$$\sum_k Y_{mk}^2(\xi) = \text{const}\,(k_m + 1)$$



Original source details: GELFAND-P000126

**Diagram and typed description.** Use the identity precedes the formula. The sum has subscript $k$; at the right, the parenthesis $(k_m + 1)$.

(on spherical fns. with a given $m$, a nontriv. irred. rep. $\lambda$ In the matrix

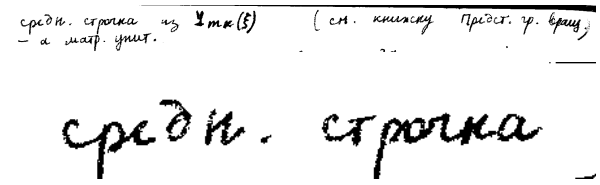


Original source details: GELFAND-P000126

**Diagram and typed description.** The bottom line ends with $\lambda$ In the matrix; there is an opening parenthesis at the left, with no closing parenthesis on this image.

GELFAND-P000127

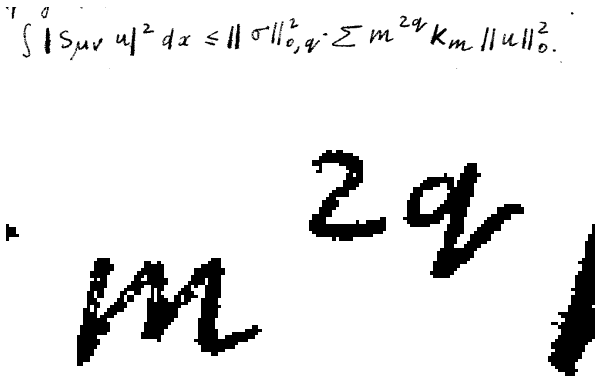
middle row of $Y_{mk}(\xi)$ (see the book Reps. of the rotation grp.) — and the matrix is unitary.



Original source details: GELFAND-P000127

**Diagram and typed description.** The first word is middle; then row of $Y_{mk}(\xi)$. At the right, the parenthesis see the book Reps. of the rotation grp.; beneath it, a separate line — and the matrix is unitary.

$$\int |S_{\mu\nu} u|^2 \, dx \le \|\sigma\|_{0,q}^2 \cdot \sum m^{2q} k_m \|u\|_0^2.$$

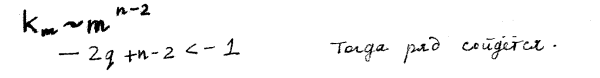


Original source details: GELFAND-P000127

**Diagram and typed description.** The exponent of $m$ is $2q$, with no visible minus before $2$. A separate multiplication dot follows the norm with subscripts $0, q$.

$$k_m \sim m^{n-2}, \qquad -2q + n - 2 < -1$$

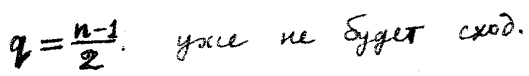
Then the series will converge.



Original source details: GELFAND-P000127

**Diagram and typed description.** The first formula has $k_m$ and exponent $n - 2$. The next line begins with a minus before $2q$, then $+n - 2 < -1$. At the right, Then the series will converge.

$q = \frac{n-1}{2}$. there will no longer be conv.



Original source details: GELFAND-P000127

**Diagram and typed description.** A period follows the fraction $(n-1)/2$; then there will no longer be conv.

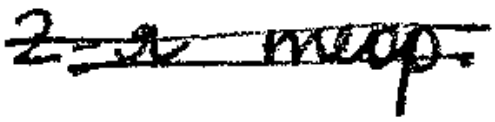
No smoothness in $\xi$ is needed.



Original source details: GELFAND-P000127

**Diagram and typed description.** After in, a dense $\xi$-shaped sign; the phrase says no smoothness is needed.

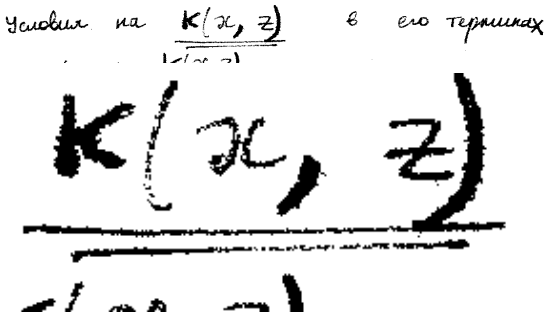
[Crossed out: 2nd thm.]



Original source details: GELFAND-P000127

**Diagram and typed description.** 2nd thm. is crossed out with two long lines.

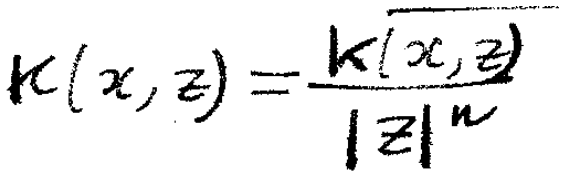
Conditions on $K(x,z)$ in its terms



Original source details: GELFAND-P000127

**Diagram and typed description.** Two separate lines beneath $K(x,z)$. Conditions on is at the left, and in its terms at the right.

$$K(x,z) = \frac{k(x,z)}{|z|^n}$$



Original source details: GELFAND-P000127

**Diagram and typed description.** At the left, $K(x,z)$; at the right, numerator $k(x,z)$ and denominator $|z|^n$.

$$\|k(X,z)\|_q \le \text{const} \qquad q > -\frac{1}{2}$$
$$(\text{Norm on } |z|=1)$$

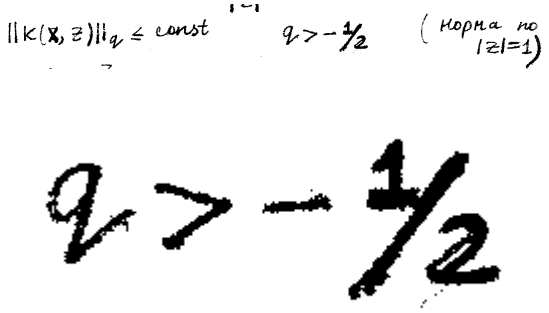


**Diagram and typed description.** The first argument of $k$ is X-shaped. A short horizontal minus precedes $1/2$; a separate parenthesis says Norm on $|z| = 1$.

Original source details: GELFAND-P000127

Example ($n = 2$)

$$K(z) = \frac{z_1}{(z_1^2 + z_2^2)^{3/2}}.$$

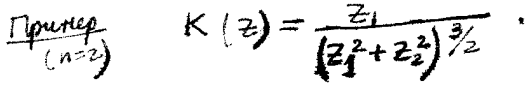


**Diagram and typed description.** Example is underlined, with the parenthesis $n = 2$ beneath it. The numerator is $z_1$; the denominator is the sum of two squares to the power $3/2$.

Original source details: GELFAND-P000127

$$\widetilde{Au}(\xi) = \sigma(\infty,\xi)\widetilde{u}(\xi) + \frac{1}{(2\pi)^{n/2}} \int \widetilde{\sigma}'(\xi-\eta,\eta)\widetilde{u}(\eta)\,d\eta$$

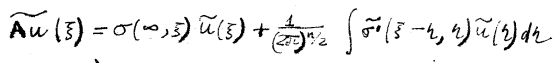


**Diagram and typed description.** Wavy lines above Au, u and $\sigma'$. The denominator is $(2\pi)^{n/2}$. The arguments of $\widetilde{\sigma}'$ are $\xi - \eta, \eta$.

Original source details: GELFAND-P000127

$\sigma'(x,\xi)$ is smooth in $X$ through order $n+1$.

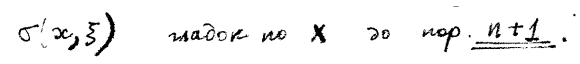


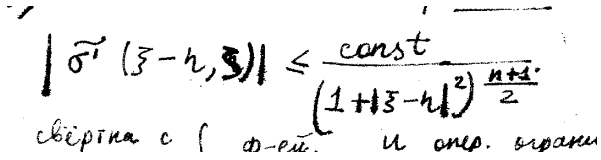
**Diagram and typed description.** After in, a separate X-shaped sign. Two lines beneath $n+1$.

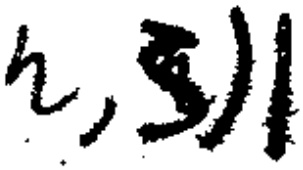
Original source details: GELFAND-P000127

$$|\widetilde{\sigma}'(\xi-\eta,\xi[?])| \le \frac{\text{const}}{(1+|\xi-\eta|^2)^{(n+1)/2}}.$$



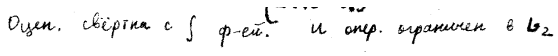
**Uncertain reading.** The last argument at the left is densely overwritten and $\xi$- / $\eta$-like. The denominator at the right is $1+|\xi-\eta|^2$ to the power $(n+1)/2$. The exact reading of the overwritten argument is undetermined.



Original source details: GELFAND-P000127

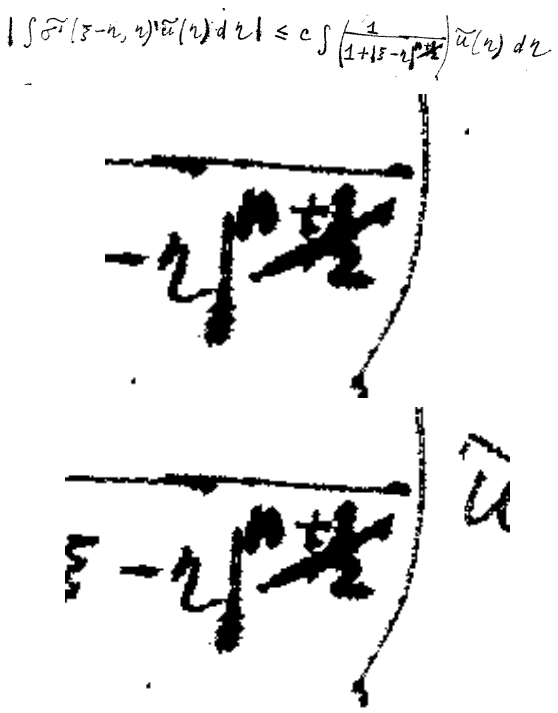
Estimate: convolution with an $\int$ fn. And the oper. is bounded in $L_2$.




Original source details: GELFAND-P000127

**Diagram and typed description.** After with, a separate integral-like sign and fn. The second phrase is And the oper. is bounded in $L_2$.

$$\left| \int \widetilde{\sigma}'(\xi-\eta,\eta)\widetilde{u}(\eta)\,d\eta \right| \le c \int \left( \frac{1}{1+|\xi-\eta|^{\text{[illegible]}}} \right) \widetilde{u}(\eta)\,d\eta$$



**Uncertain reading.** The power of $|\xi-\eta|$ in the right denominator is densely corrected: $n+1$ / $(n+1)/2$; its exact reading is undetermined. Large parentheses surround the fraction; they are followed by $\widetilde{u}(\eta)$ without a visible modulus.

Original source details: GELFAND-P000127

1. Commutation formula.

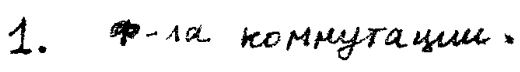


**Diagram and typed description.** Digit 1 with a period; then Commutation formula.

Original source details: GELFAND-P000127

2. $A \cdot B - \underset{\sigma_1\sigma_2}{A \circ B}$ — lower order.

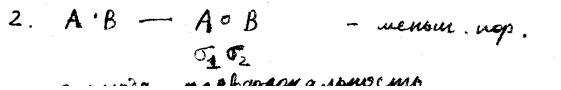


**Diagram and typed description.** A dot between the first $A$ and $B$, and a circle in the second product. Beneath $A \circ B$, a separate $\sigma_1\sigma_2$; at the right, — lower order.

Original source details: GELFAND-P000127

Hence pseudolocality.

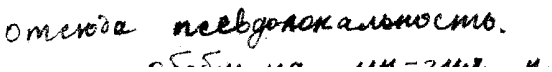
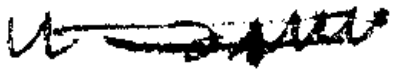


**Diagram and typed description.** A separate phrase Hence pseudolocality., with a period.

Original source details: GELFAND-P000127

[End of the expression crossed out: $u \longrightarrow$ [?].]
Generalization to manifolds and elliptic theory.



**Uncertain reading.** A u-like letter at the left, a long line with an arrow to the right and a densely cancelled endpoint. Looped and vertical strokes at the end; the complete reading is undetermined.



Original source details: GELFAND-P000127

обобщ-на ин-зия и эллипт. теория.

Original source details: GELFAND-P000127

**Diagram and typed description.** The concluding phrase is Generalization to manifolds and elliptic theory. Elliptic is written with dense strokes.

GELFAND-P000128

Let us start again.
Sing. $\int$ operator: (of order 0)

Начнём сначала.
Сингл. $\int$ оператор:
(0-го пор.)

Original source details: GELFAND-P000128

**Diagram and typed description.** Let us start again. is underlined. Below, Sing. and a separate integral-like sign; of order 0 in parentheses.

$$(1)\quad \widetilde{Au}(\xi) = F_{x\to\xi}\sigma(x,\xi)u(x)$$
$$(2)\quad \widetilde{Au}(\xi) = \sigma(x,\xi)\widetilde{u}(\xi) +$$

(C. and N.;) (early works)

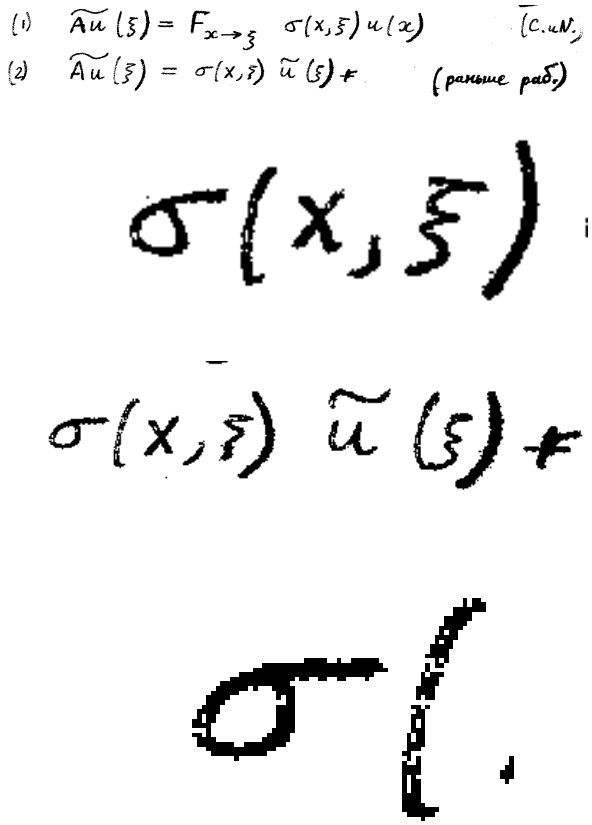
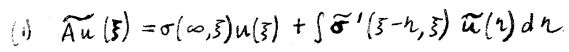


**Diagram and typed description.** In both initial definitions, $\sigma$ has no detached prime. In the second line, a plus follows $\widetilde{u}(\xi)$; no continuation is written. At the right, separate parentheses C. and N.; and early works.

Original source details: GELFAND-P000128

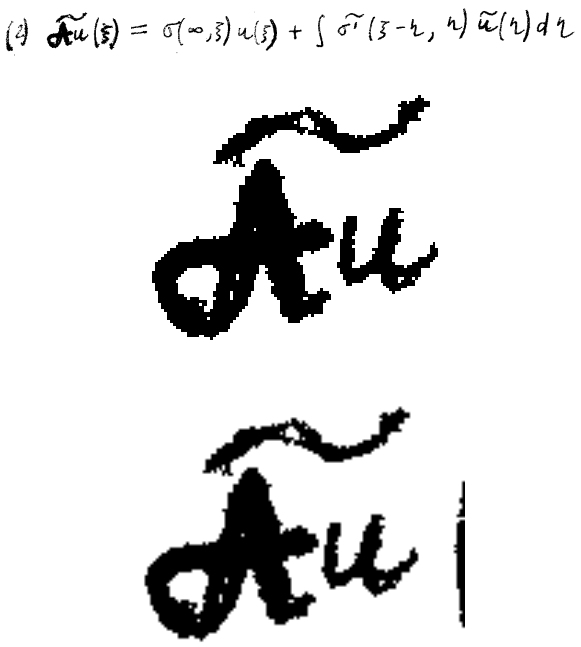
$$(1)\quad \widetilde{Au}(\xi) = \sigma(\infty,\xi)u(\xi) + \int \widetilde{\sigma}'(\xi-\eta,\xi)\widetilde{u}(\eta)\,d\eta$$



Original source details: GELFAND-P000128

**Diagram and typed description.** The first term contains $u(\xi)$ with no tilde. In the integral, $\sigma$ has a detached prime and a tilde; the second argument is $\xi$.

$$(2) \quad \widetilde{\mathcal{A}[?]u}(\xi) = \sigma(\infty, \xi)u(\xi) + \int \widetilde{\sigma}'(\xi - \eta, \eta)\widetilde{u}(\eta)\, d\eta$$



**Uncertain reading.** The second operator sign is heavily overwritten: $\mathcal{A}$- / B-like, with a tilde above it and $u$. The exact overwritten letter is unresolved. The first term again has $u(\xi)$ without a tilde.

Original source details: GELFAND-P000128

But they differ by an operator of lower order $(-1)$.

$$|\widetilde{Au} - \mathcal{A}u| = \int [\widetilde{\sigma}'(\xi - \eta, \xi) - \widetilde{\sigma}'(\xi - \eta, \eta)]\widetilde{u}(\eta)\, d\eta$$

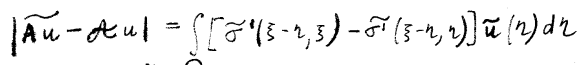
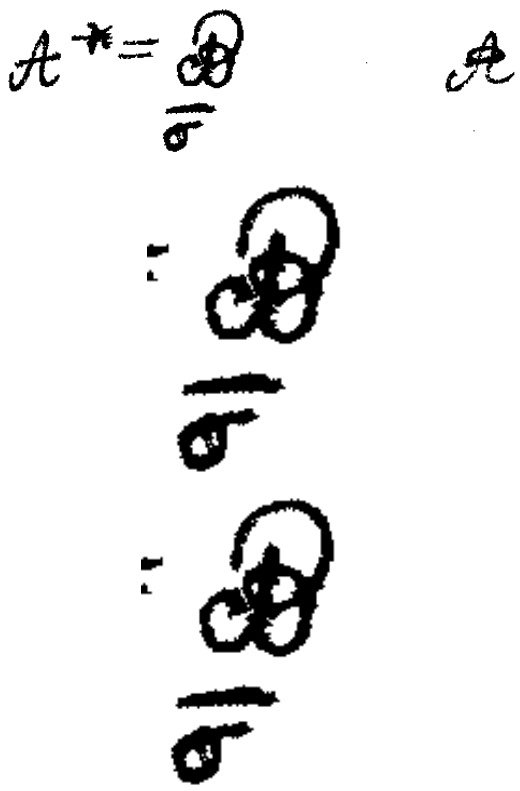


**Diagram and typed description.** At the left, vertical bars and a tilde only above the first Au; the second operator is calligraphic. At the right, an integral without an outer modulus, the difference in square brackets.

Original source details: GELFAND-P000128

$$\mathcal{A}^* = B[?] \qquad \mathcal{A}$$
$\overline{\sigma}$



**Uncertain reading.** After calligraphic $A^* =$, a heavily overwritten looped B-like sign. Beneath it, $\overline{\sigma}$. At the right, a separate calligraphic A-like letter; its connection to the equality is unresolved.

Original source details: GELFAND-P000128

(All this is done by Taylor's formula)

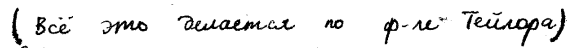
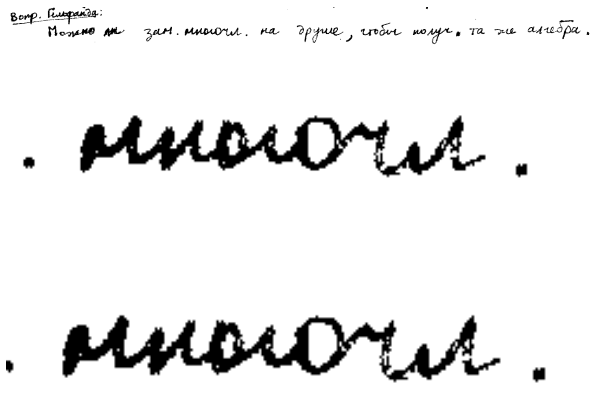


**Diagram and typed description.** The entire phrase is in parentheses; the source contraction for formula and Taylor's name.

Original source details: GELFAND-P000128

Gelfand's question:

Can polyn.[?] be replaced by others, so as to obtain the same algebra.



**Uncertain reading.** The source contraction after replaced: многочл. / многом.; the exact final part is unresolved. Gelfand's question: is underlined.

Original source details: GELFAND-P000128

The symbol is defined on the cotang. bundle.

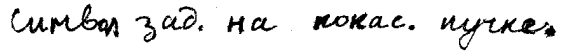
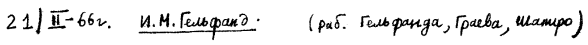


**Diagram and typed description.** A separate phrase: The symbol is defined on the cotang. bundle.

Original source details: GELFAND-P000128

21/II–66 I. M. Gelfand. (works of Gelfand, Graev, Shapiro)

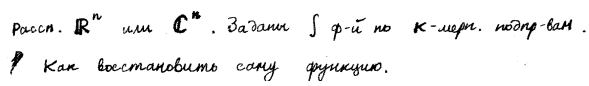


Original source details: GELFAND-P000128

**Diagram and typed description.** II has horizontal lines above and below. Date 21/II–66. The name I. M. Gelfand. is underlined; at the right, parentheses with three surnames.

Consider $R^n$ or $C^n$. The $\int$ of functions over $k$-dim. subspaces are given.

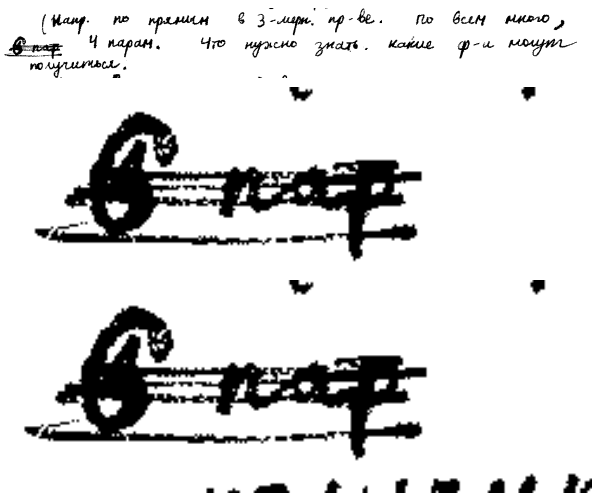
? How to recover the function itself.



Original source details: GELFAND-P000128

**Diagram and typed description.** At the left, a separate question mark. In the first line, $R^n$, $C^n$ and a separate integral-like sign before the contraction for functions.

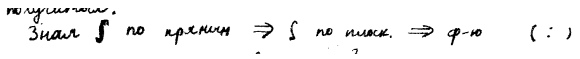
(E.g. over lines in 3-dim. space. Over all of them is too much, [Crossed out: 6 param.[?]] 4 param. What needs to be known. which functions can result.



Original source details: GELFAND-P000128

**Uncertain reading.** Before 4 param., 6 пар. / 6-пар. is crossed out; horizontal lines cross the contraction, and a separate hyphen is unresolved. The initial parenthesis in this passage has no visible closure.

Knowing $\int$ over lines $\Rightarrow$ $\int$ over planes $\Rightarrow$ the function (:)



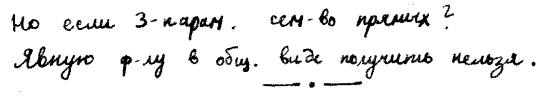
Original source details: GELFAND-P000128

**Diagram and typed description.** Two separate $\Rightarrow$ signs; after the function, parentheses containing a colon.

But if a 3-param. family of lines?

An explicit formula cannot be obtained in the general case.

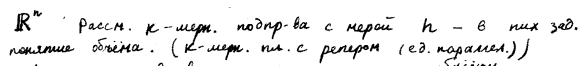
— . —



Original source details: GELFAND-P000128

**Diagram and typed description.** After the question, a separate sentence about an explicit formula. Below, two horizontal strokes with a dot between.

$R^n$ Consider $k$-dim. subspaces with measure $h$ — a notion of volume is given in them. ($k$-dim. plane with a frame (unit parallelep.))

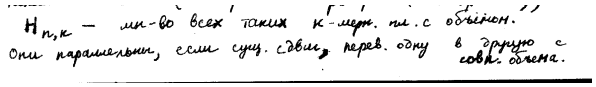


Original source details: GELFAND-P000128

**Diagram and typed description.** After measure, a separate h-like sign. At the end, two nested parentheses, frame and unit parallelep.

$H_{n,k}$ — the set of all such $k$-dim. planes with volume.

They are parallel if there exists a translation taking one into the other with volume preserved.

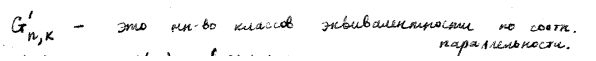


Original source details: GELFAND-P000128

**Diagram and typed description.** The subscript of $H$ is $n, k$. The final sentence ends with volume preserved.

GELFAND-P000129

$G'_{n,k}$ — this is the set of equivalence classes under the relation of parallelism.

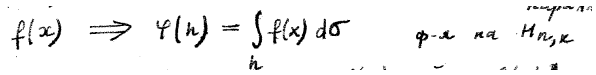


Original source details: GELFAND-P000129

**Diagram and typed description.** A detached prime above $G$, and subscripts $n, k$. The phrase ends with parallelism.

$$f(x) \Rightarrow \varphi(h) = \int_h f(x)\, d\sigma$$

a function on $H_{n,k}$.

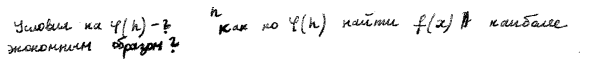


Original source details: GELFAND-P000129

**Diagram and typed description.** The integral has subscript $h$; at the right, $d\sigma$. A separate phrase: a function on $H_{n,k}$.

Conditions on $\varphi(h)$ — ?

how to find $f(x)$ from $\varphi(h)$ [Crossed-out sign: illegible] in the most economical way?



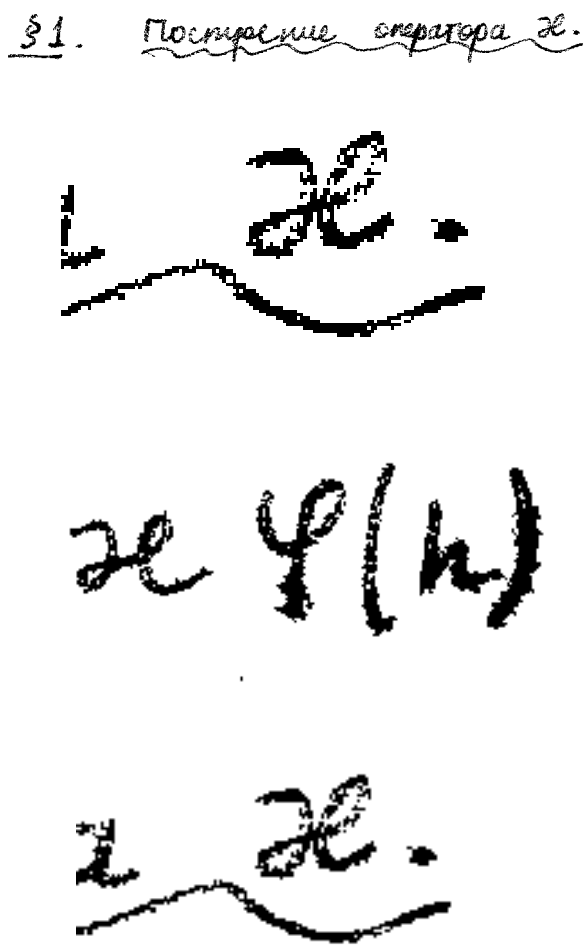




Original source details: GELFAND-P000129

**Uncertain reading.** After $f(x)$, a dense crossed-out B-like sign; its exact reading is unresolved. The active sentence continues with most.

§1. Construction of the operator $\mathcal{X}$[?].



**Uncertain reading.** The operator sign is X-like / $\chi$; the exact notation is unresolved. The same sign recurs in the concluding line. A straight underline beneath §1, a wavy one beneath the title.

Original source details: GELFAND-P000129

$S_\beta$. $\beta \in R^n$. Consider all $k$-dim. [Crossed out: subsp.] [Inserted above: hyperpl.], passing through $\beta$.

One parallel will be singled out.

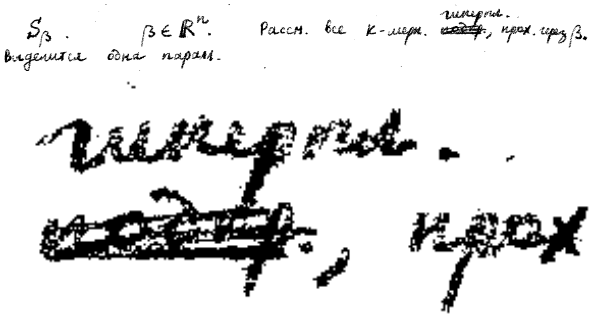


**Diagram and typed description.** Subsp. is crossed out, hyperpl. inserted above the line. The active line has $k$-dim., passing through $\beta$; below, One parallel will be singled out.

Original source details: GELFAND-P000129

$$S_\beta : G'_{n,k} \longrightarrow H_{n,k}$$

This is a global section.

consider $H_{n,k} \to G'_{n,k}$.

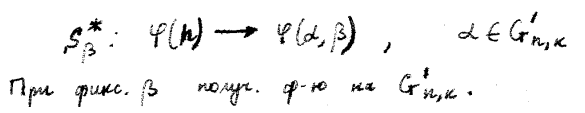


Original source details: GELFAND-P000129

**Diagram and typed description.** The subscript of $S$ is $\beta$. In the first map, the arrow goes from $G'_{n,k}$ to $H_{n,k}$; in the next line, the opposite direction.

$$S^*_\beta : \quad \varphi(h) \to \varphi(\alpha, \beta), \qquad \alpha \in G'_{n,k}$$

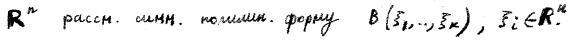
With $\beta$ fixed, we obtain a function on $G'_{n,k}$.



Original source details: GELFAND-P000129

**Diagram and typed description.** A superscript star on $S_\beta$. An arrow after $\varphi(h)$, then $\varphi(\alpha, \beta)$; at the right, $\alpha \in G'_{n,k}$.

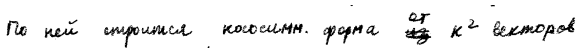
$R^n$ consider a symm. multilinear form $B(\xi_1, \ldots, \xi_k)$, $\xi_i \in R^n$.



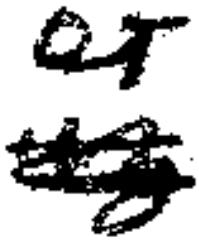
Original source details: GELFAND-P000129

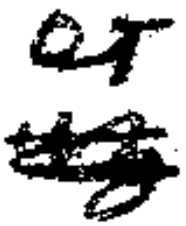
**Diagram and typed description.** Symm. multilinear form and $B(\xi_1, \ldots, \xi_k)$. The exponent on the last $R$ is $n$.

From it is constructed a skew-symm. form [Crossed out: from[?]] [Inserted above: of] $k^2$ vectors



**Uncertain reading.** Beneath the inserted of, a short heavily crossed-out word, resembling source из; its exact reading is unresolved. Then $k^2$ vectors.





Original source details: GELFAND-P000129

$$\Xi = \begin{Vmatrix} \xi_{11} & \cdots & \xi_{1k} \\ \xi_{21} & \cdots & \xi_{2k} \\ \\ \xi_{k1} & \cdots & \xi_{kk} \end{Vmatrix}.$$

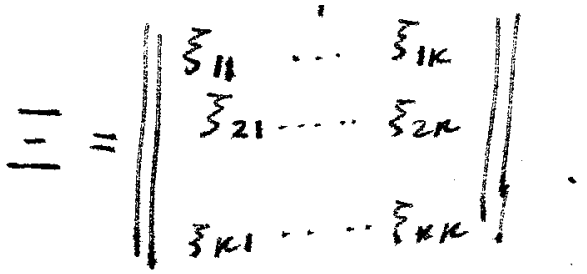
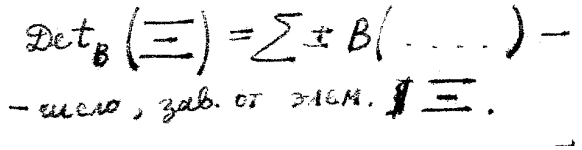


**Diagram and typed description.** Double vertical bars, rows with first indices $1$, $2$, $k$, and horizontal dots. There is a blank gap between the second and last rows; no vertical dots are visible.

Original source details: GELFAND-P000129

$$\mathrm{Det}_B(\Xi) = \sum \pm B(\ldots) —$$

— a number depending on the entries of [Crossed-out sign: illegible] $\Xi$.



**Uncertain reading.** Before the active $\Xi$, a short crossed-out vertical sign; its exact reading is unresolved. In the preceding line, $\sum \pm B(\ldots)$ and a separate stroke at the right.





Original source details: GELFAND-P000129

For vectors $\xi \in R^1$, $B = \xi_1 \cdots \xi_k$ this is the ordinary det.

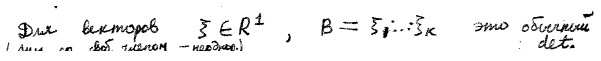
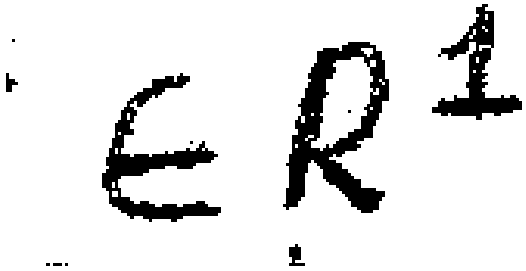
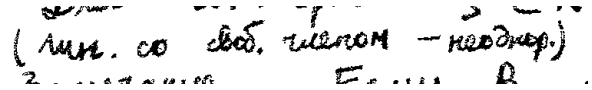


**Diagram and typed description.** The exponent on $R$ is a separate digit $1$. Then the product $\xi_1 \cdots \xi_k$ and ordinary det.



Original source details: GELFAND-P000129

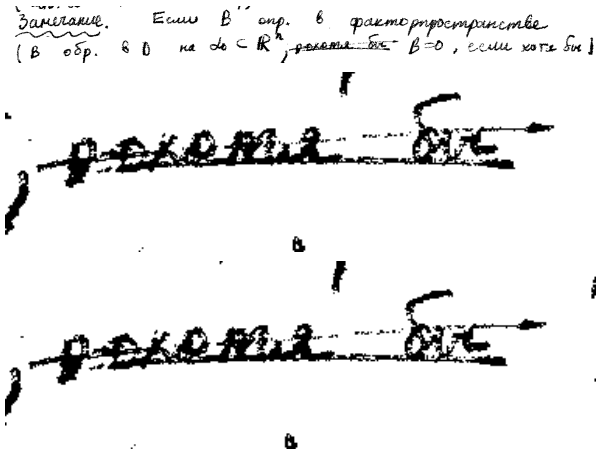
(lin. with a constant term — inhomog.)



Original source details: GELFAND-P000129

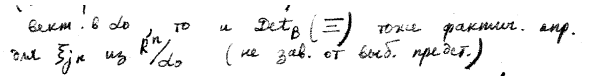
**Diagram and typed description.** A separate parenthesis: lin. with a constant term — inhomog.

Remark. If $B$ is defined on a quotient space ( $B$ vanishes on $\alpha_0 \subset R^n$, [Crossed out: would have to[?]] $B = 0$ if at least 1 vector is in $\alpha_0$, then $\mathrm{Det}_B(\Xi)$ too is in fact defined for $\xi_{jk}$ from $R^n/\alpha_0$ (does not depend on the choice of representative)



Original source details: GELFAND-P000129

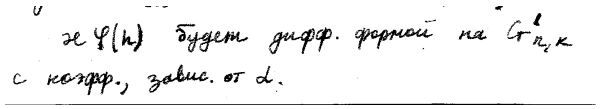
**Uncertain reading.** The source words resembling должны бы are crossed out; their exact reading is unresolved. The first word after the opening parenthesis is B-like, then vanishes on $\alpha_0 \subset R^n$. Remark. has a wavy underline.



Original source details: GELFAND-P000129

**Diagram and typed description.** After the condition $B = 0$ if at least 1 vector is in $\alpha_0$, the text continues through $R^n/\alpha_0$ and the parenthesis does not depend on the choice of representative. The initial parenthesis before $B$ has no separate visible closure.

$\mathcal{X}$[?]$\varphi(h)$ will be a differential form on $G'_{n,k}$ with coefficients depending on $\alpha$.

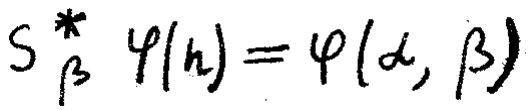


Original source details: GELFAND-P000129

**Diagram and typed description.** The operator sign before $\varphi(h)$ repeats the sign in the title. The last line reads with coefficients depending on $\alpha$.

GELFAND-P000130

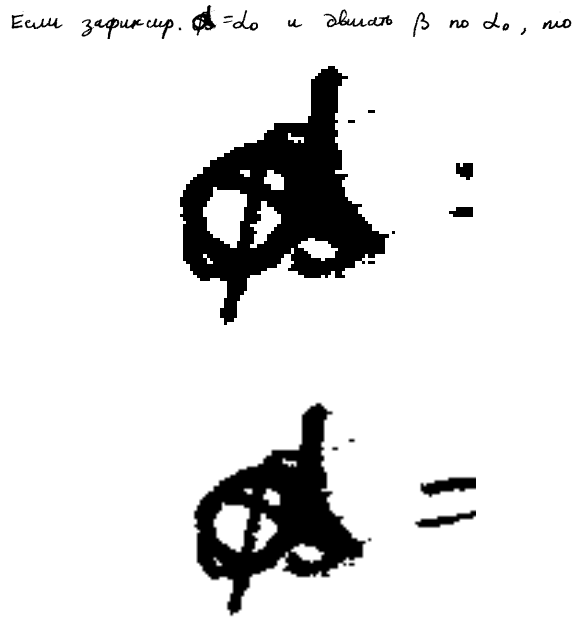
$$S^*_\beta \varphi(h) = \varphi(\alpha, \beta)$$



Original source details: GELFAND-P000130

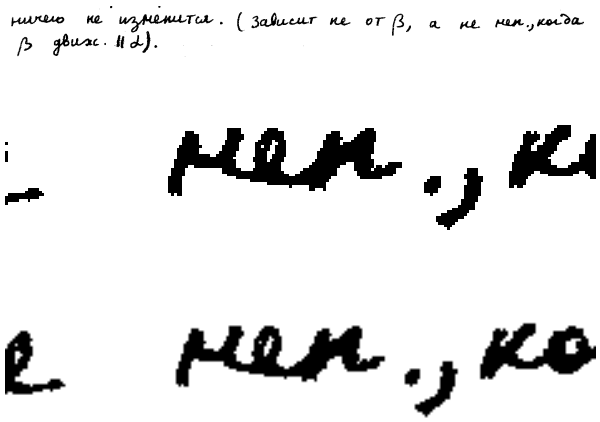
**Diagram and typed description.** Subscript $\beta$ and superscript star on $S$. At the right, $\varphi(\alpha, \beta)$.

If we fix $\alpha[?] = \alpha_0$ and move $\beta$ along $\alpha_0$, nothing changes. (Depends not on $\beta$, but not [нем./мен.?], when $\beta$ moves $\| \alpha$).



**Uncertain reading.** Before $= \alpha_0$, a heavily overwritten $\alpha$-like letter with loops and a vertical stroke; the exact reading is unresolved.

Original source details: GELFAND-P000130



**Uncertain reading.** After but not, the source contraction нем. / мен.; the exact reading and expansion are unresolved. The parenthesis begins: Depends not on $\beta$.

Original source details: GELFAND-P000130

Fix $\varphi(\alpha_0, \beta)$, consider $d^k_\beta \varphi(\alpha_0, \beta)$ — — a symm. $k$-linear form of $k$ differentials $d\beta$.

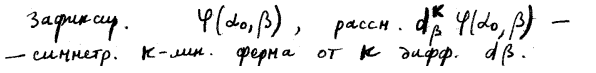


**Diagram and typed description.** Superscript $k$ and subscript $\beta$ on $d$. A separate dash after the expression, and another at the start of the next line; then $k$-linear and $k$ differentials $d\beta$.

Original source details: GELFAND-P000130

It is defined not from $\beta$, but from the quotient by $\alpha_0$. Defined on the quotient space.

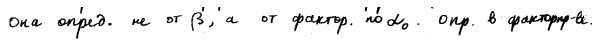
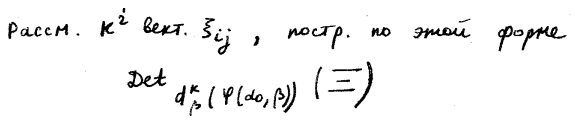


**Diagram and typed description.** The phrase not from $\beta$, but from the quotient by $\alpha_0$. Then Defined on the quotient space.

Original source details: GELFAND-P000130

Consider $k^2$ vectors $\xi_{ij}$, construct from this form

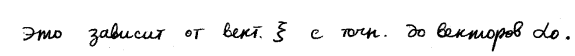
$$\mathrm{Det}_{d^k_\beta(\varphi(\alpha_0,\beta))}(\Xi)$$



Original source details: GELFAND-P000130

**Diagram and typed description.** The entire expression $d^k_\beta(\varphi(\alpha_0,\beta))$ is beneath Det; at the right, large parentheses containing $\Xi$. In the line above, $k^2$ vectors $\xi_{ij}$.

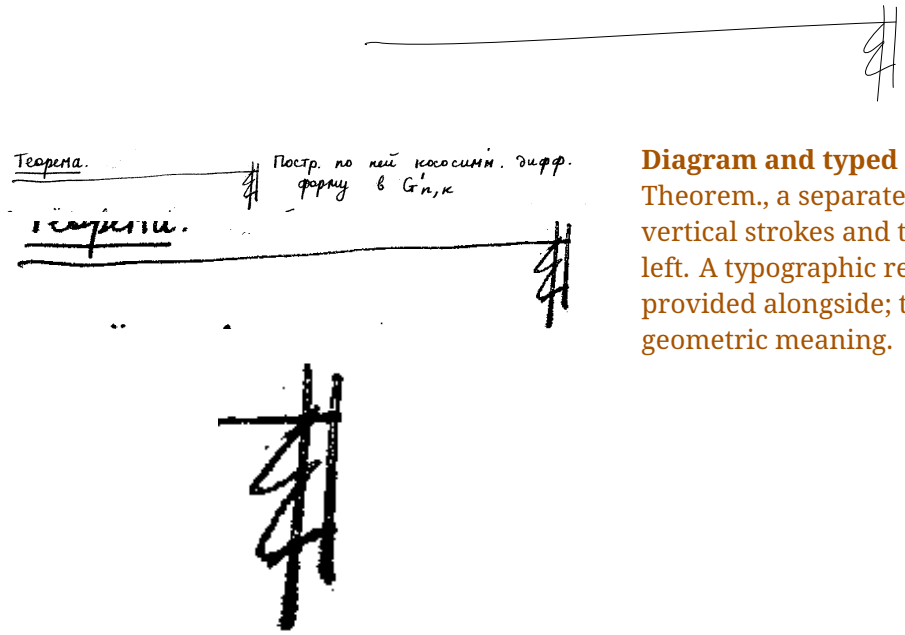
This depends on the vectors $\xi$ up to vectors of $\alpha_0$.



Original source details: GELFAND-P000130

**Diagram and typed description.** Up to vectors of $\alpha_0$ at the end of a separate line.

Theorem. Construct from it a skew-symm. differential form in $G'_{n,k}$.

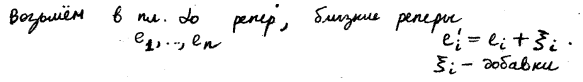


Original source details: GELFAND-P000130

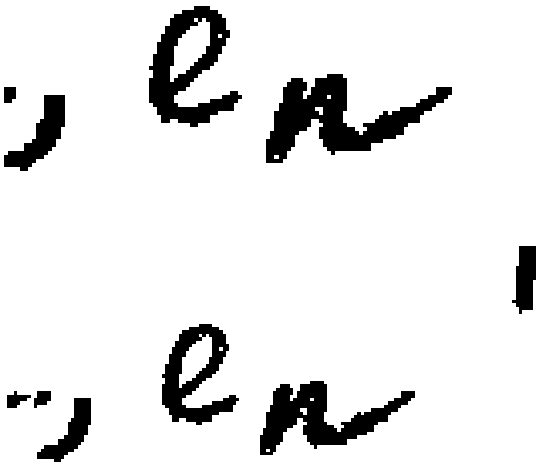
**Diagram and typed description.** Beneath Theorem., a separate long line to the right, two vertical strokes and two looped portions to their left. A typographic rendition of the strokes is provided alongside; there are no labels or stated geometric meaning.

Take a frame in the plane $\alpha_0$, nearby frames $e_1, \dots, e_{n[?]}$, $e'_i = e_i + \xi_i$.

$\xi_i$ — increments.



**Uncertain reading.** The last subscript on $e$ is n-like; its exact reading is unresolved. At the right, the formula $e'_i = e_i + \xi_i$, and beneath it, $\xi_i$ — increments.



Original source details: GELFAND-P000130

(planes up to small quantities of higher order)

They define the same plane $\Leftrightarrow \xi_i - \widetilde{\xi_i} \in \alpha_0$.

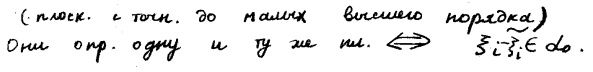


**Diagram and typed description.** After $\Leftrightarrow$, two $\xi$ with subscripts $i$ and a short minus between them; a tilde over the second. At the right, $\in \alpha_0$.



Original source details: GELFAND-P000130

$k$ nearby planes

$$e'_{ij} = e_i + \xi_{ij} \qquad j = 1, \dots, k$$

Consider as $\Xi$ a matrix defining $k$ nearby planes.

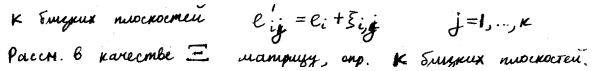
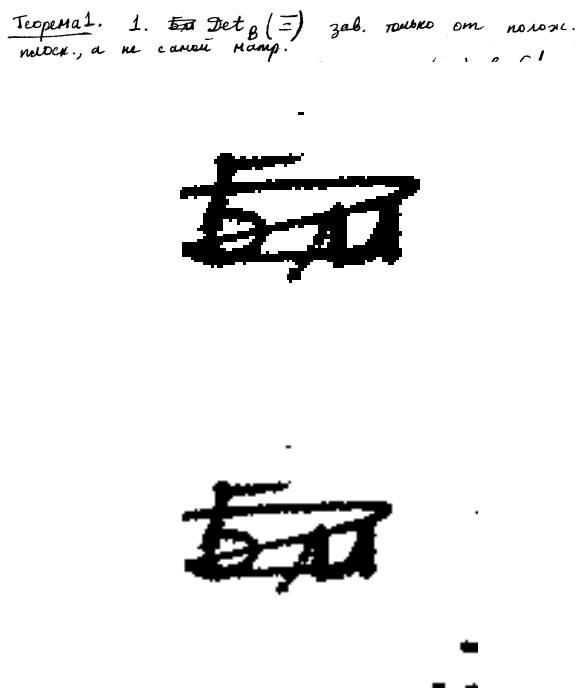


**Diagram and typed description.** Two subscripts $i, j$ on $e'$; at the right, $j = 1, \dots, k$. The next line contains $\Xi$ and $k$ nearby planes.

Original source details: GELFAND-P000130

Theorem 1.

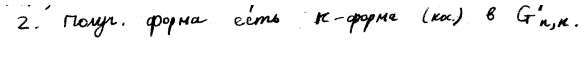
1. [Crossed-out B-like sign: illegible] $\mathrm{Det}_B(\Xi)$ depends only on the position of the planes, and not on the matrix itself.



**Uncertain reading.** Between 1. and the active Det, a dense B-like sign or short entry is crossed out; its full reading is unresolved.

Original source details: GELFAND-P000130

2. The resulting form is a $k$-form (skew.) in $G'_{n,k}$.

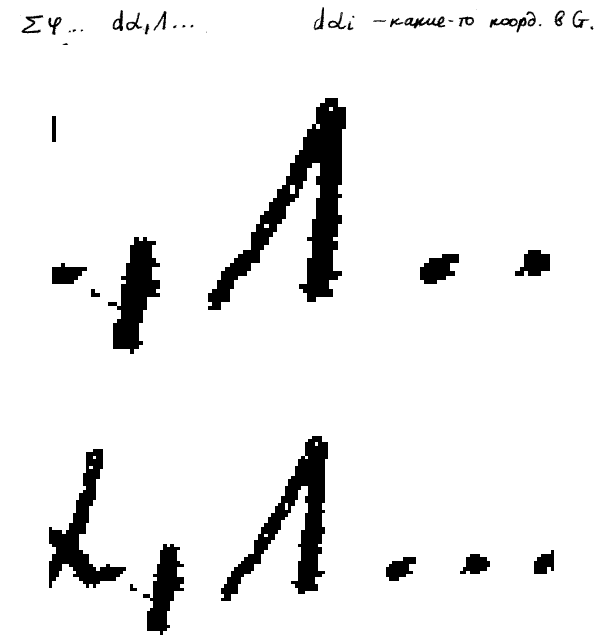


Original source details: GELFAND-P000130

**Diagram and typed description.** The second item contains $k$-form, the parenthesis skew. and $G'_{n,k}$.

$$\sum \varphi \ldots \quad d\alpha_1 \wedge [?] \ldots$$

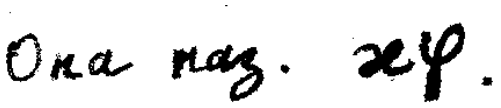
$d\alpha_i$ — some coordinates in $G$.



**Uncertain reading.** Ellipses after $\varphi$ and after $d\alpha_1$ with a $\wedge$- / $\Lambda$-like sign. The exact notation of the sign is unresolved; the remaining factors and indices are not written out.

Original source details: GELFAND-P000130

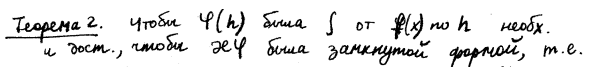
It is called $\mathcal{X}[?]\varphi$.



Original source details: GELFAND-P000130

**Uncertain reading.** After called, an X-like / $\chi$-like operator sign before $\varphi$; its exact notation is unresolved.

Theorem 2. For $\varphi(h)$ to be the $\int$ of $f(x)$ over $h$, it is necessary and sufficient that $\mathcal{X}[?]\varphi$ be a closed form, i.e.

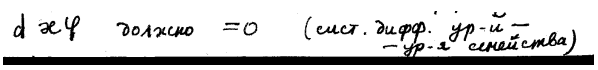


Original source details: GELFAND-P000130

**Diagram and typed description.** Theorem 2. is underlined. The full sentence contains necessary and sufficient, and a closed form, i.e.

$$d\mathcal{X}[?]\varphi \text{ must } = 0$$

(system of differential equations — — equations of the family)

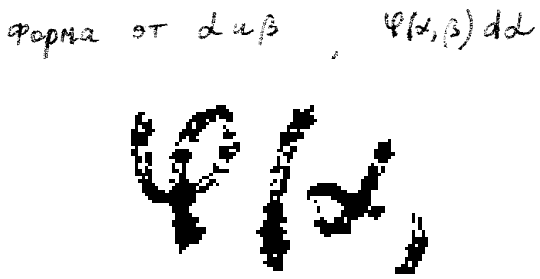


Original source details: GELFAND-P000130

**Diagram and typed description.** After the operator expression, the word must, an equals sign and 0. At the right, parentheses with system of differential equations, two dashes at the line junction, and equations of the family.

GELFAND-P000131

A form in $\alpha$ and $\beta$: $\psi[?](\alpha, \beta)\, d\alpha$.



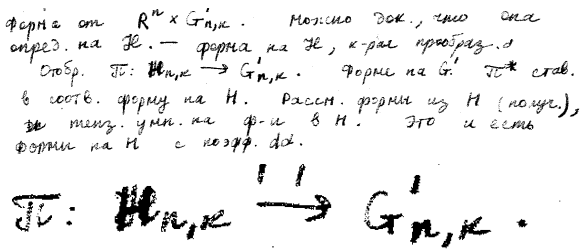
Original source details: GELFAND-P000131

**Uncertain reading.** The first letter of the form has an open left branch, a right loop and a downward stroke: $\psi$ / $\varphi$. Its exact notation is unresolved.

A form on $R^n \times G'_{n,k}$. One can prove that it is defined on $H$ — a form on $H$ which transforms with the map

$$\pi : H_{n,k} \longrightarrow G'_{n,k}.$$

Forms on $G'$: $\pi^*$ assigns a form on $H$. Consider the forms from $H$ (obtained),



Original source details: GELFAND-P000131

**Diagram and typed description.** The first product contains $R^n$ and $G'_{n,k}$. The map goes from $H_{n,k}$ to $G'_{n,k}$; below, a separate $\pi^*$, a form on $H$ and the parenthesis obtained.

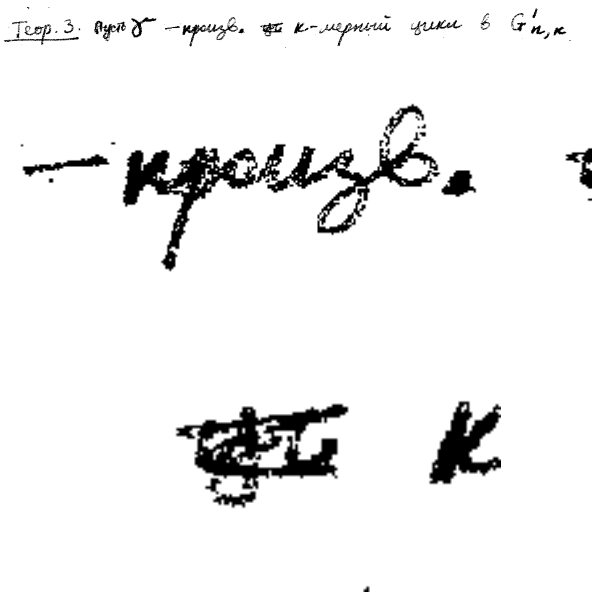
[Crossed out: a short unreadable sign.] tensor mult. by fns. on $H$. These are precisely the forms on $H$ with coeffs. $d\alpha$.



**Uncertain reading.** Before tensor mult., a short dense sign with loops and crossing strokes is cancelled; its full reading is unresolved.

Original source details: GELFAND-P000131

Thm. 3. Let $\gamma$ be a curv. [Crossed out: several unreadable letters or strokes.] $k$-dimensional cycle in $G'_{n,k}$.



**Uncertain reading.** Between curv. and $k$-dimensional, a short entry is densely crossed out. Loops are visible below; its exact reading is unresolved. Thm. 3. is underlined.

Original source details: GELFAND-P000131

$$\int_\gamma \mathcal{X}[?]\psi[?] = C_\gamma f(\beta).$$



**Uncertain reading.** The sign after the integral has crossing loops: X-like / $\chi$ / $\kappa$; its exact notation is unresolved. Below, the same sign is called a diff. oper.

Original source details: GELFAND-P000131

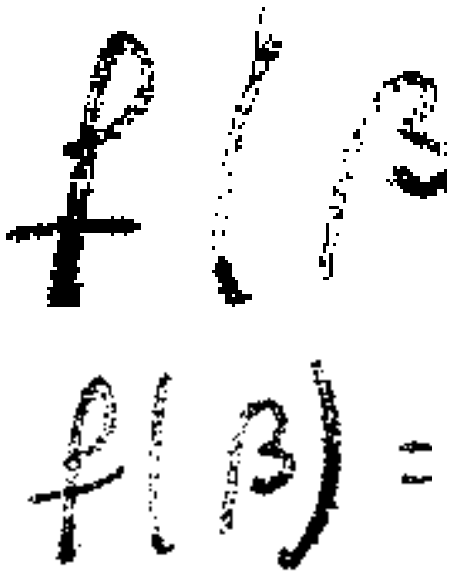


**Diagram and typed description.** In both occurrences of $C_\gamma f(\beta)$, a tall thin opening parenthesis follows $f$; no separate upper stroke on $f$ is visible.

Original source details: GELFAND-P000131

$C_\gamma$ — up to $\pi$, is the intersection index of $\gamma$ with the Euler cycle (all $k$-dim. subspaces lying in a $(k+1)$-dimensional space).

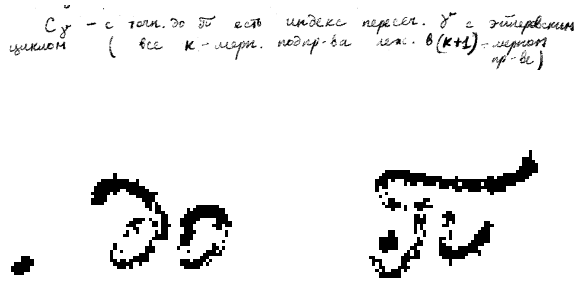
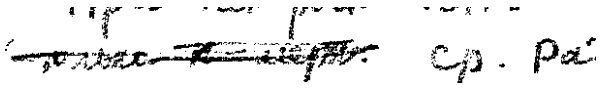


**Diagram and typed description.** After up to, a separate $\pi$ with a long upper stroke. The parenthesis contains all $k$-dim. subspaces and a $(k+1)$-dimensional space; the source word for cycle continues at the beginning of the lower line.

Original source details: GELFAND-P000131

For some parities there is no local inversion formula ([Crossed out: only $k$-dim.[?]] cf. Radon).



**Uncertain reading.** In the parentheses before cf. Radon, horizontal lines cancel an entry with parts resembling only and $k$-dim.; the complete reading of the covered strokes is unresolved.

Original source details: GELFAND-P000131

In $C^n$ there is an analytic formula, but there $C_\gamma \neq 0$ always. There is always an inversion formula there. One must know not only $\psi$[?], but also its derivatives.

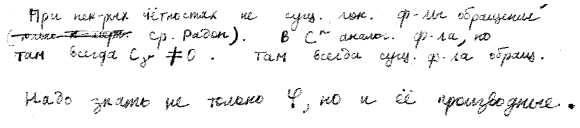


**Diagram and typed description.** The first phrase says parities. The next part contains $C^n$, then $C_\gamma \neq 0$ and a separate sentence about an inversion formula. The lower line ends with its derivatives.

Original source details: GELFAND-P000131

[Crossed out: a beginning resembling $\delta(h) =$.]

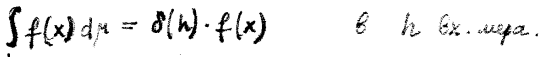


Original source details: GELFAND-P000131

**Uncertain reading.** To the left of the active integral, an earlier $\delta(h) =$ entry is crossed out by two long lines. The upper part of the opening parenthesis is thin; the covered interior strokes remain unread.

$$\int_h f(x)\, d\mu = \delta(h) \cdot f(x).$$

[At right:] in $h$, Eucl. measure.



Original source details: GELFAND-P000131

**Diagram and typed description.** The integral has subscript $h$; at the right, a dot between $\delta(h)$ and $f(x)$. A separate right-hand note: in $h$, Eucl. measure.

$$C_\gamma f(\beta) = \int_\gamma \mathcal{X}[?]\psi[?](h).$$

$$C_\gamma(\delta(x-\beta), f(x)) = \int_\gamma \mathcal{X}[?](\delta(h), f(x)).$$

$$C_\gamma \delta(x-\beta) = \int_\gamma \mathcal{X}[?]\delta(h).$$

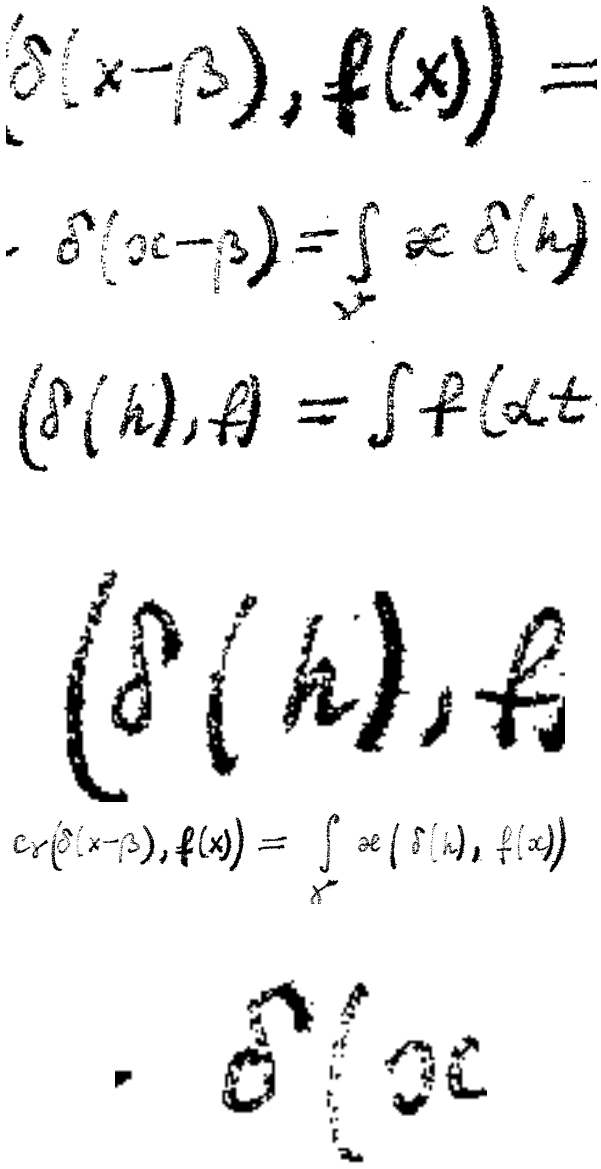


**Diagram and typed description.** In the formulas with $\delta(x-\beta)$ and $\delta(h)$, the tall thin strokes form opening parentheses; no separate apostrophe between the letter and parenthesis is visible. In the penultimate line, both pairings are in round brackets.

Original source details: GELFAND-P000131

[At right:] $\mathcal{X}$[?] — diff. oper.; $\int$ — over the cycle $\gamma$.

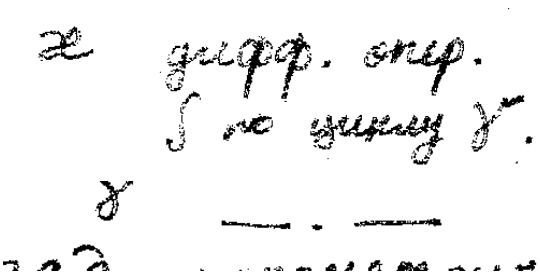


**Diagram and typed description.** The right-hand note contains the operator sign and diff. oper.; the next line is the integral over the cycle $\gamma$. Beneath the formula, two horizontal strokes with a dot between them.

Original source details: GELFAND-P000131

In $C^n$ lines are given by parametric eqns.

$$x = \alpha t + \beta.$$

$$(\delta(h), f) = \int f(\alpha t + \beta)\, dt\, d\bar{t}.$$

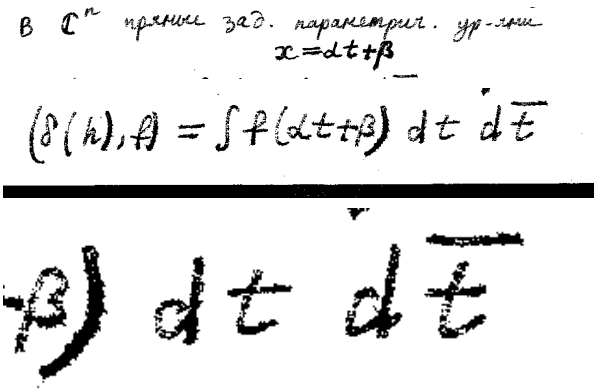
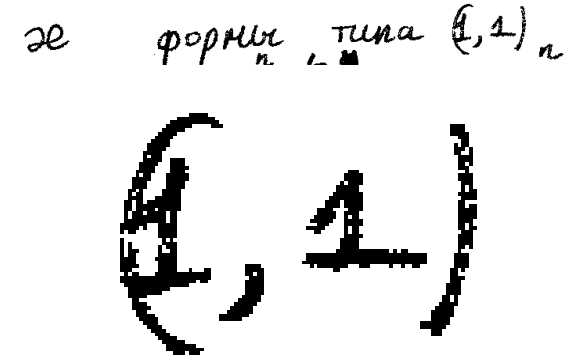


**Diagram and typed description.** The parametric equation has plus $\beta$. In the last line, the round pairing $\delta(h), f$ is at the left; the integrand is $f(\alpha t + \beta)$, followed by $dt$ and $d\bar{t}$, with a separate bar only over the final $t$.

Original source details: GELFAND-P000131

GELFAND-P000132

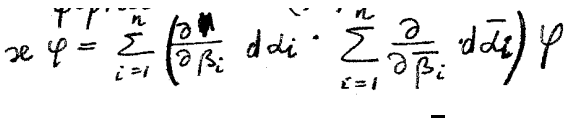
$\mathcal{X}$[?] — forms of type $(1[?], 1)$.



**Uncertain reading.** Intersecting strokes in the first numeral of type $(1, 1)$; a $1$ / $4$-like reading. The overwritten numeral has not been read definitively.

Original source details: GELFAND-P000132

$$\mathcal{X}[?]\psi[?] = \sum_{i=1}^{n} \left( \frac{\partial^{n[?]}}{\partial \beta_i} d\alpha_i \cdot \sum_{i=1}^{n} \frac{\partial}{\partial \overline{\beta}_i} d\overline{\alpha}_i \right) \psi[?].$$





Original source details: GELFAND-P000132

**Uncertain reading.** A dense raised $n$-like sign at the first $\partial$; the covered strokes have not been read.





Original source details: GELFAND-P000132

**Uncertain reading.** The intersecting looped operator is X-like / $\chi$ / $\kappa$. The form letter has an open left branch, a right loop and a descending stroke: $\psi$ / $\varphi$. The precise symbols remain uncertain. A separate dot stands between the first $d\alpha_i$ and the second sum.

Final form.:

$$C_\gamma \delta(x) = \int_\gamma \left( \sum \frac{\partial}{\partial \beta_i} d\alpha_i \wedge \sum \frac{\partial}{\partial \overline{\beta}_i} d\overline{\alpha}_i \right)$$

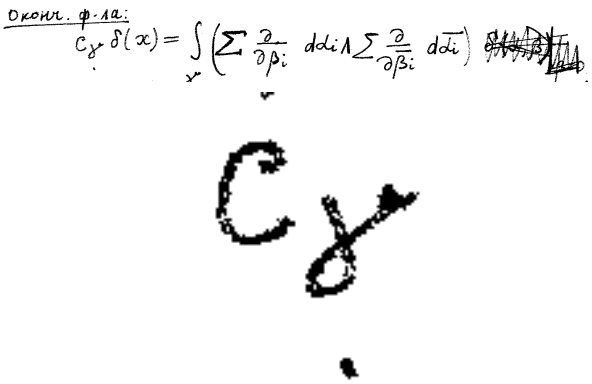


**Diagram and typed description.** The left side is $C_\gamma \delta(x)$; the integral has subscript $\gamma$. A $\wedge$ stands between the sums in parentheses. The heading Final form.: is underlined.



Original source details: GELFAND-P000132

[At right, crossed out: an entry resembling $\delta(x - \beta)$ and unread strokes.]

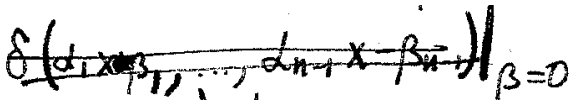


**Uncertain reading.** To the right of the parenthesis, a densely cancelled entry has a $\delta$-like beginning, parentheses and a $\beta$-like part; the complete former kernel and subsequent strokes remain unread.

Original source details: GELFAND-P000132

[Below at right, crossed out:]

$$\delta(\alpha_1 x \text{[?]} \beta_1, \ldots, \alpha_{n-1} x \text{[?]} \beta_{n-1})|_{\beta=0} .$$

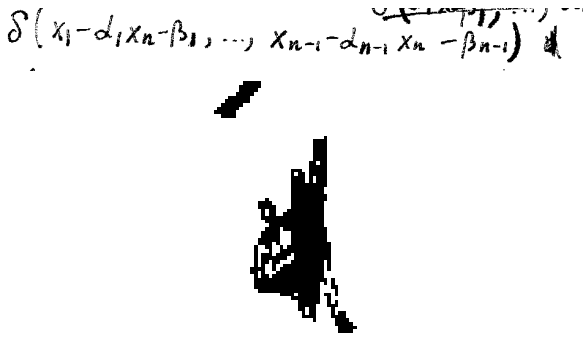


**Uncertain reading.** The cancelled line begins with $\delta(\alpha_1 x \ldots \beta_1, \ldots)$ and ends with $|_{\beta=0}$. Lines cross the intermediate signs by $\beta_i$; their precise reading remains uncertain.

Original source details: GELFAND-P000132

[Inserted below:]

$$\delta(x_1 - \alpha_1 x_n - \beta_1, \ldots, x_{n-1} - \alpha_{n-1} x_n - \beta_{n-1})\, d\text{[?]}$$



**Uncertain reading.** A dense overwritten $d$- / $x$-like sign follows the inserted kernel. Its precise reading remains uncertain.

Original source details: GELFAND-P000132

Lines[?] dep. on fewer param., not $2n$.




Original source details: GELFAND-P000132

**Uncertain reading.** The ending of the first word permits a singular or plural reading; its precise reading remains uncertain. The remainder reads dep. on fewer param., not $2n$.

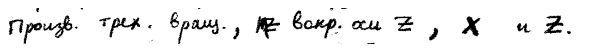
Klein's work on the top. (Karpelevich) (1896) (4 lectures at Princeton).




Original source details: GELFAND-P000132

**Diagram and typed description.** Klein's work on the top. has a straight underline. The name in parentheses is followed by 1896; farther right, 4 lectures at Princeton, with the last words below.

One point fixed. Rot. of the top — a curve in the group of rot. of 3-dim. space. Coord. in this group. 2 ways (Euler angles $\varphi, \theta, \psi$). Product of three rot., [Crossed out: a short dense sign.] about axes $Z$, $X$ and $Z$.




**Uncertain reading.** Before about axes, short dense К- / и-like strokes are crossed out. The complete reading remains uncertain.

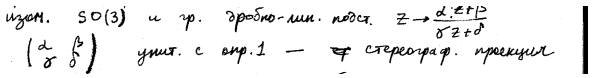


Original source details: GELFAND-P000132

Isom. of $SO(3)$ and the gr. of fractional-lin. subst.

$$z \longrightarrow \frac{\alpha z + \beta}{\gamma z + \delta}.$$

$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ unitary with det. $1$ — [Crossed out: an unread sign.] stereograph. projection.




Original source details: GELFAND-P000132

**Diagram and typed description.** The text reads Isom. of $SO(3)$ and the gr. of fractional-lin. subst.; at right, a fraction with $\alpha z + \beta$ and $\gamma z + \delta$. Below the line, a matrix and unitary with det. $1$.

**Uncertain reading.** Between the dash and stereograph. projection is a dense cancelled sign resembling $\varphi$ or an arrow. Its precise reading remains uncertain.



Original source details: GELFAND-P000132

(Rotate the sphere — fractional-lin. map of the plane.) (This is the 1st lecture.)
2nd lecture.

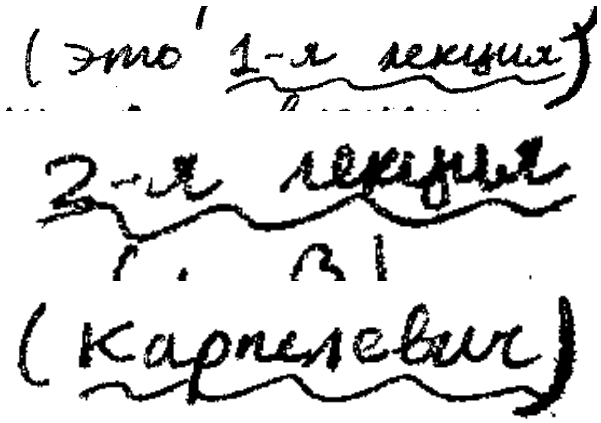


**Diagram and typed description.** Wavy lines under Karpelevich, under 1st lecture in parentheses, and under 2nd lecture.

Original source details: GELFAND-P000132

$\varphi, \psi, \theta$ sought as functions of time. $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ as functions of time are simpler. Eqs. of motion:

$$T = \frac{1}{2}(\varphi'^2 + \psi'^2 + 2\varphi'\psi' \cos\theta + \theta'^2).$$

[At right:] (All moments of inertia $= 1$ rel. to the point of attachment — sphere-top.)

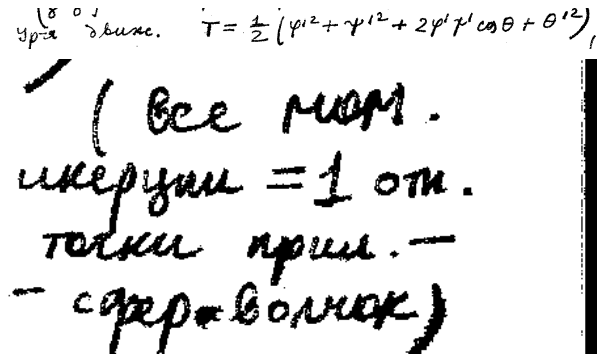


**Diagram and typed description.** Primes and squares at $\varphi$, $\psi$, $\theta$; the mixed term is $2\varphi'\psi' \cos\theta$. A separate parenthesis at right: All moments of inertia $= 1$ rel. to the point of attachment — sphere-top.

Original source details: GELFAND-P000132

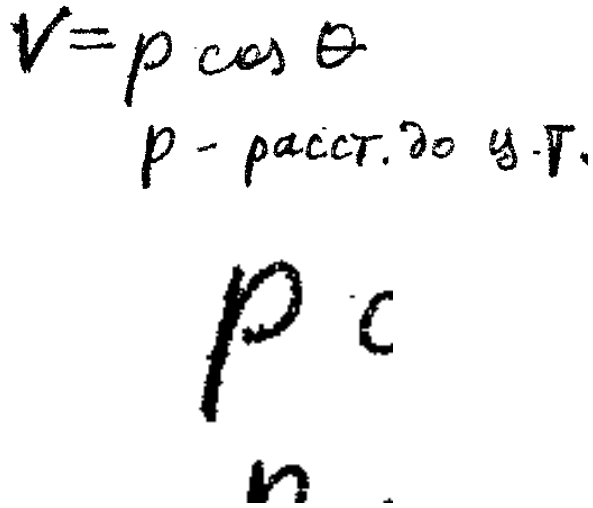
Potential energy [Crossed out: a $p$- / $\rho$-like beginning.] $V = p$[?] $\cos\theta$.



**Uncertain reading.** Before $V$, a letter with a loop and a descending stroke is cancelled with long lines: $p$ / $\rho$. The complete former beginning remains unread.

Original source details: GELFAND-P000132

$p$[?] — dist. to c. of g.




**Uncertain reading.** In $V$ and in the following line, a letter $p$ / $\rho$ with a loop and a descending stroke. The precise symbol remains uncertain.

Original source details: GELFAND-P000132

$\theta$ — axis $Z$ with c. of g. vert.

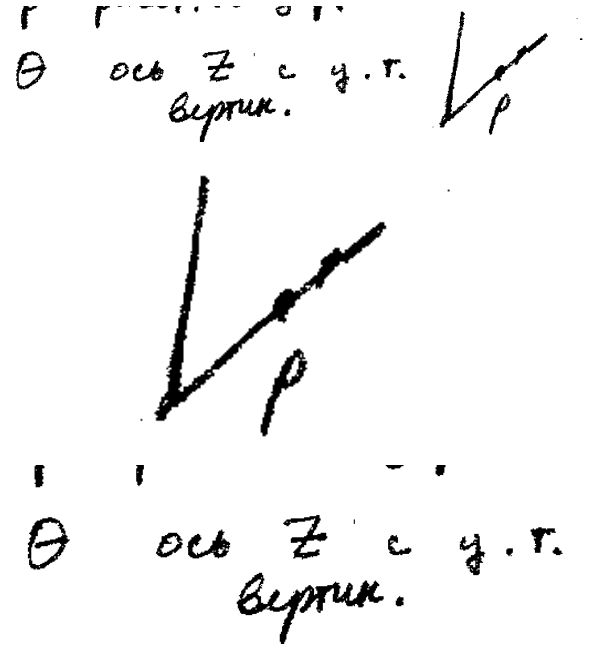
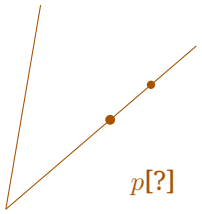



**Diagram and typed description.** Two segments from a common lower endpoint, one nearly vertical, the other sloping upward to the right. Two dense points lie on the oblique segment, with a $p$ / $\rho$ label below. The drawing has no angle arc or axis labels. Beside it: $\theta$ — axis $Z$ with c. of g. vert.




Original source details: GELFAND-P000132

$$\frac{d}{dt}\frac{\partial T}{\partial \varphi'} = 0.$$

$$\begin{cases} \varphi' + 2\psi' \cos\theta = n, \\ \psi' + 2\varphi' \cos\theta = e, \\ T + V = h. \end{cases}$$

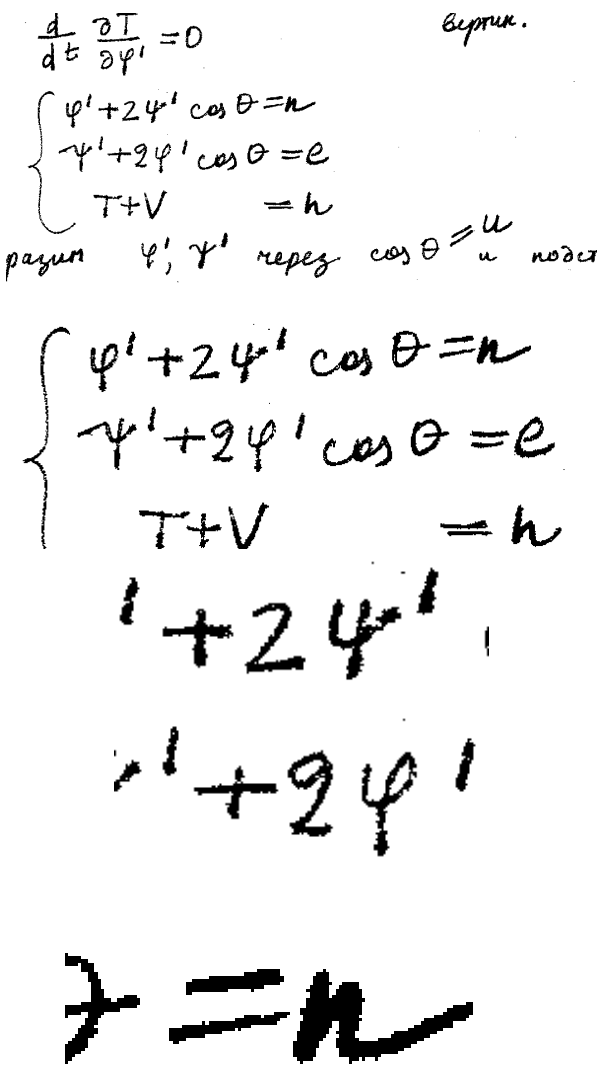



**Diagram and typed description.** In each of the first two lines, a clear 2 follows the plus. The right side of the first line is $n$, of the second $e$; the third line is $T + V = h$.

Original source details: GELFAND-P000132

Express $\varphi', \psi'$ in terms of $\cos\theta = u$ and subst. into that eq.

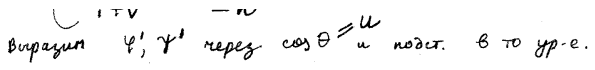
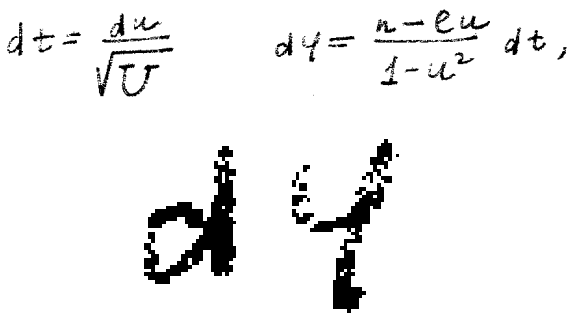



**Diagram and typed description.** The addition $= u$ is above the end of $\cos\theta$. The last phrase ends with into that eq.

Original source details: GELFAND-P000132

$$dt = \frac{du}{\sqrt{U}}, \qquad d\varphi[?] = \frac{n - eu}{1 - u^2}\, dt.$$

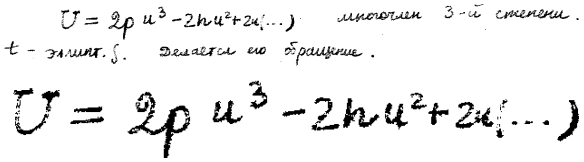


**Uncertain reading.** The letter in the second differential has an open branch and a descending stroke: $\varphi$ / $\psi$. The precise symbol remains uncertain.

Original source details: GELFAND-P000133

$$U = 2p[?]u^3 - 2hu^2 + 2u(\ldots).$$

A polynomial of degree 3. $t$ — ellipt. $\int$. Its inversion is carried out.



**Uncertain reading.** After 2, a letter with a loop and a descending stroke: $p$ / $\rho$. The first term is $2pu^3$, then a minus before $2hu^2$ and a plus before $2u(\ldots)$; the ellipsis is in parentheses.

Original source details: GELFAND-P000133

[Crossed out: a short unread beginning.] $28/\overline{\text{II}}$.



**Uncertain reading.** Before $28/\text{II}$, a short looped sign is densely cancelled; its complete reading remains uncertain.

Original source details: GELFAND-P000133

Continuation of Agranovich's talk.



**Diagram and typed description.** The Roman II in the date has upper and lower bars. The heading Continuation of Agranovich's talk. has a wavy underline.

Original source details: GELFAND-P000133

$$Au(x) = \underbrace{F^{-1}_{\xi\to x}\big(F_{x\to\xi}\,\sigma(x,\xi)u(x)\big)}.$$
$$\sigma(x,t\xi) = t^s\sigma(x,\xi).$$

This def. differs inessentially.

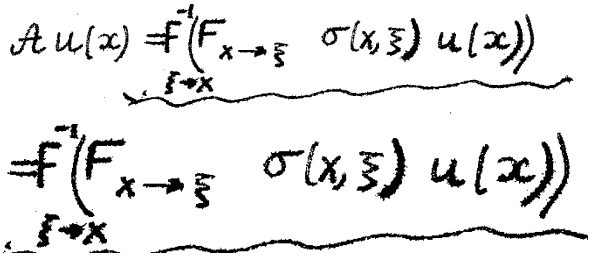
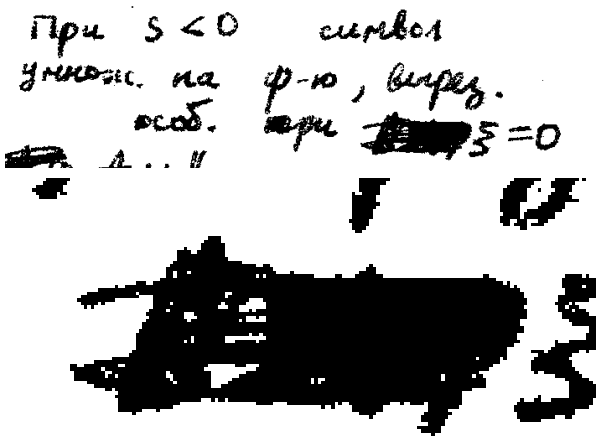


**Diagram and typed description.** A separate $\xi \to x$ below the outer $F^{-1}$; $x \to \xi$ below the inner $F$. The whole right side of the definition has a wavy underline. Below it, the condition with $t\xi$ and $t^s$.

Original source details: GELFAND-P000133

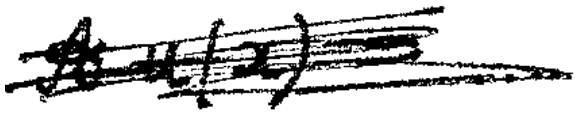
[In the right margin:] For $s < 0$, the symbol is mult. by a fn. cutting out the sing. at [Crossed out: an unread sign.] $\xi = 0$.



**Uncertain reading.** Between at and $\xi = 0$, a short densely cancelled entry; the covered strokes remain unread. The complete marginal note ends with $\xi = 0$.

Original source details: GELFAND-P000133

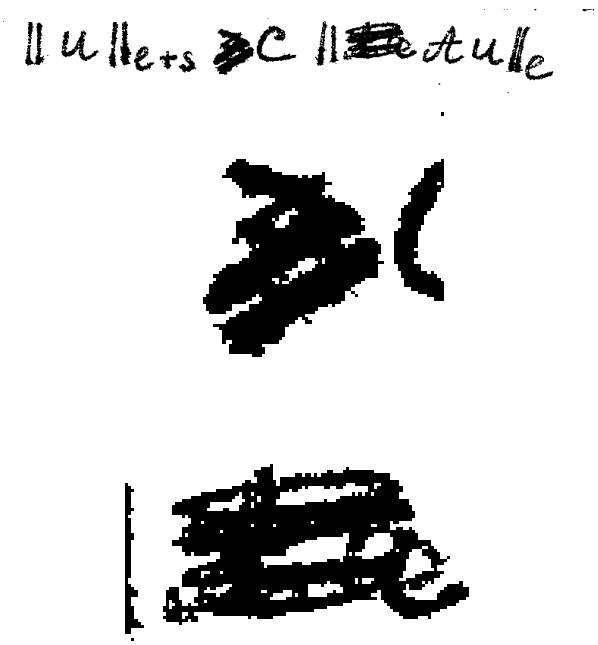
[At left, crossed out: $Au(x) =$ and an unread continuation.]



**Uncertain reading.** The beginning $Au(x) =$ of the cancelled line is visible; long lines and dense strokes cover its continuation.

Original source details: GELFAND-P000133

$\|u\|_{\ell+s} \geq$ [?] $C\|$[cancelled] $Au\|_{\ell}$.



**Uncertain reading.** The comparison sign is overwritten: $\geq$ / $\leq$. A dense sign before $Au$ inside the right norm is cancelled. The altered strokes have not been read definitively.

Original source details: GELFAND-P000133

1) Boundedness theorem.

2) Composition theorem.

$$A_1 \cdot A_2 - A_1 \circ A_2 = T.$$
$$\sigma_1 \quad \sigma_2 \qquad\quad \sigma_1 \cdot \sigma_2$$

[At right:] ord. $s_1 + s_2 - 1$.

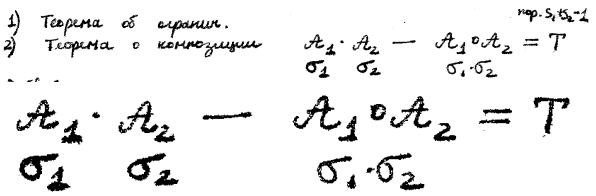
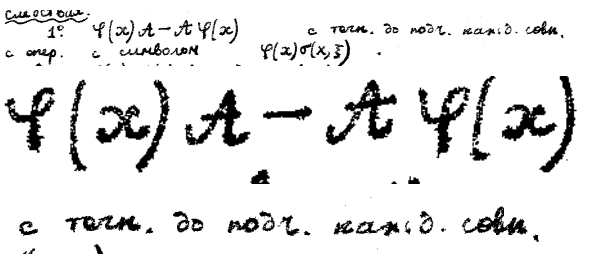


**Diagram and typed description.** A $\sigma_1$ below the first $A_1$, a $\sigma_2$ below $A_2$. Below $A_1 \circ A_2$, the entry $\sigma_1 \cdot \sigma_2$. Above at right, ord. $s_1 + s_2 - 1$.

Original source details: GELFAND-P000133

Corollaries.

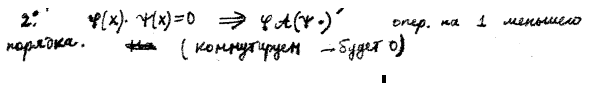
1°. $\varphi(x)A - A\varphi(x)$ up to podch. kom.d. sovp.[?] with an oper. having symbol $\varphi(x)\sigma(x,\xi)$.



**Uncertain reading.** The uncertain abbreviations are retained in transliteration: podch. kom.d. sovp. / podch. kam.d. svya.; their full reading remains uncertain. A horizontal stroke stands between $\varphi(x)A$ and $A\varphi(x)$. Corollaries. has a wavy underline.

Original source details: GELFAND-P000133

$2^\circ$. $\varphi(x)\cdot\psi(x)=0 \Rightarrow \varphi A(\psi\,\cdot)$ — an oper. of order $1$ less. [Crossed out: a short unread word.] (With the commutator — will be $0$.)



**Uncertain reading.** After order and before the parenthesis, a short densely cancelled entry. In parentheses: With the commutator — will be $0$.



Original source details: GELFAND-P000133

$3^\circ$. Let $s=0$. Suppose there is an estimate $\|Au\|_1 \le c\|u\|_0$ (smoothing oper.). Then the symbol is ident. $0$ and the operator $0$.

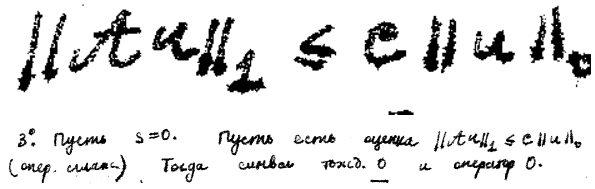
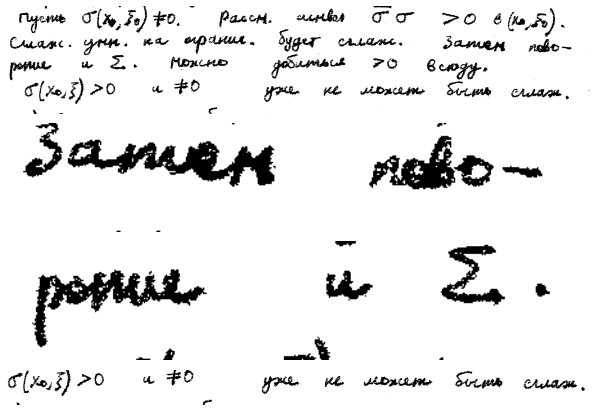


**Diagram and typed description.** The left norm has subscript $1$, the right $0$; the sign is $\le$. Then smoothing oper. in parentheses, symbol ident. $0$, and operator $0$.

Original source details: GELFAND-P000133

Let $\sigma(x,\xi_0)\neq 0$. Consider the symbol $\bar\sigma\sigma>0$ at $(x,\xi_0)$. Smoothing mult. by bounded will be smoothing. Zamen[?] levo-rymi[?] and $\Sigma$. One can obtain $>0$ everywhere. $\sigma(x_0,\xi)>0$ and $\neq 0$ can no longer be smoothing.



**Uncertain reading.** The divided word is retained in transliteration: levo-rymi / neko-rymi; the full reading remains uncertain. It is preceded by Zamen / Zatem. The final line reads $\sigma(x_0,\xi)>0$ and $\neq 0$ can no longer be smoothing.

Original source details: GELFAND-P000133

$3)$ Coordinate transformation theorem. $x=x(y)$ — inf. smooth, nondeg.

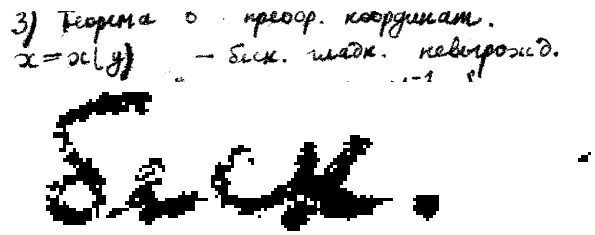
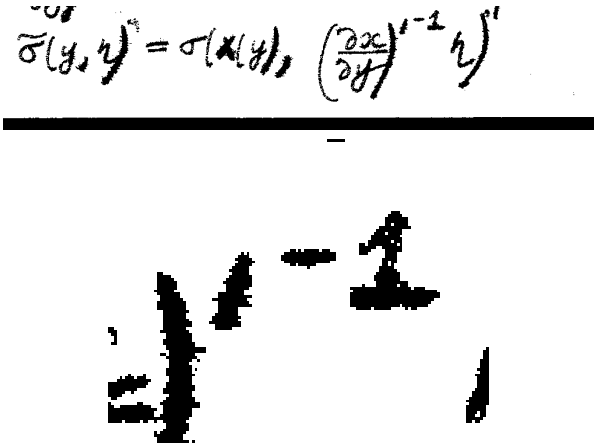


**Diagram and typed description.** Number $3)$, then Coordinate transformation theorem. After $x=x(y)$, the abbreviations inf. smooth, nondeg.

Original source details: GELFAND-P000133

$$\tilde{\sigma}(y,\eta) = \sigma\left(x(y), \left(\frac{\partial x}{\partial y}\right)^{\prime -1[?]} \eta\right).$$

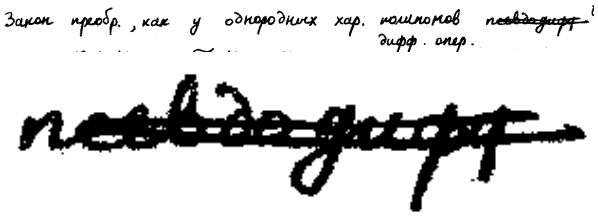


**Uncertain reading.** After the matrix's closing parenthesis, a separate raised apostrophe-like / $t$-like stroke, followed by $-1$. The precise superscript reading remains uncertain.

Original source details: GELFAND-P000133

GELFAND-P000134

Transformation law, as for homogeneous char. polynomials [Crossed out: pseudodiff.[?]]. [Below the line:] diff. oper.

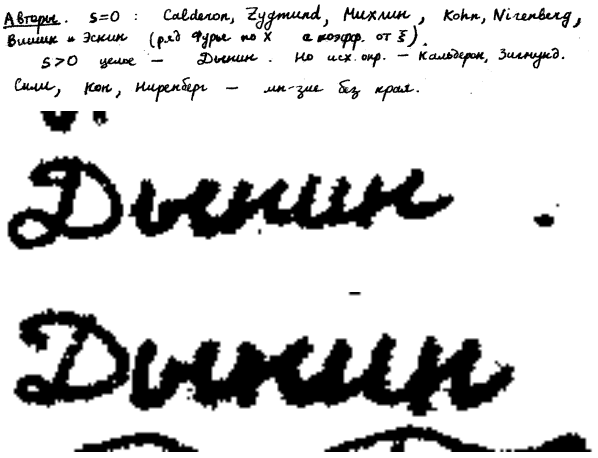


**Uncertain reading.** At the end of the first line, a word resembling pseudodiff. is crossed out; the covered letters remain incompletely read. Separately below: diff. oper.

Original source details: GELFAND-P000134

Consider $\varphi A\psi - \varphi\tilde{A}\psi = T$ — this is a subord. oper.

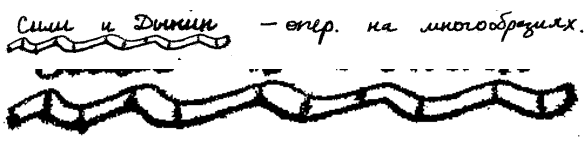
Authors. $s = 0$: Calderon, Zygmund, Mikhlin, Kohn, Nirenberg, Vishik and Eskin (Fourier series in $x$, with coeff. depending on $\xi$). $s > 0$ integer — Dynin[?]. But the orig. def. — Calderon, Zygmund. Sili, Kohn, Nirenberg — manifold without boundary.



**Uncertain reading.** In the name after $s > 0$ and in the following underlined pair, dense joined strokes: Dynin / Dynkin. The precise reading remains uncertain. In Latin script in the first line: Calderon, Zygmund, Kohn, Nirenberg.

Original source details: GELFAND-P000134

Sili and Dynin[?] — oper. on manifolds.




Original source details: GELFAND-P000134

**Diagram and typed description.** Below the pair of names, a double wavy strip with short cross-strokes.



Let $\underline{s=0}$. $A$ — sing. $\int$ oper. on the manifold $V$. Consider $\varphi$ and $\psi$ on $V$. 1) $\operatorname{supp}\varphi \cap \operatorname{supp}\psi = \varnothing \Rightarrow \varphi A(\psi u) = Tu$ ($\|Tu\|_1 \le c\|u\|_0$).

2) Let $\operatorname{supp}\varphi$ and $\psi$ lie in a coord. nbd. Then

$$\varphi A\psi = \varphi A_\nu \psi + Tu,$$

where $A_\nu$ — a sing. $\int$ oper., written in local coordinates.

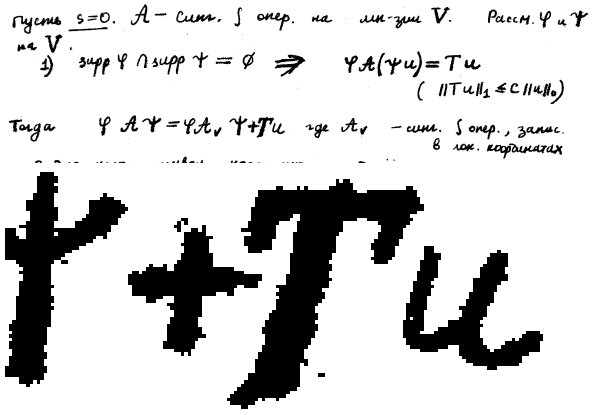


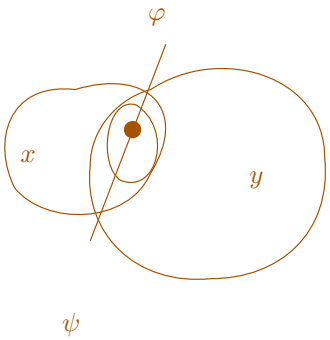

**Diagram and typed description.** In the first condition, two supports with empty intersection, then $Tu$ and an estimate with indices $1$ and $0$. The local formula has $+Tu$ at right.

Original source details: GELFAND-P000134

The symbol is evidently well-defined as a function on the cotangent bundle. ($A_\nu$ on the overlap.) Hence, in particular, the symbol transformation formula.




**Diagram and typed description.** Two overlapping rounded contours, $x$ at left and $y$ at right. Above, $\varphi$; below, $\psi$; inside the overlap, a small loop with a heavy point and an oblique stroke.




Original source details: GELFAND-P000134

$$\varphi A\psi. = \varphi A_x\psi + T_1 = \varphi A_y\psi + T_2.$$

Hence

$$\begin{array}{ll} \varphi A_x\psi & -\varphi A_y\psi \quad \text{smoothing.} \\ \| & \\ \varphi\tilde{A}_y\psi + \text{smoothing.} & \end{array}$$

[At right:] Hence smoothing symbol.

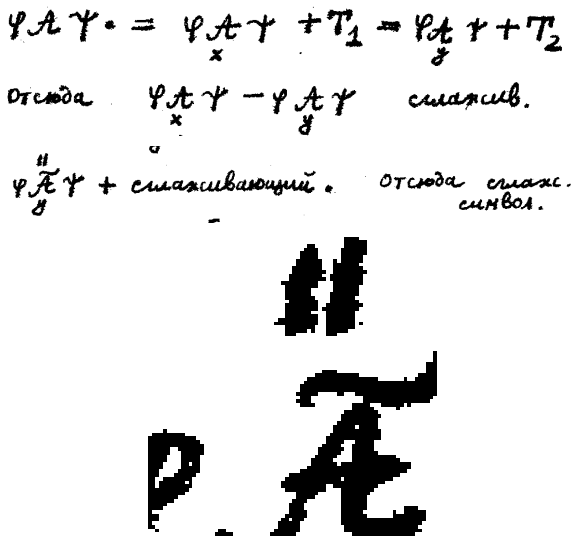


**Diagram and typed description.** A separate point after the first $\psi$ and before the equals sign. Below the first term, two vertical strokes, then $\varphi\tilde{A}_y\psi+$ smoothing. Separately at right: Hence smoothing symbol.

Original source details: GELFAND-P000134

$$A = \sum \varphi_\nu A, \qquad \psi_\nu = \begin{cases} 1 & \text{in a nbd. of supp}\,\varphi_\nu, \\ 0 & \text{outside the nbd.} \end{cases}$$

$$A = \sum \varphi_\nu A\psi_\nu + \text{smoothing.} = \sum \varphi_\nu A_\nu\psi_\nu + \text{smoothing.}$$

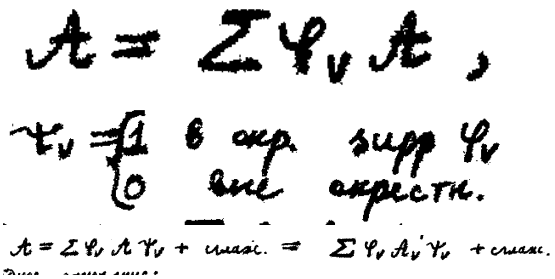
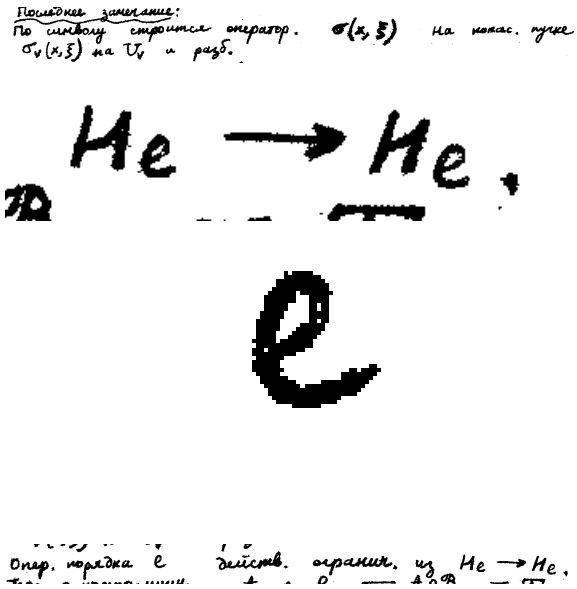


**Diagram and typed description.** In the first sum, only $A$ follows $\varphi_\nu$. The two cases for $\psi_\nu$ are $1$ in a nbd. of the support of $\varphi_\nu$ and $0$ outside the nbd. The lower line has two sums with $\psi_\nu$; in the second, $A$ also has subscript $\nu$.

Original source details: GELFAND-P000134

Final remark: An operator is constructed from the symbol. $\sigma(x,\xi)$ on the cotang. bundle, $\sigma_\nu(x,\xi)$ on $U_\nu$ and a partition. An oper. of order $\ell$[?] acts boundedly from $H_{\ell[?]} \to H_{\ell[?]}$.




**Uncertain reading.** The order letter and both indices have the same looped appearance: $\ell$ / $e$. Both spaces are written with the same index. Final remark: has a wavy underline.

Original source details: GELFAND-P000134

Composition thm.:

$$\underset{\sigma_1}{A} \cdot \underset{\sigma_2}{B} - \underset{\sigma_1\cdot\sigma_2}{A \circ B} = T.$$

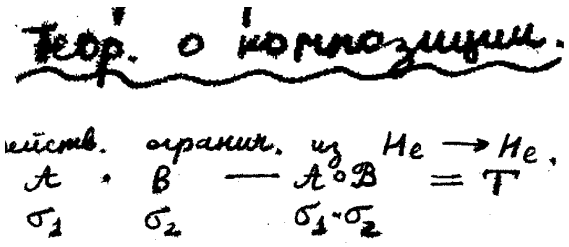



**Diagram and typed description.** A $\sigma_1$ below $A$, a $\sigma_2$ below $B$, and $\sigma_1 \cdot \sigma_2$ below $A \circ B$. The circle is between $A$ and $B$; the upper loop above this part belongs to iz (from) in the preceding line. The heading Composition thm.: has a wavy underline.

Original source details: GELFAND-P000134

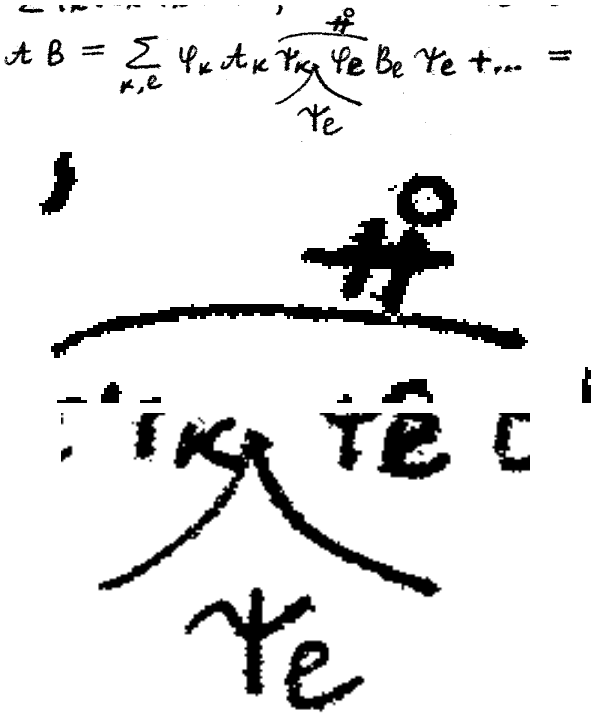
GELFAND-P000135

Proof.

$$A = \sum \varphi_k A_k \psi_k + T, \qquad B = \sum \varphi_\ell B_\ell \psi_\ell + T_2.$$

$$AB = \sum_{k,\ell} \varphi_k A_k \overbrace{\psi_k \varphi_\ell}^{A^\circ[?]} B_\ell \psi_\ell + \ldots =$$

$\psi_\ell$

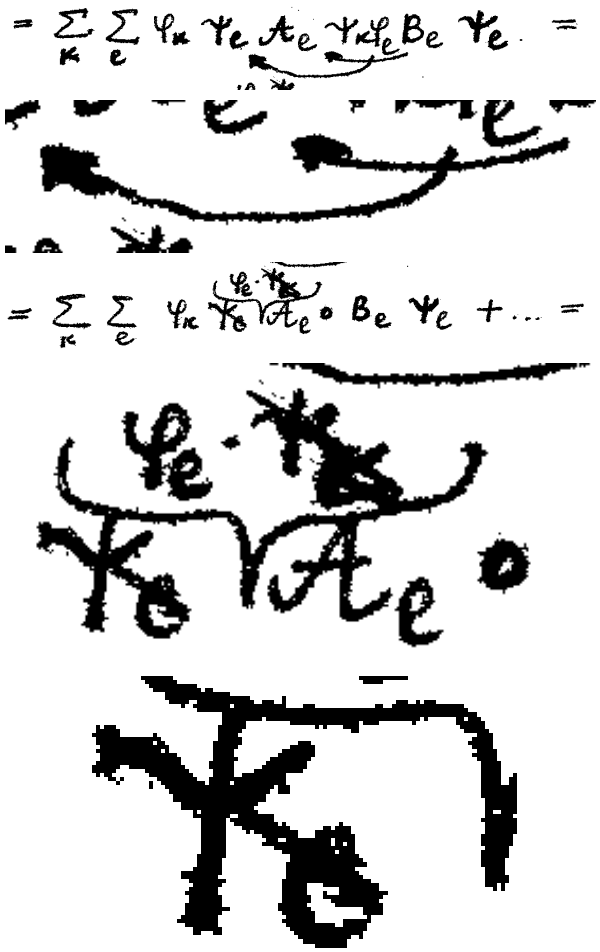


**Uncertain reading.** An arc above $\psi_k \varphi_\ell$ has an $A^\circ$- / $T^\circ$- / $t^\circ$-like label; the exact reading is unsettled. Below are two converging curved strokes and the label $\psi_\ell$.

Original source details: GELFAND-P000135

$$= \sum_k \sum_\ell \varphi_k \, \psi_\ell \, A_\ell \, \psi_k \, \varphi_\ell \, B_\ell \psi_\ell . \quad =$$

$$= \sum_k \sum_\ell \varphi_k \, \cancel{\psi_{\ell[?]}} \, \tilde{A}_\ell \cdot B_\ell \psi_\ell + \ldots =$$



**Uncertain reading.** Below the second line are two curved arrows pointing left. In the next line, $\psi_{\ell[?]}$ before $\tilde{A}_\ell$ is crossed out; $(\varphi_\ell \cdot \psi_{k[?]})$ is inserted above, with its second part heavily crossed out. Below the insertion are joined arcs and an angular insertion mark. The obscured indices remain partly unread.

Original source details: GELFAND-P000135

$$= \sum_\ell \varphi_\ell A_\ell \circ B_\ell \psi_\ell + \text{smooth.}$$

The theor. on compos. is proved.

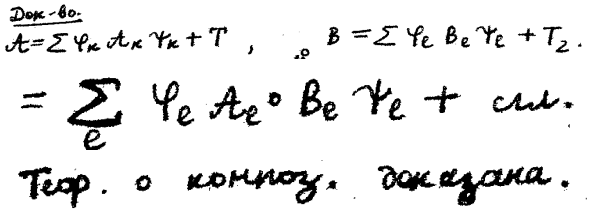


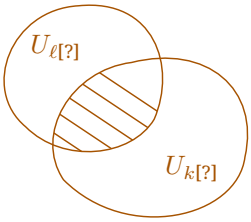
**Diagram and typed description.** The first three product lines also end with an equals sign; the following lines begin with an equals sign.

Original source details: GELFAND-P000135



Original source details: GELFAND-P000135

**Diagram and typed description.** Two intersecting contours with diagonal hatching in their overlap. The upper label is $U_{\ell[?]}$, the lower $U_{k[?]}$; the dense indices remain unsettled.




Elliptic theory. $A_s$, $\sigma(P,\xi_P)$; $\xi_P$ is a cotang. vector. Elliptic operator: $\sigma(P,\xi_P)\neq 0$ for $\xi_P\neq 0$ (if a matrix, then det). Theorem 1. Ellipticity is equivalent to being Noetherian. a) Prove that an elliptic oper. is Noetherian. Consider $\sigma^{-1}$, also a symbol, homog. of degree $\underline{\underline{-s}}$, $\sigma^{-1}\leftrightarrow B$.

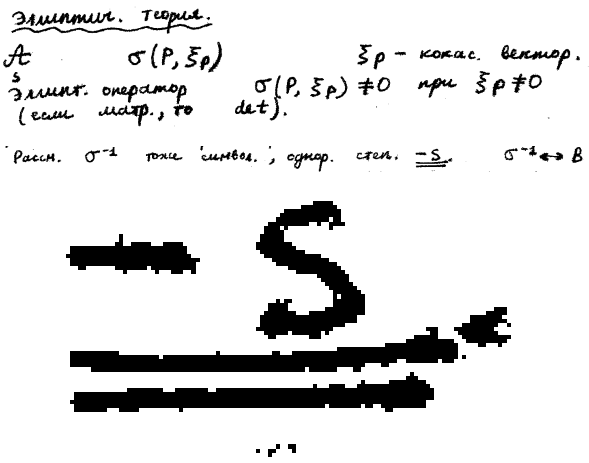
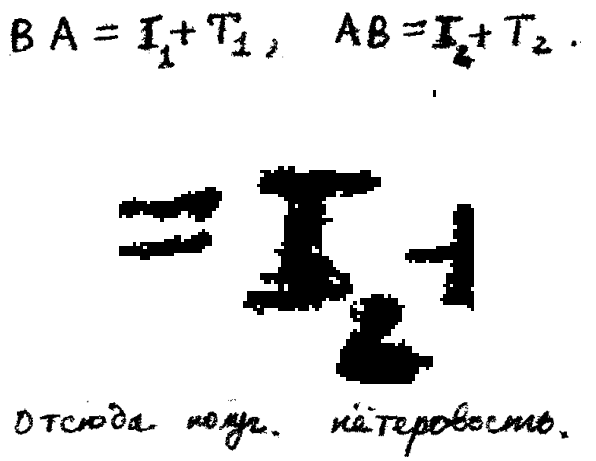



**Diagram and typed description.** $A$ has the subscript $s$. The degree $-s$ has a double underline.

Original source details: GELFAND-P000135

$$BA = I_1 + T_1, \qquad AB = I_{2[?]} + T_2.$$

Hence normal Noetherianity. $d(A) < \infty, \quad \beta(A) < \infty$.




**Uncertain reading.** The subscript below the second $I$ is heavily overwritten; a 2-like sign is visible, while the former stroke remains unread.

Original source details: GELFAND-P000135

b) Noeth. $\Rightarrow$ ell. $s = 1$.

$$\|u\|_1 \le C(\|Au\|_0 + \|u\|_0)$$

from Noetherianity. Passage to $R^n$, elim. the depend. of the symb. on $x$. Functions with small supports and a symb. indep. of $x$. $\sigma(x_0, \xi_0) = 0$ and the est. leads to a contradiction:

$$\varphi(x) e^{i\lambda\langle x, \xi_0\rangle}.$$

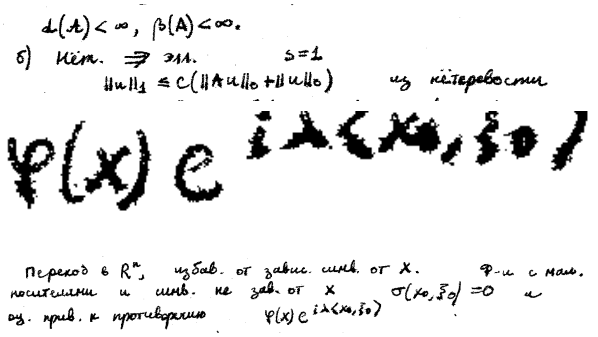



**Diagram and typed description.** The exponent is $i\lambda\langle x, \xi_0\rangle$. The line above has $\sigma(x_0, \xi_0) = 0$.

Original source details: GELFAND-P000135

Theorem 2 (on smoothness). $Au = f, \quad s = 0$. Let $u \in H_0,\ f \in H_1 \ \Rightarrow\ u \in H_1$.

$$Au = f, \quad u + \underset{H_1 \text{ (smooth.)}}{\overset{\uparrow}{T_2}} \quad f = Bf \in H_1 \ \Rightarrow\ u \in H_1.$$

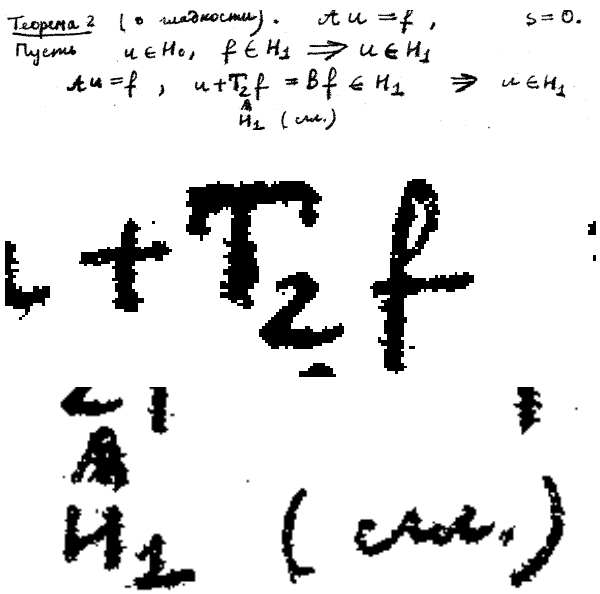


**Diagram and typed description.** $T_2$ is followed by $f$. Below $T_2$, an arrow points up from the label $H_1$ (smooth.).

Original source details: GELFAND-P000135

Theorem 3. Homotop. stability of the index $\chi(A) = d(A) - \beta(A)$.

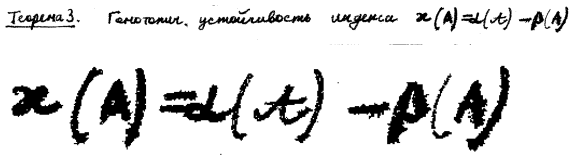
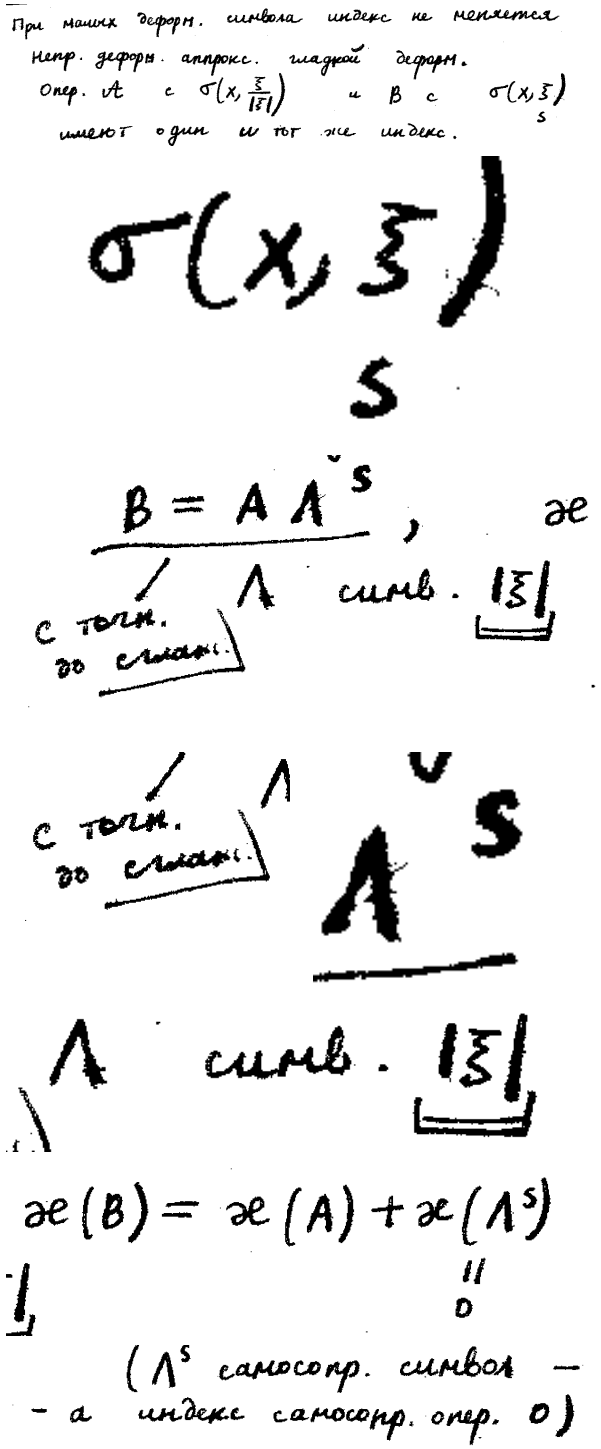


**Diagram and typed description.** The final formula is $\chi(A) = d(A) - \beta(A)$.

Original source details: GELFAND-P000135

GELFAND-P000136

Under small deformations of the symbol, the index does not change. A continuous deform. is approx. by a smooth deform. The oper. $A$ with $\sigma\left(x, \frac{\xi}{|\xi|}\right)$ and $B$ with $\underset{s}{\sigma(x,\xi)}$ have the same index.




Original source details: GELFAND-P000136

**Diagram and typed description.** An $s$ is written separately beneath $\xi$ in the second symbol. $B = A\Lambda^s$ is underlined; the margin has "up to smooth." with an oblique stroke and an angular line. Below $\chi(\Lambda^s)$ are two vertical strokes and $0$. Below $|\xi|$ are two lines with short ends.

$$\underline{B = A\Lambda^s},$$

up to smooth. $\Lambda$ symb. $\underline{\underline{|\xi|}}$

$$\chi(B) = \chi(A) + \underset{\|}{\chi(\Lambda^s)}.$$
$$0$$

($\Lambda^s$ is a selfadj. symbol—and the index of a selfadj. oper. is $0$.)

Index theorem. (Sili) $V_n$, $n \times n$ systems. $\sigma(P, \xi_P)$ is a unitary matrix.

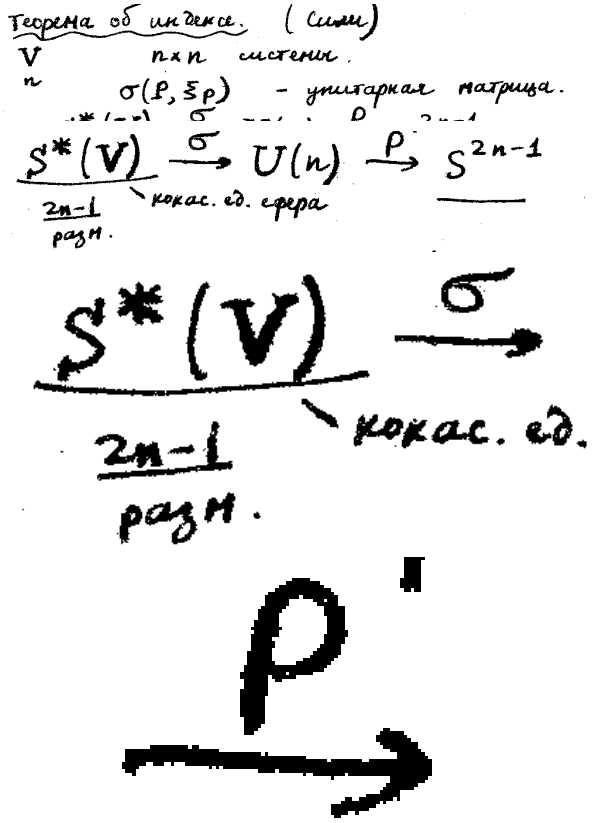



**Diagram and typed description.** $S^*(V)$ and the sphere on the right are underlined. Below the first space, $2n-1$ is underlined, with "dim." beneath; an oblique stroke leads to the label "unit cotang. sphere".

$$\underline{S^*(V)} \xrightarrow{\sigma} U(n) \xrightarrow{\rho} \underline{S^{2n-1}}$$

$\underline{2n-1}$ dim. — unit cotang. sphere

Original source details: GELFAND-P000136

$$\operatorname{ind} A = \frac{1}{(n-1)!} \deg(\rho \circ \sigma) \iff \operatorname{ind}_{(n-1)\times(n-1)} B = 0.$$

The operation $A\#B$ is used—$\operatorname{ind} A\#B = \operatorname{ind} A \cdot \operatorname{ind} B$. For boundaries of domains in $R^{n+1}$, such a formula holds. (the right side is correct) Left formula.

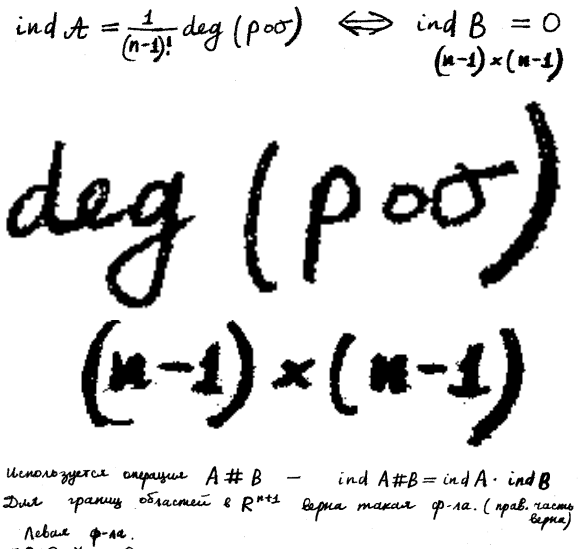
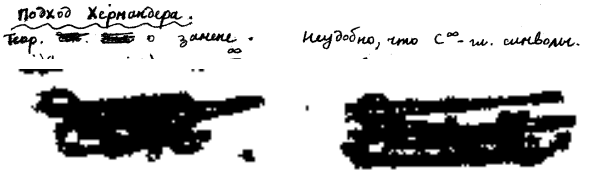



**Diagram and typed description.** Below $B$, separately, is $(n-1) \times (n-1)$. The product of indices on the right has a dot; after ind on the left is $A\#B$. The preceding line has $\rho \circ \sigma$.

Original source details: GELFAND-P000136

Hermander's approach. Theor. [Crossed out: two unread words] on change. It is inconvenient that the symbols are $C^\infty$.



**Uncertain reading.** Before "on change" are two heavily crossed-out words. The obscured letters remain unread.



Original source details: GELFAND-P000136

$$e^{-i\lambda g} A[f e^{ig\lambda}] \sim \sum_0^\infty P_j(f,g)\lambda^{s_j}, \qquad s_j \downarrow -\infty.$$

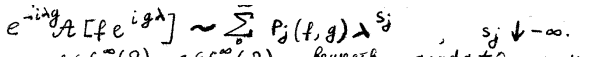
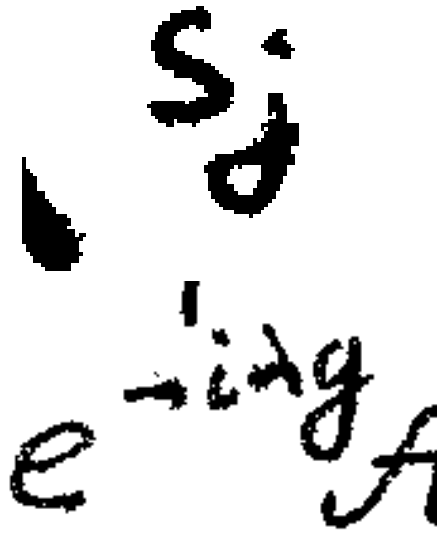


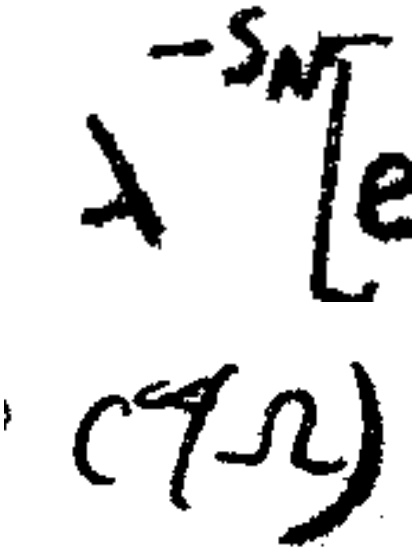
**Diagram and typed description.** The exponent on the left is $-i\lambda g$, and inside the first brackets it is $ig\lambda$; on the right are $\lambda^{s_j}$ and $s_j \downarrow -\infty$. The lower summation limit is $0$.



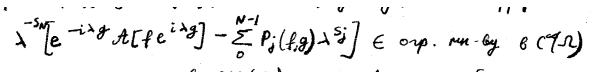
Original source details: GELFAND-P000136

$f \in C_0^\infty(\Omega), \quad g \in C^\infty(\Omega)$, real, grad $g \neq 0$ on supp $f$.

$$\lambda^{-s_N}\left[e^{-i\lambda g}A[fe^{i\lambda g}] - \sum_0^{N-1} P_j(f,g)\lambda^{s_j}\right] \in \text{a bounded set in } C^{q[?]}(\Omega).$$



**Uncertain reading.** The exponent on $C$ has an upper loop and a long descending stroke: $q$ / $\ell$. The exact reading remains unsettled. Before the bracket is $\lambda^{-s_N}$, and the upper summation limit is $N-1$.



Original source details: GELFAND-P000136

$g \in$ a compact subset of $C^\infty(\Omega)$, and $A$ must be a continuous oper. from $C_0^\infty$ to $C^\infty$.

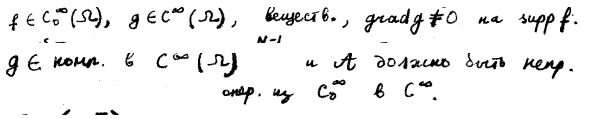


**Diagram and typed description.** $f$ belongs to $C_0^\infty(\Omega)$, and $g$ is real; the text has grad $g \neq 0$ on supp $f$. Below are "$g \in$ a compact subset of $C^\infty(\Omega)$" and the continuity condition on $A$.



Original source details: GELFAND-P000136

$\Omega \subset R^n, \quad g = \langle x, \xi \rangle$ —fund. Consider analogous diff. oper.

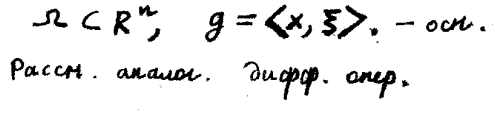
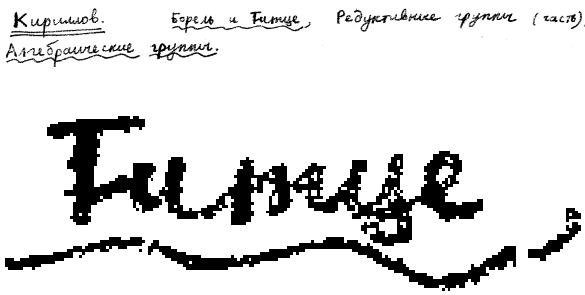


**Diagram and typed description.** The final lines have $\Omega \subset R^n$, the linear phase $g = \langle x, \xi \rangle$, and "Consider analogous diff. oper.".

Original source details: GELFAND-P000136

GELFAND-P000137

Kirillov. Borel and Titse[?]. Reductive groups (part). Algebraic groups.



**Uncertain reading.** The second surname has dense final letters: Titse / Tits. The exact reading remains unsettled.

Original source details: GELFAND-P000137

Algebr. variety—a locally aff. scheme. From the ring $A$, construct a ringed space; Spec $A$—the prime ideals of $A$. $A = C[x_1, x_2]$. A prime ideal is generated either by $\underbrace{((x_1 - a), (x_2 - b))}_{\text{(point)}}$, or by an irreduc. polynomial $I = (P(x_1, x_2))$. It is equipped with the corresponding ringed-space structure.

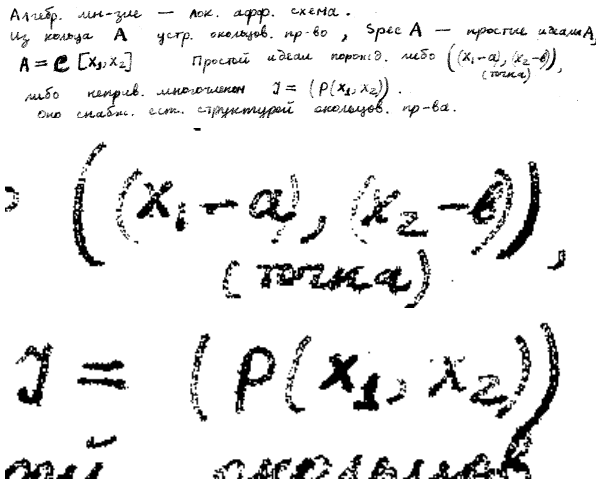


**Diagram and typed description.** The point example has outer parentheses and separate parentheses around $x_1 - a$ and $x_2 - b$, with "(point)" below. The second example has $I = (P(x_1, x_2))$.

Original source details: GELFAND-P000137

An algebr. group—an algebr. variety, with multipl. $G \times G \to G$, given everywhere by a regular (rat.) function. Chevalley's struct. theorem: in every alg. group $G$ there is a max. linear group $G_0$ such that $G/G_0$ is an abelian variety.

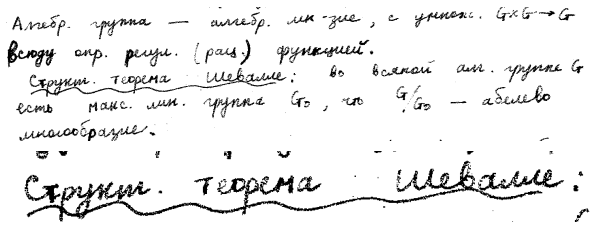
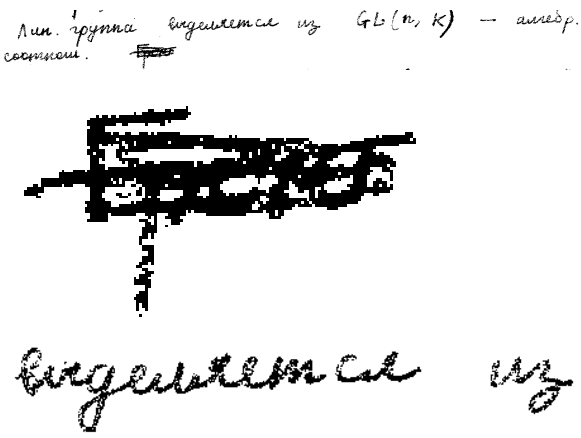


**Diagram and typed description.** The text has $G_0$ and $G/G_0$; the structural-theorem heading has a wavy underline.

Original source details: GELFAND-P000137

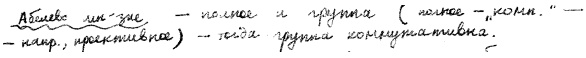
Linear groups are singled out from $GL(n, K)$—algebr. relations. [Crossed out: unread word].



Original source details: GELFAND-P000137

**Uncertain reading.** After the line about algebr. relations, a separate word is heavily crossed out; the obscured letters remain unread.

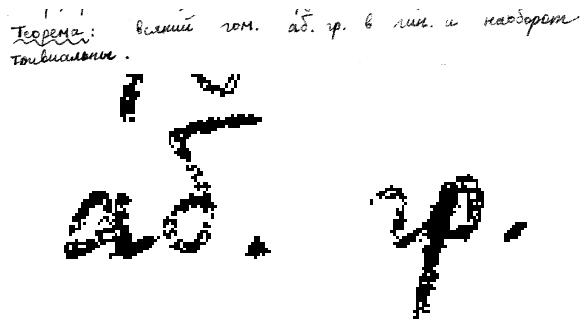
Abelian variety —complete and a group (complete—“comp.”—e.g. projective)—then the group is commutative.



Original source details: GELFAND-P000137

**Diagram and typed description.** “Complete” is explained in parentheses by “comp.” and the projective example; the phrase about commutativity follows.

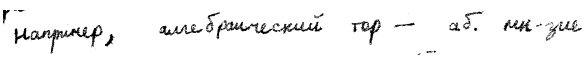
Theorem: every hom. from ab. f.[?] into linear, and conversely, is trivial.



Original source details: GELFAND-P000137

**Uncertain reading.** After “ab.” is a rounded sign with a descending stroke: f. / gr. The exact reading remains unsettled.

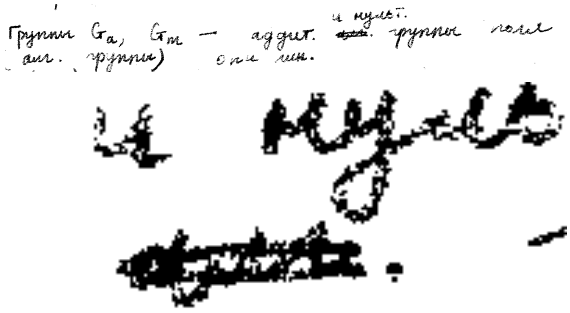
For example, an algebraic torus—an ab. variety.



Original source details: GELFAND-P000137

**Diagram and typed description.** The text reads “algebraic torus—an ab. variety”.

The groups $G_a$, $G_m$—additive [Crossed out: unread short word] [Inserted above: and multi.] groups of the field (alg. groups). They are linear.



**Uncertain reading.** “And multi.” is inserted above the crossed-out short word. The obscured word remains unread.

Original source details: GELFAND-P000137

$$\begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \qquad (x).$$

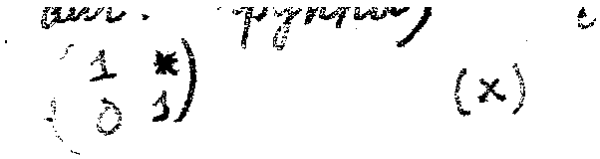


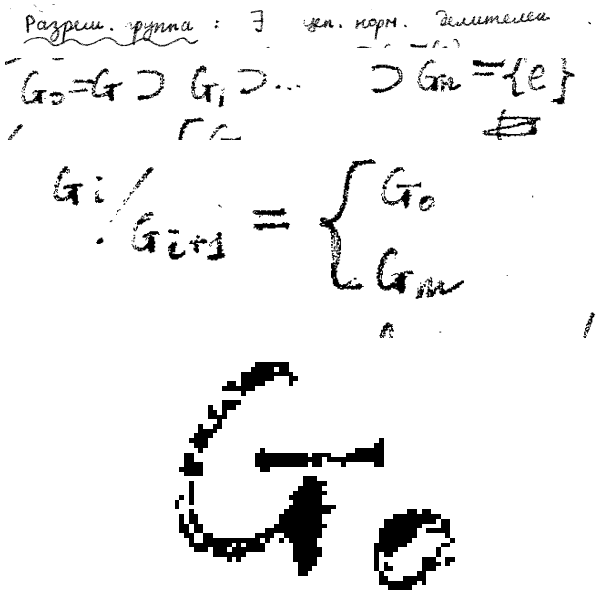
**Diagram and typed description.** On the left is a matrix with $1, *, 0, 1$; on the right, separately, is $(x)$.

Original source details: GELFAND-P000137

Solvable groups: $\exists$ chain of normal subgroups

$$G_0 = G \supset G_1 \supset \ldots \supset G_m = \{e\}.$$

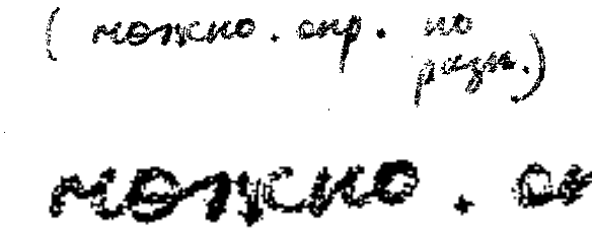
Such that $G_i/G_{i+1} = \begin{cases} G_{a[?]}, \\ G_m. \end{cases}$



**Uncertain reading.** In the first alternative, the subscript is $a$- / $0$-like; the exact reading remains unsettled. The chain has $G_0 = G$, then $G_1$, and ends with $G_m = \{e\}$.

Original source details: GELFAND-P000137

$\Longleftrightarrow$ (imprec.[?] def. but reason.)



**Uncertain reading.** The first abbreviation resembles “netchno.” / “netochno.”; its exact letters remain unsettled. The following words are “def. but reason.”.

Original source details: GELFAND-P000137

Radical—max connected (Zariski top.) [Inserted above: solvable] normal subgroup.

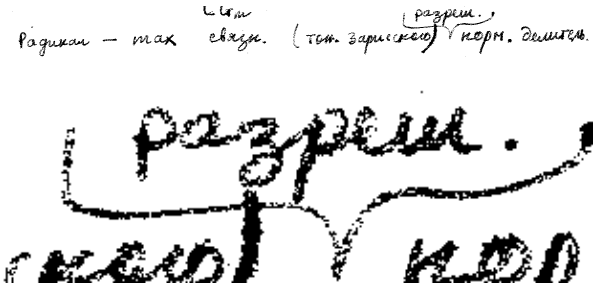
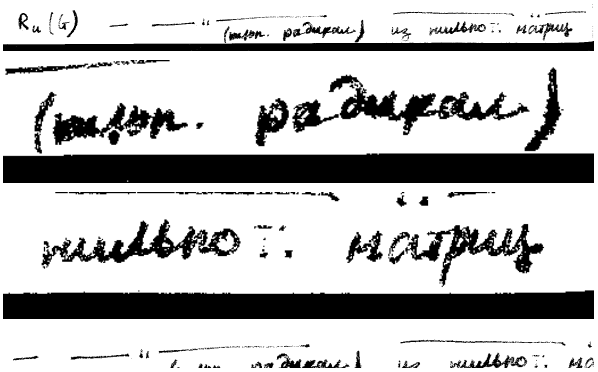


**Diagram and typed description.** “Solvable” is inserted above “normal”, with an arc below and an angular insertion mark.

solvable

normal

Original source details: GELFAND-P000137

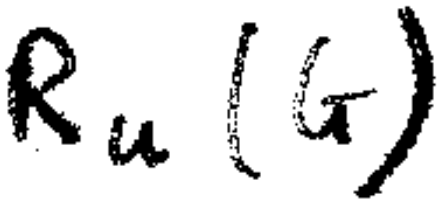


**Uncertain reading.** At the lower edge is $R_u(G)$, with two groups of long strokes and ditto marks; below the first is “(unip.[?] radical)”, with part of the first word obscured. On the right is “from nilpo:: [?] matrices”; the abbreviation and following marks remain unsettled.

$R_u(G)$ — —— " —— —— " —

(unip.[?] radical)

from nilpo:: [?] matrices



Original source details: GELFAND-P000137

GELFAND-P000138

(Nilpot. $(u-E)^k=0$.) Reductive group—its radical is a torus. (Direct product of several mult. groups of the field.)

$$G \text{ semisimple} \iff R(G)=1.$$

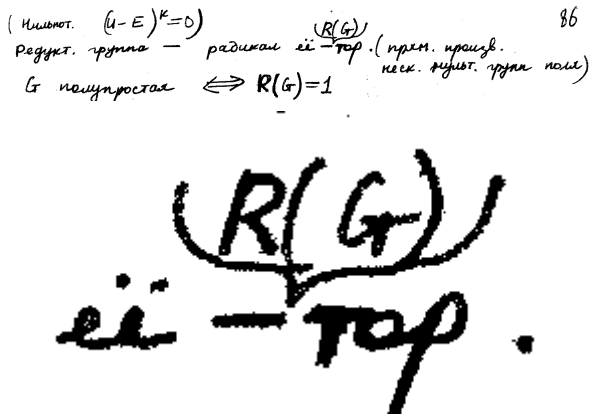



**Diagram and typed description.** $R(G)$ is written above “torus”, with an arc below and an angular stroke.

$R(G)$

torus

Original source details: GELFAND-P000138

$GL(n,K)$—reductive, $SGL(n,K)$—semisimple. Classif. of semisimple groups: $4$ infinite series and $5$ exceptional (as for Lie groups). Basic notions—max torus and a root system.

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

real matrices—isom. to the group of complex numbers under mult.—a torus. All max tori are conjugate. (A root system is associated with a torus.)

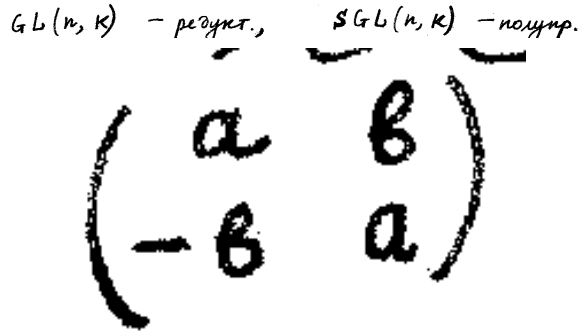
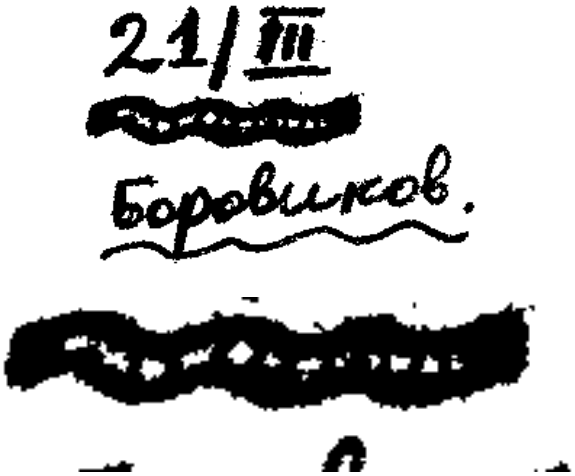



**Diagram and typed description.** The text has $SGL(n,K)$. The real matrix has $a, b$ above and $-b, a$ below.

Original source details: GELFAND-P000138

21/III.




**Uncertain reading.** Below the date is a dense wavy band: heavy underlining / a cancelled line. There are too few distinguishable letters.

Original source details: GELFAND-P000138

Borovikov.

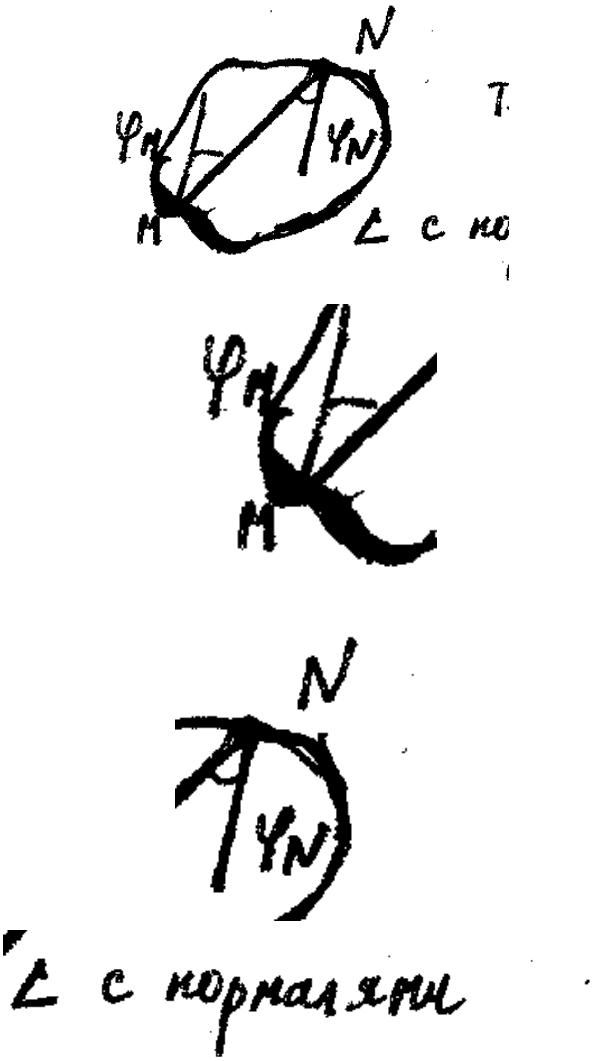
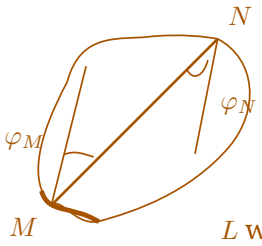



**Diagram and typed description.** A rounded contour, with $M$ below left and $N$ above right; a chord, interior strokes and angles $\varphi_M, \varphi_N$. The label is "$L$ with normals". The dense lower part and exact joins remain partly unread.




$L$ with normals

Original source details: GELFAND-P000138

Theorem.

$$\iint \cos\varphi_M \cdot \cos\varphi_N \, dM \, dN = 2\pi S.$$

($S$—area.)

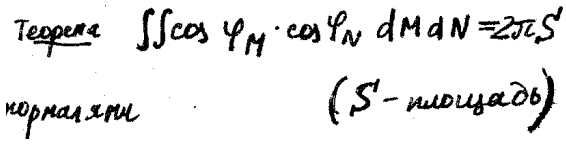



**Diagram and typed description.** The visible formula has $\cos\varphi_M \cdot \cos\varphi_N \, dM \, dN = 2\pi S$; no distance denominator is visible.

Original source details: GELFAND-P000138

Gelfand.

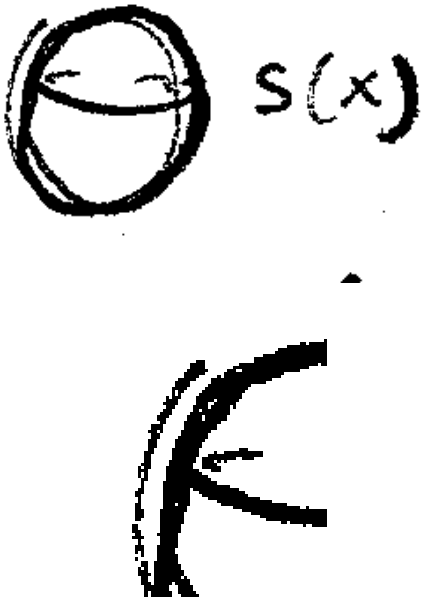
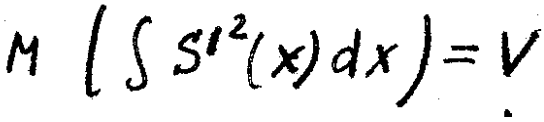



**Diagram and typed description.** A rounded contour with a repeated left stroke, an interior section arc and the label $S(x)$ on the right. Short upper arcs are visible at the ends; the dense left join remains partly unread.

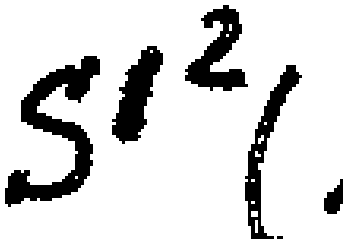



Original source details: GELFAND-P000138

$$M\left(\int S'^2(x)[?]\,dx\right) = V.$$




**Uncertain reading.** After $S$ is a tall vertical stroke, then a separate raised 2: $S'^2$ / $S^2$ with an extra stroke. The exact reading remains unsettled.




Original source details: GELFAND-P000138

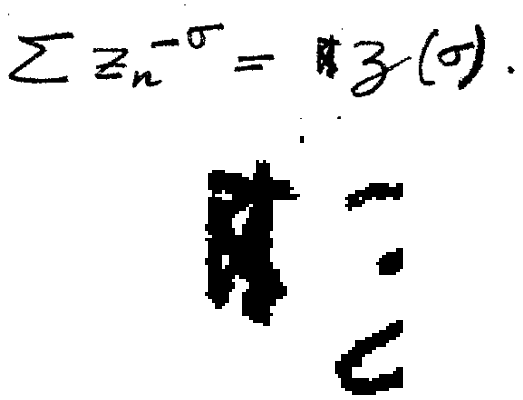
$M$—a universal constant (of the type of $\pi$).

Lidskii. (Viktor Borisovich.)

$$f(z) = \sum_{k=1}^{N} e^{\alpha_k z} \mathcal{P}_k(z).$$

$\mathcal{P}_k(z)$—polyn. and asympt. series in $\frac{1}{z}$.

$$\sum z_n^{-\sigma} = \text{[Crossed out: unread sign]}\ \zeta(\sigma).$$



**Uncertain reading.** Before $\zeta(\sigma)$ is a heavily crossed-out vertical sign; its former reading remains unsettled.

Original source details: GELFAND-P000138

$\zeta(\sigma)$ is always entire in $\sigma$. $z_n$ are the zeros of $f(z)$.

$$\mathcal{P}_k(z) \simeq p_k(z) + \sum_{s=0}^{\infty} \frac{a_s}{z^s}.$$

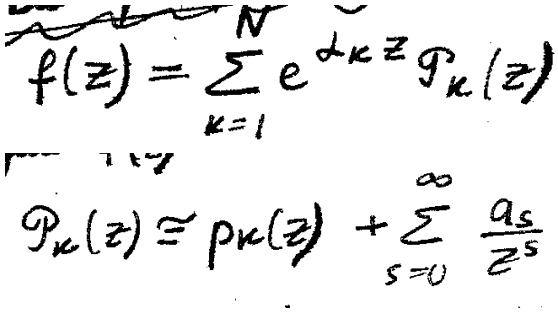
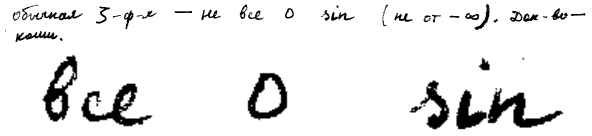


**Diagram and typed description.** The first sum runs from $k = 1$ to $N$, with exponent $\alpha_k z$ and factor $\mathcal{P}_k(z)$. The lower sum runs from $s = 0$ to $\infty$.

Original source details: GELFAND-P000138

The usual $\zeta$-function—not all $0$[?] sin (not from $-\infty$). Proof—
— $\int$ Cauchy.

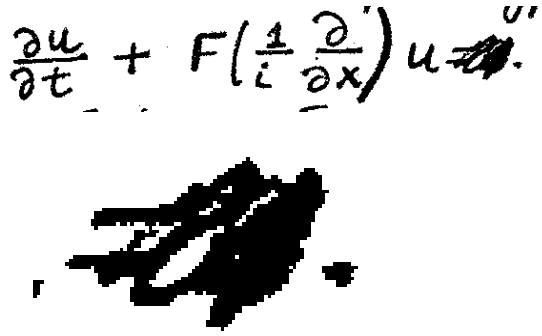


**Uncertain reading.** Between “all” and sin is a rounded sign: $0$ / $O$. The contracted wording remains unsettled. In the proof note, a dash occurs both at the end of the line and before $\int$ Cauchy.

Original source details: GELFAND-P000138

Gindikin. Generalization of a parabolic eq.

$$\frac{\partial u}{\partial t} + F\left(\frac{1}{i}\frac{\partial}{\partial x}\right) u$$ [Crossed out: unread ending]



**Uncertain reading.** After $u$ is a heavily crossed-out $= 0$-like ending; the exact reading remains unsettled.

Original source details: GELFAND-P000138

In place of $\frac{\partial u}{\partial t}$—any pseudo-operator $P\left(\frac{\partial}{\partial t}\right)$. A family of oper. is associated with $P$, such that with $\frac{\partial u}{\partial t}$ one obtains parabolic.

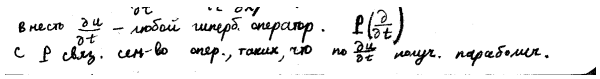


**Diagram and typed description.** The text has $P\left(\frac{\partial}{\partial t}\right)$; the final line associates $P$ with a family of operators and repeats $\frac{\partial u}{\partial t}$.

Original source details: GELFAND-P000138

GELFAND-P000139

Instead of ellipt. — $V$-ellipt., map. in a cone, corresp. to hyperb. polyn.

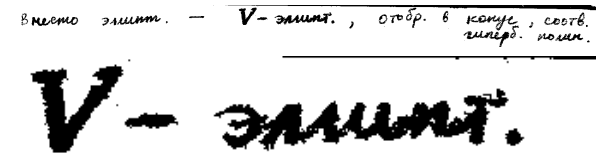


**Diagram and typed description.** Opening line: $V$-ellipt., map. in a cone, corresp. to hyperb. polyn.

Original source details: GELFAND-P000139

Walk along a line — heat eqn.

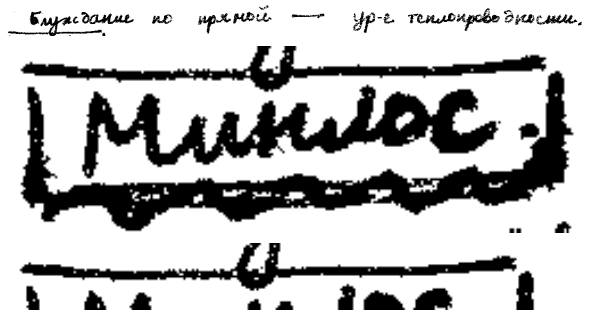


**Diagram and typed description.** The word “Walk” is underlined; this stroke forms the top of the name’s frame. The bottom of the frame is reinforced.

Minlos.

Original source details: GELFAND-P000139

Ising model.

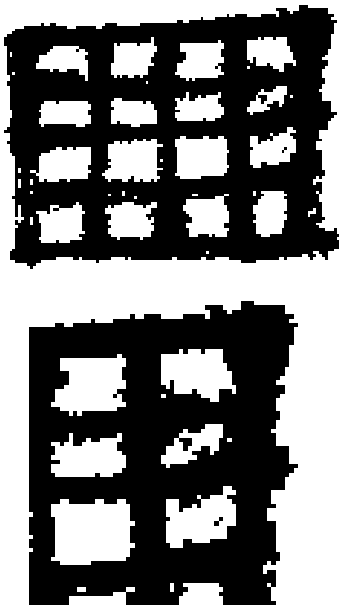
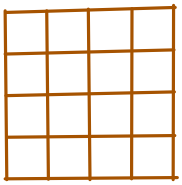
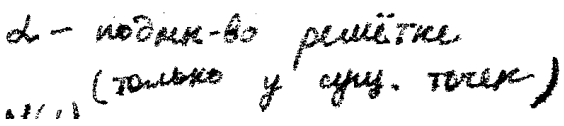


**Diagram and typed description.** Lattice: four cells in each direction. The intersections are dense; individual filled cells or state markings have not been established.



Original source details: GELFAND-P000139

$L$ — subset of the lattice. (only at exist. [?] points)



**Uncertain reading.** Middle abbreviation: “exist.” / “neighb.”; the dense joins do not permit a definite expansion.

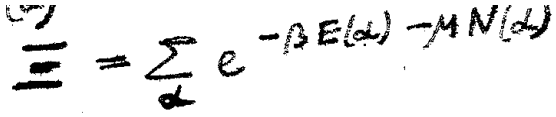


Original source details: GELFAND-P000139

$E(L)$ — number of neighbors.

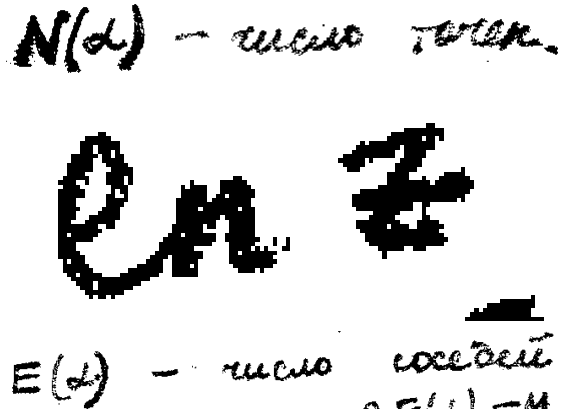
$$\Xi[?] = \sum_L e^{-\beta E(L) - \mu N(L)}.$$

$N(L)$ — number of points.



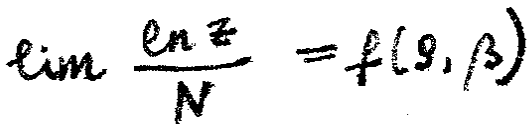
**Uncertain reading.** Three separate horizontal bands before the sum: $\Xi$ / stylized $Z$. The next formula has a $Z$-like sign.





Original source details: GELFAND-P000139

$$\lim \frac{\ln Z}{N} = f(\theta[?], \beta).$$

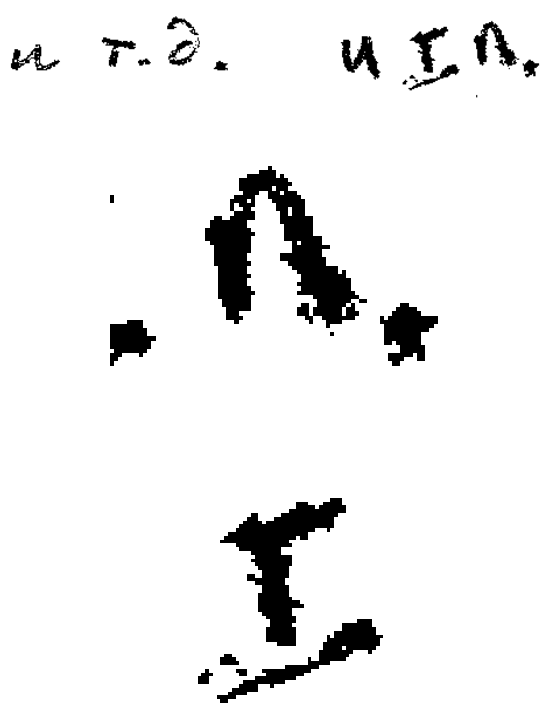


**Uncertain reading.** The first argument of $f$ is a loop with a lower stroke: $\theta$ / $\vartheta$ / $\rho$; the exact reading is unsettled.



Original source details: GELFAND-P000139

And so on. And so forth [?]



Original source details: GELFAND-P000139

**Uncertain reading.** Final short entry: “and so forth” / “and $T$, $\lambda$” / “and $T$, $\rho$”. After “and” is an underlined t-/T-like sign; the final sign has an arch and two legs, Cyrillic п-/П-/П-/n- or $\lambda$-/$\Lambda$-like, with a dot to its right.